

Study Notes

Concavity and Optimization in Economics

Graduate course

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Notes based on the following bibliography:

Aragón, Goberna, López & Rodríguez, *Nonlinear Optimization*

Boyd & Vandenberghe, *Convex Optimization*

Brinkhuis, *Convex Analysis for Optimization*

Li, *Selected Applications of Convex Optimization*

Nocedal & Wright, *Numerical Optimization*

Ruszczynski, *Nonlinear Optimization*

Snyman & Wilke, *Practical Mathematical Optimization*

Wright & Recht, *Optimization for Data Analysis*

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How to use these notes

These are study notes for *Concavity and Optimization in Economics*, written for a graduate course. They follow, week by week, the order in which the course is usually taught, and they are based on the bibliography listed below.

They are written for a specific reader: someone who comes to convex analysis and optimization **from economics**, with mathematical training at a basic to intermediate level. This explains the two style choices that govern the whole text: **nothing is assumed** (every concept is built from scratch, with all the algebraic steps explained) and intuition always comes before formalism, without giving up rigor afterwards.

Remark 0.1 (Preliminary note). This is unofficial study material: it is not the course guide and it does not replace lectures, the syllabus or the textbooks, and it is not endorsed by any institution. The proofs follow the standard developments of the cited bibliography, with the wording and the order of presentation adapted to the goals of these notes. The priority marks explained below are a personal study criterion, not an official classification of contents. Any error belongs to these notes and not to the sources.

Structure of each chapter

- **Goals and place in the course:** which week of the schedule it covers and which book of the bibliography treats it.
- **Motivation:** why the concept exists and what problem it solves.
- **Definitions and theorems:** numbered, with proofs explained *step by step* (each step says what is done and why).
- Colored boxes:
 - **Intuition:** the idea in plain words, before the formalism.
 - **Careful (common mistake):** typical errors that cost points in exams.
 - **Exam strategy:** how the topic is usually asked and how to attack it.
 - **Link with economics:** where the concept appears in economic theory.
 - **What is this used for in Microeconomics?:** the concrete link with the topics of a graduate micro course.
 - **Chapter summary:** the minimum you have to know by heart.
- **Exam-style exercises with complete solutions.**

The priority marks and what they are for

An optimization course covers much more than an economist will use later on a daily basis. Since these notes were also written as a *bridge* towards graduate microeconomics, each chapter and each section carries a **priority mark** [P1]–[P5] that answers a single question: how much is this material used in a typical **Microeconomics** course?

The calibration was made against the canonical syllabus of a graduate course based on MAS-COLELL, WHINSTON and GREEN (MWG), which is organized in five parts. The violet boxes throughout the text refer to them:

1. **Individual decision making** (MWG chs. 1–6): preference, choice and revealed preference; classical demand theory (duality, Slutsky equation, envelope theorem); aggregate demand; production and duality; choice under uncertainty (expected utility, risk aversion).
2. **Game theory** (MWG chs. 7–9): static games and Nash equilibrium in pure and mixed strategies; dynamic games; Bayesian games; repeated games and discounting.
3. **Market equilibrium and market failure** (MWG chs. 10–14): partial equilibrium and surplus; externalities and public goods; market power; adverse selection, signaling and screening; the principal-agent problem and moral hazard.
4. **General equilibrium** (MWG chs. 15–18): pure exchange and production economies, the Edgeworth box; Pareto optimum and the two welfare theorems; existence of Walrasian equilibrium; the core.
5. **Welfare and incentives** (MWG chs. 21–23): social choice and Arrow’s theorem; welfare economics; mechanism design, the revelation principle and implementation.

The marks mean the following:

- **[P1] Essential:** used directly and repeatedly in micro (separation of convex sets, concavity and quasiconcavity, Karush–Kuhn–Tucker conditions, envelope theorems).
- **[P2] Very useful:** language and tools that appear often or that support the [P1] topics.
- **[P3] Useful as training:** it gives maturity and context; it is used indirectly.
- **[P4] Peripheral:** it appears little in microeconomics (although it can be central in computational economics, econometrics or finance).
- **[P5] Marginal for micro:** it is almost never used in that course.

Careful (common mistake)

Two warnings about the marks. First: they are *relative to microeconomics*, not to the optimization course. A [P5] topic is still examinable material of the course and can still appear in the midterm or in the final presentation; the marks help you decide where to go deeper with micro in mind, not what to study for the optimization exams. Second: a specific program may weight things differently. A course with a lot of game theory or with mechanism design will raise the value of extreme points, of the simplex of mixed strategies and of duality; one centered on general equilibrium and demand theory will

lower it. It is a good idea to check the marks against the syllabus you are given.

Remark 0.2 (What these notes do not cover and micro does use). To be honest with the reader, it is worth pointing out the gaps, which come from the syllabus of the optimization course and not from these notes. Three tools that modern microeconomics uses are missing here. First, the **fixed point theorems** of Brouwer and Kakutani, which are only mentioned in passing here: they are the standard way to prove the existence of Nash equilibrium and of Walrasian equilibrium, and they are studied in a real analysis course. Second, **measure theory and probability**, needed to treat uncertainty, Bayesian games and general lottery spaces with rigor. Third, **monotone comparative statics** (lattices, supermodularity, the theorems of Topkis and Milgrom–Shannon), now standard in micro courses and mentioned here only in passing: it is the tool that gives monotonicity results *without* assuming concavity or differentiability. If you aim at micro, you should cover these three fronts separately.

Bibliography and how it is used here

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For the links with economic theory, the background reference is MAS-COLELL, A.; WHINSTON, M.; GREEN, J. *Microeconomic Theory* [9], cited as MWG. The full references of all these works are in the bibliography at the end of these notes.

Suggested study plan

The course is usually graded with a midterm exam, several written problem sets and a final presentation, the last one with a considerable weight. An effective way to use these notes:

1. **Before each class:** read the motivation and the definitions of the corresponding chapter (30–45 min). Arriving with the definitions in mind lets you follow the proofs live, which is where you really learn.
2. **After each class:** go over the proofs and try the exercises *without looking* at the solution; compare afterwards.
3. **Memorize the statements with surgical precision**, together with their exact hypotheses (what each version of the separation theorems asks for, which qualification the Karush–Kuhn–Tucker theorem requires). In this kind of course, citing a theorem without its hypotheses invalidates the argument.
4. **Before the exam and the presentation:** use the appendix with the study guide and glossary, which condenses the theorems with their hypotheses, the classic mistakes and topic proposals for the presentation.

Careful (common mistake)

The most expensive mistake in this course is confusing *necessary* conditions with *sufficient* ones, and applying the Karush–Kuhn–Tucker conditions without checking the constraint qualification. The corresponding chapter insists on this with an explicit counterexample; you should not skip it.

Chapter 1

Mathematical preliminaries: vector spaces, topology and calculus [P2]

Reading before Week 1. Bibliography: Brinkhuis [3]; Wright and Recht [8]; Aragón et al. [1].

What is this used for in Microeconomics?

This chapter builds the language on which *all* of the Microeconomics course rests. In the *general equilibrium* part (Edgeworth box, Pareto optima, first and second welfare theorems, the core; MWG chs. 15–18) consumption bundles are vectors in $\mathbb{R}_+^n$, the *continuity of preferences* is a topological condition, and the *existence* of consumer demand and of Walrasian equilibrium is proved with the Weierstrass theorem on compact sets (the budget set is compact when prices are strictly positive). In the *market failure* part (contracts: moral hazard in the style of Holmström, adverse selection of Baron–Myerson, signaling of Spence; MWG chs. 13–14) the principal’s problems are solved with first and second order conditions: the *gradient* and the *Hessian* that we study here. In *individual decision making* under uncertainty (expected utility, risk aversion; MWG ch. 6) the concavity of the Bernoulli utility, which is checked with the definiteness of the Hessian, *is* risk aversion, and envelope theorems of the Milgrom–Segal type [11] require handling derivatives and Taylor expansions with ease. In short: without this chapter you cannot read [9] or [10] with rigor. Supporting references: [1] chs. 1–2, [3] appendices, [7] ch. 1.

1.1 Real vector spaces [P2]

1.1.1 Motivation: consumption bundles as vectors

A consumer chooses quantities of n goods: x_1 kilos of rice, x_2 liters of milk, $\dots$, x_n hours of internet. The choice is described by an ordered list of n real numbers,

$$x = (x_1, x_2, \dots, x_n),$$

called a *consumption bundle*. The set of all possible bundles with nonnegative quantities is

$$\mathbb{R}_+^n = \{x \in \mathbb{R}^n : x_i \geq 0 \text{ for all } i = 1, \dots, n\},$$

the *nonnegative orthant*. With bundles we want to perform two natural operations: *add them* (put two shopping baskets together) and *scale them* (buy twice as much of everything). A *vector space* is exactly a set where these two operations behave well.

Definition 1.1 (Real vector space). A **real vector space** is a set V (whose elements we call *vectors*) equipped with two operations:

- **addition:** to each pair $u, v \in V$ it assigns a vector $u + v \in V$;
- **scalar multiplication:** to each $\lambda \in \mathbb{R}$ and $v \in V$ it assigns a vector $\lambda v \in V$;

which satisfy, for all $u, v, w \in V$ and $\lambda, \mu \in \mathbb{R}$:

- (A1) $u + v = v + u$ (*commutativity*: the order of the sum does not matter)
- (A2) $(u + v) + w = u + (v + w)$ (*associativity*: the grouping does not matter)
- (A3) there is $0 \in V$ with $v + 0 = v$ (*zero vector*: adding “nothing” changes nothing)
- (A4) for each v there is $-v$ with $v + (-v) = 0$ (*opposite*: every sum can be undone)
- (A5) $\lambda(u + v) = \lambda u + \lambda v$ (scaling a sum = adding the scaled vectors)
- (A6) $(\lambda + \mu)v = \lambda v + \mu v$ (distributivity in the scalar)
- (A7) $\lambda(\mu v) = (\lambda\mu)v$ (scaling twice = scaling by the product)
- (A8) $1 \cdot v = v$ (scaling by 1 does nothing)

The eight axioms are exactly the rules of “arrow algebra” that one uses without thinking; the definition only makes them explicit so that we can work abstractly.

Example 1.2 (The central space of the course: $\mathbb{R}^n$). The set $\mathbb{R}^n = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}\}$ with the *componentwise* operations

$$x + y = (x_1 + y_1, \dots, x_n + y_n), \quad \lambda x = (\lambda x_1, \dots, \lambda x_n),$$

is a real vector space. Let us check (A5) numerically in $\mathbb{R}^2$ with $x = (1, 2)$, $y = (3, -1)$, $\lambda = 2$:

$$\begin{aligned} \lambda(x + y) &= 2((1, 2) + (3, -1)) = 2(1 + 3, 2 + (-1)) = 2(4, 1) = (8, 2), \\ \lambda x + \lambda y &= 2(1, 2) + 2(3, -1) = (2, 4) + (6, -2) = (2 + 6, 4 - 2) = (8, 2). \end{aligned}$$

They agree, as they should. The zero vector is $0 = (0, \dots, 0)$ and the opposite of x is $-x = (-x_1, \dots, -x_n)$.

Careful (common mistake)

Three frequent mistakes in the first exams:

1. $\mathbb{R}_+^n$ is **not** a vector space. If $x = (1, 2) \in \mathbb{R}_+^2$ and $\lambda = -1$, then $\lambda x = (-1, -2) \notin \mathbb{R}_+^2$.

$\mathbb{R}_+^2$: closure under scalar multiplication fails. $\mathbb{R}_+^n$ is a *convex cone* (we will see this in the chapter on convex sets), not a subspace.

2. **Do not confuse the scalar** $0 \in \mathbb{R}$ **with the vector** $0 \in V$. They are different objects even though they are written the same way.

3. **Componentwise “multiplication of vectors” is not part of the structure.**

In a vector space there is only addition and *scalar* multiplication; the inner product (Section 1.2) returns a *number*, not a vector.

1.1.2 Subspaces, linear combinations and bases

Definition 1.3 (Subspace). A nonempty subset $S \subseteq V$ is a **linear subspace** if it is closed under the two operations: for all $u, v \in S$ and $\lambda \in \mathbb{R}$ we have $u + v \in S$ and $\lambda v \in S$. Equivalently: $\lambda u + \mu v \in S$ for all $u, v \in S$, $\lambda, \mu \in \mathbb{R}$.

Every subspace contains the vector 0 (take $\lambda = 0$). In $\mathbb{R}^3$ the subspaces are exactly: $\{0\}$, the lines through the origin, the planes through the origin, and all of $\mathbb{R}^3$. A line that does *not* pass through the origin is not a subspace (it does not contain 0); for example, the budget line $\{x \in \mathbb{R}^2 : p_1x_1 + p_2x_2 = m\}$ with $m > 0$ is not a subspace.

Definition 1.4 (Linear combination, independence, basis). Let $v_1, \dots, v_k \in V$.

1. A **linear combination** is a vector of the form $\lambda_1v_1 + \dots + \lambda_kv_k$ with $\lambda_i \in \mathbb{R}$.
2. The vectors are **linearly independent** if the *only* way to obtain the zero vector as a linear combination is with all coefficients equal to zero:

$$\lambda_1v_1 + \dots + \lambda_kv_k = 0 \implies \lambda_1 = \dots = \lambda_k = 0.$$

Otherwise they are *linearly dependent*: one of them is a linear combination of the others (it is “redundant”).

3. A **basis** of V is a family of linearly independent vectors such that *every* vector of V is a linear combination of them. If V has a basis with n elements, every basis has n elements, and we say that the **dimension** of V is n .

Example 1.5 (Standard basis and a complete computation in $\mathbb{R}^2$). The *standard basis* of $\mathbb{R}^n$ is $e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, $\dots$, $e_n = (0, \dots, 0, 1)$; every $x = (x_1, \dots, x_n)$ is written as $x = x_1e_1 + \dots + x_ne_n$, so $\dim \mathbb{R}^n = n$.

Now a computation with all the steps. Are $v_1 = (1, 2)$ and $v_2 = (3, 4)$ linearly independent? We set up $\lambda_1v_1 + \lambda_2v_2 = (0, 0)$, that is,

$$\lambda_1(1, 2) + \lambda_2(3, 4) = (\lambda_1 + 3\lambda_2, 2\lambda_1 + 4\lambda_2) = (0, 0) \iff \begin{cases} \lambda_1 + 3\lambda_2 = 0, \\ 2\lambda_1 + 4\lambda_2 = 0. \end{cases}$$

From the first equation, $\lambda_1 = -3\lambda_2$. Substituting into the second: $2(-3\lambda_2) + 4\lambda_2 = -6\lambda_2 + 4\lambda_2 = -2\lambda_2 = 0$, so $\lambda_2 = 0$ and then $\lambda_1 = -3 \cdot 0 = 0$. The only solution is the trivial one: *they are*

independent, and since they are 2 independent vectors in a space of dimension 2, they form a basis of $\mathbb{R}^2$.

By contrast, $w_1 = (1, 2)$ and $w_2 = (2, 4)$ are dependent: $w_2 = 2w_1$, that is, $2w_1 + (-1)w_2 = 0$ with coefficients that are not all zero.

Intuition

Think of a basis as a *coordinate system*: a minimal set of “reference directions” with which you can describe any vector in a unique way. The dimension counts how many “degrees of freedom” the space has. In economics: if there are n goods, the space of bundles has n degrees of freedom, and each e_i is the bundle “one unit of good i and nothing else”. That a set of vectors is dependent means that some direction is redundant: it could already be made from the others.

1.2 Inner product and norm [P1]

To do *analysis* (limits, continuity, optimization) we need to measure lengths, distances and angles. In $\mathbb{R}^n$ everything comes from a single operation: the inner product.

Definition 1.6 (Euclidean inner product). The **inner product** or **dot product** of $x, y \in \mathbb{R}^n$ is the real number

$$\langle x, y \rangle = x_1y_1 + x_2y_2 + \cdots + x_ny_n = \sum_{i=1}^n x_iy_i.$$

Immediate properties (all of them are checked by expanding the sum): for all $x, y, z \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$,

(P1) $\langle x, y \rangle = \langle y, x \rangle$ (symmetry);

(P2) $\langle \lambda x + z, y \rangle = \lambda \langle x, y \rangle + \langle z, y \rangle$ (linearity in each argument);

(P3) $\langle x, x \rangle = x_1^2 + \cdots + x_n^2 \geq 0$, with equality if and only if $x = 0$ (positivity).

Example 1.7 (Expenditure as an inner product). If $p = (p_1, \dots, p_n) \in \mathbb{R}_{++}^n$ is the price vector and $x \in \mathbb{R}_+^n$ a bundle, the consumer’s *expenditure* is exactly $\langle p, x \rangle = p_1x_1 + \cdots + p_nx_n$, and the budget constraint is $\langle p, x \rangle \leq m$. Almost all the linear expressions of microeconomics (expenditure, revenue, profit $\langle p, y \rangle$ of a production plan y) are inner products. With $p = (2, 5)$ and $x = (3, 1)$: $\langle p, x \rangle = 2 \cdot 3 + 5 \cdot 1 = 6 + 5 = 11$.

Definition 1.8 (Euclidean norm and distance). The **Euclidean norm** of $x \in \mathbb{R}^n$ is

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x_1^2 + \cdots + x_n^2},$$

and the **distance** between x and y is $d(x, y) = \|x - y\|$.

For $n = 2$ this is the Pythagorean theorem: the length of the arrow from $(0, 0)$ to $(3, 4)$ is $\|(3, 4)\| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

Definition 1.9 (Norm in general). A function $N : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **norm** if for all $x, y \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$:

- (N1) $N(x) \geq 0$, and $N(x) = 0$ if and only if $x = 0$ (positivity);
- (N2) $N(\lambda x) = |\lambda| N(x)$ (absolute homogeneity);
- (N3) $N(x + y) \leq N(x) + N(y)$ (triangle inequality).

The key to proving that the Euclidean norm satisfies (N3) is the most important inequality of analysis in $\mathbb{R}^n$:

Theorem 1.10 (Cauchy–Schwarz inequality). *For all $x, y \in \mathbb{R}^n$,*

$$|\langle x, y \rangle| \leq \|x\| \|y\|,$$

with equality if and only if x and y are linearly dependent (one is a multiple of the other).

Proof. If $y = 0$ both sides are 0 and there is nothing to prove. Assume $y \neq 0$. The idea: for any $t \in \mathbb{R}$, the vector $x - ty$ has nonnegative squared norm. We expand using bilinearity and symmetry of the inner product:

$$\begin{aligned} 0 \leq \|x - ty\|^2 &= \langle x - ty, x - ty \rangle \\ &= \langle x, x \rangle - t \langle y, x \rangle - t \langle x, y \rangle + t^2 \langle y, y \rangle \\ &= \|x\|^2 - 2t \langle x, y \rangle + t^2 \|y\|^2. \end{aligned}$$

This is a quadratic polynomial in t that is never negative. We choose the t that minimizes it. Differentiating with respect to t and setting the result to zero: $-2 \langle x, y \rangle + 2t \|y\|^2 = 0$, that is, $t^* = \langle x, y \rangle / \|y\|^2$ (well defined because $y \neq 0$). Substituting t^* :

$$0 \leq \|x\|^2 - 2 \frac{\langle x, y \rangle}{\|y\|^2} \langle x, y \rangle + \frac{\langle x, y \rangle^2}{\|y\|^4} \|y\|^2 = \|x\|^2 - \frac{\langle x, y \rangle^2}{\|y\|^2}.$$

Multiplying by $\|y\|^2 > 0$ and rearranging: $\langle x, y \rangle^2 \leq \|x\|^2 \|y\|^2$; taking the square root (both sides are nonnegative) gives $|\langle x, y \rangle| \leq \|x\| \|y\|$.

Equality: it happens exactly when $\|x - t^*y\|^2 = 0$, that is, when $x = t^*y$, so when x is a multiple of y . $\square$

Corollary 1.11 (Triangle inequality). $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in \mathbb{R}^n$.

Proof. We expand the square and apply Cauchy–Schwarz in the middle step:

$$\|x + y\|^2 = \|x\|^2 + 2 \langle x, y \rangle + \|y\|^2 \leq \|x\|^2 + 2 \|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2,$$

and we take the square root. $\square$

Definition 1.12 (Orthogonality and angle). Two vectors x, y are **orthogonal**, written $x \perp y$, if $\langle x, y \rangle = 0$. For $x, y \neq 0$, the **angle** $\theta \in [0, \pi]$ between them is defined by

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|}.$$

(The definition makes sense because, by Cauchy–Schwarz, that quotient lies in $[-1, 1]$.)

Example 1.13 (Complete computation). Let $x = (1, 2)$ and $y = (3, 1)$. Then

$$\langle x, y \rangle = 1 \cdot 3 + 2 \cdot 1 = 5, \quad \|x\| = \sqrt{1+4} = \sqrt{5}, \quad \|y\| = \sqrt{9+1} = \sqrt{10},$$

$$\cos \theta = \frac{5}{\sqrt{5}\sqrt{10}} = \frac{5}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}} \implies \theta = 45^\circ.$$

And $d(x, y) = \|x - y\| = \|(1-3, 2-1)\| = \|(-2, 1)\| = \sqrt{4+1} = \sqrt{5}$. As a check of Cauchy-Schwarz: $|\langle x, y \rangle| = 5 \leq \sqrt{5} \cdot \sqrt{10} = \sqrt{50} \approx 7.07$. It does hold.

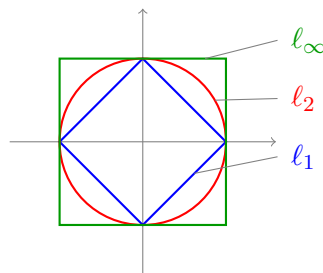
1.2.1 Other norms: ℓ_1 and ℓ_∞

The Euclidean norm is not the only useful one. We define

$$\|x\|_1 = |x_1| + \cdots + |x_n| \quad (\ell_1 \text{ or “taxicab” norm}),$$

$$\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\} \quad (\ell_\infty \text{ or “max” norm}).$$

Both satisfy (N1)–(N3); for example, the triangle inequality for $\|\cdot\|_1$ follows from $|x_i + y_i| \leq |x_i| + |y_i|$ by summing over i . For $x = (3, -4)$: $\|x\|_1 = 3 + 4 = 7$, $\|x\| = 5$, $\|x\|_\infty = 4$. The *unit ball* $\{x : N(x) \leq 1\}$ has a different shape depending on the norm:



The ℓ_1 ball is a diamond, the ℓ_2 ball a circle and the ℓ_∞ ball a square; it is always true that $\|x\|_\infty \leq \|x\| \leq \|x\|_1$. One important fact that we will use without proving it: *in $\mathbb{R}^n$ all norms are equivalent* (they generate the same notions of convergence and of open set), so in topology we can use whichever is most convenient [1, ch. 1].

Intuition

$\|x\|_1$ measures the distance “driving on a grid of streets” (only horizontal and vertical moves); $\|x\|$ measures the distance “in a straight line”; $\|x\|_\infty$ measures only the worst coordinate. In modern optimization the choice of norm matters a lot (ℓ_1 regularization produces solutions with many zeros, the basis of the LASSO method in econometrics; see [8]), but for the *topology* of $\mathbb{R}^n$ the three tell the same story.

1.3 Topology of $\mathbb{R}^n$ [P1]

Topology answers questions of the type: what does “being close” mean?, which sets “contain their edge”?, when does a maximization problem *have* a solution? This last question is the reason for this section: before computing consumer demand you have to know that it *exists*.

1.3.1 Balls, open sets and closed sets

Definition 1.14 (Balls). Let $x_0 \in \mathbb{R}^n$ and $r > 0$. The **open ball** with center x_0 and radius r is

$$B(x_0, r) = \{x \in \mathbb{R}^n : \|x - x_0\| < r\},$$

and the **closed ball** is $\bar{B}(x_0, r) = \{x \in \mathbb{R}^n : \|x - x_0\| \leq r\}$. The difference is only the edge: the open ball does not include the sphere $\|x - x_0\| = r$, the closed one does.

Definition 1.15 (Open and closed set). A set $A \subseteq \mathbb{R}^n$ is **open** if each of its points has an open ball around it entirely contained in A : for every $x \in A$ there is $r > 0$ such that $B(x, r) \subseteq A$. A set C is **closed** if its complement $\mathbb{R}^n \setminus C$ is open.

Example 1.16 (Examples and non examples in $\mathbb{R}$). • $(0, 1)$ is open: if $x \in (0, 1)$, taking $r = \min\{x, 1 - x\} > 0$ we get $(x - r, x + r) \subseteq (0, 1)$.

- $[0, 1]$ is closed: its complement $(-\infty, 0) \cup (1, \infty)$ is a union of open sets, hence open.
- $[0, 1)$ is *neither open nor closed*: it is not open because no ball centered at 0 fits inside (it always sticks out into the negatives); it is not closed because its complement contains 1 but no ball centered at 1 avoids $[0, 1)$.
- $\mathbb{R}^n$ and $\emptyset$ are *both* open and closed. “Open” and “closed” are **not** opposites: a set can be both or neither.

In $\mathbb{R}^2$: the budget set $B(p, m) = \{x \in \mathbb{R}_+^2 : \langle p, x \rangle \leq m\}$ is *closed* (it is defined by weak inequalities with continuous functions), and the set $\{x : \langle p, x \rangle < m\}$ is open.

Definition 1.17 (Interior, closure, boundary, bounded). Let $S \subseteq \mathbb{R}^n$.

- The **interior** $\text{int } S$ is the largest open set contained in S : the points of S that have a ball around them inside S .
- The **closure** $\text{cl } S$ is the smallest closed set that contains S : S together with all its “limit points”.
- The **boundary** is $\text{bd } S = \text{cl } S \setminus \text{int } S$: the points “on the edge”, arbitrarily close to S and to its complement at the same time.
- S is **bounded** if it fits inside some ball: there is $M > 0$ with $\|x\| \leq M$ for every $x \in S$.

For example, with $S = [0, 1) \subset \mathbb{R}$: $\text{int } S = (0, 1)$, $\text{cl } S = [0, 1]$, $\text{bd } S = \{0, 1\}$, and S is bounded (it fits inside $\bar{B}(0, 1)$). With the budget set in $\mathbb{R}^2$ and $p \gg 0$: its boundary includes the expenditure line $\langle p, x \rangle = m$ and the segments of the axes; its interior is $\{x \gg 0 : \langle p, x \rangle < m\}$.

1.3.2 Sequences, convergence and compactness

Definition 1.18 (Convergence of sequences). A **sequence** in $\mathbb{R}^n$ is an infinite list $x^1, x^2, x^3, \dots$ of vectors, written $(x^k)_{k \in \mathbb{N}}$. The sequence **converges** to $x^* \in \mathbb{R}^n$, written $x^k \rightarrow x^*$, if $\|x^k - x^*\| \rightarrow 0$ as a sequence of real numbers; that is, if for every $\varepsilon > 0$ there is $K \in \mathbb{N}$ such that $\|x^k - x^*\| < \varepsilon$ for every $k \geq K$. In words: however small the tolerance ε is, from some point on *all* the terms are at distance less than ε from the limit. In $\mathbb{R}^n$, $x^k \rightarrow x^*$ is equivalent to *each coordinate* converging: $x_i^k \rightarrow x_i^*$ for $i = 1, \dots, n$.

For example, $x^k = (\frac{1}{k}, 2 + \frac{3}{k}) \rightarrow (0, 2)$ because $\frac{1}{k} \rightarrow 0$ and $2 + \frac{3}{k} \rightarrow 2$.

A characterization that is used a lot: *C is closed if and only if it contains the limits of all its convergent sequences* (if $x^k \in C$ for every k and $x^k \rightarrow x^*$, then $x^* \in C$). This is the practical way to check closedness in proofs.

Definition 1.19 (Compact set). A set $K \subseteq \mathbb{R}^n$ is **compact** if every sequence in K has a convergent subsequence whose limit is in K . (A *subsequence* is obtained by keeping infinitely many terms of the original sequence, in order: for example, the even terms.)

In $\mathbb{R}^n$ there is a completely operational characterization:

Theorem 1.20 (Heine–Borel). $K \subseteq \mathbb{R}^n$ is compact if and only if K is closed and bounded.

That is: in $\mathbb{R}^n$, to check compactness it is enough to check two concrete things. (Careful: in infinite dimensional spaces, which appear in macro and in dynamic contracts, this is false; there compact is more demanding than closed and bounded.) The engine behind Heine–Borel is:

Theorem 1.21 (Bolzano–Weierstrass). Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof idea (case $n = 1$): the sequence lives in an interval $[a, b]$. We cut the interval in half; one of the two halves contains infinitely many terms of the sequence; we keep that half and repeat. We get nested intervals of length $(b - a)/2^k \rightarrow 0$ that “trap” a point x^* , and by choosing a term of the sequence in each interval we build a subsequence that converges to x^* . For $n > 1$ we apply the one dimensional case coordinate by coordinate. You can look at [1, ch. 1] or [3], the appendix on analysis.

Example 1.22 (The budget set is compact). Let $p \in \mathbb{R}_{++}^n$ (all prices strictly positive) and $m > 0$. We claim that $B(p, m) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq m\}$ is compact.

- *Closed*: if $x^k \in B(p, m)$ and $x^k \rightarrow x$, then $x_i = \lim x_i^k \geq 0$ (limits preserve weak inequalities) and $\langle p, x \rangle = \lim \langle p, x^k \rangle \leq m$. Hence $x \in B(p, m)$.
- *Bounded*: for each i , since all coordinates are nonnegative, $p_i x_i \leq p_1 x_1 + \dots + p_n x_n \leq m$, from which $0 \leq x_i \leq m/p_i$. So $\|x\| \leq \sqrt{\sum_i (m/p_i)^2} =: M < \infty$.

By Heine–Borel, $B(p, m)$ is compact. *This is the key step for demand to exist* (Theorem 1.24). Note where $p \gg 0$ was used: if some $p_i = 0$, the coordinate x_i is free and the set stops being bounded (the consumer would “demand infinity” of the free good).

1.3.3 Continuity and the Weierstrass theorem

Definition 1.23 (Continuous function). Let $f : D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $x_0 \in D$. Two equivalent definitions of f being **continuous** at x_0 :

1. *Sequential version*: for every sequence $x^k \in D$ with $x^k \rightarrow x_0$ we have $f(x^k) \rightarrow f(x_0)$. In words: “if the inputs get close, the outputs get close”; f commutes with the limit.
2. ε – δ *version*: for every $\varepsilon > 0$ there is $\delta > 0$ such that if $x \in D$ and $\|x - x_0\| < \delta$, then $\|f(x) - f(x_0)\| < \varepsilon$. In words: given any error tolerance ε in the output, there is an input tolerance δ that guarantees it.

f is continuous (without qualification) if it is continuous at every point of D . Sums, products, quotients (with nonzero denominator) and compositions of continuous functions are continuous; polynomials, exponentials, logarithms and Cobb–Douglas functions are continuous on their domains.

In consumer theory, the *continuity of preferences* (the sets “at least as good as x ” and “at most as good as x ” are closed) is exactly the condition that guarantees that preferences are represented by a *continuous* utility function (Debreu’s theorem; see [9, ch. 3] or [10, ch. 1]). And the continuity of u is what feeds the star theorem of this section:

Theorem 1.24 (Weierstrass: existence of extrema). *Let $K \subseteq \mathbb{R}^n$ be compact and nonempty, and let $f : K \rightarrow \mathbb{R}$ be continuous. Then f attains a maximum and a minimum on K : there are $x^{\max}, x^{\min} \in K$ such that*

$$f(x^{\min}) \leq f(x) \leq f(x^{\max}) \quad \text{for every } x \in K.$$

Proof (sketch, the maximum case). The proof has two acts.

Act 1: the supremum is finite. Suppose f were not bounded above on K . Then for each $k \in \mathbb{N}$ there would be $x^k \in K$ with $f(x^k) > k$. Since K is compact, the sequence (x^k) has a subsequence $x^{k_j} \rightarrow x^* \in K$. By continuity, $f(x^{k_j}) \rightarrow f(x^*)$, a finite real number; but $f(x^{k_j}) > k_j \rightarrow \infty$, a contradiction. Hence $s = \sup_{x \in K} f(x)$ is finite.

Act 2: the supremum is attained. By the definition of supremum, for each k there is $x^k \in K$ with $f(x^k) > s - \frac{1}{k}$ (a *maximizing sequence*). Again by compactness, there is a subsequence $x^{k_j} \rightarrow x^{\max} \in K$. By continuity, $f(x^{\max}) = \lim_j f(x^{k_j}) = s$. So the supremum is a maximum, attained at $x^{\max} \in K$. For the minimum apply the above to $-f$. Full details in [1, ch. 2] and [3]. $\square$

Example 1.25 (Consumer demand exists). Let $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ be continuous (for example Cobb–Douglas), $p \gg 0$, $m > 0$. The consumer’s problem is

$$\max_{x \in \mathbb{R}_+^n} u(x) \quad \text{subject to } \langle p, x \rangle \leq m.$$

The feasible set $B(p, m)$ is compact (Example 1.22) and nonempty ($0 \in B(p, m)$), and u is continuous. By Weierstrass, at least one optimal bundle $x^*(p, m)$ *exists*: the demand correspondence is nonempty. The same argument gives existence of firm supply with compact production sets, and it is an essential ingredient in the existence of Walrasian equilibrium [9, chs. 15–17]. The *uniqueness* and the *characterization* of x^* require more structure (concavity, differentiability): that is the rest of the course.

Careful (common mistake)

The Weierstrass theorem needs *all three* hypotheses; if one fails, there may be no optimum:

- *Without closedness:* $f(x) = x$ on $(0, 1)$ attains no maximum (the candidate 1 does not belong to the set).
- *Without boundedness:* $f(x) = x$ on $[0, \infty)$ attains no maximum. In economic terms:

with a free good, there is no optimal demand.

- *Without continuity:* on $[0, 1]$, the function $f(x) = x$ for $x < 1$ and $f(1) = 0$ attains no maximum: the supremum is 1 but it is never reached.

In an exam, before differentiating and setting to zero, **always** ask yourself: why does the optimum I am looking for exist? Weierstrass is usually the answer.

1.4 Differential calculus in several variables [P1]

Weierstrass says that the optimum exists; differential calculus says *where to look for it*. First order conditions (zero gradient or Lagrange/KKT conditions) and second order conditions (definiteness of the Hessian) are the daily bread of the consumer's problem, the producer's problem and the principal's problem in contracts.

1.4.1 Partial derivatives and gradient

Definition 1.26 (Partial derivative). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $x_0 \in \mathbb{R}^n$. The **partial derivative** of f with respect to x_i at x_0 is

$$\frac{\partial f}{\partial x_i}(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h e_i) - f(x_0)}{h},$$

when the limit exists: the rate of change of f when moving *only* the i th coordinate, freezing the others. In practice: differentiate with respect to x_i treating the other variables as constants.

In economics partial derivatives have their own names: if u is a utility, $\partial u / \partial x_i$ is the *marginal utility* of good i ; if f is a production function, $\partial f / \partial x_i$ is the *marginal product* of input i .

Definition 1.27 (Gradient). The **gradient** of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at x_0 is the vector of all the partial derivatives (we write it as a row, but in matrix products we treat it as a column vector):

$$\nabla f(x_0) = \left(\frac{\partial f}{\partial x_1}(x_0), \dots, \frac{\partial f}{\partial x_n}(x_0) \right).$$

We say that f is *differentiable* at x_0 if it admits a good linear approximation there:

$$f(x_0 + v) = f(x_0) + \langle \nabla f(x_0), v \rangle + r(v), \quad \text{with } \frac{r(v)}{\|v\|} \rightarrow 0 \text{ as } v \rightarrow 0.$$

(The remainder $r(v)$ becomes small *faster* than $\|v\|$.) If the partial derivatives exist and are continuous near x_0 (f of class C^1), then f is differentiable.

Definition 1.28 (Directional derivative). The **directional derivative** of f at x_0 in the direction $d \in \mathbb{R}^n$, $\|d\| = 1$, is

$$D_d f(x_0) = \lim_{h \rightarrow 0^+} \frac{f(x_0 + h d) - f(x_0)}{h}.$$

If f is differentiable, the key formula is

$$D_d f(x_0) = \langle \nabla f(x_0), d \rangle :$$

the rate of change in any direction is the inner product of the gradient with that direction.

Proposition 1.29 (The gradient points towards the steepest increase). *Let f be differentiable at x_0 with $\nabla f(x_0) \neq 0$. Among all unit directions d , the directional derivative $D_d f(x_0)$ is largest when $d = \nabla f(x_0) / \|\nabla f(x_0)\|$, and its maximum value is $\|\nabla f(x_0)\|$. The direction of steepest descent is $-\nabla f(x_0) / \|\nabla f(x_0)\|$, and the directions orthogonal to the gradient are tangent to the level curve of f (along them f does not change to first order).*

Proof. By Cauchy–Schwarz (Theorem 1.10), for every d with $\|d\| = 1$,

$$D_d f(x_0) = \langle \nabla f(x_0), d \rangle \leq \|\nabla f(x_0)\| \cdot \|d\| = \|\nabla f(x_0)\|,$$

and equality in Cauchy–Schwarz happens exactly when d is a multiple (positive, for the maximum) of $\nabla f(x_0)$, that is, when $d = \nabla f(x_0) / \|\nabla f(x_0)\|$. Substituting that d :

$$\left\langle \nabla f(x_0), \frac{\nabla f(x_0)}{\|\nabla f(x_0)\|} \right\rangle = \frac{\|\nabla f(x_0)\|^2}{\|\nabla f(x_0)\|} = \|\nabla f(x_0)\|. \quad \square$$

Intuition

Picture the graph of f as a hill and picture yourself standing at the point $(x_0, f(x_0))$. The gradient is a compass that points, on the map (the plane of the domain), in the direction of steepest climb; its norm says *how* steep it is. Walking perpendicular to the gradient is going around the hill at constant height: that is why the gradient is *orthogonal to the level curves*. In micro, the level curves of u are the *indifference curves*, and $\nabla u(x)$ points towards the better bundles; at an interior optimum of the consumer, ∇u becomes aligned with the price vector p (hence the tangency $\text{MRS} = p_1/p_2$). Besides, this proposition explains the method of *gradient descent* that dominates numerical optimization and machine learning [5, ch. 3], [8].

1.4.2 Jacobian, chain rule and Hessian

Definition 1.30 (Jacobian matrix). For a vector valued function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $F(x) = (F_1(x), \dots, F_m(x))$, the **Jacobian matrix** at x_0 is the $m \times n$ matrix whose rows are the (transposed) gradients of the components:

$$JF(x_0) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1}(x_0) & \cdots & \frac{\partial F_1}{\partial x_n}(x_0) \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1}(x_0) & \cdots & \frac{\partial F_m}{\partial x_n}(x_0) \end{pmatrix}.$$

It is the “derivative” of F : the best linear approximation $F(x_0 + v) \approx F(x_0) + JF(x_0)v$.

Theorem 1.31 (Chain rule). *If $g : \mathbb{R}^k \rightarrow \mathbb{R}^n$ is differentiable at t_0 and $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $g(t_0)$, then $f \circ g$ is differentiable at t_0 and*

$$J(f \circ g)(t_0) = Jf(g(t_0)) Jg(t_0) \quad (\text{matrix product}).$$

The case most used in the course ($k = 1, m = 1$): if $x(t)$ is a curve in $\mathbb{R}^n$ and f is scalar,

$$\frac{d}{dt}f(x(t)) = \langle \nabla f(x(t)), x'(t) \rangle = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x(t)) x'_i(t).$$

For example, with $f(x_1, x_2) = x_1 x_2$ and the curve $x(t) = (t, t^2)$: $\nabla f = (x_2, x_1)$, $x'(t) = (1, 2t)$, so $\frac{d}{dt}f(x(t)) = x_2 \cdot 1 + x_1 \cdot 2t = t^2 + t \cdot 2t = 3t^2$. Direct check: $f(x(t)) = t \cdot t^2 = t^3$, whose derivative is $3t^2$. They agree.

Definition 1.32 (Hessian matrix). If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has second partial derivatives, its **Hessian matrix** at x_0 is the $n \times n$ matrix

$$\nabla^2 f(x_0) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(x_0) \right]_{i,j=1}^n,$$

where $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right)$. If those second partials are continuous (f of class C^2), *Schwarz's theorem (also called Clairaut–Young)* guarantees that the cross partials agree, $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$, so that *the Hessian is symmetric*. The Hessian is the Jacobian of the gradient: it measures how the slope *changes*, that is, the curvature of f .

1.4.3 Taylor's theorem: the tool behind optimality conditions

Theorem 1.33 (First and second order Taylor). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $x_0 \in \mathbb{R}^n$.

1. If f is of class C^1 near x_0 , then

$$f(x_0 + v) = f(x_0) + \langle \nabla f(x_0), v \rangle + o(\|v\|),$$

where $o(\|v\|)$ denotes a remainder $r(v)$ with $r(v)/\|v\| \rightarrow 0$ as $v \rightarrow 0$: an error negligible compared to the size of the step.

2. If f is of class C^2 near x_0 , then

$$f(x_0 + v) = f(x_0) + \langle \nabla f(x_0), v \rangle + \frac{1}{2} v^\top \nabla^2 f(x_0) v + o(\|v\|^2),$$

with a remainder negligible compared to $\|v\|^2$. Version with Lagrange remainder: there is $\theta \in (0, 1)$ such that

$$f(x_0 + v) = f(x_0) + \langle \nabla f(x_0), v \rangle + \frac{1}{2} v^\top \nabla^2 f(x_0 + \theta v) v,$$

that is, the quadratic term evaluated at some intermediate point of the segment is exact, not approximate.

You can look at [1, ch. 2] and [5, ch. 2].

Intuition

Taylor says: near x_0 , every smooth function “is” a polynomial. To first order, f is a tilted plane with slope $\nabla f(x_0)$; to second order a quadratic “bowl” or “dome” governed by the Hessian is added. From this the optimality conditions follow: at an interior maximum x^* , the linear term must vanish ($\nabla f(x^*) = 0$, otherwise moving in the direction of the gradient would improve f) and the quadratic term must not be positive in any direction ($v^\top \nabla^2 f(x^*) v \leq 0$ for every v : negative semidefinite Hessian, Section 1.5). Conversely, $\nabla f(x^*) = 0$ with a *negative definite* Hessian guarantees a strict local maximum. The whole theory of second order conditions of the consumer and of the principal is order 2 Taylor read properly.

1.4.4 Fully worked example: Cobb–Douglas

Example 1.34 (Gradient and Hessian of $u(x, y) = x^a y^b$). Let $u(x, y) = x^a y^b$ with $a, b > 0$, defined for $x, y > 0$. Let us compute everything step by step.

Step 1: first order partials. For $\partial u / \partial x$ we treat y^b as a constant and differentiate x^a (power rule, $\frac{d}{dx} x^a = ax^{a-1}$):

$$\frac{\partial u}{\partial x} = ax^{a-1}y^b, \quad \frac{\partial u}{\partial y} = bx^ay^{b-1}.$$

Both are positive: more consumption gives more utility (monotonicity). Hence

$$\nabla u(x, y) = (ax^{a-1}y^b, \quad bx^ay^{b-1}).$$

Step 2: second partials. We differentiate each partial with respect to each variable.

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} (ax^{a-1}y^b) = a(a-1)x^{a-2}y^b, \\ \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} (bx^ay^{b-1}) = b(b-1)x^ay^{b-2}, \\ \frac{\partial^2 u}{\partial y \partial x} &= \frac{\partial}{\partial y} (ax^{a-1}y^b) = abx^{a-1}y^{b-1} = \frac{\partial^2 u}{\partial x \partial y} \quad (\text{Schwarz, as it should be}). \end{aligned}$$

Therefore

$$\nabla^2 u(x, y) = \begin{pmatrix} a(a-1)x^{a-2}y^b & abx^{a-1}y^{b-1} \\ abx^{a-1}y^{b-1} & b(b-1)x^ay^{b-2} \end{pmatrix}.$$

Step 3: numerical evaluation. Take $a = b = \frac{1}{2}$ (the symmetric Cobb–Douglas $u = \sqrt{xy}$) and the point $(x, y) = (1, 4)$.

$$\frac{\partial u}{\partial x}(1, 4) = \frac{1}{2} \cdot 1^{-1/2} \cdot 4^{1/2} = \frac{1}{2} \cdot 1 \cdot 2 = 1, \quad \frac{\partial u}{\partial y}(1, 4) = \frac{1}{2} \cdot 1^{1/2} \cdot 4^{-1/2} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4},$$

so $\nabla u(1, 4) = (1, \frac{1}{4})$: at that bundle, one extra unit of x is worth (marginally) four times as much as one of y ; the marginal rate of substitution is $\text{MRS} = \frac{u_x}{u_y} = \frac{1}{1/4} = 4$. For the Hessian,

with $a(a-1) = \frac{1}{2} \cdot (-\frac{1}{2}) = -\frac{1}{4}$ and $ab = \frac{1}{4}$:

$$\frac{\partial^2 u}{\partial x^2}(1,4) = -\frac{1}{4} \cdot 1^{-3/2} \cdot 4^{1/2} = -\frac{1}{4} \cdot 2 = -\frac{1}{2}, \quad \frac{\partial^2 u}{\partial y^2}(1,4) = -\frac{1}{4} \cdot 1 \cdot 4^{-3/2} = -\frac{1}{4} \cdot \frac{1}{8} = -\frac{1}{32},$$

$$\frac{\partial^2 u}{\partial x \partial y}(1,4) = \frac{1}{4} \cdot 1^{-1/2} \cdot 4^{-1/2} = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}, \quad \implies \quad \nabla^2 u(1,4) = \begin{pmatrix} -\frac{1}{2} & \frac{1}{8} \\ \frac{1}{8} & -\frac{1}{32} \end{pmatrix}.$$

In Section 1.5 we will learn to read this matrix: its definiteness will decide the concavity of u (preview: with $a+b \leq 1$ the Cobb–Douglas is concave on $\mathbb{R}_{++}^2$; here $a+b=1$ and the Hessian will turn out to be negative semidefinite).

1.5 Quadratic forms and definiteness [P1]

The second order term of Taylor, $\frac{1}{2}v^\top H v$, is a *quadratic form*. Knowing how to decide its sign is knowing how to check second order conditions and, later on, concavity and convexity. This section gives the two practical criteria: eigenvalues and principal minors.

Definition 1.35 (Symmetric matrix and quadratic form). A matrix $A \in \mathbb{R}^{n \times n}$ is **symmetric** if $A = A^\top$ (that is, $a_{ij} = a_{ji}$: the matrix equals its reflection across the diagonal). The **quadratic form** associated with A is the function $Q_A : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$Q_A(v) = v^\top A v = \sum_{i=1}^n \sum_{j=1}^n a_{ij} v_i v_j.$$

For $n=2$, writing $A = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$ and $v = (v_1, v_2)$:

$$Q_A(v) = \alpha v_1^2 + 2\beta v_1 v_2 + \gamma v_2^2.$$

(The 2 appears because the cross terms $a_{12}v_1v_2$ and $a_{21}v_2v_1$ are equal and add up.)

Definition 1.36 (Definiteness). A symmetric matrix A is:

- **positive definite** (pd) if $v^\top A v > 0$ for every $v \neq 0$;
- **positive semidefinite** (psd) if $v^\top A v \geq 0$ for every v ;
- **negative definite** (nd) if $v^\top A v < 0$ for every $v \neq 0$;
- **negative semidefinite** (nsd) if $v^\top A v \leq 0$ for every v ;
- **indefinite** if it takes positive and negative values: there are v, w with $v^\top A v > 0 > w^\top A w$.

Geometrically, the graph of Q_A in $\mathbb{R}^2$ is: a bowl opening upwards (pd), a bowl opening downwards (nd), or a *saddle* (indefinite: it goes up in some directions and down in others). Note that $\text{pd} \Rightarrow \text{psd}$ and $\text{nd} \Rightarrow \text{nsd}$, and that A is nd if and only if $-A$ is pd (useful for recycling criteria).

1.5.1 Criterion 1: eigenvalues

Definition 1.37 (Eigenvalue and eigenvector). A number $\lambda \in \mathbb{R}$ is an **eigenvalue** of the matrix A if there is a vector $v \neq 0$, called an **eigenvector**, such that

$$Av = \lambda v.$$

In words: v is a direction that the matrix does *not bend*, it only *stretches* (or shrinks, or flips) by the factor λ . Eigenvalues are computed by solving the *characteristic equation* $\det(A - \lambda I) = 0$, a polynomial of degree n in λ (here I is the identity matrix and $\det$ the determinant).

Fundamental fact (spectral theorem): if A is *symmetric*, all its eigenvalues are real and there is an orthonormal basis of eigenvectors; in those coordinates the quadratic form “decouples” as $Q_A(v) = \lambda_1 w_1^2 + \dots + \lambda_n w_n^2$, a sum of squares weighted by the eigenvalues. From that expression one reads directly:

Theorem 1.38 (Definiteness via eigenvalues). *Let A be symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$ (counted with multiplicity). Then A is $pd \Leftrightarrow$ all the $\lambda_i > 0$; $psd \Leftrightarrow$ all $\lambda_i \geq 0$; $nd \Leftrightarrow$ all $\lambda_i < 0$; $nsd \Leftrightarrow$ all $\lambda_i \leq 0$; *indefinite* $\Leftrightarrow$ there are eigenvalues of both signs.*

Example 1.39 (Full eigenvalue computation, 2×2). Let us classify $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. The characteristic equation:

$$\det(A - \lambda I) = \det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 \cdot 1 = \lambda^2 - 4\lambda + 4 - 1 = \lambda^2 - 4\lambda + 3.$$

Setting it to zero and factoring: $\lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0$, so $\lambda_1 = 1$ and $\lambda_2 = 3$. Both positive: A is *positive definite*. Direct check with the quadratic form: $v^\top Av = 2v_1^2 + 2v_1v_2 + 2v_2^2 = v_1^2 + v_2^2 + (v_1 + v_2)^2 > 0$ for every $v \neq 0$. (Writing the form as a sum of squares is another valid technique and sometimes the fastest.)

1.5.2 Criterion 2: leading principal minors (Sylvester)

Definition 1.40 (Leading principal minors). The **leading principal minor** of order k of A , written D_k , is the determinant of the $k \times k$ submatrix formed by the first k rows and the first k columns of A . So, $D_1 = a_{11}$, $D_2 = \det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $\dots$, $D_n = \det A$.

Theorem 1.41 (Sylvester criterion). *Let A be symmetric of order n with leading principal minors $D_1, \dots, D_n$.*

1. A is **positive definite** $\Leftrightarrow D_k > 0$ for every $k = 1, \dots, n$.
2. A is **negative definite** $\Leftrightarrow$ the D_k alternate in sign starting with negative: $(-1)^k D_k > 0$, that is, $D_1 < 0$, $D_2 > 0$, $D_3 < 0$, $\dots$

For semidefiniteness the criterion with leading minors is **not** enough: you have to check all the principal minors (those of submatrices obtained by deleting any rows and the same columns): $psd \Leftrightarrow$ all principal minors ≥ 0 ; $nsd \Leftrightarrow$ those of order k have sign $(-1)^k$ or are 0. You can look at [7, ch. 1] and [2, appendix].

Example 1.42 (Sylvester in 2×2 : the Cobb–Douglas of Example 1.34). Let us classify $H = \nabla^2 u(1, 4) = \begin{pmatrix} -\frac{1}{2} & \frac{1}{8} \\ \frac{1}{8} & -\frac{1}{32} \end{pmatrix}$.

$$D_1 = -\frac{1}{2} < 0, \quad D_2 = \det H = \left(-\frac{1}{2}\right)\left(-\frac{1}{32}\right) - \left(\frac{1}{8}\right)^2 = \frac{1}{64} - \frac{1}{64} = 0.$$

Since $D_2 = 0$, H is *not* negative definite (Sylvester requires $D_2 > 0$ strictly). Is it negative semidefinite? We check *all* the principal minors: those of order 1 are $-\frac{1}{2} \leq 0$ and $-\frac{1}{32} \leq 0$ (sign $(-1)^1$ or zero: fine); the one of order 2 is $D_2 = 0 \geq 0$ (sign $(-1)^2$ or zero: fine). Conclusion: H is *negative semidefinite but not negative definite*. This reflects the fact that $u = \sqrt{xy}$ is concave but not *strictly* concave along the rays through the origin (it is homogeneous of degree 1: on the line $y = 4x$ the function is linear).

Example 1.43 (Sylvester in 3×3 , solved). Let us classify

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Step 1: $D_1 = 2 > 0$.

Step 2: $D_2 = \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 2 \cdot 2 - (-1)(-1) = 4 - 1 = 3 > 0$.

Step 3: $D_3 = \det A$, expanding along the first row:

$$\begin{aligned} D_3 &= 2 \cdot \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} - (-1) \cdot \det \begin{pmatrix} -1 & -1 \\ 0 & 2 \end{pmatrix} + 0 \cdot \det \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix} \\ &= 2 \cdot (4 - 1) + 1 \cdot ((-1)(2) - (-1)(0)) + 0 = 2 \cdot 3 + 1 \cdot (-2) = 6 - 2 = 4 > 0. \end{aligned}$$

The three minors are positive: A is *positive definite*. By contrast, $-A$ has minors $-2 < 0$, $3 > 0$, $-4 < 0$ (alternating starting with negative): $-A$ is negative definite, as it should be.

Exam strategy

Exam recipe to classify a symmetric matrix A (typically a Hessian):

1. *Look at the diagonal first.* If there are entries of both signs on the diagonal, A is already indefinite (since $e_i^\top A e_i = a_{ii}$), and you are done. If some $a_{ii} = 0$, A cannot be definite (neither positive nor negative); and if in addition row i is nonzero, it is not semidefinite either.
2. *For strict definiteness, use Sylvester:* compute $D_1, D_2, \dots, D_n$ in order. All positive $\Rightarrow$ pd; alternating $-, +, -, \dots \Rightarrow$ nd. If some D_k breaks with a strict inequality *both* patterns at once (for example, $D_k < 0$ with k even: it contradicts both pd and nd), A is indefinite.
3. *If some $D_k = 0$ appears*, or if the pattern breaks without contradicting both cases, leading Sylvester does not decide: move on to all the principal minors (or to eigenvalues) to distinguish semidefinite from indefinite.

4. In 2×2 there is a shortcut: $\det A > 0$ and $a_{11} > 0 \Rightarrow$ pd; $\det A > 0$ and $a_{11} < 0 \Rightarrow$ nd; $\det A < 0 \Rightarrow$ indefinite (the determinant is the product of eigenvalues: a negative sign reveals eigenvalues of opposite signs).
5. Link with the course: nd Hessian on the whole domain (convex and open) $\Rightarrow f$ strictly concave $\Rightarrow$ the critical point of the consumer's or principal's problem is the unique global maximum. This chain is proved in the following chapters; memorize the logical flow from now.

Remark 1.44. In Microeconomics definiteness appears everywhere: the second order condition of the consumer is that the Hessian of the Lagrangian be negative definite on the tangent space of the constraint (bordered Hessian); the risk aversion of an agent with Bernoulli utility u is $u'' < 0$ (the 1×1 case of negative definiteness); the law of supply comes from the Slutsky substitution matrix being negative semidefinite [9, ch. 3]. Mastering this section pays dividends all year.

Chapter summary

Key results of the chapter, with their exact hypotheses:

- **Vector space** (Def. 1.1): a set with addition and scalar multiplication satisfying (A1)–(A8). $\mathbb{R}^n$ is one; $\mathbb{R}_+^n$ is *not* (it fails with negative scalars). $\dim \mathbb{R}^n = n$, with standard basis $e_1, \dots, e_n$.
- **Cauchy–Schwarz** (Thm. 1.10): for all $x, y \in \mathbb{R}^n$, $|\langle x, y \rangle| \leq \|x\| \|y\|$, with equality if and only if they are linearly dependent. From this follow the triangle inequality and the fact that the angle is well defined.
- **Norm** (Def. 1.9): positivity, absolute homogeneity, triangle inequality. $\|\cdot\|_1$, $\|\cdot\|$, $\|\cdot\|_\infty$ are norms and all norms on $\mathbb{R}^n$ are equivalent.
- **Heine–Borel** (Thm. 1.20): in $\mathbb{R}^n$ (only in finite dimension!), compact $\Leftrightarrow$ closed and bounded. **Bolzano–Weierstrass** (Thm. 1.21): every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.
- **Weierstrass** (Thm. 1.24): if K is compact and *nonempty* and $f : K \rightarrow \mathbb{R}$ is *continuous*, then f attains a maximum and a minimum on K . Application: with $p \gg 0$, $m > 0$ and continuous utility, consumer demand exists.
- **Gradient** (Prop. 1.29): if f is differentiable and $\nabla f(x_0) \neq 0$, the direction of steepest increase is $\nabla f(x_0) / \|\nabla f(x_0)\|$ with rate $\|\nabla f(x_0)\|$; the gradient is orthogonal to the level (indifference) curves.
- **Taylor** (Thm. 1.33): if $f \in C^1$, $f(x_0 + v) = f(x_0) + \langle \nabla f(x_0), v \rangle + o(\|v\|)$; if $f \in C^2$, we add $\frac{1}{2} v^\top \nabla^2 f(x_0) v$ with remainder $o(\|v\|^2)$ (or an exact Lagrange remainder at an intermediate point). If $f \in C^2$, the Hessian is *symmetric* (Schwarz).
- **Definiteness** (Thm. 1.38 and 1.41): a symmetric A is pd/nd if and only if all its eigenvalues are $> 0 / < 0$; equivalently (Sylvester), if its leading minors satisfy

$D_k > 0$ for every k / $(-1)^k D_k > 0$ for every k . For *semidefiniteness* *all* the principal minors are required, not only the leading ones.

1.6 Exam-style exercises [P1]

Exercise 1.45. Let $v_1 = (1, 1, 0)$, $v_2 = (0, 1, 1)$ and $v_3 = (1, 0, 1)$ in $\mathbb{R}^3$. (a) Prove that they are linearly independent. (b) Conclude that they form a basis of $\mathbb{R}^3$ and write the vector $x = (2, 3, 5)$ as a linear combination of them.

Solution. (a) We set up $\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 = 0$:

$$\lambda_1(1, 1, 0) + \lambda_2(0, 1, 1) + \lambda_3(1, 0, 1) = (\lambda_1 + \lambda_3, \lambda_1 + \lambda_2, \lambda_2 + \lambda_3) = (0, 0, 0),$$

which gives the system $\lambda_1 + \lambda_3 = 0$, $\lambda_1 + \lambda_2 = 0$, $\lambda_2 + \lambda_3 = 0$. From the first, $\lambda_3 = -\lambda_1$; from the second, $\lambda_2 = -\lambda_1$. Substituting both into the third: $-\lambda_1 - \lambda_1 = -2\lambda_1 = 0$, so $\lambda_1 = 0$ and therefore $\lambda_2 = \lambda_3 = 0$. Only the trivial solution exists: the vectors are linearly independent.

(b) Three independent vectors in a space of dimension 3 form a basis. We look for μ_1, μ_2, μ_3 with $\mu_1 v_1 + \mu_2 v_2 + \mu_3 v_3 = (2, 3, 5)$, that is,

$$\mu_1 + \mu_3 = 2, \quad \mu_1 + \mu_2 = 3, \quad \mu_2 + \mu_3 = 5.$$

Adding the three equations: $2(\mu_1 + \mu_2 + \mu_3) = 10$, that is, $\mu_1 + \mu_2 + \mu_3 = 5$. Subtracting each original equation from this one: $\mu_2 = 5 - 2 = 3$, $\mu_3 = 5 - 3 = 2$, $\mu_1 = 5 - 5 = 0$. Check: $0 \cdot (1, 1, 0) + 3 \cdot (0, 1, 1) + 2 \cdot (1, 0, 1) = (0 + 0 + 2, 0 + 3 + 0, 0 + 3 + 2) = (2, 3, 5)$. Correct: $x = 0 v_1 + 3 v_2 + 2 v_3$. $\square$

Exercise 1.46. Let $x = (3, -1, 2)$ and $y = (1, 2, -2)$ in $\mathbb{R}^3$. (a) Compute $\langle x, y \rangle$, $\|x\|$, $\|y\|$, $\|x\|_1$ and $\|y\|_\infty$. (b) Check numerically the Cauchy-Schwarz inequality and the triangle inequality for x and y . (c) Are x and y orthogonal?

Solution. (a) $\langle x, y \rangle = 3 \cdot 1 + (-1) \cdot 2 + 2 \cdot (-2) = 3 - 2 - 4 = -3$. $\|x\| = \sqrt{3^2 + (-1)^2 + 2^2} = \sqrt{9 + 1 + 4} = \sqrt{14}$. $\|y\| = \sqrt{1^2 + 2^2 + (-2)^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$. $\|x\|_1 = |3| + |-1| + |2| = 6$. $\|y\|_\infty = \max\{1, 2, 2\} = 2$.

(b) Cauchy-Schwarz: $|\langle x, y \rangle| = 3$ and $\|x\| \|y\| = 3\sqrt{14} \approx 3 \times 3.742 = 11.22$; indeed $3 \leq 11.22$. Triangle inequality: $x + y = (4, 1, 0)$, so $\|x + y\| = \sqrt{16 + 1 + 0} = \sqrt{17} \approx 4.12$, while $\|x\| + \|y\| = \sqrt{14} + 3 \approx 6.74$; indeed $4.12 \leq 6.74$.

(c) No: $x \perp y$ would require $\langle x, y \rangle = 0$, but $\langle x, y \rangle = -3 \neq 0$. (The angle between them is obtuse, since the cosine is $-3/(3\sqrt{14}) = -1/\sqrt{14} < 0$.) $\square$

Exercise 1.47. For each set, decide whether it is closed, bounded and compact, justifying with Heine-Borel:

$$(a) A = \{x \in \mathbb{R}_+^2 : 2x_1 + 3x_2 \leq 12\}; \quad (b) B = \{x \in \mathbb{R}^2 : x_1 > 0, x_2 > 0, x_1 x_2 \geq 1\};$$

$$(c) C = \{x \in \mathbb{R}_+^2 : 0 \cdot x_1 + 3x_2 \leq 12\}.$$

In which of them does Weierstrass guarantee that every continuous function attains its maximum?

Solution. (a) A is the budget set with $p = (2, 3) \gg 0$, $m = 12$. *Closed*: it is defined by the weak inequalities $x_1 \geq 0$, $x_2 \geq 0$, $2x_1 + 3x_2 \leq 12$ with continuous functions; limits preserve weak inequalities, so it contains the limits of its sequences. *Bounded*: since $x_2 \geq 0$, $2x_1 \leq 2x_1 + 3x_2 \leq 12$ gives $0 \leq x_1 \leq 6$; similarly $0 \leq x_2 \leq 4$; hence $\|x\| \leq \sqrt{36 + 16} = \sqrt{52}$. By Heine–Borel, A is compact.

(b) *Closed*: yes. Although strict inequalities $x_1 > 0$, $x_2 > 0$ appear, these are a *consequence* of $x_1 x_2 \geq 1$ with $x_1, x_2 > 0$: if $x^k \in B$ and $x^k \rightarrow x$, then $x_1 x_2 = \lim x_1^k x_2^k \geq 1$, which forces $x_1 \neq 0 \neq x_2$; together with $x_i = \lim x_i^k \geq 0$ we get $x \in B$. *Bounded*: no; the points $(k, 1/k) \in B$ have $\|(k, 1/k)\| \geq k \rightarrow \infty$. By Heine–Borel, B is *not compact* (this B is the “at least as good as” set of a Cobb–Douglas: closed but not bounded).

(c) Here the price of good 1 is zero. C is closed (same argument as in (a)), but *not bounded*: $(k, 0) \in C$ for every k . It is not compact.

Weierstrass applies with a guarantee only in (a): compact, nonempty. In (b) and (c) the existence of a maximum may fail; for example, $f(x) = x_1$ attains no maximum on B or on C . $\square$

Exercise 1.48. Let $f(x, y) = \ln x + \ln y$ (the logarithmic transformation of a Cobb–Douglas), defined for $x, y > 0$. (a) Compute $\nabla f(x, y)$ and $\nabla^2 f(x, y)$. (b) Write the second order Taylor approximation of f around the point $(1, 1)$. (c) Use it to approximate $f(1.1, 0.9)$ and compare with the exact value $f(1.1, 0.9) = \ln(0.99) \approx -0.01005$.

Solution. (a) $\frac{\partial f}{\partial x} = \frac{1}{x}$ and $\frac{\partial f}{\partial y} = \frac{1}{y}$, so $\nabla f(x, y) = (\frac{1}{x}, \frac{1}{y})$. Differentiating again: $\frac{\partial^2 f}{\partial x^2} = -\frac{1}{x^2}$, $\frac{\partial^2 f}{\partial y^2} = -\frac{1}{y^2}$, and the cross partials are 0 (in $\partial f / \partial x = 1/x$ there is no y). So

$$\nabla^2 f(x, y) = \begin{pmatrix} -1/x^2 & 0 \\ 0 & -1/y^2 \end{pmatrix}.$$

(b) At $(1, 1)$: $f(1, 1) = \ln 1 + \ln 1 = 0$, $\nabla f(1, 1) = (1, 1)$, $\nabla^2 f(1, 1) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. With $v = (h, k) = (x - 1, y - 1)$, order 2 Taylor gives

$$f(1 + h, 1 + k) \approx 0 + \langle (1, 1), (h, k) \rangle + \frac{1}{2} \begin{pmatrix} h & k \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix} = h + k - \frac{1}{2}(h^2 + k^2).$$

(c) For $(x, y) = (1.1, 0.9)$: $h = 0.1$, $k = -0.1$. Then

$$f(1.1, 0.9) \approx 0.1 + (-0.1) - \frac{1}{2}(0.1^2 + (-0.1)^2) = 0 - \frac{1}{2}(0.01 + 0.01) = -0.01.$$

The exact value is $\ln(1.1 \times 0.9) = \ln 0.99 \approx -0.01005$: the approximation error is of order 10^{-5} , consistent with a remainder $o(\|v\|^2)$ for $\|v\| \approx 0.14$. Note that the Hessian is negative definite on the whole domain ($-1/x^2 < 0$ and determinant $1/(x^2 y^2) > 0$ with $D_1 < 0$): a preview of the fact that f is strictly concave, the property that makes the consumer’s optimum unique. $\square$

Exercise 1.49. Classify by definiteness (positive/negative definite, semidefinite, indefinite):

$$(a) A = \begin{pmatrix} -2 & 1 \\ 1 & -3 \end{pmatrix}, \quad (b) B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad (c) C = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}.$$

For (b), find in addition the eigenvalues and a vector v with $v^\top B v < 0$.

Solution. (a) Leading minors: $D_1 = -2 < 0$; $D_2 = (-2)(-3) - 1 \cdot 1 = 6 - 1 = 5 > 0$. They alternate starting with negative: A is *negative definite*. (As a Hessian, it would indicate a strict local maximum at a critical point.)

(b) $D_1 = 1 > 0$, but $D_2 = 1 \cdot 1 - 2 \cdot 2 = 1 - 4 = -3 < 0$. A negative determinant in 2×2 means eigenvalues of opposite signs: B is *indefinite*. Eigenvalues: $\det(B - \lambda I) = (1 - \lambda)^2 - 4 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) = 0$, so $\lambda_1 = 3 > 0 > \lambda_2 = -1$, confirming indefiniteness. A vector with negative quadratic form: the eigenvector of $\lambda_2 = -1$ solves $(B + I)v = 0$, that is, $\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} v = 0$, for example $v = (1, -1)$. Check: the quadratic form of B is $Q_B(v) = v_1^2 + 4v_1v_2 + v_2^2$ (the 4 is $2b_{12}$), so that $Q_B(1, -1) = 1 - 4 + 1 = -2 < 0$. (And with $w = (1, 1)$: $Q_B(1, 1) = 1 + 4 + 1 = 6 > 0$, which confirms indefiniteness.)

(c) $D_1 = 1 > 0$. $D_2 = 1 \cdot 2 - 1 \cdot 1 = 1 > 0$. $D_3 = \det C$, expanding along the first row:

$$D_3 = 1 \cdot \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} - 1 \cdot \det \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} + 0 = 1(6 - 1) - 1(3 - 0) = 5 - 3 = 2 > 0.$$

The three leading minors are positive: C is *positive definite*. □

Exercise 1.50. (Existence and nonexistence of demand.) A consumer has utility $u(x_1, x_2) = x_1 + \ln(1 + x_2)$ on $\mathbb{R}_+^2$, wealth $m = 10$ and faces prices $p = (p_1, p_2)$. (a) If $p = (1, 2)$, prove that the problem $\max\{u(x) : x \in \mathbb{R}_+^2, \langle p, x \rangle \leq 10\}$ has a solution. (b) If $p = (0, 2)$ (good 1 is free), prove that it does *not* have a solution. (c) Explain which Weierstrass hypothesis fails in (b).

Solution. (a) The function u is continuous on $\mathbb{R}_+^2$: x_1 is continuous (a polynomial) and $\ln(1 + x_2)$ is the composition of the continuous polynomial $1 + x_2$ (which takes values $\geq 1 > 0$) with $\ln$, continuous on $(0, \infty)$; the sum of continuous functions is continuous. The feasible set $B((1, 2), 10) = \{x \in \mathbb{R}_+^2 : x_1 + 2x_2 \leq 10\}$ is closed (weak inequalities of continuous functions) and bounded: from $x_1, x_2 \geq 0$ and $x_1 + 2x_2 \leq 10$ we get $0 \leq x_1 \leq 10$ and $0 \leq x_2 \leq 5$. By Heine–Borel it is compact, and it is nonempty since it contains $(0, 0)$. By the Weierstrass theorem (Thm. 1.24), u attains its maximum: demand exists.

(b) With $p = (0, 2)$ the feasible set is $F = \{x \in \mathbb{R}_+^2 : 2x_2 \leq 10\} = [0, \infty) \times [0, 5]$. On F , taking the sequence $x^k = (k, 0)$ we get $u(x^k) = k + \ln 1 = k \rightarrow \infty$: utility is not bounded above on F , so there is no feasible bundle that is maximal (any candidate x is beaten by $(x_1 + 1, x_2)$, which is still feasible and gives strictly higher utility).

(c) The *compactness* of the feasible set fails: F is closed but *not bounded* in the direction of the free good, so Heine–Borel does not give compactness and Weierstrass does not apply. Economic moral: the existence of demand requires strictly positive prices (or satiable preferences); that is why in general equilibrium the existence theorems make sure that desirable goods have a positive price [9, ch. 17]. □

Chapter 2

Convex sets and affine sets [P2]

Week 1 of the course. Bibliography: Boyd and Vandenberghe [2]; Brinkhuis [3]; Ruszczyński [6].

What is this used for in Microeconomics?

This chapter is the geometric basis of almost everything you will see in Microeconomics. In **general equilibrium** (MWG chs. 15–18), convexity is *the* hypothesis that makes the theory work: the budget set $\{x \geq 0 : p \cdot x \leq m\}$ is convex (we prove it in Example 2.8); the convexity of preferences and of production sets is the key condition for the existence of Walrasian equilibrium and for the *second welfare theorem* (every Pareto optimum can be decentralized with prices), since both results rest on separating convex sets with hyperplanes (MWG, chs. 15–18, [9]). The Edgeworth box is a Cartesian product of convex sets, and the *Minkowski sum* $Y_1 + \dots + Y_J$ is exactly how the technology of J firms is aggregated in a production economy (MWG, ch. 5). In **contract theory** (MWG ch. 14), the sets of payoffs reachable through random contracts are convex combinations of pure payoffs. In **mechanism design** (MWG ch. 23), the set of lotteries over n outcomes is the probability simplex Δ^{n-1} , a compact convex set: without that fact you cannot even set up expected utility or random mechanisms. Mastering this chapter is mastering the language in which microeconomics is written.

Before optimizing functions we need to understand *over which sets* we optimize. The central idea of the course is that optimization problems become manageable when the feasible set is *convex*: a set with no “holes” or “dents”, in which the average of two feasible options is still feasible. This chapter develops that idea from scratch: segments, convex combinations, convex and affine sets, the operations that preserve convexity and the topological properties (interior, closure, relative interior) that we will use later in the separation theorems. The main references are [3] (ch. 2), [2] (ch. 2, sections 2.1–2.3), [6] (ch. 2) and [1] (ch. 1). All we need from the preliminaries chapter is: the Euclidean norm $\|x\|$, the inner product $\langle x, y \rangle$, the triangle inequality, and the notions of open ball, interior, closure and boundary in $\mathbb{R}^n$.

2.1 Segments and convex combinations [P1]

2.1.1 The segment between two points

Let us start with the most elementary question possible: given two points x and y in $\mathbb{R}^n$, how do we describe algebraically the line segment that joins them? The idea is to “walk” from x towards y : we start at x , and the vector pointing from x to y is $y - x$. If we move forward a fraction λ of the way (with $0 \leq \lambda \leq 1$), we reach the point

$$x + \lambda(y - x) = x + \lambda y - \lambda x = (1 - \lambda)x + \lambda y.$$

Look at the algebraic step: we distribute $\lambda(y - x) = \lambda y - \lambda x$ and then collect the terms with x , obtaining $(1 - \lambda)x$. When $\lambda = 0$ we are at x ; when $\lambda = 1$ we are at y ; when $\lambda = 1/2$ we are exactly at the midpoint.

Definition 2.1 (Segment). Given $x, y \in \mathbb{R}^n$, the *line segment* joining x with y is the set

$$[x, y] = \{ (1 - \lambda)x + \lambda y : \lambda \in [0, 1] \}.$$

We write $[x, y]$ for the segment that includes x but not y , that is, taking $\lambda \in [0, 1)$.

Example 2.2. Let $x = (1, 2)$ and $y = (5, 4)$ in $\mathbb{R}^2$. The point of the segment corresponding to $\lambda = 1/4$ is

$$(1 - \frac{1}{4})(1, 2) + \frac{1}{4}(5, 4) = \frac{3}{4}(1, 2) + \frac{1}{4}(5, 4) = (\frac{3}{4} + \frac{5}{4}, \frac{6}{4} + \frac{4}{4}) = (2, 2.5).$$

Let us check each coordinate: the first one is $\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 5 = \frac{3}{4} + \frac{5}{4} = \frac{8}{4} = 2$; the second one is $\frac{3}{4} \cdot 2 + \frac{1}{4} \cdot 4 = \frac{6}{4} + \frac{4}{4} = \frac{10}{4} = 2.5$. The point $(2, 2.5)$ is a quarter of the way from x to y .

2.1.2 Convex combinations: weighted averages and lotteries

The expression $(1 - \lambda)x + \lambda y$ is a *weighted average* of x and y : the weights are $1 - \lambda$ and λ , both nonnegative and adding up to 1. The natural generalization to k points is immediate.

Definition 2.3 (Convex combination). A *convex combination* of the points $x_1, \dots, x_k \in \mathbb{R}^n$ is any point of the form

$$\lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_k x_k, \quad \text{with } \lambda_i \geq 0 \text{ for all } i \text{ and } \sum_{i=1}^k \lambda_i = 1.$$

The numbers $\lambda_1, \dots, \lambda_k$ are called *weights*.

Intuition

A convex combination is a *weighted average*, and in economics weighted averages with nonnegative weights that add up to one have a name: **lotteries**. If $x_1, \dots, x_k$ are k bundles (or payoffs, or allocations) and λ_i is the probability of receiving bundle x_i , then the convex combination $\sum_i \lambda_i x_i$ is the *expected bundle* of the lottery. So when in Microeconomics (choice under uncertainty, MWG ch. 6) you say “the set of lotteries over k outcomes”,

you are saying, in the language of this course, “the set of all convex combinations of k points”. You can also think of it physically: $\sum_i \lambda_i x_i$ is the *center of mass* of a system of particles located at the points x_i with masses λ_i .

Example 2.4 (A lottery over three payoffs). A random contract pays the agent one of three amounts: $x_1 = 100$, $x_2 = 400$ or $x_3 = 900$ (in local currency), with probabilities $\lambda_1 = 0.5$, $\lambda_2 = 0.3$, $\lambda_3 = 0.2$. First we check that the weights are valid: all of them are nonnegative and $0.5 + 0.3 + 0.2 = 1$. The expected payoff is the convex combination

$$0.5 \cdot 100 + 0.3 \cdot 400 + 0.2 \cdot 900 = 50 + 120 + 180 = 350.$$

Note that 350 lies between the smallest (100) and the largest (900) of the payoffs: every convex combination of real numbers falls between the smallest and the largest of them. This is the one dimensional version of a general fact: convex combinations never “leave” the set of original points in any direction.

2.2 Convex sets: definition and examples [P1]

2.2.1 The definition

Definition 2.5 (Convex set). A set $C \subseteq \mathbb{R}^n$ is *convex* if for every pair of points $x, y \in C$ and every $\lambda \in [0, 1]$ we have

$$(1 - \lambda)x + \lambda y \in C,$$

that is, if $[x, y] \subseteq C$ for all $x, y \in C$: the segment that joins any pair of points of the set is contained in the set. The empty set and one point sets are convex: the condition holds trivially because there are no pairs to check, or because the only possible segment is the point itself.

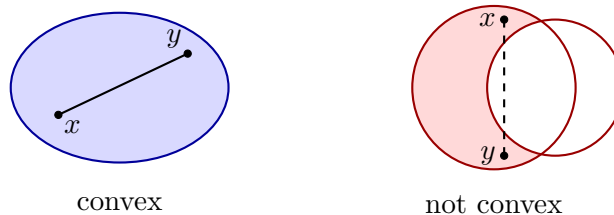


Figure 2.1: On the left, a convex set: the segment between any two points stays inside. On the right, a moon shaped set: the dashed segment between x and y leaves the set.

Exam strategy

To **prove** that a set C is convex, the recipe is always the same and it is worth memorizing it as a three step ritual:

- (1) Take two *arbitrary* points $x, y \in C$ and an arbitrary $\lambda \in [0, 1]$. Write explicitly what it means for x and y to belong to C (the inequalities or equations that define C).
- (2) Form the point $z = (1 - \lambda)x + \lambda y$ and *check that z satisfies the definition of C ,*

using the properties of x and y from step (1). The typical tools are: linearity of the inner product, the triangle inequality, and the fact that adding inequalities or multiplying them by nonnegative numbers preserves them.

(3) Conclude: since x, y, λ were arbitrary, C is convex.

To **refute** convexity, *one* concrete counterexample is enough: two points of the set and a value of λ (almost always $\lambda = 1/2$ works) such that the intermediate point does not belong to the set. In an exam, showing the complete numerical counterexample earns all the points; saying “you can see it in the picture” earns none.

2.2.2 Fundamental examples, solved step by step

Example 2.6 (The ball is convex). Let $\bar{B}(c, r) = \{x \in \mathbb{R}^n : \|x - c\| \leq r\}$ be the closed ball with center c and radius $r > 0$ (same notation as in the preliminaries chapter: B for the open ball, $\bar{B}$ for the closed one). We claim that $\bar{B}(c, r)$ is convex.

Step 1. Take arbitrary $x, y \in \bar{B}(c, r)$ and $\lambda \in [0, 1]$. By the definition of the ball, we know that $\|x - c\| \leq r$ and $\|y - c\| \leq r$.

Step 2. Let $z = (1 - \lambda)x + \lambda y$. We must prove that $\|z - c\| \leq r$. The key trick is to write c as a convex combination of itself: $c = (1 - \lambda)c + \lambda c$ (this is true because $(1 - \lambda) + \lambda = 1$). Then

$$z - c = (1 - \lambda)x + \lambda y - [(1 - \lambda)c + \lambda c] = (1 - \lambda)(x - c) + \lambda(y - c),$$

where in the last step we simply collected the terms that have factor $(1 - \lambda)$ and those that have factor λ . Now we apply the triangle inequality ($\|u + v\| \leq \|u\| + \|v\|$) and the homogeneity of the norm ($\|\alpha u\| = |\alpha| \|u\|$, recalling that $1 - \lambda \geq 0$ and $\lambda \geq 0$, so the absolute values are not needed):

$$\|z - c\| = \|(1 - \lambda)(x - c) + \lambda(y - c)\| \leq (1 - \lambda)\|x - c\| + \lambda\|y - c\| \leq (1 - \lambda)r + \lambda r = r.$$

In the last inequality we used $\|x - c\| \leq r$ and $\|y - c\| \leq r$, multiplied by the nonnegative numbers $1 - \lambda$ and λ (which preserves the inequalities), and then $(1 - \lambda)r + \lambda r = r[(1 - \lambda) + \lambda] = r \cdot 1 = r$.

Step 3. Since $\|z - c\| \leq r$, we have $z \in \bar{B}(c, r)$. We conclude that $\bar{B}(c, r)$ is convex. The same argument, changing $\leq$ to $<$, shows that the *open* ball is convex too.

Example 2.7 (Hyperplanes and half spaces). Let $a \in \mathbb{R}^n$ with $a \neq 0$ and let $b \in \mathbb{R}$. The *hyperplane* with normal vector a and level b is

$$H = \{x \in \mathbb{R}^n : \langle a, x \rangle = b\},$$

and the associated *half spaces* are $H^- = \{x : \langle a, x \rangle \leq b\}$ and $H^+ = \{x : \langle a, x \rangle \geq b\}$. In $\mathbb{R}^2$, a hyperplane is a line; in $\mathbb{R}^3$, a plane. Let us prove that H^- is convex (the argument for H and H^+ is identical).

Let $x, y \in H^-$ and $\lambda \in [0, 1]$. Then $\langle a, x \rangle \leq b$ and $\langle a, y \rangle \leq b$. Using the *linearity of the inner*

product (seen in the preliminaries chapter: $\langle a, \alpha u + \beta v \rangle = \alpha \langle a, u \rangle + \beta \langle a, v \rangle$):

$$\langle a, (1 - \lambda)x + \lambda y \rangle = (1 - \lambda) \langle a, x \rangle + \lambda \langle a, y \rangle \leq (1 - \lambda)b + \lambda b = b,$$

where again we multiplied the inequalities $\langle a, x \rangle \leq b$, $\langle a, y \rangle \leq b$ by the nonnegative weights and added them. Therefore $(1 - \lambda)x + \lambda y \in H^-$: the half space is convex. For H we get equality instead of inequality, so H is convex too. *Economic reading*: if $a = p$ is a price vector and $b = m$ is income, H^- is the set of bundles whose expenditure does not exceed income; the hyperplane H is the *budget line*, the bundles that exhaust income exactly.

Example 2.8 (The budget set is convex). Let $p \in \mathbb{R}^n$ be a price vector with $p \gg 0$ (all coordinates positive) and $m > 0$ the consumer's income. The *budget set* is

$$B(p, m) = \{x \in \mathbb{R}^n : x \geq 0, p \cdot x \leq m\},$$

where $x \geq 0$ means $x_i \geq 0$ for each good i , and $p \cdot x = \langle p, x \rangle = \sum_{i=1}^n p_i x_i$ is total expenditure. *Proof of convexity*. Let $x, y \in B(p, m)$ and $\lambda \in [0, 1]$; write $z = (1 - \lambda)x + \lambda y$. We must check the two conditions that define $B(p, m)$.

(i) *Nonnegativity*. For each coordinate i : since $x_i \geq 0$, $y_i \geq 0$, $1 - \lambda \geq 0$ and $\lambda \geq 0$, the number $z_i = (1 - \lambda)x_i + \lambda y_i$ is a sum of two products of nonnegative numbers, hence $z_i \geq 0$. So $z \geq 0$.

(ii) *Expenditure constraint*. By linearity of the inner product,

$$p \cdot z = p \cdot [(1 - \lambda)x + \lambda y] = (1 - \lambda)(p \cdot x) + \lambda(p \cdot y) \leq (1 - \lambda)m + \lambda m = m,$$

using $p \cdot x \leq m$ and $p \cdot y \leq m$. By (i) and (ii), $z \in B(p, m)$, and the budget set is convex. In fact, we have just checked in a concrete case a general principle that we will prove in Theorem 2.18: $B(p, m)$ is the intersection of $n + 1$ half spaces ($x_i \geq 0$ for each i , plus $p \cdot x \leq m$), and the intersection of convex sets is convex. The economic reading is important: if two bundles are affordable, any “mixture” (weighted average) of them is affordable too. Without this property, the consumer's problem of MWG ch. 3 [9] would lose almost all its structure.

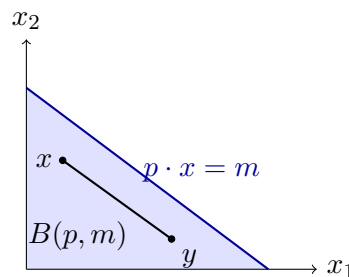


Figure 2.2: The budget set with two goods: a triangle, which is the intersection of three half spaces ($x_1 \geq 0$, $x_2 \geq 0$, $p \cdot x \leq m$). The segment between two affordable bundles stays inside the set.

Example 2.9 (Polyhedra). A *polyhedron* is a finite intersection of half spaces and hyperplanes:

$$P = \{x \in \mathbb{R}^n : \langle a_j, x \rangle \leq b_j, j = 1, \dots, s; \langle c_k, x \rangle = d_k, k = 1, \dots, t\}.$$

Since each half space and each hyperplane is convex (Example 2.7) and the intersection of convex sets is convex (Theorem 2.18 below), every polyhedron is convex. The budget set of Example 2.8 is a polyhedron; so are the feasible sets of linear programming problems [2].

Example 2.10 (The probability simplex). The *probability simplex* in $\mathbb{R}^n$ is

$$\Delta^{n-1} = \left\{ \pi \in \mathbb{R}^n : \pi_i \geq 0 \text{ for all } i, \sum_{i=1}^n \pi_i = 1 \right\}.$$

Each element of Δ^{n-1} is a *lottery*: a probability distribution over n outcomes. It is a polyhedron (n half spaces $\pi_i \geq 0$ and one hyperplane $\sum_i \pi_i = 1$), hence it is convex. Let us also check it directly to fix the method: if $\pi, \sigma \in \Delta^{n-1}$ and $\lambda \in [0, 1]$, then each coordinate of $\rho = (1 - \lambda)\pi + \lambda\sigma$ satisfies $\rho_i = (1 - \lambda)\pi_i + \lambda\sigma_i \geq 0$, and

$$\sum_{i=1}^n \rho_i = (1 - \lambda) \sum_{i=1}^n \pi_i + \lambda \sum_{i=1}^n \sigma_i = (1 - \lambda) \cdot 1 + \lambda \cdot 1 = 1.$$

Reading: mixing two lotteries with weights $(1 - \lambda, \lambda)$ produces another lottery (a “compound lottery”). This is the reason why in mechanism design (MWG ch. 23) random mechanisms live in a convex set, and why the axioms of expected utility are stated over mixtures of lotteries [9].

2.2.3 Non examples

Example 2.11 (Two sets that are not convex). (a) *The annulus*. Let $A = \{x \in \mathbb{R}^2 : 1 \leq \|x\| \leq 2\}$ (a circular ring). Take $x = (1, 0)$ and $y = (-1, 0)$; both satisfy $\|x\| = \|y\| = 1$, so $x, y \in A$. With $\lambda = 1/2$:

$$\frac{1}{2}(1, 0) + \frac{1}{2}(-1, 0) = (0, 0), \quad \|(0, 0)\| = 0 < 1,$$

so the midpoint does not belong to A . The annulus is not convex: the central “hole” breaks convexity.

(b) *Union of two balls*. Let $U = B((0, 0), 1) \cup B((4, 0), 1)$ in $\mathbb{R}^2$. The points $x = (0, 0)$ and $y = (4, 0)$ are in U (each one is the center of a ball). With $\lambda = 1/2$, the midpoint is $(2, 0)$. But $\|(2, 0) - (0, 0)\| = 2 > 1$ and $\|(2, 0) - (4, 0)\| = 2 > 1$, so $(2, 0)$ is in neither ball and therefore is not in U . The union of convex sets, in general, is *not* convex (we will come back to this in Section 2.4).

2.2.4 Convexity and combinations of many points

The definition of convexity only asks the set to contain the convex combinations of *two* points. The next result says that this is enough for it to contain those of *any* finite number of points. It is the geometric analogue of a fact you will use a thousand times: if mixtures of two lotteries are admissible, mixtures of k lotteries are admissible too.

Theorem 2.12 (Characterization by convex combinations). *A set $C \subseteq \mathbb{R}^n$ is convex if and only if every convex combination of finitely many points of C belongs to C ; that is, for every $k \in \mathbb{N}$, all $x_1, \dots, x_k \in C$ and all $\lambda_1, \dots, \lambda_k \geq 0$ with $\sum_{i=1}^k \lambda_i = 1$ we have $\sum_{i=1}^k \lambda_i x_i \in C$.*

Proof. ($\Leftarrow$) This is immediate: the case $k = 2$ is exactly the definition of convexity.

($\Rightarrow$) Assume C is convex and proceed by induction on k . *Base case $k = 1$:* the only combination is $1 \cdot x_1 = x_1 \in C$. *Case $k = 2$:* this is the definition. *Inductive step:* assume the result holds for $k - 1$ points ($k \geq 3$) and take $x_1, \dots, x_k \in C$ with weights $\lambda_1, \dots, \lambda_k \geq 0$ adding up to 1. If $\lambda_k = 1$, then all the other weights are zero and the combination is $x_k \in C$; done. If $\lambda_k < 1$, define $\mu = 1 - \lambda_k = \lambda_1 + \dots + \lambda_{k-1} > 0$ and write

$$\sum_{i=1}^k \lambda_i x_i = \mu \underbrace{\sum_{i=1}^{k-1} \frac{\lambda_i}{\mu} x_i}_{=: y} + \lambda_k x_k.$$

Let us check that y is a convex combination of $k - 1$ points of C : the new weights λ_i/μ are nonnegative (a quotient of nonnegative numbers with positive denominator) and they add up to

$$\sum_{i=1}^{k-1} \frac{\lambda_i}{\mu} = \frac{1}{\mu} \sum_{i=1}^{k-1} \lambda_i = \frac{\mu}{\mu} = 1.$$

By the inductive hypothesis, $y \in C$. Finally, $\sum_i \lambda_i x_i = \mu y + \lambda_k x_k = (1 - \lambda_k)y + \lambda_k x_k$ is a convex combination of the *two* points $y, x_k \in C$, which belongs to C by the definition of convexity. This completes the induction. $\square$

Intuition

The trick in the previous proof appears everywhere in decision theory: a lottery over k prizes can be “taken apart” as a two stage lottery (first you decide between “prize k ” with probability λ_k and “the rest” with probability $1 - \lambda_k$; if “the rest” comes up, you play the reduced lottery over the remaining $k - 1$ prizes with renormalized probabilities $\lambda_i/(1 - \lambda_k)$). The reduction of compound lotteries axiom of expected utility is exactly this algebraic identity.

Definition 2.13 (Convex hull). The *convex hull* of a set $S \subseteq \mathbb{R}^n$, written $\text{conv } S$, is the set of all convex combinations of points of S :

$$\text{conv } S = \left\{ \sum_{i=1}^k \lambda_i x_i : k \in \mathbb{N}, x_i \in S, \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1 \right\}.$$

It is the smallest convex set containing S (you check this in Exercise 2.38). For example, $\Delta^{n-1} = \text{conv}\{e_1, \dots, e_n\}$, where e_i are the standard basis vectors: every lottery is a convex combination of the degenerate lotteries (those that give one outcome with certainty).

2.3 Affine sets and dimension [P2]

2.3.1 Lines instead of segments

If in the definition of convexity we replace the *segment* by the *whole line*, allowing λ outside $[0, 1]$, we get a more rigid class of sets.

Definition 2.14 (Affine set). A set $M \subseteq \mathbb{R}^n$ is *affine* if for all $x, y \in M$ and every $\lambda \in \mathbb{R}$ we have $(1 - \lambda)x + \lambda y \in M$: the line through any two points of M is contained in M . An *affine combination* of $x_1, \dots, x_k$ is $\sum_i \lambda_i x_i$ with $\sum_i \lambda_i = 1$ but *without* requiring $\lambda_i \geq 0$.

Every affine set is convex (if it contains the line, it contains in particular the segment), but not the other way around: the ball is convex and is not affine. The typical examples of affine sets are: all of $\mathbb{R}^n$, a point, a line, a plane in $\mathbb{R}^3$, and in general the hyperplanes $H = \{x : \langle a, x \rangle = b\}$ (repeat the computation of Example 2.7 with $\lambda \in \mathbb{R}$: the equality is preserved for any λ , not only for λ in $[0, 1]$).

Recall from the preliminaries chapter that a *linear subspace* $V \subseteq \mathbb{R}^n$ is a set that contains 0 and is closed under sums and scalar multiplication. The relation between affine sets and subspaces is exactly the one suggested by geometry: an affine set is a “shifted” subspace that no longer has to pass through the origin.

Theorem 2.15 (Affine = translated subspace). *A nonempty set $M \subseteq \mathbb{R}^n$ is affine if and only if there are a linear subspace $V \subseteq \mathbb{R}^n$ and a point $x_0 \in M$ such that*

$$M = x_0 + V := \{x_0 + v : v \in V\}.$$

Moreover, the subspace $V = M - x_0 = \{x - x_0 : x \in M\}$ is the same no matter which point $x_0 \in M$ is chosen; it is called the *direction subspace* of M .

Proof. ($\Leftarrow$) Assume $M = x_0 + V$ with V a subspace. Take $x = x_0 + v$, $y = x_0 + w$ with $v, w \in V$, and $\lambda \in \mathbb{R}$. Then

$$(1 - \lambda)x + \lambda y = (1 - \lambda)(x_0 + v) + \lambda(x_0 + w) = \underbrace{[(1 - \lambda) + \lambda]}_{=1} x_0 + \underbrace{(1 - \lambda)v + \lambda w}_{\in V},$$

where the vector $(1 - \lambda)v + \lambda w$ belongs to V because V is closed under linear combinations. Hence $(1 - \lambda)x + \lambda y \in x_0 + V = M$, and M is affine.

($\Rightarrow$) Assume M is affine, fix any $x_0 \in M$ and define $V = M - x_0$. We must check that V is a subspace; we verify the three properties.

(i) $0 \in V$: because $0 = x_0 - x_0$ and $x_0 \in M$.

(ii) *Closed under scalars*: let $v = x - x_0 \in V$ (with $x \in M$) and $\alpha \in \mathbb{R}$. We want to see that $\alpha v \in V$, that is, that $x_0 + \alpha v \in M$. We write

$$x_0 + \alpha v = x_0 + \alpha(x - x_0) = (1 - \alpha)x_0 + \alpha x,$$

which is a point of the line through $x_0, x \in M$; since M is affine, it belongs to M . Done.

(iii) *Closed under sums*: let $v = x - x_0$ and $w = y - x_0$ be in V , with $x, y \in M$. We want $v + w \in V$, that is, $x_0 + v + w \in M$. The trick is to go through the midpoint and then stretch.

First, the midpoint $z = \frac{1}{2}x + \frac{1}{2}y$ is in M (it is an affine combination with $\lambda = 1/2$). Now note that

$$x_0 + v + w = x + y - x_0 = 2z - x_0 = (1 - 2)x_0 + 2z,$$

which is an affine combination of $x_0, z \in M$ (the coefficients are -1 and 2 , adding up to 1 ; here it is essential that affine combinations allow negative weights). Since M is affine, $x_0 + v + w \in M$, that is, $v + w \in V$.

By (i)–(iii), V is a subspace and $M = x_0 + V$. *Uniqueness of V* : if $x_1 \in M$ is another point, then for each $x \in M$, $x - x_1 = (x - x_0) - (x_1 - x_0)$ is the difference of two elements of V , hence it is in V ; this gives $M - x_1 \subseteq V$, and the symmetric argument gives the other inclusion. So $M - x_1 = M - x_0 = V$. $\square$

Definition 2.16 (Affine hull and dimension). The *affine hull* of $S \subseteq \mathbb{R}^n$, written $\text{aff } S$, is the set of all affine combinations of points of S ; it is the smallest affine set that contains S . The *dimension* of an affine set $M = x_0 + V$ is the dimension of the direction subspace V (well defined by Theorem 2.15). The *dimension of a convex set C* is defined as $\dim C = \dim(\text{aff } C)$.

Example 2.17. Let us compute $\dim \Delta^2$, the simplex of lotteries over 3 outcomes, which lives in $\mathbb{R}^3$. Its affine hull is the hyperplane $H = \{\pi \in \mathbb{R}^3 : \pi_1 + \pi_2 + \pi_3 = 1\}$: indeed, every affine combination of lotteries still adds up to 1 (same computation as in Example 2.10), so $\text{aff } \Delta^2 \subseteq H$; and conversely you can check that every point of H is an affine combination of the vertices e_1, e_2, e_3 (you do this in Exercise 2.35). Taking $x_0 = e_1 = (1, 0, 0) \in H$, the direction subspace is $V = H - e_1 = \{v \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 0\}$, which has dimension 2 (one independent linear equation in $\mathbb{R}^3$ leaves $3 - 1 = 2$ degrees of freedom). Therefore $\dim \Delta^2 = 2$: the three outcome simplex is a two dimensional triangle “floating” in $\mathbb{R}^3$. In general, $\dim \Delta^{n-1} = n - 1$, and that is the reason for the superscript $n - 1$ in the notation. Keep this example: it will be the main character when we talk about relative interior in Section 2.5.

2.4 Operations that preserve convexity [P1]

In practice we almost never check convexity “by hand” with the definition: we recognize the set as the result of applying operations that preserve convexity to basic blocks (balls, half spaces, hyperplanes). This is the “convex set calculus” of [2], section 2.3. We prove every result of this section in full: they are short proofs and they are exactly the kind of exam question you will get.

Theorem 2.18 (Arbitrary intersection). *Let $\{C_i\}_{i \in I}$ be any family (finite or infinite) of convex sets in $\mathbb{R}^n$. Then the intersection $C = \bigcap_{i \in I} C_i$ is convex.*

Proof. Let $x, y \in C$ and $\lambda \in [0, 1]$. By the definition of intersection, x and y belong to C_i for every $i \in I$. Fix an arbitrary i : since C_i is convex and $x, y \in C_i$, we have $(1 - \lambda)x + \lambda y \in C_i$. Since this holds for every $i \in I$, the point $(1 - \lambda)x + \lambda y$ belongs to all the C_i at once, that is, to C . Hence C is convex. (Note that we never used the fact that I is finite: the result holds for intersections of infinite families, even uncountable ones.) $\square$

Careful (common mistake)

The **union** of convex sets is **not** convex in general. We already saw this in Example 2.11(b): the union of two disjoint balls does not contain the segment joining their centers. A frequent exam mistake is to “prove” that a union is convex by taking x, y in the *same* C_i ; the definition requires considering also $x \in C_1, y \in C_2$ with $C_1 \neq C_2$, and that is where the argument breaks. Another symmetric mistake: believing that the intersection can fail; it cannot, the intersection of convex sets is *always* convex (even if it is empty: the empty set is convex by convention). Rule of thumb: intersecting *cuts* (and cutting with linear or convex conditions does not create holes); taking unions *glues separate pieces* (and gluing can create holes).

Definition 2.19 (Minkowski sum and scaling). Given $A, B \subseteq \mathbb{R}^n$ and $\alpha \in \mathbb{R}$, we define the *Minkowski sum* and the *scaling* by

$$A + B = \{a + b : a \in A, b \in B\}, \quad \alpha A = \{\alpha a : a \in A\}.$$

Theorem 2.20 (Sum and scaling preserve convexity). *If $A, B \subseteq \mathbb{R}^n$ are convex and $\alpha \in \mathbb{R}$, then $A + B$ and αA are convex.*

Proof. Sum. Let $z_1, z_2 \in A + B$ and $\lambda \in [0, 1]$. By the definition of the sum, there are $a_1, a_2 \in A$ and $b_1, b_2 \in B$ such that $z_1 = a_1 + b_1$ and $z_2 = a_2 + b_2$. Then

$$(1 - \lambda)z_1 + \lambda z_2 = (1 - \lambda)(a_1 + b_1) + \lambda(a_2 + b_2) = \underbrace{[(1 - \lambda)a_1 + \lambda a_2]}_{\in A} + \underbrace{[(1 - \lambda)b_1 + \lambda b_2]}_{\in B},$$

where we have only distributed and regrouped: the terms with a together and the terms with b together. The first bracket is in A by convexity of A ; the second one is in B by convexity of B . Hence the combination is (element of A) + (element of B), which belongs to $A + B$ by definition.

Scaling. Let $\alpha a_1, \alpha a_2 \in \alpha A$ with $a_1, a_2 \in A$, and $\lambda \in [0, 1]$. Then

$$(1 - \lambda)\alpha a_1 + \lambda\alpha a_2 = \alpha[(1 - \lambda)a_1 + \lambda a_2] \in \alpha A,$$

because the bracket belongs to A by convexity. (This holds for any α , positive, negative or zero.) $\square$

Example 2.21 (Aggregation of technologies, MWG ch. 5). In an economy with J firms, the technology of firm j is described by its *production set* $Y_j \subseteq \mathbb{R}^n$: each $y \in Y_j$ is a production plan, with the sign convention of MWG [9] (ch. 5): negative coordinates are inputs and positive ones are outputs. The *aggregate* production set is what the economy as a whole can produce by splitting activity among the firms:

$$Y = Y_1 + Y_2 + \cdots + Y_J = \{y_1 + \cdots + y_J : y_j \in Y_j \text{ for each } j\},$$

exactly a Minkowski sum! By Theorem 2.20 (applied $J - 1$ times), if each Y_j is convex, the

aggregate Y is convex. This fact is what allows, in the welfare theorems of MWG ch. 16, treating “the production of the economy” as a single convex set and separating it with a price hyperplane.

Complete numerical example with one input and one output ($n = 2$; we write plans as (input, output)). Suppose two firms with bounded linear technologies:

$$Y_1 = \{(-t, t) : t \in [0, 1]\}, \quad Y_2 = \{(-s, 2s) : s \in [0, 1]\}.$$

Firm 1 turns each unit of input into 1 of output (up to 1 unit of input); firm 2 is more productive: it turns each unit of input into 2 of output (up to 1 unit). Each Y_j is a segment, hence convex. The sum is

$$Y_1 + Y_2 = \{(-t - s, t + 2s) : t \in [0, 1], s \in [0, 1]\}.$$

Let us compute some points: with $(t, s) = (0, 0)$ we get $(0, 0)$ (do nothing); with $(t, s) = (1, 0)$, the point $(-1, 1)$; with $(t, s) = (0, 1)$, the point $(-1, 2)$; with $(t, s) = (1, 1)$, the point $(-2, 3)$. The set $Y_1 + Y_2$ is the parallelogram with vertices $(0, 0)$, $(-1, 1)$, $(-1, 2)$, $(-2, 3)$: convex, as the theorem predicts. Economic note: using one unit of input in total, the aggregate economy can produce any amount between 1 (everything in firm 1) and 2 (everything in firm 2), and also any intermediate value by splitting the input; the *efficient frontier* of the aggregate uses the more productive technology first. The Minkowski sum does this “accounting” of splitting activity among firms automatically.

Theorem 2.22 (Cartesian product). *If $C \subseteq \mathbb{R}^n$ and $D \subseteq \mathbb{R}^m$ are convex, then the Cartesian product $C \times D = \{(x, u) \in \mathbb{R}^{n+m} : x \in C, u \in D\}$ is convex in $\mathbb{R}^{n+m}$.*

Proof. Let $(x_1, u_1), (x_2, u_2) \in C \times D$ and $\lambda \in [0, 1]$. The convex combination is computed coordinate by coordinate (by blocks):

$$(1 - \lambda)(x_1, u_1) + \lambda(x_2, u_2) = ((1 - \lambda)x_1 + \lambda x_2, (1 - \lambda)u_1 + \lambda u_2).$$

The first block belongs to C (convexity of C , since $x_1, x_2 \in C$) and the second one to D (convexity of D). Hence the pair belongs to $C \times D$. $\square$

Definition 2.23 (Affine map). A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *affine* if $T(x) = Ax + b$ for some $m \times n$ matrix A and some vector $b \in \mathbb{R}^m$. (If $b = 0$, T is linear.) The key property, which we will use constantly, is that affine functions *respect convex combinations*: if $\lambda \in [0, 1]$,

$$T((1 - \lambda)x + \lambda y) = A((1 - \lambda)x + \lambda y) + b = (1 - \lambda)Ax + \lambda Ay + \underbrace{[(1 - \lambda) + \lambda]}_{=1} b = (1 - \lambda)T(x) + \lambda T(y).$$

The middle step uses the linearity of the matrix vector product and the by now familiar trick of writing $b = (1 - \lambda)b + \lambda b$.

Theorem 2.24 (Image and preimage under affine maps). *Let $T(x) = Ax + b$ be an affine map from $\mathbb{R}^n$ to $\mathbb{R}^m$.*

- (a) *If $C \subseteq \mathbb{R}^n$ is convex, its image $T(C) = \{T(x) : x \in C\}$ is convex in $\mathbb{R}^m$.*
- (b) *If $D \subseteq \mathbb{R}^m$ is convex, its preimage $T^{-1}(D) = \{x \in \mathbb{R}^n : T(x) \in D\}$ is convex in $\mathbb{R}^n$.*

Proof. (a) Let $v_1, v_2 \in T(C)$ and $\lambda \in [0, 1]$. There are $x_1, x_2 \in C$ with $v_1 = T(x_1)$, $v_2 = T(x_2)$. Using the property of Definition 2.23 “backwards”:

$$(1 - \lambda)v_1 + \lambda v_2 = (1 - \lambda)T(x_1) + \lambda T(x_2) = T((1 - \lambda)x_1 + \lambda x_2).$$

The point $(1 - \lambda)x_1 + \lambda x_2$ belongs to C by convexity, so its image under T belongs to $T(C)$. Hence $T(C)$ is convex.

(b) Let $x_1, x_2 \in T^{-1}(D)$ and $\lambda \in [0, 1]$; this means $T(x_1) \in D$ and $T(x_2) \in D$. Then

$$T((1 - \lambda)x_1 + \lambda x_2) = (1 - \lambda)T(x_1) + \lambda T(x_2) \in D,$$

where the final membership holds by convexity of D applied to the points $T(x_1), T(x_2) \in D$. By the definition of preimage, $(1 - \lambda)x_1 + \lambda x_2 \in T^{-1}(D)$. $\square$

Remark 2.25. These results give a very powerful toolbox. Examples of immediate use: (i) the *projection* of a convex set onto some coordinates is convex, because projecting is applying the linear transformation that deletes the other coordinates; (ii) a set defined by linear constraints $\{x : Ax \leq c\}$ is convex because it is the preimage of the set $\{u : u \leq c\}$ (a translated orthant, convex because it is an intersection of half spaces) under the linear map $x \mapsto Ax$, which proves again that polyhedra are convex; (iii) the sum $A + B$ can be seen as the image of the product $A \times B$ (convex by Theorem 2.22) under the linear map $(a, b) \mapsto a + b$, which gives a second proof of Theorem 2.20. You can look at [2], section 2.3, and [3], ch. 2, for broader catalogs.

2.5 Topological aspects of convex sets [P2]

Let us recall from the preliminaries chapter the topological notions we will use: the *interior* $\text{int } C$ (points that have a small ball around them contained in C), the *closure* $\text{cl } C$ (points that are limits of sequences in C , equivalently, C together with its boundary) and the *boundary* $\text{bd } C = \text{cl } C \setminus \text{int } C$. The question of this section: do $\text{int } C$ and $\text{cl } C$ inherit the convexity of C ? The answer is yes, and the central tool is a lemma so useful that it has its own name.

Lemma 2.26 (Line segment lemma, accessibility of the interior). *Let $C \subseteq \mathbb{R}^n$ be convex. If $x \in \text{int } C$ and $y \in \text{cl } C$, then the whole segment except possibly the endpoint y is in the interior:*

$$[x, y) \subseteq \text{int } C,$$

that is, $(1 - \lambda)y + \lambda x \in \text{int } C$ for every $\lambda \in (0, 1]$.

Proof. Fix $\lambda \in (0, 1]$ and call $z = \lambda x + (1 - \lambda)y$ the point of the segment (note the order of the weights: weight λ on the interior point x). If $\lambda = 1$ then $z = x \in \text{int } C$ and there is nothing to prove; so assume $\lambda \in (0, 1)$. We want to find a radius $\rho > 0$ such that $B(z, \rho) \subseteq C$.

Available data. Since $x \in \text{int } C$, there is $r > 0$ with $B(x, r) \subseteq C$. Since $y \in \text{cl } C$, there are points of C as close to y as we want: for any $\delta > 0$ there is $y' \in C$ with $\|y' - y\| < \delta$. We will choose δ and ρ at the end; first let us see what we need.

The geometric idea. We want to write any point u close to z as a convex combination $u = \lambda x' + (1 - \lambda)y'$ with $x' \in B(x, r) \subseteq C$ and $y' \in C$; the convexity of C will then give $u \in C$.

Given u and given the point $y' \in C$ close to y , the equation $u = \lambda x' + (1 - \lambda)y'$ has a unique solution for x' (we solve for it):

$$x' = \frac{u - (1 - \lambda)y'}{\lambda}.$$

It only remains to check that this x' falls inside $B(x, r)$ if u is close enough to z and y' close enough to y .

The estimate. Let us compute $\|x' - x\|$. Since $z = \lambda x + (1 - \lambda)y$, we can solve for $x = (z - (1 - \lambda)y)/\lambda$. Subtracting the two expressions:

$$x' - x = \frac{u - (1 - \lambda)y'}{\lambda} - \frac{z - (1 - \lambda)y}{\lambda} = \frac{(u - z) + (1 - \lambda)(y - y')}{\lambda}.$$

Taking norms and applying the triangle inequality:

$$\|x' - x\| \leq \frac{\|u - z\| + (1 - \lambda)\|y - y'\|}{\lambda}.$$

Choice of the parameters. Choose

$$\rho = \frac{\lambda r}{2} \quad \text{and} \quad y' \in C \text{ with } \|y - y'\| < \frac{\lambda r}{2(1 - \lambda)}.$$

(Such a y' exists because $y \in \text{cl } C$.) Then, for any $u \in B(z, \rho)$:

$$\|x' - x\| < \frac{1}{\lambda} \left(\frac{\lambda r}{2} + (1 - \lambda) \cdot \frac{\lambda r}{2(1 - \lambda)} \right) = \frac{1}{\lambda} \left(\frac{\lambda r}{2} + \frac{\lambda r}{2} \right) = \frac{1}{\lambda} \cdot \lambda r = r.$$

So $x' \in B(x, r) \subseteq C$. Since also $y' \in C$ and $\lambda \in (0, 1)$, the convexity of C gives

$$u = \lambda x' + (1 - \lambda)y' \in C.$$

We have proved that *every* point u of the ball $B(z, \rho)$ belongs to C , that is, $B(z, \rho) \subseteq C$, so $z \in \text{int } C$. Since $\lambda \in (0, 1]$ was arbitrary, $[x, y) \subseteq \text{int } C$. $\square$

Intuition

Think of the interior point x as a *light source* surrounded by a free ball $B(x, r) \subseteq C$, and of y as a point on the edge (or almost). The lemma says that the “light cone” going from the ball towards y stays inside C : as we move from y towards x , even a little (λ small but positive), we immediately gain a margin of slack proportional to λr : the ball around x “shrinks” towards z but it does not vanish while $\lambda > 0$. This picture explains why the endpoint y has to be excluded: at $\lambda = 0$ the margin $\lambda r/2$ shrinks to zero. This lemma is the workhorse of the theory: from it we will get, in a later chapter, the separation theorems, which in turn are the heart of the second welfare theorem [9].

Theorem 2.27 (The interior and the closure of a convex set are convex). *If $C \subseteq \mathbb{R}^n$ is convex, then $\text{int } C$ and $\text{cl } C$ are convex.*

Proof. Interior. Let $x, y \in \text{int } C$ and $\lambda \in (0, 1)$ (the cases $\lambda \in \{0, 1\}$ are trivial because they return x or y). Since in particular $y \in \text{int } C \subseteq \text{cl } C$, Lemma 2.26 with interior point x and

closure point y gives $[x, y] \subseteq \text{int } C$, and the point $(1 - \lambda)x + \lambda y$ with $\lambda \in (0, 1)$ belongs to $[x, y]$. Hence $\text{int } C$ is convex.

Closure. Let $x, y \in \text{cl } C$ and $\lambda \in [0, 1]$. By the sequential characterization of the closure (preliminaries chapter), there are sequences (x_k) and (y_k) of points of C with $x_k \rightarrow x$ and $y_k \rightarrow y$. By convexity of C , each point $z_k = (1 - \lambda)x_k + \lambda y_k$ belongs to C . By the rules for limits (the limit of a linear combination is the linear combination of the limits):

$$z_k = (1 - \lambda)x_k + \lambda y_k \longrightarrow (1 - \lambda)x + \lambda y.$$

So $(1 - \lambda)x + \lambda y$ is the limit of a sequence of points of C , hence it belongs to $\text{cl } C$. Therefore $\text{cl } C$ is convex. $\square$

2.5.1 The relative interior

There is a practical problem with the “ordinary” interior: many important convex sets in economics have *empty* interior, not because they are degenerate, but because they are “flat” inside a space of higher dimension.

Example 2.28 (A segment in the plane has no interior). Let $S = \{(x_1, 0) : x_1 \in [0, 1]\} \subseteq \mathbb{R}^2$, a horizontal segment. Take any point $z = (z_1, 0) \in S$ and any radius $\varepsilon > 0$. The ball $B(z, \varepsilon)$ contains the point $(z_1, \varepsilon/2)$, whose second coordinate is not zero, so it does not belong to S . That is, *no* ball centered at a point of S fits inside S : $\text{int } S = \emptyset$. However, our intuition says that the “middle” points of the segment, such as $(1/2, 0)$, are qualitatively different from the endpoints $(0, 0)$ and $(1, 0)$. The ordinary interior does not capture that difference; the *relative* interior does. The same happens with the simplex $\Delta^{n-1} \subseteq \mathbb{R}^n$ (Example 2.17): since it lives inside the hyperplane $\sum_i \pi_i = 1$, its interior in $\mathbb{R}^n$ is empty, even though the lotteries with all probabilities positive are clearly “interior” in the relevant sense.

Definition 2.29 (Relative interior). The *relative interior* of a convex set $C \subseteq \mathbb{R}^n$, written $\text{ri } C$, is its interior seen inside its affine hull:

$$\text{ri } C = \{x \in C : \exists \varepsilon > 0 \text{ such that } B(x, \varepsilon) \cap \text{aff } C \subseteq C\}.$$

That is: instead of requiring a whole ball to fit inside C (impossible if C is flat), we require the *slice* of the ball that lives in the natural affine plane of C to fit.

Example 2.30. For the segment S of Example 2.28, the affine hull is the line $\text{aff } S = \{(x_1, 0) : x_1 \in \mathbb{R}\}$ (the horizontal axis). A point $z = (z_1, 0)$ with $z_1 \in (0, 1)$ is in $\text{ri } S$: taking $\varepsilon = \min\{z_1, 1 - z_1\} > 0$, the slice $B(z, \varepsilon) \cap \text{aff } S = \{(x_1, 0) : |x_1 - z_1| < \varepsilon\}$ stays inside S (since ε was chosen as the distance to the nearest endpoint). The endpoints are not in $\text{ri } S$: any slice around $(0, 0)$ contains points $(x_1, 0)$ with $x_1 < 0$, which are not in S . So,

$$\text{ri } S = \{(x_1, 0) : x_1 \in (0, 1)\} \neq \emptyset = \text{int } S.$$

Similarly, $\text{ri } \Delta^{n-1} = \{\pi \in \Delta^{n-1} : \pi_i > 0 \text{ for all } i\}$: the *completely mixed* lotteries. This set will show up in Microeconomics every time an argument requires “perturbing” a lottery in all directions without leaving the simplex.

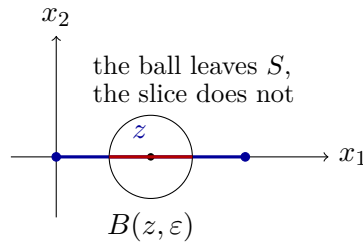


Figure 2.3: The segment $S \subseteq \mathbb{R}^2$ has empty interior: no two dimensional ball fits inside it. But the intersection of the ball with $\text{aff } S$ (the thick trace on the axis) does fit: $z \in \text{ri } S$.

The next result, which we state without a complete proof, says that for nonempty convex sets the relative interior is never empty and “fills” the set when we take the closure. It is the technical reason why the separation theorems at the end of the course are stated with ri and not with int .

Theorem 2.31 (Properties of the relative interior). *Let $C \subseteq \mathbb{R}^n$ be convex and nonempty. Then:*

- (a) $\text{ri } C \neq \emptyset$ (and moreover $\dim(\text{ri } C) = \dim C$);
- (b) $\text{cl}(\text{ri } C) = \text{cl } C$ and $\text{ri}(\text{cl } C) = \text{ri } C$;
- (c) the relative version of the line segment lemma holds: if $x \in \text{ri } C$ and $y \in \text{cl } C$, then $[x, y] \subseteq \text{ri } C$.

Proof sketch. The idea of (a): if $\dim C = d$, inside C you can choose $d + 1$ points $x_0, x_1, \dots, x_d$ that are “affinely independent” (that is, such that the vectors $x_1 - x_0, \dots, x_d - x_0$ are linearly independent; they exist because otherwise C would fit inside an affine set of smaller dimension). The average $\bar{x} = \frac{1}{d+1} \sum_i x_i$, called the *barycenter*, belongs to $\text{ri } C$: inside $\text{aff } C$ you can move from $\bar{x}$ a small distance in any direction without leaving the convex hull of the x_i , which is contained in C by Theorem 2.12. For (b) and (c), the argument is the same as in Lemma 2.26 and Theorem 2.27, working inside $\text{aff } C$ instead of $\mathbb{R}^n$ (all the balls are replaced by slices $B \cap \text{aff } C$). The inclusion $\text{cl}(\text{ri } C) \subseteq \text{cl } C$ is immediate ($\text{ri } C \subseteq C$); for the converse, given $y \in \text{cl } C$, take any $x \in \text{ri } C$ (it exists by (a)) and use (c): the points of the segment $[x, y]$ are in $\text{ri } C$ and converge to y as $\lambda \rightarrow 0$, hence $y \in \text{cl}(\text{ri } C)$. The detailed proofs are in [6], ch. 2 (theorems 2.26–2.27), and in [3], ch. 2. $\square$

Remark 2.32. Another topological fact that we will use often (and that connects with the Weierstrass theorem of the preliminaries chapter): the budget set $B(p, m)$ with $p \gg 0$ and the simplex Δ^{n-1} are, besides convex, *compact* (closed and bounded). Closed, because they are defined by weak inequalities ($\leq, \geq$) and equalities of continuous functions; bounded, because $0 \leq x_i \leq m/p_i$ in the first case and $0 \leq \pi_i \leq 1$ in the second. The combination “convex + compact” is the star hypothesis of the existence theorems (of equilibrium, of fixed points) in MWG chs. 15–17 [9].

2.6 Three economic constructions with convex sets [P1]

We close the chapter by developing in detail three constructions of Microeconomics that are, seen through the lens of this chapter, pure handling of convex sets.

2.6.1 The Edgeworth box as a product of convex sets

Consider a pure exchange economy with two consumers (A and B) and two goods, with aggregate endowments $\omega = (\omega_1, \omega_2) \gg 0$. An *allocation* for consumer A is a bundle $x^A = (x_1^A, x_2^A)$ with $0 \leq x_i^A \leq \omega_i$: consumer A cannot consume negative quantities nor more than what exists in the economy. The set of such bundles is the *Edgeworth box*:

$$E = [0, \omega_1] \times [0, \omega_2] \subseteq \mathbb{R}^2.$$

Each interval $[0, \omega_i]$ is a convex set of $\mathbb{R}$ (it is a segment), so by Theorem 2.22 the box E is convex (and it is compact, being a product of compact sets). The allocation of consumer B is determined by that of A through $x^B = \omega - x^A$, which is an affine transformation of x^A (with $A = -I$ and $b = \omega$ in the notation of Definition 2.23); by Theorem 2.24(a), the set of possible bundles for B is the affine image of E , again convex. This double reading, in which each point of the box is simultaneously a bundle for A (measured from its origin) and one for B (measured from the opposite origin), is the central graphical device of MWG ch. 15 [9].

2.6.2 Convex preferences and upper contour sets

Definition 2.33 (Convex preferences). Let $\succsim$ be a preference relation on a convex $X \subseteq \mathbb{R}^n$. For each bundle x , the *upper contour set* is

$$U(x) = \{y \in X : y \succsim x\},$$

the bundles at least as good as x . We say that $\succsim$ is *convex* if $U(x)$ is a convex set for every $x \in X$. If preferences are represented by a utility function u , then $U(x) = \{y \in X : u(y) \geq u(x)\}$, and the convexity of all these sets is exactly the definition of u being *quasiconcave*, the connection that we will develop in the chapter on quasiconcavity.

Reading: taste for diversification. If $y \succsim x$ and $z \succsim x$ (two bundles at least as good as x), the convexity of $U(x)$ says that any mixture $(1-\lambda)y + \lambda z$ is also at least as good as x : averages are not worse than the extremes. *Worked example.* Let $u(x_1, x_2) = \min\{x_1, x_2\}$ (Leontief preferences, perfect complements) on $X = \mathbb{R}_+^2$. Let us prove that $U(x) = \{y \geq 0 : \min\{y_1, y_2\} \geq c\}$ is convex, where $c = \min\{x_1, x_2\}$. First we rewrite the set without the minimum: $\min\{y_1, y_2\} \geq c$ is equivalent to requiring *both* inequalities $y_1 \geq c$ and $y_2 \geq c$ (the minimum of two numbers is $\geq c$ exactly when both are). Therefore

$$U(x) = \{y : y_1 \geq c\} \cap \{y : y_2 \geq c\} \cap \mathbb{R}_+^2,$$

an intersection of three half spaces (the first two with normal vectors e_1 and e_2 , and $\mathbb{R}_+^2$ which is the intersection of two more). By Theorem 2.18, $U(x)$ is convex, hence Leontief preferences are convex. Note the method: we did not touch the definition of convexity; we only *recognized* the set as an intersection of convex pieces. In MWG chs. 16–17 [9], the convexity of the preferences of *all* consumers (together with that of the production sets) is the hypothesis that allows applying the separating hyperplane theorem and proving the second welfare theorem; without it, there are Pareto optima that no price system can support.

2.6.3 The set of feasible allocations

In an exchange economy with I consumers and n goods, an *allocation* is a list $x = (x^1, \dots, x^I)$ of bundles, one per consumer, that is, a point of $\mathbb{R}^{nI}$. The allocation is *feasible* if nobody consumes negative quantities and the totals do not exceed the aggregate endowment $\omega \in \mathbb{R}^n$:

$$F = \left\{ (x^1, \dots, x^I) \in \mathbb{R}^{nI} : x^i \geq 0 \text{ for all } i, \sum_{i=1}^I x^i \leq \omega \right\}.$$

Claim: F is convex. We can see this in two ways. *Via operations:* the conditions $x_j^i \geq 0$ are nI half spaces in $\mathbb{R}^{nI}$, and each of the n resource conditions $\sum_i x_j^i \leq \omega_j$ is also a half space (the function $x \mapsto \sum_i x_j^i$ is linear in the long vector x); F is the intersection of all of them, convex by Theorem 2.18. *Via the definition,* for practice: if $x = (x^1, \dots, x^I)$ and $\tilde{x} = (\tilde{x}^1, \dots, \tilde{x}^I)$ are feasible and $\lambda \in [0, 1]$, then the mixture assigns to each consumer i the bundle $(1-\lambda)x^i + \lambda\tilde{x}^i \geq 0$, and adding over consumers:

$$\sum_{i=1}^I [(1-\lambda)x^i + \lambda\tilde{x}^i] = (1-\lambda) \sum_{i=1}^I x^i + \lambda \sum_{i=1}^I \tilde{x}^i \leq (1-\lambda)\omega + \lambda\omega = \omega,$$

where the inequality is coordinate by coordinate and uses the feasibility of x and $\tilde{x}$. The convexity of F (together with its compactness) is a permanent ingredient of the existence arguments and of the welfare theorems in MWG chs. 16–17 [9]; in economies with production, the aggregate set $Y = \sum_j Y_j$ of Example 2.21 is added to F , again via a Minkowski sum.

2.7 Summary and solved exercises [P1]

Chapter summary

- A set C is **convex** if it contains the segment between any pair of its points: $(1-\lambda)x + \lambda y \in C$ for $x, y \in C$, $\lambda \in [0, 1]$. Equivalently (Theorem 2.12), it contains all finite convex combinations of its points. Convex combination = weighted average = lottery.
- **Basic examples:** balls (via the triangle inequality), hyperplanes and half spaces (via linearity of $\langle a, \cdot \rangle$), polyhedra, the budget set $B(p, m)$, the lottery simplex Δ^{n-1} . **Non examples:** the annulus, the union of disjoint balls.
- An **affine** set contains the *line* through each pair of points; it is exactly a translated subspace, $M = x_0 + V$ (Theorem 2.15). $\dim C := \dim(\text{aff } C)$.
- **Operations that preserve convexity:** arbitrary intersections, the Minkowski sum $A + B$ (aggregation of technologies, MWG ch. 5), scaling αA , the Cartesian product $C \times D$ (Edgeworth box), image and preimage under affine maps. The **union does not** preserve convexity.
- **Topology:** if C is convex, $\text{int } C$ and $\text{cl } C$ are convex; the key piece is the *line segment lemma*: $x \in \text{int } C$, $y \in \text{cl } C$ imply $[x, y] \subseteq \text{int } C$ (Lemma 2.26). When $\text{int } C = \emptyset$ because C is “flat”, the correct substitute is the **relative interior** $\text{ri } C$ (interior inside

aff C), which is never empty for a nonempty convex C , and satisfies $\text{cl}(\text{ri } C) = \text{cl } C$ (Theorem 2.31).

- **In Microeconomics:** convex preferences = convex upper contour sets (taste for diversification); feasible allocations and the Edgeworth box are convex and compact; the welfare theorems separate convex sets with price hyperplanes (MWG chs. 15–18 [9]; see also [10], ch. 5).

Exercise 2.34. Let $C = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0, x_2 > 0, x_1 x_2 \geq 1\}$, the upper contour set at level 1 of the Cobb–Douglas utility $u(x_1, x_2) = x_1 x_2$. Prove that C is convex using the definition directly.

Solution. Let $a = (a_1, a_2)$ and $b = (b_1, b_2)$ be in C , and $\lambda \in [0, 1]$; write $z = (1 - \lambda)a + \lambda b$. *Positivity:* $z_1 = (1 - \lambda)a_1 + \lambda b_1 > 0$ because it is a combination of positive numbers with weights that are not both zero (if $\lambda \in \{0, 1\}$, z is a or b and there is nothing to prove; we assume $\lambda \in (0, 1)$). Similarly $z_2 > 0$. *The product inequality:* we must prove $z_1 z_2 \geq 1$. We expand the product with the distributive property:

$$z_1 z_2 = [(1 - \lambda)a_1 + \lambda b_1][(1 - \lambda)a_2 + \lambda b_2] = (1 - \lambda)^2 a_1 a_2 + \lambda^2 b_1 b_2 + \lambda(1 - \lambda)(a_1 b_2 + a_2 b_1).$$

We know that $a_1 a_2 \geq 1$ and $b_1 b_2 \geq 1$. For the cross term we use the inequality between the arithmetic and geometric means (AM–GM: for $u, v \geq 0$, $u + v \geq 2\sqrt{uv}$, which follows from $(\sqrt{u} - \sqrt{v})^2 \geq 0$ by expanding the square):

$$a_1 b_2 + a_2 b_1 \geq 2\sqrt{a_1 b_2 \cdot a_2 b_1} = 2\sqrt{(a_1 a_2)(b_1 b_2)} \geq 2\sqrt{1 \cdot 1} = 2.$$

Substituting the three bounds into the expansion:

$$z_1 z_2 \geq (1 - \lambda)^2 \cdot 1 + \lambda^2 \cdot 1 + \lambda(1 - \lambda) \cdot 2 = [(1 - \lambda) + \lambda]^2 = 1,$$

where in the next to last step we recognized the expansion of the binomial $(u + v)^2 = u^2 + 2uv + v^2$ with $u = 1 - \lambda$, $v = \lambda$. Hence $z \in C$ and C is convex. (This is a first look at the *quasiconcavity* of the Cobb–Douglas, the topic of the corresponding chapter.) $\square$

Exercise 2.35. (a) Prove that $\text{aff } \Delta^{n-1} = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 1\}$ and conclude that $\dim \Delta^{n-1} = n - 1$. (b) Describe $\text{ri } \Delta^{n-1}$ and justify that the uniform lottery $\bar{\pi} = (1/n, \dots, 1/n)$ belongs to it.

Solution. (a) Call $H = \{x : \sum_i x_i = 1\}$. *Inclusion $\subseteq$:* every affine combination $\sum_k \mu_k \pi^{(k)}$ of lotteries (with $\sum_k \mu_k = 1$, weights of any sign) satisfies

$$\sum_{i=1}^n \left(\sum_k \mu_k \pi_i^{(k)} \right) = \sum_k \mu_k \left(\sum_{i=1}^n \pi_i^{(k)} \right) = \sum_k \mu_k \cdot 1 = 1,$$

(we swapped the order of the finite sums, which is always allowed), so $\text{aff } \Delta^{n-1} \subseteq H$. *Inclusion $\supseteq$:* let $x \in H$. We write x as an affine combination of the vertices $e_1, \dots, e_n \in \Delta^{n-1}$: taking

$\mu_i = x_i$, we have $\sum_i \mu_i e_i = x$ (since e_i has a 1 in coordinate i) and $\sum_i \mu_i = \sum_i x_i = 1$; the μ_i may be negative, but that is allowed in an affine combination. Hence $x \in \text{aff } \Delta^{n-1}$. With $x_0 = e_1$, the direction subspace is $V = H - e_1 = \{v : \sum_i v_i = 0\}$, defined by one nontrivial linear equation in $\mathbb{R}^n$, hence $\dim V = n - 1$ and $\dim \Delta^{n-1} = n - 1$.

(b) $\text{ri } \Delta^{n-1} = \{\pi \in \Delta^{n-1} : \pi_i > 0 \ \forall i\}$. Justification for $\bar{\pi}$: we must find $\varepsilon > 0$ such that every $x \in H$ with $\|x - \bar{\pi}\| < \varepsilon$ belongs to Δ^{n-1} . If $\|x - \bar{\pi}\| < \varepsilon$ then each coordinate satisfies $|x_i - 1/n| \leq \|x - \bar{\pi}\| < \varepsilon$ (the absolute value of a coordinate never exceeds the Euclidean norm), from which $x_i > 1/n - \varepsilon$. It is enough to choose $\varepsilon = 1/n$ to guarantee $x_i > 0$ for all i ; and the condition $\sum_i x_i = 1$ is already given by $x \in H$. Hence the slice $B(\bar{\pi}, 1/n) \cap H$ is contained in Δ^{n-1} , that is, $\bar{\pi} \in \text{ri } \Delta^{n-1}$. The same argument, with $\varepsilon = \min_i \pi_i$, works for any lottery with all probabilities positive. $\square$

Exercise 2.36. Let $Y_1 = \{(-t, t) : t \in [0, 2]\}$ and $Y_2 = \{(-s, 3s) : s \in [0, 1]\}$ be the production sets of two firms (input, output). (a) Compute $Y_1 + Y_2$ explicitly and draw it (or describe it with precision). (b) What is the maximum output attainable by the aggregate economy if it has 2 units of input in total? Justify using the set $Y_1 + Y_2$.

Solution. (a) By the definition of the Minkowski sum,

$$Y_1 + Y_2 = \{(-t - s, t + 3s) : t \in [0, 2], s \in [0, 1]\}.$$

It is the image of the rectangle $[0, 2] \times [0, 1]$ (convex, a product of segments) under the linear map $(t, s) \mapsto (-t - s, t + 3s)$, hence it is convex by Theorem 2.24(a): a parallelogram (possibly degenerate; here it is not, because the generating vectors $(-1, 1)$ and $(-1, 3)$ are linearly independent, since $(-1) \cdot 3 - 1 \cdot (-1) = -2 \neq 0$). Its vertices are the images of the vertices of the rectangle: $(t, s) = (0, 0) \mapsto (0, 0)$; $(2, 0) \mapsto (-2, 2)$; $(0, 1) \mapsto (-1, 3)$; $(2, 1) \mapsto (-3, 5)$.

(b) Having 2 units of input means restricting to the aggregate plans with first coordinate ≥ -2 , that is, $t + s \leq 2$. We want to maximize output $t + 3s$ subject to $t \in [0, 2]$, $s \in [0, 1]$, $t + s \leq 2$. Since the coefficient of s in the objective (3) exceeds that of t (1), it is better to use firm 2 first at full capacity: $s = 1$; this leaves $t \leq 2 - s = 1$, and we take $t = 1$. The maximum output is $t + 3s = 1 + 3 = 4$, with aggregate plan $(-2, 4)$. Reading: aggregation via Minkowski lets you read the productive efficiency of the economy directly from the boundary of the sum set, just as in MWG ch. 5 [9]. $\square$

Exercise 2.37. Let $C \subseteq \mathbb{R}^n$ and let $\alpha, \beta \geq 0$. (a) Prove that if C is convex, then $\alpha C + \beta C = (\alpha + \beta)C$. (b) Give a counterexample showing that the equality can fail if C is not convex. (c) Which inclusion always holds, whether or not C is convex?

Solution. (c) *first (the easy inclusion):* $(\alpha + \beta)C \subseteq \alpha C + \beta C$ always holds: if $x \in C$, then $(\alpha + \beta)x = \alpha x + \beta x$, which is an element of αC plus an element of βC (the same x in both!).

(a) What is missing is the inclusion $\alpha C + \beta C \subseteq (\alpha + \beta)C$ when C is convex. If $\alpha + \beta = 0$, then $\alpha = \beta = 0$ and both sides are $\{0\}$ (if $C \neq \emptyset$); assume $\alpha + \beta > 0$. A generic element of the left hand side is $\alpha x + \beta y$ with $x, y \in C$. We rewrite it factoring out $\alpha + \beta$:

$$\alpha x + \beta y = (\alpha + \beta) \left(\frac{\alpha}{\alpha + \beta} x + \frac{\beta}{\alpha + \beta} y \right).$$

The weights $\frac{\alpha}{\alpha+\beta}$ and $\frac{\beta}{\alpha+\beta}$ are nonnegative and add up to $\frac{\alpha+\beta}{\alpha+\beta} = 1$: the parenthesis is a convex combination of $x, y \in C$, which belongs to C by convexity. Hence $\alpha x + \beta y \in (\alpha + \beta)C$.

(b) Take $C = \{0, 1\} \subseteq \mathbb{R}$ (two points: not convex, since $1/2 \notin C$) and $\alpha = \beta = 1$. Then $C + C = \{0 + 0, 0 + 1, 1 + 0, 1 + 1\} = \{0, 1, 2\}$, while $(\alpha + \beta)C = 2C = \{0, 2\}$. Since $1 \in C + C$ but $1 \notin 2C$, the equality fails. Moral: for nonempty sets, the identity $\alpha C + \beta C = (\alpha + \beta)C$ for all $\alpha, \beta \geq 0$ is a *characterization* of convexity. Indeed, if it holds for all pairs, taking $\alpha = \lambda$ and $\beta = 1 - \lambda$ with $\lambda \in [0, 1]$ we get $\lambda x + (1 - \lambda)y \in (\alpha + \beta)C = C$ for all $x, y \in C$, which is the definition. *Careful*: the case $\alpha = \beta = \frac{1}{2}$ is not enough. The inclusion $\frac{1}{2}C + \frac{1}{2}C \subseteq C$ says only that C is convex *at the midpoint*, and that does not imply convexity without an additional hypothesis (for example, that C is closed): the set $C = \mathbb{Q} \subseteq \mathbb{R}$ satisfies $\frac{1}{2}\mathbb{Q} + \frac{1}{2}\mathbb{Q} = \mathbb{Q}$ and is not convex, since $\frac{\sqrt{2}}{2} \cdot 0 + (1 - \frac{\sqrt{2}}{2}) \cdot 1 \notin \mathbb{Q}$. Iterating the midpoint only reaches dyadic weights $\lambda (k/2^j)$; to go from those to all of $[0, 1]$ a limit argument is needed. $\square$

Exercise 2.38. Prove that $\text{conv } S$ (Definition 2.13) is the smallest convex set containing S ; that is: (a) $\text{conv } S$ is convex and contains S ; (b) if D is convex and $S \subseteq D$, then $\text{conv } S \subseteq D$.

Solution. (a) $S \subseteq \text{conv } S$: each $x \in S$ is the trivial convex combination $1 \cdot x$. Convexity of $\text{conv } S$: let $u = \sum_{i=1}^k \lambda_i x_i$ and $v = \sum_{j=1}^l \mu_j y_j$ be two convex combinations of points of S , and $\theta \in [0, 1]$. Then

$$(1 - \theta)u + \theta v = \sum_{i=1}^k (1 - \theta)\lambda_i x_i + \sum_{j=1}^l \theta\mu_j y_j,$$

which is a combination of the $k + l$ points $x_1, \dots, x_k, y_1, \dots, y_l \in S$ with coefficients $(1 - \theta)\lambda_i \geq 0$ and $\theta\mu_j \geq 0$ whose sum is

$$(1 - \theta) \sum_{i=1}^k \lambda_i + \theta \sum_{j=1}^l \mu_j = (1 - \theta) \cdot 1 + \theta \cdot 1 = 1.$$

Hence $(1 - \theta)u + \theta v \in \text{conv } S$.

(b) Let D be convex with $S \subseteq D$. Every element of $\text{conv } S$ is a finite convex combination of points of S , in particular of points of D ; by Theorem 2.12, that combination belongs to D . So $\text{conv } S \subseteq D$. Combining (a) and (b): $\text{conv } S$ is convex, contains S , and is contained in any other convex set that contains S ; in that precise sense it is “the smallest”. (Equivalently, $\text{conv } S$ is the intersection of all the convex sets that contain S , an intersection that is convex by Theorem 2.18.) $\square$

Exercise 2.39. Let $C \subseteq \mathbb{R}^n$ be convex with $\text{int } C \neq \emptyset$. Prove, using Lemma 2.26, that $\text{cl}(\text{int } C) = \text{cl } C$. (Comment: this is the “ordinary interior” analogue of the identity $\text{cl}(\text{ri } C) = \text{cl } C$ of Theorem 2.31, and the argument of this exercise is exactly the one used there.)

Solution. *Inclusion $\subseteq$* : since $\text{int } C \subseteq C$, taking the closure preserves the inclusion (the closure is monotone: if $A \subseteq B$ then $\text{cl } A \subseteq \text{cl } B$, since every limit of points of A is a limit of points of B), so $\text{cl}(\text{int } C) \subseteq \text{cl } C$.

Inclusion $\supseteq$: let $y \in \text{cl } C$; we want to see that y is a limit of points of $\text{int } C$. Fix a point

$x \in \text{int } C$ (it exists by hypothesis). By Lemma 2.26, for each $\lambda \in (0, 1]$ the point

$$z_\lambda = \lambda x + (1 - \lambda)y$$

belongs to $\text{int } C$. Now take the sequence $\lambda_k = 1/k$ and compute the limit of z_{λ_k} as $k \rightarrow \infty$:

$$\|z_{\lambda_k} - y\| = \left\| \frac{1}{k}x + \left(1 - \frac{1}{k}\right)y - y \right\| = \left\| \frac{1}{k}(x - y) \right\| = \frac{1}{k} \|x - y\| \rightarrow 0,$$

where we simply grouped $(1 - \frac{1}{k})y - y = -\frac{1}{k}y$ and factored out $\frac{1}{k}$. So (z_{λ_k}) is a sequence of points of $\text{int } C$ that converges to y , hence $y \in \text{cl}(\text{int } C)$. Both inclusions give $\text{cl}(\text{int } C) = \text{cl } C$. Geometrically: a convex set with nonempty interior is the closure of its interior; it has no “hairs” or “antennas” of lower dimension glued to the main body (compare with [6], ch. 2, and [3], ch. 2). $\square$

Chapter 3

Conic structure, convex hull and affine hull [P2]

Week 2 of the course. Bibliography: Aragón et al. [1].

What is this used for in Microeconomics?

This chapter gives you three tools that come back in Microeconomics:

- **Individual decision making and mechanism design.** A *lottery* over a set of prizes is exactly a *convex combination* of those prizes, the basic object of choice under uncertainty (MWG ch. 6). The **Carathéodory theorem** (Theorem 3.15) says that any expected outcome in $\mathbb{R}^n$ reachable with a complicated lottery can be replicated with a lottery over at most $n + 1$ prizes. This is why optimal random mechanisms, in the welfare and incentives part (MWG ch. 23), can be searched for among “small” lotteries. Randomization in contracts and mechanisms lives, mathematically, inside the convex hull.
- **General equilibrium (MWG chs. 15–18).** A technology with *constant returns to scale* is a **cone** (Example 3.7); *free disposal* is modeled by adding the cone $-\mathbb{R}_+^n$ to the production set; and the **recession cone** (Section 3.6) shows up in the proofs of equilibrium existence, where you must guarantee that the sets of feasible allocations are bounded [9, ch. 17].
- **Extreme points.** Core allocations and optimal *bang-bang* contracts (all or nothing) are extreme points of convex sets: a linear objective on a compact convex set always attains its optimum at an extreme point (Theorem 3.21).

In the previous chapter we studied convex sets: those that contain the segment between any two of their points. This chapter studies three related structures. First, *cones*, sets closed under scaling of their points: they are the natural language of every “proportional” phenomenon, such as a technology with constant returns to scale. Second, the *hulls* (convex, affine and conic): given an arbitrary set S , which may be ugly (finite, disconnected, full of holes), the hull is the smallest “good” set containing it. Convexifying a set is a constant operation in economics (lotteries, convexification of technologies, the core of an economy). Third, *extreme points*: the

points of a convex set that are not an average of others. The Minkowski theorem will say that a compact convex set is fully reconstructed from them, and from there we get the most used result of the whole chapter: *a linear objective on a compact convex set is optimized at an extreme point.*

3.1 Cones and convex cones [P2]

Motivation: scaling without changing nature

Think of a firm whose technology can produce 1 ton of steel using 2 tons of ore and 1 unit of labor. If the technology has *constant returns to scale*, then it can also produce 10 tons using 20 of ore and 10 of labor: multiplying the production plan by any positive number gives another feasible plan. The set of feasible plans is then a set that “contains every ray that leaves the origin and passes through its points”. That is exactly the notion of a cone.

Definition 3.1 (Cone). A set $C \subseteq \mathbb{R}^n$ is a **cone** if it is closed under multiplication by positive scalars, that is, if

$$x \in C, \lambda > 0 \implies \lambda x \in C.$$

If in addition C is a convex set, we say that C is a **convex cone**.

Remark 3.2. Some texts, such as [2, ch. 2], require $\lambda \geq 0$ (so every nonempty cone contains the origin 0, since $0 \cdot x = 0$); others ask only for $\lambda > 0$. The difference is minor: with $\lambda > 0$, a cone may or may not contain the origin. In these notes we use $\lambda > 0$ and we will say explicitly when we need $0 \in C$. In the important examples ($\mathbb{R}_+^n$, the second-order cone, generated cones with $\lambda \geq 0$) the origin is always there.

Geometrically, a cone is a union of *rays* (half lines) starting at the origin: if $x \in C$, the whole ray $\{\lambda x : \lambda > 0\}$ is in C . A cone need not be convex: the union of the two coordinate axes of $\mathbb{R}^2$, $C = \{(t, 0) : t \in \mathbb{R}\} \cup \{(0, t) : t \in \mathbb{R}\}$, is a cone (scaling a point of an axis leaves it on that axis), but it is not convex: $(1, 0)$ and $(0, 1)$ are in C , but their midpoint $(1/2, 1/2)$ is not.

The following characterization is the central result about convex cones: it replaces the convexity condition (which involves combinations with weights λ and $1 - \lambda$) by two much more manageable algebraic conditions.

Proposition 3.3 (Characterization of convex cones). *A nonempty set $C \subseteq \mathbb{R}^n$ is a convex cone if and only if it is closed under sums and under multiplication by positive scalars:*

$$x, y \in C \implies x + y \in C \quad \text{and} \quad x \in C, \lambda > 0 \implies \lambda x \in C.$$

Proof. We prove the two implications separately.

($\Rightarrow$) Suppose C is a convex cone. Closure under positive scalars is the definition of a cone, so only closure under sums is left. Let $x, y \in C$. Since C is convex, it contains the convex combination with weights $\frac{1}{2}$ and $\frac{1}{2}$:

$$\frac{1}{2}x + \frac{1}{2}y \in C.$$

Since C is a cone, we can scale this point by $\lambda = 2 > 0$ and stay inside C :

$$2\left(\frac{1}{2}x + \frac{1}{2}y\right) = x + y \in C.$$

Notice the trick: convexity only gives *averages*, but the conic structure lets you scale the average up into the sum.

($\Leftarrow$) Suppose now that C is closed under sums and under positive scalars. The second property says directly that C is a cone. Let us see that it is convex. Take $x, y \in C$ and $\lambda \in (0, 1)$ (the cases $\lambda = 0$ and $\lambda = 1$ give y or x , which are already in C). Since $\lambda > 0$ and $1 - \lambda > 0$, closure under positive scalars gives

$$\lambda x \in C \quad \text{and} \quad (1 - \lambda)y \in C,$$

and closure under sums gives

$$\lambda x + (1 - \lambda)y \in C.$$

This is exactly the definition of convexity. □

Intuition

Proposition 3.3 says that a convex cone is as close to a vector subspace as you can get without being one: in a subspace you can add and scale by *any* scalar; in a convex cone you can add and scale only by *positive* scalars. This is why convex cones are the natural model for economic situations where quantities cannot be negative or where direction matters: producing twice as much makes sense, “producing -1 times” does not. In fact, a convex cone C that also satisfies $C = -C$ is exactly a vector subspace.

An immediate consequence (applying the proposition repeatedly) is that a convex cone is closed under **conic combinations**: if $x_1, \dots, x_k \in C$ and $\lambda_1, \dots, \lambda_k > 0$, then $\lambda_1 x_1 + \dots + \lambda_k x_k \in C$. Note the difference with convex combinations: here the weights are positive but they do *not* have to add up to 1.

Basic examples

Example 3.4 (The nonnegative orthant). The **nonnegative orthant**

$$\mathbb{R}_+^n = \{x \in \mathbb{R}^n : x_i \geq 0 \text{ for all } i = 1, \dots, n\}$$

is a convex cone. Let us check it with Proposition 3.3. If $x, y \in \mathbb{R}_+^n$, each coordinate of $x + y$ is $x_i + y_i \geq 0 + 0 = 0$, so $x + y \in \mathbb{R}_+^n$. If $\lambda > 0$, each coordinate of λx is $\lambda x_i \geq 0$, so $\lambda x \in \mathbb{R}_+^n$. In economics, $\mathbb{R}_+^n$ is the space of consumption bundles: nonnegative quantities of n goods.

Example 3.5 (The second-order cone). The **second-order cone**, also called the *Lorentz cone* or the *ice cream cone*, is

$$\mathcal{K}_n = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} : \|x\| \leq t\} \subseteq \mathbb{R}^{n+1},$$

where $\|\cdot\|$ is the Euclidean norm. In $\mathbb{R}^3$ (that is, $n = 2$), it is literally an ice cream cone resting on the origin and opening upward: at height t , the horizontal “slice” is the disk of radius t .

Let us check that it is a convex cone using Proposition 3.3.

- *Positive scalars*: if $\|x\| \leq t$ and $\lambda > 0$, then $\|\lambda x\| = \lambda \|x\| \leq \lambda t$ (we use the homogeneity of the norm, seen in the preliminaries), so $(\lambda x, \lambda t) \in \mathcal{K}_n$.
- *Sums*: if $\|x\| \leq t$ and $\|y\| \leq s$, the triangle inequality gives

$$\|x + y\| \leq \|x\| + \|y\| \leq t + s,$$

so $(x + y, t + s) \in \mathcal{K}_n$.

This cone is the basic piece of *second-order cone programming*, a class of optimization problems that generalizes linear programming [2, sec. 2.2.3].

Example 3.6 (Cone generated by a set). Given any nonempty set $S \subseteq \mathbb{R}^n$, the **generated cone** of S is the set of all nonnegative multiples of points of S :

$$\{\lambda x : x \in S, \lambda \geq 0\}.$$

For example, if $S = \{(1, 1)\} \subset \mathbb{R}^2$, the generated cone is the ray leaving the origin in the direction $(1, 1)$. If $S = \{(1, 0), (0, 1)\}$, the generated cone is the union of the two positive half axes: a cone, but *not* convex. To get a convex cone you also have to close under sums; that is the *conic hull* $\text{cone}(S)$ that we define in Section 3.2.

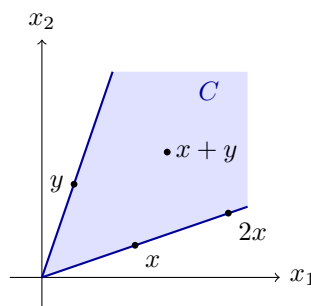


Figure 3.1: A convex cone $C \subseteq \mathbb{R}^2$ generated by two rays. It contains all positive multiples of its points ($2x$) and all sums of its points ($x + y$), as guaranteed by Proposition 3.3.

Economic example: constant returns to scale and free disposal

Example 3.7 (Technology with constant returns to scale). In producer theory [9, ch. 5], a technology is described by a **production set** $Y \subseteq \mathbb{R}^n$: each $y \in Y$ is a *production plan* where negative coordinates are inputs and positive ones are outputs. For example, $y = (-2, -1, 1)$ means “use 2 units of good 1 and 1 unit of good 2 to produce 1 unit of good 3”.

The technology has **constant returns to scale** if

$$y \in Y, \lambda \geq 0 \implies \lambda y \in Y,$$

that is, exactly when Y is a cone (containing the origin: producing nothing is feasible). If in addition Y is convex (the economic reading: production processes can be combined without loss), then Y is a convex cone and, by Proposition 3.3, two feasible plans can be *run in parallel*: $y, y' \in Y \Rightarrow y + y' \in Y$. This property, called *additivity*, is what makes maximum profit equal to zero in a competitive equilibrium with constant returns: if a plan y gave profit $p \cdot y > 0$, scaling it by $\lambda \rightarrow \infty$ would give unbounded profits, which is incompatible with the existence of a maximum [9, sec. 5.B].

Free disposal is the assumption that goods can always be thrown away at no cost: if $y \in Y$ and $y' \leq y$ coordinate by coordinate (it produces less or uses more inputs), then $y' \in Y$. In set notation:

$$Y - \mathbb{R}_+^n \subseteq Y, \quad \text{where } Y - \mathbb{R}_+^n = \{y - z : y \in Y, z \in \mathbb{R}_+^n\}.$$

That is, free disposal is equivalent to Y absorbing the cone $-\mathbb{R}_+^n$ under sums. This apparently harmless assumption is key in the proofs of general equilibrium existence [9, ch. 17]: it guarantees that equilibrium prices are nonnegative.

3.2 Convex, affine and conic hull [P2]

Motivation: the smallest convex set containing S

Many sets that appear in practice are not convex: a finite set of points (the possible prizes of a lottery), a technology with indivisibilities, the union of two regions. However, almost all the machinery of optimization and economic theory requires convexity. The standard way out is to *convexify*: replace S by the smallest convex set containing it. The pleasant surprise is that this abstract set (“the smallest convex set containing S ”) has a completely concrete description: it is the set of all convex combinations of points of S . Proving that equivalence is the main goal of this section.

Recall from the previous chapter: a **convex combination** of the points $x_1, \dots, x_k \in \mathbb{R}^n$ is any point of the form

$$\lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_k x_k, \quad \text{with } \lambda_i \geq 0 \text{ for all } i \text{ and } \sum_{i=1}^k \lambda_i = 1.$$

The weights λ_i have a natural reading as *probabilities*: a convex combination of prizes is the expected value of a lottery over those prizes.

Definition 3.8 (Convex hull). Let $S \subseteq \mathbb{R}^n$ be a nonempty set. The **convex hull** of S , written $\text{conv}(S)$, is the intersection of all convex sets containing S :

$$\text{conv}(S) = \bigcap \{D \subseteq \mathbb{R}^n : D \text{ is convex and } S \subseteq D\}.$$

Two comments so that the definition does not look like a trick. First, the family we intersect is not empty: $\mathbb{R}^n$ itself is convex and contains S . Second, in the previous chapter we proved that *the intersection of any family of convex sets is convex*; therefore $\text{conv}(S)$ is convex, contains S (all the intersected sets contain it), and is the *smallest* set with those properties: if D is convex

and $S \subseteq D$, then D is one of the sets in the family, so $\text{conv}(S) \subseteq D$.

The definition as an intersection is elegant but useless for computation. The next theorem gives the working description.

Theorem 3.9 (The convex hull is the set of convex combinations). *For every nonempty set $S \subseteq \mathbb{R}^n$,*

$$\text{conv}(S) = \left\{ \sum_{i=1}^k \lambda_i x_i : k \in \mathbb{N}, x_i \in S, \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1 \right\}.$$

Proof. Call M the set on the right (all convex combinations of finitely many points of S). We must prove two inclusions.

Step 1: M is convex and contains S ; therefore $\text{conv}(S) \subseteq M$.

That $S \subseteq M$ is immediate: each $x \in S$ is the “trivial” convex combination of a single point ($k = 1, \lambda_1 = 1$).

Let us see that M is convex. Take two points of M :

$$u = \sum_{i=1}^k \lambda_i x_i, \quad v = \sum_{j=1}^m \mu_j y_j,$$

with $x_i, y_j \in S$, nonnegative weights, $\sum_i \lambda_i = 1$ and $\sum_j \mu_j = 1$. Let $\theta \in [0, 1]$. Then

$$\theta u + (1 - \theta)v = \sum_{i=1}^k (\theta \lambda_i) x_i + \sum_{j=1}^m ((1 - \theta) \mu_j) y_j.$$

This expression is again a combination of points of S (the $k + m$ points $x_1, \dots, x_k, y_1, \dots, y_m$) with weights $\theta \lambda_i \geq 0$ and $(1 - \theta) \mu_j \geq 0$. Let us check that the weights add up to 1:

$$\sum_{i=1}^k \theta \lambda_i + \sum_{j=1}^m (1 - \theta) \mu_j = \theta \underbrace{\sum_{i=1}^k \lambda_i}_{=1} + (1 - \theta) \underbrace{\sum_{j=1}^m \mu_j}_{=1} = \theta + (1 - \theta) = 1.$$

So $\theta u + (1 - \theta)v \in M$, and M is convex. Since $\text{conv}(S)$ is the *smallest* convex set containing S , we conclude $\text{conv}(S) \subseteq M$.

Step 2: every convex set containing S contains M ; therefore $M \subseteq \text{conv}(S)$.

Let D be convex with $S \subseteq D$. We prove by induction on k that every convex combination of k points of S is in D .

Base ($k = 1$): the combination is a point of S , which is in D by hypothesis. (The case $k = 2$ is the definition of convexity of D .)

Inductive step: suppose the result holds for $k - 1$ points, and let

$$z = \sum_{i=1}^k \lambda_i x_i, \quad x_i \in S, \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1.$$

If $\lambda_k = 1$, then all the other weights are zero and $z = x_k \in D$. Suppose $\lambda_k < 1$, so that $1 - \lambda_k = \sum_{i=1}^{k-1} \lambda_i > 0$. The trick is to *regroup*: we separate the last term and renormalize the

first $k - 1$ weights so that they add up to 1:

$$z = (1 - \lambda_k) \underbrace{\sum_{i=1}^{k-1} \frac{\lambda_i}{1 - \lambda_k} x_i}_{=: w} + \lambda_k x_k.$$

The new weights $\lambda_i/(1 - \lambda_k)$ are nonnegative and add up to

$$\sum_{i=1}^{k-1} \frac{\lambda_i}{1 - \lambda_k} = \frac{1 - \lambda_k}{1 - \lambda_k} = 1,$$

so w is a convex combination of $k - 1$ points of S and, by the inductive hypothesis, $w \in D$. Finally, $z = (1 - \lambda_k)w + \lambda_k x_k$ is a convex combination of *two* points of D (w and x_k), so $z \in D$ by convexity of D . This closes the induction.

Applying Step 2 with $D = \text{conv}(S)$ (which is convex and contains S) we get $M \subseteq \text{conv}(S)$. The two inclusions give equality. $\square$

Intuition

Think of $\text{conv}(S)$ as “the set of everything reachable through lotteries over S ”. If S is the set of prizes that a mechanism can deliver (in money, in bundles, in utility vectors), then $\text{conv}(S)$ is the set of *expected values* that the designer can induce by randomizing between prizes. The proof of Theorem 3.9 also has a nice probabilistic reading: the inductive step says that a lottery over k prizes can be run in two stages: first you draw “ x_k or not x_k ” with probabilities λ_k and $1 - \lambda_k$, and if the answer is “not”, you play the conditional lottery over the other $k - 1$ prizes. Compound lotteries reduce to simple lotteries: this is the same argument as the reduction axiom in expected utility theory [9, ch. 6].

Example 3.10 (Convex hull of three points). Let $S = \{(0, 0), (2, 0), (0, 2)\} \subset \mathbb{R}^2$. By Theorem 3.9,

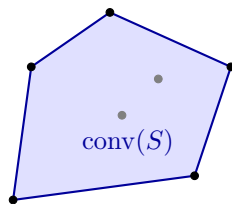
$$\begin{aligned} \text{conv}(S) &= \{\lambda_1(0, 0) + \lambda_2(2, 0) + \lambda_3(0, 2) : \lambda_i \geq 0, \lambda_1 + \lambda_2 + \lambda_3 = 1\} \\ &= \{(2\lambda_2, 2\lambda_3) : \lambda_2, \lambda_3 \geq 0, \lambda_2 + \lambda_3 \leq 1\}, \end{aligned}$$

which is the (filled) triangle with vertices at the three points: $\text{conv}(S) = \{(u, v) \in \mathbb{R}_+^2 : u + v \leq 2\}$. Let us check one concrete point: $(1, \frac{1}{2})$ is obtained with $\lambda_2 = \frac{1}{2}$, $\lambda_3 = \frac{1}{4}$, $\lambda_1 = \frac{1}{4}$. In general, the convex hull of finitely many points is a *polytope* (Section 3.5), and the weights λ_i are its “barycentric coordinates”.

Affine hull and dimension

The convex hull uses nonnegative weights that add up to 1. If we keep the restriction that they add up to 1 but *allow negative weights*, we get affine combinations and the affine hull, which already appeared in the previous chapter when we defined the relative interior. We collect the definition here to have the full picture.

Definition 3.11 (Affine hull and dimension). An **affine combination** of $x_1, \dots, x_k \in \mathbb{R}^n$ is a point $\sum_{i=1}^k \lambda_i x_i$ with $\sum_{i=1}^k \lambda_i = 1$ (weights of any sign). The **affine hull** of $S \subseteq \mathbb{R}^n$, written



S = the 7 points; its convex hull is the pentagon

Figure 3.2: The convex hull of a finite set S of points in $\mathbb{R}^2$. The five black points are the *extreme points* (Section 3.4); the two gray ones, being convex combinations of the others, add nothing to the hull.

$\text{aff}(S)$, is the smallest affine set containing S ; as in Theorem 3.9 (same proof, without the sign restriction), it coincides with the set of all affine combinations of points of S . The **dimension** of a convex set S is defined as the dimension of $\text{aff}(S)$ as a translate of a subspace: if $\text{aff}(S) = x_0 + V$ with V a vector subspace, then $\dim S = \dim V$.

Example 3.12. With $S = \{(0, 0), (2, 0), (0, 2)\}$ from Example 3.10: $\text{conv}(S)$ is the triangle, but $\text{aff}(S) = \mathbb{R}^2$ entirely, because with weights that may be negative you reach any point of the plane; for example, $(-2, 0) = 2 \cdot (0, 0) + (-1) \cdot (2, 0) + 0 \cdot (0, 2)$, with weights $2 - 1 + 0 = 1$. So the triangle has dimension 2. In contrast, if $S' = \{(0, 0), (2, 0)\}$, then $\text{conv}(S')$ is the segment between the two points, $\text{aff}(S')$ is the whole horizontal axis $\{(t, 0) : t \in \mathbb{R}\}$, and $\dim \text{conv}(S') = 1$: the segment is a one dimensional object even though it lives in $\mathbb{R}^2$. The affine hull is the “natural universe” of the set, and it is with respect to it that the relative interior $\text{ri}(S)$ of the previous chapter is defined.

Conic hull

Definition 3.13 (Conic hull). The **conic hull** of a nonempty set $S \subseteq \mathbb{R}^n$, written $\text{cone}(S)$, is the smallest convex cone containing S together with the origin. It coincides with the set of all *conic combinations* of points of S :

$$\text{cone}(S) = \left\{ \sum_{i=1}^k \lambda_i x_i : k \in \mathbb{N}, x_i \in S, \lambda_i \geq 0 \right\}.$$

(The proof of the equivalence copies that of Theorem 3.9, dropping the restriction $\sum_i \lambda_i = 1$; with all $\lambda_i = 0$ you get the origin.)

Example 3.14. $\text{cone}(\{(1, 0), (0, 1)\}) = \mathbb{R}_+^2$: every combination $\lambda_1(1, 0) + \lambda_2(0, 1) = (\lambda_1, \lambda_2)$ with $\lambda_1, \lambda_2 \geq 0$ sweeps exactly the orthant. Compare with Example 3.6: the *generated cone* of those two points (scaling only, no sums) was the union of the two half axes; the *conic hull* fills the whole quadrant. In general, $\text{cone}(S) = \{\lambda x : x \in \text{conv}(S), \lambda \geq 0\}$: first convexify, then scale.

Careful (common mistake)

Three frequent mistakes with hulls.

1. **Confusing the three families of weights.** Convex: $\lambda_i \geq 0$ and $\sum \lambda_i = 1$. Conic: $\lambda_i \geq 0$ with no restriction on the sum. Affine: $\sum \lambda_i = 1$ with no sign restriction. With both restrictions at once you get the smallest set (conv); with none, the largest one (the generated subspace, span). In an exam, identify *first* which restrictions the weights have.
2. **Believing that $\text{conv}(S)$ of a closed set is closed.** This is false in general: if $S = \{(t, 1/t) : t > 0\} \cup \{(0, 0)\} \subset \mathbb{R}^2$ (a branch of a hyperbola plus the origin), S is closed, but $\text{conv}(S)$ contains points $(t, 1/t)$ combined with $(0, 0)$ that accumulate on the open horizontal half axis $\{(t, 0) : t > 0\}$, which does *not* belong to $\text{conv}(S)$: the hull is not closed. What is true: the convex hull of a *compact* set is compact (Exercise 3.32).
3. **Forgetting the origin in $\text{cone}(S)$.** The conic hull always contains 0 (take all weights equal to zero), even if $0 \notin S$.

3.3 The Carathéodory theorem [P2]

Theorem 3.9 describes $\text{conv}(S)$ through combinations of *finitely many* points of S , but it does not say how many: in principle, a point of the hull could require combining a million points. The Carathéodory theorem gives a remarkably small bound that depends only on the dimension of the space: in $\mathbb{R}^n$ *you never need more than $n + 1$ points*. In $\mathbb{R}^2$, every point of $\text{conv}(S)$ is a combination of at most 3 points of S (it lies inside a triangle with vertices in S); in $\mathbb{R}^3$, of at most 4; and so on.

Theorem 3.15 (Carathéodory). *Let $S \subseteq \mathbb{R}^n$ be nonempty. Every point of $\text{conv}(S)$ can be written as a convex combination of at most $n + 1$ points of S .*

Proof. Let $z \in \text{conv}(S)$. By Theorem 3.9, there is *some* representation

$$z = \sum_{i=1}^k \lambda_i x_i, \quad x_i \in S, \quad \lambda_i > 0, \quad \sum_{i=1}^k \lambda_i = 1$$

(we may assume all weights are strictly positive: terms with weight zero are dropped without changing anything). Among all possible representations, take one with the *smallest* number of points k . The strategy is to prove: *if $k \geq n + 2$, then another representation with $k - 1$ points can be built*, contradicting minimality. From this it follows that $k \leq n + 1$.

Step 1: find an affine dependence. Suppose $k \geq n + 2$. Consider the $k - 1 \geq n + 1$ vectors

$$x_2 - x_1, x_3 - x_1, \dots, x_k - x_1 \in \mathbb{R}^n.$$

These are at least $n + 1$ vectors in a space of dimension n , so they are *linearly dependent*: there

are scalars $\mu_2, \dots, \mu_k$, not all zero, with

$$\sum_{i=2}^k \mu_i (x_i - x_1) = 0.$$

Setting $\mu_1 = -\sum_{i=2}^k \mu_i$, the previous equation can be rewritten as

$$\sum_{i=1}^k \mu_i x_i = 0 \quad \text{with} \quad \sum_{i=1}^k \mu_i = 0,$$

and not all the μ_i are zero. This is called an *affine dependence* among the x_i . Since the μ_i add up to zero and are not all zero, *at least one is strictly positive*.

Step 2: perturb the weights without moving the point. For any $t \in \mathbb{R}$, using $\sum_i \mu_i x_i = 0$:

$$z = \sum_{i=1}^k \lambda_i x_i - t \underbrace{\sum_{i=1}^k \mu_i x_i}_{=0} = \sum_{i=1}^k (\lambda_i - t\mu_i) x_i.$$

Moreover the new weights still add up to 1, because $\sum_i (\lambda_i - t\mu_i) = \sum_i \lambda_i - t \sum_i \mu_i = 1 - t \cdot 0 = 1$. In other words: we have a *family of representations* of z , one for each t , and all that is left is to choose t so that all the weights stay nonnegative and *one of them vanishes*.

Step 3: choose the best t . The weights $\lambda_i - t\mu_i$ with $\mu_i \leq 0$ do not decrease as $t \geq 0$ grows; those with $\mu_i > 0$ decrease linearly and at some moment hit zero. We choose t as the first instant at which some weight vanishes:

$$t^* = \min_{i: \mu_i > 0} \frac{\lambda_i}{\mu_i} > 0,$$

which is well defined because there is at least one $\mu_i > 0$ (Step 1), and positive because the λ_i are. Let j be an index where the minimum is attained. Then, for each i :

- if $\mu_i \leq 0$: $\lambda_i - t^* \mu_i \geq \lambda_i > 0 \geq 0$;
- if $\mu_i > 0$: $\lambda_i - t^* \mu_i = \mu_i \left(\frac{\lambda_i}{\mu_i} - t^* \right) \geq 0$, by the definition of t^* as the minimum of the ratios;
- and for $i = j$: $\lambda_j - t^* \mu_j = \mu_j \left(\frac{\lambda_j}{\mu_j} - \frac{\lambda_j}{\mu_j} \right) = 0$.

So $z = \sum_{i \neq j} (\lambda_i - t^* \mu_i) x_i$ is a convex combination of *at most* $k - 1$ points of S : we have removed the point x_j . This contradicts the minimality of k . Therefore $k \leq n + 1$. $\square$

Intuition

The proof is a *pruning algorithm*: while the lottery uses “too many” prizes ($k \geq n + 2$), its prizes are affinely dependent, and that dependence lets you shift probability from some prizes to others (t is how much you shift) *without moving the expected value* z , until the probability of some prize reaches exactly zero and that prize is removed. Repeating this, you go down to $n + 1$ prizes. This same scheme of “perturb along a direction of dependence until a variable vanishes” is the heart of the simplex method of linear programming and

will come back in the corresponding chapter. See [3, ch. 2] and [6, lemma 2.5] for other presentations of this proof.

The version for conic hulls is analogous and the bound improves by one, because the restriction $\sum_i \lambda_i = 1$ disappears (one equation less to satisfy).

Theorem 3.16 (Carathéodory, conic version). *Let $S \subseteq \mathbb{R}^n$ be nonempty. Every point of $\text{cone}(S)$ is a conic combination of at most n points of S .*

Proof idea. Repeat the previous argument with a minimal representation $z = \sum_{i=1}^k \lambda_i x_i$, $\lambda_i > 0$. If $k \geq n + 1$, the vectors $x_1, \dots, x_k$ (themselves, not their differences) are linearly dependent: $\sum_i \mu_i x_i = 0$ with some $\mu_i \neq 0$; changing the sign of all the μ_i if needed, one of them is positive. You no longer need $\sum_i \mu_i = 0$, because conic weights do not have to add up to 1. The same choice $t^* = \min_{\mu_i > 0} \lambda_i / \mu_i$ makes one weight vanish while keeping the others nonnegative, and you conclude $k \leq n$. See [3, ch. 2]. $\square$

Economic application: how many prizes a lottery needs

Example 3.17 (Reduction of lotteries in mechanism design). In mechanism design, the designer often gains from *randomizing*: instead of assigning a deterministic outcome, they assign a lottery over outcomes (think of auctions with random tie breaking, contracts with stochastic bonuses, or school seat assignment by draw). Let us use Carathéodory to formalize why that randomization never needs to be complicated.

Suppose each possible outcome is summarized by a vector in $\mathbb{R}^n$: for example, n could be the number of agents and each outcome $x \in S \subseteq \mathbb{R}^n$ the vector of utilities (or transfers) that this outcome generates for the agents. A lottery with finite support that assigns probability p_i to outcome $x_i \in S$, $i = 1, \dots, k$, induces the expected vector

$$\bar{x} = \sum_{i=1}^k p_i x_i, \quad p_i \geq 0, \quad \sum_{i=1}^k p_i = 1.$$

If the agents are risk neutral in these variables (the usual assumption when x are expected monetary transfers or assignment probabilities), all that matters about the mechanism is $\bar{x}$. Now, $\bar{x}$ is by construction a point of $\text{conv}(S)$, and the set of expected vectors implementable by *some* lottery is exactly $\text{conv}(S)$. Theorem 3.15 then says:

any expected vector reachable with an arbitrary lottery (even with huge support) is reachable with a lottery over at most $n + 1$ prizes.

Practical consequences:

1. **Simplicity without loss.** When looking for the optimal random mechanism, the designer can restrict attention, without loss of generality, to lotteries with support of size $n + 1$. The design problem goes from an optimization over a space of probability measures (infinite dimensional) to one over finitely many weights and prizes.

2. **With one extra expected constraint, one more prize.** If the lottery must also satisfy a constraint in expectation, for example a fixed expected cost $\sum_i p_i c(x_i) = \bar{c}$, it is enough to apply Carathéodory in $\mathbb{R}^{n+1}$ to the augmented vectors $(x_i, c(x_i))$: you replicate the expected vector and the expected cost at the same time with at most $(n+1) + 1 = n+2$ prizes. This trick of “augmenting the dimension with the constraints” is standard in the literature on mechanisms with randomization and in applied information theory (Bayesian persuasion, where posterior distributions are convex combinations that average to the prior).
3. **Numerical example.** A principal wants to induce an expected payment $\bar{x} = 50$ for an agent (here $n = 1$). Any lottery over payments with mean 50, however complex, is replicated with at most $n+1 = 2$ prizes: for example, pay 0 with probability 1/2 and 100 with probability 1/2. That a binary lottery is enough to replicate any mean is the simplest case of Carathéodory, and it explains why so many optimal random contracts in the literature are binary (“bonus or no bonus”).

3.4 Extreme points and the Minkowski theorem [P2]

Motivation: the points that are nobody’s average

Look again at Figure 3.2: of the seven points of S , only five “hold up” the convex hull; the two gray ones are redundant because they are averages of the others. The points that hold up a convex set (those that cannot be obtained by averaging other points of the set) are called extreme points, and they are the main characters of linear optimization.

Definition 3.18 (Extreme point). Let $C \subseteq \mathbb{R}^n$ be convex. A point $x \in C$ is an **extreme point** of C if it cannot be written as a strict convex combination of two distinct points of C ; that is, if

$$x = \lambda y + (1 - \lambda)z, \quad y, z \in C, \quad \lambda \in (0, 1) \quad \implies \quad y = z = x.$$

We write $\text{ext}(C)$ for the set of extreme points of C .

Put the other way around: x is *not* extreme if it lies strictly inside some segment contained in C . An extreme point is necessarily on the boundary of C (if $x \in \text{int}(C)$, there is a small ball around x inside C , and x is the midpoint of a diameter of that ball).

Example 3.19 (Extreme points of familiar figures). 1. **A (filled) triangle** in $\mathbb{R}^2$ has exactly 3 extreme points: its vertices. A point in the interior of an edge is not extreme (it is the average of the two vertices of the edge); an interior point is not extreme either. In general, the extreme points of a polytope are its *vertices* (Section 3.5).

2. **The closed Euclidean ball** $B = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ has as extreme points exactly *all* of its boundary sphere $\{x : \|x\| = 1\}$: infinitely many extreme points. Let us check that x with $\|x\| = 1$ is extreme: if $x = \lambda y + (1 - \lambda)z$ with $\lambda \in (0, 1)$, $\|y\| \leq 1$ and $\|z\| \leq 1$, then by the triangle inequality $1 = \|x\| \leq \lambda \|y\| + (1 - \lambda) \|z\| \leq 1$; all the inequalities are equalities, so $\|y\| = \|z\| = 1$ and there is equality in the triangle inequality. In the

Euclidean norm this forces λy and $(1 - \lambda)z$ to be nonnegative multiples of each other (equality in Cauchy–Schwarz), that is, y and z point in the same direction; having also the same norm, $y = z$, and therefore $y = z = x$.

3. **A half space** $\{x : \langle a, x \rangle \leq b\}$ with $a \neq 0$ has no extreme point at all: every one of its points lies in the middle of a segment contained in it (just move in a direction orthogonal to a). Unbounded sets can end up with no extreme points; this is why the next theorem asks for compactness.

Theorem 3.20 (Minkowski; Krein–Milman in finite dimension). *Let $C \subseteq \mathbb{R}^n$ be a nonempty convex and compact (closed and bounded) set. Then $\text{ext}(C) \neq \emptyset$ and*

$$C = \text{conv}(\text{ext}(C)) :$$

every point of C is a convex combination of extreme points of C (by Carathéodory, of at most $n + 1$ of them).

Proof idea. We argue by induction on the dimension of C (Definition 3.11). If $\dim C = 0$, C is a point, which is extreme in itself. For the inductive step, let $x \in C$.

Case 1: x is on the relative boundary of C . By the supporting hyperplane theorem (proved in the separation chapter; here we borrow it), there is a hyperplane H that “supports” C at x : C stays on one side of H and $x \in H$. The *face* $F = C \cap H$ is convex, compact and of dimension strictly smaller than that of C . By the inductive hypothesis, $x \in F$ is a convex combination of extreme points of F ; and a direct argument shows that every extreme point of a face $F = C \cap H$ is also an extreme point of C (if a point of F were the average of two points of C , both would have to be in H , hence in F).

Case 2: x is in the relative interior of C . Draw any line through x inside $\text{aff}(C)$; by compactness of C , the line meets the relative boundary at two points u, v , and x is a convex combination of them: $x = \lambda u + (1 - \lambda)v$. Each of u, v falls in Case 1, so both are convex combinations of extreme points, and therefore so is x .

The complete proof, with the details of the supporting hyperplane, is in [3, ch. 3] and [6, theorem 2.30]; the infinite dimensional version is the Krein–Milman theorem. $\square$

Intuition

The Minkowski theorem says that a compact convex set is *completely encoded* by its extreme points: they are its “DNA”. In economics this has two readings. First: if the set of feasible allocations (or of attainable utility vectors) is compact and convex, then every allocation is a lottery over extreme allocations; extreme allocations are the “bricks” with which everything else is built. Second: when analyzing the core of an economy or the set of equilibrium payoffs of a repeated game, it is often enough to characterize the extreme points, because the rest of the set is recovered by convexification.

The key consequence for optimization

Theorem 3.21 (A linear objective is optimized at an extreme point). *Let $C \subseteq \mathbb{R}^n$ be convex, compact and nonempty, and let $f(x) = \langle c, x \rangle$ be a linear function ($c \in \mathbb{R}^n$). Then the problem $\max_{x \in C} f(x)$ has a solution, and at least one solution is an extreme point of C :*

$$\max_{x \in C} \langle c, x \rangle = \max_{x \in \text{ext}(C)} \langle c, x \rangle.$$

The same holds for the minimum.

Proof. Existence. f is continuous (it is linear) and C is compact and nonempty; by the Weierstrass theorem (seen in the preliminaries), the maximum is attained at some $x^* \in C$.

The maximum is attained at an extreme point. By the Minkowski theorem (Theorem 3.20), x^* is a convex combination of extreme points: there are $v_1, \dots, v_k \in \text{ext}(C)$ and weights $\lambda_i > 0$ with $\sum_{i=1}^k \lambda_i = 1$ such that $x^* = \sum_{i=1}^k \lambda_i v_i$. By linearity of f :

$$f(x^*) = \langle c, \sum_i \lambda_i v_i \rangle = \sum_{i=1}^k \lambda_i \langle c, v_i \rangle = \sum_{i=1}^k \lambda_i f(v_i).$$

That is, the optimal value $f(x^*)$ is a *weighted average* of the values $f(v_i)$. An average never exceeds the largest of its terms, so there is an index j with $f(v_j) \geq f(x^*)$. Formally: if $f(v_i) < f(x^*)$ for all i , then

$$f(x^*) = \sum_{i=1}^k \lambda_i f(v_i) < \sum_{i=1}^k \lambda_i f(x^*) = f(x^*),$$

a contradiction. But $f(x^*)$ is the maximum value on C and $v_j \in C$, so $f(v_j) \leq f(x^*)$. We conclude $f(v_j) = f(x^*)$: the extreme point v_j is optimal too. For the minimum, apply the above to $-c$. $\square$

This result, with such a short proof, is a bridge to two larger topics:

1. **Linear programming.** In a linear problem $\max\{\langle c, x \rangle : Ax \leq b\}$ with compact feasible region, it is enough to examine the vertices. The *simplex method* [5, ch. 13] is precisely a smart walk over the extreme points of the feasible polyhedron, jumping from vertex to neighboring vertex while the objective improves. We come back to this in the linear programming chapter.
2. **Bang-bang and corner solutions in contracts.** In many contract and incentive problems, the principal's objective is linear (or affine) in the choice instrument (for example, in assignment probabilities or expected transfers) and the feasible set (determined by participation, incentive and feasibility constraints) is compact and convex. Theorem 3.21 implies that the optimum can be taken at an extreme point of the feasible set: the optimal contract is *extreme*, of **bang-bang** type (all or nothing): pay the full bonus or pay nothing, assign the object with probability 1 or 0, punish with the maximum penalty or do not punish. When an optimal contract in Microeconomics “jumps” between corners instead of

interpolating smoothly, the underlying mathematical reason is usually this: linear objective + convex compact feasible set $\Rightarrow$ extreme optimum. See [9, ch. 14] and [10, ch. 9] for the contract models, and [11] for the envelope machinery that goes with them.

Exam strategy

In the course exams (and in the problem sets), the typical requests from this chapter are four, and it pays to have a reflex for each one.

1. “*Prove that C is a convex cone*”: do not use the definition of convexity; use Proposition 3.3 and check only two things: $x + y \in C$ and $\lambda x \in C$ for $\lambda > 0$. It is usually a two line computation (imitate Examples 3.4 and 3.5).
2. “*Compute $\text{conv}(S)$ / $\text{cone}(S)$ / $\text{aff}(S)$* ”: write the generic combination with the right weights (check the warning box), parametrize and eliminate the weights to get a description by inequalities. Always verify the two inclusions: your candidate contains S and is convex/conic/affine (this gives one inclusion), and each point of the candidate is written as a combination (this gives the other).
3. “*Find the extreme points of C* ”: first discard the whole interior (there are never extreme points there); on the boundary, a point is extreme if the system of *active* constraints at it determines it uniquely. To prove that a point is NOT extreme, exhibit explicit $y \neq z$ in C with $x = \frac{1}{2}y + \frac{1}{2}z$: it is the fastest and most gradable route.
4. “*Show that the optimum is attained at a vertex*”: cite Weierstrass for existence and reproduce the averaging argument of Theorem 3.21; it is a short proof that graders ask for often because it combines three ideas of the course.

3.5 Polyhedra and polytopes [P3]

The most important convex sets in practice are described by finitely many linear inequalities: budget, feasibility and nonnegativity constraints. They deserve their own name.

Definition 3.22 (Polyhedron and polytope). A **polyhedron** is the solution set of a finite system of linear inequalities:

$$P = \{x \in \mathbb{R}^n : \langle a_i, x \rangle \leq b_i, i = 1, \dots, m\} = \{x : Ax \leq b\},$$

where A is an $m \times n$ matrix and the inequality $Ax \leq b$ is understood coordinate by coordinate. A **polytope** is a *bounded* polyhedron. The extreme points of a polyhedron are called **vertices**.

Every polyhedron is convex and closed: it is a finite intersection of closed half spaces $\{x : \langle a_i, x \rangle \leq b_i\}$, each convex and closed, and both properties are preserved under intersections (previous chapter). Examples: $\mathbb{R}_+^n$ is a polyhedron (with $A = -I$, $b = 0$), but not a polytope (it is not bounded); the budget set $B(p, m) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq m\}$ with prices $p \gg 0$ and wealth

$m > 0$ is a polytope; the *probability simplex*

$$\Delta^{n-1} = \left\{ \pi \in \mathbb{R}_+^n : \sum_{i=1}^n \pi_i = 1 \right\}$$

(the set of all lotteries over n outcomes) is a polytope whose vertices are the canonical vectors $e_1, \dots, e_n$: the degenerate lotteries that give one prize with certainty.

There are then two natural ways to describe a polytope: “from outside”, by inequalities ($\mathcal{H}$ representation, for half spaces), and “from inside”, as the convex hull of its vertices ($\mathcal{V}$ representation, for vertices). The fundamental theorem of polyhedral theory says that both describe the same class of objects.

Theorem 3.23 (Minkowski–Weyl, bounded case). *A set $P \subseteq \mathbb{R}^n$ is a polytope (a bounded polyhedron) if and only if it is the convex hull of a finite set of points:*

$$P = \{x : Ax \leq b\} \text{ bounded} \iff P = \text{conv}(\{v_1, \dots, v_N\}).$$

In the general (unbounded) version, a set is a polyhedron if and only if it can be written as $\text{conv}(\{v_1, \dots, v_N\}) + \text{cone}(\{d_1, \dots, d_M\})$: a “vertex” part plus a conic part of escape directions.

We will not prove this theorem (modern proofs use Fourier–Motzkin elimination or duality; see [3, ch. 4] and [6, sec. 2.1]), but its content is worth digesting. The direction “polytope $\Rightarrow$ hull of finitely many points” is almost the Minkowski theorem (Theorem 3.20) plus the nontrivial fact that a polyhedron has *finitely many* extreme points: each vertex is determined by a subset of n linearly independent active constraints, and there are only finitely many possible subsets. The converse direction says that the “wrapping” of finitely many points can always be cut out by finitely many linear inequalities.

Remark 3.24. The two representations have different computational uses. Checking whether a given point belongs to P is immediate in the $\mathcal{H}$ form (evaluate m inequalities) but costly in the $\mathcal{V}$ form; optimizing a linear objective is immediate in the $\mathcal{V}$ form (evaluate the objective at the N vertices, by Theorem 3.21) but requires an algorithm (simplex, interior point) in the $\mathcal{H}$ form. Moreover, going from one form to the other can be exponentially costly: the cube $[0, 1]^n$ has $2n$ inequalities but 2^n vertices.

3.6 Recession cone [P3]

Motivation: in which directions does a convex set escape to infinity?

To know whether an optimization problem has a solution, the crucial question is usually whether the feasible set is bounded and, if it is not, *in which directions* it is unbounded and how the objective behaves in those directions. The recession cone is the object that catalogues those escape directions.

Definition 3.25 (Recession cone). Let $C \subseteq \mathbb{R}^n$ be convex, closed and nonempty. The **recession cone** of C is

$$C^\infty = \{d \in \mathbb{R}^n : x + \lambda d \in C \text{ for all } x \in C \text{ and all } \lambda \geq 0\},$$

the set of *directions* d such that, starting from any point of the set, you can travel indefinitely in direction d without leaving C . It is easy to check that C^∞ is a convex cone containing 0; for C convex and *closed* it is enough to verify the condition from a *single* point $x_0 \in C$ (the property spreads to all the others; see [6, sec. 2.2]).

- Example 3.26** (Concrete recession cones). 1. If C is **bounded** (a ball, a polytope), then $C^\infty = \{0\}$: there is no direction in which you can travel forever. The converse also holds for closed convex sets: $C^\infty = \{0\}$ implies C bounded; this equivalence is what makes the recession cone useful as a *compactness test*.
2. If C is a **closed convex cone** (with $0 \in C$), then $C^\infty = C$: the escape directions of a cone are the cone itself.
3. $C = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \geq x_1^2\}$ (the epigraph of the parabola): $C^\infty = \{(0, t) : t \geq 0\}$, the vertical half axis. From any point on the parabola you can go straight up forever, but any direction with a nonzero horizontal component ends up hitting the parabolic wall.
4. For a nonempty polyhedron $P = \{x : Ax \leq b\}$: $P^\infty = \{d : Ad \leq 0\}$. Indeed, $A(x + \lambda d) = Ax + \lambda Ad \leq b$ for all $\lambda \geq 0$ if and only if $Ad \leq 0$. Note that the recession cone “forgets” the right hand side b : it depends only on the slopes of the constraints.

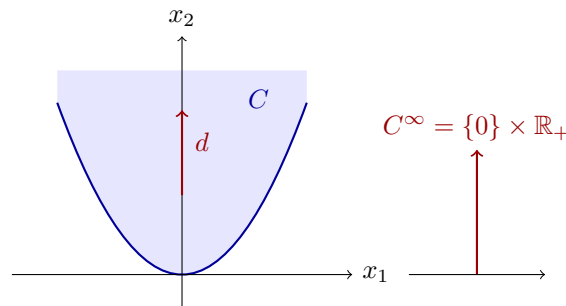


Figure 3.3: The set $C = \{x_2 \geq x_1^2\}$ is unbounded, but it only “escapes to infinity” vertically: its recession cone is the vertical half axis.

What it is for: existence of optima and general equilibrium

The recession cone answers two existence questions that will come back throughout the course.

(i) Existence of minimizers without compactness. The Weierstrass theorem requires a compact feasible set, but many economic problems have unbounded feasible sets (there is no a priori upper bound on production or transfers). The substitute is a recession argument: if C is closed convex and the linear objective $\langle c, x \rangle$ satisfies $\langle c, d \rangle > 0$ for every recession direction $d \in C^\infty \setminus \{0\}$, then $\min_{x \in C} \langle c, x \rangle$ is attained: the objective grows along every escape direction, so the search for the minimum can be confined to a bounded (hence compact) portion of C , where Weierstrass does apply. If instead there is $d \in C^\infty$ with $\langle c, d \rangle < 0$, the infimum is $-\infty$: traveling in direction d improves the objective indefinitely. The recession cone thus turns the question “does the optimum exist?” into the often mechanical computation of $\{d : Ad \leq 0\}$ and of the signs $\langle c, d \rangle$; the systematic treatment is in [6, sec. 2.2 and ch. 3] and [3, ch. 4].

(ii) **Boundedness of the set of feasible allocations in general equilibrium.** In a production economy, the existence of Walrasian equilibrium is proved by applying a fixed point theorem on the set of feasible allocations, and for that this set must be *compact*. Aggregate feasibility is $\sum_j y_j + \omega \geq \sum_i x_i \geq 0$ with $y_j \in Y_j$: nothing prevents a priori two firms from taking huge plans that cancel each other ($y_1 = tv$, $y_2 = -tv$ with $t \rightarrow \infty$), leaving the sum bounded but the individual plans unbounded. The assumption that rules this out is stated with recession cones: one requires that the recession directions of the aggregate technology $Y = \sum_j Y_j$ that are “mutually cancelable” be trivial, essentially $Y^\infty \cap (-Y^\infty) \subseteq \{0\}$ together with $Y^\infty \cap \mathbb{R}_+^n = \{0\}$ (you cannot produce something out of nothing at arbitrary scale: the asymptotic version of “no free lunch”). With this, the set of feasible allocations turns out to be compact and the fixed point argument goes through. This is exactly the machinery of the appendix of chapter 17 of [9] (existence of equilibrium with production); the recession cone is the tool used there, sometimes under the name *asymptotic cone*.

3.7 Summary and solved exercises [P1]

Chapter summary

- **Cone:** C such that $x \in C, \lambda > 0 \Rightarrow \lambda x \in C$. **Convex cone** $\iff$ closed under sums and positive scalars (Proposition 3.3). Examples: $\mathbb{R}_+^n$, the second-order cone $\|x\| \leq t$. Economics: technology with constant returns to scale = conic production set; free disposal = $Y - \mathbb{R}_+^n \subseteq Y$.
- **Hulls:** $\text{conv}(S)$ (weights ≥ 0 adding to 1) = smallest convex set containing S = all convex combinations (Theorem 3.9); $\text{aff}(S)$ (weights adding to 1, any sign), defines the *dimension*; $\text{cone}(S)$ (weights ≥ 0 , free sum).
- **Carathéodory:** in $\mathbb{R}^n$, every point of $\text{conv}(S)$ uses at most $n+1$ points of S ; every point of $\text{cone}(S)$, at most n (Theorems 3.15 and 3.16). Application: every lottery with expected value in $\mathbb{R}^n$ is replicated with at most $n+1$ prizes (simple random mechanisms without loss of generality).
- **Extreme points:** those that are not a strict average of two distinct points of the set. Minkowski (Theorem 3.20): compact convex set = conv of its extreme points. Consequence (Theorem 3.21): a linear objective on a compact convex set attains its optimum at an extreme point $\Rightarrow$ linear programming through vertices, bang-bang contracts.
- **Polyhedra and polytopes:** $\{x : Ax \leq b\}$; polytope = bounded polyhedron. Minkowski–Weyl (Theorem 3.23): representation by inequalities $\iff$ representation by vertices (+ a cone in the unbounded case).
- **Recession cone:** C^∞ = directions of infinite travel inside C ; $C^\infty = \{0\} \iff C$ (closed convex) bounded. For polyhedra, $P^\infty = \{d : Ad \leq 0\}$. Uses: existence of optima on unbounded feasible sets; compactness of the set of allocations in general equilibrium existence [9, ch. 17].

Main references of the chapter: [3, chs. 2–4], [6, ch. 2], [2, ch. 2], [1].

Exercise 3.27. Let $a \in \mathbb{R}^n$, $a \neq 0$. Prove that the half space $H = \{x \in \mathbb{R}^n : \langle a, x \rangle \leq 0\}$ is a convex cone, but that the translated half space $H_b = \{x : \langle a, x \rangle \leq b\}$ with $b > 0$ is convex and is *not* a cone.

Solution. H is a convex cone. We use Proposition 3.3 and the linearity of the inner product. Sums: if $\langle a, x \rangle \leq 0$ and $\langle a, y \rangle \leq 0$, then $\langle a, x + y \rangle = \langle a, x \rangle + \langle a, y \rangle \leq 0$, so $x + y \in H$. Positive scalars: if $\lambda > 0$, $\langle a, \lambda x \rangle = \lambda \langle a, x \rangle \leq 0$ (a positive number times a nonpositive one), so $\lambda x \in H$. By the proposition, H is a convex cone.

H_b is convex: if $\langle a, x \rangle \leq b$, $\langle a, y \rangle \leq b$ and $\theta \in [0, 1]$, then $\langle a, \theta x + (1 - \theta)y \rangle = \theta \langle a, x \rangle + (1 - \theta) \langle a, y \rangle \leq \theta b + (1 - \theta)b = b$.

H_b is not a cone if $b > 0$. One scaling counterexample is enough. Since $a \neq 0$, the point $x_0 = \frac{b}{\|a\|^2} a$ satisfies $\langle a, x_0 \rangle = \frac{b}{\|a\|^2} \langle a, a \rangle = b \leq b$, so $x_0 \in H_b$. But for $\lambda = 2$: $\langle a, 2x_0 \rangle = 2b > b$ (because $b > 0$), so $2x_0 \notin H_b$. Closure under positive scalars fails, and H_b is not a cone. (Moral: cones are objects “anchored at the origin”; translating destroys the conic structure but not convexity.) $\square$

Exercise 3.28. Let $S = \{(1, 0), (0, 1), (1, 1)\} \subset \mathbb{R}^2$. Compute and describe through equalities and inequalities: (a) $\text{conv}(S)$; (b) $\text{aff}(S)$; (c) $\text{cone}(S)$. State the dimension of $\text{conv}(S)$.

Solution. (a) A generic convex combination is

$$z = \lambda_1(1, 0) + \lambda_2(0, 1) + \lambda_3(1, 1) = (\lambda_1 + \lambda_3, \lambda_2 + \lambda_3), \quad \lambda_i \geq 0, \lambda_1 + \lambda_2 + \lambda_3 = 1.$$

Writing $z = (u, v)$: $u + v = \lambda_1 + \lambda_2 + 2\lambda_3 = 1 + \lambda_3 \in [1, 2]$ (since $\lambda_3 \in [0, 1]$), $u = \lambda_1 + \lambda_3 \in [0, 1]$ and $v = \lambda_2 + \lambda_3 \in [0, 1]$. Conversely, if $(u, v) \in [0, 1]^2$ with $1 \leq u + v \leq 2$, taking $\lambda_3 = u + v - 1 \geq 0$, $\lambda_1 = u - \lambda_3 = 1 - v \geq 0$ and $\lambda_2 = v - \lambda_3 = 1 - u \geq 0$ you check that the weights are nonnegative and add up to $(1 - v) + (1 - u) + (u + v - 1) = 1$. Therefore

$$\text{conv}(S) = \{(u, v) : 0 \leq u \leq 1, 0 \leq v \leq 1, u + v \geq 1\} :$$

the triangle with vertices at the three points of S .

(b) The three points are not aligned: $(0, 1) - (1, 0) = (-1, 1)$ and $(1, 1) - (1, 0) = (0, 1)$ are linearly independent. So $\text{aff}(S) = \mathbb{R}^2$ (with two independent directions and free weights adding to 1 you reach the whole plane), and $\dim \text{conv}(S) = 2$.

(c) $\text{cone}(S) = \{\lambda_1(1, 0) + \lambda_2(0, 1) + \lambda_3(1, 1) : \lambda_i \geq 0\} = \{(\lambda_1 + \lambda_3, \lambda_2 + \lambda_3) : \lambda_i \geq 0\}$. Every coordinate is nonnegative, so $\text{cone}(S) \subseteq \mathbb{R}_+^2$; and conversely every $(u, v) \in \mathbb{R}_+^2$ is obtained with $\lambda_3 = 0$, $\lambda_1 = u$, $\lambda_2 = v$. Therefore $\text{cone}(S) = \mathbb{R}_+^2$: the point $(1, 1)$ is redundant for the conic hull (it is a conic combination of the other two), although it was not redundant for the convex hull (it is a vertex of the triangle). This contrast shows that being a “minimal generator” depends on the structure you consider. $\square$

Exercise 3.29. The point $z = \left(\frac{1}{2}, \frac{1}{2}\right) \in \mathbb{R}^2$ is written as a convex combination of *four* points:

$$z = \frac{1}{4}(1, 0) + \frac{1}{4}(0, 1) + \frac{1}{4}(1, 1) + \frac{1}{4}(0, 0).$$

Following the algorithm in the proof of the Carathéodory theorem (Theorem 3.15), reduce the representation to one with at most $3 = n + 1$ points.

Solution. Label $x_1 = (1, 0)$, $x_2 = (0, 1)$, $x_3 = (1, 1)$, $x_4 = (0, 0)$, with weights $\lambda_i = \frac{1}{4}$. Since $k = 4 \geq n + 2 = 4$, the theorem guarantees an affine dependence. We find it: we look for μ_i , not all zero, with $\sum_i \mu_i x_i = 0$ and $\sum_i \mu_i = 0$. Noting that $(1, 0) + (0, 1) = (1, 1) + (0, 0)$, we take

$$\mu_1 = 1, \quad \mu_2 = 1, \quad \mu_3 = -1, \quad \mu_4 = -1,$$

which satisfy $\sum_i \mu_i x_i = (1, 0) + (0, 1) - (1, 1) - (0, 0) = (0, 0)$ and $\sum_i \mu_i = 0$. The indices with $\mu_i > 0$ are $i = 1, 2$, with ratios $\lambda_1/\mu_1 = \lambda_2/\mu_2 = \frac{1}{4}$; the step of the algorithm is

$$t^* = \min \left\{ \frac{1/4}{1}, \frac{1/4}{1} \right\} = \frac{1}{4}.$$

The new weights are $\lambda_i - t^* \mu_i$:

$$\lambda'_1 = \frac{1}{4} - \frac{1}{4} = 0, \quad \lambda'_2 = \frac{1}{4} - \frac{1}{4} = 0, \quad \lambda'_3 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, \quad \lambda'_4 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

(They add up to 1, as they should.) *Two* weights vanish at once (the minimum was attained at two indices), and what is left is

$$z = \frac{1}{2}(1, 1) + \frac{1}{2}(0, 0),$$

which is indeed $(\frac{1}{2}, \frac{1}{2})$: a representation with only 2 points, even better than the bound $n + 1 = 3$. Lottery reading: the uniform lottery over 4 prizes with mean $(\frac{1}{2}, \frac{1}{2})$ is replicated with a fair coin between the prizes $(1, 1)$ and $(0, 0)$. $\square$

Exercise 3.30. Consider the budget set $B = \{(x_1, x_2) \in \mathbb{R}_+^2 : 2x_1 + x_2 \leq 4\}$ (prices $p = (2, 1)$, wealth 4). (a) Find all extreme points of B . (b) A consumer has *linear* utility $u(x_1, x_2) = 3x_1 + 2x_2$. Use Theorem 3.21 to find the optimal demand. (c) Explain why the solution is a “corner” one.

Solution. (a) B is a polytope defined by three inequalities: $x_1 \geq 0$, $x_2 \geq 0$, $2x_1 + x_2 \leq 4$. Its extreme points are the points where two linearly independent constraints are active (hold with equality) and which belong to B :

$$x_1 = 0, x_2 = 0 \Rightarrow (0, 0); \quad x_1 = 0, 2x_1 + x_2 = 4 \Rightarrow (0, 4); \quad x_2 = 0, 2x_1 + x_2 = 4 \Rightarrow (2, 0).$$

All three belong to B , so $\text{ext}(B) = \{(0, 0), (0, 4), (2, 0)\}$: B is the triangle with those vertices. To confirm that, for example, $(1, 1) \in B$ is not extreme: $(1, 1) = \frac{1}{2}(0, 0) + \frac{1}{2}(2, 2)$ with both points in B and distinct.

(b) B is compact (closed because it is a polyhedron; bounded because $0 \leq x_1 \leq 2$ and $0 \leq x_2 \leq 4$) and convex, and u is linear. By Theorem 3.21, it is enough to evaluate u at the three extreme points:

$$u(0, 0) = 0, \quad u(0, 4) = 3 \cdot 0 + 2 \cdot 4 = 8, \quad u(2, 0) = 3 \cdot 2 + 2 \cdot 0 = 6.$$

The optimal demand is $x^* = (0, 4)$, with utility 8: the consumer spends all wealth on good 2.

(c) With linear utility the goods are perfect substitutes and the consumer compares “utility per unit of money spent”: good 1 yields $3/2 = 1.5$ per unit of spending and good 2 yields $2/1 = 2$. Since the yields are constant (there is no decreasing marginal utility to stop specialization), it is best to spend *everything* on the good with the higher yield: a corner solution. This is the bang-bang mechanism in miniature: linear objective + compact convex feasible set $\Rightarrow$ optimum at a vertex. If the two yields were tied ($3/2 = 2/1$ does not happen here, but for instance with $u = 4x_1 + 2x_2$), the whole edge between two vertices would be optimal: the extreme optimum still exists, but it is not unique. $\square$

Exercise 3.31. Compute the recession cone of the following closed convex sets of $\mathbb{R}^2$ and decide in each case whether the set is bounded: (a) $C_1 = \{(x_1, x_2) : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \geq 1\}$; (b) $C_2 = \{(x_1, x_2) : x_1^2 + x_2^2 \leq 9\}$; (c) $C_3 = \{(x_1, x_2) : x_2 \geq e^{x_1}\}$.

Solution. (a) C_1 is a polyhedron $\{x : Ax \leq b\}$ with rows $-x_1 \leq 0, -x_2 \leq 0, -x_1 - x_2 \leq -1$. By Example 3.26(4), $C_1^\infty = \{d : Ad \leq 0\} = \{(d_1, d_2) : -d_1 \leq 0, -d_2 \leq 0, -d_1 - d_2 \leq 0\} = \mathbb{R}_+^2$ (the third inequality is redundant given the first two). Since $C_1^\infty \neq \{0\}$, C_1 is unbounded; its escape directions are all those of the first quadrant. Note how the recession cone ignored the constant term -1 : translating the walls does not change where the set escapes to.

(b) C_2 is the closed ball of radius 3: it is bounded, so $C_2^\infty = \{0\}$. Directly: if $d \neq 0$, then $\|x + \lambda d\| \rightarrow \infty$ as $\lambda \rightarrow \infty$, and the point leaves the ball; no nonzero direction is a recession direction.

(c) We look for $d = (d_1, d_2)$ such that $(x_1 + \lambda d_1, x_2 + \lambda d_2) \in C_3$ for all $\lambda \geq 0$, starting for instance from $(0, 1) \in C_3$ (recall that for closed convex sets one base point is enough). The condition is

$$1 + \lambda d_2 \geq e^{\lambda d_1} \quad \text{for all } \lambda \geq 0.$$

If $d_1 > 0$, the right hand side grows exponentially and the left hand side linearly: impossible for large λ . If $d_1 \leq 0$, the right hand side is bounded by 1 (since $e^{\lambda d_1} \leq e^0 = 1$): with $d_2 \geq 0$ the left hand side is $\geq 1 \geq e^{\lambda d_1}$ and the condition always holds; with $d_2 < 0$ the left hand side becomes negative for large λ while the right hand side is positive: impossible. Conclusion:

$$C_3^\infty = \{(d_1, d_2) : d_1 \leq 0, d_2 \geq 0\},$$

the (closed) second quadrant. C_3 is unbounded: it escapes upward and to the left (where the exponential flattens toward 0), but never to the right, where the exponential wall grows faster than any ray. $\square$

Exercise 3.32. Prove, using the Carathéodory theorem, that if $S \subseteq \mathbb{R}^n$ is compact then $\text{conv}(S)$ is compact. (Hint: write $\text{conv}(S)$ as the continuous image of a compact set.)

Solution. By Theorem 3.15, every point of $\text{conv}(S)$ is a convex combination of at most $n + 1$ points of S (if it uses fewer, complete with zero weights). Define the set of “data” of such combinations:

$$D = \Delta^n \times S^{n+1} = \left\{ (\lambda, x_1, \dots, x_{n+1}) : \lambda \in \mathbb{R}_+^{n+1}, \sum_{i=1}^{n+1} \lambda_i = 1, x_i \in S \right\},$$

and the map

$$\Phi : D \rightarrow \mathbb{R}^n, \quad \Phi(\lambda, x_1, \dots, x_{n+1}) = \sum_{i=1}^{n+1} \lambda_i x_i.$$

Three observations close the proof.

1. *D is compact.* The simplex $\Delta^n = \{\lambda \in \mathbb{R}_+^{n+1} : \sum_i \lambda_i = 1\}$ (the one of lotteries over $n+1$ prizes, with the same index convention as the previous chapter) is closed (an intersection of closed half spaces and a hyperplane) and bounded ($0 \leq \lambda_i \leq 1$), hence compact; S is compact by hypothesis; and a finite cartesian product of compact sets is compact (seen in the preliminaries).
2. *Φ is continuous,* since it is a polynomial (sums and products) in the coordinates of its arguments.
3. *$\Phi(D) = \text{conv}(S)$.* The inclusion $\subseteq$ is clear: Φ produces convex combinations of points of S , which are in $\text{conv}(S)$ by Theorem 3.9. The inclusion $\supseteq$ is exactly Carathéodory: every $z \in \text{conv}(S)$ admits a representation with at most $n+1$ points, that is, $z = \Phi(\lambda, x_1, \dots, x_{n+1})$ for some datum in D .

The continuous image of a compact set is compact (preliminaries), so $\text{conv}(S) = \Phi(D)$ is compact. Note the essential role of Carathéodory: without the uniform bound $n+1$ on the number of points, there would be no compact domain of *fixed* dimension from which to parametrize all the combinations. Compare with the warning box: for S only *closed* (unbounded) the result is false. $\square$

Chapter 4

Hyperplanes and separation theorems [P1]

Week 3 of the course. Bibliography: Ruszczyński [6]; Boyd and Vandenberghe [2]; Nocedal and Wright [5].

What is this used for in Microeconomics?

If you had to choose *one single* chapter of these notes to master before Microeconomics, this would be it. The **Second Welfare Theorem** (general equilibrium, MWG section 16.D, [9]) says that every Pareto optimum can be decentralized as a market equilibrium with transfers; its proof consists, literally, of separating two convex sets with a hyperplane, and the **normal vector of that separating hyperplane IS the price vector** that supports the optimum. In other words: equilibrium prices are not “postulated”, they are *built* with the separation theorem that we prove here. The same machinery produces the *shadow prices* (multipliers) of any constrained optimization problem: the **Farkas lemma**, which we prove through separation, is the hidden engine behind the Karush–Kuhn–Tucker (KKT) conditions that govern contract problems with incentive compatibility (IC) and participation (IR) constraints in principal agent models and mechanism design (MWG chs. 14 and 23). Separation also supports *duality* in optimization (later chapters), the existence of subjective probabilities in decision theory, and even classification with *support vector machines* (SVM) in computational econometrics: separating data with a hyperplane is exactly the geometric problem of this chapter. In short: this is where prices are made.

In chapter 2 we learned what a convex set is and studied its topological structure (interior, closure, relative interior). In this chapter we collect the first big dividend of that investment: the *separation theorems*, which say that two convex sets that do not touch can be “put on opposite sides” of a hyperplane. This result, geometrically obvious in a picture, is surprisingly deep: it is the finite dimensional version of the Hahn–Banach theorem, and it is the bridge between convex geometry and the linear algebra of prices. The plan of the chapter is constructive and builds up step by step: first we review what a hyperplane is and what its normal vector means (Section 4.1); then we prove the *projection theorem* onto a closed convex set (Section 4.2), which

is the technical tool from which everything else follows; with the projection in hand, we separate a point from a closed convex set (Section 4.3), we build supporting hyperplanes at boundary points (Section 4.4), and we separate two convex sets from each other (Section 4.5); as an algebraic application we prove the Farkas lemma (Section 4.6), and as the star economic application we develop in full detail the second welfare theorem (Section 4.7). The main references are [6] (sections 2.2–2.3), [3] (ch. 4), [2] (section 2.5) and [1]. For the economic application we follow [9] (section 16.D) and [10] (ch. 5).

4.1 Hyperplanes and half spaces [P1]

4.1.1 Motivation: the level sets of a linear function

The simplest functions from $\mathbb{R}^n$ to $\mathbb{R}$ are the *linear* ones. Fix a vector $a \in \mathbb{R}^n$ with $a \neq 0$ and define

$$f(x) = \langle a, x \rangle = a_1x_1 + a_2x_2 + \cdots + a_nx_n.$$

(Recall from the preliminaries chapter that $\langle a, x \rangle$ denotes the Euclidean inner product.) A natural question: what does the set of points where f equals exactly some number β look like? In $\mathbb{R}^2$ with $a = (2, 3)$ and $\beta = 12$, the set $\{(x_1, x_2) : 2x_1 + 3x_2 = 12\}$ is a line. In $\mathbb{R}^3$, the set $\{x : \langle a, x \rangle = \beta\}$ is a plane. In $\mathbb{R}^n$ it is a flat object of dimension $n - 1$ that we call a *hyperplane*. Hyperplanes are, then, the level sets of nonzero linear functions, and that double nature, geometric object and linear equation, is the key to the whole chapter.

Definition 4.1 (Hyperplane and half spaces). Let $a \in \mathbb{R}^n$, $a \neq 0$, and $\beta \in \mathbb{R}$. The *hyperplane* with *normal vector* a and level β is the set

$$H(a, \beta) = \{x \in \mathbb{R}^n : \langle a, x \rangle = \beta\}.$$

The hyperplane determines two *closed half spaces*

$$H^-(a, \beta) = \{x : \langle a, x \rangle \leq \beta\}, \quad H^+(a, \beta) = \{x : \langle a, x \rangle \geq \beta\},$$

and two *open half spaces*, defined with the strict inequalities $<$ and $>$.

Remark 4.2. The pair (a, β) is not unique: $H(a, \beta) = H(\lambda a, \lambda \beta)$ for every $\lambda \neq 0$. This is why we often *normalize* by taking $\|a\| = 1$, which fixes a up to the sign. Multiplying by $\lambda < 0$ swaps the roles of H^- and H^+ : the “positive side” depends on the orientation chosen for the normal. In economics this freedom of scale is exactly the *homogeneity of degree zero* of demands in prices: only relative prices matter, and normalizing $\|p\| = 1$ (or $p_1 = 1$, the numeraire good) is choosing a scale for the budget hyperplane.

4.1.2 Geometry of the normal vector

Why do we call a the “normal” vector? Because a is perpendicular to the hyperplane, in the following precise sense.

Proposition 4.3 (The normal vector is orthogonal to the hyperplane). *Let $H = H(a, \beta)$ and let $x_0 \in H$ be any point of the hyperplane. Then:*

- (a) $H = x_0 + \{v \in \mathbb{R}^n : \langle a, v \rangle = 0\}$; that is, H is the translate by x_0 of the subspace of vectors orthogonal to a . In particular, H is an affine set of dimension $n - 1$.
- (b) For any $x, y \in H$ we have $\langle a, x - y \rangle = 0$: the vector a is orthogonal to every vector “contained in” H .
- (c) The function $x \mapsto \langle a, x \rangle$ grows in the direction of a : if we start at $x_0 \in H$ and move to $x_0 + ta$ with $t > 0$, then $\langle a, x_0 + ta \rangle = \beta + t\|a\|^2 > \beta$. The normal vector points toward the upper half space H^+ .

Proof. (a) If $x \in H$, then $\langle a, x - x_0 \rangle = \langle a, x \rangle - \langle a, x_0 \rangle = \beta - \beta = 0$, so $v = x - x_0$ satisfies $\langle a, v \rangle = 0$ and $x = x_0 + v$. Conversely, if $\langle a, v \rangle = 0$ then $\langle a, x_0 + v \rangle = \beta + 0 = \beta$, so $x_0 + v \in H$. The set $\{v : \langle a, v \rangle = 0\}$ is the kernel of the linear map $v \mapsto \langle a, v \rangle$, which is onto $\mathbb{R}$ (because $a \neq 0$); by the rank nullity theorem its kernel has dimension $n - 1$.

(b) This is immediate from (a): $x - y = (x - x_0) - (y - x_0)$ is a difference of two vectors orthogonal to a , and by linearity of the inner product $\langle a, x - y \rangle = 0$.

(c) Direct computation: $\langle a, x_0 + ta \rangle = \langle a, x_0 \rangle + t\langle a, a \rangle = \beta + t\|a\|^2$, and $\|a\|^2 > 0$ because $a \neq 0$. $\square$

Intuition

Think of the hyperplane as a “contour line” on a topographic map where the height of the point x is $\langle a, x \rangle$. The normal vector a is the direction of steepest ascent: it points perpendicular to the contour lines, toward where the “height” $\langle a, x \rangle$ grows fastest (in fact, a is the gradient of the linear function $x \mapsto \langle a, x \rangle$). When in the separation theorems we say that a hyperplane leaves the set A “below” and the set B “above”, we are saying that the linear function $\langle a, \cdot \rangle$ is smaller on all of A than on all of B . To separate is to compare with a linear ruler.

4.1.3 Economic reading: the price is the normal

The budget line is the most famous hyperplane in economics. If $p = (p_1, \dots, p_L) \in \mathbb{R}^L$ is a price vector and $m > 0$ is the wealth of the consumer, the set of bundles that exhaust the budget is the hyperplane

$$H(p, m) = \{x \in \mathbb{R}^L : \langle p, x \rangle = m\},$$

and the budget set $\{x \geq 0 : \langle p, x \rangle \leq m\}$ is the intersection of the lower half space $H^-(p, m)$ with the nonnegative orthant. Proposition 4.3 then says that **the price vector is the normal vector of the budget line**: p points perpendicular to it, toward the more expensive bundles. Similarly, for a firm with net production vector y (inputs with negative sign, outputs with positive sign), the *isoprofit* lines $\{y : \langle p, y \rangle = \pi\}$ are hyperplanes with normal p , and maximizing profit is pushing the hyperplane in the direction of the normal as far as the technology allows. This picture (prices as normals of hyperplanes) is what will make the Second Welfare Theorem transparent in Section 4.7.

Example 4.4 (Budget line and its normal). Let $L = 2$, $p = (2, 3)$ and $m = 12$. The budget line is $\{(x_1, x_2) : 2x_1 + 3x_2 = 12\}$, which meets the axes at $(6, 0)$ and $(0, 4)$. Let us check with all the steps that $p = (2, 3)$ is orthogonal to it. Take the two intercepts: the vector going from $(6, 0)$ to $(0, 4)$ is $v = (0, 4) - (6, 0) = (-6, 4)$. Then

$$\langle p, v \rangle = 2 \cdot (-6) + 3 \cdot 4 = -12 + 12 = 0,$$

as Proposition 4.3(b) predicts. Moreover, the bundle $(3, 3)$ costs $2 \cdot 3 + 3 \cdot 3 = 15 > 12$: it is in the upper half space, the side toward which p points (the unaffordable bundles). The bundle $(3, 1)$ costs $9 < 12$: it is in the lower half space (the ones affordable with slack). Note the reading of the slope: the line has slope $-p_1/p_2 = -2/3$, the relative price, while the normal $(2, 3)$ has “slope” $3/2$, perpendicular to the previous one (the product of the slopes is -1).

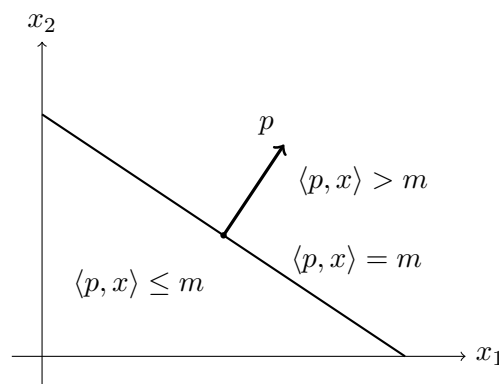


Figure 4.1: The budget line is a hyperplane in $\mathbb{R}^2$; the price vector p is its normal and points toward the unaffordable bundles.

Remark 4.5 (Half spaces and convexity). Every half space (open or closed) is convex, and every hyperplane is convex: they are sets defined by a *linear* equality or inequality, and linear inequalities are preserved when taking convex combinations. Indeed, if $\langle a, x \rangle \leq \beta$ and $\langle a, y \rangle \leq \beta$, then for $\lambda \in [0, 1]$,

$$\langle a, \lambda x + (1 - \lambda)y \rangle = \lambda \langle a, x \rangle + (1 - \lambda) \langle a, y \rangle \leq \lambda \beta + (1 - \lambda)\beta = \beta.$$

Since the intersection of convex sets is convex (chapter 2), any intersection of half spaces is convex; *polyhedra*, finite intersections of closed half spaces, are the central example in linear programming [2]. One of the star results of this chapter (Corollary 4.12) is the “infinite” converse: every closed convex set is an intersection of the half spaces that contain it. Closed convexity is, exactly, “being describable by linear inequalities”.

4.2 Projection onto a closed convex set [P1]

4.2.1 Motivation: the nearest point

The whole theory of separation rests on a single construction: given a set C and a point x outside it, find *the point of C closest to x* . The intuition is physical: if you connect x with its nearest

point $z \in C$ by a taut string, the direction of the string $x - z$ is perpendicular to the “surface” of C at z , and the hyperplane perpendicular to the string through the midpoint leaves x on one side and all of C on the other. For this plan to work we need three things: that the nearest point *exists*, that it is *unique* (so that we can speak of “the” projection), and a manageable algebraic characterization of “being the nearest one”. All three things require hypotheses: existence asks that C be *closed* (if C is the open ball and x is outside, the infimum of the distance is not attained), and uniqueness asks that C be *convex* (if C is a circle and x its center, all points of C are equally close). This is the content of the following theorem, the workhorse of the chapter ([6], Theorem 2.10; [3], ch. 4; [1]).

Theorem 4.6 (Projection theorem). *Let $C \subseteq \mathbb{R}^n$ be a nonempty, closed and convex set, and let $x \in \mathbb{R}^n$ be any point. Then:*

- (i) **Existence:** *there is at least one point $z \in C$ that minimizes the distance to x , that is, $\|x - z\| \leq \|x - y\|$ for all $y \in C$.*
- (ii) **Uniqueness:** *that minimizer is unique. We write it $z = \text{proj}_C(x)$ and call it the projection of x onto C .*
- (iii) **Variational characterization:** *a point $z \in C$ is the projection of x if and only if*

$$\langle x - z, y - z \rangle \leq 0 \quad \text{for all } y \in C. \quad (4.1)$$

Before the proof, let us understand condition (4.1), called the *variational inequality*. Recall from the preliminaries chapter that $\langle u, v \rangle = \|u\| \|v\| \cos \theta$, where θ is the angle between u and v . Condition (4.1) says that the angle between the “residual” vector $x - z$ (which goes from the projection toward the original point) and any vector $y - z$ (which goes from the projection toward any other point of the set) is *obtuse or right*: greater than or equal to 90 degrees. Geometrically: seen from z , the whole set C stays “on the other side” of the hyperplane perpendicular to $x - z$ through z . You can already see separation coming.

Proof. Part (i): existence, through Weierstrass. The function $f(y) = \|x - y\|$ is continuous on $\mathbb{R}^n$ (it is the composition of the norm, which is continuous by the reverse triangle inequality $|||u| - |v||| \leq \|u - v\|$, with the translation $y \mapsto x - y$). We would like to apply the Weierstrass theorem from the preliminaries chapter, but C may fail to be bounded (think of $C = \mathbb{R}_+^n$). The standard trick is to *compactify*: fix any point $y_0 \in C$ (it exists because $C \neq \emptyset$) and let $r = \|x - y_0\|$. Consider the truncated set

$$C_0 = C \cap \bar{B}(x, r) = \{y \in C : \|x - y\| \leq r\}.$$

This set is closed (an intersection of two closed sets: C by hypothesis and the closed ball $\bar{B}(x, r)$), bounded (it lies inside the ball of radius r) and nonempty (it contains y_0). By the Heine–Borel theorem, C_0 is compact, and by Weierstrass the continuous function f attains a minimum at some $z \in C_0$. It remains to see that z minimizes over *all* of C , not only over C_0 : if $y \in C \setminus C_0$, then by the definition of C_0 we have $\|x - y\| > r = \|x - y_0\| \geq \|x - z\|$ (the last inequality because $y_0 \in C_0$ and z minimizes on C_0). So no point outside the truncation can beat z , and z

minimizes the distance over all of C . Note that in this part *we did not use the convexity of C* : every nonempty closed set admits a nearest point.

Part (iii) first, direction “ $\Rightarrow$ ”: every minimizer satisfies (4.1). (We prove (iii) before (ii) because uniqueness will come for free from the characterization; this order is the one in [6].) Let z be a minimizer and let $y \in C$ be arbitrary. Here convexity enters: for each $\lambda \in (0, 1]$, the point

$$y_\lambda = z + \lambda(y - z) = (1 - \lambda)z + \lambda y$$

belongs to C (it is a convex combination of $z, y \in C$). Since z is a minimizer, $\|x - y_\lambda\|^2 \geq \|x - z\|^2$. We expand the left hand side using $\|u - v\|^2 = \|u\|^2 - 2\langle u, v \rangle + \|v\|^2$ with $u = x - z$ and $v = \lambda(y - z)$:

$$\|x - y_\lambda\|^2 = \|(x - z) - \lambda(y - z)\|^2 = \|x - z\|^2 - 2\lambda\langle x - z, y - z \rangle + \lambda^2\|y - z\|^2.$$

Substituting into $\|x - y_\lambda\|^2 \geq \|x - z\|^2$ and cancelling $\|x - z\|^2$ on both sides:

$$-2\lambda\langle x - z, y - z \rangle + \lambda^2\|y - z\|^2 \geq 0.$$

Since $\lambda > 0$, we can divide by λ without flipping the inequality:

$$-2\langle x - z, y - z \rangle + \lambda\|y - z\|^2 \geq 0.$$

This inequality holds for *every* $\lambda \in (0, 1]$; letting $\lambda \rightarrow 0^+$ (the left hand side is a continuous function, in fact affine, of λ), we get $-2\langle x - z, y - z \rangle \geq 0$, that is, $\langle x - z, y - z \rangle \leq 0$. Since $y \in C$ was arbitrary, (4.1) is proved. Note the logic of the limit step: if we had only evaluated at $\lambda = 1$ we would get an inequality contaminated by the quadratic term $\|y - z\|^2$; sending $\lambda \rightarrow 0^+$ isolates the first order term, exactly as in a first order condition in calculus.

Part (iii), direction “ $\Leftarrow$ ”: condition (4.1) implies that z is a minimizer. Suppose that $z \in C$ satisfies (4.1) and let $y \in C$ be arbitrary. We write $x - y = (x - z) + (z - y)$ and expand:

$$\begin{aligned} \|x - y\|^2 &= \|(x - z) + (z - y)\|^2 \\ &= \|x - z\|^2 + 2\langle x - z, z - y \rangle + \|z - y\|^2 \\ &= \|x - z\|^2 + \underbrace{-2\langle x - z, y - z \rangle}_{\geq 0 \text{ by (4.1)}} + \underbrace{\|z - y\|^2}_{\geq 0} \geq \|x - z\|^2. \end{aligned}$$

(In the third line we used $\langle x - z, z - y \rangle = -\langle x - z, y - z \rangle$, and that (4.1) says precisely that $-\langle x - z, y - z \rangle \geq 0$.) Taking square roots, $\|x - y\| \geq \|x - z\|$ for all $y \in C$: z is a minimizer. Note that this direction gives something stronger than we needed, the quantitative bound

$$\|x - y\|^2 \geq \|x - z\|^2 + \|y - z\|^2 \quad \text{for all } y \in C, \tag{4.2}$$

a kind of “Pythagorean theorem with an inequality” that we will use in the exercises.

Part (ii): uniqueness. We give two proofs, because both techniques come back in the course.

Proof A (through the characterization). Suppose $z_1, z_2 \in C$ are both minimizers. By part (iii), both satisfy (4.1). Apply (4.1) at z_1 with $y = z_2$, and at z_2 with $y = z_1$:

$$\langle x - z_1, z_2 - z_1 \rangle \leq 0, \quad \langle x - z_2, z_1 - z_2 \rangle \leq 0.$$

In the second one, factor the sign: $\langle x - z_2, z_1 - z_2 \rangle = -\langle x - z_2, z_2 - z_1 \rangle \leq 0$, that is, $\langle x - z_2, z_2 - z_1 \rangle \geq 0$. Subtracting the first inequality from this last one (that is, adding $\langle z_2 - x, z_2 - z_1 \rangle \geq 0$ and $\langle x - z_1, z_2 - z_1 \rangle \leq 0$ with the right sign):

$$\begin{aligned} 0 &\leq \langle x - z_2, z_2 - z_1 \rangle - \langle x - z_1, z_2 - z_1 \rangle \\ &= \langle (x - z_2) - (x - z_1), z_2 - z_1 \rangle = \langle z_1 - z_2, z_2 - z_1 \rangle = -\|z_1 - z_2\|^2. \end{aligned}$$

We obtained $0 \leq -\|z_1 - z_2\|^2$, and since a squared norm is always ≥ 0 , the only possibility is $\|z_1 - z_2\| = 0$, that is, $z_1 = z_2$. Uniqueness is proved.

Proof B (through the parallelogram law). Recall the parallelogram identity, valid in any inner product space (you check it by expanding both sides):

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2.$$

Let z_1, z_2 be two minimizers, both at distance $d = \|x - z_1\| = \|x - z_2\|$ from x . Apply the identity with $u = x - z_1$ and $v = x - z_2$; then $u + v = 2x - z_1 - z_2 = 2(x - \frac{z_1 + z_2}{2})$ and $u - v = z_2 - z_1$:

$$4\left\|x - \frac{z_1 + z_2}{2}\right\|^2 + \|z_2 - z_1\|^2 = 2d^2 + 2d^2 = 4d^2,$$

from which, solving,

$$\left\|x - \frac{z_1 + z_2}{2}\right\|^2 = d^2 - \frac{1}{4}\|z_2 - z_1\|^2.$$

Here convexity enters: the midpoint $\frac{z_1 + z_2}{2}$ belongs to C (a convex combination with $\lambda = \frac{1}{2}$), so its distance to x is at least d , the minimum distance: $\left\|x - \frac{z_1 + z_2}{2}\right\|^2 \geq d^2$. Combining,

$$d^2 \leq d^2 - \frac{1}{4}\|z_2 - z_1\|^2 \implies \|z_2 - z_1\|^2 \leq 0 \implies z_1 = z_2. \quad \square$$

Intuition

Proof B has a beautiful geometric moral: if there were *two* distinct nearest points, the midpoint between them would be *strictly closer* to x (the hypotenuse is longer than the leg), and by convexity that midpoint would still be in C : a contradiction with minimality. Convexity is what guarantees that “averaging candidates” does not take you out of the set. This is the same logic behind the fact that *strictly convex* functions have a unique minimizer, as we will see later: in fact, the function $y \mapsto \|x - y\|^2$ is strictly convex, and proof B is a particular case of that general principle.

Example 4.7 (Projection onto a ball and onto the orthant). Two projections that can be computed by hand and are worth memorizing.

(a) *Closed ball* $C = \bar{B}(0, r) = \{y : \|y\| \leq r\}$. If $\|x\| \leq r$, then $x \in C$ and $\text{proj}_C(x) = x$ (the minimum distance, zero, is attained at the point itself). If $\|x\| > r$, we claim that $\text{proj}_C(x) =$

$rx/\|x\|$ (the point of the sphere in the direction of x). Let us check it with the variational characterization, which is the cleanest route: with $z = rx/\|x\|$,

$$x - z = \left(1 - \frac{r}{\|x\|}\right)x,$$

and for any $y \in C$,

$$\langle x - z, y - z \rangle = \left(1 - \frac{r}{\|x\|}\right)(\langle x, y \rangle - \langle x, z \rangle) = \left(1 - \frac{r}{\|x\|}\right)(\langle x, y \rangle - r\|x\|).$$

The first factor is positive (since $\|x\| > r$). For the second, Cauchy–Schwarz gives $\langle x, y \rangle \leq \|x\| \|y\| \leq \|x\| r$, so $\langle x, y \rangle - r\|x\| \leq 0$. The product is ≤ 0 : condition (4.1) holds and, by uniqueness, z is the projection.

(b) *Nonnegative orthant* $C = \mathbb{R}_+^n$. We claim that the projection is computed coordinate by coordinate: $\text{proj}_C(x) = x^+$, where $(x^+)_i = \max\{x_i, 0\}$. Variational check: $x - x^+$ has coordinates $(x - x^+)_i = \min\{x_i, 0\} \leq 0$, and moreover $(x - x^+)_i = 0$ whenever $x_i \geq 0$. For $y \in \mathbb{R}_+^n$,

$$\langle x - x^+, y - x^+ \rangle = \sum_{i=1}^n (x - x^+)_i (y_i - x_i^+) = \sum_{i: x_i < 0} x_i y_i \leq 0,$$

because in the coordinates with $x_i \geq 0$ the first factor is zero, and in the coordinates with $x_i < 0$ we have $x_i^+ = 0$, first factor $x_i < 0$ and second factor $y_i \geq 0$. Again (4.1) holds and the formula is proved. This projection appears in economics every time nonnegativity constraints are imposed (quantities, prices, KKT multipliers).

4.2.2 Nonexpansiveness of the projection

The projection not only exists and is unique: it is an extraordinarily well behaved operator. The following property, called *nonexpansiveness*, says that projecting never pushes points apart: the distance between the projections of two points is at most the distance between the original points. It is the basis of alternating projection algorithms and of the analysis of projected gradient methods ([5], ch. 16; [8]).

Proposition 4.8 (Nonexpansiveness). *Let $C \subseteq \mathbb{R}^n$ be nonempty, closed and convex. For all $x, x' \in \mathbb{R}^n$:*

- (a) (*Firm nonexpansiveness*) $\|\text{proj}_C(x) - \text{proj}_C(x')\|^2 \leq \langle x - x', \text{proj}_C(x) - \text{proj}_C(x') \rangle$.
- (b) (*Nonexpansiveness*) $\|\text{proj}_C(x) - \text{proj}_C(x')\| \leq \|x - x'\|$. In particular, proj_C is a continuous function (in fact, Lipschitz with constant 1).

Proof. Write $z = \text{proj}_C(x)$ and $z' = \text{proj}_C(x')$ to lighten the notation.

(a) We apply the variational inequality (4.1) twice, crossing the points: at x with test point $y = z' \in C$, and at x' with test point $y = z \in C$:

$$\langle x - z, z' - z \rangle \leq 0, \quad \langle x' - z', z - z' \rangle \leq 0.$$

The strategy is to add them and group terms; the signs must be handled with care. We add the

two inequalities as they are:

$$\langle x - z, z' - z \rangle + \langle x' - z', z - z' \rangle \leq 0.$$

In the second summand, $\langle x' - z', z - z' \rangle = -\langle x' - z', z' - z \rangle$, so

$$\langle x - z, z' - z \rangle - \langle x' - z', z' - z \rangle \leq 0 \iff \langle (x - x') - (z - z'), z' - z \rangle \leq 0.$$

Distributing the inner product:

$$\langle x - x', z' - z \rangle - \langle z - z', z' - z \rangle \leq 0.$$

Since $\langle z - z', z' - z \rangle = -\|z - z'\|^2$, this is

$$\langle x - x', z' - z \rangle + \|z - z'\|^2 \leq 0 \iff \|z - z'\|^2 \leq \langle x - x', z - z' \rangle,$$

which is exactly (a).

(b) If $z = z'$ there is nothing to prove. If $z \neq z'$, we apply the Cauchy–Schwarz inequality to the right hand side of (a):

$$\|z - z'\|^2 \leq \langle x - x', z - z' \rangle \leq \|x - x'\| \|z - z'\|.$$

Dividing both sides by $\|z - z'\| > 0$ we get $\|z - z'\| \leq \|x - x'\|$. Lipschitz continuity is immediate from this bound. $\square$

Remark 4.9. The “firm” version (a) is stronger than (b): it says that the displacement of the projections is not only short, but *positively aligned* with the displacement of the original points. In the algorithm chapters this property guarantees the convergence of projected gradient; in consumer theory, it is the technical reason why demands (which are generalized projections onto the budget set) respond continuously to prices when there is convexity.

4.3 Separation of a point and a convex set [P1]

4.3.1 The vocabulary of separation

Before the theorems, let us fix precisely what “to separate” means, because there are several notions of increasing strength and the statements in Microeconomics and in textbooks mix them if you are not careful.

Definition 4.10 (Types of separation). Let $A, B \subseteq \mathbb{R}^n$ be nonempty and let $H(a, \beta)$ be a hyperplane ($a \neq 0$). We say that $H(a, \beta)$:

- (a) *separates* A and B if $\langle a, u \rangle \leq \beta \leq \langle a, v \rangle$ for all $u \in A, v \in B$; equivalently, $A \subseteq H^-$ and $B \subseteq H^+$;
- (b) *separates them properly* if in addition $A \cup B \not\subseteq H$, that is, it does not happen that both sets live inside the hyperplane itself (which would make the separation trivial and useless);

- (c) separates them *strictly* if $\langle a, u \rangle < \beta < \langle a, v \rangle$ for all $u \in A, v \in B$;
- (d) separates them *strongly* if there is $\varepsilon > 0$ such that $\langle a, u \rangle \leq \beta - \varepsilon < \beta + \varepsilon \leq \langle a, v \rangle$ for all $u \in A, v \in B$; equivalently, $\sup_{u \in A} \langle a, u \rangle < \inf_{v \in B} \langle a, v \rangle$.

Clearly strong $\Rightarrow$ strict $\Rightarrow$ proper $\Rightarrow$ plain separation (the second implication holds because, if every point of A satisfies $\langle a, u \rangle < \beta$, then $A \cup B \not\subseteq H$). The converse implications fail, and part of the work of this chapter is to understand *which hypotheses buy which level of separation*; Section 4.5 gives the counterexamples.

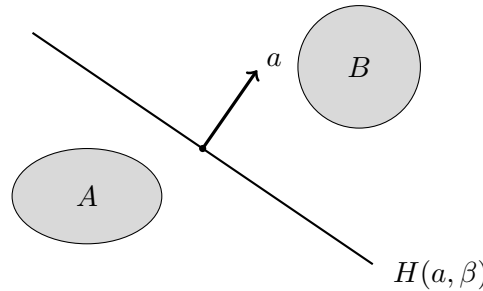


Figure 4.2: Two convex sets separated by the hyperplane $H(a, \beta)$: the linear function $\langle a, \cdot \rangle$ is at most β on A and at least β on B . The normal a points to the side of B .

4.3.2 Strong separation of a point and a closed convex set

With the projection theorem, the first separation theorem comes out almost by itself: the taut string $x - \text{proj}_C(x)$ hands us the normal.

Theorem 4.11 (Strong point set separation). *Let $C \subseteq \mathbb{R}^n$ be nonempty, closed and convex, and let $x \notin C$. Then there are $a \in \mathbb{R}^n, a \neq 0$, and $\beta \in \mathbb{R}$ such that*

$$\langle a, y \rangle < \beta < \langle a, x \rangle \quad \text{for all } y \in C,$$

and in fact the separation is strong: $\sup_{y \in C} \langle a, y \rangle < \beta < \langle a, x \rangle$. Concretely, $a = x - \text{proj}_C(x)$ works, with $\beta = \langle a, \text{proj}_C(x) \rangle + \frac{1}{2} \|a\|^2$.

Proof. Let $z = \text{proj}_C(x)$, which exists and is unique by Theorem 4.6, and define $a = x - z$.

Step 1: $a \neq 0$. If $a = 0$ we would have $x = z \in C$, contradicting $x \notin C$. (Here it is essential that C be closed: this is what guarantees that the projection is *in* C and that a point outside C is at positive distance, $\|a\| > 0$.)

Step 2: the linear function $\langle a, \cdot \rangle$ is bounded on C by its value at z . Take any $y \in C$. The variational inequality (4.1) says $\langle x - z, y - z \rangle \leq 0$, that is, $\langle a, y - z \rangle \leq 0$. By linearity,

$$\langle a, y \rangle \leq \langle a, z \rangle \quad \text{for all } y \in C.$$

In words: the whole set C lives in the half space $\{w : \langle a, w \rangle \leq \langle a, z \rangle\}$.

Step 3: the value at x is strictly larger, with quantified slack. We compute

$$\langle a, x \rangle - \langle a, z \rangle = \langle a, x - z \rangle = \langle a, a \rangle = \|a\|^2 > 0.$$

That is, $\langle a, x \rangle = \langle a, z \rangle + \|a\|^2$.

Step 4: choose the level β in the middle. We take $\beta = \langle a, z \rangle + \frac{1}{2} \|a\|^2$, the midpoint between the largest possible value on C and the value at x . Then, for all $y \in C$,

$$\langle a, y \rangle \leq \langle a, z \rangle = \beta - \frac{1}{2} \|a\|^2 < \beta < \beta + \frac{1}{2} \|a\|^2 = \langle a, x \rangle.$$

The chain shows separation with slack $\varepsilon = \frac{1}{2} \|a\|^2 > 0$ on each side: strong separation. Moreover $\sup_{y \in C} \langle a, y \rangle \leq \langle a, z \rangle < \beta$, as claimed. $\square$

Intuition

The proof is a miniature price construction algorithm: (1) project the point onto the set; (2) the direction of the “projection error” $x - \text{proj}_C(x)$ is the separating normal; (3) the level β is placed halfway. In the Second Welfare Theorem we will do exactly this with C equal to a set of allocations “better than the optimum”, and the resulting normal will be the price vector. When you ask yourself “where do prices come from?”, the technical answer is: from a projection.

4.3.3 Consequence: closed convex sets are intersections of half spaces

Corollary 4.12 (External representation of a closed convex set). *Every nonempty closed convex set $C \subseteq \mathbb{R}^n$ is the intersection of all closed half spaces containing it:*

$$C = \bigcap \{H^-(a, \beta) : C \subseteq H^-(a, \beta)\}.$$

Proof. Call D the intersection on the right.

($\subseteq$) If $x \in C$, then x belongs to every half space containing C , so $x \in D$. This inclusion is trivial and uses no hypothesis.

($\supseteq$) We prove the contrapositive: if $x \notin C$, we exhibit a half space that contains C but not x , so that $x \notin D$. By Theorem 4.11 there are $a \neq 0$ and β with $\langle a, y \rangle < \beta$ for all $y \in C$ and $\langle a, x \rangle > \beta$. Then $C \subseteq H^-(a, \beta)$ (even with strict inequality), so $H^-(a, \beta)$ is one of the half spaces of the family, but $x \notin H^-(a, \beta)$ because $\langle a, x \rangle > \beta$. Therefore $x \notin D$. $\square$

Remark 4.13 (Duality in embryo). This corollary is conceptually huge: it says that a closed convex set admits two equivalent descriptions, an *internal* one (as the set of its points, or of convex combinations of them) and an *external* one (as an intersection of linear inequalities). Going from one description to the other is the essence of *duality* in convex optimization ([2], sections 2.5 and 5.1; [6], ch. 2). In economics, the same duality connects the utility function with the expenditure function, and the production set with the profit function: to know “all the supporting hyperplanes” of a convex set is to know the set. The envelope theorem and the Shephard and Hotelling identities (which you will see in Microeconomics and in chapter 11, [11]) live precisely in this correspondence.

Careful (common mistake)

The hypotheses of Theorem 4.11 are the two essential ones and neither can be dropped.

Without closedness: take $C = B(0,1)$ the open ball in $\mathbb{R}^2$ and $x = (1,0) \notin C$. Every hyperplane that “separates” must satisfy $\langle a, y \rangle \leq \beta \leq \langle a, x \rangle$, but since there are points of C arbitrarily close to x , by continuity $\sup_{y \in C} \langle a, y \rangle \geq \langle a, x \rangle$: the separation exists (not strict, with equal supremum and infimum) but *strong* separation is impossible.

Without convexity: take $C = \{y \in \mathbb{R}^2 : \|y\| = 1\}$ the circle (closed but not convex) and $x = 0$. Any hyperplane through the origin leaves points of C strictly on both sides, and a hyperplane not through the origin cannot leave the whole circle on one side and the origin on the other (the circle “surrounds” x). There is no separation of any kind. Moral: when an exam asks you to separate, *check first* closedness and convexity, and if it asks for strong separation between two sets, look also for compactness (Theorem 4.19).

4.4 Supporting hyperplanes [P1]

4.4.1 Motivation and definition

Theorem 4.11 separates a point *outside* the set. But in economics the interesting points are usually on the *boundary*: the optimal bundle is on the budget line, the optimal production plan is on the boundary of the production set, the Pareto optimum is on the boundary of the set of feasible utilities. For those points there can be no strong separation (the point belongs to the closure of the set), but there is something very useful: a hyperplane that *touches* the set at that point and leaves all of it on one side, like a ruler resting against the rim of a plate.

Definition 4.14 (Supporting hyperplane). Let $C \subseteq \mathbb{R}^n$ and $x_0 \in \text{bd } C$ a point of its boundary. A hyperplane $H(a, \beta)$ is a *supporting hyperplane* of C at x_0 if

$$\langle a, x_0 \rangle = \beta \quad \text{and} \quad \langle a, y \rangle \leq \beta \quad \text{for all } y \in C.$$

That is: the hyperplane passes through x_0 and the whole set stays in the lower half space. The vector a is called the *outer normal* of C at x_0 , and we say that H *supports* C at x_0 .

Note the connection with maximization: $H(a, \beta)$ supports C at x_0 exactly when $x_0 \in \arg \max_{y \in C} \langle a, y \rangle$, with maximum value β . A supporting hyperplane is, literally, the certificate that x_0 maximizes a linear function on C : for a firm, “the plan y^* maximizes profit at prices p ” means “the isoprofit hyperplane with normal p supports the production set at y^* ”.

Theorem 4.15 (Existence of a supporting hyperplane). *Let $C \subseteq \mathbb{R}^n$ be convex and nonempty, and let $x_0 \in \text{bd } C$ be a boundary point of C . Then there is a supporting hyperplane of C at x_0 : there is $a \neq 0$ such that $\langle a, y \rangle \leq \langle a, x_0 \rangle$ for all $y \in C$.*

Proof. The idea: approximate x_0 by *exterior* points $x_k \rightarrow x_0$, separate each x_k from the set with Theorem 4.11, and pass to the limit in the normals. The limit step requires a compactness argument that we spell out.

Step 0: reduction to the case with nonempty interior. If $\text{int } C = \emptyset$, chapter 2 tells us that C is contained in a proper affine set $\text{aff } C \neq \mathbb{R}^n$, and therefore in some hyperplane $H(a, \beta)$ through x_0 (every proper affine set containing C is contained in a hyperplane containing C ; since $x_0 \in \text{cl } C \subseteq H$ because the hyperplane is closed, we can take $\beta = \langle a, x_0 \rangle$). A hyperplane that *contains* C supports it trivially at any of its points: $\langle a, y \rangle = \beta$ for all $y \in C$. From now on we assume $\text{int } C \neq \emptyset$.

Step 1: existence of the exterior sequence. We work with $D = \text{cl } C$, which is convex (the closure of a convex set, ch. 2) and closed. Since $\text{int } C \neq \emptyset$, we know from chapter 2 that $\text{int}(\text{cl } C) = \text{int } C$ and $\text{bd}(\text{cl } C) = \text{bd } C$; in particular $x_0 \in \text{bd } D$, so every ball $B(x_0, 1/k)$ contains points of the complement of D . For each $k \in \mathbb{N}$ we choose a point $x_k \notin D$ with $\|x_k - x_0\| < 1/k$; by construction $x_k \rightarrow x_0$.

Step 2: separate each x_k . Since D is closed, convex, nonempty and $x_k \notin D$, Theorem 4.11 produces a vector $\tilde{a}_k \neq 0$ with $\langle \tilde{a}_k, y \rangle < \langle \tilde{a}_k, x_k \rangle$ for all $y \in D$. We normalize (Remark 4.2): define $a_k = \tilde{a}_k / \|\tilde{a}_k\|$, which satisfies the same inequality (dividing by a positive number preserves it):

$$\langle a_k, y \rangle < \langle a_k, x_k \rangle \quad \text{for all } y \in D, \quad \|a_k\| = 1. \quad (4.3)$$

Step 3: compactness of the unit sphere. Here is the heart of the argument. The sequence (a_k) lives on the unit sphere $S^{n-1} = \{a \in \mathbb{R}^n : \|a\| = 1\}$, which is a closed and bounded subset of $\mathbb{R}^n$, hence *compact* (Heine–Borel). By the Bolzano–Weierstrass theorem (preliminaries chapter), every sequence in a compact set has a convergent subsequence *with limit in the compact set*: there are a subsequence (a_{k_j}) and a vector $a \in S^{n-1}$ with $a_{k_j} \rightarrow a$. The crucial point is that $\|a\| = 1$, in particular $a \neq 0$: the normalization of Step 2 is what prevents the normals from “degenerating” to zero in the limit, which would produce the empty statement $0 \leq 0$. Without compactness there would be no way to guarantee a limit; this pattern (normalize plus Bolzano–Weierstrass) is a technique that comes back throughout convex analysis and in general equilibrium theory.

Step 4: pass to the limit. Fix $y \in D$ and take limits in (4.3) along the subsequence. The inner product is continuous in each argument, and $a_{k_j} \rightarrow a$, $x_{k_j} \rightarrow x_0$; therefore

$$\langle a_{k_j}, y \rangle \rightarrow \langle a, y \rangle, \quad \langle a_{k_j}, x_{k_j} \rangle \rightarrow \langle a, x_0 \rangle$$

(for the second convergence: $|\langle a_{k_j}, x_{k_j} \rangle - \langle a, x_0 \rangle| \leq |\langle a_{k_j}, x_{k_j} - x_0 \rangle| + |\langle a_{k_j} - a, x_0 \rangle| \leq \|x_{k_j} - x_0\| + \|a_{k_j} - a\| \|x_0\| \rightarrow 0$, using Cauchy–Schwarz and $\|a_{k_j}\| = 1$). Strict inequalities become non strict in the limit (general rule: $u_j < v_j$ and $u_j \rightarrow u$, $v_j \rightarrow v$ only guarantee $u \leq v$), and we obtain

$$\langle a, y \rangle \leq \langle a, x_0 \rangle \quad \text{for all } y \in D \supseteq C.$$

With $\beta = \langle a, x_0 \rangle$, the hyperplane $H(a, \beta)$ passes through x_0 and leaves C in the lower half space: it is a supporting hyperplane. $\square$

Remark 4.16 (Why only non strict separation?). In the limit we lost the strict inequality, and this is not a defect of the proof: it is impossible to do better. Since $x_0 \in \text{cl } C$, there are points $y \in C$ arbitrarily close to x_0 , and by continuity $\sup_{y \in C} \langle a, y \rangle = \langle a, x_0 \rangle$ exactly. The supporting

hyperplane *touches* the set; it separates, but neither strictly nor strongly. This phenomenon will come back in the Second Welfare Theorem: the separation obtained there is also non strict, and this is why the theorem first delivers a “quasi equilibrium” that then has to be improved with an extra argument (Section 4.7).

Example 4.17 (Supporting hyperplanes of the disk and of the square). (a) Let $C = \bar{B}(0, 1) \subset \mathbb{R}^2$ and $x_0 = (\frac{3}{5}, \frac{4}{5}) \in \text{bd } C$ (it is on the unit circle since $(3/5)^2 + (4/5)^2 = 1$). We claim that $a = x_0$ is an outer normal. Indeed, for $y \in C$, Cauchy–Schwarz gives

$$\langle x_0, y \rangle \leq \|x_0\| \|y\| \leq 1 \cdot 1 = 1 = \langle x_0, x_0 \rangle,$$

so the hyperplane $\{y : \frac{3}{5}y_1 + \frac{4}{5}y_2 = 1\}$ supports C at x_0 and is unique (the circle is “smooth”: at each boundary point there is exactly one tangent line).

(b) Let $C = [0, 1]^2$ and $x_0 = (1, 1)$, a corner. Here there are *many* supporting hyperplanes: any $a = (a_1, a_2)$ with $a_1, a_2 \geq 0$, $a \neq 0$, satisfies, for $y \in [0, 1]^2$, $\langle a, y \rangle = a_1y_1 + a_2y_2 \leq a_1 + a_2 = \langle a, x_0 \rangle$. At “corner” points the supporting hyperplane exists but is not unique. This multiplicity is not a curiosity: it is exactly what happens with kinked indifference curves (for example, Leontief preferences $u(x) = \min\{x_1, x_2\}$), where there is a fan of prices that support the same optimal bundle, and it is the seed of the notion of *subdifferential* of chapter 7.

4.5 Separation of two convex sets [P1]

4.5.1 The Minkowski difference trick

To separate two sets A and B the master trick is to reduce the problem to the point set case that we already control, using the *Minkowski difference*

$$A - B = \{u - v : u \in A, v \in B\}.$$

The key observation is purely logical:

$$A \cap B = \emptyset \iff 0 \notin A - B,$$

because $0 = u - v$ with $u \in A, v \in B$ is the same as $u = v \in A \cap B$. Moreover, if A and B are convex, $A - B = A + (-B)$ is convex (Minkowski sum of convex sets, chapter 2). And separating the point 0 from $A - B$ with normal a , that is $\langle a, u - v \rangle \leq 0$ for all u, v , is exactly saying $\langle a, u \rangle \leq \langle a, v \rangle$: A and B are separated. The whole problem then reduces to one question: which properties does $A - B$ inherit from A and B ? Here a famous topological subtlety appears.

Lemma 4.18 (Compact minus closed is closed). *If $A \subseteq \mathbb{R}^n$ is compact and $B \subseteq \mathbb{R}^n$ is closed, then $A - B$ is closed. In contrast, the difference of two closed sets may fail to be closed.*

Proof. Let (w_k) be a sequence in $A - B$ with $w_k \rightarrow w$; we must prove $w \in A - B$. Write $w_k = u_k - v_k$ with $u_k \in A, v_k \in B$. Since A is compact, by Bolzano–Weierstrass there is a subsequence $u_{k_j} \rightarrow u \in A$. Then

$$v_{k_j} = u_{k_j} - w_{k_j} \longrightarrow u - w,$$

because both u_{k_j} and w_{k_j} converge (here the compactness of A is what “anchors” the sequence v_{k_j} , which a priori could escape to infinity). Since B is closed and $v_{k_j} \in B$, the limit satisfies $u - w \in B$. Therefore $w = u - (u - w) \in A - B$.

For the counterexample with two closed sets: in $\mathbb{R}^2$ take

$$A = \{(t, 0) : t \in \mathbb{R}\} \quad (\text{the horizontal axis}), \quad B = \{(t, 1/t) : t > 0\} \quad (\text{a branch of a hyperbola}),$$

both closed and, in fact, the first one convex (the second is not convex, but it can be replaced by its convex “epigraph” $\{(t, s) : t > 0, s \geq 1/t\}$, also closed, and the argument is identical). The difference $A - B$ contains the points $(t - t, 0 - 1/t) = (0, -1/t)$ for every $t > 0$, which converge to $(0, 0)$; but $(0, 0) \notin A - B$, since $(0, 0) = (t, 0) - (s, 1/s)$ would require $1/s = 0$, which is impossible. The origin is an accumulation point that does not belong: $A - B$ is not closed. $\square$

4.5.2 Strong separation: disjoint compact and closed sets

Theorem 4.19 (Strong separation of two convex sets). *Let $A, B \subseteq \mathbb{R}^n$ be convex, nonempty and disjoint, with A compact and B closed. Then there are $a \neq 0$ and $\beta_1 < \beta_2$ such that*

$$\langle a, u \rangle \leq \beta_1 < \beta_2 \leq \langle a, v \rangle \quad \text{for all } u \in A, v \in B :$$

strong separation, $\sup_{u \in A} \langle a, u \rangle < \inf_{v \in B} \langle a, v \rangle$.

Proof. Consider $D = B - A$. By what we said above: D is convex (Minkowski sum $B + (-A)$ of convex sets), it is closed (by Lemma 4.18, applied to the difference of the closed set B and the compact set A ; note that $B - A = -(A - B)$ and the negative of a closed set is closed), it is nonempty, and $0 \notin D$ because $A \cap B = \emptyset$.

We are in the exact hypotheses of Theorem 4.11 with $C = D$ and $x = 0$: the point set theorem delivers $\tilde{a} \neq 0$ (concretely $\tilde{a} = 0 - \text{proj}_D(0) = -\text{proj}_D(0)$) and a level β with

$$\langle \tilde{a}, d \rangle < \beta < \langle \tilde{a}, 0 \rangle = 0 \quad \text{for all } d \in D.$$

The *strong* separation conclusion of that theorem says more: $\sup_{d \in D} \langle \tilde{a}, d \rangle < \beta < 0$, so there is $\varepsilon = -\beta > 0$ such that

$$\langle \tilde{a}, v - u \rangle \leq -\varepsilon \quad \text{for all } u \in A, v \in B.$$

We define $a = -\tilde{a}$ to orient the normal toward B ; the inequality becomes $\langle a, v - u \rangle \geq \varepsilon$, that is

$$\langle a, v \rangle \geq \langle a, u \rangle + \varepsilon \quad \text{for all } u \in A, v \in B.$$

Taking the supremum over $u \in A$ on the right hand side (the supremum is finite: A is compact and $\langle a, \cdot \rangle$ is continuous, so by Weierstrass the supremum is attained) and then the infimum over $v \in B$ on the left:

$$\inf_{v \in B} \langle a, v \rangle \geq \sup_{u \in A} \langle a, u \rangle + \varepsilon > \sup_{u \in A} \langle a, u \rangle.$$

It is enough to choose $\beta_1 = \sup_{u \in A} \langle a, u \rangle$ and $\beta_2 = \beta_1 + \varepsilon$; any level $\beta \in (\beta_1, \beta_2)$ defines a hyperplane that separates strongly. $\square$

Remark 4.20. The proof shows why we ask for *one* compact set and not two closed ones: without compactness, $B - A$ could fail to be closed and the origin could be an accumulation point of $B - A$ without belonging to it, which kills strong separation. The canonical counterexample, which you should know how to draw and cite in the exam, is:

$$A = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \leq 0\}, \quad B = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0, x_2 \geq 1/x_1\}.$$

Both are closed, convex (the second is the epigraph of the convex function $x_1 \mapsto 1/x_1$ on the domain $x_1 > 0$, and it is easy to check that it is closed) and disjoint. The only separating hyperplane is the horizontal axis $\{x_2 = 0\}$: it separates, even properly, but *not strongly*, because $\inf_{v \in B} v_2 = 0 = \sup_{u \in A} u_2$ (the hyperbola approaches the axis asymptotically without touching it). The distance between A and B is zero even though they are disjoint. *Another* counterexample, even simpler, for “strict”: two closed disks tangent from outside, such as $\bar{B}((-1, 0), 1)$ and $\bar{B}((1, 0), 1)$, which touch at the origin; the vertical line $\{x_1 = 0\}$ separates them, but no hyperplane can separate them strictly because the common point $(0, 0)$ would have to be strictly on both sides at once.

4.5.3 Separation of arbitrary disjoint convex sets

What if there is no compactness? You can still separate, although only in the weak sense. The next theorem requires neither closedness nor boundedness: only convexity and disjointness.

Theorem 4.21 (Separation of disjoint convex sets). *Let $A, B \subseteq \mathbb{R}^n$ be convex, nonempty and disjoint. Then there is $a \neq 0$ such that*

$$\langle a, u \rangle \leq \langle a, v \rangle \quad \text{for all } u \in A, v \in B;$$

that is, $\sup_{u \in A} \langle a, u \rangle \leq \inf_{v \in B} \langle a, v \rangle$, and any β between those two values defines a separating hyperplane.

Proof. Let $D = B - A$, convex and nonempty, with $0 \notin D$ (disjointness). We distinguish two cases according to the position of the origin relative to the closure $\text{cl } D$, and in both we produce $a \neq 0$ with $\langle a, d \rangle \geq 0$ for all $d \in D$.

Case 1: $0 \notin \text{cl } D$. The set $\text{cl } D$ is closed, convex (the closure of a convex set) and nonempty, and the origin is outside. Theorem 4.11 (applied with the right orientation of the normal, as in the previous proof) gives $a \neq 0$ with $\langle a, d \rangle > \langle a, 0 \rangle = 0$ for all $d \in \text{cl } D \supseteq D$. Done (even with strict inequality).

Case 2: $0 \in \text{cl } D$ but $0 \notin D$. Then the origin is in the closure without being in the set, so $0 \in \text{bd } D$ (a point of $\text{cl } D \setminus D$ cannot be interior, since $\text{int } D \subseteq D$). By the supporting hyperplane theorem (Theorem 4.15, applied to the convex set D and its boundary point 0 ; note that the theorem does not require the point to belong to the set, only to be on the boundary), there is $a' \neq 0$ with $\langle a', d \rangle \leq \langle a', 0 \rangle = 0$ for all $d \in D$. Taking $a = -a'$: $\langle a, d \rangle \geq 0$ for all $d \in D$.

In both cases we have $a \neq 0$ with $\langle a, v - u \rangle \geq 0$, that is $\langle a, u \rangle \leq \langle a, v \rangle$, for all $u \in A, v \in B$. Taking supremum and infimum gives the conclusion. $\square$

Remark 4.22 (The relative interior version). The sharper statement, the one cited by [6] (Theorem 2.15) and Rockafellar, uses the *relative interior* of chapter 2: two nonempty convex sets A and B admit *proper* separation (Definition 4.10(b)) if and only if

$$\text{ri } A \cap \text{ri } B = \emptyset.$$

This condition is weaker than disjointness: it allows A and B to share boundary points. For example, the two tangent disks above touch at one point, but their relative interiors are disjoint, and indeed the common tangent line separates them properly. The need to ask for *proper* separation (and not plain separation) is seen with $A = B =$ a segment inside a line: the hyperplane containing the line “separates” trivially ($\langle a, u \rangle = \beta = \langle a, v \rangle$ for all points) without giving any information; proper separation excludes that triviality. We will not prove the general sufficiency here (it requires a careful analysis of $\text{ri}(A - B)$; see [6], sec. 2.2), but it is worth memorizing the statement because it is the one used in serious proofs of general equilibrium. A useful particular case that does follow from what we proved: if A is *open* and disjoint from B , the separation of Theorem 4.21 is automatically strict on the side of A , that is $\langle a, u \rangle < \beta$ for all $u \in A$ (if some point of an open set attained the supremum level β , we could move a little in the direction a without leaving A and exceed β , a contradiction); this is Exercise 4.34.

Exam strategy

Facing a separation problem (in the exam or in a paper), ask yourself these questions in order. (1) Are the two sets convex? If not, no theorem applies: look for a way to convexify (convex hull, randomization, aggregation in the style of Aumann). (2) Which strength of separation do I need? For *existence of prices* non strict separation is usually enough (Theorem 4.21 or supporting hyperplanes); for *uniqueness or robustness* you look for strong separation. (3) Which topology do I have? point outside a closed set $\Rightarrow$ strong (Theorem 4.11); disjoint compact and closed sets $\Rightarrow$ strong (Theorem 4.19); merely disjoint $\Rightarrow$ weak (Theorem 4.21); they touch on the boundary $\Rightarrow$ supporting hyperplane or disjoint relative interiors (Theorem 4.15, Remark 4.22). (4) To *build* the hyperplane in concrete examples: project and use $a = x - \text{proj}_C(x)$, or in the case of two sets minimize $\|u - v\|$ over $A \times B$. (5) Always check the conclusion by evaluating the linear function at concrete points of each set: it is the fastest way to detect a sign error.

4.6 The Farkas lemma [P1]

4.6.1 Motivation: linear systems with signs

Many questions in optimization and in economics reduce to: does this system of linear equations and inequalities have a solution? For example: is there a portfolio with nonnegative returns in all states and negative cost (an arbitrage)? Are there nonnegative multipliers that express the gradient of the objective as a combination of the gradients of the active constraints (KKT)? The *Farkas lemma* answers this type of question with a very elegant logical structure called an *alternative*: of two systems, **exactly one** has a solution. When the “primal” system has no

solution, the alternative system produces a verifiable *certificate* of that impossibility. The proof is a direct application of strong separation, with the cones of chapter 3 in the leading role.

Recall from chapter 3 the key object: given a matrix A of size $m \times n$ with columns $A_1, \dots, A_n \in \mathbb{R}^m$, the *finitely generated cone*

$$K = \{Ax : x \in \mathbb{R}^n, x \geq 0\} = \left\{ \sum_{j=1}^n x_j A_j : x_j \geq 0 \right\} = \text{cone}\{A_1, \dots, A_n\}$$

is the set of *nonnegative* linear combinations of the columns. In chapter 3 we saw that K is a convex cone. We also need a second fact, which we only stated there when presenting the Minkowski–Weyl theorem and which we admit here without proof: every finitely generated cone is *closed* (it is proved with a Carathéodory type argument, reducing to combinations of linearly independent columns; see [6, ch. 2] and [3, ch. 4]). That fact, which is far from obvious (the linear image of a closed set may fail to be closed), is what allows us to use strong separation.

Theorem 4.23 (Farkas lemma). *Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Exactly one of the two following systems has a solution:*

- (I) $Ax = b, \quad x \geq 0 \quad (\text{with } x \in \mathbb{R}^n);$
- (II) $A^\top y \leq 0, \quad \langle b, y \rangle > 0 \quad (\text{with } y \in \mathbb{R}^m).$

Before the proof, let us read the two systems. System (I) says “ b is a nonnegative combination of the columns of A ”, that is, $b \in K$. System (II) says “there is a vector y that makes an acute angle with b (since $\langle b, y \rangle > 0$) and an obtuse or right angle with each column A_j (since $(A^\top y)_j = \langle A_j, y \rangle \leq 0$)”. Such a y is the normal of a hyperplane through the origin that leaves the whole cone K on one side and b strictly on the other: a *geometric certificate* that $b \notin K$. The alternative says: either b is in the cone, or there is a witness hyperplane that exposes it; there is no third possibility, and never both.

Proof. Step 1: (I) and (II) cannot hold at the same time. Suppose $x \geq 0$ solves (I) and y solves (II). Then

$$0 < \langle b, y \rangle = \langle Ax, y \rangle = \left\langle x, A^\top y \right\rangle = \sum_{j=1}^n x_j (A^\top y)_j \leq 0,$$

where the second equality is the definition of the transpose matrix (adjoint property of the inner product, chapter 1) and the last inequality uses that each summand is (nonnegative) $\times$ (nonpositive) ≤ 0 . The chain $0 < 0$ is absurd; at most one of the systems is solvable.

Step 2: if (I) has no solution, then (II) does. That (I) has no solution means exactly $b \notin K$, with $K = \{Ax : x \geq 0\}$ the cone of columns. By chapter 3, K is convex, closed and nonempty ($0 = A \cdot 0 \in K$). We apply strong point set separation (Theorem 4.11) to the set K and the point $b \notin K$: there are $y \neq 0$ and β such that

$$\langle y, k \rangle < \beta < \langle y, b \rangle \quad \text{for all } k \in K. \tag{4.4}$$

(We called the normal y , oriented toward b .) Now we exploit the cone structure of K in two observations.

Observation (a): $\beta > 0$, and therefore $\langle b, y \rangle > 0$. Since $0 \in K$, inequality (4.4) with $k = 0$ gives $0 = \langle y, 0 \rangle < \beta$. And then $\langle y, b \rangle > \beta > 0$. We already have the second condition of (II).

Observation (b): $\langle y, k \rangle \leq 0$ for all $k \in K$, and therefore $A^\top y \leq 0$. Here is the typical argument with cones: suppose, looking for a contradiction, that some $k_0 \in K$ had $\langle y, k_0 \rangle > 0$. Since K is a cone, $t k_0 \in K$ for every $t > 0$, and then

$$\langle y, t k_0 \rangle = t \langle y, k_0 \rangle \xrightarrow{t \rightarrow \infty} +\infty,$$

which violates the upper bound $\langle y, k \rangle < \beta$ of (4.4) for large t . A contradiction. Therefore $\langle y, k \rangle \leq 0$ for all $k \in K$. In particular, each column $A_j = A e_j$ (with e_j the j -th canonical vector, $e_j \geq 0$) belongs to K , so $(A^\top y)_j = \langle A_j, y \rangle \leq 0$ for all j : this is $A^\top y \leq 0$, the first condition of (II).

With (a) and (b), the vector y solves (II), and the alternative is proved. $\square$

Intuition

Note how the cone structure “improves the separation for free”: a hyperplane that separates a point from a *cone* can always be taken through the origin (level 0), because a linear bound on a cone is sustainable only if the linear function is ≤ 0 on the whole cone. Cones turn inequalities with level β into homogeneous inequalities with level 0; that homogenization is the reason why KKT multipliers and arbitrage prices appear “without absolute units”, defined up to scale.

Remark 4.24 (Farkas is the key to KKT). In chapter 9 we will prove the Karush–Kuhn–Tucker conditions for $\max f(x)$ subject to $g_i(x) \leq 0$. The central step will be: “at an optimum x^* , there is no direction d that improves the objective ($\langle \nabla f(x^*), d \rangle > 0$) and keeps first order feasibility ($\langle \nabla g_i(x^*), d \rangle \leq 0$ for the active constraints)”. That nonexistence is exactly a system (II) with no solution, with $b = \nabla f(x^*)$ and columns $\nabla g_i(x^*)$, so by Farkas the corresponding system (I) has a solution. And the solution $x \geq 0$ of (I) is precisely the set of *multipliers* $\lambda_i \geq 0$ with $\nabla f(x^*) = \sum_i \lambda_i \nabla g_i(x^*)$. In contract problems (principal agent with IC and IR constraints, MWG ch. 14), the multipliers of the IC constraints delivered by Farkas and KKT measure how much the need to provide incentives “costs” in terms of the objective: they are shadow prices of information. See [5] (ch. 12), [7], [4] and [8] for the algorithmic versions.

Theorem 4.25 (Gordan alternative). *Let $A \in \mathbb{R}^{m \times n}$ with rows $a_1^\top, \dots, a_m^\top$. Exactly one of the two systems has a solution:*

$$(I) \quad Ax < 0 \quad (\text{all inequalities strict; } x \in \mathbb{R}^n);$$

$$(II) \quad A^\top y = 0, \quad y \geq 0, \quad y \neq 0 \quad (y \in \mathbb{R}^m).$$

Proof. Not both. If $Ax < 0$ and $A^\top y = 0$ with $y \geq 0$, $y \neq 0$, then

$$0 = \langle A^\top y, x \rangle = \langle y, Ax \rangle = \sum_{i=1}^m y_i (Ax)_i < 0,$$

because each $(Ax)_i < 0$, each $y_i \geq 0$ and at least one $y_i > 0$ makes that summand strictly negative. Absurd.

If (I) fails, (II) holds. Consider the two convex subsets of $\mathbb{R}^m$

$$\begin{aligned} P &= \{Ax : x \in \mathbb{R}^n\} && \text{(the image subspace, convex),} \\ N &= \{w \in \mathbb{R}^m : w < 0\} && \text{(the open negative orthant, convex).} \end{aligned}$$

That (I) has no solution means $P \cap N = \emptyset$. By Theorem 4.21 there is $y \neq 0$ with

$$\langle y, w \rangle \leq \langle y, Ax \rangle \quad \text{for all } w \in N, x \in \mathbb{R}^n.$$

First, the right hand side: the function $x \mapsto \langle y, Ax \rangle = \langle A^\top y, x \rangle$ is linear and bounded below on all of $\mathbb{R}^n$ (by the constant $\langle y, w_0 \rangle$, fixing any $w_0 \in N$, for example $w_0 = -\mathbf{1}$); a linear function bounded below on a subspace must be identically zero on it (if $\langle A^\top y, x_0 \rangle \neq 0$ for some x_0 , evaluating at tx_0 with $t \rightarrow \pm\infty$ the function takes arbitrarily negative values). We conclude $A^\top y = 0$. Second, the left hand side: with $A^\top y = 0$ the inequality reads $\langle y, w \rangle \leq 0$ for all $w < 0$. Taking $w = -e_i - \epsilon \mathbf{1}$ with $\epsilon \rightarrow 0^+$ (here $\mathbf{1}$ is the vector of ones, to keep w strictly negative) we get in the limit $\langle y, -e_i \rangle \leq 0$, that is $y_i \geq 0$, for each i . So $y \geq 0$, $y \neq 0$: (II) has a solution. $\square$

Remark 4.26. Gordan is the alternative “without a right hand side” and it is the one used to prove the Fritz John condition (KKT without constraint qualification, [5]). In financial economics, the fundamental theorem of asset pricing is a Farkas statement: *either* there is an arbitrage portfolio, *or* there are positive state prices (risk neutral probabilities, up to normalization) that price all the assets; the vector $y \geq 0$ of the alternative is literally the vector of state prices. The same logic of “separation $\Rightarrow$ positive weights that rationalize decisions” gives the existence of subjective probabilities in the expected utility theorems: the bets accepted and rejected by a coherent decision maker form disjoint convex cones, and the separating normal, normalized to add up to one, is their subjective probability.

4.7 Star application: the Second Welfare Theorem [P1]

This section develops in detail the application that gives economic meaning to the whole chapter: the *decentralization* of Pareto optima through prices. We follow the structure of [9] (section 16.D) and [10] (ch. 5), simplifying to a pure exchange economy so that each mathematical step is transparent.

4.7.1 The setting: an exchange economy

There are L goods and I consumers. Each consumer $i \in \{1, \dots, I\}$ has consumption set $X_i = \mathbb{R}_+^L$ and a preference relation $\succsim_i$ on X_i that we assume *rational* (complete and transitive) and *continuous*. We write $x_i \succ_i x'_i$ for strict preference. The aggregate endowment of the economy is $\bar{\omega} \in \mathbb{R}_+^L$, $\bar{\omega} \neq 0$. An *allocation* is a vector $x = (x_1, \dots, x_I)$ with $x_i \in X_i$; it is *feasible* if $\sum_{i=1}^I x_i \leq \bar{\omega}$ (with equality if you want full exhaustion; we work with equality to simplify).

Definition 4.27 (Pareto optimum). A feasible allocation $x^* = (x_1^*, \dots, x_I^*)$ is a *Pareto optimum* if there is no other feasible allocation x such that $x_i \succsim_i x_i^*$ for all i and $x_j \succ_j x_j^*$ for at least one j . That is: you cannot make someone better off without making someone else worse off.

Definition 4.28 (Equilibrium with transfers). A feasible allocation x^* together with a price vector $p \in \mathbb{R}^L$, $p \neq 0$, is a *price equilibrium with transfers* if, giving each consumer i the wealth $w_i = \langle p, x_i^* \rangle$, each x_i^* is *maximal* for $\succsim_i$ in the budget set $\{x_i \in X_i : \langle p, x_i \rangle \leq w_i\}$; equivalently,

$$x_i \succ_i x_i^* \implies \langle p, x_i \rangle > \langle p, x_i^* \rangle \quad \text{for all } i :$$

everything strictly better is strictly more expensive. If we only guarantee the weak implication $x_i \succ_i x_i^* \implies \langle p, x_i \rangle \geq \langle p, x_i^* \rangle$ (what is better is not cheaper), we speak of a *price quasi-equilibrium with transfers*.

The First Welfare Theorem says that (under local nonsatiation) every equilibrium is Pareto optimal: the market does not waste. The *Second* Theorem is the converse, much deeper: *every* Pareto optimum can be supported as an equilibrium, by choosing the wealth transfers well. It is the theorem that justifies separating efficiency from distribution: if society prefers another point on the Pareto frontier, it does not need to abolish the market, only to redistribute wealth and let prices do the rest. And the prices that do the rest come from a separating hyperplane.

We need two hypotheses on preferences, both already familiar from chapter 2:

Definition 4.29. The preference $\succsim_i$ is *convex* if the upper contour set $\{x_i \in X_i : x_i \succsim_i \hat{x}_i\}$ is convex for every $\hat{x}_i$; it is *locally nonsatiated* (LNS) if for every $\hat{x}_i \in X_i$ and every $\varepsilon > 0$ there is $x_i \in X_i$ with $\|x_i - \hat{x}_i\| < \varepsilon$ and $x_i \succ_i \hat{x}_i$: arbitrarily close to any bundle there is a strictly better one.

4.7.2 The theorem and its proof by separation

Theorem 4.30 (Second Welfare Theorem, exchange version). *Suppose each $\succsim_i$ is rational, continuous, convex and locally nonsatiated. Let x^* be a Pareto optimum with $\sum_i x_i^* = \bar{\omega}$. Then there is $p \in \mathbb{R}^L$, $p \neq 0$, such that (x^*, p) is a price quasi-equilibrium with transfers:*

$$x_i \succ_i x_i^* \implies \langle p, x_i \rangle \geq \langle p, x_i^* \rangle \quad \text{for all } i.$$

If in addition each consumer has in the budget a bundle strictly cheaper than x_i^* (the cheaper point condition; for example, if $\langle p, x_i^* \rangle > 0$ for all i), then (x^*, p) is a price equilibrium with transfers.

Proof. The proof has five steps; the heart is Step 3, where we separate.

Step 1: build the set of “aggregate improvements”. For each consumer we define the set of bundles strictly preferred to the optimal allocation:

$$V_i = \{x_i \in X_i : x_i \succ_i x_i^*\} \subseteq \mathbb{R}^L,$$

and we add them up (Minkowski sum, chapter 2):

$$V = V_1 + V_2 + \cdots + V_I = \left\{ \sum_i x_i : x_i \succ_i x_i^* \forall i \right\}.$$

V is the set of *aggregate resource demands* that would allow giving *each* consumer something strictly better. Each V_i is nonempty by LNS (near x_i^* there is something better), so $V \neq \emptyset$.

Step 2: V is convex and does not contain $\bar{\omega}$. *Convexity:* we first prove that each V_i is convex; here the convexity of preferences enters, together with continuity and transitivity. Let $x_i, x'_i \in V_i$ and $\lambda \in [0, 1]$; without loss of generality $x'_i \succsim_i x_i$ (completeness). Then x_i and x'_i both belong to the upper contour set $\{z : z \succsim_i x_i\}$, which is convex by hypothesis, so $\lambda x_i + (1 - \lambda)x'_i \succsim_i x_i \succ_i x_i^*$, and by transitivity (of the mixed $\succsim$ - $\succ$ part, which follows from rationality plus continuity) $\lambda x_i + (1 - \lambda)x'_i \succ_i x_i^*$: the combination is in V_i . Hence V , a Minkowski sum of convex sets, is convex (chapter 2). $\bar{\omega} \notin V$: if $\bar{\omega} \in V$, there would be $x_i \succ_i x_i^*$ for *all* i with $\sum_i x_i = \bar{\omega}$: a feasible allocation that makes everyone strictly better off, contradicting that x^* is a Pareto optimum. (Note: the definition of Pareto forbids improving *one* agent without hurting others; here we have only used the weak version (improving everyone), so up to this point weak Pareto optimality would be enough. LNS will later turn the conclusion into the right one.)

Step 3: separate. We have a nonempty convex set V and a point $\bar{\omega} \notin V$. The set $\{\bar{\omega}\}$ is convex, nonempty and disjoint from V ; by the separation theorem for disjoint convex sets (Theorem 4.21) there is $p \neq 0$ such that

$$\langle p, z \rangle \geq \langle p, \bar{\omega} \rangle \quad \text{for all } z \in V. \quad (4.5)$$

(We orient the normal toward V : aggregate improvements cost at least as much as the endowment.) **This p is the price vector of the statement:** the normal of the separating hyperplane. Note that we did not need V to be closed or bounded (it is neither), which is why it was important to prove weak separation without topological hypotheses.

Step 4: from the aggregate inequality to the individual one. We want: if $x_j \succ_j x_j^*$ then $\langle p, x_j \rangle \geq \langle p, x_j^* \rangle$. Fix j and such an x_j . The difficulty: (4.5) talks about sums $\sum_i x_i$ with *all* agents improving, and we only have j improving. The solution is local nonsatiation: for $\varepsilon > 0$ and each $i \neq j$, LNS produces $x_i^\varepsilon \in X_i$ with $\|x_i^\varepsilon - x_i^*\| < \varepsilon$ and $x_i^\varepsilon \succ_i x_i^*$. Then $x_j + \sum_{i \neq j} x_i^\varepsilon \in V$, and (4.5) gives

$$\langle p, x_j \rangle + \sum_{i \neq j} \langle p, x_i^\varepsilon \rangle \geq \langle p, \bar{\omega} \rangle = \sum_{i=1}^I \langle p, x_i^* \rangle,$$

using $\bar{\omega} = \sum_i x_i^*$ in the equality. Rearranging,

$$\langle p, x_j \rangle - \langle p, x_j^* \rangle \geq \sum_{i \neq j} \langle p, x_i^* - x_i^\varepsilon \rangle \geq - \sum_{i \neq j} \|p\| \|x_i^* - x_i^\varepsilon\| \geq -(I-1) \|p\| \varepsilon,$$

where the second inequality is Cauchy–Schwarz. Letting $\varepsilon \rightarrow 0^+$ we get $\langle p, x_j \rangle \geq \langle p, x_j^* \rangle$: exactly the quasi-equilibrium condition. (Incidentally, taking x_j close to x_j^* with $x_j \succ_j x_j^*$ (LNS again) and passing to the limit, you see that x_j^* is in the closure of V_j , and the hyperplane “supports” the upper contour set of each consumer at their optimal bundle: every consumer is tangent to the same price line.)

Step 5: from quasi-equilibrium to equilibrium. Suppose that consumer i has a cheaper point: there is $x'_i \in X_i$ with $\langle p, x'_i \rangle < \langle p, x_i^* \rangle = w_i$. We claim that then the weak implication becomes strict. Let $x_i \succ_i x_i^*$ and suppose, for contradiction, that $\langle p, x_i \rangle = w_i$ (by Step 4 it cannot be $<$; we rule out $=$). For $\lambda \in (0, 1)$ consider $x_i^\lambda = \lambda x_i + (1 - \lambda)x'_i$. Its cost is $\lambda \langle p, x_i \rangle + (1 - \lambda) \langle p, x'_i \rangle < w_i$: strictly cheaper than w_i for every $\lambda \in (0, 1)$. By *continuity* of $\succsim_i$, the set $\{z : z \succ_i x_i^*\}$ is open

(relative to X_i); since x_i belongs to it and $x_i^\lambda \rightarrow x_i$ as $\lambda \rightarrow 1^-$, for λ close enough to 1 we have $x_i^\lambda \succ_i x_i^*$ and $\langle p, x_i^\lambda \rangle < w_i$. But this contradicts Step 4 (quasi-equilibrium: what is better is never strictly cheaper). Therefore $\langle p, x_i \rangle > w_i$: equilibrium with transfers for consumer i . If everyone has a cheaper point (for example if $p \geq 0$, $p \neq 0$ and every consumer has positive wealth $w_i > 0$, since then the bundle 0 or a proportional reduction is cheaper), the pair (x^*, p) is a full price equilibrium with transfers. The transfers that implement it are $T_i = w_i - \langle p, \omega_i \rangle$ if the original individual endowments were ω_i : wealth is redistributed, not goods. $\square$

Remark 4.31 (Sign of prices). If in addition preferences are *monotone* (more is better), the separating prices come out nonnegative: $p \geq 0$. The argument is a cone argument: by monotonicity, $V + \mathbb{R}_+^L \subseteq V$ (if an aggregate bundle allows improving everyone, adding resources does too). If some $p_\ell < 0$, taking $z \in V$ and adding te_ℓ with $t \rightarrow +\infty$, the cost $\langle p, z + te_\ell \rangle = \langle p, z \rangle + tp_\ell \rightarrow -\infty$ would violate (4.5). This is the same “cones that force signs” argument that we used in Farkas (Theorem 4.23).

4.7.3 What does each hypothesis buy, and what fails without convexity?

Let us do an autopsy of the proof, hypothesis by hypothesis:

- **Convexity of preferences** $\Rightarrow V$ convex $\Rightarrow$ separation is possible. Without it the theorem is false: with non convex preferences, the upper contour set of a consumer can have a non convex “bay”, and the Pareto optimal bundle can lie at the bottom of the bay. Any price line through that bundle *cuts* the upper contour set: at those prices the consumer can buy something strictly better and not more expensive, and decentralization fails. Graphically: an indifference curve with a concave stretch cannot “rest” on a budget line from the right side at the points of the concave stretch (Figure 4.3). The optimum is still an optimum; what does not exist is a *price* that supports it.
- **Local nonsatiation** enters twice: to make $V_i \neq \emptyset$ and, crucially, in Step 4, to “isolate” one consumer by perturbing the others with arbitrarily small improvements. Without LNS (think of a consumer satiated at x_i^*), the aggregate inequality does not come down to the individual level and different prices may be needed for different consumers, which destroys the market reading.
- **Continuity** supports Step 5 (the sets of strict preference are open) and the limit step of Step 4.
- **The cheaper point** avoids the quasi-equilibrium trap: if at x_i^* the consumer has the minimum possible wealth (for example $w_i = 0$ with $p \geq 0$ and $X_i = \mathbb{R}_+^L$), there may be better bundles that cost exactly the same and the consumer does not really maximize. The classic example of Arrow: on the boundary of the orthant, quasi-equilibrium without equilibrium. In Microeconomics exams this distinction shows up often ([9], section 16.D, Example 16.D.3).

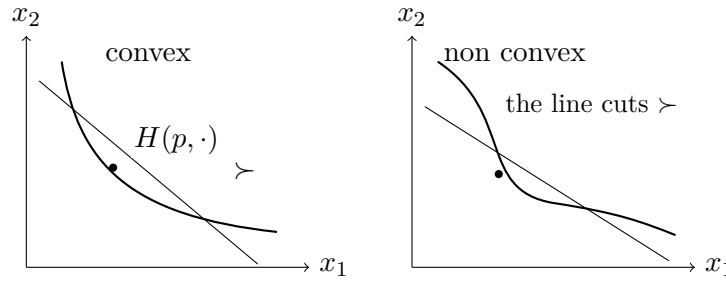


Figure 4.3: Left: with convex preferences, the price line supports the upper contour set at the optimal bundle (supporting hyperplane). Right: with a non convex stretch, every line through the marked bundle cuts the set of strictly preferred bundles: there is no price that decentralizes that point.

Intuition

It is worth pausing on the conceptual moral. The Second Welfare Theorem is, mathematically, *only* the separating hyperplane theorem applied to the set of aggregate improvements. The sentence “prices are signals of scarcity” is formalized like this: the hyperplane $\{z : \langle p, z \rangle = \langle p, \bar{\omega} \rangle\}$ puts on the same side everything that the economy *can* pay for, and on the other side everything it *would like* to consume in order to make everyone better off; prices are the normal that certifies that both wishes are incompatible except on the boundary. And since the hyperplane is *common* to all consumers (the same p for everyone), decentralization works: each agent, looking only at p and their wealth, reproduces the social optimum on their own. Convexity is the price of this magic; with non convexities (indivisibilities, increasing returns), separation fails and the classic problems of pricing with fixed costs appear. The same mathematics, in different clothes, appears in the *shadow prices* of KKT (Remark 4.24), in Lagrangian duality (chapter 10) and in the expected utility representation theorems.

4.8 Chapter summary [P1]

Chapter summary

- **Vocabulary.** $H(a, \beta) = \{x : \langle a, x \rangle = \beta\}$ with $a \neq 0$ (Definition 4.1); the half spaces $H^-(a, \beta)$, $H^+(a, \beta)$ are convex and closed, and the normal a can be normalized ($\|a\| = 1$) without losing anything (Remark 4.2). The strength scale, from least to most demanding, is: plain separation, proper, strict, strong; each one implies all the previous ones, that is, strong $\Rightarrow$ strict $\Rightarrow$ proper $\Rightarrow$ separation (Definition 4.10). *Which hypothesis buys which strength* is the central question of the chapter.
- **Projection theorem** (Theorem 4.6). *Hypotheses:* $C \subseteq \mathbb{R}^n$ nonempty, closed and convex; $x \in \mathbb{R}^n$ arbitrary. *Conclusion:* the point $\text{proj}_C(x)$ that minimizes $\|x - y\|$ over C exists (Weierstrass plus closedness) and is unique (convexity), characterized by the

variational inequality

$$\langle x - \text{proj}_C(x), y - \text{proj}_C(x) \rangle \leq 0 \quad \text{for all } y \in C$$

and with the Pythagorean type bound $\|x - y\|^2 \geq \|x - \text{proj}_C(x)\|^2 + \|y - \text{proj}_C(x)\|^2$ (4.2). Moreover proj_C is *nonexpansive* (1-Lipschitz, hence continuous), Proposition 4.8. Dropping closedness kills existence; dropping convexity kills uniqueness.

- **Strong point convex set separation** (Theorem 4.11). *Hypotheses:* C nonempty, closed and convex, $x \notin C$. *Conclusion:* with $a = x - \text{proj}_C(x) \neq 0$ and $\beta = \langle a, \text{proj}_C(x) \rangle + \frac{1}{2} \|a\|^2$,

$$\sup_{y \in C} \langle a, y \rangle < \beta < \langle a, x \rangle,$$

strong separation with slack $\frac{1}{2} \|a\|^2$ on each side. Without closedness (open ball and a point of its boundary) there is only non strict separation; without convexity (circle and its center) there is none. Corollary: every nonempty closed convex set is the *intersection of the closed half spaces that contain it* (Corollary 4.12): external description = internal description, the seed of duality.

- **Supporting hyperplane** (Theorem 4.15). *Hypotheses:* C convex and nonempty (closedness and boundedness are *not* required) and $x_0 \in \text{bd } C$. *Conclusion:* there is $a \neq 0$ with $\langle a, y \rangle \leq \langle a, x_0 \rangle$ for all $y \in C$; equivalently $x_0 \in \arg \max_{y \in C} \langle a, y \rangle$. The separation is necessarily *non strict* (Remark 4.16): the hyperplane touches the set. Proof technique to memorize: approximate by exterior points, separate each one, *normalize* the normals and use Bolzano–Weierstrass on the unit sphere. The support is unique at “smooth” points and multiple at corners (Example 4.17), which is the seed of the subdifferential (chapter 7).
- **Separation of two convex sets: two versions, two prices.** (i) *Strong* (Theorem 4.19): A, B convex, nonempty, *disjoint*, with A **compact** and B **closed** $\Rightarrow \sup_A \langle a, \cdot \rangle < \inf_B \langle a, \cdot \rangle$. Compactness is used so that $B - A$ is closed (Lemma 4.18); with *two closed sets* it is not enough: the epigraph of $1/x_1$ and the half plane $\{x_2 \leq 0\}$ are closed, convex and disjoint but they are at distance zero (Exercise 4.34). (ii) *Weak* (Theorem 4.21): A, B convex, nonempty and disjoint, **with no topological hypothesis** $\Rightarrow$ there is $a \neq 0$ with $\sup_A \langle a, \cdot \rangle \leq \inf_B \langle a, \cdot \rangle$. (iii) *Sharp version* (Remark 4.22): nonempty convex sets A, B admit *proper* separation if and only if $\text{ri } A \cap \text{ri } B = \emptyset$: they need not be disjoint, only not overlapping in their relative interiors. If in addition one of them is *open*, the separation is strict on that side.
- **Farkas lemma** (Theorem 4.23). *Hypotheses:* $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and nothing else (the hidden hypothesis is a theorem that we admit here without proof: the finitely generated cone $K = \{Ax : x \geq 0\}$ is *closed*, in the line of Weyl–Minkowski). *Conclusion:* exactly one of the systems has a solution,

$$(I) \ Ax = b, \ x \geq 0 \quad \text{or} \quad (II) \ A^\top y \leq 0, \ \langle b, y \rangle > 0.$$

The y of (II) is the *certificate* of infeasibility: the normal of a hyperplane through the origin that leaves the cone on one side and b on the other. Homogeneous variant: the **Gordan alternative** (Theorem 4.25): either $Ax < 0$ has a solution, or there is $y \geq 0$, $y \neq 0$, with $A^\top y = 0$. Farkas is the key to KKT (Remark 4.24) and to the fundamental theorem of asset pricing.

- **Second Welfare Theorem** (Theorem 4.30). *Hypotheses:* exchange economy with $X_i = \mathbb{R}_+^L$; each $\succsim_i$ rational, continuous, convex and locally nonsatiated; x^* a Pareto optimum with $\sum_i x_i^* = \bar{\omega}$. *Conclusion:* there is $p \neq 0$ such that (x^*, p) is a price quasi-equilibrium with transfers ($x_i \succsim_i x_i^* \Rightarrow \langle p, x_i \rangle \geq \langle p, x_i^* \rangle$); if in addition each consumer has a *cheaper point* in the budget (for example $\langle p, x_i^* \rangle > 0$), it is a price equilibrium with transfers (strict implication). *Role of each hypothesis:* **convexity** makes the set of aggregate improvements $V = \sum_i V_i$ convex and is the only thing that allows separation, since without it the theorem is false, not merely unprovable (Figure 4.3); **local nonsatiation** guarantees $V_i \neq \emptyset$ and allows going down from the aggregate inequality to the individual one by perturbing the other consumers; **continuity** gives the step from quasi-equilibrium to equilibrium; **monotonicity** (if added) gives $p \geq 0$ (Remark 4.31). Prices *are* the normal of the hyperplane that separates V from $\bar{\omega}$: they are not data, they are a theorem.

4.9 Solved exercises [P1]

Exercise 4.32 (Explicit projections and variational check). Compute, in $\mathbb{R}^2$, the projection of the point $x = (3, 3)$ onto each of the following sets, and check in each case the variational characterization (4.1):

- (a) the half space $S = \{y \in \mathbb{R}^2 : y_1 + 2y_2 \leq 4\}$;
- (b) the hyperplane $H = \{y \in \mathbb{R}^2 : y_1 + 2y_2 = 4\}$;
- (c) the closed ball $\bar{B}(0, 2) = \{y : \|y\| \leq 2\}$.

Deduce also the general formula for the distance from a point to a hyperplane.

Solution. (a) *Half space.* Write $a = (1, 2)$, $\beta = 4$, so that $S = H^-(a, \beta)$; note $\|a\|^2 = 1^2 + 2^2 = 5$. First we check that $x \notin S$: $\langle a, x \rangle = 3 + 6 = 9 > 4 = \beta$. *Candidate:* we go down from x in the direction $-a$ (the only direction in which the linear function $\langle a, \cdot \rangle$ decreases fastest) just enough to reach the border. We look for $t \geq 0$ with $\langle a, x - ta \rangle = \beta$, that is $\langle a, x \rangle - t\|a\|^2 = \beta$, that is

$$t = \frac{\langle a, x \rangle - \beta}{\|a\|^2} = \frac{9 - 4}{5} = 1, \quad z = x - ta = (3, 3) - (1, 2) = (2, 1).$$

Check: $z \in S$ because $\langle a, z \rangle = 2 + 2 = 4 \leq 4$. And for any $y \in S$,

$$\langle x - z, y - z \rangle = \langle ta, y - z \rangle = t(\langle a, y \rangle - \langle a, z \rangle) = 1 \cdot (\underbrace{\langle a, y \rangle}_{\leq 4} - 4) \leq 0.$$

Condition (4.1) holds, and by the uniqueness in Theorem 4.6, $\text{proj}_S(x) = (2, 1)$. The distance is $\|x - z\| = \|(1, 2)\| = \sqrt{5}$. In general, for $\langle a, x \rangle > \beta$,

$$\text{proj}_{H^-(a, \beta)}(x) = x - \frac{\langle a, x \rangle - \beta}{\|a\|^2} a, \quad d(x, H^-(a, \beta)) = \frac{\langle a, x \rangle - \beta}{\|a\|},$$

and $\text{proj}_{H^-(a, \beta)}(x) = x$ when $\langle a, x \rangle \leq \beta$ (the point is already in the set, the minimum distance is zero and is attained at the point itself).

(b) *Hyperplane*. The same computation works, with one important difference in the check. The candidate is again $z = (2, 1)$, which now satisfies $\langle a, z \rangle = 4$ with equality, so $z \in H$. For $y \in H$ we have $\langle a, y \rangle = 4$ exactly, so

$$\langle x - z, y - z \rangle = t(\langle a, y \rangle - \langle a, z \rangle) = 1 \cdot (4 - 4) = 0 \quad \text{for all } y \in H.$$

Condition (4.1) holds (with equality), so $\text{proj}_H(x) = (2, 1)$. The equality is not an accident: when C is an *affine* set, the vector $y - z$ sweeps the whole direction subspace and can be taken in both directions $\pm v$; the inequalities $\langle x - z, v \rangle \leq 0$ and $\langle x - z, -v \rangle \leq 0$ force $\langle x - z, v \rangle = 0$. That is, for affine sets the variational characterization becomes the classical *orthogonality*: the residual is perpendicular to the set. From the general formula you read off the well known point to hyperplane distance

$$d(x, H(a, \beta)) = \frac{|\langle a, x \rangle - \beta|}{\|a\|} \quad \left(\text{here } \frac{|9 - 4|}{\sqrt{5}} = \sqrt{5}\right),$$

valid now in both directions (hence the absolute value: the hyperplane can be reached from either side, while the half space is only “fallen into” from outside).

(c) *Ball*. Since $\|x\| = \|(3, 3)\| = 3\sqrt{2} > 2$, the point is outside and the natural candidate is the point of the sphere in the direction of x :

$$z = 2 \frac{x}{\|x\|} = \frac{2}{3\sqrt{2}} (3, 3) = (\sqrt{2}, \sqrt{2}), \quad \|z\| = \sqrt{2+2} = 2 \quad (\text{it is on the border}).$$

Check: $x - z = (3 - \sqrt{2})(1, 1)$ with $3 - \sqrt{2} > 0$, and for $y \in \bar{B}(0, 2)$,

$$\langle x - z, y - z \rangle = (3 - \sqrt{2})(\langle (1, 1), y \rangle - \langle (1, 1), z \rangle) = (3 - \sqrt{2})(y_1 + y_2 - 2\sqrt{2}).$$

By Cauchy–Schwarz, $y_1 + y_2 = \langle (1, 1), y \rangle \leq \|(1, 1)\| \|y\| \leq \sqrt{2} \cdot 2 = 2\sqrt{2}$, so the parenthesis is ≤ 0 and so is the product: (4.1) holds and $\text{proj}_{\bar{B}(0, 2)}(x) = (\sqrt{2}, \sqrt{2})$, at distance $\|x - z\| = (3 - \sqrt{2})\sqrt{2} = 3\sqrt{2} - 2$. (Consistent with Example 4.7: $\|x\| - r = 3\sqrt{2} - 2$.) For a ball centered at c , the same translated computation gives $\text{proj}_{\bar{B}(c, r)}(x) = c + r(x - c) / \|x - c\|$ when $\|x - c\| > r$.

Working moral. In the three cases the method was the same: (1) guess the candidate with geometry; (2) check that it belongs to the set; (3) verify (4.1); (4) invoke uniqueness. You never need to solve the minimization problem. $\square$

Exercise 4.33 (Explicit construction of a separating hyperplane). Consider the polyhedron

$$P = \{y \in \mathbb{R}^2 : y_1 + 2y_2 \geq 4, \ y_1 \geq 0, \ y_2 \geq 0\}.$$

- (a) Check that P is convex, closed and nonempty. For $x = (1, 0)$, verify that $x \notin P$, compute $\text{proj}_P(x)$ and build explicitly the hyperplane that strongly separates x from P following the recipe of Theorem 4.11.
- (b) Repeat the exercise with $x' = (-3, -1)$, where the projection falls on a different *face* of the polyhedron.

Solution. *Structure of P .* P is the intersection of three closed half spaces: $\{y : \langle (1, 2), y \rangle \geq 4\} = H^+((1, 2), 4)$, $\{y : y_1 \geq 0\}$ and $\{y : y_2 \geq 0\}$. Each half space is convex and closed (Remark 4.5), and the intersection of closed convex sets is convex and closed; $P \neq \emptyset$ because, for example, $(4, 0) \in P$. (It is not bounded: it contains the half line $\{(4 + s, 0) : s \geq 0\}$. Compactness is not needed: Theorem 4.11 only asks for closed and convex.)

(a) *The point $x = (1, 0)$.* It does not belong to P : $\langle (1, 2), x \rangle = 1 + 0 = 1 < 4$. *Projection candidate:* the set is a polyhedron, so the projection is on its boundary; we start by ignoring the sign constraints and projecting onto the line $L = \{y : y_1 + 2y_2 = 4\}$ with the formula of the previous exercise. With $a = (1, 2)$, $\|a\|^2 = 5$:

$$z = x - \frac{\langle a, x \rangle - 4}{\|a\|^2} a = (1, 0) - \frac{1 - 4}{5} (1, 2) = (1, 0) + \frac{3}{5} (1, 2) = \left(\frac{8}{5}, \frac{6}{5}\right).$$

Is it valid? We must check that this z also satisfies the constraints we ignored: $z_1 = 8/5 \geq 0$ and $z_2 = 6/5 \geq 0$, and $\langle a, z \rangle = 8/5 + 12/5 = 4$. Yes: $z \in P$. *Variational check (this is what legitimizes the shortcut):* for all $y \in P$ we have $\langle a, y \rangle \geq 4 = \langle a, z \rangle$, and since

$$x - z = -\frac{3}{5} (1, 2) = -\frac{3}{5} a,$$

we get

$$\langle x - z, y - z \rangle = -\frac{3}{5} (\langle a, y \rangle - \langle a, z \rangle) = -\frac{3}{5} \underbrace{(\langle a, y \rangle - 4)}_{\geq 0} \leq 0.$$

So $\text{proj}_P(x) = (8/5, 6/5)$, with distance $\|x - z\| = \frac{3}{5} \|(1, 2)\| = \frac{3\sqrt{5}}{5} = \frac{3}{\sqrt{5}}$.

The hyperplane. The recipe of Theorem 4.11 (reading the theorem with the role of the “set” played by P and the “point” by x) gives the normal

$$\tilde{a} = x - \text{proj}_P(x) = -\frac{3}{5} (1, 2), \quad \|\tilde{a}\|^2 = \frac{9}{25} \cdot 5 = \frac{9}{5},$$

and the level $\tilde{\beta} = \langle \tilde{a}, z \rangle + \frac{1}{2} \|\tilde{a}\|^2 = -\frac{3}{5} \cdot 4 + \frac{9}{10} = -\frac{12}{5} + \frac{9}{10} = -\frac{3}{2}$. Check of the strong separation chain: for $y \in P$, $\langle \tilde{a}, y \rangle = -\frac{3}{5} \langle a, y \rangle \leq -\frac{12}{5} = -2.4$, while $\langle \tilde{a}, x \rangle = -\frac{3}{5} \cdot 1 = -0.6$; indeed

$$-2.4 < -1.5 < -0.6.$$

Since the normal is defined up to a positive scale (Remark 4.2), it is convenient to rescale by multiplying by $-5/3 < 0$ and *flip the inequalities*, or equivalently to present the result with the normal oriented toward P : with $p = (1, 2)$ and $\beta = 5/2$,

$$\langle p, x \rangle = 1 < \frac{5}{2} < 4 \leq \langle p, y \rangle \quad \text{for all } y \in P.$$

The separating hyperplane is the line $\{y : y_1 + 2y_2 = 5/2\}$, parallel to the active face of the polyhedron and placed halfway. It is *strong* separation: the value of $\langle p, \cdot \rangle$ at x is 1 and its infimum over P is 4, with a slack of $3/2$ on each side of the chosen level.

(b) The point $x' = (-3, -1)$. Now the shortcut fails: projecting onto the line L ,

$$\hat{z} = x' - \frac{\langle a, x' \rangle - 4}{5} a = (-3, -1) - \frac{-5 - 4}{5}(1, 2) = (-3, -1) + \frac{9}{5}(1, 2) = \left(-\frac{6}{5}, \frac{13}{5}\right),$$

which has a negative first coordinate: $\hat{z} \notin P$. This tells us that the minimum over P is attained on another face, the vertical half line $F = \{(0, y_2) : y_2 \geq 2\}$ (the part of the boundary $y_1 = 0$ that survives the constraint $y_1 + 2y_2 \geq 4$, that is $y_2 \geq 2$). Projecting onto F is easy: the point of F closest to $(-3, -1)$ has first coordinate 0 and second coordinate equal to -1 truncated to the allowed interval $[2, \infty)$, that is 2. Candidate: $z' = (0, 2)$.

Variational check (the one that really decides). We have $x' - z' = (-3, -1) - (0, 2) = (-3, -3)$. For $y \in P$:

$$\langle x' - z', y - z' \rangle = -3(y_1 - 0) - 3(y_2 - 2) = -3(y_1 + y_2 - 2).$$

We must see that $y_1 + y_2 \geq 2$ on all of P . Let $y \in P$. If $y_2 \geq 2$, then $y_1 + y_2 \geq 0 + 2 = 2$. If $y_2 < 2$, the constraint $y_1 \geq 4 - 2y_2$ gives $y_1 + y_2 \geq 4 - 2y_2 + y_2 = 4 - y_2 > 4 - 2 = 2$. In both cases $y_1 + y_2 \geq 2$, so $\langle x' - z', y - z' \rangle \leq 0$ and, by uniqueness, $\text{proj}_P(x') = (0, 2)$: the projection is the *vertex* of the polyhedron. The distance is $\|(-3, -3)\| = 3\sqrt{2}$.

The hyperplane. Normal $\tilde{a}' = x' - z' = (-3, -3)$, or after rescaling and orienting toward P , $p' = (1, 1) = -\frac{1}{3}\tilde{a}'$. The level given by the theorem is $\tilde{\beta}' = \langle \tilde{a}', z' \rangle + \frac{1}{2}\|\tilde{a}'\|^2 = -6 + 9 = 3$, and the chain is $\langle \tilde{a}', y \rangle \leq -6 < 3 < 12 = \langle \tilde{a}', x' \rangle$. Dividing by -3 (a negative number) flips the inequalities and the level becomes $\beta' = 3/(-3) = -1$; with $p' = (1, 1)$ we then have $\langle p', y \rangle \geq 2$ for all $y \in P$ (we just proved it) and $\langle p', x' \rangle = -4$, that is

$$\langle p', x' \rangle = -4 < -1 < 2 \leq \langle p', y \rangle \quad \text{for all } y \in P.$$

Comment. Note the phenomenon: as the exterior point moves, the separating normal jumps from $(1, 2)$ (parallel to a face) to $(1, 1)$ (one of the infinitely many supporting normals of the vertex $(0, 2)$). It is the multiplicity of supporting hyperplanes at corners of Example 4.17(b): at a vertex there is a *fan* of prices that support it, and which one appears depends on where you look from. In linear programming (chapter 13) that fan is the set of optimal dual solutions. $\square$

Exercise 4.34 (When strong separation is impossible; and when strict separation is free). (a)

Let

$$A = \{u \in \mathbb{R}^2 : u_2 \leq 0\}, \quad B = \{v \in \mathbb{R}^2 : v_1 > 0, v_2 \geq 1/v_1\}.$$

Prove that A and B are convex, closed, nonempty and disjoint, but that they do *not* admit strong separation (not even strict); identify the only hyperplane that separates them and explain exactly which hypothesis of Theorem 4.19 fails and where the proof breaks down.

(b) Prove the claim of Remark 4.22: if A and B are nonempty disjoint convex sets and *in addition* A is open, then the separation of Theorem 4.21 is strict on the side of A : $\langle a, u \rangle < \beta$ for all $u \in A$, with $\beta = \sup_A \langle a, \cdot \rangle$.

Solution. (a1) *A is convex and closed.* It is the half space $H^-((0,1),0)$, hence convex and closed (Remark 4.5); it contains $(0,0)$.

(a2) *B is convex.* B is the epigraph of the function $f(t) = 1/t$ on $t > 0$. Let $u, v \in B$ and $\lambda \in [0,1]$, and let $w = \lambda u + (1-\lambda)v$. First, $w_1 = \lambda u_1 + (1-\lambda)v_1 > 0$ (a combination of positive numbers with weights not both zero). Second, we use that $f(t) = 1/t$ is convex on $(0, \infty)$: since $f''(t) = 2/t^3 > 0$, we have $f(\lambda u_1 + (1-\lambda)v_1) \leq \lambda f(u_1) + (1-\lambda)f(v_1)$. Then

$$w_2 = \lambda u_2 + (1-\lambda)v_2 \geq \lambda \frac{1}{u_1} + (1-\lambda) \frac{1}{v_1} \geq \frac{1}{\lambda u_1 + (1-\lambda)v_1} = \frac{1}{w_1},$$

where the first inequality uses $u, v \in B$ and the second is the convexity of $1/t$. So $w \in B$.

(a3) *B is closed.* Let $v^{(k)} \in B$ with $v^{(k)} \rightarrow v$. Since $v_1^{(k)} > 0$, in the limit $v_1 \geq 0$. If $v_1 = 0$, then $v_2^{(k)} \geq 1/v_1^{(k)} \rightarrow +\infty$, contradicting that $v_2^{(k)} \rightarrow v_2 \in \mathbb{R}$ (a convergent sequence is bounded). So $v_1 > 0$, and passing to the limit in $v_2^{(k)} \geq 1/v_1^{(k)}$ (the function $t \mapsto 1/t$ is continuous at $v_1 > 0$) we get $v_2 \geq 1/v_1$. Thus $v \in B$ and B is closed.

(a4) *They are disjoint.* If $v \in B$ then $v_2 \geq 1/v_1 > 0$, while every $u \in A$ has $u_2 \leq 0$. No point satisfies both.

(a5) *The distance between them is zero.* For each $t > 0$, the point $v^t = (t, 1/t)$ is in B and the point $u^t = (t, 0)$ is in A , and $\|v^t - u^t\| = 1/t \rightarrow 0$ as $t \rightarrow \infty$. Therefore $\inf\{\|u - v\| : u \in A, v \in B\} = 0$ although the infimum is not attained (the sets are disjoint). *This already rules out strong separation:* if there were $a \neq 0$ and $\varepsilon > 0$ with $\langle a, u \rangle \leq \beta - \varepsilon$ and $\langle a, v \rangle \geq \beta + \varepsilon$, then for all $u \in A, v \in B$,

$$2\varepsilon \leq \langle a, v - u \rangle \leq \|a\| \|v - u\| \implies \|v - u\| \geq \frac{2\varepsilon}{\|a\|} > 0,$$

(Cauchy-Schwarz), contradicting that the distance is zero. Strong separation *always* implies positive distance; that is its geometric content.

(a6) *The unique separating hyperplane, and why the separation is not even strict.* Suppose $H(a, \beta)$ separates: $\langle a, u \rangle \leq \beta \leq \langle a, v \rangle$ for $u \in A, v \in B$. First, $a_1 = 0$: the points $(s, 0)$ belong to A for every $s \in \mathbb{R}$, so $a_1 s \leq \beta$ for every $s \in \mathbb{R}$, which is possible only if $a_1 = 0$ (if $a_1 \neq 0$, taking s of the same sign as a_1 and large, the left hand side exceeds any β). Second, $a_2 > 0$: since $a \neq 0$, we have $a_2 \neq 0$; the points $(0, u_2)$ with $u_2 \leq 0$ are in A , and $a_2 u_2 \leq \beta$ for all $u_2 \leq 0$ forces $a_2 > 0$ (if $a_2 < 0$, with $u_2 \rightarrow -\infty$ the product $a_2 u_2 \rightarrow +\infty$). Third, $\beta = 0$: from A we get $\beta \geq \sup_{u \in A} a_2 u_2 = 0$ (the supremum is attained at $u_2 = 0$); from B we get $\beta \leq \inf_{v \in B} a_2 v_2 = a_2 \inf_{v \in B} v_2 = 0$, because $v_2 = 1/t$ can be made arbitrarily small and positive (infimum 0, not attained). So $\beta = 0$ and, up to a positive scale, $a = (0, 1)$: the only separating hyperplane is the horizontal axis $\{x_2 = 0\}$. And the separation *is not strict* because the point $(0, 0) \in A$ satisfies $\langle a, (0, 0) \rangle = 0 = \beta$: there is contact with the hyperplane on the side of A .

(a7) *Which hypothesis fails.* Theorem 4.19 asks that *one of the two sets be compact*; here both are closed but *neither is bounded*. The exact place where the proof breaks is Lemma 4.18: without compactness, the difference set $D = B - A$ may fail to be closed. Indeed,

$$(t, 1/t) - (t, 0) = (0, 1/t) \in D \quad \text{for all } t > 0, \quad (0, 1/t) \rightarrow (0, 0),$$

so $0 \in \text{cl } D$ although $0 \notin D$ (by disjointness). We are exactly in “Case 2” of the proof of Theorem 4.21: the origin is a *boundary* point of D , and the only thing that can be invoked is the supporting hyperplane, which gives non strict separation. Everything fits: the weak theorem does apply (two disjoint convex sets always separate) and delivers precisely the horizontal axis.

(b) *With an open set the separation becomes strict on that side.* By Theorem 4.21 there is $a \neq 0$ with $\langle a, u \rangle \leq \langle a, v \rangle$ for all $u \in A$, $v \in B$; let $\beta = \sup_{u \in A} \langle a, u \rangle$, which is finite (it is bounded by $\langle a, v_0 \rangle$ for any fixed $v_0 \in B$). Suppose, for contradiction, that there is $u_0 \in A$ with $\langle a, u_0 \rangle = \beta$. Since A is open, there is $\varepsilon > 0$ with $B(u_0, \varepsilon) \subseteq A$. Consider the point

$$u_* = u_0 + \frac{\varepsilon}{2} \frac{a}{\|a\|}, \quad \|u_* - u_0\| = \frac{\varepsilon}{2} < \varepsilon \implies u_* \in A.$$

But then

$$\langle a, u_* \rangle = \langle a, u_0 \rangle + \frac{\varepsilon}{2} \frac{\langle a, a \rangle}{\|a\|} = \beta + \frac{\varepsilon}{2} \|a\| > \beta,$$

which contradicts that β is the supremum of $\langle a, \cdot \rangle$ over A . Therefore no point of A attains the level β : $\langle a, u \rangle < \beta \leq \langle a, v \rangle$ for all $u \in A$, $v \in B$.

Reading. The trick is always the same: *in an open set you can walk a little in the direction a* , and walking in the direction of the normal strictly increases the linear function. This is why sets with nonempty interior (sets of strict improvement of continuous preferences, for example) give stronger separations, and this is why in the second welfare theorem the continuity of preferences (which makes the sets $\{z \succ_i x_i^*\}$ *open*) is what allows the jump from quasi-equilibrium to equilibrium. $\square$

Exercise 4.35 (Farkas: numerical certificate and the alternative for inequalities). (a) Let

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 1 & 1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

Prove that the system $Ax = b$, $x \geq 0$, has no solution by exhibiting a *certificate* y of system (II) of the Farkas lemma (Theorem 4.23), and explain how that y “is” the linear combination of equations that reveals the impossibility.

(b) Using the Farkas lemma, prove the following alternative for systems of *inequalities* (Gale theorem): given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, exactly one of the systems has a solution:

$$(I') \ Ax \leq b \ (x \in \mathbb{R}^n \text{ free}) \quad \text{or} \quad (II') \ A^\top w = 0, \ w \geq 0, \ \langle b, w \rangle < 0 \ (w \in \mathbb{R}^m).$$

Solution. (a) *Looking for the certificate.* The columns of A are $A_1 = (1, 1)$, $A_2 = (-1, 1)$ and $A_3 = (2, 0)$, and system (I) asks whether $b = (-2, 1)$ belongs to the cone $K = \text{cone}\{A_1, A_2, A_3\}$. A geometric glance: the nonnegative combinations of $(1, 1)$ and $(-1, 1)$ fill the upper half plane $\{y_2 \geq 0\}$ “tilted” upward, and adding $(2, 0)$ pushes to the right; concretely you check that $K = \{y : y_2 \geq 0, y_1 + y_2 \geq 0\}$. Since $b_1 + b_2 = -2 + 1 = -1 < 0$, the point b is outside K , and the violated inequality, $y_1 + y_2 \geq 0$, suggests the normal.

We look for y with $A^\top y \leq 0$ and $\langle b, y \rangle > 0$. The condition $A^\top y \leq 0$ means $\langle A_j, y \rangle \leq 0$ for

each column:

$$y_1 + y_2 \leq 0, \quad -y_1 + y_2 \leq 0, \quad 2y_1 \leq 0.$$

The vector $y = (-1, -1)$ satisfies all three: $-2 \leq 0$, $1 - 1 = 0 \leq 0$, $-2 \leq 0$. And moreover

$$\langle b, y \rangle = (-2)(-1) + (1)(-1) = 2 - 1 = 1 > 0.$$

So $y = (-1, -1)$ solves (II) and, by the Farkas lemma, (I) has no solution.

Reading the certificate as a combination of equations. System (I) written row by row is

$$(\text{row 1}) \quad x_1 - x_2 + 2x_3 = -2, \quad (\text{row 2}) \quad x_1 + x_2 = 1.$$

Multiplying row i by y_i and adding is exactly computing $\langle y, Ax \rangle = \langle A^\top y, x \rangle$ against $\langle y, b \rangle$.

With $y = (-1, -1)$: $(-1) \times (\text{row 1}) + (-1) \times (\text{row 2})$ gives

$$-2x_1 - 2x_3 = 2 - 1 = 1 \quad \Longleftrightarrow \quad x_1 + x_3 = -\frac{1}{2}.$$

But $x \geq 0$ forces $x_1 + x_3 \geq 0$: a contradiction. The certificate y is, literally, *the recipe for which combination of equations to take* so that the left hand side has coefficients ≤ 0 (that is $A^\top y \leq 0$) while the right hand side stays > 0 (that is $\langle b, y \rangle > 0$). Checking infeasibility then becomes an arithmetic check of three inequalities, with no need to sweep the infinitely many $x \geq 0$: that is the practical value of an alternative. Geometrically, y is the normal of the hyperplane $\{z : \langle y, z \rangle = 0\}$, which leaves the whole cone K in the half space $\langle y, \cdot \rangle \leq 0$ and b strictly on the other side.

(b) *From Farkas to Gale.* The idea is to rewrite (I') as a system in *standard form* (equalities and nonnegative variables) and apply Farkas to the augmented matrix.

Step 1: rewriting. Every $x \in \mathbb{R}^n$ splits as $x = x^+ - x^-$ with $x^+, x^- \geq 0$ (for example $x_i^+ = \max\{x_i, 0\}$, $x_i^- = \max\{-x_i, 0\}$), and conversely any pair $x^+, x^- \geq 0$ defines a free x . Moreover $Ax \leq b$ is equivalent to the existence of a *slack* $s \geq 0$ with $Ax + s = b$ (take $s = b - Ax \geq 0$). Therefore

$$(I') \text{ has a solution } \Longleftrightarrow \exists \bar{x} = (x^+, x^-, s) \geq 0 : \tilde{A}\bar{x} = b, \quad \tilde{A} = [A \quad -A \quad I_m] \in \mathbb{R}^{m \times (2n+m)}.$$

The system on the right is exactly system (I) of the Farkas lemma for the matrix $\tilde{A}$ and the vector b .

Step 2: write the alternative (II) for $\tilde{A}$. By Theorem 4.23, that system is infeasible if and only if there is $y \in \mathbb{R}^m$ with $\tilde{A}^\top y \leq 0$ and $\langle b, y \rangle > 0$. Now we break $\tilde{A}^\top y \leq 0$ block by block:

$$A^\top y \leq 0, \quad (-A)^\top y = -A^\top y \leq 0, \quad I_m^\top y = y \leq 0.$$

The first two together say $A^\top y \leq 0$ and $A^\top y \geq 0$, that is $A^\top y = 0$ (here you see the effect of having a *free* variable: it turns a dual inequality into an equality). The third says $y \leq 0$. So:

$$(I') \text{ infeasible } \Longleftrightarrow \exists y : A^\top y = 0, y \leq 0, \langle b, y \rangle > 0.$$

Step 3: cosmetic sign change. Setting $w = -y$, the conditions $y \leq 0$, $A^\top y = 0$, $\langle b, y \rangle > 0$ become $w \geq 0$, $A^\top w = 0$, $\langle b, w \rangle < 0$: exactly (II'). Since Farkas guarantees that *exactly one* of the two systems of the alternative applied to $\tilde{A}$ has a solution, the same holds for (I') and (II').

Direct check of “not both” (useful as a control). If $Ax \leq b$ and $w \geq 0$ with $A^\top w = 0$, then multiplying $Ax \leq b$ by $w \geq 0$ component by component and adding (multiplying an inequality by a nonnegative number preserves it):

$$\langle b, w \rangle \geq \langle Ax, w \rangle = \langle x, A^\top w \rangle = \langle x, 0 \rangle = 0,$$

which is incompatible with $\langle b, w \rangle < 0$. The economic reading of (II') is transparent: w is a nonnegative combination of constraints whose part in x cancels ($A^\top w = 0$) and whose aggregate right hand side is negative ($\langle b, w \rangle < 0$): the list of constraints itself says “ $0 \leq$ something negative”, that is, it contradicts itself. This version is the one used for duality in linear programming (chapter 13) and for the theorem on the existence of arbitrage free prices. $\square$

Exercise 4.36 (The support function determines the closed convex set). Let $C \subseteq \mathbb{R}^n$ be nonempty. Its *support function* is

$$\sigma_C : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}, \quad \sigma_C(a) = \sup_{x \in C} \langle a, x \rangle.$$

- (a) Prove that σ_C is positively homogeneous ($\sigma_C(ta) = t\sigma_C(a)$ for $t > 0$) and subadditive ($\sigma_C(a + a') \leq \sigma_C(a) + \sigma_C(a')$); deduce that it is convex. Check also that $\sigma_C = \sigma_{\text{cl conv } C}$.
- (b) Prove that if C is in addition convex and closed, then

$$C = \{x \in \mathbb{R}^n : \langle a, x \rangle \leq \sigma_C(a) \text{ for all } a \in \mathbb{R}^n\},$$

and conclude that two nonempty closed convex sets with the same support function are equal.

- (c) Compute σ_C for $C = \bar{B}(0, r)$ and for $C = \text{conv}\{v_1, \dots, v_m\}$, and check the representation of (b) in the first case.

Solution. (a) *Positive homogeneity.* For $t > 0$,

$$\sigma_C(ta) = \sup_{x \in C} \langle ta, x \rangle = \sup_{x \in C} t \langle a, x \rangle = t \sup_{x \in C} \langle a, x \rangle = t \sigma_C(a),$$

where in the third step we use that multiplying by a *positive* constant commutes with the supremum (if $t > 0$, $tu \leq tM$ is equivalent to $u \leq M$). The identity also holds if $\sigma_C(a) = +\infty$, with the convention $t \cdot (+\infty) = +\infty$. Note that $\sigma_C(0) = 0$ (the supremum of the zero function over a nonempty set).

Subadditivity. For each $x \in C$,

$$\langle a + a', x \rangle = \langle a, x \rangle + \langle a', x \rangle \leq \sigma_C(a) + \sigma_C(a'),$$

because each summand is bounded by the corresponding supremum (evaluated at the *same* x). Taking the supremum over $x \in C$ on the left hand side, $\sigma_C(a + a') \leq \sigma_C(a) + \sigma_C(a')$. The

inequality can be strict: the key point is that on the right hand side each supremum may be attained at a different point.

Convexity. Positive homogeneity plus subadditivity imply convexity: for $\lambda \in [0, 1]$,

$$\sigma_C(\lambda a + (1 - \lambda)a') \leq \sigma_C(\lambda a) + \sigma_C((1 - \lambda)a') = \lambda \sigma_C(a) + (1 - \lambda) \sigma_C(a'),$$

(using subadditivity and then homogeneity; the cases $\lambda \in \{0, 1\}$ are trivial). A positively homogeneous and subadditive function is called *sublinear*, and the deep result (which we will see in chapter 12) is that *every* closed sublinear function is the support function of some closed convex set.

$\sigma_C = \sigma_{\text{cl conv } C}$. Since $C \subseteq \text{cl conv } C$, the supremum over the larger set is at least the one over the smaller. For the other inequality: if $x = \sum_k \lambda_k x_k$ is a convex combination of points of C , then $\langle a, x \rangle = \sum_k \lambda_k \langle a, x_k \rangle \leq \sum_k \lambda_k \sigma_C(a) = \sigma_C(a)$; and if $x = \lim_j x_j$ with $x_j \in \text{conv } C$, the continuity of $\langle a, \cdot \rangle$ gives $\langle a, x \rangle = \lim_j \langle a, x_j \rangle \leq \sigma_C(a)$. Hence $\sigma_{\text{cl conv } C}(a) \leq \sigma_C(a)$. *Conceptual consequence:* the support function does not distinguish a set from its closed convex hull; it can only reconstruct the latter, which is exactly what (b) states.

(b) *Representation.* Call D the set on the right.

($\subseteq$) If $x \in C$, then by the definition of supremum, $\langle a, x \rangle \leq \sigma_C(a)$ for all a ; so $x \in D$. This inclusion uses no hypothesis.

($\supseteq$) Contrapositive: let $x \notin C$. Since C is nonempty, closed and convex, Theorem 4.11 gives $a \neq 0$ and β with

$$\sup_{y \in C} \langle a, y \rangle < \beta < \langle a, x \rangle, \quad \text{that is} \quad \sigma_C(a) < \langle a, x \rangle.$$

That a certifies that $x \notin D$. Therefore $D \subseteq C$, and with the other inclusion, $D = C$.

This is Corollary 4.12 in its most economical form: instead of taking *all* the half spaces containing C , one per direction a is enough, the tightest possible one, that is the one at level $\sigma_C(a)$. *Uniqueness:* if C and C' are nonempty closed convex sets with $\sigma_C = \sigma_{C'}$, then by the representation

$$C = \{x : \langle a, x \rangle \leq \sigma_C(a) \ \forall a\} = \{x : \langle a, x \rangle \leq \sigma_{C'}(a) \ \forall a\} = C'.$$

Knowing all the supporting hyperplanes *is* knowing the set (Remark 4.13).

(c) *Examples.* *Ball:* for $C = \bar{B}(0, r)$ and $a \neq 0$, Cauchy-Schwarz gives $\langle a, x \rangle \leq \|a\| \|x\| \leq r \|a\|$, and the value $r \|a\|$ is attained at $x = ra / \|a\| \in C$; so $\sigma_{\bar{B}(0, r)}(a) = r \|a\|$ (and $\sigma_C(0) = 0$, consistent with the formula). *Check of (b):* the set $D = \{x : \langle a, x \rangle \leq r \|a\| \ \forall a\}$ clearly contains the ball; and if $x \in D$, taking the particular direction $a = x$ we get $\|x\|^2 = \langle x, x \rangle \leq r \|x\|$, hence $\|x\| \leq r$ (dividing by $\|x\| > 0$; if $x = 0$ there is nothing to prove). So $D = \bar{B}(0, r)$: *a single well chosen direction is enough* to recover the set in this case.

Polytope: if $C = \text{conv}\{v_1, \dots, v_m\}$, then by what we proved in (a) $\sigma_C = \sigma_{\{v_1, \dots, v_m\}}$, that is

$$\sigma_C(a) = \max_{1 \leq k \leq m} \langle a, v_k \rangle,$$

a maximum of finitely many linear functions: finite for every a (the set is bounded) and *polyhe-*

dual. In general, σ_C is finite on all of $\mathbb{R}^n$ if and only if C is bounded.

Economic reading. The support function is an object you already know under other names: the *profit function* $\pi(p) = \sup_{y \in Y} \langle p, y \rangle$ is the support function of the production set Y , and by part (b) it determines $\text{cl conv } Y$: two technologies with the same profit function are indistinguishable ([9], ch. 5; [10], ch. 3). Similarly, the *expenditure function* $e(p, u) = \inf \{ \langle p, x \rangle : x \succsim u \}$ is (up to a sign) the support function of the upper contour set, and recovering the set from it is utility expenditure duality. All of this reappears in a formalized way in Fenchel conjugation (chapter 12) and in [2], sec. 2.5. $\square$

Exercise 4.37 (Decentralizing a Pareto optimum in the Edgeworth box). An exchange economy with $L = 2$ goods and $I = 2$ consumers has aggregate endowment $\bar{\omega} = (10, 10)$ and preferences represented by

$$u_A(x_1, x_2) = x_1^{1/2} x_2^{1/2}, \quad u_B(x_1, x_2) = x_1^{1/3} x_2^{2/3}, \quad x \in \mathbb{R}_{++}^2.$$

Consider the allocation

$$x_A^* = \left(4, \frac{5}{2}\right), \quad x_B^* = \left(6, \frac{15}{2}\right).$$

- Check that it is feasible and prove that it is a Pareto optimum.
- Find a price vector p that decentralizes it as a price equilibrium with transfers, and check directly that each consumer maximizes utility in the budget set.
- If the initial endowments were $\omega_A = \omega_B = (5, 5)$, which transfers implement the optimum?
- Identify precisely at which step of the argument separation was used, exhibiting the separated set and the hyperplane.

Solution. (a) *Feasibility.* $x_A^* + x_B^* = (4 + 6, \frac{5}{2} + \frac{15}{2}) = (10, 10) = \bar{\omega}$: the allocation exhausts resources and is feasible with equality.

Pareto optimality through Negishi weights. The clean (and rigorous) strategy is: find weights $\lambda \in (0, 1)$ such that x^* maximizes the weighted sum

$$W(x_A) = \lambda \ln u_A(x_A) + (1 - \lambda) \ln u_B(\bar{\omega} - x_A)$$

over feasible allocations, and note that *every maximizer of a weighted sum with strictly positive weights of utilities (or of strictly increasing transformations of them) is a Pareto optimum*. This last fact is immediate: if there were a feasible allocation y with $u_A(y_A) \geq u_A(x_A^*)$, $u_B(y_B) \geq u_B(x_B^*)$ and at least one strict inequality, then, since $\ln$ is strictly increasing and the weights are positive, we would have $W(y_A) > W(x_A^*)$, contradicting maximality.

Writing $x_A = (a_1, a_2)$ and $x_B = (10 - a_1, 10 - a_2)$,

$$W(a) = \lambda \left(\frac{1}{2} \ln a_1 + \frac{1}{2} \ln a_2 \right) + (1 - \lambda) \left(\frac{1}{3} \ln(10 - a_1) + \frac{2}{3} \ln(10 - a_2) \right).$$

W is *concave* in a on the open rectangle $(0, 10)^2$ (it is a sum of concave functions: $\ln$ is concave and $a \mapsto \ln(10 - a)$ is the composition of $\ln$ with an affine function), so a point where the

gradient vanishes is a global maximum (chapter 8). The first order conditions are

$$\frac{\partial W}{\partial a_1} = \frac{\lambda}{2a_1} - \frac{1-\lambda}{3(10-a_1)} = 0, \quad \frac{\partial W}{\partial a_2} = \frac{\lambda}{2a_2} - \frac{2(1-\lambda)}{3(10-a_2)} = 0.$$

Let us evaluate at the candidate $a = (4, \frac{5}{2})$, where $10 - a_1 = 6$ and $10 - a_2 = \frac{15}{2}$. The first equation asks for

$$\frac{\lambda}{8} = \frac{1-\lambda}{18} \iff 18\lambda = 8 - 8\lambda \iff \lambda = \frac{8}{26} = \frac{4}{13}.$$

We check that this same λ satisfies the second equation:

$$\frac{\lambda}{2 \cdot \frac{5}{2}} = \frac{4/13}{5} = \frac{4}{65}, \quad \frac{2(1-\lambda)}{3 \cdot \frac{15}{2}} = \frac{2 \cdot \frac{9}{13}}{\frac{45}{2}} = \frac{18}{13} \cdot \frac{2}{45} = \frac{36}{585} = \frac{4}{65}.$$

They agree. Therefore, with $\lambda = 4/13$, the point x^* makes the gradient of the concave function W vanish and is its global maximum over the interior feasible allocations; the points on the boundary of the rectangle give $W = -\infty$ (some utility is zero), so the global maximum is interior. Conclusion: x^* is a Pareto optimum. (Equivalently, and faster for the exam: the marginal rates of substitution are equal: for A , $MRS_A = \frac{\partial_1 u_A}{\partial_2 u_A} = \frac{x_2}{x_1} = \frac{5/2}{4} = \frac{5}{8}$; for B , $MRS_B = \frac{(1/3)u_B/x_1}{(2/3)u_B/x_2} = \frac{x_2}{2x_1} = \frac{15/2}{12} = \frac{5}{8}$; and with convex and monotone preferences the equality of MRS in the interior characterizes the contract curve.)

(b) *The prices.* The relative price must equal the common MRS: $p_1/p_2 = 5/8$. Normalizing, we take

$$p = (5, 8).$$

The wealth levels that support the allocation are

$$w_A = \langle p, x_A^* \rangle = 5 \cdot 4 + 8 \cdot \frac{5}{2} = 20 + 20 = 40, \quad w_B = \langle p, x_B^* \rangle = 5 \cdot 6 + 8 \cdot \frac{15}{2} = 30 + 60 = 90,$$

which add up to $130 = \langle p, \bar{\omega} \rangle = 50 + 80$, as they should.

Check 1 (through demands). For a Cobb–Douglas $u(x) = x_1^\alpha x_2^{1-\alpha}$, the Marshallian demand is $x_1 = \alpha w/p_1$, $x_2 = (1-\alpha)w/p_2$ (spend the fraction α of wealth on good 1). Then:

$$\begin{aligned} \text{consumer } A: \quad x_1 &= \frac{1}{2} \cdot \frac{40}{5} = 4, & x_2 &= \frac{1}{2} \cdot \frac{40}{8} = \frac{5}{2}; \\ \text{consumer } B: \quad x_1 &= \frac{1}{3} \cdot \frac{90}{5} = 6, & x_2 &= \frac{2}{3} \cdot \frac{90}{8} = \frac{15}{2}. \end{aligned}$$

Each consumer demands exactly their allocation, and markets clear ($4 + 6 = 10$, $\frac{5}{2} + \frac{15}{2} = 10$): (x^*, p) is a price equilibrium with transfers.

Check 2 (through the supporting hyperplane, which is the one that connects with the chapter). Let us verify *without using calculus* that the budget supports the upper contour set of each consumer, that is, that everything at least as good costs at least as much. For A : if $x \in \mathbb{R}_{++}^2$ satisfies $u_A(x) = \sqrt{x_1 x_2} \geq \sqrt{4 \cdot \frac{5}{2}} = \sqrt{10}$, then $x_1 x_2 \geq 10$ and, by the inequality between the arithmetic and geometric means applied to the numbers $5x_1$ and $8x_2$,

$$5x_1 + 8x_2 \geq 2\sqrt{(5x_1)(8x_2)} = 2\sqrt{40x_1x_2} \geq 2\sqrt{400} = 40 = w_A,$$

with equality if and only if $5x_1 = 8x_2$ and $x_1x_2 = 10$, that is at $x = x_A^*$. For B we use the *weighted* AM–GM inequality (which follows from the concavity of the logarithm: $\theta \ln u + (1 - \theta) \ln v \leq \ln(\theta u + (1 - \theta)v)$, and then you exponentiate), with $\theta = 1/3$, $u = 15x_1$, $v = 12x_2$:

$$5x_1 + 8x_2 = \frac{1}{3}(15x_1) + \frac{2}{3}(12x_2) \geq (15x_1)^{1/3}(12x_2)^{2/3} = 15^{1/3}12^{2/3}x_1^{1/3}x_2^{2/3}.$$

If $u_B(x) \geq u_B(x_B^*) = 6^{1/3}(15/2)^{2/3}$, substituting:

$$5x_1 + 8x_2 \geq (15 \cdot 6)^{1/3}(12 \cdot \frac{15}{2})^{2/3} = 90^{1/3}90^{2/3} = 90 = w_B,$$

again with equality exactly at $x = x_B^*$ (there $15x_1 = 90 = 12x_2$). We have proved that the budget lines $\{x : \langle p, x \rangle = w_i\}$ are *supporting hyperplanes* of the upper contour sets $\{x : u_i(x) \geq u_i(x_i^*)\}$ at x_i^* (Definition 4.14), which is literally the maximization condition of Definition 4.28: everything strictly better is strictly more expensive.

(c) *Transfers*. With $\omega_A = \omega_B = (5, 5)$, the value of each endowment is $\langle p, (5, 5) \rangle = 25 + 40 = 65$. The target wealth levels are $w_A = 40$ and $w_B = 90$, so

$$T_A = w_A - \langle p, \omega_A \rangle = 40 - 65 = -25, \quad T_B = w_B - \langle p, \omega_B \rangle = 90 - 65 = +25,$$

and indeed $T_A + T_B = 0$: the transfer is budget neutral, as the second welfare theorem requires (Theorem 4.30). Consumer A gives up 25 units of account and they are given to B ; *goods are not redistributed, only purchasing power*, and the market does the rest.

(d) *Where was the separation?* In Step 3 of the proof of Theorem 4.30. Here we can exhibit it explicitly. The sets of strict improvement are

$$V_A = \{x \in \mathbb{R}_{++}^2 : \sqrt{x_1x_2} > \sqrt{10}\}, \quad V_B = \{x \in \mathbb{R}_{++}^2 : x_1^{1/3}x_2^{2/3} > 6^{1/3}(15/2)^{2/3}\},$$

both convex (they are the strict upper contour sets of quasiconcave functions: this is the *convexity of preferences*), and the set of aggregate improvements is the Minkowski sum $V = V_A + V_B$. The two AM–GM inequalities of part (b), with *strict* inequality when utility is strictly larger, give for every $z = x_A + x_B \in V$:

$$\langle p, z \rangle = \langle p, x_A \rangle + \langle p, x_B \rangle > 40 + 90 = 130 = \langle p, \bar{\omega} \rangle.$$

That is, the hyperplane

$$H = \{z \in \mathbb{R}^2 : 5z_1 + 8z_2 = 130\}$$

leaves $\bar{\omega} = (10, 10)$ on it and the whole set of aggregate improvements V strictly above: this is exactly inequality (4.5), and its normal $p = (5, 8)$ is the price vector. That $\bar{\omega} \notin V$ is the translation of the fact that x^* is Pareto optimal; that V is convex is what allows invoking Theorem 4.21 to *produce* such a hyperplane in general.

Two final remarks on the role of the hypotheses in this concrete example. First, local nonsatiation (here, strict monotonicity of the Cobb–Douglas functions) was used twice: so that the V_i are nonempty and, in Step 4 of the theorem, to go from the aggregate inequality to the individual one; it also guarantees $p > 0$ (Remark 4.31). Second, the *cheaper point* condition holds

effortlessly: $w_A = 40 > 0$ and $w_B = 90 > 0$, and with positive prices there are always bundles cheaper than the optimal one (reduce any coordinate), so the quasi-equilibrium delivered by separation is a genuine equilibrium. If instead preferences had a non convex stretch, separation could fail and *no* price would decentralize this optimum (Figure 4.3); and if the allocation were on the boundary of the orthant with zero wealth, we would have the classic quasi-equilibrium without equilibrium of [9], sec. 16.D. Compare with the treatment in [10], ch. 5, where the same Edgeworth box is solved with calculus: calculus gives the prices when the functions are differentiable and interior, but only separation guarantees that they always exist. $\square$

Chapter 5

Concave and convex functions [P1]

Week 4 of the course. Bibliography: Boyd and Vandenberghe [2]; Ruszczyński [6]; Aragón et al. [1].

What is this used for in Microeconomics?

No structural property is invoked as often in Microeconomics as concavity, and this is the chapter that builds it. In **decision theory under uncertainty**, the concavity of the Bernoulli utility function *is*, literally, risk aversion: the Jensen inequality (Theorem 5.9) says that $u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)]$ when u is concave, that is, the agent prefers the sure expected value to the lottery; from there come the certainty equivalent, the *risk premium* and the Arrow–Pratt coefficients of absolute and relative aversion that you will use in every expected utility problem (MWG, ch. 6, [9]). In **general equilibrium** (MWG, chs. 15–18), the concavity of the production function makes the production set convex (it is the hypograph of the function, Section 5.3), and that convexity is the hypothesis on which equilibrium existence and the welfare theorems rest. In the **duality theory** of the consumer and the firm, the expenditure function $e(p, u)$ is *concave* in prices and the profit function $\pi(p)$ is *convex* in prices, not by accident, but because both are support functions of a set (Section 5.9), and from that concavity and convexity come the Shephard and Hotelling lemmas with which Hicksian demands and net supplies are computed (MWG, chs. 3 and 5; [10], chs. 1–3). Finally, in **contract theory** (MWG, ch. 14), the concavity of the agent’s utility is the source of the *trade-off* between insurance and incentives in the moral hazard model of Holmström: since the agent is risk averse, loading risk on them to provide incentives is costly, and the optimal contract balances both forces. All of that is built on the definitions and theorems of this chapter. Study it until you can reproduce every proof.

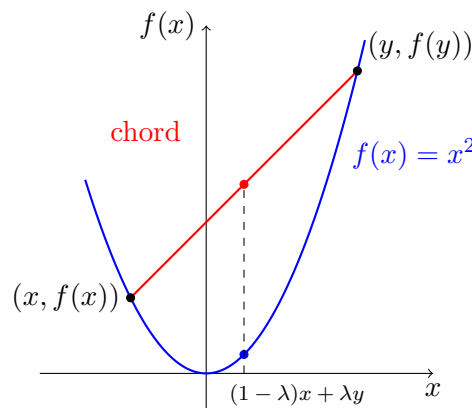
So far we have studied convex *sets*: segments, hulls, cones, and the separation theorems. This chapter takes the jump to *functions*. A convex function is, in essence, a function whose graph “curves upward”: the chord joining two points of the graph stays *above* the graph. A concave function is the opposite: the chord stays below. We will see that this simple geometric idea has huge consequences: convex functions can be characterized by their epigraph (a convex set, which connects this chapter with the previous ones), by their gradient (the tangent underestimates)

and by their Hessian (positive semidefinite); they are continuous in the interior of their domain without asking for it; they are preserved under natural operations such as sums and suprema; and they satisfy the Jensen inequality, which is the cornerstone of decision theory under uncertainty. We close with the *support function* of a set, the bridge between convex geometry and economic duality. Main references: [2] (ch. 3, sections 3.1–3.2), [3] (ch. 2), [6] (sections 2.4–2.5) and [1] (chs. 3–4). From the preliminaries chapter we will use: gradient, Hessian, the Taylor theorem and the definiteness criteria by principal minors; from the chapter on convex sets, the definition of a convex set and its operations.

5.1 Definitions and graphical characterization [P1]

5.1.1 The idea: the chord and the graph

Take the parabola $f(x) = x^2$ and two points of its graph, say $(-1, 1)$ and $(2, 4)$. Draw the line segment (the *chord*) that joins them. If you look at the picture, you will see that the chord stays completely *above* the parabola on the stretch between -1 and 2 . That is exactly the property that defines a convex function. For a concave function, such as $f(x) = \ln x$, the opposite happens: the chord stays *below* the graph.



• height of the chord: $(1 - \lambda)f(x) + \lambda f(y)$ • height of the graph: $f((1 - \lambda)x + \lambda y)$

The figure shows the algebraic translation of “the chord is above the graph”. Fix two points x and y of the domain and a weight $\lambda \in [0, 1]$. The point of the domain “at fraction λ of the way from x to y ” is $(1 - \lambda)x + \lambda y$ (a convex combination, as in Chapter 2). Above that point there are two heights to compare:

- the height of the *graph*: $f((1 - \lambda)x + \lambda y)$;
- the height of the *chord*: $(1 - \lambda)f(x) + \lambda f(y)$, which is the weighted average of the heights at the endpoints (the chord is a straight line, and on a straight line the intermediate value is the weighted average of the end values).

“Chord above the graph” then means $f((1 - \lambda)x + \lambda y) \leq (1 - \lambda)f(x) + \lambda f(y)$. That inequality is the definition.

5.1.2 Formal definitions

Before defining, a warning about the domain: for the inequality to make sense we need f to be defined at the intermediate point $(1 - \lambda)x + \lambda y$. That is guaranteed exactly when the domain is a *convex set*. This is why the whole theory of convex functions is done on convex domains.

Definition 5.1 (Convex function and concave function). Let $C \subseteq \mathbb{R}^n$ be a nonempty convex set and $f : C \rightarrow \mathbb{R}$.

- (i) f is *convex* on C if for all $x, y \in C$ and every $\lambda \in [0, 1]$,

$$f((1 - \lambda)x + \lambda y) \leq (1 - \lambda)f(x) + \lambda f(y). \quad (5.1)$$

- (ii) f is *strictly convex* if inequality (5.1) is strict ($<$) whenever $x \neq y$ and $\lambda \in (0, 1)$.

- (iii) f is *concave* if the inequality is reversed: $f((1 - \lambda)x + \lambda y) \geq (1 - \lambda)f(x) + \lambda f(y)$ for all $x, y \in C$, $\lambda \in [0, 1]$; and *strictly concave* if it is strict when $x \neq y$ and $\lambda \in (0, 1)$.

Remark 5.2. Directly from the definition: f is concave if and only if $-f$ is convex. Indeed, multiplying the inequality $f((1 - \lambda)x + \lambda y) \geq (1 - \lambda)f(x) + \lambda f(y)$ by -1 reverses the direction and produces exactly (5.1) for $-f$. This trivial observation is very important in practice: *every* theorem about convex functions translates automatically into a theorem about concave ones by replacing f with $-f$ and reversing inequalities. In this chapter we prove almost everything for convex functions; you should be able to state the concave version of each result, because in microeconomics the main characters (utility, production, expenditure) are usually concave.

Remark 5.3. In (5.1) the cases $\lambda = 0$ and $\lambda = 1$ are trivial (they give $f(x) \leq f(x)$ and $f(y) \leq f(y)$), so the substantive condition is for $\lambda \in (0, 1)$. Note also that in the definition of *strict* convexity we require $x \neq y$: if $x = y$, both sides are equal no matter what.

Intuition

Three readings of the same inequality, and it is worth having all three at hand: **(1) Geometric:** the chord is above the graph (convex) or below it (concave). **(2) In terms of averages:** “the function of the average is less than or equal to the average of the function” (convex); for concave functions, “the function of the average is greater than or equal to the average of the function”. **(3) Economic:** if x and y are two bundles and λ a probability, the concave version says that the utility of the *expected bundle* exceeds the *expected utility*: the consumer prefers the sure thing to the random average. That third reading is the Jensen inequality with two points, and it is the seed of the whole theory of risk aversion (Section 5.2).

5.1.3 Examples checked from the definition

Checking convexity directly from the definition is laborious, and in practice we will almost always use the first and second order criteria (Sections 5.5 and 5.6). But doing it “by hand” at least once is the best training to understand what the definition says.

Example 5.4 ($f(x) = x^2$ is strictly convex). Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$. Take $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$, and compare the two sides of (5.1). We compute the difference “chord minus graph”:

$$D := (1 - \lambda)x^2 + \lambda y^2 - ((1 - \lambda)x + \lambda y)^2.$$

We expand the square of the binomial:

$$((1 - \lambda)x + \lambda y)^2 = (1 - \lambda)^2 x^2 + 2\lambda(1 - \lambda)xy + \lambda^2 y^2.$$

Substituting and grouping the terms in x^2 , in y^2 and the cross term:

$$\begin{aligned} D &= [(1 - \lambda) - (1 - \lambda)^2]x^2 + [\lambda - \lambda^2]y^2 - 2\lambda(1 - \lambda)xy \\ &= (1 - \lambda)[1 - (1 - \lambda)]x^2 + \lambda(1 - \lambda)y^2 - 2\lambda(1 - \lambda)xy \\ &= \lambda(1 - \lambda)x^2 + \lambda(1 - \lambda)y^2 - 2\lambda(1 - \lambda)xy \\ &= \lambda(1 - \lambda)(x - y)^2, \end{aligned}$$

where in the second step we used $1 - (1 - \lambda) = \lambda$ and in the last one we recognized the perfect square $x^2 - 2xy + y^2 = (x - y)^2$. Since $\lambda(1 - \lambda) \geq 0$ for $\lambda \in [0, 1]$ and $(x - y)^2 \geq 0$, we conclude $D \geq 0$: the chord is above, f is convex. Even more: if $x \neq y$ and $\lambda \in (0, 1)$, then $\lambda(1 - \lambda) > 0$ and $(x - y)^2 > 0$, so $D > 0$: f is *strictly* convex.

Example 5.5 (Affine functions are convex and concave at the same time). Let $f(x) = ax + b$ with $a, b \in \mathbb{R}$ (and in general $f(x) = \langle a, x \rangle + b$ in $\mathbb{R}^n$). We compute directly:

$$f((1 - \lambda)x + \lambda y) = a((1 - \lambda)x + \lambda y) + b = (1 - \lambda)(ax) + \lambda(ay) + (1 - \lambda)b + \lambda b,$$

where we used $b = (1 - \lambda)b + \lambda b$ (the weights add up to one). Grouping, this is exactly $(1 - \lambda)f(x) + \lambda f(y)$. Inequality (5.1) holds *with equality*, so f is convex; and since the reversed inequality also holds, it is concave. In fact, affine functions are the *only* ones that are convex and concave at the same time. And no affine function is strictly convex or strictly concave: the equality in the definition prevents it.

Example 5.6 ($f(x) = |x|$ is convex but not strictly convex). Let $f(x) = |x|$. By the triangle inequality in $\mathbb{R}$ and the homogeneity of the absolute value ($|ct| = |c||t|$, and $|1 - \lambda| = 1 - \lambda$, $|\lambda| = \lambda$ for $\lambda \in [0, 1]$):

$$|(1 - \lambda)x + \lambda y| \leq |(1 - \lambda)x| + |\lambda y| = (1 - \lambda)|x| + \lambda|y|.$$

That is exactly (5.1): f is convex. It is not strictly convex: taking $x = 1$, $y = 2$ and $\lambda = 1/2$, the left hand side is $|3/2| = 3/2$ and the right hand side is $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 2 = 3/2$; there is equality with $x \neq y$, because f is linear on the stretch $[1, 2]$. The same argument with the triangle inequality of the norm shows that $f(x) = \|x\|$ is convex on $\mathbb{R}^n$. Note also that $|x|$ is convex *without being differentiable* at 0: convexity does not require derivatives.

Example 5.7 (e^x is strictly convex; $\ln x$ is strictly concave). For these functions the direct check from the definition is possible but cumbersome (it requires the weighted arithmetic and

geometric mean inequality). We give the argument for $\ln$ because we will use it: we want to see that for $x, y > 0$ and $\lambda \in (0, 1)$,

$$\ln((1 - \lambda)x + \lambda y) \geq (1 - \lambda)\ln x + \lambda \ln y.$$

The right hand side is $\ln(x^{1-\lambda}y^\lambda)$ (properties of the logarithm: $a \ln t = \ln t^a$ and $\ln s + \ln t = \ln st$). Since $\ln$ is increasing, the inequality is equivalent to

$$(1 - \lambda)x + \lambda y \geq x^{1-\lambda}y^\lambda,$$

which is the *weighted arithmetic mean geometric mean inequality* (AM–GM). And this one, in turn, is proved using the convexity of e^t : writing $x = e^s$, $y = e^t$, the inequality becomes $(1 - \lambda)e^s + \lambda e^t \geq e^{(1-\lambda)s + \lambda t}$, which is exactly the convexity of the exponential. Moral: in practice, the clean way to check that e^x is strictly convex and $\ln x$ strictly concave is the second derivative criterion of Section 5.6: $(e^x)'' = e^x > 0$ for all x , and $(\ln x)'' = -1/x^2 < 0$ for all $x > 0$. Keep this example as an illustration that “from the definition” can be hard, and that the calculus tools we will develop are worth gold.

Exam strategy

Facing the question “is f convex or concave?”, this is the order of attack I recommend in an exam:

1. **Is it affine?** Then it is both (Example 5.5). Done.
2. **Is it C^2 ?** Compute the Hessian and apply the second order criterion (Theorem 5.25) with the principal minors of the preliminaries chapter. It is the most mechanical route.
3. **Is it built from known convex functions?** Use the catalogue of operations that preserve convexity (Section 5.7): sums with nonnegative weights, suprema, composition with affine maps, monotone convex composition. This saves the non differentiable cases such as $\|x\|$ or $\max_i x_i$.
4. **Is it a supremum of affine functions in the variable of interest?** (profit functions, support functions, conjugates): convex automatically, no matter how ugly the other part of the problem is (Theorem 5.33).
5. **Only as a last resort**, check the definition by hand, as in Example 5.4.

To *refute* convexity a numerical counterexample is enough: two points and a λ that violate the inequality, or a direction in which the Hessian has negative curvature.

5.2 Jensen inequality and risk aversion [P1]

The definition of convexity compares the function at an average of *two* points with the average of the values. The Jensen inequality says that the same holds for averages of *any* number of points, and in the limit, for expectations of random variables. It is, without exaggeration, the

most important inequality in the economic theory of uncertainty.

Theorem 5.8 (Jensen inequality, finite version). *Let $C \subseteq \mathbb{R}^n$ be convex and $f : C \rightarrow \mathbb{R}$ convex. Then for any points $x_1, \dots, x_k \in C$ and weights $\lambda_1, \dots, \lambda_k \geq 0$ with $\sum_{i=1}^k \lambda_i = 1$,*

$$f\left(\sum_{i=1}^k \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i).$$

If f is concave, the inequality is reversed.

Proof. By induction on k . **Base case $k = 2$:** it is exactly the definition of convexity (5.1) with $\lambda_1 = 1 - \lambda$, $\lambda_2 = \lambda$. (The case $k = 1$ is trivial: $\lambda_1 = 1$ and both sides are $f(x_1)$.)

Inductive step: suppose the result holds for $k - 1$ points (inductive hypothesis) and let us prove it for k points. Let $x_1, \dots, x_k \in C$ and weights $\lambda_1, \dots, \lambda_k \geq 0$ that add up to 1. If $\lambda_k = 1$, then all the other weights are zero and the inequality is trivial ($f(x_k) \leq f(x_k)$). So suppose $\lambda_k < 1$, so that $\mu := 1 - \lambda_k = \sum_{i=1}^{k-1} \lambda_i > 0$. The idea is to *regroup*: we separate the last point and treat the first $k - 1$ as a single block. We define the point

$$z := \sum_{i=1}^{k-1} \frac{\lambda_i}{\mu} x_i.$$

Note that the weights λ_i/μ are nonnegative and add up to $\frac{1}{\mu} \sum_{i=1}^{k-1} \lambda_i = \mu/\mu = 1$: they are legitimate convex weights on $k - 1$ points, so $z \in C$ (a convex set contains all convex combinations of its points, Chapter 2). Moreover, by construction,

$$\sum_{i=1}^k \lambda_i x_i = \mu z + \lambda_k x_k, \quad \mu + \lambda_k = 1.$$

Now we first apply the definition of convexity (case $k = 2$) to the points z and x_k with weights μ and λ_k :

$$f\left(\sum_{i=1}^k \lambda_i x_i\right) = f(\mu z + \lambda_k x_k) \leq \mu f(z) + \lambda_k f(x_k),$$

and then the inductive hypothesis to $f(z)$:

$$f(z) = f\left(\sum_{i=1}^{k-1} \frac{\lambda_i}{\mu} x_i\right) \leq \sum_{i=1}^{k-1} \frac{\lambda_i}{\mu} f(x_i).$$

Combining the two:

$$f\left(\sum_{i=1}^k \lambda_i x_i\right) \leq \mu \sum_{i=1}^{k-1} \frac{\lambda_i}{\mu} f(x_i) + \lambda_k f(x_k) = \sum_{i=1}^{k-1} \lambda_i f(x_i) + \lambda_k f(x_k) = \sum_{i=1}^k \lambda_i f(x_i),$$

which is what we wanted. The concave version follows by applying the above to $-f$ (Remark 5.2). $\square$

When the weights λ_i are probabilities, the sum $\sum_i \lambda_i x_i$ is the *expectation* of a discrete random variable and $\sum_i \lambda_i f(x_i)$ is the expectation of f of that variable. This gives immediately

the probabilistic version:

Theorem 5.9 (Jensen inequality, expectation version). *Let X be a random variable taking values in a convex set $C \subseteq \mathbb{R}^n$ with finite expectation $\mathbb{E}[X] \in C$. If $f : C \rightarrow \mathbb{R}$ is convex (and $\mathbb{E}[f(X)]$ exists), then*

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

If f is concave, $f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$. If f is strictly convex (resp. strictly concave) and X is not constant (that is, it is genuinely random), the inequality is strict.

Proof idea. If X is discrete with finite support, this is exactly Theorem 5.8 with $\lambda_i = \Pr(X = x_i)$. For the general case one uses the first order characterization that we will see in Section 5.5 (or, without differentiability, the existence of a supporting line, which we will prove in the chapter on subdifferentials): there is a vector g such that $f(x) \geq f(\mathbb{E}[X]) + \langle g, x - \mathbb{E}[X] \rangle$ for all $x \in C$. Evaluating at $x = X$ and taking expectations on both sides (the expectation preserves inequalities and is linear):

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) + \langle g, \mathbb{E}[X] - \mathbb{E}[X] \rangle = f(\mathbb{E}[X]). \quad \square$$

5.2.1 Worked application: risk aversion

This subsection is a mini course on the theory of risk aversion that you will see in Microeconomics (decision under uncertainty; MWG ch. 6, [9]; [10] ch. 2). Consider an agent with random wealth: a *lottery* X that pays x_i with probability λ_i . The agent evaluates lotteries with *expected utility*: welfare is $\mathbb{E}[u(X)] = \sum_i \lambda_i u(x_i)$, where $u : \mathbb{R} \rightarrow \mathbb{R}$ is the Bernoulli utility function over money, which we assume strictly increasing.

Risk aversion. The agent is said to be *risk averse* if for every lottery they prefer to receive the expectation $\mathbb{E}[X]$ for sure rather than play the lottery:

$$u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)] \quad \text{for every lottery } X.$$

By the Jensen inequality (Theorem 5.9), *this happens if and only if u is concave*. The direction “concave $\Rightarrow$ risk averse” is Jensen; the converse follows by taking two point lotteries: if the inequality holds for every binary lottery (x with probability $1 - \lambda$, y with probability λ), then $u((1 - \lambda)x + \lambda y) \geq (1 - \lambda)u(x) + \lambda u(y)$ for all x, y, λ , which is the definition of concavity. The concavity of utility *is* risk aversion: they are not two assumptions, they are the same one.

Certainty equivalent and risk premium. The *certainty equivalent* of the lottery X , written $c(X)$, is the sure amount that leaves the agent indifferent to the lottery:

$$u(c(X)) = \mathbb{E}[u(X)] \quad \Longleftrightarrow \quad c(X) = u^{-1}(\mathbb{E}[u(X)]),$$

where u^{-1} exists because u is strictly increasing. The *risk premium* is

$$\rho(X) := \mathbb{E}[X] - c(X),$$

the most the agent would pay to get rid of the risk and keep a sure payment. If u is concave,

Jensen gives $u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)] = u(c(X))$, and since u is increasing, $\mathbb{E}[X] \geq c(X)$: *the risk premium of a risk averse agent is nonnegative*, and it is strictly positive if u is strictly concave and the lottery has genuine risk. This is the foundation of the insurance business: the risk averse insured is willing to pay above the expected loss.

Example 5.10 (Full computation of a certainty equivalent). An agent with $u(x) = \ln x$ (strictly concave, Example 5.7) faces the lottery: wealth 100 with probability 1/2 and wealth 25 with probability 1/2.

- *Expected wealth*: $\mathbb{E}[X] = \frac{1}{2}(100) + \frac{1}{2}(25) = 62.5$.
- *Expected utility*: $\mathbb{E}[u(X)] = \frac{1}{2} \ln 100 + \frac{1}{2} \ln 25 = \ln(100^{1/2} \cdot 25^{1/2}) = \ln(\sqrt{2500}) = \ln 50$.
- *Certainty equivalent*: $c(X)$ solves $\ln c = \ln 50$, that is, $c(X) = 50$. (For logarithmic utility, the certainty equivalent is the *geometric mean* of the payoffs.)
- *Risk premium*: $\rho(X) = 62.5 - 50 = 12.5$.

Jensen check: $u(\mathbb{E}[X]) = \ln 62.5 \approx 4.135 > \ln 50 \approx 3.912 = \mathbb{E}[u(X)]$. The agent gives up as much as 12.5 units of expected value in order to remove the risk: a risk neutral insurer can offer a contract that both prefer.

Arrow–Pratt coefficients. How do we measure *how much* risk aversion an agent has? The curvature of u is the natural candidate ($u'' \leq 0$ is concavity, as we will see in Section 5.6), but u'' alone does not work as a measure: expected utility is invariant to positive affine transformations ($u \mapsto au + b$ with $a > 0$ represents the same preferences), and u'' changes with a . The right normalization divides by u' :

Definition 5.11 (Arrow–Pratt coefficients). Let u be twice differentiable with $u' > 0$. The *coefficient of absolute risk aversion* at wealth x is

$$A(x) := -\frac{u''(x)}{u'(x)},$$

and the *coefficient of relative risk aversion* is

$$R(x) := -\frac{x u''(x)}{u'(x)} = x A(x).$$

Both are invariant to positive affine transformations of u (the factor a cancels in the ratio), and $A(x) \geq 0$ exactly where u is concave. A Taylor approximation (which you will see in Microeconomics) shows that for small risks the risk premium is approximately $\rho \approx \frac{1}{2}A(x)\sigma^2$, where σ^2 is the variance of the lottery: A measures, literally, how much premium per unit of variance.

Example 5.12 (CARA and CRRA, full computation). The two families of utilities most used in macro, finance and contracts:

(a) *CARA (constant absolute risk aversion)*: $u(x) = -e^{-\alpha x}$ with $\alpha > 0$. We compute the derivatives step by step:

$$u'(x) = -e^{-\alpha x} \cdot (-\alpha) = \alpha e^{-\alpha x} > 0, \quad u''(x) = \alpha e^{-\alpha x} \cdot (-\alpha) = -\alpha^2 e^{-\alpha x} < 0.$$

(Concave on all of $\mathbb{R}$, by the second order criterion.) Then

$$A(x) = -\frac{u''(x)}{u'(x)} = -\frac{-\alpha^2 e^{-\alpha x}}{\alpha e^{-\alpha x}} = \alpha \quad \text{constant}, \quad R(x) = \alpha x.$$

Absolute aversion does not depend on wealth: a CARA agent pays the same premium for an additive risk whether rich or poor. This is why CARA is the standard utility in contract and moral hazard models (MWG, ch. 14): combined with normal noise it produces the famous certainty equivalent “mean minus $\frac{\alpha}{2}$ times variance”.

(b) *CRRA (constant relative risk aversion)*: $u(x) = \frac{x^{1-\gamma}}{1-\gamma}$ for $\gamma > 0$, $\gamma \neq 1$, with $x > 0$ (and $u(x) = \ln x$ for $\gamma = 1$, which is the limit of $(x^{1-\gamma} - 1)/(1 - \gamma)$ as $\gamma \rightarrow 1$; subtracting the constant $1/(1 - \gamma)$ does not change preferences). We differentiate:

$$u'(x) = \frac{(1-\gamma)x^{-\gamma}}{1-\gamma} = x^{-\gamma} > 0, \quad u''(x) = -\gamma x^{-\gamma-1} < 0.$$

Then

$$A(x) = -\frac{-\gamma x^{-\gamma-1}}{x^{-\gamma}} = \frac{\gamma}{x}, \quad R(x) = x \cdot \frac{\gamma}{x} = \gamma \quad \text{constant}.$$

Relative aversion is constant: the agent pays the same *fraction* of wealth to avoid a risk *proportional* to wealth, whether rich or poor. For $\gamma = 1$ ($u = \ln$): $u' = 1/x$, $u'' = -1/x^2$, $A(x) = 1/x$, $R(x) = 1$, consistent with the limit. Check that in Example 5.10 the agent had $R \equiv 1$.

Intuition

The full bridge, from geometry to economics: *chord below the graph* (concavity) $\Rightarrow$ *the function of the average exceeds the average of the function* (Jensen) $\Rightarrow$ *the sure thing is preferred to the random one* (risk aversion) $\Rightarrow$ *there is a positive premium for removing risk* (insurance markets, the insurance versus incentives trade-off in contracts). Each arrow of this chain is a theorem of this chapter. When in Microeconomics you are told “assume u concave”, you should *see* the chord under the graph and *hear* “the agent hates variance”.

5.3 Epigraph, hypograph and level sets [P1]

This section connects the chapter with the previous ones: the whole theory of convex *functions* can be reduced to the theory of convex *sets* by looking at the set of points that lie *above* the graph.

Definition 5.13 (Epigraph and hypograph). Let $f : C \rightarrow \mathbb{R}$ with $C \subseteq \mathbb{R}^n$. The *epigraph* (“what lies above the graph”) of f is the subset of $\mathbb{R}^{n+1}$

$$\text{epi } f = \{ (x, t) \in C \times \mathbb{R} : t \geq f(x) \},$$

and the *hypograph* is $\text{hyp } f = \{ (x, t) \in C \times \mathbb{R} : t \leq f(x) \}$.

Think of the graph of f as a “surface” in $\mathbb{R}^{n+1}$: the epigraph is everything above it (including

the surface itself), the hypograph everything below. For $f(x) = x^2$, the epigraph is the region inside the parabola (upwards); for a concave production function f , the hypograph $\{(x, q) : q \leq f(x)\}$ is, up to sign conventions, the *production set*: the feasible input–output pairs.

Theorem 5.14 (Characterization through the epigraph). *Let $C \subseteq \mathbb{R}^n$ be convex and $f : C \rightarrow \mathbb{R}$. Then:*

$$f \text{ is convex} \iff \text{epi } f \text{ is a convex subset of } \mathbb{R}^{n+1}.$$

In the same way, f is concave if and only if its hypograph is convex.

Proof. ($\Rightarrow$) Assume f is convex. Take two points of the epigraph, (x, s) and (y, t) with $x, y \in C$, $s \geq f(x)$, $t \geq f(y)$, and a weight $\lambda \in [0, 1]$. We must show that the convex combination

$$(1 - \lambda)(x, s) + \lambda(y, t) = ((1 - \lambda)x + \lambda y, (1 - \lambda)s + \lambda t)$$

also lies in $\text{epi } f$. Its first component is in C (because C is convex). For the second one, we chain two inequalities: first the convexity of f , and then the bounds $s \geq f(x)$, $t \geq f(y)$ multiplied by the nonnegative weights $(1 - \lambda)$ and λ :

$$f((1 - \lambda)x + \lambda y) \leq (1 - \lambda)f(x) + \lambda f(y) \leq (1 - \lambda)s + \lambda t.$$

That is, the second component of the combined point lies above f evaluated at the first component: the point is in $\text{epi } f$. Therefore $\text{epi } f$ is convex.

($\Leftarrow$) Assume $\text{epi } f$ is convex. Take $x, y \in C$ and $\lambda \in [0, 1]$. The points $(x, f(x))$ and $(y, f(y))$ are in $\text{epi } f$ (with equality $t = f$). By convexity of the epigraph, their combination

$$((1 - \lambda)x + \lambda y, (1 - \lambda)f(x) + \lambda f(y)) \in \text{epi } f,$$

and belonging to the epigraph means, by definition, that the second coordinate dominates f of the first one:

$$(1 - \lambda)f(x) + \lambda f(y) \geq f((1 - \lambda)x + \lambda y),$$

which is exactly the convexity of f . The statement for concave functions follows by applying the above to $-f$ and noting that $\text{hyp } f = \{(x, t) : (x, -t) \in \text{epi}(-f)\}$, and that the map $(x, t) \mapsto (x, -t)$ is linear and therefore preserves convexity of sets. $\square$

Intuition

This theorem is a *dictionary*: every theorem about convex sets can be translated into functions through the epigraph, and the other way round. Examples we will see: the intersection of convex sets is convex $\leftrightarrow$ the supremum of convex functions is convex (Theorem 5.33, because the epigraph of the supremum is the intersection of the epigraphs); the supporting hyperplanes of $\text{epi } f \leftrightarrow$ the tangents or subgradients that minorize f (chapter on subdifferentials). In economics: “concave production function” $\leftrightarrow$ “convex production set”, which is the hypothesis of the welfare theorems. The bridge between sets and functions is not a curiosity: it is the working method of convex analysis.

5.3.1 Sublevel sets

Definition 5.15 (Sublevel set). For $f : C \rightarrow \mathbb{R}$ and $\alpha \in \mathbb{R}$, the *sublevel set* at height α is

$$S_\alpha(f) = \{x \in C : f(x) \leq \alpha\}.$$

The *upper contour set* is $U_\alpha(f) = \{x \in C : f(x) \geq \alpha\}$; for a utility u , $U_\alpha(u)$ is the set of bundles “at least as good as” the level α .

Proposition 5.16. *If f is convex, all its sublevel sets $S_\alpha(f)$ are convex. If f is concave, all its upper contour sets $U_\alpha(f)$ are convex.*

Proof. Let $x, y \in S_\alpha(f)$, that is, $f(x) \leq \alpha$ and $f(y) \leq \alpha$, and let $\lambda \in [0, 1]$. By convexity of f and then by the two bounds:

$$f((1 - \lambda)x + \lambda y) \leq (1 - \lambda)f(x) + \lambda f(y) \leq (1 - \lambda)\alpha + \lambda\alpha = \alpha.$$

Therefore $(1 - \lambda)x + \lambda y \in S_\alpha(f)$, and $S_\alpha(f)$ is convex. For concave f , apply the above to $-f$ with level $-\alpha$: $U_\alpha(f) = S_{-\alpha}(-f)$. $\square$

Careful (common mistake)

The converse is false, and getting confused here costs points in micro exams. The fact that *all* sublevel sets are convex does **not** imply that the function is convex. Counterexample: $f(x) = x^3$ on $\mathbb{R}$. Its sublevel sets are

$$S_\alpha = \{x : x^3 \leq \alpha\} = (-\infty, \alpha^{1/3}],$$

always an interval, always convex. But x^3 is *not* convex: with $x = -1$, $y = 0$ and $\lambda = 1/2$, the left side of (5.1) is $f(-1/2) = -1/8$ and the right side is $\frac{1}{2}(-1) + \frac{1}{2}(0) = -1/2$; since $-1/8 > -1/2$, the inequality fails. (Also: $f''(x) = 6x < 0$ for $x < 0$.) Functions whose sublevel sets are all convex are called *quasiconvex*, and those with convex upper contour sets, *quasiconcave*: they form a strictly larger class that we study in the corresponding chapter. In micro this matters a lot: convex preferences require a *quasiconcave* utility, not a concave one; concavity is stronger and is not an ordinal property.

5.4 Extended valued functions and the indicator [P2]

This section introduces a convention that looks strange at first but greatly simplifies modern convex analysis: allowing functions to take the value $+\infty$.

Motivation. Consider the problem of minimizing $f(x)$ subject to $x \in C$. These are two objects: a function and a constraint. The extended value convention merges them into one: define

$$\tilde{f}(x) = \begin{cases} f(x) & \text{if } x \in C, \\ +\infty & \text{if } x \notin C. \end{cases}$$

Minimizing $\tilde{f}$ over *all* of $\mathbb{R}^n$ is the same as minimizing f over C : the points outside C have value $+\infty$ and will never be chosen by a minimizer. The constraint has been “encoded” inside the function. This lets us state theorems only once, without carrying the domain as a separate object.

Definition 5.17 (Extended function, effective domain, proper function). An *extended valued function* is a function $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$. Its *effective domain* is

$$\text{dom } f = \{x \in \mathbb{R}^n : f(x) < +\infty\},$$

the points where f takes real values. We say that f is *convex* if inequality (5.1) holds for all $x, y \in \mathbb{R}^n$, with the natural arithmetic conventions ($t + \infty = \infty$, $\lambda \cdot \infty = \infty$ for $\lambda > 0$, $0 \cdot \infty = 0$). Equivalently, f is convex if $\text{epi } f$ is convex. A convex function is *proper* if it is not identically $+\infty$, that is, if $\text{dom } f \neq \emptyset$.

Remark 5.18. If f is convex in the extended sense, then $\text{dom } f$ is a convex set. Indeed, if $x, y \in \text{dom } f$ (finite values), the convexity inequality gives $f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y) < +\infty$, so the intermediate point is also in $\text{dom } f$. The $+\infty$ convention does not create strange domains: they are automatically convex.

Definition 5.19 (Indicator function). The *indicator function* of a set $C \subseteq \mathbb{R}^n$ is

$$\delta_C(x) = \begin{cases} 0 & \text{if } x \in C, \\ +\infty & \text{if } x \notin C. \end{cases}$$

Proposition 5.20. δ_C is convex if and only if C is convex. Moreover, if $C \neq \emptyset$, for any $f : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\min_{x \in C} f(x) \quad \text{and} \quad \min_{x \in \mathbb{R}^n} [f(x) + \delta_C(x)]$$

have the same optimal value and the same minimizers.

Proof. For the first part: the epigraph of δ_C is $\text{epi } \delta_C = \{(x, t) : x \in C, t \geq 0\} = C \times [0, \infty)$, a Cartesian product. A Cartesian product is convex if and only if each factor is ($[0, \infty)$ is always convex), so $\text{epi } \delta_C$ is convex if and only if C is; by Theorem 5.14 (in its extended version), this is equivalent to the convexity of δ_C . For the second part: if $x \in C$, then $f(x) + \delta_C(x) = f(x) + 0 = f(x)$; if $x \notin C$, then $f(x) + \delta_C(x) = +\infty$, which is never a minimum when $C \neq \emptyset$. The two problems agree point by point where it matters. $\square$

Remark 5.21 (Careful with the name). Do not confuse the *indicator* of convex analysis (0 inside, $+\infty$ outside) with the *indicator* of probability ($\mathbf{1}_C$: 1 inside, 0 outside). They are different objects with similar names; the one here is designed so that *adding* it imposes the constraint $x \in C$ in minimization problems.

This machinery reappears in the chapters on subdifferentials and duality: for example, the support function of Section 5.9 turns out to be the *conjugate* of the indicator. For now, it is enough that the $+\infty$ convention feels natural to you. From here on, when we say “let f be convex” without specifying a domain, we allow extended values and $\text{dom } f$ is its effective domain (see [6], sec. 2.4, and [1]).

5.5 First order characterization [P1]

If f is differentiable, convexity can be read off the *gradient*. The geometric idea: in a convex function, the tangent plane at any point stays *below* the graph everywhere. The tangent is the best local linear approximation; for a convex function, that approximation is also a *global lower bound*.

Theorem 5.22 (First order characterization). *Let $C \subseteq \mathbb{R}^n$ be convex and open, and let $f : C \rightarrow \mathbb{R}$ be differentiable. Then f is convex on C if and only if*

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad \text{for all } x, y \in C. \quad (5.2)$$

Moreover, f is strictly convex if and only if the inequality is strict whenever $y \neq x$. For concave f , the inequality is reversed: $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle$.

Before the proof, let us understand what it says. The right side of (5.2) is the *first order Taylor approximation* of f centered at x , that is, the hyperplane tangent to the graph at $(x, f(x))$. The theorem states that for a convex function this tangent *underestimates* f on the whole domain, not only near x : local information (the gradient at one point) produces a global bound. This is the property that makes first order conditions *sufficient* for global optimality in convex optimization, as we see later in the course.

Proof. The strategy is classical and you should master it: first the case $n = 1$, and then the general case by *restricting f to lines*.

Step 1: case $n = 1$. Let $f : I \rightarrow \mathbb{R}$ be differentiable on an open interval I .

($\Rightarrow$) Assume f is convex and let $x, y \in I$ with $x \neq y$. For $\lambda \in (0, 1]$, convexity at the point $x + \lambda(y - x) = (1 - \lambda)x + \lambda y$ gives

$$f(x + \lambda(y - x)) \leq (1 - \lambda)f(x) + \lambda f(y) = f(x) + \lambda(f(y) - f(x)).$$

Subtract $f(x)$ and divide by $\lambda > 0$:

$$\frac{f(x + \lambda(y - x)) - f(x)}{\lambda} \leq f(y) - f(x).$$

The left side, after multiplying and dividing by $(y - x)$, is

$$\frac{f(x + \lambda(y - x)) - f(x)}{\lambda(y - x)} (y - x),$$

and when $\lambda \downarrow 0$ the quotient converges, by the definition of derivative (with increment $h = \lambda(y - x) \rightarrow 0$), to $f'(x)$. Taking the limit in the inequality (limits preserve non strict inequalities):

$$f'(x)(y - x) \leq f(y) - f(x),$$

which is exactly (5.2) in one dimension.

($\Leftarrow$) Assume now that $f(y) \geq f(x) + f'(x)(y - x)$ for all $x, y \in I$. Let $u, v \in I$, $\lambda \in (0, 1)$, and call $z = (1 - \lambda)u + \lambda v$ the intermediate point. We apply the hypothesis *twice, both times*

centered at z : once with $y = u$ and once with $y = v$:

$$f(u) \geq f(z) + f'(z)(u - z), \quad (5.3)$$

$$f(v) \geq f(z) + f'(z)(v - z). \quad (5.4)$$

We multiply (5.3) by $(1 - \lambda) \geq 0$ and (5.4) by $\lambda \geq 0$ (multiplying by nonnegative numbers preserves the direction) and add:

$$(1 - \lambda)f(u) + \lambda f(v) \geq f(z) + f'(z)[(1 - \lambda)(u - z) + \lambda(v - z)].$$

The bracket is zero: $(1 - \lambda)u + \lambda v - [(1 - \lambda) + \lambda]z = z - z = 0$. We are left with $(1 - \lambda)f(u) + \lambda f(v) \geq f(z)$, the definition of convexity. Note how neat the argument is: the two tangents at z “lift” the chord above the graph.

Step 2: general case by restriction to lines. The tool is the following equivalence, useful on its own: *f is convex on C if and only if its restriction to every line is convex*; formally, if and only if for all $x, y \in C$ the one variable function

$$\varphi_{x,y}(\lambda) := f(x + \lambda(y - x)), \quad \lambda \in [0, 1],$$

is convex. The equivalence is immediate: the convexity inequality of f between x and y with weight λ is exactly the convexity inequality of $\varphi_{x,y}$ between 0 and 1 with weight λ ; and conversely, any pair of points of a segment in C lives on one of these lines. (Check the detail: if $s, t \in [0, 1]$ and $\mu \in [0, 1]$, then $\varphi_{x,y}((1 - \mu)s + \mu t)$ is f evaluated at the combination with weight μ of the points $x + s(y - x)$ and $x + t(y - x)$.)

By the chain rule, $\varphi_{x,y}$ is differentiable with

$$\varphi'_{x,y}(\lambda) = \langle \nabla f(x + \lambda(y - x)), y - x \rangle.$$

($\Rightarrow$) If f is convex, each $\varphi_{x,y}$ is convex and differentiable in one variable; by Step 1 applied at $\lambda = 0$ with comparison point 1:

$$\varphi_{x,y}(1) \geq \varphi_{x,y}(0) + \varphi'_{x,y}(0)(1 - 0),$$

which translates into $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$: we obtained (5.2).

($\Leftarrow$) If (5.2) holds, fix x, y and prove that $\varphi := \varphi_{x,y}$ satisfies the first order inequality of Step 1. Let $\lambda, \mu \in [0, 1]$, and call $x_\lambda = x + \lambda(y - x)$, $x_\mu = x + \mu(y - x)$. Applying (5.2) to the points x_λ (center) and x_μ :

$$f(x_\mu) \geq f(x_\lambda) + \langle \nabla f(x_\lambda), x_\mu - x_\lambda \rangle = f(x_\lambda) + (\mu - \lambda) \langle \nabla f(x_\lambda), y - x \rangle,$$

where we used $x_\mu - x_\lambda = (\mu - \lambda)(y - x)$. In the notation of φ : $\varphi(\mu) \geq \varphi(\lambda) + \varphi'(\lambda)(\mu - \lambda)$. By Step 1 ($\Leftarrow$), φ is convex; since this holds for all x, y , f is convex.

Strict case. If f is strictly convex and $y \neq x$, we repeat ($\Rightarrow$) more carefully: by strict convexity at $\lambda = 1/2$,

$$f\left(\frac{x+y}{2}\right) < \frac{1}{2}f(x) + \frac{1}{2}f(y),$$

and applying (5.2) (the non strict version, already proved) between x and the midpoint:

$$f\left(\frac{x+y}{2}\right) \geq f(x) + \left\langle \nabla f(x), \frac{x+y}{2} - x \right\rangle = f(x) + \frac{1}{2} \langle \nabla f(x), y - x \rangle.$$

Combining both: $f(x) + \frac{1}{2} \langle \nabla f(x), y - x \rangle < \frac{1}{2} f(x) + \frac{1}{2} f(y)$, and multiplying by 2 and rearranging, $f(y) > f(x) + \langle \nabla f(x), y - x \rangle$: the strict inequality. Conversely, if the first order inequality is strict for $y \neq x$, the argument of Step 1 ($\Leftarrow$) with the inequalities (5.3)–(5.4) strict (note that $u \neq z \neq v$ when $u \neq v$ and $\lambda \in (0, 1)$) produces strict convexity. $\square$

Intuition

“The tangent underestimates.” In economic terms: if u is a *concave* utility and you linearize it around x (as one does when approximating), the concave version of (5.2) says $u(y) \leq u(x) + \langle \nabla u(x), y - x \rangle$: *the linearization overestimates the true utility at any other point*. A planner who evaluates policies with a linear approximation of a concave utility will be systematically too optimistic. The same inequality, read backwards, is what makes first order conditions global: if $\nabla f(x^*) = 0$ in a convex function, then $f(y) \geq f(x^*) + 0$ for every y : *every critical point is a global minimum*. That single sentence justifies the whole course.

Example 5.23 (Direct use). We know that e^x is convex. The tangent at $x = 0$ is $e^0 + e^0(y - 0) = 1 + y$. The theorem gives, for free, the global inequality

$$e^y \geq 1 + y \quad \text{for every } y \in \mathbb{R},$$

with equality only at $y = 0$ (strict convexity). In the same way, the concavity of $\ln$ with tangent at $x = 1$ gives $\ln y \leq y - 1$ for every $y > 0$. These two inequalities appear again and again in information theory, in estimation and in macro; now you know where they come from: they are the “tangent that under or overestimates”.

Remark 5.24 (Monotonicity of the gradient). Adding (5.2) written at (x, y) and at (y, x) :

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle, \quad f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle,$$

the terms $f(x) + f(y)$ cancel and we are left with $0 \geq \langle \nabla f(x) - \nabla f(y), y - x \rangle$, that is,

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0 \quad \text{for all } x, y \in C.$$

This property is called *monotonicity* of the gradient: in one dimension it simply says that f' is nondecreasing. It is also equivalent to convexity (we do not prove it here; see [6], sec. 2.5). In micro, the monotonicity of the gradient of the profit function is the *law of supply*: price changes and net supply changes “go in the same direction”.

5.6 Second order characterization [P1]

For twice differentiable functions, convexity can be read off the *Hessian*. In practice, this is the criterion you will use in 90% of the exercises. Recall from the preliminaries chapter: the Hessian $\nabla^2 f(x)$ is the symmetric matrix of second derivatives $[\partial^2 f / \partial x_i \partial x_j]$; a symmetric matrix A is *positive semidefinite* (PSD, $A \succeq 0$) if $\langle v, Av \rangle \geq 0$ for every v , and *positive definite* (PD, $A \succ 0$) if $\langle v, Av \rangle > 0$ for every $v \neq 0$; the minor criteria: PD is equivalent to all leading principal minors being positive (Sylvester), and PSD is equivalent to *all* principal minors (not only the leading ones) being nonnegative.

Theorem 5.25 (Second order characterization). *Let $C \subseteq \mathbb{R}^n$ be convex and open, and let $f : C \rightarrow \mathbb{R}$ be of class C^2 . Then:*

(i) *f is convex on C if and only if $\nabla^2 f(x) \succeq 0$ (PSD) for every $x \in C$.*

(ii) *If $\nabla^2 f(x) \succ 0$ (PD) for every $x \in C$, then f is strictly convex. **The converse is false.***

For concavity: f is concave iff $\nabla^2 f(x) \preceq 0$ (NSD) on all of C , and a ND Hessian at every point implies strictly concave.

Proof. As in the previous theorem, everything reduces to one variable through lines. Fix $x \in C$ and a direction $v \in \mathbb{R}^n$, and consider $\varphi(\lambda) = f(x + \lambda v)$ for those λ such that $x + \lambda v \in C$ (an open interval containing 0, because C is open). By the chain rule applied twice:

$$\varphi'(\lambda) = \langle \nabla f(x + \lambda v), v \rangle, \quad \varphi''(\lambda) = \langle v, \nabla^2 f(x + \lambda v) v \rangle.$$

The second equality deserves a comment: differentiating $\langle \nabla f(x + \lambda v), v \rangle = \sum_i \partial_i f(x + \lambda v) v_i$ with respect to λ gives $\sum_i \sum_j \partial_{ij}^2 f(x + \lambda v) v_j v_i$, which is the quadratic form of the Hessian at v .

($\Leftarrow$ of (i)) Assume $\nabla^2 f \succeq 0$ on all of C . Then $\varphi''(\lambda) = \langle v, \nabla^2 f(x + \lambda v) v \rangle \geq 0$ on its interval. We use Taylor's theorem with Lagrange remainder (preliminaries): for λ in the interval there exists θ between 0 and λ such that

$$\varphi(\lambda) = \varphi(0) + \varphi'(0)\lambda + \frac{1}{2}\varphi''(\theta)\lambda^2 \geq \varphi(0) + \varphi'(0)\lambda,$$

because the second order term is ≥ 0 . Translated with $y = x + \lambda v$ (that is, $v = y - x$, $\lambda = 1$, taking any $y \in C$; the segment $[x, y]$ lies in C by convexity):

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle.$$

This is the first order characterization (5.2), and by Theorem 5.22, f is convex.

($\Rightarrow$ of (i)) Assume f is convex and, for contradiction, that there exist $x \in C$ and $v \neq 0$ with $\langle v, \nabla^2 f(x) v \rangle < 0$. Since f is C^2 , the function $\lambda \mapsto \langle v, \nabla^2 f(x + \lambda v) v \rangle$ is continuous, so $\varphi'' < 0$ on a whole interval $(-\varepsilon, \varepsilon)$. By Taylor centered at 0 with remainder at $\theta \in (0, \lambda)$, for $0 < \lambda < \varepsilon$:

$$\varphi(\lambda) = \varphi(0) + \varphi'(0)\lambda + \frac{1}{2}\varphi''(\theta)\lambda^2 < \varphi(0) + \varphi'(0)\lambda,$$

which violates the first order inequality (5.2) between x and $y = x + \lambda v$ (which convexity guarantees). Contradiction; therefore the Hessian is PSD at every point.

(ii) If $\nabla^2 f \succ 0$ on all of C and $y \neq x$, the same Taylor expansion gives, with $v = y - x \neq 0$,

$$f(y) = f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2} \langle y - x, \nabla^2 f(z)(y - x) \rangle$$

for some z in the segment $(x, y) \subseteq C$; the last term is strictly positive by PD, so $f(y) > f(x) + \langle \nabla f(x), y - x \rangle$, and by the strict part of Theorem 5.22, f is strictly convex. $\square$

Careful (common mistake)

Strictly convex does NOT imply a positive definite Hessian. The canonical counterexample is $f(x) = x^4$ on $\mathbb{R}$: it is strictly convex (check it: its derivative $f'(x) = 4x^3$ is strictly increasing, which in one variable is equivalent to strict convexity; or do it from the definition), but $f''(x) = 12x^2$ *vanishes* at $x = 0$: there the Hessian is only PSD, not PD. The correct logic is:

$$\nabla^2 f \succ 0 \text{ at every point} \implies f \text{ strictly convex} \implies \nabla^2 f \succeq 0 \text{ at every point,}$$

and neither arrow can be reversed. In exams: to *prove* strict convexity it is enough to have PD at every point; but if the Hessian is PD except at isolated points, do not conclude that strictness fails (it may still hold, as in x^4), and do not conclude that it holds either: you need a finer look.

5.6.1 Worked examples with the Hessian and minors

Example 5.26 (Quadratic function). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$f(x) = \frac{1}{2} x^\top Q x + b^\top x + c,$$

with Q symmetric ($n \times n$), $b \in \mathbb{R}^n$, $c \in \mathbb{R}$. Let us compute the gradient component by component: $f(x) = \frac{1}{2} \sum_{i,j} Q_{ij} x_i x_j + \sum_i b_i x_i + c$, and differentiating with respect to x_k (using the symmetry $Q_{ij} = Q_{ji}$ to merge the two terms where x_k appears):

$$\frac{\partial f}{\partial x_k} = \frac{1}{2} \left(\sum_j Q_{kj} x_j + \sum_i Q_{ik} x_i \right) + b_k = \sum_j Q_{kj} x_j + b_k,$$

that is, $\nabla f(x) = Qx + b$. Differentiating once more, $\nabla^2 f(x) = Q$ for every x : the Hessian is constant. By Theorem 5.25:

$$f \text{ is convex} \iff Q \succeq 0, \quad f \text{ is strictly convex} \iff Q \succ 0.$$

(For quadratics the “if and only if” does hold in the strict case: if Q is not PD, there exists $v \neq 0$ with $v^\top Q v \leq 0$, and along the line $\{x + \lambda v\}$ the function is affine or concave in λ , killing strictness.) Numerical case: $Q = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Leading principal minors: $\Delta_1 = 2 > 0$, $\Delta_2 = \det Q = 2 \cdot 3 - 1 \cdot 1 = 5 > 0$; by Sylvester, $Q \succ 0$ and f is strictly convex on all of $\mathbb{R}^2$.

Example 5.27 (Cobb–Douglas: concave iff nonincreasing returns). Let $f : \mathbb{R}_{++}^2 \rightarrow \mathbb{R}$ (the open positive orthant, which is convex and open),

$$f(x, y) = x^a y^b, \quad a, b > 0.$$

Claim: f is concave on $\mathbb{R}_{++}^2$ if and only if $a + b \leq 1$ (with $a, b > 0$); that is, exactly when returns to scale are nonincreasing.

Computing the derivatives. First order:

$$f_x = a x^{a-1} y^b, \quad f_y = b x^a y^{b-1}.$$

Second order (differentiate again, with care in the exponents):

$$f_{xx} = a(a-1) x^{a-2} y^b, \quad f_{yy} = b(b-1) x^a y^{b-2}, \quad f_{xy} = f_{yx} = ab x^{a-1} y^{b-1}.$$

The Hessian is

$$\nabla^2 f(x, y) = \begin{pmatrix} a(a-1) x^{a-2} y^b & ab x^{a-1} y^{b-1} \\ ab x^{a-1} y^{b-1} & b(b-1) x^a y^{b-2} \end{pmatrix}.$$

Criterion for NSD in 2×2 (preliminaries): a symmetric 2×2 matrix is negative semidefinite if and only if its diagonal entries are ≤ 0 and its determinant is ≥ 0 (equivalently: all principal minors of order 1 are ≤ 0 and the one of order 2 is ≥ 0).

Minors of order 1 (the diagonal). Since $x, y > 0$, the factors $x^{a-2} y^b$ and $x^a y^{b-2}$ are positive, so

$$f_{xx} \leq 0 \iff a(a-1) \leq 0 \iff a \leq 1, \quad f_{yy} \leq 0 \iff b \leq 1.$$

Minor of order 2 (the determinant). We compute:

$$\begin{aligned} \det \nabla^2 f &= a(a-1) x^{a-2} y^b \cdot b(b-1) x^a y^{b-2} - (ab x^{a-1} y^{b-1})^2 \\ &= ab(a-1)(b-1) x^{2a-2} y^{2b-2} - a^2 b^2 x^{2a-2} y^{2b-2} \\ &= ab x^{2a-2} y^{2b-2} [(a-1)(b-1) - ab]. \end{aligned}$$

We expand the bracket: $(a-1)(b-1) - ab = ab - a - b + 1 - ab = 1 - a - b$. Therefore,

$$\det \nabla^2 f(x, y) = ab x^{2a-2} y^{2b-2} (1 - a - b),$$

and since $ab x^{2a-2} y^{2b-2} > 0$ on $\mathbb{R}_{++}^2$,

$$\det \nabla^2 f \geq 0 \iff a + b \leq 1.$$

Conclusion. If $a + b \leq 1$ (and $a, b > 0$, which forces $a \leq 1$ and $b \leq 1$ automatically, since $a \leq a + b \leq 1$), the three minors satisfy the conditions and the Hessian is NSD at every point of $\mathbb{R}_{++}^2$: f is concave. Conversely, if $a + b > 1$, the determinant is negative at every point; a 2×2 matrix with negative determinant has one positive and one negative eigenvalue (indefinite), so the Hessian is not NSD and f is not concave (there is a direction of positive curvature). If moreover $a + b < 1$, you can check that the Hessian is ND (leading minors: $f_{xx} < 0$ and $\det > 0$)

and f is *strictly* concave on $\mathbb{R}_{++}^2$; with $a + b = 1$ (constant returns) the determinant is zero: the function is concave but not strictly, since it is linear along rays from the origin, as corresponds to a homogeneous function of degree 1. In Microeconomics: a Cobb–Douglas technology gives a convex production set exactly when $a + b \leq 1$; with increasing returns ($a + b > 1$) convexity fails, and with it the welfare theorems and the existence of competitive equilibrium.

Example 5.28 (Log-sum-exp and the norm). Two more members of the catalogue worth knowing:

- (a) *Log-sum-exp*: $f(x) = \ln(e^{x_1} + \cdots + e^{x_n})$ is convex on $\mathbb{R}^n$ (we state it without proof; the proof computes the Hessian and checks PSD using the Cauchy–Schwarz inequality; see [2], sec. 3.1.5). It is the “smoothing” of the maximum: $\max_i x_i \leq f(x) \leq \max_i x_i + \ln n$, and it appears in the logit models of discrete choice you will see in structural econometrics: the expected utility of the best of n alternatives with Gumbel shocks is exactly a log-sum-exp.
- (b) *Norms*: every norm on $\mathbb{R}^n$ (in particular the Euclidean $\|x\|$, $\|x\|_1$, $\|x\|_\infty$) is convex: the argument of Example 5.6 uses only the triangle inequality and absolute homogeneity, which every norm satisfies. No norm is strictly convex as a function (along a ray from the origin it is linear: $\|\lambda x\| = \lambda \|x\|$ for $\lambda \geq 0$). Note that the Euclidean norm is not differentiable at 0: another example of a convex function that is not smooth.

5.7 Operations that preserve convexity [P1]

Just as in Chapter 2 we built a catalogue of operations that preserve the convexity of *sets* (intersections, affine images, sums), here we build the catalogue for *functions*. It is the toolbox that lets you certify convexity of complicated functions without computing a single Hessian. Reference: [2], sec. 3.2.

Theorem 5.29 (Nonnegative linear combinations). *If $f_1, \dots, f_m$ are convex on a convex set C and $\alpha_1, \dots, \alpha_m \geq 0$, then $g = \alpha_1 f_1 + \cdots + \alpha_m f_m$ is convex on C . If moreover some f_i with $\alpha_i > 0$ is strictly convex, then g is strictly convex. The same holds with “concave” instead of “convex”.*

Proof. The case $m = 2$ is enough (the general one follows by iteration). For $x, y \in C$ and $\lambda \in [0, 1]$, we apply the convexity of f_1 and of f_2 and multiply by $\alpha_1, \alpha_2 \geq 0$ (which preserves the direction of the inequalities) before adding:

$$\begin{aligned} g((1 - \lambda)x + \lambda y) &= \alpha_1 f_1((1 - \lambda)x + \lambda y) + \alpha_2 f_2((1 - \lambda)x + \lambda y) \\ &\leq \alpha_1 [(1 - \lambda)f_1(x) + \lambda f_1(y)] + \alpha_2 [(1 - \lambda)f_2(x) + \lambda f_2(y)] \\ &= (1 - \lambda)g(x) + \lambda g(y). \end{aligned}$$

If f_1 is strictly convex, $\alpha_1 > 0$, $x \neq y$ and $\lambda \in (0, 1)$, the first inequality used is strict and so is the sum. $\square$

Remark 5.30. Be careful with what is **not** in the theorem: the *difference* of convex functions need not be convex ($x^2 - 2x^2 = -x^2$), and neither does the *product* of convex functions. Sometimes

the product works out ($x \cdot x = x^2$), but in general it fails: $f(x) = x^2$ and $g(x) = (x - 1)^2$ are convex and their product $f(x)g(x) = x^2(x - 1)^2$ is not, since it has two minima (at $x = 0$ and $x = 1$) with a local maximum between them. Only nonnegative weights.

Theorem 5.31 (Composition with an affine function). *Let A be an $m \times n$ matrix, $b \in \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}$ convex. Then $g(x) = f(Ax + b)$ is convex on $\mathbb{R}^n$ (on $\{x : Ax + b \in \text{dom } f\}$, which is convex).*

Proof. The key is that the affine map *commutes* with convex combinations:

$$A((1 - \lambda)x + \lambda y) + b = (1 - \lambda)(Ax) + \lambda(Ay) + [(1 - \lambda) + \lambda]b = (1 - \lambda)(Ax + b) + \lambda(Ay + b).$$

Then

$$g((1 - \lambda)x + \lambda y) = f((1 - \lambda)(Ax + b) + \lambda(Ay + b)) \leq (1 - \lambda)f(Ax + b) + \lambda f(Ay + b) = (1 - \lambda)g(x) + \lambda g(y),$$

using the convexity of f at the points $Ax + b$ and $Ay + b$. $\square$

Example 5.32. $g(x_1, x_2) = e^{3x_1 - 2x_2 + 1}$ is convex: it is $f(t) = e^t$ (convex) composed with the affine map $t = 3x_1 - 2x_2 + 1$. In the same way, $g(x) = \|Ax - b\|$ (the error of a linear system) is convex: a norm composed with an affine map. No Hessians needed.

5.7.1 The pointwise supremum: the star operation

Theorem 5.33 (Pointwise supremum). *Let $\{f_i\}_{i \in I}$ be an arbitrary family (finite or infinite) of convex functions on a convex set C . Then*

$$g(x) = \sup_{i \in I} f_i(x)$$

is convex on C (as an extended function: it may equal $+\infty$).

Proof. We give two proofs, because each one teaches something.

Through epigraphs (the elegant one). A point (x, t) is in $\text{epi } g$ if and only if $t \geq \sup_i f_i(x)$, if and only if $t \geq f_i(x)$ for every i , if and only if $(x, t) \in \text{epi } f_i$ for every i . That is,

$$\text{epi } g = \bigcap_{i \in I} \text{epi } f_i.$$

Each $\text{epi } f_i$ is convex (Theorem 5.14) and an arbitrary intersection of convex sets is convex (Chapter 2); therefore $\text{epi } g$ is convex and g is convex.

Through the definition (the direct one). Let $x, y \in C$, $\lambda \in [0, 1]$. For each $i \in I$:

$$f_i((1 - \lambda)x + \lambda y) \leq (1 - \lambda)f_i(x) + \lambda f_i(y) \leq (1 - \lambda)g(x) + \lambda g(y),$$

where the second inequality uses $f_i \leq g$ pointwise and the nonnegative weights. The right side is an upper bound of $f_i((1 - \lambda)x + \lambda y)$ that is *uniform in i* ; taking the supremum over i on the

left side (the supremum of a family bounded by M is $\leq M$):

$$g((1 - \lambda)x + \lambda y) \leq (1 - \lambda)g(x) + \lambda g(y). \quad \square$$

Intuition

Maximum of linear functions. The most important case of the theorem: if each $f_i(x) = \langle a_i, x \rangle + b_i$ is *affine* (convex and concave at the same time), the supremum $g(x) = \sup_i \{ \langle a_i, x \rangle + b_i \}$ is convex, no matter how infinite the family is. Geometrically, g is the “upper envelope” of a family of lines or hyperplanes, and its graph is a “polyhedral bowl” (or a smooth one, if the family is continuous). What is remarkable is the converse, which we see when we discuss conjugates: *every* (closed) convex function is a supremum of affine functions, namely its tangents or affine minorants. Convex functions are, neither more nor less, the suprema of families of affine functions. Very many functions in economics are defined as suprema of affine functions of prices: profit, expenditure (with a sign), cost, and so their convexity or concavity is *free*, without any assumption on the underlying technology or preferences. This is one of the most powerful and least appreciated results in all of microeconomics.

Example 5.34 (The profit function is convex in prices, always). A firm chooses a production plan $y \in Y \subseteq \mathbb{R}^n$ (positive components: outputs; negative: inputs) or, in the formulation with a cost function, a level $y \geq 0$ with cost $c(y)$. Its profit function is

$$\pi(p) = \sup_y \{ p \cdot y - c(y) \} \quad (\text{or } \pi(p) = \sup_{y \in Y} p \cdot y \text{ in the set version}).$$

Claim: π is convex in p *without any assumption on c* (no convexity, no continuity, nothing) or on Y .

Proof. For each fixed y , the function

$$p \longmapsto p \cdot y - c(y)$$

is *affine in p* : linear in p (with coefficient y) plus the constant $-c(y)$, which does not depend on p . Then π is a supremum of affine functions of p , indexed by y , and Theorem 5.33 concludes that π is convex. Done.

Note what just happened: the convexity of π in prices says nothing about the technology; it is a pure reflection of the fact that the firm *optimizes*. The structure $\sup_y \{ p \cdot y - c(y) \}$ is exactly that of the *Fenchel conjugate* $c^*(p)$ of the cost function, which we study in the duality chapter: the profit function *is* the conjugate of the cost. From the convexity of π we obtain, through differentiability, *Hotelling’s lemma* ($\nabla \pi(p) = y(p)$, the net supply) and the law of supply (monotonicity of the gradient, Remark 5.24); see MWG, ch. 5 [9].

5.7.2 Monotone convex composition

When is a composition $g(x) = h(f(x))$ convex? Not always: composing convex functions can fail. The exact rules:

Theorem 5.35 (Scalar composition rules). *Let $f : C \rightarrow \mathbb{R}$ (C convex) and $h : D \rightarrow \mathbb{R}$ with $D \subseteq \mathbb{R}$ an interval containing $f(C)$. Then $g = h \circ f$ is convex on C if one of the following holds:*

- (i) f convex and h convex and nondecreasing;
- (ii) f concave and h convex and nonincreasing.

And g is concave if:

- (iii) f concave and h concave and nondecreasing;
- (iv) f convex and h concave and nonincreasing.

Proof. We prove (i); the others are identical after changing signs and directions. Let $x, y \in C$, $\lambda \in [0, 1]$. First, the convexity of f :

$$f((1 - \lambda)x + \lambda y) \leq (1 - \lambda)f(x) + \lambda f(y).$$

Now we apply h to both sides. This is where *monotonicity* comes in: since h is nondecreasing, it preserves the inequality:

$$h(f((1 - \lambda)x + \lambda y)) \leq h((1 - \lambda)f(x) + \lambda f(y)).$$

Finally, the *convexity* of h applied to the numbers $f(x)$ and $f(y)$ with weight λ :

$$h((1 - \lambda)f(x) + \lambda f(y)) \leq (1 - \lambda)h(f(x)) + \lambda h(f(y)).$$

Chaining: $g((1 - \lambda)x + \lambda y) \leq (1 - \lambda)g(x) + \lambda g(y)$. □

Careful (common mistake)

The monotonicity conditions are not decorative. If you violate them, the composition may fail to be convex even though f and h are. The cheapest counterexample is $f(x) = x^2$ (convex) with $h(t) = -t$, which is convex (it is affine) but *decreasing*: the composition $h(f(x)) = -x^2$ is concave, not convex. And so that it does not look like the problem is only the sign, take $h(t) = (t - 1)^2$, convex but *not monotone* (decreasing for $t < 1$, increasing for $t > 1$). The composition is

$$g(x) = h(f(x)) = (x^2 - 1)^2 = x^4 - 2x^2 + 1.$$

We compute $g''(x) = 12x^2 - 4$, which is *negative* on the interval $(-1/\sqrt{3}, 1/\sqrt{3})$: g is not convex (its graph has a “W” shape, with two minima at $x = \pm 1$ and a local maximum at 0). Both pieces were convex; what failed was the monotonicity of h . An economic example of the same phenomenon: the indirect utility composed with non monotone functions of price indices can lose all curvature structure. Memorize the slogan of [2]: “convex nondecreasing of convex is convex; convex nonincreasing of concave is convex”, and *always* check monotonicity on the relevant range.

Example 5.36. Applications of the rules, without Hessians:

- (a) $g(x) = e^{\|x\|}$: $\|x\|$ convex, e^t convex increasing $\Rightarrow$ convex (rule (i)).
- (b) $g(x) = \frac{1}{f(x)}$ with f concave and positive: $h(t) = 1/t$ is convex and *decreasing* on $t > 0$, and f is concave $\Rightarrow g$ convex (rule (ii)). Example: $1/(xy)^{1/3}$ on $\mathbb{R}_{++}^2$ is convex because $(xy)^{1/3}$ is a concave Cobb–Douglas (Example 5.27, $a + b = 2/3 \leq 1$).
- (c) $g(x) = \ln f(x)$ with f concave and positive: $\ln$ is concave increasing, f concave $\Rightarrow g$ concave (rule (iii)). So the logarithm of a concave Cobb–Douglas is concave; in fact $\ln(x^a y^b) = a \ln x + b \ln y$ is concave for *all* $a, b \geq 0$, another sign that concavity is not ordinal.

5.7.3 Partial minimization

Theorem 5.37 (Partial minimization). *Let $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ be jointly convex in (x, y) , and let $D \subseteq \mathbb{R}^m$ be convex and nonempty. Then*

$$g(x) = \inf_{y \in D} f(x, y)$$

is convex on $\mathbb{R}^n$ (wherever it is $> -\infty$; in general it is convex as an extended function if $g > -\infty$ at some point).

Proof. Let $x_1, x_2 \in \mathbb{R}^n$ and $\lambda \in [0, 1]$; write $x_\lambda = (1 - \lambda)x_1 + \lambda x_2$. Fix an arbitrary $\varepsilon > 0$. By the definition of infimum, there exist “almost optimal” $y_1, y_2 \in D$:

$$f(x_1, y_1) \leq g(x_1) + \varepsilon, \quad f(x_2, y_2) \leq g(x_2) + \varepsilon.$$

(This ε trick is needed because the infimum may not be attained.) The point $y_\lambda = (1 - \lambda)y_1 + \lambda y_2$ is in D by convexity of D , so it is an admissible candidate in the minimization that defines $g(x_\lambda)$:

$$\begin{aligned} g(x_\lambda) &\leq f(x_\lambda, y_\lambda) = f((1 - \lambda)(x_1, y_1) + \lambda(x_2, y_2)) \\ &\leq (1 - \lambda)f(x_1, y_1) + \lambda f(x_2, y_2) && \text{(joint convexity of } f) \\ &\leq (1 - \lambda)[g(x_1) + \varepsilon] + \lambda[g(x_2) + \varepsilon] && \text{(almost optimality)} \\ &= (1 - \lambda)g(x_1) + \lambda g(x_2) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, letting $\varepsilon \downarrow 0$ gives $g(x_\lambda) \leq (1 - \lambda)g(x_1) + \lambda g(x_2)$. □

Remark 5.38. Note the contrast between the two theorems of this section that involve optimization: the *supremum* of convex functions is convex *with no conditions* on how they depend on the index (Theorem 5.33); the *infimum*, on the other hand, requires *joint* convexity in both variables. An infimum of functions convex in x (without joint convexity) may fail to be convex: for example $g(x) = \inf_{y \in \{-1, 1\}} (x - y)^2 = (|x| - 1)^2$ is not convex (it equals 0 at $x = \pm 1$ and 1 at $x = 0$, violating the chord inequality between -1 and 1); here the failure is because $D = \{-1, 1\}$ is not convex. In economics, Theorem 5.37 is what makes the *cost function* $c(w, q) = \min_z \{w \cdot z : F(z) \geq q\}$ jointly convex when the problem has convex structure, and in general it is behind the fact that “optimizing preserves convexity” in the parameter variable.

5.8 Continuity and differentiability of convex functions [P2]

Convex functions are surprisingly regular: convexity, which is a purely algebraic condition (an inequality), implies continuity without asking for it.

Theorem 5.39 (Continuity in the interior). *Let $f : C \rightarrow \mathbb{R}$ be convex, with $C \subseteq \mathbb{R}^n$ convex. Then f is continuous on $\text{int } C$ (the interior of C). In particular, a convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined on the whole space is continuous everywhere. Moreover, f is locally Lipschitz on $\text{int } C$.*

Proof idea. We give the skeleton of the argument (details in [6], theorem 2.35, or [1]). Let $x_0 \in \text{int } C$. **Step 1: f is bounded above near x_0 .** Since x_0 is interior, a small cube (or simplex) with vertices $v_1, \dots, v_{2^n}$ around x_0 fits inside C . Every point x of the cube is a convex combination of the vertices, $x = \sum_i \lambda_i v_i$, and by the finite Jensen inequality (Theorem 5.8),

$$f(x) \leq \sum_i \lambda_i f(v_i) \leq \max_i f(v_i) =: M.$$

Step 2: the upper bound forces a lower bound. For x near x_0 , the reflected point $x' = 2x_0 - x$ is also in the cube, and $x_0 = \frac{1}{2}x + \frac{1}{2}x'$; convexity gives $f(x_0) \leq \frac{1}{2}f(x) + \frac{1}{2}f(x') \leq \frac{1}{2}f(x) + \frac{1}{2}M$, hence $f(x) \geq 2f(x_0) - M$: bounded below. **Step 3: local boundedness + convexity $\Rightarrow$ locally Lipschitz.** With f trapped in a band $[m, M]$ near x_0 , an argument with chords (comparing slopes: the slope of a chord of a convex function increases with the base point) shows that $|f(x) - f(y)| \leq L \|x - y\|$ for a constant L that depends on $M - m$ and on the radius; Lipschitz implies continuity. $\square$

Example 5.40 (It can fail on the boundary). The “interior” hypothesis is essential. Let $f : [0, 1] \rightarrow \mathbb{R}$,

$$f(x) = \begin{cases} x^2 & \text{if } 0 < x \leq 1, \\ 1 & \text{if } x = 0. \end{cases}$$

Let us check that it is convex: its epigraph is the one of x^2 over $(0, 1]$ together with the half line $\{0\} \times [1, \infty)$, and one checks that the chord inequality holds (raising the value at an *extreme point* of the domain never breaks convexity: the chords that go from $(0, 1)$ to points (x, x^2) stay above the parabola). But f is discontinuous at $x = 0$: $\lim_{x \downarrow 0} f(x) = 0 \neq 1 = f(0)$. The discontinuity could only occur on the boundary of the domain, and only by “jumping upwards”. In $\mathbb{R}^n$ with $n \geq 2$ the boundary pathologies can be worse; this is why textbooks work with *closed* convex functions (lower semicontinuous), a technical point we return to when discussing conjugates.

Remark 5.41 (Differentiability almost everywhere). Although a convex function may fail to be differentiable (like $|x|$ at 0 or $\|x\|$ at the origin), it cannot be “very” irregular: a theorem of Rademacher and Alexandrov guarantees that a convex function is differentiable at *almost every* point of $\text{int dom } f$ (except on a set of measure zero), and twice differentiable at almost every point. The points of non differentiability are handled with the *subdifferential*, the set of slopes of the supporting lines, which is the topic of Chapter 7 if you follow the order of the notebook. In micro, non differentiability is not exotic: the profit function of a linear technology has “kinks”

exactly at the prices where the firm is indifferent between plans, and the subdifferential there is the set of optimal supplies.

5.9 The support function [P1]

We close the chapter with the object that ties everything above to economic duality: the support function of a set. The question it answers is: given a set C and a direction p , how far does C reach in the direction p ?

Definition 5.42 (Support function). Let $C \subseteq \mathbb{R}^n$ be nonempty. The *support function* of C is $\sigma_C : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$,

$$\sigma_C(p) = \sup_{x \in C} \langle p, x \rangle.$$

Geometric interpretation: for $\|p\| = 1$, $\langle p, x \rangle$ is the “height” of x measured in the direction p , and $\sigma_C(p)$ is the height of the supporting hyperplane of C with normal p (from the separation chapter: the hyperplane $\{x : \langle p, x \rangle = \sigma_C(p)\}$ “touches” C from outside, if the supremum is attained). Immediate economic interpretation: if $C = Y$ is a production set and p a price vector, $\langle p, y \rangle$ is the profit of the plan y , and $\sigma_Y(p)$ is *the maximum profit*: the support function of the production set *is* the profit function.

Theorem 5.43 (Basic properties of the support function). *For any $C \neq \emptyset$ (it need not be convex, closed or bounded):*

- (i) σ_C is convex on $\mathbb{R}^n$.
- (ii) σ_C is positively homogeneous of degree 1: $\sigma_C(tp) = t\sigma_C(p)$ for every $t > 0$ (and $\sigma_C(0) = 0$).
- (iii) $\sigma_C = \sigma_{\text{cl } C} = \sigma_{\text{conv } C}$: the set, its closure and its convex hull have the same support function.

Proof. (i) For each fixed $x \in C$, the function $p \mapsto \langle p, x \rangle$ is *linear* in p (the variable is now p , not x). Then σ_C is a supremum of linear functions of p , indexed by $x \in C$, and Theorem 5.33 gives convexity. Note that this is the same argument as in Example 5.34.

(ii) Let $t > 0$. We use that the supremum “pulls out positive scalars”:

$$\sigma_C(tp) = \sup_{x \in C} \langle tp, x \rangle = \sup_{x \in C} t \langle p, x \rangle = t \sup_{x \in C} \langle p, x \rangle = t \sigma_C(p),$$

where the central step is valid because multiplying by $t > 0$ is an increasing bijection of $\mathbb{R}$ onto $\mathbb{R}$ and therefore commutes with suprema. At $p = 0$: $\sigma_C(0) = \sup_x \langle 0, x \rangle = \sup_x 0 = 0$.

(iii) $\sigma_{\text{cl } C} = \sigma_C$ because $\langle p, \cdot \rangle$ is continuous and the supremum over a set and over its closure agree. For the hull: clearly $\sigma_{\text{conv } C} \geq \sigma_C$ (the supremum over a larger set is greater or equal). Conversely, every $z \in \text{conv } C$ is $z = \sum_i \lambda_i x_i$ with $x_i \in C$ and convex weights; then

$$\langle p, z \rangle = \sum_i \lambda_i \langle p, x_i \rangle \leq \sum_i \lambda_i \sigma_C(p) = \sigma_C(p),$$

and taking the supremum over z , $\sigma_{\text{conv } C} \leq \sigma_C$. □

Remark 5.44. Part (iii) has a deep economic reading: since the support function does not distinguish between C and $\text{cl conv } C$, *maximum profit only sees the closed convex hull of the technology*. Two production sets with the same closed convex hull are indistinguishable for an observer who only sees profits at different prices. Conversely (and this is the duality theorem that makes “recoverability” work in micro), a closed convex set is *completely determined* by its support function:

$$\text{cl conv } C = \{x \in \mathbb{R}^n : \langle p, x \rangle \leq \sigma_C(p) \text{ for every } p \in \mathbb{R}^n\}.$$

The inclusion $\subseteq$ is immediate from the definition; the $\supseteq$ one uses the separation theorem of the previous chapter: if $x_0 \notin \text{cl conv } C$, there is a hyperplane that strictly separates x_0 from the closed convex set $\text{cl conv } C$, that is, there exists p with $\langle p, x_0 \rangle > \sup_{x \in C} \langle p, x \rangle = \sigma_C(p)$, so x_0 does not satisfy the system of inequalities. Conclusion: *knowing σ_C is knowing $\text{cl conv } C$* . In micro: from the profit function one recovers the (convexified) technology, and from the expenditure function one recovers preferences; see MWG, ch. 5 (sections 5.C and 5.E) [9] and [10], ch. 2.

5.9.1 Computed examples

Example 5.45 (Support of the ball, the box and a cone). (a) *Euclidean unit ball* $B = \{x : \|x\| \leq 1\}$. By the Cauchy–Schwarz inequality, $\langle p, x \rangle \leq \|p\| \|x\| \leq \|p\|$ for $x \in B$, with equality at $x = p/\|p\|$ (if $p \neq 0$), which is in B . Therefore

$$\sigma_B(p) = \|p\|.$$

The support function of the ball is the norm. (Convex and homogeneous of degree 1, as it had to be.)

(b) *Box* $Q = [-1, 1]^n = \{x : |x_i| \leq 1 \ \forall i\}$. The product $\langle p, x \rangle = \sum_i p_i x_i$ is maximized coordinate by coordinate: the optimal choice is $x_i = \text{sign}(p_i)$ (that is, $x_i = 1$ if $p_i > 0$, $x_i = -1$ if $p_i < 0$, anything if $p_i = 0$), which gives $p_i x_i = |p_i|$. Then

$$\sigma_Q(p) = \sum_{i=1}^n |p_i| = \|p\|_1.$$

(The support of the ball of the ∞ norm is the 1 norm: norms are “dual” in pairs.)

(c) *A cone*. Let $K \subseteq \mathbb{R}^n$ be a cone (closed under multiplication by scalars ≥ 0). If $\langle p, x_0 \rangle > 0$ for some $x_0 \in K$, then $\langle p, tx_0 \rangle = t \langle p, x_0 \rangle \rightarrow +\infty$ as $t \rightarrow \infty$: the supremum is $+\infty$. If instead $\langle p, x \rangle \leq 0$ for every $x \in K$, the supremum is 0 (attained at $x = 0$ if $0 \in K$, or as a limit). Conclusion:

$$\sigma_K(p) = \begin{cases} 0 & \text{if } p \in K^\circ := \{p : \langle p, x \rangle \leq 0 \ \forall x \in K\}, \\ +\infty & \text{otherwise;} \end{cases}$$

that is, $\sigma_K = \delta_{K^\circ}$, the *indicator* (Definition 5.19) of the *polar cone* K° from the chapter on cones. The three examples show the pattern: the support function encodes the geometry of the set in the language of prices, and extended functions appear naturally (one more reason for the $+\infty$ convention of Section 5.4). Economic example of case (c): a production set with

constant returns to scale is a cone, and its profit function equals 0 at “sustainable” prices and $+\infty$ where profits would be unbounded: exactly the zero profit condition of equilibrium with constant returns.

5.9.2 Economic application: profit and expenditure as support functions

(1) The profit function. As anticipated, if $Y \subseteq \mathbb{R}^n$ is the production set (feasible net plans), the profit function is

$$\pi(p) = \sup_{y \in Y} \langle p, y \rangle = \sigma_Y(p).$$

Theorem 5.43 hands over the properties that MWG (ch. 5, proposition 5.C.1) lists for π : *convex* in p and *homogeneous of degree 1* in p (doubling all prices doubles maximum profit: the scale of prices is nominal). And the duality remark gives recovery: $\text{cl conv } Y = \{y : \langle p, y \rangle \leq \pi(p) \ \forall p\}$, the technology is reconstructed from profit data.

(2) The expenditure function is concave in prices. Let u be a utility and $V_u = \{x \in \mathbb{R}_+^n : u(x) \geq u\}$ the upper contour set of level u (the bundles that reach at least utility u). The *expenditure function* is

$$e(p, u) = \inf_{x \in V_u} \langle p, x \rangle,$$

the minimum cost of reaching utility u at prices p .

Proposition 5.46. *For each fixed level u , the function $p \mapsto e(p, u)$ is concave and homogeneous of degree 1 in p , whatever the set $V_u \neq \emptyset$ is (with no assumptions on the utility).*

Proof. The trick is to recognize an infimum as “minus a supremum”:

$$e(p, u) = \inf_{x \in V_u} \langle p, x \rangle = - \sup_{x \in V_u} \langle -p, x \rangle = - \sigma_{V_u}(-p).$$

(We use $\inf_x g(x) = -\sup_x [-g(x)]$ and $-\langle p, x \rangle = \langle -p, x \rangle$.) Now we chain operations that preserve structure: σ_{V_u} is convex (Theorem 5.43(i)); composing with the linear map $p \mapsto -p$ preserves convexity (Theorem 5.31); and putting a minus sign in front turns convex into concave (Remark 5.2). Therefore $e(\cdot, u)$ is concave. The direct proof is also instructive and worth seeing once: for p_1, p_2 and $\lambda \in [0, 1]$, and any $x \in V_u$,

$$\langle (1 - \lambda)p_1 + \lambda p_2, x \rangle = (1 - \lambda) \langle p_1, x \rangle + \lambda \langle p_2, x \rangle \geq (1 - \lambda) e(p_1, u) + \lambda e(p_2, u),$$

because $\langle p_i, x \rangle \geq e(p_i, u)$ (nobody spends less than the minimum expenditure). Taking the infimum over $x \in V_u$ on the left side:

$$e((1 - \lambda)p_1 + \lambda p_2, u) \geq (1 - \lambda)e(p_1, u) + \lambda e(p_2, u) :$$

concavity. Homogeneity works as in Theorem 5.43(ii): $\inf_x \langle tp, x \rangle = t \inf_x \langle p, x \rangle$ for $t > 0$. $\square$

Intuition

Why is expenditure *concave* in prices? Economic intuition from the direct proof: fix the bundle $\bar{x}$ that was optimal at the average prices. If expenditure could not reoptimize, it would be *linear* in p (spending $\langle p, \bar{x} \rangle$ at each p). But the consumer *does* reoptimize: when a price rises, they substitute towards relatively cheaper goods, and so their actual expenditure grows *less* than linearly. “Minimum of linear functions” = “linear minus the substitution gains” = concave. Symmetrically, profit is *convex* in prices because the firm reoptimizes in its favor: “maximum of linear functions”. From the concavity of e in p come two gems of Microeconomics: *Shephard’s lemma* ($\nabla_p e(p, u) = h(p, u)$, the Hicksian demand, which is the point where the hyperplane with slope p supports V_u) and the *negative semidefiniteness of the Slutsky matrix* (which is exactly $\nabla_p^2 e \preceq 0$, the second order characterization of concavity, Theorem 5.25). The law of compensated demand is not a psychological assumption: it is the Hessian of a concave function. See MWG, sections 3.E–3.G [9], and [10], ch. 1.

5.10 Chapter summary [P1]**Chapter summary**

- **Definition (Definition 5.1).** Let $C \subseteq \mathbb{R}^n$ be *convex* (the hypothesis on the domain is not optional: without it the point $(1 - \lambda)x + \lambda y$ may fail to be in C and the inequality cannot even be written). $f : C \rightarrow \mathbb{R}$ is **convex** if

$$f((1 - \lambda)x + \lambda y) \leq (1 - \lambda)f(x) + \lambda f(y) \quad \forall x, y \in C, \forall \lambda \in [0, 1],$$

and **strictly convex** if the inequality is *strict* for every $x \neq y$ and every $\lambda \in (0, 1)$ (the endpoints $\lambda \in \{0, 1\}$ and the case $x = y$ must be excluded: there is always equality there). f is concave iff $-f$ is convex (Remark 5.2); affine functions are the only ones that are convex and concave at the same time (Example 5.5).

- **Jensen (Theorems 5.8 and 5.9).** If f is convex, then $f(\sum_i \lambda_i x_i) \leq \sum_i \lambda_i f(x_i)$ for weights $\lambda_i \geq 0$ with $\sum_i \lambda_i = 1$ and points $x_i \in C$; in probabilistic form, $f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$ for X with values in C and integrable. For concave f the inequalities are reversed. *Economic reading:* u concave $\iff$ risk aversion; risk premium $\rho(X) = \mathbb{E}[X] - c(X) \geq 0$ with $u(c(X)) = \mathbb{E}[u(X)]$; Arrow–Pratt $A = -u''/u'$, $R = xA$ (Definition 5.11); CARA has constant A , CRRA has constant R (Example 5.12).
- **Epigraph (Theorem 5.14).** f is convex if and only if $\text{epi } f = \{(x, t) \in C \times \mathbb{R} : f(x) \leq t\}$ is a convex subset of $\mathbb{R}^{n+1}$; f is concave iff its hypograph is convex. This bridge translates the whole previous chapter into results about functions.
- **Level sets (Proposition 5.16).** If f is convex, then $S_\alpha = \{x \in C : f(x) \leq \alpha\}$ is convex for every $\alpha \in \mathbb{R}$. **The converse is false:** $f(x) = \sqrt{|x|}$ or $f(x) = x^3$ on $\mathbb{R}$ have all their sublevel sets convex (they are intervals) without being convex. The class that

characterizes that property is that of *quasiconvex* functions (Chapter 6).

- **First order (Theorem 5.22).** Hypotheses: C convex and *open* and f *differentiable* on C . Then f convex $\iff f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all $x, y \in C$ (the graph lies above every tangent plane); and f strictly convex $\iff$ the inequality is strict for $y \neq x$. Consequences: if $\nabla f(x^*) = 0$, then x^* is a *global* minimum; and the gradient is monotone, $\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0$ (Remark 5.24).
- **Second order (Theorem 5.25).** Hypotheses: C convex and *open*, f of class C^2 . Then (i) f convex $\iff \nabla^2 f(x) \succeq 0$ (PSD) **at every point of C** (it is not enough at one point or on part of the domain); (ii) $\nabla^2 f \succ 0$ (PD) at every point $\Rightarrow f$ strictly convex, **but the converse is false** ($f(x) = x^4$ is strictly convex and $f''(0) = 0$). For concavity, NSD and ND respectively. *Minors*: Sylvester's criterion with *leading* principal minors characterizes PD (all > 0) and ND (alternating signs starting with < 0); for PSD or NSD you must use *all* principal minors, not only the leading ones.
- **Operations that preserve convexity.** With f, f_i convex: (1) $\sum_i \alpha_i f_i$ with $\alpha_i \geq 0$ (Theorem 5.29); (2) $x \mapsto f(Ax + b)$, composition with an *affine* map (Theorem 5.31); (3) pointwise supremum $\sup_{i \in I} f_i$ of an *arbitrary* family, with no conditions on I (Theorem 5.33); (4) scalar composition $h \circ f$ (Theorem 5.35) *only* under the exact combinations: h convex **nondecreasing** with f convex, or h convex **nonincreasing** with f concave (for concavity: h concave nondecreasing with f concave, or h concave nonincreasing with f convex), where monotonicity must hold on an interval containing the range of f ; (5) partial minimization $g(x) = \inf_{y \in D} f(x, y)$ requires *joint* convexity in (x, y) and D convex (Theorem 5.37). They do *not* preserve convexity: the pointwise minimum of convex functions, the product of convex functions, composition without monotonicity.
- **Continuity (Theorem 5.39).** A convex function with *finite values* on a convex set C is continuous, and locally Lipschitz, on the *interior* $\text{int } C$. On the boundary it can jump (Example 5.40); this is why one works with *closed* convex functions. It is also differentiable at almost every point of $\text{int dom } f$.
- **Support function (Theorem 5.43).** For *any* $C \neq \emptyset$ (no convexity, no closedness, no boundedness), $\sigma_C(p) = \sup_{x \in C} \langle p, x \rangle$ is *convex* and *positively homogeneous of degree 1* ($\sigma_C(tp) = t\sigma_C(p)$ for $t > 0$, $\sigma_C(0) = 0$), with values in $\mathbb{R} \cup \{+\infty\}$. Moreover $\sigma_C = \sigma_{\text{cl } C} = \sigma_{\text{conv } C}$, and a closed convex set is recovered from its support: $\text{cl conv } C = \{x : \langle p, x \rangle \leq \sigma_C(p) \ \forall p\}$.
- **Micro (Section 5.9.2).** The profit function $\pi(p) = \sup_{y \in Y} \langle p, y \rangle$ is *convex* and homogeneous of degree 1 in prices *with no assumption* on the technology (Example 5.34); the expenditure function $e(p, u) = \inf_{x \in V_u} \langle p, x \rangle$ is *concave* and homogeneous of degree 1 in prices with no assumptions on preferences (Proposition 5.46). From there come Hotelling, Shephard and the negative semidefiniteness of the Slutsky matrix, which is nothing but $\nabla_p^2 e \preceq 0$ through Theorem 5.25. See [9] chs. 3 and 5, [10] chs. 1–3; for the

convex machinery, [2] ch. 3, [6] ch. 2, [3] and [1].

5.11 Solved exercises [P1]

The six exercises below go, in order, through the tools of the chapter: minors and the Hessian, the algebra of operations, risk aversion, the pointwise supremum, the support function and, at the end, the typical exam traps. All solutions are complete: there is no “left to the reader”.

Exercise 5.47 (Hessian with a parameter and minors). Let $\alpha \in \mathbb{R}$ and $f_\alpha : \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined by

$$f_\alpha(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + 2\alpha(x_1x_2 + x_1x_3 + x_2x_3).$$

(a) Compute $\nabla^2 f_\alpha$. (b) Determine exactly for which values of α the function is convex, and for which it is *strictly* convex. (c) For which α is it concave? (d) Explain why the criterion of *leading* principal minors (Sylvester) is not enough to answer (b).

Solution. (a) *The Hessian.* We write $f_\alpha(x) = x^\top M(\alpha)x$ with

$$M(\alpha) = \begin{pmatrix} 1 & \alpha & \alpha \\ \alpha & 1 & \alpha \\ \alpha & \alpha & 1 \end{pmatrix},$$

since $x^\top Mx = \sum_i x_i^2 + 2\alpha \sum_{i < j} x_i x_j$ (each cross product $x_i x_j$ appears twice, in positions (i, j) and (j, i)). By Example 5.26 with $Q = 2M(\alpha)$, $b = 0$, $c = 0$:

$$\nabla f_\alpha(x) = 2M(\alpha)x, \quad \nabla^2 f_\alpha(x) = 2M(\alpha) \quad \text{for every } x,$$

a *constant* Hessian. The domain $\mathbb{R}^3$ is convex and open and f_α is a polynomial (class C^∞), so the hypotheses of Theorem 5.25 hold and we can use it without hesitation.

(b) *The values of α .* The cleanest route here is to diagonalize. Let J be the 3×3 matrix of all ones. Then

$$M(\alpha) = (1 - \alpha)I + \alpha J.$$

The matrix J has rank 1: its image is the line generated by $\mathbf{1} = (1, 1, 1)^\top$ and its kernel is the plane $\{v : \sum_i v_i = 0\}$. Its eigenvalues are therefore 3 (with eigenvector $\mathbf{1}$, since $J\mathbf{1} = 3\mathbf{1}$) and 0 (double). Since adding $(1 - \alpha)I$ shifts all eigenvalues by $1 - \alpha$, the eigenvalues of $M(\alpha)$ are

$$\mu_1 = (1 - \alpha) + 3\alpha = 1 + 2\alpha \quad (\text{eigenvector } \mathbf{1}), \quad \mu_2 = \mu_3 = 1 - \alpha \quad (\text{the plane } \sum_i v_i = 0).$$

A symmetric matrix is PSD iff all its eigenvalues are ≥ 0 , and PD iff all are > 0 . Therefore

$$M(\alpha) \succeq 0 \iff 1 + 2\alpha \geq 0 \text{ and } 1 - \alpha \geq 0 \iff -\frac{1}{2} \leq \alpha \leq 1,$$

$$M(\alpha) \succ 0 \iff -\frac{1}{2} < \alpha < 1.$$

By Theorem 5.25(i), f_α is **convex if and only if** $\alpha \in [-1/2, 1]$, and by (ii) it is strictly convex

when $\alpha \in (-1/2, 1)$. In this case the converse of (ii) also holds, because the function is quadratic (Example 5.26): at the endpoints strictness really fails. Let us check it:

- If $\alpha = 1$: $M = J$ and $f_1(x) = x^\top Jx = (x_1 + x_2 + x_3)^2$. On any line contained in the plane $x_1 + x_2 + x_3 = 0$ the function is identically 0: it is affine (constant) there, so it is convex but *not* strictly.
- If $\alpha = -1/2$: the vector $\mathbf{1}$ satisfies $M\mathbf{1} = (1 + 2\alpha)\mathbf{1} = 0$, and on the line $t \mapsto t\mathbf{1}$ we have $f_{-1/2}(t\mathbf{1}) = t^2\mathbf{1}^\top M\mathbf{1} = 0$: again constant on a line, hence not strictly convex.

An alternative check, without eigenvalues, comes from the identity

$$x^\top M(\alpha)x = (1 - \alpha) \|x\|^2 + \alpha \left(\sum_{i=1}^3 x_i \right)^2,$$

which is verified by expanding the square. For $0 \leq \alpha \leq 1$ both coefficients are ≥ 0 and f_α is a nonnegative sum of two convex functions (Theorem 5.29); for α outside $[-1/2, 1]$ it is enough to evaluate along the direction that breaks everything: if $\alpha > 1$, at $v = (1, -1, 0)$ we get $v^\top Mv = 2 - 2\alpha < 0$; if $\alpha < -1/2$, at $v = \mathbf{1}$ we get $v^\top Mv = 3 + 6\alpha < 0$. In both cases the Hessian is not PSD and f_α is not convex.

(c) *Concavity.* f_α would be concave iff $\nabla^2 f_\alpha \preceq 0$, which requires in particular that the diagonal entries be ≤ 0 (taking $v = e_i$ gives $e_i^\top M e_i = M_{ii} = 1 \leq 0$, false). Since $M_{ii} = 1 > 0$ for every α , *there is no α that makes f_α concave.*

(d) *Why leading Sylvester is not enough.* The leading principal minors of $M(\alpha)$ are

$$\Delta_1 = 1, \quad \Delta_2 = \det \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix} = 1 - \alpha^2,$$

$$\Delta_3 = \det M(\alpha) = 1 - 3\alpha^2 + 2\alpha^3 = (1 - \alpha)^2(1 + 2\alpha),$$

where the factorization is checked by noting that $\alpha = 1$ makes the polynomial vanish ($1 - 3 + 2 = 0$) and dividing twice by $(\alpha - 1)$. Sylvester's criterion says: $M \succ 0$ iff $\Delta_1, \Delta_2, \Delta_3 > 0$, that is, $1 - \alpha^2 > 0$ and $(1 - \alpha)^2(1 + 2\alpha) > 0$, so $-1/2 < \alpha < 1$: this agrees with what we found, *for the strict case*. But the “non strict” version of the leading minor criterion is **false**: requiring $\Delta_1, \Delta_2, \Delta_3 \geq 0$ does not guarantee PSD. The minimal counterexample is

$$N = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}, \quad \Delta_1 = 0, \quad \Delta_2 = \det N = 0,$$

both ≥ 0 , and yet N is not PSD, since $e_2^\top N e_2 = -1 < 0$; what gives the problem away is the *non leading* principal minor obtained by deleting the first row and column, which equals $-1 < 0$. The correct rule is: $M \succeq 0$ iff all principal minors (not only the leading ones) are ≥ 0 . For $M(\alpha)$ those minors are: the ones of order 1, all equal to $1 > 0$; the ones of order 2, all equal to $1 - \alpha^2$, which requires $|\alpha| \leq 1$; and the one of order 3, $(1 - \alpha)^2(1 + 2\alpha) \geq 0$, which within that range requires $\alpha \geq -1/2$. Intersecting: $-1/2 \leq \alpha \leq 1$, the same result the eigenvalues gave. Exam lesson: *leading minors for the strict case (PD/ND); all principal minors for the non strict one (PSD/NSD).* $\square$

Exercise 5.48 (Convexity without Hessians). Prove that the following functions are convex using *only* the operations that preserve convexity, without computing any second derivative. In each case identify the domain, check that it is convex and cite the exact result you use.

- (a) $f(x) = (\max\{0, \langle a, x \rangle - b\})^2$ on $\mathbb{R}^n$, with $a \in \mathbb{R}^n$, $b \in \mathbb{R}$ (quadratic cost for exceeding a threshold).
- (b) $g(x) = \ln \left(\sum_{j=1}^m e^{\langle a_j, x \rangle + b_j} \right)$ on $\mathbb{R}^n$ (the *inclusive value* of the logit model).
- (c) $h(x) = -\ln(b - \langle a, x \rangle)$ on $D = \{x \in \mathbb{R}^n : \langle a, x \rangle < b\}$ (logarithmic barrier of a budget constraint).

Solution. (a) We decompose $f = h \circ u$ with

$$u(x) = \max\{0, \langle a, x \rangle - b\}, \quad h(t) = t^2.$$

Step 1: u is convex. The constant function 0 is affine, and $x \mapsto \langle a, x \rangle - b$ is affine; both are convex (Example 5.5). The pointwise maximum of two convex functions is convex: it is the finite case of Theorem 5.33 with $I = \{1, 2\}$. Therefore u is convex on $\mathbb{R}^n$ (which is convex).

Step 2: the range of u . By construction $u(x) \geq 0$ for every x , so $u(\mathbb{R}^n) \subseteq [0, \infty) =: D_h$, an interval.

Step 3: h is convex and nondecreasing on D_h . On $[0, \infty)$, $h(t) = t^2$ is convex (Example 5.4) and also *nondecreasing*, since for $0 \leq s \leq t$ we have $t^2 - s^2 = (t - s)(t + s) \geq 0$. Here is the fine point: t^2 is *not* nondecreasing on all of $\mathbb{R}$, but Theorem 5.35 only asks for monotonicity on an interval containing the range of the inner function, and that range is $[0, \infty)$.

Step 4: conclusion. By Theorem 5.35(i) (u convex, h convex nondecreasing on an interval containing $u(\mathbb{R}^n)$), $f = h \circ u$ is convex on $\mathbb{R}^n$.

(b) We decompose $g = \text{lse} \circ A$ where $\text{lse}(z) = \ln \left(\sum_{j=1}^m e^{z_j} \right)$ and $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the *affine* map $A(x) = (\langle a_1, x \rangle + b_1, \dots, \langle a_m, x \rangle + b_m)$, which in matrix form is $A(x) = Ax + b$ with rows $a_j^\top$. Indeed,

$$g(x) = \ln \left(\sum_j e^{\langle a_j, x \rangle + b_j} \right) = \text{lse}(Ax + b).$$

The log-sum-exp function is convex on $\mathbb{R}^m$ (Example 5.28), and composition with an affine map preserves convexity (Theorem 5.31). Therefore g is convex on $\mathbb{R}^n$. *Economic note:* in the discrete choice model with utilities $\langle a_j, x \rangle + b_j$ and Gumbel shocks, this g is exactly the expected utility of the maximum (the “inclusive value”), and its gradient gives the logit choice probabilities; its convexity in the parameters is what makes maximum likelihood estimation tractable.

(c) **The domain.** $D = \{x : \langle a, x \rangle < b\}$ is an open half space, and half spaces are convex (Chapter 2); moreover $\langle a, x \rangle < b$ guarantees $b - \langle a, x \rangle > 0$, so the logarithm is well defined. **The decomposition.** Let $\varphi(t) = -\ln t$ on $t > 0$ and $u(x) = b - \langle a, x \rangle$, which is affine (hence convex and concave) with range contained in $(0, \infty)$. There are two routes, both legitimate:

- *Through affine composition:* $\varphi(t) = -\ln t$ is convex on $(0, \infty)$ because $\ln$ is concave (Example 5.7) and changing the sign swaps convexity and concavity (Remark 5.2); composing φ with the affine map $x \mapsto b - \langle a, x \rangle$ preserves convexity (Theorem 5.31). Therefore $h = \varphi \circ u$ is convex.

- *Through scalar composition:* u is concave (being affine) and φ is convex and *nonincreasing* on $(0, \infty)$ (since $\varphi(s) - \varphi(t) = \ln(t/s) \geq 0$ for $s \leq t$); Theorem 5.35(ii) gives the convexity of $\varphi \circ u$ directly.

In both cases h is convex on D . *Note:* this is the logarithmic barrier used by interior point methods for the constraint $\langle a, x \rangle \leq b$ [2]; its convexity is precisely what allows the penalized problem to be solved with descent methods. $\square$

Exercise 5.49 (Certainty equivalent, risk premium and Arrow–Pratt). (a) An agent with CARA utility $u(w) = -e^{-\alpha w}$, $\alpha = 0.1$, faces the lottery X that pays 0 soles with probability 1/2 and 20 soles with probability 1/2. Compute $\mathbb{E}[X]$, $\mathbb{E}[u(X)]$, the certainty equivalent $c(X)$, the risk premium $\rho(X)$ and the coefficients $A(w)$ and $R(w)$. Compare $\rho(X)$ with the approximation $\frac{1}{2}A\sigma^2$. Prove also that if the agent starts from an initial wealth W (the lottery is added to W), the premium does not change.

(b) An agent with CRRA utility $u(w) = w^{1-\gamma}/(1-\gamma)$, $\gamma = 2$, faces the lottery that pays 100 or 25 with probability 1/2 each. Compute $c(X)$ and $\rho(X)$, compare them with the logarithmic agent of Example 5.10 and prove that for CRRA the certainty equivalent is homogeneous of degree 1 in the payoffs.

Solution. (a) *CARA agent.* First the elementary part:

$$\mathbb{E}[X] = \frac{1}{2}(0) + \frac{1}{2}(20) = 10.$$

Expected utility. With $\alpha = 0.1$:

$$\mathbb{E}[u(X)] = \frac{1}{2}(-e^{-0.1 \cdot 0}) + \frac{1}{2}(-e^{-0.1 \cdot 20}) = -\frac{1}{2}(1 + e^{-2}).$$

Numerically, $e^{-2} = 0.135335$, so $\mathbb{E}[u(X)] = -\frac{1}{2}(1.135335) = -0.567668$.

Certainty equivalent. We solve $u(c) = \mathbb{E}[u(X)]$:

$$-e^{-0.1c} = -0.567668 \implies e^{-0.1c} = 0.567668 \implies -0.1c = \ln(0.567668) \implies c = -10 \ln(0.567668).$$

Since $\ln(0.567668) = -0.566221$, we obtain

$$c(X) = 5.6622 \text{ soles (approx.)}.$$

(Note that $c(X) < \mathbb{E}[X] = 10$, as Jensen requires for an averse agent: u is concave because $u''(w) = -\alpha^2 e^{-\alpha w} < 0$, Example 5.12.)

Risk premium.

$$\rho(X) = \mathbb{E}[X] - c(X) = 10 - 5.6622 = 4.3378 \text{ soles}.$$

The agent would pay up to 4.34 soles to turn this lottery into its sure expected value; equivalently, they would sell the lottery for any price above 5.66.

Arrow–Pratt coefficients. With $u'(w) = \alpha e^{-\alpha w}$ and $u''(w) = -\alpha^2 e^{-\alpha w}$:

$$A(w) = -\frac{u''(w)}{u'(w)} = \alpha = 0.1 \quad (\text{constant}), \quad R(w) = w A(w) = 0.1 w.$$

Comparison with the approximation. The variance of a symmetric binary lottery with payoffs 0 and 20 is

$$\sigma^2 = \frac{1}{2}(0 - 10)^2 + \frac{1}{2}(20 - 10)^2 = 100,$$

so $\frac{1}{2}A\sigma^2 = \frac{1}{2}(0.1)(100) = 5$, against the exact premium 4.34. The approximation overestimates by about 15 percent, and this should not surprise you: the Arrow–Pratt formula is a Taylor expansion valid for *small* risks, and here the standard deviation ($\sigma = 10$) is of the same order as the wealth at stake. With the same α but payoffs ± 1 around the mean, the approximation would be excellent.

Invariance to initial wealth. Let W be the initial wealth and consider the lottery $W + X$. Its certainty equivalent c_W solves

$$-e^{-\alpha c_W} = \mathbb{E}[-e^{-\alpha(W+X)}] = -e^{-\alpha W} \mathbb{E}[e^{-\alpha X}],$$

where we pulled $e^{-\alpha W}$ out of the expectation because it is constant. Taking logarithms in the equality $e^{-\alpha c_W} = e^{-\alpha W} \mathbb{E}[e^{-\alpha X}]$:

$$-\alpha c_W = -\alpha W + \ln \mathbb{E}[e^{-\alpha X}] \implies c_W = W - \frac{1}{\alpha} \ln \mathbb{E}[e^{-\alpha X}] = W + c(X),$$

since $c(X) = -\frac{1}{\alpha} \ln \mathbb{E}[e^{-\alpha X}]$ is exactly the computation above with $W = 0$. Then

$$\rho(W + X) = \mathbb{E}[W + X] - c_W = (W + \mathbb{E}[X]) - (W + c(X)) = \rho(X) :$$

the premium *does not depend on wealth*. This is exactly what “constant absolute risk aversion” means, and it is the reason CARA is the standard utility in contract models, where one does not want wealth effects to contaminate incentives.

(b) *CRRA agent with $\gamma = 2$.* Here $u(w) = w^{-1}/(-1) = -1/w$ for $w > 0$. We compute:

$$\mathbb{E}[u(X)] = \frac{1}{2} \left(-\frac{1}{100} \right) + \frac{1}{2} \left(-\frac{1}{25} \right) = -\frac{1}{2}(0.01 + 0.04) = -0.025.$$

The certainty equivalent solves $-1/c = -0.025$, that is,

$$c(X) = \frac{1}{0.025} = 40.$$

(Note that $c(X)$ is the *harmonic mean* of the payoffs: $c = [\frac{1}{2} \cdot \frac{1}{100} + \frac{1}{2} \cdot \frac{1}{25}]^{-1} = 40$.) With $\mathbb{E}[X] = \frac{1}{2}(100) + \frac{1}{2}(25) = 62.5$:

$$\rho(X) = 62.5 - 40 = 22.5.$$

Comparison. Ordering by relative aversion γ :

Agent	γ	$c(X)$	$\rho(X)$
neutral ($u(w) = w$)	0	62.5	0
logarithmic (Example 5.10)	1	50	12.5
CRRA	2	40	22.5

The certainty equivalent goes from the arithmetic mean ($\gamma = 0$) to the geometric mean ($\gamma = 1$) and to the harmonic mean ($\gamma = 2$), which is the classical chain of inequalities between means: the larger γ , the smaller c and the larger the premium. The Arrow–Pratt coefficients of the agent with $\gamma = 2$ are $A(w) = -u''/u' = 2/w$ and $R(w) = 2$ (Example 5.12).

Homogeneity of the certainty equivalent. Let $t > 0$ and consider the rescaled lottery tX . Since

$$u(tw) = \frac{(tw)^{1-\gamma}}{1-\gamma} = t^{1-\gamma} u(w),$$

taking expectations gives $\mathbb{E}[u(tX)] = t^{1-\gamma} \mathbb{E}[u(X)]$. If c is the certainty equivalent of X , then

$$u(tc) = t^{1-\gamma} u(c) = t^{1-\gamma} \mathbb{E}[u(X)] = \mathbb{E}[u(tX)],$$

and since u is strictly increasing, the certainty equivalent is unique: $c(tX) = tc(X)$. Therefore $\rho(tX) = t\rho(X)$, and the premium as a *fraction* of the wealth at stake is the same. Numerical check: doubling the payoffs to 200 and 50, $\mathbb{E}[u] = -\frac{1}{2}(1/200 + 1/50) = -0.0125$ and $c = 80 = 2 \times 40$, indeed. That is “constant relative risk aversion”: the agent evaluates *proportional* risks in the same way, rich or poor, which makes CRRA the natural specification in growth and portfolio models. $\square$

Exercise 5.50 (The pointwise maximum, from the definition, and profit). (a) Let $f_1, \dots, f_m : C \rightarrow \mathbb{R}$ be convex on a convex set $C \subseteq \mathbb{R}^n$. Prove *from the definition* (without using epigraphs) that $g(x) = \max_{1 \leq j \leq m} f_j(x)$ is convex. (b) Explain what has to change if the family is infinite and $g = \sup_j f_j$. (c) A firm has three production plans available: produce nothing (profit 0), plan A (produces 3 units at a total cost of 6) and plan B (produces 5 units at a total cost of 14). Write the profit function $\pi(p)$, describe its explicit piecewise form, check its convexity and deduce the optimal supply and its monotonicity.

Solution. (a) *Direct proof.* Let $x, y \in C$ and $\lambda \in [0, 1]$, and write $x_\lambda = (1 - \lambda)x + \lambda y$, which belongs to C since C is convex. Because the maximum is over a *finite* set of indices, it is attained: there exists $k \in \{1, \dots, m\}$ such that

$$g(x_\lambda) = f_k(x_\lambda).$$

Now we chain two steps. First, the convexity of that particular f_k :

$$f_k(x_\lambda) \leq (1 - \lambda)f_k(x) + \lambda f_k(y).$$

Second, the definition of g as a maximum implies that g dominates every member of the family at every point; in particular $f_k(x) \leq g(x)$ and $f_k(y) \leq g(y)$. Since the weights $1 - \lambda$ and

λ are *nonnegative*, we can substitute preserving the direction of the inequality (this is where nonnegativity is used: multiplying an inequality by a negative number would reverse it):

$$(1 - \lambda)f_k(x) + \lambda f_k(y) \leq (1 - \lambda)g(x) + \lambda g(y).$$

Putting the three relations together:

$$g((1 - \lambda)x + \lambda y) = f_k(x_\lambda) \leq (1 - \lambda)f_k(x) + \lambda f_k(y) \leq (1 - \lambda)g(x) + \lambda g(y),$$

which is exactly the convexity inequality for g . Note that the index k attaining the maximum *depends on* x_λ , and that this causes no trouble at all: the proof only needs that *some* index exists.

(b) *Infinite family.* With $g = \sup_{j \in I} f_j$ and I arbitrary, the supremum may not be attained, so the step “there exists k with $g(x_\lambda) = f_k(x_\lambda)$ ” is no longer valid. The repair is standard: for each $j \in I$,

$$f_j(x_\lambda) \leq (1 - \lambda)f_j(x) + \lambda f_j(y) \leq (1 - \lambda)g(x) + \lambda g(y),$$

that is, the number $(1 - \lambda)g(x) + \lambda g(y)$ is an *upper bound* of the set $\{f_j(x_\lambda) : j \in I\}$. By the definition of supremum (the least of the upper bounds),

$$g(x_\lambda) = \sup_{j \in I} f_j(x_\lambda) \leq (1 - \lambda)g(x) + \lambda g(y).$$

This is the argument of Theorem 5.33; one must also allow the value $+\infty$ (working with extended functions, Section 5.4) because the supremum can be infinite at some points.

(c) *The profit function.* Each plan gives a profit that is *affine* in the output price p :

$$\pi_0(p) = 0, \quad \pi_A(p) = 3p - 6, \quad \pi_B(p) = 5p - 14.$$

The firm picks the best plan at each price, so

$$\pi(p) = \max\{0, 3p - 6, 5p - 14\},$$

a maximum of three affine functions: convex by part (a), and also by Example 5.34, with $Y = \{0, 3, 5\}$ and costs $c(0) = 0$, $c(3) = 6$, $c(5) = 14$. **Piecewise form.** We look for the crossing points:

$$0 = 3p - 6 \iff p = 2, \quad 3p - 6 = 5p - 14 \iff 2p = 8 \iff p = 4.$$

Comparing in each region (for instance, evaluating at $p = 1$: the values are 0, -3 , -9 ; at $p = 3$: 0, 3, 1; at $p = 5$: 0, 9, 11):

$$\pi(p) = \begin{cases} 0 & \text{if } p \leq 2, \\ 3p - 6 & \text{if } 2 \leq p \leq 4, \\ 5p - 14 & \text{if } p \geq 4. \end{cases}$$

Convexity, checked by hand. The function is continuous (at $p = 2$ both pieces equal 0; at

$p = 4$ both equal 6) and piecewise linear with slopes $0 < 3 < 5$, *increasing* from left to right: a piecewise linear function is convex if and only if its slopes are nondecreasing, which is the discrete version of “the derivative increases” (Theorem 5.22 and Remark 5.24).

Supply and the law of supply. Where π is differentiable, Hotelling’s lemma says that $\pi'(p)$ is the quantity supplied:

$$y(p) = \pi'(p) = \begin{cases} 0 & \text{if } p < 2, \\ 3 & \text{if } 2 < p < 4, \\ 5 & \text{if } p > 4, \end{cases}$$

which is *nondecreasing* in p : the law of supply is not a behavioral assumption, it is the monotonicity of the derivative of a convex function. At the kink prices $p = 2$ and $p = 4$ the firm is *indifferent* between two plans and π is not differentiable; there the supply is multivalued ($\{0, 3\}$ and $\{3, 5\}$ respectively, and their convex combinations if mixing is allowed), and the correct object is the subdifferential $\partial\pi(p) = [0, 3]$ and $[3, 5]$ (Chapter 7). $\square$

Exercise 5.51 (Support functions of the simplex and a box). Compute step by step, and check in each case convexity and homogeneity of degree 1:

- (a) $\sigma_\Delta(p)$ for the simplex $\Delta = \{x \in \mathbb{R}^n : x_i \geq 0 \ \forall i, \ \sum_{i=1}^n x_i = 1\}$.
- (b) $\sigma_Q(p)$ for the box $Q = \prod_{i=1}^n [a_i, b_i]$ with $a_i \leq b_i$.

Interpret (a) when Δ is the set of lotteries over n scenarios and (b) when Q describes feasible ranges of activities.

Solution. (a) *The simplex.* We claim that

$$\sigma_\Delta(p) = \max_{1 \leq i \leq n} p_i.$$

Upper bound ($\leq$). Let $x \in \Delta$ and write $p_{\max} = \max_i p_i$. Since $x_i \geq 0$ for every i , each term satisfies $p_i x_i \leq p_{\max} x_i$ (we multiply the inequality $p_i \leq p_{\max}$ by the *nonnegative* number x_i). Adding up and using $\sum_i x_i = 1$:

$$\langle p, x \rangle = \sum_{i=1}^n p_i x_i \leq p_{\max} \sum_{i=1}^n x_i = p_{\max} \cdot 1 = p_{\max}.$$

Since this holds for every $x \in \Delta$, the supremum satisfies $\sigma_\Delta(p) \leq p_{\max}$.

Lower bound ($\geq$) and attainment. Let i^* be an index where the maximum is attained, $p_{i^*} = p_{\max}$, and take the vertex $x = e_{i^*}$ (the vector with a 1 in position i^* and zeros elsewhere). Clearly $e_{i^*} \in \Delta$ (nonnegative coordinates adding up to 1), and

$$\langle p, e_{i^*} \rangle = p_{i^*} = p_{\max}.$$

Therefore $\sigma_\Delta(p) \geq p_{\max}$. Both inequalities give equality, and moreover the supremum is *attained* (at a vertex).

Checking the properties. $\sigma_\Delta(p) = \max_i p_i$ is a finite maximum of linear functions $p \mapsto$

$\langle e_i, p \rangle$: convex by Exercise 5.50(a), as Theorem 5.43(i) guarantees. And for $t > 0$, $\max_i(tp_i) = t \max_i p_i$, the homogeneity of degree 1 of Theorem 5.43(ii). It is also never $+\infty$, because Δ is compact.

Interpretation. If Δ is the set of lotteries over n scenarios and p_i is the payoff in scenario i , then $\langle p, x \rangle$ is the expected payoff under the lottery x , and $\sigma_\Delta(p) = \max_i p_i$ says that the best expected payoff is obtained by putting all the probability on the best scenario: *optimizing over a polytope leads to a vertex*, the principle behind the simplex method (Chapter 13). The same with the sign changed: $\min_{x \in \Delta} \langle p, x \rangle = \min_i p_i$ is the *maximin* criterion of robust decisions, and its concavity in p is Proposition 5.46 in miniature. In production: if $Y = \text{conv}\{y^1, \dots, y^n\}$ is the convex hull of finitely many plans, $\pi(p) = \sigma_Y(p) = \max_j \langle p, y^j \rangle$, which is exactly Exercise 5.50(c).

(b) *The box.* We claim that

$$\sigma_Q(p) = \sum_{i=1}^n \max\{a_i p_i, b_i p_i\}.$$

Separability. The key is that both the objective and the constraint split coordinate by coordinate:

$$\sup_{x \in Q} \langle p, x \rangle = \sup_{\substack{x_i \in [a_i, b_i] \\ i=1, \dots, n}} \sum_{i=1}^n p_i x_i = \sum_{i=1}^n \sup_{x_i \in [a_i, b_i]} p_i x_i,$$

because each summand depends on a single variable and the variables are chosen *independently* (maximizing a sum of terms with no crossed constraints is the same as maximizing each term separately).

The one dimensional problem. The function $x_i \mapsto p_i x_i$ is linear on a closed interval, so its maximum is attained at an endpoint:

$$\sup_{x_i \in [a_i, b_i]} p_i x_i = \begin{cases} b_i p_i & \text{if } p_i \geq 0 \quad (\text{take } x_i = b_i), \\ a_i p_i & \text{if } p_i < 0 \quad (\text{take } x_i = a_i), \end{cases} = \max\{a_i p_i, b_i p_i\}.$$

Adding over i gives the formula. **Alternative form (center and radius).** Writing $m_i = (a_i + b_i)/2$ and $r_i = (b_i - a_i)/2 \geq 0$,

$$\max\{a_i p_i, b_i p_i\} = m_i p_i + r_i |p_i|,$$

which is checked by cases: if $p_i \geq 0$, $m_i p_i + r_i p_i = (m_i + r_i) p_i = b_i p_i$; if $p_i < 0$, $m_i p_i - r_i p_i = (m_i - r_i) p_i = a_i p_i$. Therefore

$$\sigma_Q(p) = \langle p, m \rangle + \sum_{i=1}^n r_i |p_i|.$$

In the symmetric case $a_i = -1$, $b_i = 1$ we get $m = 0$, $r_i = 1$ and $\sigma_Q(p) = \sum_i |p_i| = \|p\|_1$, recovering Example 5.45(b).

Verification. σ_Q is convex: it is a sum with nonnegative coefficients of the linear function $\langle p, m \rangle$ and of the functions $|p_i|$, which are convex (Example 5.6) and composed with the linear map $p \mapsto p_i$ (Theorems 5.29 and 5.31). And it is homogeneous of degree 1, because both $\langle \cdot, m \rangle$

and $|\cdot|$ are. *Numerical check:* with $n = 2$, $Q = [0, 2] \times [1, 3]$ and $p = (1, -2)$, the direct formula gives $\max\{0, 2\} + \max\{-2, -6\} = 2 - 2 = 0$; the center-radius formula, with $m = (1, 2)$ and $r = (1, 1)$, gives $(1)(1) + (-2)(2) + 1 \cdot |1| + 1 \cdot |-2| = -3 + 3 = 0$. They agree.

Interpretation. If x_i is the level of activity i , confined to a technical range $[a_i, b_i]$, and p_i is its unit margin, maximum profit is obtained by pushing every profitable activity to its ceiling b_i and every activity with a negative margin to its floor a_i : a “corner” solution (*bang-bang*). The non differentiability of σ_Q at the points where some $p_i = 0$ reflects exactly the indifference price at which the firm can jump from one endpoint to the other. $\square$

Exercise 5.52 (Three classic traps). (a) Let $f(x, y) = x^3 + y^3$ on $\mathbb{R}^2$. Compute its Hessian, determine on which sets it is PSD and decide whether f is convex on $\mathbb{R}^2$, exhibiting an explicit numerical counterexample. (b) Prove that the sum of two concave functions is concave, and show with an example that the *product* of two positive concave functions may fail to be concave. (c) State the general lesson about “checking the Hessian at a point”.

Solution. (a) *PSD only on part of the domain.* The partial derivatives are

$$f_x = 3x^2, \quad f_y = 3y^2, \quad f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = f_{yx} = 0,$$

so that

$$\nabla^2 f(x, y) = \begin{pmatrix} 6x & 0 \\ 0 & 6y \end{pmatrix}.$$

Being diagonal, its eigenvalues are $6x$ and $6y$; therefore

$$\nabla^2 f(x, y) \succeq 0 \iff x \geq 0 \text{ and } y \geq 0.$$

The set where the Hessian is PSD is the closed orthant $\mathbb{R}_+^2$, which *is* convex, so Theorem 5.25 allows the conclusion: **f is convex on $\mathbb{R}_+^2$** (applying it on the interior $\mathbb{R}_{++}^2$ and extending by continuity to the boundary). But it is **not** convex on $\mathbb{R}^2$: there the Hessian is not PSD at any point with a negative coordinate.

Explicit counterexample. It is enough to work on the line $y = 0$, where $f(x, 0) = x^3$. Take $u = (-1, 0)$ and $v = (0, 0)$ with $\lambda = 1/2$, whose midpoint is $w = (-1/2, 0)$:

$$f(w) = \left(-\frac{1}{2}\right)^3 = -\frac{1}{8} = -0.125, \quad \frac{1}{2}f(u) + \frac{1}{2}f(v) = \frac{1}{2}(-1) + \frac{1}{2}(0) = -\frac{1}{2} = -0.5.$$

The convexity inequality would require $-0.125 \leq -0.5$, which is **false**. Therefore f is not convex on $\mathbb{R}^2$ (not even restricted to the whole line $y = 0$). Note that it *is* concave on $\mathbb{R}_-^2$ (where $\nabla^2 f \preceq 0$): the same function is convex on one piece and concave on another, and it is neither of the two globally.

(b) *Sum yes, product no.* **The sum.** Let $f, g : C \rightarrow \mathbb{R}$ be concave, C convex. For $x, y \in C$ and $\lambda \in [0, 1]$,

$$f((1 - \lambda)x + \lambda y) \geq (1 - \lambda)f(x) + \lambda f(y), \quad g((1 - \lambda)x + \lambda y) \geq (1 - \lambda)g(x) + \lambda g(y).$$

Adding the two inequalities term by term (legitimate: if $A \geq B$ and $C \geq D$, then $A + C \geq B + D$)

and grouping:

$$(f + g)((1 - \lambda)x + \lambda y) \geq (1 - \lambda)(f + g)(x) + \lambda(f + g)(y),$$

which is the concavity of $f + g$. It is Theorem 5.29 applied to $-f$ and $-g$ with coefficients $1, 1 \geq 0$. (The hypothesis of *nonnegative* coefficients is essential: $f - g$ with g concave need not be concave.)

The product. Minimal counterexample: on $C = (0, \infty)$ take $f(x) = g(x) = x$. Both are affine, hence concave (Example 5.5), and positive on C . Their product is

$$(fg)(x) = x^2,$$

which is *strictly convex* (Example 5.4) and therefore the opposite of concave. If you want a counterexample with *strictly* concave factors, take $f(x) = g(x) = \ln x$ on $x > 1$ (where $\ln x > 0$): the product is $\varphi(x) = (\ln x)^2$, and

$$\varphi'(x) = \frac{2 \ln x}{x}, \quad \varphi''(x) = \frac{\frac{2}{x} \cdot x - 2 \ln x}{x^2} = \frac{2(1 - \ln x)}{x^2},$$

which is *positive* for $1 < x < e$: φ is convex on $(1, e)$ even though it is the product of two strictly concave and positive functions there. *Economic lesson*: if u and v are concave utilities, $u + v$ is concave (this is why utilitarian welfare functions inherit concavity), but a multiplicative utility of the type $u(x)v(y)$ does *not* inherit concavity: the Cobb–Douglas $x^a y^b$ is the paradigmatic case, since with $0 < a, b \leq 1$ it is a product of two concave functions that is concave only if $a + b \leq 1$ (Example 5.27). What is always preserved when multiplying two *positive concave* functions is *quasiconcavity*: the logarithm of the product is the sum of two concave functions, hence concave, and the upper contour sets of the product coincide with those of that logarithm. This is the topic of Chapter 6.

(c) *The lesson.* Convexity is a *global property on a convex set*, not a pointwise property. The three rules to internalize:

- (i) *One point says nothing.* That $\nabla^2 f(x_0) \succeq 0$ at a point x_0 does not imply convexity on any neighborhood with economic content; Theorem 5.25 requires PSD at *all* points of the open convex set where convexity is claimed. Example (a) shows this: $\nabla^2 f(1, 1) \succ 0$ and yet f is not convex on $\mathbb{R}^2$.
- (ii) *The domain must be convex.* If the set where the Hessian is PSD is not convex, nothing global can be concluded; you must restrict to a convex subset of it (in (a), $\mathbb{R}_+^2$ works).
- (iii) *The operations have hypotheses.* Adding with weights ≥ 0 , taking suprema and composing with affine maps is always safe; composing without checking monotonicity, multiplying functions or taking pointwise minima is not. When in doubt, go back to the definition or to the epigraph.

□

Exam strategy**Protocol for “is this function convex?” in an exam.**

1. *Look at the domain first.* If it is not convex, the question is badly posed or you must restrict (Exercise 5.52).
2. *Try the algebra of operations before derivatives.* Sums with weights ≥ 0 , affine composition, pointwise supremum, monotone composition: if the function is built from those pieces, you are done in three lines and with no computational errors (Exercise 5.48).
3. *If it is a supremum or an infimum of affine functions, you are done.* Supremum of affine functions $\Rightarrow$ convex; infimum of affine functions $\Rightarrow$ concave. Every value function of a linear optimization problem in the parameter falls here: profit, expenditure, cost, inclusive value (Exercises 5.50 and 5.51).
4. *Only then, the Hessian.* Check that f is C^2 , compute $\nabla^2 f$ and use *leading* minors for PD/ND and *all* principal minors for PSD/NSD (Exercise 5.47). If there are parameters, solve the inequalities in them.
5. *To deny convexity, a numerical counterexample.* Two points and $\lambda = 1/2$ are usually enough; or a line along which the restriction is not convex. It is faster and more convincing than arguing with Hessians.
6. *Strict convexity:* PD at every point guarantees it, but its absence does not deny it (x^4). To deny it, look for a segment on which f is affine.

This closes the study of “full” convexity. The next chapter (Chapter 6) relaxes the requirement: we will see that for most of demand theory the utility does not need to be concave (it is enough that its upper contour sets be convex, that is, *quasiconcavity*), the property that recovers the failed converse of Proposition 5.16 and that is, unlike concavity, *ordinal*.

Chapter 6

Generalized concavity: quasiconcave and quasiconvex functions [P1]

Week 4 of the course. Bibliography: Aragón et al. [1]; Brinkhuis [3].

What is this used for in Microeconomics?

No assumption on preferences is used as often in Microeconomics as quasiconcavity, and this is the chapter that builds it. The reason is a deep conceptual fact: in consumer theory utility is **ordinal**, not cardinal. The function u only encodes an *order* of preferences, so u and $v = g \circ u$ (with g strictly increasing) represent exactly the same preferences and must be economically indistinguishable. But concavity does *not* survive increasing monotone transformations (we will check this with $u(x, y) = xy$ and $v = \ln u$), while **quasiconcavity** does: it is the notion of “curvature” that is *ordinally invariant*, the only one it makes sense to require of a utility function. In **individual decision making** (consumer theory, MWG chs. 1–3, [9]), convex preferences are exactly quasiconcave utility, and *strict quasiconcavity* is what guarantees that the consumer problem has a *unique* solution, that is, that there is a well defined demand *function* $x(p, m)$ and not just a correspondence. Without it, comparative statics, the Slutsky equation and the whole demand apparatus become much more delicate. The **bordered Hessian** is the working tool used to check quasiconcavity in exams: in this chapter we build it carefully, we separate the necessary condition from the sufficient one (a point where many texts are imprecise) and we compute it step by step in 2×2 and 3×3 examples. Finally, the **Spence–Mirrlees** or *single crossing* condition, a direct relative of this family of ordinal properties, is the engine of the Spence signaling models and of the price discrimination and screening models of Mussa–Rosen and Maskin–Riley (**market failures**, MWG chs. 13–14): it orders the indifference curves of the different agent types and makes the contracts that separate types work. Everything you do in contracts and mechanisms will use, sometimes without saying so, the concepts of this chapter.

The previous chapters developed the theory of concave functions: the Jensen inequality, the first and second order characterizations, the support function. Concavity is a *cardinal* property: it depends on the numerical *values* the function takes, not only on the order it induces between

points. In economics, however, utility is ordinal, and we need a notion of “preference for mixing” that depends only on the *level curves* of the function and not on the numerical labels we attach to them. That notion is quasiconcavity, and this chapter develops it from scratch: equivalent definitions, the hierarchy with concavity, differential criteria (gradient and bordered Hessian), the intermediate class of pseudoconcave functions, and the economic applications that justify the whole effort. The main references are [1] (chapter on generalized convexity), [9] (chs. 1–3 and appendix M), [10] (chs. 1–2) and [3]. From this notebook we will use: convex sets and intersections (ch. 2), and the first and second order characterizations of concavity.

6.1 Utility is ordinal: why concavity is not enough [P1]

6.1.1 Preferences, utility and monotone transformations

Let us recall the starting point of consumer theory. A consumer has a preference relation $\succsim$ on a set of bundles $X \subseteq \mathbb{R}_+^n$: the expression $x \succsim y$ reads “ x is at least as good as y ”. A function $u: X \rightarrow \mathbb{R}$ represents $\succsim$ if

$$x \succsim y \iff u(x) \geq u(y) \quad \text{for all } x, y \in X.$$

All that u does is *order* bundles. The specific numbers mean nothing: saying that $u(x) = 10$ and $u(y) = 5$ does *not* say that x “gives twice as much happiness” as y ; it only says that $x \succ y$. This is why utility is called *ordinal* and not *cardinal*.

This observation has an immediate consequence. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a *strictly increasing* function: $s > t$ implies $g(s) > g(t)$. Then the composition $v = g \circ u$, that is $v(x) = g(u(x))$, satisfies

$$v(x) \geq v(y) \iff g(u(x)) \geq g(u(y)) \iff u(x) \geq u(y) \iff x \succsim y,$$

where the second equivalence uses that g is strictly increasing (if $u(x) \geq u(y)$ then $g(u(x)) \geq g(u(y))$; and if $u(x) < u(y)$ then $g(u(x)) < g(u(y))$, so $g(u(x)) \geq g(u(y))$ forces $u(x) \geq u(y)$). Conclusion: u and $v = g \circ u$ represent exactly the same preferences. We say that v is a *monotone increasing transformation* of u .

Intuition

Think of u as a thermometer that instead of temperature measures “how high” a bundle sits in the consumer ranking. Moving from u to $g \circ u$ is like moving from Celsius to Fahrenheit, or to any odd scale, even a nonlinear one: the numbers change, but the order “it is hotter here than there” is the same. Any property of u that claims to have *economic meaning* must be invariant under these changes of scale. The properties that survive are called *ordinal*; those that do not are *cardinal*. The main lesson of this chapter: concavity is cardinal, quasiconcavity is ordinal.

6.1.2 A revealing example: $u(x, y) = xy$

Let us see in full detail that concavity can appear or disappear when we apply a monotone transformation, even though preferences do not change.

Example 6.1 ($u(x, y) = xy$ is not concave, but $\ln u$ is). Consider $u(x, y) = xy$ on the open quadrant $\mathbb{R}_{++}^2 = \{(x, y) : x > 0, y > 0\}$, a convex and open set.

Step 1: u is not concave. Since u is of class C^2 , we can use the second order characterization: u is concave on a convex open set if and only if its Hessian matrix is negative semidefinite at every point. We compute the partial derivatives:

$$\frac{\partial u}{\partial x} = y, \quad \frac{\partial u}{\partial y} = x, \quad \frac{\partial^2 u}{\partial x^2} = 0, \quad \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial^2 u}{\partial x \partial y} = 1.$$

The Hessian is therefore constant:

$$H_u(x, y) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For H_u to be negative semidefinite we need, in particular, its determinant (the product of the eigenvalues in the 2×2 case) to be ≥ 0 . But $\det H_u = 0 \cdot 0 - 1 \cdot 1 = -1 < 0$: the eigenvalues have opposite signs (they are $+1$ and -1 , with eigenvectors $(1, 1)$ and $(1, -1)$), so H_u is *indefinite* and u is not concave (nor convex) on $\mathbb{R}_{++}^2$. Concretely, along the diagonal $x = y = t$ we have $u = t^2$, which is *convex* in t ; along the antidiagonal the restriction is concave. The surface $z = xy$ is a saddle.

Step 2: $v = \ln u$ is concave. Apply the strictly increasing transformation $g(t) = \ln t$ (defined on the values $u > 0$ that u takes on $\mathbb{R}_{++}^2$):

$$v(x, y) = \ln(xy) = \ln x + \ln y.$$

Its derivatives are $v_x = 1/x$, $v_y = 1/y$, and the Hessian is

$$H_v(x, y) = \begin{pmatrix} -1/x^2 & 0 \\ 0 & -1/y^2 \end{pmatrix},$$

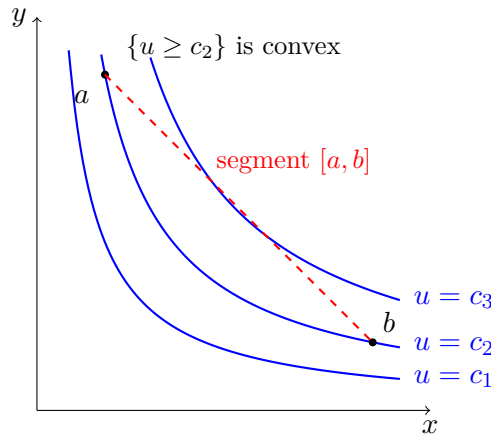
which is diagonal with strictly negative entries: negative definite at every point of $\mathbb{R}_{++}^2$. By the second order characterization, v is (strictly) concave.

Conclusion. The functions $u = xy$ and $v = \ln x + \ln y$ represent *the same preferences* (one is a monotone increasing transformation of the other), they have *the same indifference curves* (the hyperbolas $xy = c$), but one is concave and the other is not. Concavity is therefore not a property of preferences: it is an artifact of the scale we chose. $\square$

This example raises the central question of the chapter: what geometric property do $u = xy$ and $v = \ln x + \ln y$ share, and more generally all representations of the same “well behaved” preferences? The answer lies in the *level curves*. Both functions have the same sets

$$\{(x, y) \in \mathbb{R}_{++}^2 : u(x, y) \geq c\} = \{(x, y) : xy \geq c\},$$

and these sets, the region “above” each hyperbola, are *convex*. That is exactly the definition of quasiconcavity.



The figure shows indifference curves that bend toward the origin (the hyperbolas $xy = c$ with $c_1 < c_2 < c_3$). The points a and b lie on the curve $u = c_2$; the whole segment $[a, b]$ stays *above* that curve, that is, inside the set $\{u \geq c_2\}$. That this happens for *any* pair of points and *any* level is quasiconcavity.

6.2 Quasiconcavity: definitions and equivalence [P1]

Throughout the chapter, $S \subseteq \mathbb{R}^n$ denotes a nonempty *convex* set (the domain of our functions). Recall from ch. 2 that this means: if $x, y \in S$ and $\lambda \in [0, 1]$, then $\lambda x + (1 - \lambda)y \in S$.

Definition 6.2 (Upper and lower contour sets). Let $f: S \rightarrow \mathbb{R}$ and $\alpha \in \mathbb{R}$. The *upper contour set* (also called upper level set) at level α is

$$S_\alpha(f) = \{x \in S : f(x) \geq \alpha\},$$

and the *lower contour set* is $I_\alpha(f) = \{x \in S : f(x) \leq \alpha\}$.

In consumer theory, if u represents $\succsim$, the set $S_{u(x^0)}(u) = \{x : u(x) \geq u(x^0)\}$ is exactly the set of bundles *at least as good as* x^0 : the upper contour set of preferences at x^0 .

Definition 6.3 (Quasiconcave function). A function $f: S \rightarrow \mathbb{R}$ on a convex set $S \subseteq \mathbb{R}^n$ is *quasiconcave* if its upper contour set $S_\alpha(f)$ is convex for *every* $\alpha \in \mathbb{R}$.

Definition 6.4 (Definition by inequality). A function $f: S \rightarrow \mathbb{R}$ satisfies the *quasiconcavity inequality* if for all $x, y \in S$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}.$$

Before proving that the two definitions agree, let us compare with concavity. A concave function satisfies $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$: the value at the intermediate point beats the *weighted average* of the extreme values. Quasiconcavity asks for something much weaker: to beat the *worse* of the two values. Since the weighted average always lies between the

minimum and the maximum, the quasiconcave requirement is weaker, and this anticipates that every concave function will be quasiconcave.

Theorem 6.5 (Equivalence of the two definitions). *Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$. The following are equivalent:*

- (i) $S_\alpha(f)$ is convex for every $\alpha \in \mathbb{R}$;
- (ii) $f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$ for all $x, y \in S$ and $\lambda \in [0, 1]$.

Proof. (ii) $\Rightarrow$ (i). Fix an arbitrary $\alpha \in \mathbb{R}$ and take any two points $x, y \in S_\alpha(f)$; we want to see that the whole segment $[x, y]$ stays inside $S_\alpha(f)$. By definition of $S_\alpha(f)$ we have $f(x) \geq \alpha$ and $f(y) \geq \alpha$, and therefore $\min\{f(x), f(y)\} \geq \alpha$ (the minimum of two numbers that are $\geq \alpha$ is $\geq \alpha$). Let $\lambda \in [0, 1]$. Since S is convex, the point $z = \lambda x + (1 - \lambda)y$ belongs to S , and applying (ii):

$$f(z) \geq \min\{f(x), f(y)\} \geq \alpha,$$

that is, $z \in S_\alpha(f)$. Since x, y, λ were arbitrary, $S_\alpha(f)$ is convex; and since α was arbitrary, all upper contour sets are convex.

(i) $\Rightarrow$ (ii). Take arbitrary $x, y \in S$ and $\lambda \in [0, 1]$. The key idea is to *choose the level α well*: we set

$$\alpha^* = \min\{f(x), f(y)\}.$$

With this choice, *both* points belong to $S_{\alpha^*}(f)$: indeed, $f(x) \geq \alpha^*$ (since α^* is the minimum of $f(x)$ and another number) and likewise $f(y) \geq \alpha^*$. By (i), $S_{\alpha^*}(f)$ is convex, so the point $z = \lambda x + (1 - \lambda)y$ also belongs to $S_{\alpha^*}(f)$. But belonging to $S_{\alpha^*}(f)$ means exactly

$$f(z) \geq \alpha^* = \min\{f(x), f(y)\},$$

which is inequality (ii). □

Intuition

The definition by level sets is the *economic* one: it says that the sets “at least as good as” are convex, that is, that indifference curves bend toward the origin and the consumer (weakly) prefers mixtures. The definition by inequality is the *working* one: it says that when you walk in a straight line from x to y , the value of f never falls below the worse of the two endpoints: there are no “valleys” along the way. The trick in the proof of (i) $\Rightarrow$ (ii) is worth memorizing: when you want to use convexity of level sets to compare two points, evaluate the level at $\alpha = \min\{f(x), f(y)\}$, because that is the highest level at which both points belong at the same time.

6.2.1 Quasiconvexity, quasilinearity and the strict version

Definition 6.6 (Quasiconvex function). $f: S \rightarrow \mathbb{R}$ is *quasiconvex* if $-f$ is quasiconcave; equivalently, if its *lower* contour sets $I_\alpha(f) = \{x \in S : f(x) \leq \alpha\}$ are convex for every α , or if

$$f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\} \quad \text{for all } x, y \in S, \lambda \in [0, 1].$$

The equivalence of the three formulations follows by applying Theorem 6.5 to $-f$ and using that $\{x : -f(x) \geq -\alpha\} = \{x : f(x) \leq \alpha\}$ and that $\min\{-a, -b\} = -\max\{a, b\}$ (check it with two numbers). In economics, *cost* or *expenditure* functions seen as functions of quantities are usually handled with quasiconvexity; and the *indirect* utility function $v(p, m)$ is quasiconvex in p (a classical result of consumer theory, MWG ch. 3, which you can now prove).

Definition 6.7 (Quasilinear function). $f: S \rightarrow \mathbb{R}$ is *quasilinear* (also called quasimonotone) if it is both quasiconcave and quasiconvex; equivalently, if *all* its upper and lower contour sets are convex.

Example 6.8 (Every monotone function on $\mathbb{R}$ is quasilinear). Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be nondecreasing (the nonincreasing case is analogous). Then $S_\alpha(f) = \{x : f(x) \geq \alpha\}$ is an interval of the form $[c, \infty)$ or (c, ∞) (or empty, or all of $\mathbb{R}$): indeed, if $f(x) \geq \alpha$ and $x' > x$, monotonicity gives $f(x') \geq f(x) \geq \alpha$, so $S_\alpha(f)$ contains every point to the right of any of its elements, and a subset of $\mathbb{R}$ with that property is an interval unbounded to the right. Intervals are convex, so f is quasiconcave. The same argument with reversed inequalities shows that the lower contour sets are intervals, so f is quasiconvex as well: f is quasilinear. In particular, $f(x) = x^3$ and $f(x) = e^x$ are quasilinear on $\mathbb{R}$, even though x^3 is neither concave nor convex and e^x is convex but not concave. $\square$

Remark 6.9 (On the term “quasilinear utility” in micro). In microeconomics, *quasilinear utility* means $u(x, m) = \varphi(x) + m$, linear in the numeraire m (“money”). The name refers to that partial linearity, and it does *not* agree with Definition 6.7: if φ is concave, then $u(x, m) = \varphi(x) + m$ is concave (a sum of a concave and a linear function), hence quasiconcave, but in general it is *not* quasiconvex, so it is not quasilinear in the sense of generalized convexity. The canonical example that does join both worlds is *linear* utility $u(x) = \langle a, x \rangle$, which is concave and convex at the same time and therefore quasilinear: its upper and lower contour sets are halfspaces. Keep this double use of the term in mind so that you do not get confused in Microeconomics.

Definition 6.10 (Strict quasiconcavity). $f: S \rightarrow \mathbb{R}$ is *strictly quasiconcave* if for all $x, y \in S$ with $x \neq y$ and all $\lambda \in (0, 1)$,

$$f(\lambda x + (1 - \lambda)y) > \min\{f(x), f(y)\}.$$

Likewise, f is *strictly quasiconvex* if $-f$ is strictly quasiconcave.

The strict version rules out “flat pieces” in the indifference curves: if two distinct bundles are indifferent to each other, any proper mixture is *strictly better*. Geometrically, the indifference curves contain no line segments. This is the hypothesis that in Section 6.8 will give us *uniqueness* of demand.

Careful (common mistake)

Watch out for two frequent slips in exams. *First*: in Definition 6.10 the strict inequality is required for *every* pair $x \neq y$, even when $f(x) = f(y)$; some texts use a weaker version (only when $f(x) \neq f(y)$), but that weak version does *not* guarantee uniqueness

of the maximizer, which is exactly what we want in micro. Always use Definition 6.10 unless told otherwise. *Second*: “strictly quasiconcave” does *not* imply “strictly concave”, and “concave” does not imply “strictly quasiconcave” either: in $\mathbb{R}^n$ with $n \geq 2$, the linear function $f(x) = \langle a, x \rangle$ with $a \neq 0$ is concave but not strictly quasiconcave (on an indifference hyperplane, mixtures improve nothing); and $f(x) = x^3$ is strictly quasiconcave on $\mathbb{R}$ (it is strictly increasing, see Exercise 6.49) without being concave. These are parallel hierarchies, not nested ones.

6.3 Hierarchy and ordinal properties [P1]

6.3.1 Concave implies quasiconcave (but not the converse)

Proposition 6.11 (Concavity implies quasiconcavity). *If $f: S \rightarrow \mathbb{R}$ is concave on the convex set S , then f is quasiconcave. If f is strictly concave, then it is strictly quasiconcave.*

Proof. Let $x, y \in S$ and $\lambda \in [0, 1]$. Write $m = \min\{f(x), f(y)\}$; by definition of minimum, $f(x) \geq m$ and $f(y) \geq m$. Concavity of f gives the first inequality in the chain

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y) \geq \lambda m + (1 - \lambda)m = m,$$

and the second inequality uses that $\lambda \geq 0$ and $1 - \lambda \geq 0$ (multiplying an inequality by a nonnegative number preserves it, and then we add the two resulting inequalities). This is exactly the quasiconcavity inequality. For the strict version: if $x \neq y$ and $\lambda \in (0, 1)$, strict concavity makes the first inequality strict, and we conclude $f(\lambda x + (1 - \lambda)y) > m$. $\square$

The converse is false, and the counterexamples matter because they show how much weaker quasiconcavity is.

Example 6.12 (Quasiconcave functions that are not concave). (a) $f(x) = x^3$ on $\mathbb{R}$ is strictly increasing, hence quasiconcave (Example 6.8); but it is not concave: $f''(x) = 6x > 0$ for $x > 0$, and a twice differentiable concave function must have $f'' \leq 0$ everywhere.

(b) $f(x) = e^x$ on $\mathbb{R}$ is increasing, hence quasiconcave; but it is strictly *convex* ($f'' = e^x > 0$). A function can be convex and quasiconcave at the same time: quasiconcavity imposes no downward curvature, only the shape of the level sets.

(c) Any monotone function of one variable, however steplike or irregular, is quasiconcave. On $\mathbb{R}$, quasiconcavity of f is the same as f being “unimodal” in a broad sense (nondecreasing up to a point and nonincreasing after it, with the two monotone cases included as limits); it says nothing about curvature.

(d) $u(x, y) = xy$ on $\mathbb{R}_{++}^2$: as we saw in Example 6.1, it is not concave; in Section 6.4 we will prove that it is quasiconcave. $\square$

6.3.2 The key ordinal property

The next theorem is the reason this chapter exists: quasiconcavity survives ordinal changes of scale.

Theorem 6.13 (Invariance under monotone transformations). *Let $f: S \rightarrow \mathbb{R}$ be quasiconcave and let $g: T \rightarrow \mathbb{R}$ be nondecreasing, where $T \subseteq \mathbb{R}$ contains the image of f . Then $g \circ f$ is quasiconcave. If f is strictly quasiconcave and g is strictly increasing, then $g \circ f$ is strictly quasiconcave.*

Proof. We use the definition by inequality. Let $x, y \in S$ and $\lambda \in [0, 1]$, and set $z = \lambda x + (1 - \lambda)y \in S$. By quasiconcavity of f ,

$$f(z) \geq \min\{f(x), f(y)\}.$$

We apply g , which is nondecreasing, to both sides (this preserves the inequality):

$$g(f(z)) \geq g(\min\{f(x), f(y)\}).$$

One small but crucial step is missing: for any nondecreasing function g and numbers s, t ,

$$g(\min\{s, t\}) = \min\{g(s), g(t)\}.$$

Indeed, assume without loss of generality that $s \leq t$; then $\min\{s, t\} = s$ and, since g is nondecreasing, $g(s) \leq g(t)$, so $\min\{g(s), g(t)\} = g(s) = g(\min\{s, t\})$. Combining:

$$(g \circ f)(z) \geq \min\{(g \circ f)(x), (g \circ f)(y)\},$$

which is quasiconcavity of $g \circ f$. For the strict version: if $x \neq y$ and $\lambda \in (0, 1)$, strict quasiconcavity of f gives $f(z) > \min\{f(x), f(y)\}$ and *strict* monotonicity of g turns that strict inequality into $g(f(z)) > g(\min\{f(x), f(y)\}) = \min\{g(f(x)), g(f(y))\}$. $\square$

Intuition

Compare with concavity: if f is concave and g is increasing, $g \circ f$ may lose concavity, because the concavity inequality involves the *average* $\lambda f(x) + (1 - \lambda)f(y)$, and increasing functions do not commute with averages (only affine ones do; increasing *concave* functions at least preserve the inequality in the right direction, which is why “a concave increasing function of a concave function is concave”). The quasiconcavity inequality, by contrast, involves the *minimum*, and any increasing function commutes with the minimum: applying g does not change which of the two values is smaller. This is why quasiconcavity is ordinal and concavity is not. A concrete example of the loss of concavity: $f(x) = \sqrt{x}$ is concave on $\mathbb{R}_+$ and $g(t) = t^4$ is increasing on $\mathbb{R}_+$, but $(g \circ f)(x) = x^2$ is strictly convex.

6.3.3 Operations that do not preserve quasiconcavity

With concave functions we have a convenient “algebra”: the sum of concave functions is concave, linear combinations with positive coefficients are concave, the pointwise minimum of concave

functions is concave. With quasiconcavity almost all of that algebra *is lost*. The classic exam trap:

Careful (common mistake)

The sum of quasiconcave functions is NOT, in general, quasiconcave. Explicit counterexample on $\mathbb{R}$: let $f(x) = x^3$ and $g(x) = -x$. Both are monotone, hence quasiconcave (Example 6.8). Their sum is $h(x) = x^3 - x$. Evaluate: $h(-1) = -1 + 1 = 0$, $h(0) = 0$, $h(1) = 1 - 1 = 0$ and $h(1/2) = 1/8 - 1/2 = -3/8 < 0$. If h were quasiconcave, the set $S_0(h) = \{x : h(x) \geq 0\}$ would be convex; but $-1 \in S_0(h)$ and $1 \in S_0(h)$ while the intermediate point $1/2 = \frac{1}{4}(-1) + \frac{3}{4}(1)$ is a convex combination of -1 and 1 and satisfies $h(1/2) < 0$, so $1/2 \notin S_0(h)$. The upper contour set $S_0(h) = [-1, 0] \cup [1, \infty)$ is not convex: h is not quasiconcave. Exam lesson: if you are asked “prove or disprove: the sum of quasiconcave functions is quasiconcave”, the answer is *disprove*, with this counterexample. The same holds for differences, products and linear combinations in general.

Which operations *do* preserve quasiconcavity? Those that act on level sets in a way compatible with convexity:

- Proposition 6.14** (Operations that preserve quasiconcavity). (a) (Pointwise minimum) If $f_1, \dots, f_k : S \rightarrow \mathbb{R}$ are quasiconcave, then $f(x) = \min\{f_1(x), \dots, f_k(x)\}$ is quasiconcave.
- (b) (Monotone transformation) If f is quasiconcave and g is nondecreasing, then $g \circ f$ is quasiconcave (Theorem 6.13).
- (c) (Composition with an affine map) If f is quasiconcave on a convex set $D \subseteq \mathbb{R}^m$ and $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, then $h(x) = f(Ax + b)$ is quasiconcave on the convex set $\{x \in \mathbb{R}^n : Ax + b \in D\}$.
- (d) (Restriction to convex sets) If f is quasiconcave on S and $C \subseteq S$ is convex, the restriction of f to C is quasiconcave.

Proof. (a) For each α ,

$$\{x : \min_i f_i(x) \geq \alpha\} = \bigcap_{i=1}^k \{x : f_i(x) \geq \alpha\} = \bigcap_{i=1}^k S_\alpha(f_i),$$

because the minimum of several numbers is $\geq \alpha$ exactly when all of them are. Each $S_\alpha(f_i)$ is convex by hypothesis, and the intersection of convex sets is convex (ch. 2); hence the upper contour set of f is convex for every α . (c) $\{x : f(Ax + b) \geq \alpha\}$ is the preimage of the convex set $S_\alpha(f)$ under the affine map $x \mapsto Ax + b$, and affine preimages of convex sets are convex (ch. 2). (d) $S_\alpha(f|_C) = S_\alpha(f) \cap C$, an intersection of convex sets. $\square$

Note the contrast with the *maximum*: the pointwise maximum of quasiconcave functions is *not* quasiconcave in general (Exercise 6.46 asks for the counterexample), while for *convex* functions the maximum does preserve convexity. The symmetry is: quasiconcave $\leftrightarrow$ minimum and intersection of upper contour sets; quasiconvex $\leftrightarrow$ maximum and intersection of lower contour sets.

6.4 Fundamental economic examples [P1]

This section works out in full detail the four utility function examples that appear again and again in Microeconomics.

6.4.1 Cobb–Douglas

Proposition 6.15 (Quasiconcavity of Cobb–Douglas). *Let $u(x, y) = x^a y^b$ with $a, b > 0$, defined on $\mathbb{R}_+^2$. Then:*

- (a) *u is quasiconcave on $\mathbb{R}_+^2$ for all values $a, b > 0$, and strictly quasiconcave on $\mathbb{R}_{++}^2$;*
- (b) *u is concave on $\mathbb{R}_+^2$ if and only if $a + b \leq 1$.*

Proof. (a) *Step 1: the interior, via the logarithmic transformation.* On $\mathbb{R}_{++}^2$ the values of u are positive and we can write

$$u(x, y) = \exp(a \ln x + b \ln y) = (g \circ v)(x, y), \quad v(x, y) = a \ln x + b \ln y, \quad g(t) = e^t.$$

The function v is concave: it is a linear combination with positive coefficients a, b of the concave functions $\ln x$ and $\ln y$ (each is concave because its second derivative, $-1/x^2$ and $-1/y^2$ respectively, is negative; and weighted sums with positive weights of concave functions are concave). Moreover, v is *strictly* concave on $\mathbb{R}_{++}^2$: its Hessian $\text{diag}(-a/x^2, -b/y^2)$ is negative definite. By Proposition 6.11, v is strictly quasiconcave; and since $g(t) = e^t$ is strictly increasing, Theorem 6.13 gives that $u = g \circ v$ is strictly quasiconcave on $\mathbb{R}_{++}^2$.

Step 2: extension to the boundary of $\mathbb{R}_+^2$. We check directly that the upper contour sets of u on $\mathbb{R}_+^2$ are convex. If $\alpha \leq 0$, then $S_\alpha(u) = \mathbb{R}_+^2$ (since $u \geq 0$ always), which is convex. If $\alpha > 0$, then $u(x, y) \geq \alpha$ forces $x > 0$ and $y > 0$, so $S_\alpha(u) = \{(x, y) \in \mathbb{R}_{++}^2 : u(x, y) \geq \alpha\}$, which is convex by Step 1 (it is an upper contour set of a quasiconcave function on the convex set $\mathbb{R}_{++}^2$). Hence u is quasiconcave on all of $\mathbb{R}_+^2$.

(b) Let us compute the Hessian on $\mathbb{R}_{++}^2$. The partial derivatives are

$$\begin{aligned} u_x &= ax^{a-1}y^b = \frac{a}{x}u, & u_y &= bx^ay^{b-1} = \frac{b}{y}u, \\ u_{xx} &= a(a-1)x^{a-2}y^b, & u_{yy} &= b(b-1)x^ay^{b-2}, & u_{xy} &= abx^{a-1}y^{b-1}. \end{aligned}$$

The determinant of the Hessian is

$$\begin{aligned} \det H_u &= u_{xx}u_{yy} - u_{xy}^2 \\ &= a(a-1)b(b-1)x^{2a-2}y^{2b-2} - a^2b^2x^{2a-2}y^{2b-2} \\ &= abx^{2a-2}y^{2b-2}[(a-1)(b-1) - ab] \\ &= abx^{2a-2}y^{2b-2}[ab - a - b + 1 - ab] = abx^{2a-2}y^{2b-2}(1 - a - b). \end{aligned}$$

If $a + b > 1$, then $\det H_u < 0$ at every point: the Hessian is indefinite and u is not concave (this generalizes the case $a = b = 1$, $u = xy$, of Example 6.1). If $a + b \leq 1$, then $\det H_u \geq 0$ and moreover $u_{xx} = a(a-1)x^{a-2}y^b \leq 0$ (because $a \leq a + b \leq 1$ implies $a - 1 \leq 0$): the Hessian is

negative semidefinite on $\mathbb{R}_{++}^2$ and u is concave there; concavity extends to $\mathbb{R}_+^2$ by continuity of u (the concavity inequality, being weak, passes to the limit on the boundary). $\square$

Intuition

The economic message of Proposition 6.15 is exactly the one of Section 6.1: Cobb–Douglas preferences are the same for $u = x^a y^b$ as for its monotone transformation $u^{1/(a+b)} = x^{a/(a+b)} y^{b/(a+b)}$, whose exponents add up to 1 and which therefore *is* concave. The failure of concavity when $a + b > 1$ is a pure artifact of scale; quasiconcavity, which is what matters for preferences and for demand, always holds. This technique of concavifying by taking logarithms or roots will come back in Section 6.8 as “hidden concavity”.

6.4.2 CES, Leontief and linear utility

Example 6.16 (CES). The constant elasticity of substitution utility is

$$u(x, y) = (ax^\rho + by^\rho)^{1/\rho}, \quad a, b > 0, \quad \rho \leq 1, \quad \rho \neq 0,$$

on $\mathbb{R}_{++}^2$. Let us prove that it is quasiconcave, splitting into cases.

Case $0 < \rho \leq 1$. The inner function $w(x, y) = ax^\rho + by^\rho$ is concave: $t \mapsto t^\rho$ has second derivative $\rho(\rho - 1)t^{\rho-2} \leq 0$ for $t > 0$ (since $0 < \rho \leq 1$ makes $\rho - 1 \leq 0$), so x^ρ and y^ρ are concave and w is a positive weighted sum of concave functions. The outer transformation $g(t) = t^{1/\rho}$ is strictly increasing for $t > 0$ (positive exponent). Then $u = g \circ w$ is an increasing transformation of a concave function (in particular quasiconcave), hence quasiconcave by Theorem 6.13. (Note: g is convex if $\rho < 1$, so this argument does *not* give concavity of u directly; concavity of the CES is true but needs the homogeneity argument: a nonnegative, quasiconcave and positively homogeneous of degree 1 function on a convex cone is concave.)

Case $\rho < 0$. Now $t \mapsto t^\rho$ is *convex* and decreasing for $t > 0$ (its second derivative $\rho(\rho - 1)t^{\rho-2} > 0$ because $\rho < 0$ and $\rho - 1 < 0$ have positive product), so $w = ax^\rho + by^\rho$ is convex, and therefore *quasiconvex*. The outer transformation $g(t) = t^{1/\rho}$ is strictly *decreasing* for $t > 0$ (negative exponent). A decreasing transformation swaps upper and lower contour sets: $\{u \geq \alpha\} = \{g(w) \geq \alpha\} = \{w \leq g^{-1}(\alpha)\}$ for α in the relevant range (the inequality reverses when we apply g^{-1} , also decreasing), and the *lower* contour sets of the convex function w are convex. Hence the upper contour sets of u are convex: u is quasiconcave.

The limit cases complete the family: when $\rho \rightarrow 0$ the CES converges to Cobb–Douglas (with exponents $a/(a + b)$, $b/(a + b)$), and when $\rho \rightarrow -\infty$ it converges to the Leontief function $\min\{x, y\}$; both quasiconcave, as expected. $\square$

Example 6.17 (Leontief or fixed proportions). Let $u(x, y) = \min\{ax, by\}$ with $a, b > 0$ on $\mathbb{R}_+^2$ (perfect complements: left and right shoes). Each of the functions ax and by is linear, hence concave and quasiconcave; by Proposition 6.14(a), the minimum u is quasiconcave. You can also see it directly: for $\alpha \leq 0$ the upper contour set is all of $\mathbb{R}_+^2$; for $\alpha > 0$,

$$\{(x, y) : \min\{ax, by\} \geq \alpha\} = \{x \geq \alpha/a\} \cap \{y \geq \alpha/b\},$$

an intersection of two halfspaces, which is convex. (In fact u is *concave*: the pointwise minimum of concave functions is concave.) But u is *not* strictly quasiconcave: the points $(1, 2)$ and $(1, 3)$ with $a = b = 1$ satisfy $u = 1$ at both, and at any intermediate point $(1, y)$ with $2 \leq y \leq 3$ we still have $u = 1$, not > 1 : the L shaped indifference curves have flat pieces. $\square$

Example 6.18 (Linear utility). $u(x, y) = ax + by$ with $a, b > 0$ (perfect substitutes) is concave and convex at the same time (it is affine), hence quasiconcave and quasiconvex: quasilinear in the sense of Definition 6.7. Its indifference curves are lines with slope $-a/b$. It is not strictly quasiconcave: two distinct points on the same indifference line have mixtures that are exactly indifferent. We will see in Section 6.8 that this failure is what makes demand with perfect substitutes possibly *multivalued* (the whole budget segment is optimal when $p_1/p_2 = a/b$). $\square$

6.4.3 Convex preferences and quasiconcave utility

Definition 6.19 (Convex preferences). A preference relation $\succsim$ on a convex set $X \subseteq \mathbb{R}^n$ is *convex* if for every $x^0 \in X$ the upper contour set $\{y \in X : y \succsim x^0\}$ is convex. It is *strictly convex* if, in addition, $y \succ x^0$, $z \succ x^0$ and $y \neq z$ imply $\lambda y + (1 - \lambda)z \succ x^0$ for every $\lambda \in (0, 1)$.

Theorem 6.20 (Convex preferences = quasiconcave utility). *Let $\succsim$ be a preference relation on the convex set X represented by the utility function $u: X \rightarrow \mathbb{R}$. Then $\succsim$ is convex if and only if u is quasiconcave; and $\succsim$ is strictly convex if and only if u is strictly quasiconcave.*

Proof. ($\Leftarrow$) If u is quasiconcave, then for each x^0 the set $\{y : y \succsim x^0\} = \{y : u(y) \geq u(x^0)\} = S_{u(x^0)}(u)$ is an upper contour set of u , hence convex.

($\Rightarrow$) Assume $\succsim$ is convex, that is, that $S_{u(x^0)}(u)$ is convex for levels of the form $\alpha = u(x^0)$ (levels attained by u). Careful: this is not directly Definition 6.3, which asks for convexity at every $\alpha \in \mathbb{R}$, including levels that u never attains. So we use the definition by inequality. Let $x, y \in X$ and $\lambda \in [0, 1]$; without loss of generality $u(y) \leq u(x)$, so that $\min\{u(x), u(y)\} = u(y)$, a level that is *attained* (at the point y). Both points belong to $\{z : z \succsim y\}$: the point y trivially (reflexivity: $u(y) \geq u(y)$) and x because $u(x) \geq u(y)$. By convexity of preferences, $\lambda x + (1 - \lambda)y \in \{z : z \succsim y\}$, that is

$$u(\lambda x + (1 - \lambda)y) \geq u(y) = \min\{u(x), u(y)\} :$$

the quasiconcavity inequality. The strict case is identical, using the strict part of Definition 6.19 with $x^0 = y$ and Definition 6.10. $\square$

Intuition

Convex preference means *taste for diversification*: if two bundles are acceptable, so are their mixtures; in the strict version, mixing indifferent bundles is strictly better (“better a bit of bread and a bit of cheese than a lot of just one”). Theorem 6.20 says that this purely *behavioral* property, stated without mentioning numbers, is exactly quasiconcavity of *any* utility representation. This is the final confirmation that quasiconcavity is the right notion: a property of preferences had to correspond to an ordinal property of u . See MWG, ch. 3.B–3.C [9] and [10], ch. 1.

6.5 First order differential characterization [P1]

Recall the first order characterization of concavity: a differentiable f is concave if and only if $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle$ for all x, y (the graph stays below every tangent plane). The quasiconcave version replaces that *cardinal* inequality by a purely *directional* condition: the gradient must “point toward” the better points.

Theorem 6.21 (First order characterization of quasiconcavity). *Let $S \subseteq \mathbb{R}^n$ be convex and open, and let $f: S \rightarrow \mathbb{R}$ be differentiable. Then f is quasiconcave if and only if*

$$\text{for all } x, y \in S : \quad f(y) \geq f(x) \implies \langle \nabla f(x), y - x \rangle \geq 0.$$

Proof. ($\Rightarrow$) Full proof. Assume f is quasiconcave and take $x, y \in S$ with $f(y) \geq f(x)$. For each $\lambda \in (0, 1]$, the point $x + \lambda(y - x) = (1 - \lambda)x + \lambda y$ lies in S , and the quasiconcavity inequality gives

$$f(x + \lambda(y - x)) \geq \min\{f(x), f(y)\} = f(x),$$

where the last equality uses the hypothesis $f(y) \geq f(x)$. Subtracting $f(x)$ and dividing by $\lambda > 0$ (which preserves the direction of the inequality):

$$\frac{f(x + \lambda(y - x)) - f(x)}{\lambda} \geq 0 \quad \text{for all } \lambda \in (0, 1].$$

The left hand side is the difference quotient of the one variable function $\varphi(\lambda) = f(x + \lambda(y - x))$ at $\lambda = 0$. Since f is differentiable, the chain rule guarantees that the limit as $\lambda \rightarrow 0^+$ exists and equals the directional derivative

$$\lim_{\lambda \rightarrow 0^+} \frac{f(x + \lambda(y - x)) - f(x)}{\lambda} = \langle \nabla f(x), y - x \rangle.$$

Limits preserve weak inequalities, so $\langle \nabla f(x), y - x \rangle \geq 0$, as required.

($\Leftarrow$) *Idea and reference.* The converse direction is more laborious; we state it with its outline and leave the detailed proof to [1] (chapter on generalized convexity) or [9], Theorem M.C.3. Outline: if f were not quasiconcave, there would be x, y and a point $z^* = (1 - \lambda^*)x + \lambda^*y$ of the segment with $f(z^*) < \min\{f(x), f(y)\}$. Take z^* to be a minimum of f over the (compact) segment $[x, y]$, attained in the interior of the segment. Since $f(x) \geq f(z^*)$ and $f(y) \geq f(z^*)$, the gradient condition applied at z^* against x and against y gives $\langle \nabla f(z^*), x - z^* \rangle \geq 0$ and $\langle \nabla f(z^*), y - z^* \rangle \geq 0$. But $x - z^*$ and $y - z^*$ are *negative* multiples of each other (they point in opposite directions along the segment), so the two inner products have opposite signs: the only way out is that both are zero, that is, the derivative of f along the segment vanishes at z^* . A further argument (applying the same to *all* points where the minimum is attained and using the mean value theorem) shows that f would then be constant on a subsegment containing z^* and a point with value $> f(z^*)$, a contradiction. $\square$

Intuition

Geometric reading, which connects with the chapter on separation and support functions: fix x with $\nabla f(x) \neq 0$. The theorem says that the *whole* upper contour set $S_{f(x)}(f) = \{y : f(y) \geq f(x)\}$ is contained in the halfspace

$$H^+ = \{y : \langle \nabla f(x), y - x \rangle \geq 0\},$$

whose boundary is the hyperplane through x with normal $\nabla f(x)$. In other words: **the gradient supports the upper contour set at x** . In consumer theory this is the classical tangency: the hyperplane is the tangent line to the indifference curve, the gradient $\nabla u(x)$ (the vector of marginal utilities) is normal to it, and all better bundles lie on the side the gradient points to. At the consumer optimum, that supporting hyperplane is the budget line and $\nabla u(x^*) = \mu p$: the Lagrange first order condition is a *separation* statement between the “better than” set and the budget set.

6.6 The bordered Hessian [P1]

For concave functions, the second order tool is the Hessian (negative semidefinite). For quasiconcave functions, the tool is the Hessian *bordered* with the gradient. This section is the most practical of the chapter: it is the one you will use in exams.

6.6.1 Building the matrix

Definition 6.22 (Bordered Hessian). Let $f: S \rightarrow \mathbb{R}$ be of class C^2 on an open set $S \subseteq \mathbb{R}^n$. Write $f_i = \partial f / \partial x_i$ and $f_{ij} = \partial^2 f / \partial x_i \partial x_j$. For each $r = 1, \dots, n$, the *bordered Hessian* of order r at the point x is the $(r+1) \times (r+1)$ matrix

$$B_r(x) = \begin{pmatrix} 0 & f_1 & f_2 & \cdots & f_r \\ f_1 & f_{11} & f_{12} & \cdots & f_{1r} \\ f_2 & f_{21} & f_{22} & \cdots & f_{2r} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f_r & f_{r1} & f_{r2} & \cdots & f_{rr} \end{pmatrix},$$

with all derivatives evaluated at x : it is the upper left $r \times r$ corner of the Hessian, bordered by the truncated gradient in the first row and the first column, with a 0 in the corner. The determinants $\det B_1(x), \dots, \det B_n(x)$ are called *bordered leading principal minors*.

Note that $\det B_1 = \det \begin{pmatrix} 0 & f_1 \\ f_1 & f_{11} \end{pmatrix} = -f_1^2 \leq 0$ *always*, for any function: the first bordered minor tells us nothing by itself (we return to this when we discuss the necessary condition).

Theorem 6.23 (Bordered Hessian test). *Let $S \subseteq \mathbb{R}^n$ be convex and open, and let $f: S \rightarrow \mathbb{R}$ be of class C^2 .*

(a) (**Necessary condition.**) If f is quasiconcave on S , then

$$(-1)^r \det B_r(x) \geq 0 \quad \text{for all } x \in S \text{ and } r = 1, \dots, n,$$

that is, $\det B_1 \leq 0$, $\det B_2 \geq 0$, $\det B_3 \leq 0$, and so on, alternating.

(b) (**Sufficient condition.**) If

$$(-1)^r \det B_r(x) > 0 \quad \text{for all } x \in S \text{ and } r = 1, \dots, n$$

(strict inequalities: $\det B_1 < 0$, $\det B_2 > 0$, $\det B_3 < 0$, ...), then f is quasiconcave on S ; moreover, it is pseudoconcave and strictly quasiconcave on S .

For quasiconvexity the roles are reversed: the necessary condition is $\det B_r(x) \leq 0$ for every r (all bordered minors nonpositive), and the sufficient one is $\det B_r(x) < 0$ for every r and every x .

We will not prove this theorem (the proof uses the theory of quadratic forms restricted to a subspace: the condition says that the Hessian is negative definite *on the hyperplane tangent to the level curve*, $\{v : \langle \nabla f(x), v \rangle = 0\}$; see [9], Theorem M.C.4 and section M.D, and [1]), but we do insist on the logic, because this is where books, and students, get confused.

Careful (common mistake)

The necessary and the sufficient conditions are NOT the same, and neither is an equivalence. There is a gap when some determinant vanishes, exactly as with the ordinary Hessian (semidefinite vs definite). Two typical exam mistakes:

- (i) *Believing that the weak signs imply quasiconcavity.* False. Minimal counterexample: $f(x) = x^2$ on $\mathbb{R}$ ($n = 1$). Here $\det B_1(x) = -f'(x)^2 = -4x^2 \leq 0$ at every point, so the necessary condition $(-1)^1 \det B_1 \geq 0$ holds on all of $\mathbb{R}$; but x^2 is *not* quasiconcave: its upper contour set $\{x : x^2 \geq 1\} = (-\infty, -1] \cup [1, \infty)$ is not convex. The weak condition (a) is only good for *ruling out*: if it fails at some point, f is not quasiconcave.
- (ii) *Believing that a quasiconcave function must satisfy the strict signs.* False: $f(x, y) = x$ (linear, quasiconcave) has $f_2 = 0$ and $\det B_2 = 0$. The strict conditions (b) fail where the gradient has zeros or the curvature degenerates, without this refuting quasiconcavity. The strict condition is only good for *asserting*.

In short: strict signs on the whole domain $\Rightarrow$ quasiconcave; quasiconcave $\Rightarrow$ weak signs; and between the two there are cases this method cannot decide (you then need the definition or a monotone transformation).

One more warning about the *order of the variables*. The B_r are built from the upper left corner of the Hessian, so they depend on how you number the variables. The necessary condition (a) does not suffer from this: quasiconcavity is invariant under permutations of the variables, so the weak signs hold *for any ordering* (equivalently, for all bordered principal minors, not only the leading ones). The sufficient condition (b), by contrast,

requires $\det B_1 = -f_1^2 < 0$, that is $f_1 \neq 0$: if your first variable happens to have zero derivative, the strict test fails for a purely notational reason. Remember the variables before concluding anything.

6.6.2 A full 2×2 example: Cobb–Douglas

Example 6.24 (Bordered minors of $u = x^a y^b$, step by step). Let $u(x, y) = x^a y^b$ with $a, b > 0$, on $\mathbb{R}_{++}^2$. We already computed in Proposition 6.15:

$$u_x = \frac{au}{x}, \quad u_y = \frac{bu}{y}, \quad u_{xx} = \frac{a(a-1)u}{x^2}, \quad u_{yy} = \frac{b(b-1)u}{y^2}, \quad u_{xy} = \frac{abu}{xy},$$

where we write everything in terms of $u = x^a y^b > 0$ to lighten the notation (check, for example, that $a(a-1)x^{a-2}y^b = a(a-1)u/x^2$).

First bordered minor ($r = 1$):

$$\det B_1 = \det \begin{pmatrix} 0 & u_x \\ u_x & u_{xx} \end{pmatrix} = 0 \cdot u_{xx} - u_x \cdot u_x = -u_x^2 = -\frac{a^2 u^2}{x^2} < 0,$$

strictly negative on $\mathbb{R}_{++}^2$ because $a > 0$ and $u > 0$. So $(-1)^1 \det B_1 > 0$. ✓

Second bordered minor ($r = 2$): the matrix is

$$B_2 = \begin{pmatrix} 0 & u_x & u_y \\ u_x & u_{xx} & u_{xy} \\ u_y & u_{xy} & u_{yy} \end{pmatrix}.$$

We expand the determinant along the first row (cofactors):

$$\begin{aligned} \det B_2 &= 0 \cdot \det \begin{pmatrix} u_{xx} & u_{xy} \\ u_{xy} & u_{yy} \end{pmatrix} - u_x \det \begin{pmatrix} u_x & u_{xy} \\ u_y & u_{yy} \end{pmatrix} + u_y \det \begin{pmatrix} u_x & u_{xx} \\ u_y & u_{xy} \end{pmatrix} \\ &= -u_x(u_x u_{yy} - u_{xy} u_y) + u_y(u_x u_{xy} - u_{xx} u_y) \\ &= 2u_x u_y u_{xy} - u_x^2 u_{yy} - u_y^2 u_{xx}. \end{aligned}$$

This final formula, $\det B_2 = 2f_x f_y f_{xy} - f_x^2 f_{yy} - f_y^2 f_{xx}$, holds for any function of two variables and is worth memorizing. Substituting the Cobb–Douglas values, term by term:

$$\begin{aligned} 2u_x u_y u_{xy} &= 2 \cdot \frac{au}{x} \cdot \frac{bu}{y} \cdot \frac{abu}{xy} = \frac{2a^2 b^2 u^3}{x^2 y^2}, \\ u_x^2 u_{yy} &= \frac{a^2 u^2}{x^2} \cdot \frac{b(b-1)u}{y^2} = \frac{a^2 b(b-1)u^3}{x^2 y^2}, \quad u_y^2 u_{xx} = \frac{b^2 u^2}{y^2} \cdot \frac{a(a-1)u}{x^2} = \frac{ab^2(a-1)u^3}{x^2 y^2}. \end{aligned}$$

Then

$$\begin{aligned}\det B_2 &= \frac{u^3}{x^2y^2} [2a^2b^2 - a^2b(b-1) - ab^2(a-1)] = \frac{ab u^3}{x^2y^2} [2ab - a(b-1) - b(a-1)] \\ &= \frac{ab u^3}{x^2y^2} [2ab - ab + a - ab + b] = \frac{ab(a+b)u^3}{x^2y^2} > 0\end{aligned}$$

for all $a, b > 0$ and every $(x, y) \in \mathbb{R}_{++}^2$. So $(-1)^2 \det B_2 > 0$. ✓

Since both bordered minors have the required strictly alternating signs on all of $\mathbb{R}_{++}^2$, Theorem 6.23(b) confirms that u is quasiconcave (in fact strictly quasiconcave and pseudoconcave) on $\mathbb{R}_{++}^2$, for all $a, b > 0$, in perfect agreement with the proof by logarithmic transformation of Proposition 6.15, and with no restriction of the form $a + b \leq 1$. □

6.6.3 A 3×3 example

Example 6.25 (Three goods: $u(x, y, z) = xyz$). Let $u(x, y, z) = xyz$ on $\mathbb{R}_{++}^3$. (We know in advance that it must be quasiconcave: $\ln u = \ln x + \ln y + \ln z$ is concave and $u = e^{\ln u}$; the point of the example is to practice the bordered mechanics in dimension 3.) The derivatives:

$$u_x = yz, \quad u_y = xz, \quad u_z = xy; \quad u_{xx} = u_{yy} = u_{zz} = 0, \quad u_{xy} = z, \quad u_{xz} = y, \quad u_{yz} = x.$$

$$\text{Order } r = 1: \quad \det B_1 = -u_x^2 = -(yz)^2 < 0. \quad \checkmark$$

Order $r = 2$:

$$\det B_2 = \det \begin{pmatrix} 0 & yz & xz \\ yz & 0 & z \\ xz & z & 0 \end{pmatrix} = -yz \det \begin{pmatrix} yz & z \\ xz & 0 \end{pmatrix} + xz \det \begin{pmatrix} yz & 0 \\ xz & z \end{pmatrix},$$

expanding along the first row. The cofactors: $\det \begin{pmatrix} yz & z \\ xz & 0 \end{pmatrix} = 0 - xz^2 = -xz^2$ and $\det \begin{pmatrix} yz & 0 \\ xz & z \end{pmatrix} = yz \cdot z - 0 = yz^2$. Hence

$$\det B_2 = -yz \cdot (-xz^2) + xz \cdot yz^2 = xyz^3 + xyz^3 = 2xyz^3 > 0. \checkmark$$

Order $r = 3$: the full matrix is

$$B_3 = \begin{pmatrix} 0 & yz & xz & xy \\ yz & 0 & z & y \\ xz & z & 0 & x \\ xy & y & x & 0 \end{pmatrix}.$$

We expand along the first row: $\det B_3 = -yz \cdot M_1 + xz \cdot M_2 - xy \cdot M_3$, with

$$M_1 = \det \begin{pmatrix} yz & z & y \\ xz & 0 & x \\ xy & x & 0 \end{pmatrix}, \quad M_2 = \det \begin{pmatrix} yz & 0 & y \\ xz & z & x \\ xy & y & 0 \end{pmatrix}, \quad M_3 = \det \begin{pmatrix} yz & 0 & z \\ xz & z & 0 \\ xy & y & x \end{pmatrix}.$$

We compute each one by the Sarrus rule or by cofactors along the first row:

$$\begin{aligned} M_1 &= yz(0 \cdot 0 - x \cdot x) - z(xz \cdot 0 - x \cdot xy) + y(xz \cdot x - 0 \cdot xy) \\ &= -x^2yz + x^2yz + x^2yz = x^2yz; \\ M_2 &= yz(z \cdot 0 - x \cdot y) - 0 + y(xz \cdot y - z \cdot xy) = -xy^2z + y \cdot 0 = -xy^2z; \\ M_3 &= yz(z \cdot x - 0 \cdot y) - 0 + z(xz \cdot y - z \cdot xy) = xyz^2 + z \cdot 0 = xyz^2. \end{aligned}$$

Therefore

$$\det B_3 = -yz \cdot x^2yz + xz \cdot (-xy^2z) - xy \cdot xyz^2 = -x^2y^2z^2 - x^2y^2z^2 - x^2y^2z^2 = -3x^2y^2z^2 < 0. \checkmark$$

The three signs alternate strictly: $\det B_1 < 0$, $\det B_2 > 0$, $\det B_3 < 0$, that is $(-1)^r \det B_r > 0$ for $r = 1, 2, 3$ on all of $\mathbb{R}_{++}^3$. By Theorem 6.23(b), $u = xyz$ is quasiconcave (and pseudoconcave) on $\mathbb{R}_{++}^3$. $\square$

Exam strategy

Exam recipe for the bordered Hessian. When asked “determine whether u is quasiconcave”:

1. *Before differentiating*, try an ordinal shortcut: is u a monotone increasing transformation of something concave or already known? (logarithm for products of powers, the $1/\rho$ root for CES). If so, conclude by Theorem 6.13 and save yourself the determinants. Is it a minimum of quasiconcave functions, or a composition with an affine map? Use Proposition 6.14.
2. If there is no shortcut, compute the gradient and the Hessian, and form the B_r by adding the border (a row and a column with the gradient, 0 in the corner).
3. Compute $\det B_1 = -f_1^2, \det B_2, \dots, \det B_n$ (for $n = 2$ use the memorable formula $\det B_2 = 2f_x f_y f_{xy} - f_x^2 f_{yy} - f_y^2 f_{xx}$).
4. If the signs alternate *strictly* starting from negative $(-, +, -, \dots)$ on the *whole* domain: conclude “quasiconcave (and pseudoconcave)” citing the sufficient condition.
5. If some sign *violates* the weak pattern $(\leq, \geq, \leq, \dots)$ at some point: conclude “not quasiconcave” citing the necessary condition.
6. If there are zeros (boundary case): the test does not decide; go back to the definition by level sets, to the minimum inequality or to a monotone transformation. Write down explicitly in the exam which condition you are using and in which direction: confusing necessary with sufficient costs points.

For *quasiconvexity*, same steps with the sign pattern “all ≤ 0 ” (necessary) and “all < 0 ” (sufficient), or apply the quasiconcave test to $-f$.

6.7 Pseudoconcave functions [P2]

Between concave and quasiconcave functions there is an intermediate class, defined only for differentiable functions, which keeps the most valuable property of concavity in optimization: *stationary points are global maxima*.

Motivation: quasiconcavity is too weak for first order conditions. The example is $f(x) = x^3$: it is quasiconcave (monotone) and $f'(0) = 0$, so $x = 0$ is a stationary point; but it clearly is not a maximum (not even a local one): $f(1) = 1 > 0 = f(0)$. With general quasiconcave utilities, “solving the first order condition” is not enough to have found the optimum. Pseudoconcavity repairs exactly that defect.

Definition 6.26 (Pseudoconcave function). Let $S \subseteq \mathbb{R}^n$ be convex and open, and let $f: S \rightarrow \mathbb{R}$ be differentiable. f is *pseudoconcave* if for all $x, y \in S$,

$$f(y) > f(x) \implies \langle \nabla f(x), y - x \rangle > 0.$$

Equivalently (contrapositive): if $\langle \nabla f(x), y - x \rangle \leq 0$, then $f(y) \leq f(x)$. f is *pseudoconvex* if $-f$ is pseudoconcave.

Compare with Theorem 6.21: quasiconcavity gives $f(y) \geq f(x) \implies \langle \nabla f(x), y - x \rangle \geq 0$; pseudoconcavity requires a *strict* improvement to be detected *strictly* by the gradient. The difference looks cosmetic but it is not: for $f(x) = x^3$, the point $y = 1$ strictly improves on $x = 0$ and yet $f'(0)(1 - 0) = 0$, not > 0 : so x^3 is *not* pseudoconcave, even though it is quasiconcave.

Theorem 6.27 (Hierarchy for differentiable functions). *Let f be differentiable on the convex open set S . Then:*

$$f \text{ concave} \implies f \text{ pseudoconcave} \implies f \text{ quasiconcave},$$

and no implication can be reversed.

Proof. *Concave $\implies$ pseudoconcave.* If f is concave and differentiable, the first order characterization of concavity gives, for all $x, y \in S$,

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle.$$

If in addition $f(y) > f(x)$, subtracting $f(x)$:

$$\langle \nabla f(x), y - x \rangle \geq f(y) - f(x) > 0,$$

which is pseudoconcavity.

Pseudoconcave $\implies$ quasiconcave. Assume, for contradiction, that f is pseudoconcave but not quasiconcave: there are $x, y \in S$ and a point of the segment with value smaller than $m = \min\{f(x), f(y)\}$. The function $\varphi(\lambda) = f((1 - \lambda)x + \lambda y)$ is continuous on the compact interval $[0, 1]$, so it attains its minimum (Weierstrass) at some $\lambda^* \in [0, 1]$, with $\varphi(\lambda^*) < m \leq \min\{\varphi(0), \varphi(1)\}$; in particular $\lambda^* \in (0, 1)$ (it cannot be an endpoint, where $\varphi \geq m$). Let

$z^* = (1 - \lambda^*)x + \lambda^*y$. Since $f(x) \geq m > f(z^*)$, pseudoconcavity applied to the pair (z^*, x) gives

$$\langle \nabla f(z^*), x - z^* \rangle > 0.$$

But $x - z^* = \lambda^*(x - y)$: it is a positive multiple of $x - y$, so $\langle \nabla f(z^*), x - y \rangle > 0$ and therefore the derivative of φ at λ^* , which by the chain rule equals $\varphi'(\lambda^*) = \langle \nabla f(z^*), y - x \rangle$, is strictly negative. This contradicts the fact that λ^* is an *interior* minimum of φ (at an interior minimum of a differentiable function the derivative vanishes: $\varphi'(\lambda^*) = 0$). Contradiction; hence f is quasiconcave.

The converses fail. x^3 is quasiconcave and not pseudoconcave (seen above). On the other step, $f(x) = x^3 + x$ is pseudoconcave on $\mathbb{R}$ (Exercise 6.49: it is strictly increasing with $f' > 0$ everywhere) but it is not concave ($f''(x) = 6x > 0$ for $x > 0$). $\square$

Proposition 6.28 (Why they matter: stationary $\Rightarrow$ global maximum). *If f is pseudoconcave on the convex open set S and $\nabla f(x^*) = 0$ for some $x^* \in S$, then x^* is a global maximizer of f on S .*

Proof. Suppose not: there is $y \in S$ with $f(y) > f(x^*)$. By pseudoconcavity, $\langle \nabla f(x^*), y - x^* \rangle > 0$. But $\nabla f(x^*) = 0$ makes that inner product equal to 0, a contradiction. $\square$

The same logic extends to constrained problems: pseudoconcavity makes the Karush–Kuhn–Tucker (KKT) conditions *sufficient* for global optimality.

Theorem 6.29 (Sufficiency of KKT; Arrow–Enthoven). *Consider the problem*

$$\max_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad g_j(x) \geq 0, \quad j = 1, \dots, k,$$

with f and $g_1, \dots, g_k$ differentiable, and assume that x^ satisfies the KKT conditions: there are multipliers $\mu_1, \dots, \mu_k \geq 0$ such that*

$$\nabla f(x^*) + \sum_{j=1}^k \mu_j \nabla g_j(x^*) = 0, \quad \mu_j g_j(x^*) = 0, \quad g_j(x^*) \geq 0 \quad \forall j.$$

- (a) *If f is pseudoconcave and each g_j is quasiconcave, then x^* is a global maximum of the problem (Mangasarian).*
- (b) *(Arrow–Enthoven, 1961.) The conclusion still holds if f is only quasiconcave, at the cost of an extra nondegeneracy condition; the most usable version: it is enough that $\nabla f(x^*) \neq 0$ and that f be twice differentiable in a neighborhood of x^* (there are variants with $x \geq 0$ and conditions on the partial derivatives, which we collect in Theorem 6.37).*

We do not prove this theorem (see [1] and the classical references cited there; case (a) is proved in a few lines by combining Definition 6.26 with quasiconcavity of the feasible region), but record the message:

Intuition

In Microeconomics you will write Lagrangians constantly. The uncomfortable question “and how do I know that the point I found with the first order conditions really is the maximum?” has this clean answer: if utility is pseudoconcave (for example, if it passes the strict bordered Hessian test, which by Theorem 6.23(b) implies pseudoconcavity) and the constraints define a convex set (a linear budget constraint is quasiconcave), then *KKT is sufficient* and the first order conditions close the problem. Quasiconcavity alone is not enough, as x^3 taught us, but almost all the utility functions in exercises (Cobb–Douglas, CES in the interior, and so on) are pseudoconcave, with a gradient that never vanishes. This is the “quality guarantee” that generalized concavity sells to the economist.

6.8 Applications of generalized concavity [P1]

So far we have built the theory: definitions, equivalences, hierarchy, differential criteria. This section collects the dividends. All the results that follow are the ones used in economics (sometimes without naming them) every time one writes “let $x(p, w)$ be consumer demand” or “assume preferences are strictly convex”. The common thread is this: *quasiconcavity is exactly the amount of structure needed for the maximization problems of microeconomics to be well posed*, and not one gram more.

6.8.1 The solution set is convex

Theorem 6.30 (Convexity of the set of maximizers). *Let $C \subseteq \mathbb{R}^n$ be a nonempty convex set and let $f: C \rightarrow \mathbb{R}$ be quasiconcave. Then the set of maximizers*

$$M = \arg \max_{x \in C} f(x) = \{x^* \in C : f(x^*) \geq f(x) \text{ for all } x \in C\}$$

is convex (possibly empty).

Proof. If $M = \emptyset$ or M has a single point, it is trivially convex (the empty set and singletons are convex: the segment condition holds vacuously or because the segment reduces to a point). So assume $x, y \in M$ and let $\lambda \in [0, 1]$; write $z = \lambda x + (1 - \lambda)y$.

Step 1: z is feasible. Since $x, y \in M \subseteq C$ and C is convex, $z \in C$. (Here we use convexity of the feasible set; without it the statement does not even make sense.)

Step 2: $f(z)$ cannot be smaller than the optimal value. Call $v^* = f(x) = f(y)$ the common value of the maximizers (it is the same number: both are global maxima). By quasiconcavity of f in the form of Definition 6.4, justified by Theorem 6.5,

$$f(z) \geq \min\{f(x), f(y)\} = \min\{v^*, v^*\} = v^*.$$

Step 3: $f(z)$ cannot be larger. Since $z \in C$ and $x^* = x$ is a maximizer over C , we have $f(z) \leq f(x) = v^*$.

Combining steps 2 and 3, $f(z) = v^*$, that is, z attains the maximum value and therefore $z \in M$. Since $x, y \in M$ and $\lambda \in [0, 1]$ were arbitrary, M is convex. $\square$

Note how few hypotheses we used: no continuity, no differentiability, no compactness. Only convexity of the feasible set and quasiconcavity of the objective. An alternative, more geometric reading of the same fact: if the maximum v^* is attained, then

$$M = \{x \in C : f(x) \geq v^*\} = C \cap S_{v^*}(f),$$

the intersection of the feasible set with the upper contour set at the optimal level; it is convex as an intersection of two convex sets (Definition 6.3 and ch. 2). This second proof is one line long, but the first one teaches more about the mechanism.

Intuition

Economic translation: *consumer demand is a convex valued correspondence*. If at prices p and wealth w the consumer is indifferent between two optimal bundles x and y , then any mixture of the two is also optimal. This is not a technicality: convexity of the values of the demand correspondence is *exactly* the hypothesis required by the Kakutani fixed point theorem, the tool used to prove existence of Walrasian equilibrium when demand is not a function (MWG [9], ch. 17 and the mathematical appendix M.I). Without convex preferences, that is, without quasiconcavity, the existence argument breaks down.

6.8.2 Uniqueness: demand is a function

If we strengthen quasiconcavity to the strict version, the convex set M collapses to a point.

Theorem 6.31 (Uniqueness of the maximizer). *Let $C \subseteq \mathbb{R}^n$ be nonempty and convex and let $f: C \rightarrow \mathbb{R}$ be strictly quasiconcave (Definition 6.10). Then f has at most one maximizer on C : the set $M = \arg \max_{x \in C} f(x)$ is empty or a singleton. If in addition C is compact and f is continuous, M is exactly one point.*

Proof. Assume, seeking a contradiction, that there are two distinct maximizers $x, y \in M$, $x \neq y$, with common value $v^* = f(x) = f(y)$. Take the midpoint $z = \frac{1}{2}x + \frac{1}{2}y$, which belongs to C by convexity. Strict quasiconcavity applies because $x \neq y$ and $\lambda = \frac{1}{2} \in (0, 1)$:

$$f(z) > \min\{f(x), f(y)\} = v^*.$$

But z is feasible and v^* is the maximum value on C , so $f(z) \leq v^*$. The two inequalities together give $v^* < f(z) \leq v^*$, which is absurd. Hence there cannot be two distinct maximizers.

For the last claim: if C is nonempty and compact and f is continuous, the Weierstrass theorem guarantees that the maximum is attained, so $M \neq \emptyset$; together with the above, M is a single point. $\square$

Corollary 6.32 (Marshallian demand is a function). *Let $u: \mathbb{R}_+^n \rightarrow \mathbb{R}$ be continuous and strictly quasiconcave, and consider the consumer problem*

$$\max_{x \in \mathbb{R}_+^n} u(x) \quad \text{subject to} \quad \langle p, x \rangle \leq w,$$

with prices $p \in \mathbb{R}_{++}^n$ and wealth $w > 0$. Then the budget set $B(p, w) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq w\}$ is nonempty, convex and compact, and the problem has exactly one solution $x(p, w)$. That is, demand is a function $x: \mathbb{R}_{++}^n \times \mathbb{R}_{++} \rightarrow \mathbb{R}_+^n$, not a correspondence.

Proof. $B(p, w)$ is nonempty: it contains $x = 0$ (since $\langle p, 0 \rangle = 0 \leq w$). *Convex:* it is the intersection of the orthant $\mathbb{R}_+^n$ (convex) with the halfspace $\{x : \langle p, x \rangle \leq w\}$ (convex), and the intersection of convex sets is convex (ch. 2). *Closed:* it is an intersection of preimages of closed sets under continuous functions. *Bounded:* if $x \in B(p, w)$ then, since all coordinates are nonnegative and $p_i > 0$,

$$p_i x_i \leq \sum_{j=1}^n p_j x_j \leq w \quad \implies \quad 0 \leq x_i \leq \frac{w}{p_i} \quad \text{for each } i,$$

so $B(p, w) \subseteq \prod_i [0, w/p_i]$, a bounded box. By Heine–Borel, $B(p, w)$ is compact. Applying Theorem 6.31 with $C = B(p, w)$ and $f = u$ gives existence (Weierstrass) and uniqueness (strict quasiconcavity). $\square$

Remark 6.33 (The ordinal version: strictly convex preferences). By Theorem 6.20, quasiconcavity of u is equivalent to convexity of the represented preferences. The strict parallel is exact: u is strictly quasiconcave if and only if preferences $\succsim$ are *strictly convex* in the sense of MWG [9] (Definition 3.B.1 and the discussion that follows): if $x \neq y$, $x \succsim z$ and $y \succsim z$, then $\lambda x + (1 - \lambda)y \succ z$ for every $\lambda \in (0, 1)$. Translated: mixing two equally good bundles produces something *strictly* better, not merely equally good. The nonstrict version allows indifference curves with straight pieces (like the linear utility of Example 6.18, where demand is a correspondence that can jump from one budget vertex to the other); the strict version rules them out.

Intuition

Why does an economist care so much that demand be a function and not a correspondence? Three reasons, all from MWG ch. 3:

- (i) **Comparative statics.** You can only differentiate a function: $\partial x_i / \partial p_j$, the Slutsky matrix, the signs of the substitution and income effects. With a correspondence, the question “how does demand change if the price rises?” has no single answer.
- (ii) **Empirical identification.** When estimating a demand system, one assumes that each (p, w) maps to one observed bundle. Multiplicity would break the link between data and parameters.
- (iii) **Existence of equilibrium.** With single valued continuous demand, aggregate excess demand is a continuous function and the *Brouwer* fixed point theorem is enough; with correspondences you need *Kakutani*, which is more delicate. Both routes exist, but the first one is the one used in a first course.

The price of this convenience is a substantive assumption on tastes (strict convexity) that rules out, for example, perfect substitutes.

Careful (common mistake)

Strict quasiconcavity is *not* the same as “quasiconcave and strictly increasing”, it does not imply strict concavity, and it does not follow from the bordered Hessian inequality holding weakly. Two frequent traps:

- Cobb–Douglas $u = x^a y^b$ with $a, b > 0$ is strictly quasiconcave on $\mathbb{R}_{++}^2$ (by Theorem 6.23(b), checked in Example 6.24) for *any* $a, b > 0$, even though it is concave only if $a + b \leq 1$. Uniqueness of demand does not require concavity.
- Leontief $u = \min\{x, y\}$ (Example 6.17) is concave, hence quasiconcave, but *not* strictly quasiconcave: on the horizontal piece of an indifference curve, $u(\lambda x + (1 - \lambda)y) = \min\{u(x), u(y)\}$ with equality. Even so its demand is unique, because the optimum always sits at the kink: strict quasiconcavity is *sufficient*, not necessary, for uniqueness.

6.8.3 The Berge maximum theorem

Corollary 6.32 tells us that demand is well defined *for each* (p, w) . The other half is missing: that it *varies continuously* with (p, w) . That is the content of the maximum theorem of Claude Berge (1959), the standard tool for propagating continuity from the data of a problem to its optimal value and its solution.

The statement needs to talk about *correspondences* (set valued maps), because the feasible set depends on parameters and in general is not a single point.

Definition 6.34 (Correspondence and hemicontinuity). Let $\Theta \subseteq \mathbb{R}^m$ (the parameter space) and $X \subseteq \mathbb{R}^n$. A *correspondence* $\Gamma: \Theta \rightrightarrows X$ assigns to each $\theta \in \Theta$ a subset $\Gamma(\theta) \subseteq X$. We say that:

- Γ is *upper hemicontinuous* (uhc) at θ_0 if for every open set $V \subseteq X$ with $\Gamma(\theta_0) \subseteq V$ there is a neighborhood U of θ_0 such that $\Gamma(\theta) \subseteq V$ for all $\theta \in U$. If Γ has compact values and X is compact, this is equivalent to the *sequential* characterization: if $\theta_k \rightarrow \theta_0$, $x_k \in \Gamma(\theta_k)$ and $x_k \rightarrow x_0$, then $x_0 \in \Gamma(\theta_0)$ (closed graph).
- Γ is *lower hemicontinuous* (lhc) at θ_0 if for every open set V with $\Gamma(\theta_0) \cap V \neq \emptyset$ there is a neighborhood U of θ_0 such that $\Gamma(\theta) \cap V \neq \emptyset$ for all $\theta \in U$. Sequentially: for every $x_0 \in \Gamma(\theta_0)$ and every sequence $\theta_k \rightarrow \theta_0$ there are $x_k \in \Gamma(\theta_k)$ with $x_k \rightarrow x_0$.
- Γ is *continuous* if it is both uhc and lhc.

The intuition behind the two concepts: upper hemicontinuity forbids the set from *exploding* in the limit (new points appearing all at once); lower hemicontinuity forbids it from *imploding* (points that were there disappearing all at once). An ordinary function seen as a single valued correspondence is $\text{uhc} \Leftrightarrow \text{lhc} \Leftrightarrow \text{continuous}$.

Theorem 6.35 (Maximum theorem, Berge). Let $\Theta \subseteq \mathbb{R}^m$ and $X \subseteq \mathbb{R}^n$, and consider

$$v(\theta) = \max_{x \in \Gamma(\theta)} f(x, \theta), \quad x^*(\theta) = \arg \max_{x \in \Gamma(\theta)} f(x, \theta).$$

Assume that

(H1) $f: X \times \Theta \rightarrow \mathbb{R}$ is continuous (jointly in (x, θ));

(H2) $\Gamma: \Theta \rightrightarrows X$ is continuous (uhc and lhc) at θ ;

(H3) $\Gamma(\theta)$ is nonempty and compact for each θ .

Then:

- (a) the value function $v: \Theta \rightarrow \mathbb{R}$ is well defined (the maximum is attained) and is continuous;
- (b) the solution correspondence x^* has nonempty compact values and is upper hemicontinuous;
- (c) if in addition $x^*(\theta)$ is a singleton for every θ , then x^* is a continuous function.

Outline of the proof. We do not give it in full (see [1] and, in economic terms, MWG [9], appendix M.K), but the mechanism can be seen in two halves, and it is worth looking at because it explains why both hemicontinuities are needed.

v is upper semicontinuous, thanks to Γ being uhc and compact valued. Let $\theta_k \rightarrow \theta_0$ and choose $x_k \in x^*(\theta_k)$, so that $v(\theta_k) = f(x_k, \theta_k)$; passing at the start to a subsequence that realizes $\limsup_k v(\theta_k)$, we may assume that $v(\theta_k)$ converges to that $\limsup$. By compactness (the x_k live in a compact set for large k , by uhc) there is a subsequence $x_{k_j} \rightarrow x_0$, and uhc gives $x_0 \in \Gamma(\theta_0)$. By continuity of f , $v(\theta_{k_j}) = f(x_{k_j}, \theta_{k_j}) \rightarrow f(x_0, \theta_0) \leq v(\theta_0)$, since x_0 is feasible at θ_0 . This gives $\limsup_k v(\theta_k) \leq v(\theta_0)$.

v is lower semicontinuous, thanks to Γ being lhc. Let $x_0 \in x^*(\theta_0)$, that is $f(x_0, \theta_0) = v(\theta_0)$. By lhc there are $x_k \in \Gamma(\theta_k)$ with $x_k \rightarrow x_0$. Since x_k is feasible (not necessarily optimal) at θ_k , $v(\theta_k) \geq f(x_k, \theta_k) \rightarrow f(x_0, \theta_0) = v(\theta_0)$, so $\liminf_k v(\theta_k) \geq v(\theta_0)$. Putting both halves together, $v(\theta_k) \rightarrow v(\theta_0)$: v is continuous.

x^* is uhc. With v already continuous, the first argument is recycled: if $\theta_k \rightarrow \theta_0$, $x_k \in x^*(\theta_k)$ and $x_k \rightarrow x_0$, then $x_0 \in \Gamma(\theta_0)$ (uhc of Γ) and $f(x_0, \theta_0) = \lim_k f(x_k, \theta_k) = \lim_k v(\theta_k) = v(\theta_0)$, so $x_0 \in x^*(\theta_0)$: the graph is closed. Finally, (c) is immediate: a single valued uhc correspondence is a continuous function. $\square$

Careful (common mistake)

Berge does **not** say that x^* is lower hemicontinuous, and in general it is not: the solution can “jump”. Example: maximize $f(x, \theta) = \theta x$ over $\Gamma(\theta) = [-1, 1]$. For $\theta > 0$ the solution is $x^* = 1$; for $\theta < 0$ it is $x^* = -1$; at $\theta = 0$ it is all of $[-1, 1]$. The correspondence is uhc (at 0 the set “grows” but no points appear outside the limit) but not lhc. Microeconomic translation: with perfect substitutes, demand jumps from one end of the budget line to the other when the price ratio crosses the threshold. Strict quasiconcavity is exactly what removes the jump: it makes x^* single valued and, by (c), continuous.

Corollary 6.36 (Continuity of Marshallian demand). *Let $u: \mathbb{R}_+^n \rightarrow \mathbb{R}$ be continuous and strictly quasiconcave. Then the demand $x(p, w)$ of Corollary 6.32 is a continuous function of (p, w) on $\mathbb{R}_{++}^n \times \mathbb{R}_{++}$, and the indirect utility function $v(p, w) = u(x(p, w))$ is continuous.*

Proof. We apply Theorem 6.35 with $\theta = (p, w)$, $\Theta = \mathbb{R}_{++}^n \times \mathbb{R}_{++}$, $f(x, \theta) = u(x)$ (continuous, and trivially jointly so because it does not depend on θ) and $\Gamma(p, w) = B(p, w)$. Hypothesis

(H3) was already checked in Corollary 6.32: $B(p, w)$ is nonempty and compact. What is missing is (H2), continuity of the budget correspondence, which we sketch.

B is *uhc* at (p_0, w_0) with $p_0 \gg 0$, $w_0 > 0$: take $(p_k, w_k) \rightarrow (p_0, w_0)$ and $x_k \in B(p_k, w_k)$ with $x_k \rightarrow x_0$. Then $x_0 \geq 0$ (the orthant is closed) and, passing to the limit in $\langle p_k, x_k \rangle \leq w_k$ (the inner product is continuous), $\langle p_0, x_0 \rangle \leq w_0$: hence $x_0 \in B(p_0, w_0)$. Moreover, for (p_k, w_k) near (p_0, w_0) all the sets $B(p_k, w_k)$ fit inside one bounded box, which legitimizes the sequential argument.

B is *lhc* at (p_0, w_0) with $p_0 \gg 0$, $w_0 > 0$: let $x_0 \in B(p_0, w_0)$ and $(p_k, w_k) \rightarrow (p_0, w_0)$. If $\langle p_0, x_0 \rangle < w_0$ (interior of the budget set), then for large k we have $\langle p_k, x_0 \rangle < w_k$ and it is enough to take $x_k = x_0$. If $\langle p_0, x_0 \rangle = w_0$, define

$$x_k = \min \left\{ 1, \frac{w_k}{\langle p_k, x_0 \rangle} \right\} x_0,$$

which is feasible at (p_k, w_k) (it shrinks the bundle just enough to fit) and converges to x_0 because $w_k / \langle p_k, x_0 \rangle \rightarrow w_0 / w_0 = 1$. The hypothesis $w_0 > 0$ is essential: it is what prevents dividing by zero and what makes the budget set have nonempty interior.

With (H1)–(H3) verified, Berge gives continuity of v and upper hemicontinuity of x^* ; strict quasiconcavity makes x^* single valued (Theorem 6.31), and part (c) of the theorem turns that into continuity. $\square$

Careful (common mistake)

The hypotheses $p \gg 0$ and $w > 0$ are not decorative, and it is worth seeing exactly what fails without them. If $p_i = 0$ is allowed, the set $B(p, w)$ stops being *bounded* (the free good can be consumed without limit): compactness in (H3) and upper hemicontinuity both fail, and demand can explode. If in addition $w = 0$, *lower* hemicontinuity fails too: with p_0 having a zero coordinate, $B(p_0, 0)$ contains nonzero bundles, while $B(p_k, 0) = \{0\}$ for any $p_k \gg 0$, so those bundles cannot be approximated from the limit. (By contrast, with $p_0 \gg 0$ and $w_0 = 0$ we have $B(p_0, 0) = \{0\}$ and there is no lhc problem: just take $x_k = 0$.) In general equilibrium this is the reason for the *survival assumptions* (each consumer has an endowment with strictly positive value, $\langle p, \omega_i \rangle > 0$) and for the demand truncation tricks that appear in MWG ch. 17: without them, Berge cannot be applied on the boundary of the price simplex and the fixed point argument collapses.

Intuition

The full chain, worth memorizing as a single story:

$$\begin{array}{c} \underbrace{u \text{ quasiconcave}}_{\text{convex contour sets}} \implies \underbrace{x(p, w) \text{ convex valued}}_{\text{Kakutani}} \\ \\ \underbrace{u \text{ strictly quasiconcave}}_{\text{no flat pieces}} \implies \underbrace{x(p, w) \text{ unique}}_{\text{a function}} \xrightarrow{\text{Berge}} \underbrace{x(p, w) \text{ continuous}}_{\text{Brouwer}}. \end{array}$$

And continuous demand is the input of the fixed point theorem that produces Walrasian

equilibrium. The whole existence of general equilibrium rests, at its mathematical layer, on convexity of feasible sets + quasiconcavity of utilities + Berge.

6.8.4 Arrow–Enthoven: when KKT is enough

We already stated the short version in Theorem 6.29. We now give the statement with the extra conditions written out, because the delicate point of the result is exactly that fine print: *quasiconcavity alone is not enough* (remember x^3), and a nondegeneracy condition must be added.

Theorem 6.37 (Arrow–Enthoven, 1961). *Consider the program*

$$\max_{x \geq 0} f(x) \quad \text{subject to} \quad g_j(x) \geq 0, \quad j = 1, \dots, k,$$

with $f, g_1, \dots, g_k: \mathbb{R}_+^n \rightarrow \mathbb{R}$ differentiable, f quasiconcave and each g_j quasiconcave. Assume that $x^* \geq 0$ satisfies the Karush–Kuhn–Tucker conditions: there are $\mu_1, \dots, \mu_k \geq 0$ with

$$\frac{\partial f}{\partial x_i}(x^*) + \sum_{j=1}^k \mu_j \frac{\partial g_j}{\partial x_i}(x^*) \leq 0, \quad x_i^* \left[\frac{\partial f}{\partial x_i}(x^*) + \sum_{j=1}^k \mu_j \frac{\partial g_j}{\partial x_i}(x^*) \right] = 0, \quad \mu_j g_j(x^*) = 0, \quad g_j(x^*) \geq 0.$$

If in addition at least one of the following Arrow–Enthoven nondegeneracy conditions holds:

- (i) $\partial f / \partial x_i(x^*) < 0$ for some variable i ;
- (ii) $\partial f / \partial x_i(x^*) > 0$ for some variable i that can be increased without leaving the feasible region (that is, x_i is not at its implicit upper bound);
- (iii) $\nabla f(x^*) \neq 0$ and f is twice differentiable in a neighborhood of x^* ;
- (iv) f is concave,

then x^* is a global maximum of the program.

Remark 6.38 (How to read the extra conditions). All the variants (i)–(iv) say the same thing in different disguises: *prevent the KKT point from being a “false stationary point” like the $x = 0$ of x^3* . In that counterexample, $f(x) = x^3$ is quasiconcave and $x^* = 0$ trivially satisfies KKT (no active constraints, with $f'(0) = 0$), but it is not a maximum; none of the conditions (i)–(iv) holds there, because $\nabla f(x^*) = 0$ and f is not concave. In practice, condition (iii), a nonzero gradient, is the one always used, and it is automatic in microeconomics because of monotonicity of preferences (“more is better” $\Rightarrow$ positive marginal utility $\Rightarrow \nabla u \neq 0$). Note also the link with the previous section: under (iii) plus regularity, quasiconcavity is in fact upgraded to pseudoconcavity (Definition 6.26), and then the Mangasarian version (a) of Theorem 6.29 suffices.

Intuition

The role of Arrow–Enthoven in the curriculum is that of an *official permit*. In ch. 9 we will prove that, under a constraint qualification, KKT is *necessary*: every optimum satisfies it. That is what allows you to write the Lagrangian. Arrow–Enthoven closes the circle from

the other side: under quasiconcavity plus nondegeneracy, KKT is also *sufficient*, so the point you solve for *is* the global optimum and there is no need to compare candidates or check second order conditions. Without this theorem, every consumer problem would end with an uncomfortable “and now one would have to verify that this point is a maximum”. With it, as soon as you check quasiconcavity (for example, with the bordered Hessian, Theorem 6.23) and see that the gradient does not vanish, the problem is closed.

6.8.5 The single crossing condition (Spence–Mirrlees)

The last application is the most modern one and the one that will pay off most in courses on contracts, signaling and mechanism design. So far we have imposed ordinal conditions on *one* indifference map (quasiconcavity: convex upper contour sets). The single crossing condition imposes an ordinal condition on a *family* of indifference maps indexed by a type θ : it orders them.

The setting

An agent of type $\theta \in \Theta \subseteq \mathbb{R}$ (ability, valuation, cost) chooses a pair $(q, t) \in Q \times T \subseteq \mathbb{R}^2$, where q is a “real” variable (quality, quantity, education, effort) and t is a monetary transfer (price paid, wage received). Utility is $U(q, t, \theta)$. The type is *private information*: the principal (firm, regulator, seller) only knows the distribution of θ .

Definition 6.39 (Single crossing condition, differential version). Let $U: Q \times T \times \Theta \rightarrow \mathbb{R}$ be of class C^2 with $U_t \neq 0$. The *marginal rate of substitution* between q and t for type θ at the point (q, t) is

$$\text{MRS}(q, t, \theta) = -\frac{U_q(q, t, \theta)}{U_t(q, t, \theta)},$$

that is, the slope dt/dq of the indifference curve of type θ in the (q, t) plane. We say that U satisfies the *single crossing condition* or *Spence–Mirrlees condition* if the MRS is *strictly monotone in the type*:

$$\frac{\partial}{\partial \theta} \left[-\frac{U_q(q, t, \theta)}{U_t(q, t, \theta)} \right] \neq 0 \quad \text{with constant sign, for all } (q, t, \theta).$$

Definition 6.40 (Single crossing, ordinal version). U satisfies the *single crossing property* if for all $q' > q$, all t', t and all $\theta' > \theta$:

$$U(q', t', \theta) \geq U(q, t, \theta) \quad \implies \quad U(q', t', \theta') > U(q, t, \theta').$$

In words: *if the low type is willing to accept the package with higher q , the high type strictly prefers it*. This version needs no differentiability and is the one used in the general theorems on monotone comparative statics.

Why “single crossing”: the graphical reading

Draw the (q, t) plane with q on the horizontal axis. Each type has its family of indifference curves. The condition in Definition 6.39 says that, *at every point of the plane*, the slope of the curve of type θ' is strictly different (say, always smaller) than that of type θ when $\theta' > \theta$. Now take one indifference curve of the low type and one of the high type and superimpose them: if they touched at two points A and B , then between A and B one of them would have to be above and then below, which forces the slopes to become equal or to reverse their order at some intermediate point, exactly what strict monotonicity of the MRS forbids. Conclusion: *two indifference curves of different types cross at most once*. Hence the name. And from that elementary geometric fact comes the whole machinery of information economics: the principal, when offering a menu of contracts, can rely on “more q ” identifying “higher θ ”, because the tastes of the types are *ordered* by slope and do not get tangled.

Example 6.41 (Spence education signaling). The canonical model: $q = e \geq 0$ is education (which does not raise productivity, it only costs), $t = w$ is the wage, $\theta > 0$ is productivity or ability, and

$$u(w, e, \theta) = w - \frac{e^2}{\theta}.$$

The MRS between education and wage is

$$\left. \frac{dw}{de} \right|_{u \text{ constant}} = -\frac{u_e}{u_w} = -\frac{-2e/\theta}{1} = \frac{2e}{\theta}, \quad \frac{\partial}{\partial \theta} \left(\frac{2e}{\theta} \right) = -\frac{2e}{\theta^2} < 0 \quad (e > 0).$$

The MRS is *strictly decreasing* in θ : single crossing holds. Interpretation: the indifference curves of high types are *flatter*; to give up one dollar of wage, a high type asks for a smaller reduction in education than a low type, because education is cheaper for that type. That asymmetry of slopes is what makes a *separating* equilibrium credible: there is a level of education e^* that the high type finds worth reaching and the low type does not (Exercise 6.48). See MWG [9], ch. 13 (sections 13.C and 13.D) and [10], ch. 8.

Example 6.42 (Price discrimination: Mussa–Rosen and Maskin–Riley). In screening the order of play is reversed: the principal (a monopolist) moves first by offering a menu $\{(q, t)\}$ and the informed agent chooses. In Mussa–Rosen (1978) q is *quality* and consumer utility is

$$U(q, t, \theta) = \theta v(q) - t, \quad v' > 0, \quad v'' < 0,$$

with θ the marginal willingness to pay for quality. Here $U_q = \theta v'(q)$, $U_t = -1$, so $\text{MRS} = \theta v'(q)$, *strictly increasing* in θ : single crossing again (with the opposite sign to Spence, because now t is a payment and not an income). In Maskin–Riley (1984) q is *quantity* and the result is second degree price discrimination: nonlinear tariffs, volume discounts. The central technical consequence in both cases is the *characterization theorem for incentive compatibility*: a menu $\{(q(\theta), t(\theta))\}$ is incentive compatible if and only if (a) $q(\cdot)$ is nondecreasing and (b) the envelope condition $\frac{d}{d\theta} U(q(\theta), t(\theta), \theta) = U_\theta(q(\theta), t(\theta), \theta)$ holds. Without single crossing, the local condition (the envelope) does *not* imply the global one (that no type wants to imitate a distant type), and the “first order approach” collapses. See MWG [9], ch. 14 (section 14.C, the principal agent

model with adverse selection) and [10], ch. 8.

Remark 6.43 (What this has to do with generalized concavity). Three connections, in order of importance:

- (a) **It is an ordinal condition, like quasiconcavity.** If you replace $U(\cdot, \cdot, \theta)$ by $\varphi_\theta(U(\cdot, \cdot, \theta))$ with φ_θ strictly increasing (a monotone transformation that is *different for each type*), the MRS does not change: the factors φ'_θ cancel in the ratio $-U_q/U_t$. This is the same phenomenon as in Theorem 6.13: we are talking about the *shape of the indifference curves*, not about the numbers that label them. This is why single crossing belongs fully to the generalized concavity family.
- (b) **Quasiconcavity orders *within* one map; single crossing orders *across* maps.** Quasiconcavity says that the set $\{(q, t) : U(q, t, \theta) \geq \alpha\}$ is convex (indifference curves with the “right” curvature), which guarantees that each type’s problem has a unique, well behaved solution (Theorem 6.31). Single crossing adds that those solutions are ordered monotonically in θ . The two conditions are complementary and both are ordinal invariants.
- (c) **It is the “single crossing” version of supermodularity.** If $U(q, t, \theta) = \theta v(q) - t$, then $U_{q\theta} = v'(q) > 0$: *increasing differences* in (q, θ) . Single crossing is the ordinal generalization of that cardinal condition, just as quasiconcavity is the ordinal generalization of concavity. In the Milgrom–Shannon theorem on monotone comparative statics, the exact hypothesis is “quasisupermodularity + single crossing”, and it is *necessary and sufficient* for the $\arg \max$ to increase with the parameter. The parallel with this chapter is complete: cardinal hypotheses (concavity, supermodularity) are replaced by their ordinal counterparts (quasiconcavity, single crossing), which is all that economic theory can legitimately assume.

6.8.6 A note on concavification

One natural question remains, and it closes the chapter. If quasiconcavity is all that economics can assume, but concavity is more convenient to work with, could we always *manufacture* concavity by choosing the utility representation well? That is: given a quasiconcave f , is there a strictly increasing $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ such that $\varphi \circ f$ is concave? Such a φ is called a *concavification* of f .

Sometimes yes, and spectacularly so: the Cobb–Douglas function $f(x, y) = xy$ is not concave, but $\varphi = \ln$ gives $\ln f = \ln x + \ln y$, which is (this was Example 6.1, with which we opened the chapter). The same happens with every Cobb–Douglas $x^a y^b$ and with many families used in applications. But the general answer is **no**: Fenchel (1953) built continuous quasiconcave functions that admit no concavification at all, and the problem of characterizing those that do admit one (studied by de Finetti, Debreu (1976, “Least concave utility functions”) and Kannai (1977)) is genuinely hard. A classical sufficient condition: if f is *pseudoconcave* of class C^2 with a nonvanishing gradient on a *compact* convex set, then it is concavifiable.

Intuition

Why does it almost never matter in economics that concavification may fail? Because *all the results we need were already proved directly from quasiconcavity*: convex upper contour sets, convexity of the set of optima (Theorem 6.30), uniqueness of demand (Theorem 6.31), continuity via Berge (Corollary 6.36), sufficiency of KKT (Theorem 6.37). None of them required concavity. Concavity only becomes indispensable when the model is *cardinal* and we are therefore no longer free to transform utility: under uncertainty, von Neumann–Morgenstern utility is determined up to affine transformations, and there concavity *means* risk aversion, a real economic content and not an artifact of the representation. Rule of thumb: *in deterministic consumer theory, quasiconcave is enough and concave is a luxury; under risk, concave is an assumption with content and cannot be manufactured.*

Chapter summary

Key results of the chapter, with exact hypotheses.

- **Two equivalent definitions.** For $S \subseteq \mathbb{R}^n$ convex and $f: S \rightarrow \mathbb{R}$ (with no further hypotheses: neither continuity nor differentiability), the following are equivalent (Theorem 6.5): (i) the upper contour set $S_\alpha(f) = \{x \in S : f(x) \geq \alpha\}$ is convex for every $\alpha \in \mathbb{R}$; (ii) $f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$ for all $x, y \in S$, $\lambda \in [0, 1]$. The *strict* version (Definition 6.10) requires strict inequality for $x \neq y$ and $\lambda \in (0, 1)$.
- **Concave $\Rightarrow$ quasiconcave, and the converse is false.** Every concave function is quasiconcave (Proposition 6.11); $f(x) = x^3$ and $f(x, y) = xy$ on $\mathbb{R}_{++}^2$ are quasiconcave and not concave (Example 6.12). The inclusion is *strict*, and that is why this chapter exists.
- **Ordinal invariance.** If f is quasiconcave and g is *nondecreasing* on a set T containing the image of f , then $g \circ f$ is quasiconcave (Theorem 6.13); if f is strictly quasiconcave and g is *strictly increasing*, so is $g \circ f$. Concavity does *not* have this property: $\ln(xy)$ is concave and xy is not. Economic corollary: quasiconcavity is a property of *preferences*, not of the representation (Theorem 6.20).
- **The sum does NOT preserve quasiconcavity.** If f and g are quasiconcave, $f + g$ may fail to be: $f(x) = x^3$ and $g(x) = -x$ are both monotone (hence quasiconcave) but $f + g = x^3 - x$ is not (Exercise 6.46). What *is* preserved: the pointwise *minimum*, composition with affine maps, restriction to convex sets and monotone transformations (Proposition 6.14). The *maximum* does not preserve it either.
- **First order characterization.** If S is convex and open and f is *differentiable* on S : f is quasiconcave if and only if for all $x, y \in S$,

$$f(y) \geq f(x) \implies \langle \nabla f(x), y - x \rangle \geq 0$$

(Theorem 6.21). Reading: the gradient at x makes a nonobtuse angle with every

direction pointing to points that are at least as good; the gradient “supports” the upper contour set.

- **Bordered Hessian: necessary vs sufficient.** Let S be convex and *open* and $f \in C^2(S)$, and let $B_1, \dots, B_n$ be the bordered minors of Definition 6.22. Then (Theorem 6.23):

- *Necessary (weak).* If f is quasiconcave on S , then $(-1)^r \det B_r(x) \geq 0$ for every $r = 1, \dots, n$ and every $x \in S$; that is $\det B_1 \leq 0$, $\det B_2 \geq 0$, $\det B_3 \leq 0, \dots$ (alternating signs starting from negative, with weak inequality). It is only good for **ruling out**: if it fails at a single point, f is not quasiconcave.
- *Sufficient (strict).* If $(-1)^r \det B_r(x) > 0$ for every r and every $x \in S$ ($\det B_1 < 0$, $\det B_2 > 0$, $\det B_3 < 0, \dots$), then f is quasiconcave, and also pseudoconcave and strictly quasiconcave. It is only good for **asserting**.
- The domain matters: the signs must be checked *on the whole convex open set* S , not at one point. And there is a gap between the two conditions (when some determinant vanishes) where the test does not decide.

For quasiconvexity: necessary $\det B_r \leq 0$ for every r ; sufficient $\det B_r < 0$ for every r .

- **Hierarchy (for f differentiable on a convex open set).**

$$\text{concave} \Rightarrow \text{pseudoconcave} \Rightarrow \text{quasiconcave},$$

and none reverses (Theorem 6.27): $x^3 + x$ is pseudoconcave and not concave; x^3 is quasiconcave and not pseudoconcave. What the intermediate class buys: under pseudoconcavity, $\nabla f(x^*) = 0 \Rightarrow x^*$ is a *global* maximum (Proposition 6.28) and KKT is sufficient (Theorems 6.29 and 6.37; see ch. 9).

- **Applications to demand.** With C nonempty and convex: if f is quasiconcave, $\arg \max_{x \in C} f$ is *convex* (Theorem 6.30); if f is *strictly* quasiconcave, it has *at most one* element (Theorem 6.31), and if in addition C is compact and f is continuous, exactly one. With u continuous and strictly quasiconcave, $p \gg 0$ and $w > 0$: Marshallian demand is a *function* (Corollary 6.32) and, by the Berge maximum theorem (Theorem 6.35: continuous objective, continuous constraint correspondence with nonempty compact values), it is *continuous*, as is indirect utility (Corollary 6.36). That continuity is the input of the fixed point argument that gives existence of Walrasian equilibrium.
- **Single crossing (Spence–Mirrlees).** With types $\theta \in \Theta \subseteq \mathbb{R}$ and utility $U(q, t, \theta)$: the MRS $-U_q/U_t$ is strictly monotone in θ (Definition 6.39). It orders indifference curves by type, guarantees that two curves of different types cross at most once, and it is the hypothesis that makes the first order approach valid in signaling and screening (MWG [9], chs. 13 and 14; [10], ch. 8). Like quasiconcavity, it is an *ordinal* condition.

6.9 Solved exercises [P1]

Exercise 6.44 (Bordered Hessian, all the minors). Let $f: \mathbb{R}_{++}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = xy + x + y.$$

(a) Compute all the bordered leading principal minors $\det B_1$ and $\det B_2$ and decide whether f is quasiconcave on $\mathbb{R}_{++}^2$, stating which part of Theorem 6.23 you use. (b) Is f concave? (c) Is it strictly quasiconcave? Pseudoconcave?

Solution. (a) *Derivatives.* Directly:

$$f_1 = y + 1, \quad f_2 = x + 1, \quad f_{11} = 0, \quad f_{22} = 0, \quad f_{12} = f_{21} = 1.$$

The first bordered minor is

$$\det B_1 = \det \begin{pmatrix} 0 & f_1 \\ f_1 & f_{11} \end{pmatrix} = 0 \cdot f_{11} - f_1^2 = -(y + 1)^2.$$

On $\mathbb{R}_{++}^2$ we have $y > 0$, so $(y + 1)^2 > 0$ and therefore $\det B_1 < 0$ *strictly*. The sign required by the test is $(-1)^1 \det B_1 > 0$, that is $\det B_1 < 0$: **it holds**.

The second minor is the 3×3 determinant

$$\det B_2 = \det \begin{pmatrix} 0 & f_1 & f_2 \\ f_1 & f_{11} & f_{12} \\ f_2 & f_{21} & f_{22} \end{pmatrix} = 2f_1f_2f_{12} - f_1^2f_{22} - f_2^2f_{11},$$

using the memorable formula for the case $n = 2$ (expansion along the first row; see Section 6.6). Substituting:

$$\det B_2 = 2(y + 1)(x + 1)(1) - (y + 1)^2 \cdot 0 - (x + 1)^2 \cdot 0 = 2(x + 1)(y + 1).$$

On $\mathbb{R}_{++}^2$ we have $x + 1 > 0$ and $y + 1 > 0$, so $\det B_2 > 0$ *strictly*. The required sign is $(-1)^2 \det B_2 > 0$, that is $\det B_2 > 0$: **it holds**.

Since the two signs alternate *strictly* $(-, +)$ on the *whole* convex open set $\mathbb{R}_{++}^2$, the **sufficient condition** of Theorem 6.23(b) lets us conclude that f is quasiconcave on $\mathbb{R}_{++}^2$. (We insist on the logical direction: we used (b), the sufficient part; part (a) of the theorem would only have served to rule out.)

(b) *Concave?* No. The ordinary Hessian is

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \det H = -1 < 0,$$

and a symmetric 2×2 matrix with negative determinant is *indefinite* (its eigenvalues are $+1$ and -1 : opposite signs). To be concave we would need H negative semidefinite, which requires $\det H \geq 0$. Direct alternative: along the line $y = x$, $\varphi(t) = f(t, t) = t^2 + 2t$ has $\varphi''(t) = 2 > 0$, that is, f is *strictly convex* on that line; a concave function must be concave on every line

contained in its domain.

(c) Yes to both: the sufficient condition of Theorem 6.23(b) delivers, in the same package, *strict* quasiconcavity and pseudoconcavity. An independent and instructive check: note that

$$f(x, y) + 1 = xy + x + y + 1 = (x + 1)(y + 1),$$

so $f = \psi((x + 1)(y + 1))$ with $\psi(s) = s - 1$ strictly increasing, and $(x + 1)(y + 1)$ is a shifted Cobb–Douglas function, strictly quasiconcave by Proposition 6.15 together with composition with the affine map $(x, y) \mapsto (x + 1, y + 1)$ (Proposition 6.14(c)). Along the way, this exhibits a concavification: $\ln(f(x, y) + 1) = \ln(x + 1) + \ln(y + 1)$ is concave, and $s \mapsto \ln(s + 1)$ is strictly increasing. $\square$

Exercise 6.45 (Quasiconcave but not concave, and its concavification). Let $f: \mathbb{R}_{++}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^2y^2$. (a) Prove that f is quasiconcave. (b) Prove that f is *not* concave. (c) Exhibit a strictly increasing transformation φ such that $\varphi \circ f$ is concave, and check it.

Solution. (a) We write f as a monotone transformation of a known function. Let

$$g(x, y) = x^{1/2}y^{1/2} \quad \text{on } \mathbb{R}_{++}^2, \quad \text{so that} \quad f(x, y) = (g(x, y))^4,$$

since $(x^{1/2}y^{1/2})^4 = x^2y^2$. The function $\varphi_0(s) = s^4$ is strictly increasing for $s > 0$ (its derivative $4s^3$ is positive there), and $g > 0$ on $\mathbb{R}_{++}^2$.

Now, g is *concave*: it is the Cobb–Douglas function $x^a y^b$ with $a = b = 1/2$, and its Hessian has

$$g_{11} = a(a - 1)x^{a-2}y^b = -\frac{1}{4}x^{-3/2}y^{1/2} < 0, \quad \det H_g = ab(1 - a - b)x^{2a-2}y^{2b-2} = 0$$

(the Cobb–Douglas determinant formula, with $a + b = 1$, gives exactly 0). A symmetric 2×2 matrix with $g_{11} < 0$ and $\det H_g = 0$ is negative semidefinite, so g is concave. By Proposition 6.11, g is quasiconcave, and by Theorem 6.13 (composition with the nondecreasing φ_0), $f = \varphi_0 \circ g$ is quasiconcave. Since φ_0 is strictly increasing and g is in fact strictly quasiconcave on $\mathbb{R}_{++}^2$ (Example 6.24), f is strictly quasiconcave too.

(b) f is not concave. Restrict it to the diagonal: $\varphi(t) = f(t, t) = t^4$ for $t > 0$, with $\varphi''(t) = 12t^2 > 0$. That is, f is *strictly convex* along the line $y = x$, and therefore it cannot be concave (a concave function is concave on every line of its domain). Structural reason: f is homogeneous of degree $4 > 1$, and a positive, not identically zero function that is homogeneous of degree > 1 is never concave, because $t \mapsto f(tx) = t^4 f(x)$ grows faster than linearly.

(c) Take $\varphi(s) = s^{1/4}$, strictly increasing for $s > 0$ (derivative $\frac{1}{4}s^{-3/4} > 0$). Then

$$(\varphi \circ f)(x, y) = (x^2y^2)^{1/4} = x^{1/2}y^{1/2} = g(x, y),$$

which we just proved concave. Another valid concavification is $\varphi(s) = \ln s$: $\ln f = 2 \ln x + 2 \ln y$, a sum of two concave functions, hence concave. The moral of the exercise: the *same* preference (the same upper contour sets, the same indifference curves, the same demand) has both concave and nonconcave representations; concavity is a property of the numerical label, quasiconcavity

is a property of the preference. $\square$

Exercise 6.46 (Sum (and maximum) do not preserve quasiconcavity). (a) Build explicitly two quasiconcave functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ whose sum is not quasiconcave, exhibiting a nonconvex upper contour set. (b) Do the same with the pointwise *maximum*, and explain why the *minimum* does work.

Solution. (a) Take

$$f(x) = x^3, \quad g(x) = -x, \quad x \in \mathbb{R}.$$

Both are quasiconcave. Every monotone function on $\mathbb{R}$ is quasiconcave (Example 6.8): f is strictly increasing and g is strictly decreasing, and in both cases the upper contour sets are half lines, $\{x : x^3 \geq \alpha\} = [\alpha^{1/3}, \infty)$ and $\{x : -x \geq \alpha\} = (-\infty, -\alpha]$, which are convex.

The sum is not. Let $h = f + g$, $h(x) = x^3 - x$. Factor it:

$$h(x) = x(x^2 - 1) = x(x - 1)(x + 1),$$

whose zeros are $-1, 0, 1$. Checking signs: $h(x) \geq 0$ exactly when $x \in [-1, 0] \cup [1, \infty)$. Indeed, for $x < -1$ the three factors are negative and the product is negative; on $(-1, 0)$ two factors are negative and one positive, so the product is positive; on $(0, 1)$ one is negative and two positive, so the product is negative; and for $x > 1$ all three are positive. You can confirm this with test values: $h(-2) = -6 < 0$, $h(-1/2) = 3/8 > 0$, $h(1/2) = -3/8 < 0$, $h(2) = 6 > 0$. Therefore the upper contour set at level $\alpha = 0$ is

$$S_0(h) = \{x \in \mathbb{R} : x^3 - x \geq 0\} = [-1, 0] \cup [1, \infty),$$

which is **not convex**: it contains $x = 0$ and $y = 1$ but not their midpoint $z = 1/2$, since $h(1/2) = 1/8 - 1/2 = -3/8 < 0$. By Definition 6.3, h is not quasiconcave. Direct check with the minimum inequality:

$$h\left(\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 1\right) = -\frac{3}{8} < 0 = \min\{h(0), h(1)\}.$$

(b) *The maximum.* Now let $u(x) = x$ and $v(x) = -x$, both monotone and therefore quasiconcave. Their pointwise maximum is $m(x) = \max\{x, -x\} = |x|$, and

$$S_1(m) = \{x : |x| \geq 1\} = (-\infty, -1] \cup [1, \infty),$$

which is not convex (it contains -1 and 1 but not 0 , where $m(0) = 0 < 1$). So m is not quasiconcave; in fact $|x|$ is convex, and therefore *quasiconvex*.

Why does the minimum work? Because quasiconcavity is a property of *upper contour sets*, and the minimum turns upper contour sets into *intersections*:

$$\{x : \min\{f_1(x), f_2(x)\} \geq \alpha\} = \{x : f_1(x) \geq \alpha\} \cap \{x : f_2(x) \geq \alpha\} = S_\alpha(f_1) \cap S_\alpha(f_2),$$

an intersection of convex sets, which is convex (Proposition 6.14(a)). The maximum, by contrast, produces *unions* of upper contour sets, and the union of convex sets is not convex in general.

The sum has no simple description in terms of upper contour sets: that is why it fails, and why quasiconcavity, unlike concavity, does not define a *convex cone* of functions. $\square$

Exercise 6.47 (Cobb–Douglas demand and uniqueness). A consumer has utility $u(x, y) = x^\alpha y^{1-\alpha}$ with $\alpha \in (0, 1)$, prices $p, q > 0$ and wealth $w > 0$. (a) Derive Marshallian demand. (b) Justify rigorously that the optimum exists, that it is interior and that it is *unique*, stating which theorem of the chapter you use at each step. (c) Check that demand is continuous in (p, q, w) and comment on the link with the Berge theorem.

Solution. (a) *Derivation.* Since $u > 0$ on $\mathbb{R}_{++}^2$ and $\ln$ is strictly increasing, maximizing u is equivalent to maximizing

$$U(x, y) = \ln u(x, y) = \alpha \ln x + (1 - \alpha) \ln y,$$

which is concave (a sum of concave functions). Changing the objective is legitimate precisely because of Theorem 6.13: the monotone transformation changes neither the set of maximizers nor the contour sets.

The budget constraint binds ($px + qy = w$) because u is strictly increasing in each argument: if wealth were left over you could buy more x and do better. We form the Lagrangian $\mathcal{L} = \alpha \ln x + (1 - \alpha) \ln y + \lambda(w - px - qy)$ and take first order conditions:

$$\frac{\alpha}{x} = \lambda p, \quad \frac{1 - \alpha}{y} = \lambda q, \quad px + qy = w.$$

From the first two, $\alpha = \lambda px$ and $1 - \alpha = \lambda qy$; adding them, $1 = \lambda(px + qy) = \lambda w$, so $\lambda = 1/w$. Substituting back:

$$\boxed{x(p, q, w) = \frac{\alpha w}{p}, \quad y(p, q, w) = \frac{(1 - \alpha)w}{q}.$$

This is the classical result: with Cobb–Douglas the consumer spends a fixed fraction α of wealth on the first good, whatever the price is.

(b) *Existence, interiority, uniqueness.*

- *Existence:* $B(p, q, w)$ is nonempty, convex and compact (proved in Corollary 6.32) and u is continuous on $\mathbb{R}_{++}^2$; Weierstrass guarantees that the maximum is attained.
- *Interiority:* on the axes ($x = 0$ or $y = 0$) we have $u = 0$, while the interior point $(\frac{w}{2p}, \frac{w}{2q})$ is feasible and gives $u > 0$. So no boundary point with a zero coordinate can be optimal, and we may work on the open set $\mathbb{R}_{++}^2$, where the first order conditions make sense.
- *Uniqueness:* u is *strictly quasiconcave* on $\mathbb{R}_{++}^2$, as checked with the bordered Hessian in Example 6.24 (the minors alternate strictly for any positive exponents), and Theorem 6.23(b) delivers, along with strict quasiconcavity, pseudoconcavity. Then:

- by Theorem 6.31, the maximizer over the convex set $B(p, q, w)$ is unique: demand is a *function*;
- by pseudoconcavity of u and quasiconcavity (in fact, linearity) of the constraint, Theorem 6.29(a) guarantees that the point we solved for with the first order conditions is the *global* maximum, with no need for second order conditions.

Note that $u = x^\alpha y^{1-\alpha}$ is concave here (because $\alpha + (1 - \alpha) = 1$), but the argument above did not use that fact, and this is why it works just as well for $u = x^a y^b$ with $a + b > 1$, where concavity fails and strict quasiconcavity does not.

(c) *Continuity.* The formulas $\alpha w/p$ and $(1 - \alpha)w/q$ are quotients of continuous functions with nonzero denominators on $\mathbb{R}_{++}^2 \times \mathbb{R}_{++}$: continuous. This is exactly what Corollary 6.36 predicts with no computation at all: continuous objective + continuous budget correspondence with compact values + unique solution $\Rightarrow$ continuous demand. Indirect utility turns out to be

$$v(p, q, w) = \left(\frac{\alpha w}{p} \right)^\alpha \left(\frac{(1 - \alpha)w}{q} \right)^{1-\alpha} = \alpha^\alpha (1 - \alpha)^{1-\alpha} w p^{-\alpha} q^{-(1-\alpha)},$$

also continuous, as part (a) of Berge ensures.

Appendix: the CES case. With $u(x, y) = (x^\rho + y^\rho)^{1/\rho}$, $\rho < 1$, $\rho \neq 0$ (Example 6.16), the first order conditions give $\rho x^{\rho-1} = \lambda p$, $\rho y^{\rho-1} = \lambda q$, so $(x/y)^{\rho-1} = p/q$ and, with the elasticity of substitution $\sigma = 1/(1 - \rho)$,

$$x(p, q, w) = \frac{p^{-\sigma} w}{p^{1-\sigma} + q^{1-\sigma}}, \quad y(p, q, w) = \frac{q^{-\sigma} w}{p^{1-\sigma} + q^{1-\sigma}}.$$

Uniqueness is justified in the same way: the CES function is strictly quasiconcave for $\rho < 1$, $\rho \neq 0$. It is also concave, but only thanks to an extra argument (it is nonnegative, quasiconcave and homogeneous of degree 1); and if you raise it to a power greater than 1 it stops being concave and still gives exactly the same demand, which is precisely the ordinal moral of the chapter. $\square$

Exercise 6.48 (Single crossing in the Spence signaling model). A worker of type (productivity) $\theta > 0$ chooses education $e \geq 0$ and receives a wage w , with utility

$$u(w, e, \theta) = w - \frac{e^2}{\theta}.$$

(a) Compute the marginal rate of substitution between education and wage and determine its monotonicity in θ . (b) Verify the single crossing condition in its ordinal version (Definition 6.40). (c) With two types $0 < \theta_L < \theta_H$ and wages $w = \theta$ in a separating equilibrium, find the range of education levels e^* that separate the types.

Solution. (a) *MRS.* The partial derivatives are

$$u_w = 1, \quad u_e = -\frac{2e}{\theta}.$$

Along an indifference curve, $du = u_w dw + u_e de = 0$, so

$$\text{MRS}(e, \theta) = \frac{dw}{de} \Big|_{u \text{ const.}} = -\frac{u_e}{u_w} = \frac{2e}{\theta} > 0 \quad (e > 0).$$

It is positive: to keep utility constant, more education requires more compensating wage (education is a “bad”). Its monotonicity in the type:

$$\frac{\partial}{\partial \theta} \left(\frac{2e}{\theta} \right) = -\frac{2e}{\theta^2} < 0 \quad \text{for } e > 0.$$

The MRS is *strictly decreasing* in θ : the Spence–Mirrlees condition of Definition 6.39 holds. Graphical reading in the (e, w) plane with e on the horizontal axis: the indifference curves of the high type are *flatter* at every point. Since the slope of one type dominates that of the other *at every point*, two indifference curves of different types can cross at most once.

Equivalently, in cardinal form: $\partial^2 u / \partial e \partial \theta = 2e / \theta^2 > 0$, that is, u has *increasing differences* in (e, θ) : the marginal cost of education is less burdensome the higher the type.

(b) *Ordinal version.* Let $e' > e \geq 0$, wages w', w and types $\theta_H > \theta_L > 0$. Assume the low type (weakly) accepts the package with more education:

$$u(w', e', \theta_L) \geq u(w, e, \theta_L) \iff (w' - w) \geq \frac{e'^2 - e^2}{\theta_L}.$$

We want to conclude that the high type prefers it *strictly*. We compute the difference in utility and add and subtract the low type term:

$$u(w', e', \theta_H) - u(w, e, \theta_H) = (w' - w) - \frac{e'^2 - e^2}{\theta_H} = \underbrace{\left[(w' - w) - \frac{e'^2 - e^2}{\theta_L} \right]}_{\geq 0 \text{ by hypothesis}} + \underbrace{(e'^2 - e^2)}_{> 0 \text{ since } e' > e \geq 0} \underbrace{\left(\frac{1}{\theta_L} - \frac{1}{\theta_H} \right)}_{> 0 \text{ since } \theta_L < \theta_H}.$$

The first bracket is nonnegative and the second term is strictly positive, so the sum is > 0 :

$$u(w', e', \theta_H) > u(w, e, \theta_H).$$

This is exactly Definition 6.40. Note that the proof used no differentiability, only algebra: the ordinal version is more robust than the differential one.

(c) *Separating equilibrium.* Consider the wage schedule

$$w(e) = \begin{cases} \theta_H & \text{if } e \geq e^*, \\ \theta_L & \text{if } e < e^*, \end{cases}$$

with the conjecture that the low type chooses $e_L = 0$ and earns θ_L , and the high type chooses $e_H = e^*$ and earns θ_H . For this to be an equilibrium we need two incentive compatibility constraints.

The low type does not want to imitate the high type:

$$\underbrace{\theta_L - \frac{0^2}{\theta_L}}_{\text{being truthful}} \geq \underbrace{\theta_H - \frac{(e^*)^2}{\theta_L}}_{\text{imitating}} \iff (e^*)^2 \geq \theta_L (\theta_H - \theta_L).$$

The high type prefers to get educated rather than pass as low:

$$\underbrace{\theta_H - \frac{(e^*)^2}{\theta_H}}_{\text{being truthful}} \geq \underbrace{\theta_L - 0}_{\text{pretending}} \iff (e^*)^2 \leq \theta_H (\theta_H - \theta_L).$$

Combining, any

$$e^* \in \left[\sqrt{\theta_L (\theta_H - \theta_L)}, \sqrt{\theta_H (\theta_H - \theta_L)} \right]$$

supports a separating equilibrium. The interval is *nonempty* because $\theta_L < \theta_H$ implies $\theta_L(\theta_H - \theta_L) < \theta_H(\theta_H - \theta_L)$, and that strict inequality is the single crossing condition at work: if the marginal cost of education did not depend on the type (if single crossing failed), the two thresholds would coincide and there would be no room to separate. The existence of a continuum of separating equilibria, and the resulting multiplicity, is the starting point of the discussions about refinements in MWG [9], ch. 13. $\square$

Exercise 6.49 (Classification: concave, pseudoconcave or only quasiconcave). For each function, decide whether it is concave, pseudoconcave, quasiconcave or none of these, justifying each answer:

- (i) $f(x) = x^3 + x$ on $\mathbb{R}$;
- (ii) $f(x) = x^3$ on $\mathbb{R}$;
- (iii) $f(x, y) = x^{1/2}y^{1/2}$ on $\mathbb{R}_{++}^2$;
- (iv) $f(x, y) = x^2y^2$ on $\mathbb{R}_{++}^2$;
- (v) $f(x) = x^2$ on $\mathbb{R}$;
- (vi) $f(x, y) = \min\{x, y\}$ on $\mathbb{R}^2$;
- (vii) $f(x) = e^x$ on $\mathbb{R}$.

Solution. It helps to have a *working lemma*, which we prove first because it settles several cases at once.

Lemma. If g is concave and differentiable on a convex open set S and φ is differentiable with $\varphi' > 0$ on an interval containing the image of g , then $f = \varphi \circ g$ is pseudoconcave.

Proof. Let $x, y \in S$ with $f(y) > f(x)$. Since φ is strictly increasing, $g(y) > g(x)$. The first order characterization of concavity gives $\langle \nabla g(x), y - x \rangle \geq g(y) - g(x) > 0$. By the chain rule, $\nabla f(x) = \varphi'(g(x)) \nabla g(x)$ with $\varphi'(g(x)) > 0$, so $\langle \nabla f(x), y - x \rangle = \varphi'(g(x)) \langle \nabla g(x), y - x \rangle > 0$, which is Definition 6.26. $\square$

(i) $f(x) = x^3 + x$: *pseudoconcave, not concave*. Its derivative is $f'(x) = 3x^2 + 1 \geq 1 > 0$ everywhere, so f is strictly increasing and in particular quasiconcave (Example 6.8). It is pseudoconcave: if $f(y) > f(x)$, strict monotonicity gives $y > x$, and then $f'(x)(y - x) > 0$ because $f'(x) > 0$ and $y - x > 0$. It is not concave: $f''(x) = 6x > 0$ for $x > 0$. (This is the example promised in Theorem 6.27 to show that “pseudoconcave $\Rightarrow$ concave” is false.) In fact it is also pseudoconvex, by the same argument applied to $-f$.

(ii) $f(x) = x^3$: *quasiconcave, not pseudoconcave, not concave*. It is strictly increasing, hence quasiconcave. But $f'(0) = 0$ while $f(1) = 1 > 0 = f(0)$: there is a strict improvement that the gradient does not detect, contradicting Definition 6.26. Equivalently, by Proposition 6.28, if it were pseudoconcave the stationary point $x = 0$ would have to be a global maximum, and it is not. It is not concave because $f''(x) = 6x > 0$ for $x > 0$. This is *the* counterexample to remember: it separates quasiconcavity from pseudoconcavity and explains why Arrow–Enthoven needs extra conditions.

(iii) $f(x, y) = x^{1/2}y^{1/2}$: *concave* (hence pseudoconcave and quasiconcave). It is Cobb–Douglas with $a+b = 1 \leq 1$; its Hessian satisfies $f_{11} < 0$ and $\det H = ab(1-a-b)x^{2a-2}y^{2b-2} = 0$, so it is negative semidefinite. The other two properties follow from Theorem 6.27.

(iv) $f(x, y) = x^2y^2$: *pseudoconcave and strictly quasiconcave, not concave*. Nonconcavity was proved in Exercise 6.45(b). Pseudoconcavity follows from the lemma with $g = x^{1/2}y^{1/2}$ (concave, case (iii)) and $\varphi(s) = s^4$, whose derivative $4s^3$ is positive for $s > 0$. An alternative check with bordered minors: $f_1 = 2xy^2$, $f_2 = 2x^2y$, $f_{11} = 2y^2$, $f_{22} = 2x^2$, $f_{12} = 4xy$, so

$$\begin{aligned}\det B_1 &= -4x^2y^4 < 0, \\ \det B_2 &= 2(2xy^2)(2x^2y)(4xy) - (4x^2y^4)(2x^2) - (4x^4y^2)(2y^2) \\ &= 32x^4y^4 - 16x^4y^4 = 16x^4y^4 > 0,\end{aligned}$$

strictly alternating signs on all of $\mathbb{R}_{++}^2$: Theorem 6.23(b) gives strict quasiconcavity and pseudoconcavity.

(v) $f(x) = x^2$: *none of the three* (it is convex, not concave; and it is *quasiconvex*, not quasiconcave). The upper contour set $\{x : x^2 \geq 1\} = (-\infty, -1] \cup [1, \infty)$ is not convex. Remember the trap in the warning box of Section 6.6: here $\det B_1 = -4x^2 \leq 0$ satisfies the weak *necessary* condition at every point, and yet the function is not quasiconcave; this is why the necessary condition can never be used to assert.

(vi) $f(x, y) = \min\{x, y\}$: *concave* (hence quasiconcave); *pseudoconcavity is not even defined* because f is not differentiable on the diagonal $x = y$. Concavity: for $z = \lambda a + (1 - \lambda)b$,

$$\min\{z_1, z_2\} \geq \lambda \min\{a_1, a_2\} + (1 - \lambda) \min\{b_1, b_2\}$$

follows by noting that $z_i = \lambda a_i + (1 - \lambda)b_i \geq \lambda \min\{a_1, a_2\} + (1 - \lambda) \min\{b_1, b_2\}$ for $i = 1, 2$, and taking the minimum over i on the left hand side. Pedagogical point: pseudoconcavity is a *differentiable* notion; for nonsmooth functions the right generalization goes through the subdifferential (ch. 7).

(vii) $f(x) = e^x$: *pseudoconcave and quasiconcave, not concave* (it is convex). Quasiconcave by monotonicity; pseudoconcave because $f'(x) = e^x > 0$ everywhere and the argument of (i) repeats. It is also pseudoconvex and quasiconvex: a monotone function on $\mathbb{R}$ belongs to all the “quasi” classes at once. Moral: quasiconcave does *not* mean “bell shaped”; it means only “convex upper contour sets”, and monotone functions have them. $\square$

Exam strategy

Exam protocol for classifying a function. When asked “is f concave, pseudoconcave or quasiconcave?”, follow this order, which goes from cheap to expensive:

1. **Is it monotone in one variable, or an obvious composition?** If $f = \varphi \circ g$ with φ increasing and g known (linear, concave, Cobb–Douglas), close the question with Theorem 6.13 and the lemma of Exercise 6.49. This is the fastest route and the one that earns the most points.

2. **Try concavity first** (negative semidefinite Hessian): if it works, you are done, because concave $\Rightarrow$ pseudoconcave $\Rightarrow$ quasiconcave (Theorem 6.27).
3. **If it is not concave, go to the bordered Hessian** and look for *strictly* alternating signs on the whole domain: that gives strict quasiconcavity + pseudoconcavity at once (Theorem 6.23(b)).
4. **To disprove**, look for a nonconvex upper contour set (two points of the set whose midpoint falls outside) or a point where the weak sign of the necessary condition fails.
5. **If the gradient has zeros**, be suspicious about pseudoconcavity: check it directly against Proposition 6.28 (every stationary point must be a global maximum).

And always write down which domain you are working on: almost every statement in this chapter is true on $\mathbb{R}_{++}^n$ and false on $\mathbb{R}^n$.

Chapter 7

Subdifferentiability [P3]

Week 5 of the course. Bibliography: Ruszczyński [6]; Brinkhuis [3]; Boyd and Vandenberghe [2].

What is this used for in Microeconomics?

In Microeconomics the central value functions of the course (the expenditure function $e(p, u)$, the profit function $\pi(p)$ and the cost function $c(w, y)$) are concave or convex in prices, but *not always differentiable*: when the consumer or firm problem has multiple solutions, the value function has a kink and the classical derivative does not exist. The object that always exists is the *subdifferential*, and the general version of the Shephard lemma and of the Hotelling lemma says exactly that *demand is the subdifferential of the value function* (the *superdifferential* when the value function is concave, as with expenditure or cost) [9, 10]. In contract theory and mechanism design (the market failures and the welfare and incentives parts, MWG chs. 13–14 and 23), the indirect utility of the agent is a maximum of linear functions of that type: convex and with kinks. The envelope theorem of Milgrom and Segal [11], a standard tool in those topics, is stated precisely in terms of one sided derivatives and subgradients. Finally, budget kinks (bracketed taxes, tiered electricity or water tariffs) generate demands characterized by first order conditions with subdifferentials. This chapter builds that tool from scratch.

7.1 Motivation: when the derivative does not exist [P3]

In ch. 5 we saw the *first order characterization* of differentiable convex functions: if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable at x , then

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad \text{for all } y \in \mathbb{R}^n. \quad (7.1)$$

Geometrically: the graph of f stays *above* its tangent plane at x ; the linear approximation is a *global lower bound*, not just a local approximation. This is the property that makes first order conditions *sufficient* in convex optimization.

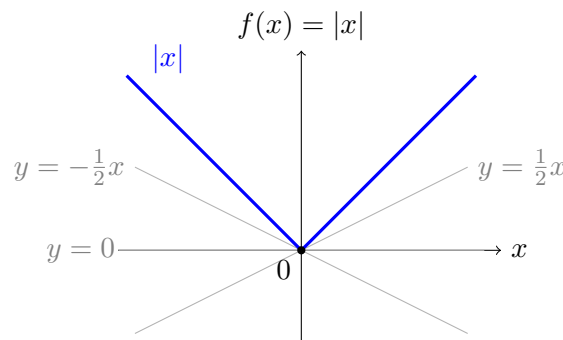
The problem is that many important convex functions are *not differentiable* everywhere, and these are not strange pathologies but completely natural examples:

- **The absolute value** $f(x) = |x|$: convex on $\mathbb{R}$, but at $x = 0$ the graph has a corner and

there is no unique tangent line.

- **Maxima of linear functions** $f(x) = \max\{\langle a_1, x \rangle + b_1, \dots, \langle a_m, x \rangle + b_m\}$: convex (a pointwise maximum of convex functions), but not differentiable where two or more “pieces” tie. The indirect utilities of agents in mechanisms have exactly this form.
- **Norms**: $\|x\|$, $\|x\|_1 = \sum_i |x_i|$, $\|x\|_\infty = \max_i |x_i|$ are convex but not differentiable at the origin (and $\|\cdot\|_1$, $\|\cdot\|_\infty$ are not differentiable at many other points either).
- **In economics**: a tiered tariff (the first 100 kWh at one price, the rest at another) generates a convex expenditure with a kink; the expenditure function $e(p, u)$ has kinks at prices where Hicksian demand is not unique; in econometrics, least absolute deviations regression minimizes $\sum_i |y_i - x_i^\top \beta|$, which is not differentiable.

How can we rescue inequality (7.1) without a derivative? The key is to *read the inequality backwards*. In (7.1) the vector $\nabla f(x)$ is the slope of *one* line (or hyperplane) that touches the graph at $(x, f(x))$ and stays below it everywhere: a *supporting line*. When f is differentiable, that supporting line is unique. When there is a corner, there is no unique supporting line: there are many. The right generalization is then this: instead of *the* tangent, consider *all* the slopes of supporting lines at the point. That set of slopes is the subdifferential.



In the figure, each line $y = gx$ with $g \in [-1, 1]$ passes through the origin and stays below $|x|$ on all of $\mathbb{R}$: each one is a supporting line of the graph at 0, and each slope $g \in [-1, 1]$ will be a *subgradient* of $|x|$ at 0.

7.2 Subgradient and subdifferential [P3]

We work with proper convex functions $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ (that is, $f \not\equiv +\infty$), with effective domain $\text{dom } f = \{x \in \mathbb{R}^n : f(x) < +\infty\}$. Allowing the value $+\infty$ is convenient: it lets us encode constraints inside the function (through the indicator, as we will see) without changing the theory.

Definition 7.1 (Subgradient and subdifferential). Let $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex and $x \in \text{dom } f$. A vector $g \in \mathbb{R}^n$ is a **subgradient** of f at x if

$$f(y) \geq f(x) + \langle g, y - x \rangle \quad \text{for all } y \in \mathbb{R}^n. \quad (7.2)$$

The set of all subgradients of f at x ,

$$\partial f(x) = \{ g \in \mathbb{R}^n : f(y) \geq f(x) + \langle g, y - x \rangle \quad \forall y \in \mathbb{R}^n \},$$

is called the **subdifferential** of f at x . If $\partial f(x) \neq \emptyset$ we say that f is **subdifferentiable** at x .

Note three things about the definition. First, it is a *global* condition: the inequality must hold for *every* y , not only near x (for convex functions this is not restrictive: a local inequality of this kind propagates to the whole space). Second, $\partial f(x)$ is a *set*, not a vector: subdifferentiability replaces a pointwise object (the gradient) by a set valued one. Third, inequality (7.2) says that the affine function $y \mapsto f(x) + \langle g, y - x \rangle$ is an *affine minorant* of f that is *exact* at x (it agrees with f there).

Intuition

Think of $\partial f(x)$ as the “fan of slopes” of all the lines (hyperplanes) you can lean against the graph of f at the point $(x, f(x))$ without cutting it. At a smooth point the fan collapses to a single slope: the gradient. At a corner, the fan opens up: there is a whole range of admissible slopes. The sharper the corner, the larger the subdifferential. Equivalently, in the language of ch. 4: $g \in \partial f(x)$ if and only if the (nonvertical) hyperplane of $\mathbb{R}^{n+1}$ determined by $(g, -1)$ *supports* the epigraph $\text{epi } f$ at the point $(x, f(x))$. Subdifferentiability is separation theory applied to the epigraph.

Let us now compute subdifferentials *directly from the definition*. It is the best possible training: it forces you to handle inequality (7.2) carefully.

Example 7.2 (The absolute value). Let $f(x) = |x|$ on $\mathbb{R}$. We claim that

$$\partial f(x) = \begin{cases} \{-1\} & \text{if } x < 0, \\ [-1, 1] & \text{if } x = 0, \\ \{1\} & \text{if } x > 0. \end{cases}$$

Let us check it in full detail in the interesting case, $x = 0$. By definition, $g \in \partial f(0)$ if and only if

$$|y| \geq |0| + g(y - 0) = gy \quad \text{for all } y \in \mathbb{R}.$$

(*Necessity.*) Suppose g satisfies $|y| \geq gy$ for all y . Taking $y = 1$: $1 \geq g$. Taking $y = -1$: $1 \geq -g$, that is $g \geq -1$. Therefore $g \in [-1, 1]$.

(*Sufficiency.*) Conversely, let $g \in [-1, 1]$, that is $|g| \leq 1$. For any $y \in \mathbb{R}$,

$$gy \leq |gy| = |g| |y| \leq 1 \cdot |y| = |y|,$$

which is exactly the required inequality. We conclude $\partial f(0) = [-1, 1]$.

For $x > 0$: if $g \in \partial f(x)$, taking y near x (where $f(y) = y$) the inequality $y \geq x + g(y - x)$ with $y > x$ gives $g \leq 1$ and with $y < x$ (still positive) gives $g \geq 1$; hence $g = 1$. And $g = 1$ does work globally: $|y| \geq y = x + 1 \cdot (y - x)$ for all y . The case $x < 0$ is symmetric.

Note the one dimensional pattern that will show up again and again: at a kink, the subdifferential is the *interval between the slope from the left and the slope from the right*.

Example 7.3 (Maximum of two affine functions). Let $f(x) = \max\{a_1x + b_1, a_2x + b_2\}$ on $\mathbb{R}$, with $a_1 < a_2$, and suppose the two lines meet at the point $\bar{x}$ (that is, $a_1\bar{x} + b_1 = a_2\bar{x} + b_2$). Then

$$\partial f(x) = \begin{cases} \{a_1\} & \text{if } x < \bar{x}, \\ [a_1, a_2] & \text{if } x = \bar{x}, \\ \{a_2\} & \text{if } x > \bar{x}. \end{cases}$$

Indeed, for $x \neq \bar{x}$ only one piece is active, f is affine in a neighborhood and globally stays above that piece, so the active slope is the only subgradient. At $\bar{x}$, let $g \in [a_1, a_2]$; write $g = \lambda a_1 + (1 - \lambda)a_2$ with $\lambda \in [0, 1]$. For every y :

$$f(y) \geq \lambda(a_1y + b_1) + (1 - \lambda)(a_2y + b_2) = f(\bar{x}) + g(y - \bar{x}),$$

where the first inequality uses that $f(y)$ dominates each piece (and hence any average of them), and the equality uses that both pieces equal $f(\bar{x})$ at $\bar{x}$. Hence $[a_1, a_2] \subseteq \partial f(\bar{x})$. For the opposite inclusion, the subgradient inequality evaluated at $y \rightarrow \bar{x}^+$ and $y \rightarrow \bar{x}^-$ forces $a_1 \leq g \leq a_2$, as in Example 7.2. Note that $\partial f(\bar{x}) = \text{conv}\{a_1, a_2\}$: the convex hull of the active slopes. This anticipates the pointwise maximum theorem of Section 7.6.

Example 7.4 (The Euclidean norm at the origin). Let $f(x) = \|x\|$ (Euclidean norm) on $\mathbb{R}^n$. We claim that

$$\partial f(0) = \{g \in \mathbb{R}^n : \|g\| \leq 1\} = \bar{B}(0, 1),$$

the closed unit ball. Indeed, $g \in \partial f(0)$ if and only if $\|y\| \geq \langle g, y \rangle$ for all $y \in \mathbb{R}^n$.

(*Sufficiency.*) If $\|g\| \leq 1$, the Cauchy–Schwarz inequality gives $\langle g, y \rangle \leq \|g\| \|y\| \leq \|y\|$ for all y .

(*Necessity.*) If $\|y\| \geq \langle g, y \rangle$ for all y , we choose the least favorable y : $y = g$. Then $\|g\| \geq \langle g, g \rangle = \|g\|^2$, so $\|g\|(\|g\| - 1) \leq 0$ and therefore $\|g\| \leq 1$.

(For $x \neq 0$, the norm is differentiable and $\partial f(x) = \{x / \|x\|\}$, by Proposition 7.10 below.)

The next example is the most important one for constrained optimization: it connects the subdifferential with the geometry of convex sets and defines an object that will star in the Karush–Kuhn–Tucker (KKT) conditions.

Definition 7.5 (Indicator function and normal cone). Let $C \subseteq \mathbb{R}^n$ be a nonempty convex set.

- The **indicator function** of C is

$$\delta_C(x) = \begin{cases} 0 & \text{if } x \in C, \\ +\infty & \text{if } x \notin C. \end{cases}$$

It is convex (its epigraph is $C \times [0, +\infty)$, a convex set) and it encodes the constraint “ $x \in C$ ”: minimizing f over C is the same as minimizing $f + \delta_C$ over all of $\mathbb{R}^n$.

- The **normal cone** to C at $x \in C$ is

$$N_C(x) = \{g \in \mathbb{R}^n : \langle g, y - x \rangle \leq 0 \quad \forall y \in C\}.$$

Read the definition of $N_C(x)$ slowly: g is normal to C at x if g makes an obtuse (or right) angle with *every* direction $y - x$ pointing from x into the set. That is, g “points outward” from C at x . It is immediate to check that $N_C(x)$ is a closed convex cone containing 0.

Example 7.6 (Subdifferential of the indicator). For $C \subseteq \mathbb{R}^n$ convex and $x \in C$:

$$\partial\delta_C(x) = N_C(x).$$

Proof from the definition. $g \in \partial\delta_C(x)$ means

$$\delta_C(y) \geq \delta_C(x) + \langle g, y - x \rangle = \langle g, y - x \rangle \quad \forall y \in \mathbb{R}^n.$$

If $y \notin C$, the left hand side is $+\infty$ and the inequality is automatic: points outside C impose no condition. If $y \in C$, the left hand side is 0 and the condition becomes $\langle g, y - x \rangle \leq 0$. Therefore $g \in \partial\delta_C(x)$ if and only if $\langle g, y - x \rangle \leq 0$ for all $y \in C$, which is exactly the definition of $N_C(x)$. $\square$

Concrete cases. (i) If $x \in \text{int } C$, then $N_C(x) = \{0\}$: for any $g \neq 0$ we can move a little from x in the direction of g without leaving C , and that y violates the inequality. (ii) If $C = [a, b] \subset \mathbb{R}$ and $x = b$, then $y - b \leq 0$ for all $y \in [a, b]$, so $N_C(b) = [0, +\infty)$: at the right endpoint, the normals point to the right. Similarly $N_C(a) = (-\infty, 0]$ and $N_C(x) = \{0\}$ for $x \in (a, b)$.

Intuition

The identity $\partial\delta_C = N_C$ is a small accounting miracle: it turns a *geometric constraint* (x must lie in C) into an *object of calculus* (a set of “generalized gradients”). When the condition $0 \in \nabla f(x^*) + N_C(x^*)$ shows up in KKT, it will literally be the generalized Fermat rule applied to $f + \delta_C$: the Lagrange multipliers are the coordinates of an element of the normal cone. Keep this example: it is the bridge between this chapter and the one on optimality.

7.3 Basic properties of the subdifferential [P3]

Proposition 7.7 (Structure of the subdifferential). *For every convex $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ and every $x \in \text{dom } f$, the set $\partial f(x)$ is convex and closed (possibly empty).*

Proof. For each fixed $y \in \mathbb{R}^n$, the set

$$H_y = \{g \in \mathbb{R}^n : \langle g, y - x \rangle \leq f(y) - f(x)\}$$

is a closed halfspace in the variable g (or all of $\mathbb{R}^n$, if $y - x = 0$ or if $f(y) = +\infty$): the function $g \mapsto \langle g, y - x \rangle$ is linear in g , and a set of the form $\{g : \text{linear}(g) \leq \text{constant}\}$ is a closed halfspace,

which is convex and closed. By definition,

$$\partial f(x) = \bigcap_{y \in \mathbb{R}^n} H_y,$$

and an arbitrary intersection of convex closed sets is convex and closed (results from ch. 2). $\square$

The delicate question is *nonemptiness*: is there always at least one subgradient? In the interior of the domain, yes, and the proof is a direct application of the supporting hyperplane theorem of ch. 4, applied to the epigraph. This is the central result of the chapter.

Theorem 7.8 (Existence of subgradients in the interior). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper convex, and let $x \in \text{int}(\text{dom } f)$. Then $\partial f(x) \neq \emptyset$.*

Proof. The idea: the point $(x, f(x))$ lies on the boundary of the convex set $\text{epi } f$; we lean a supporting hyperplane there; we check that it is not vertical; its slope is the subgradient we want.

Step 1: supporting hyperplane to the epigraph. The epigraph $\text{epi } f = \{(y, t) \in \mathbb{R}^n \times \mathbb{R} : t \geq f(y)\}$ is convex (ch. 5) and the point $(x, f(x))$ lies on its boundary: it belongs to $\text{epi } f$, but the points $(x, f(x) - \varepsilon)$ with $\varepsilon > 0$ do not. By the supporting hyperplane theorem (ch. 4), there is $(a, \beta) \in \mathbb{R}^n \times \mathbb{R}$ with $(a, \beta) \neq (0, 0)$ such that

$$\langle a, y \rangle + \beta t \leq \langle a, x \rangle + \beta f(x) \quad \text{for all } (y, t) \in \text{epi } f. \quad (7.3)$$

Step 2: $\beta \leq 0$. Fix $y = x$ and let $t \rightarrow +\infty$ (the points (x, t) with $t \geq f(x)$ all lie in $\text{epi } f$: the epigraph is “infinite upward”). If we had $\beta > 0$, the left hand side of (7.3) would tend to $+\infty$, violating the bound. Hence $\beta \leq 0$.

Step 3: $\beta \neq 0$ (here interiority is used). Suppose for contradiction that $\beta = 0$. Then (7.3) reduces to $\langle a, y - x \rangle \leq 0$ for all $y \in \text{dom } f$. Since $x \in \text{int}(\text{dom } f)$, there is $r > 0$ such that $x + z \in \text{dom } f$ for every $\|z\| \leq r$; taking $y = x + \frac{r}{2} \frac{a}{\|a\|}$ (if $a \neq 0$) we get $\frac{r}{2} \|a\| \leq 0$, absurd. Hence $a = 0$, but then $(a, \beta) = (0, 0)$, contradicting the support theorem. We conclude $\beta < 0$. Geometrically: a *vertical* supporting hyperplane can only occur at the boundary of the domain; in the interior, the hyperplane must “tilt”.

Step 4: building the subgradient. Define $g = a/(-\beta)$ (well defined since $\beta < 0$). Dividing (7.3) by $-\beta > 0$ and taking $t = f(y)$ for $y \in \text{dom } f$:

$$\langle g, y \rangle - f(y) \leq \langle g, x \rangle - f(x),$$

which rearranged is $f(y) \geq f(x) + \langle g, y - x \rangle$ for all $y \in \text{dom } f$; for $y \notin \text{dom } f$ the inequality is trivial because $f(y) = +\infty$. Therefore $g \in \partial f(x)$. $\square$

Careful (common mistake)

On the *boundary* of the domain the subdifferential can be empty even if the function is finite there. The canonical example: $f(x) = -\sqrt{x}$ for $x \geq 0$ and $f(x) = +\infty$ for $x < 0$. It is convex and $f(0) = 0$ is finite, but $\partial f(0) = \emptyset$: a subgradient g at 0 would have to

satisfy $-\sqrt{y} \geq gy$ for all $y > 0$, that is $g \leq -1/\sqrt{y}$, and letting $y \rightarrow 0^+$ we get $g \leq -\infty$, which is impossible. The “slope” at 0 is vertical and no nonvertical hyperplane supports the epigraph there. Moral: the statements in this chapter usually require $x \in \text{int}(\text{dom } f)$ or some qualification condition; these are not technical decorations, they are necessary. This same pathology will come back when we study duality and stability.

Remark 7.9. In the interior of the domain you can say more: $\partial f(x)$ is also *bounded* (and therefore compact, being closed). The underlying reason is that a convex function is locally Lipschitz in the interior of its domain, and the slopes of the exact affine minorants at x are controlled by the local Lipschitz constant. See [6], sections 2.5–2.6, or [3] for the details.

The next result closes the circle with ch. 5: the subdifferential *generalizes* the gradient, it does not replace it.

Proposition 7.10 (The differentiable case). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex and differentiable at $x \in \text{int}(\text{dom } f)$. Then*

$$\partial f(x) = \{\nabla f(x)\}.$$

Proof. ($\supseteq$) The first order characterization of ch. 5 says exactly that $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all y ; comparing with (7.2), this is $\nabla f(x) \in \partial f(x)$.

($\subseteq$) Let $g \in \partial f(x)$ and fix an arbitrary direction $d \in \mathbb{R}^n$. Applying (7.2) with $y = x + td$, $t > 0$ small:

$$f(x + td) \geq f(x) + t \langle g, d \rangle \quad \implies \quad \frac{f(x + td) - f(x)}{t} \geq \langle g, d \rangle.$$

Letting $t \rightarrow 0^+$, differentiability gives $\langle \nabla f(x), d \rangle \geq \langle g, d \rangle$, that is $\langle \nabla f(x) - g, d \rangle \geq 0$. Since d is arbitrary, we can also take $-d$, which gives the opposite inequality. Hence $\langle \nabla f(x) - g, d \rangle = 0$ for every d ; choosing $d = \nabla f(x) - g$ we get $\|\nabla f(x) - g\|^2 = 0$, that is $g = \nabla f(x)$. $\square$

The converse also holds: if $\partial f(x)$ is a single point at an interior point, then f is differentiable at x (see [6]). So “single valued subdifferential” and “differentiable” are synonyms for convex functions in the interior of the domain.

7.4 Directional derivatives [P2]

Even if a convex function has no gradient, we can always ask for its rate of change *in a given direction*, approaching from one side only.

Definition 7.11 (Directional derivative). Let $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex, $x \in \text{dom } f$ and $d \in \mathbb{R}^n$. The **directional derivative** of f at x in the direction d is

$$f'(x; d) = \lim_{t \rightarrow 0^+} \frac{f(x + td) - f(x)}{t},$$

provided the limit exists in $\mathbb{R} \cup \{\pm\infty\}$.

For arbitrary functions this limit may fail to exist. For convex functions it *always* exists, and the reason is an elementary but powerful monotonicity of the difference quotient.

Lemma 7.12 (Monotonicity of the difference quotient). *Let f be convex, $x \in \text{dom } f$ and $d \in \mathbb{R}^n$. The function*

$$t \mapsto \varphi(t) = \frac{f(x + td) - f(x)}{t}$$

is nondecreasing for $t > 0$. Consequently, the limit in Definition 7.11 exists and

$$f'(x; d) = \inf_{t > 0} \frac{f(x + td) - f(x)}{t} \in \mathbb{R} \cup \{\pm\infty\}.$$

If $x \in \text{int}(\text{dom } f)$, the value $f'(x; d)$ is finite for every d .

Proof. Let $0 < s < t$. The point $x + sd$ is a convex combination of x and $x + td$: indeed,

$$x + sd = \left(1 - \frac{s}{t}\right)x + \frac{s}{t}(x + td), \quad \frac{s}{t} \in (0, 1).$$

Applying the definition of convexity of f to this combination:

$$f(x + sd) \leq \left(1 - \frac{s}{t}\right)f(x) + \frac{s}{t}f(x + td).$$

Subtracting $f(x)$ from both sides:

$$f(x + sd) - f(x) \leq \frac{s}{t}(f(x + td) - f(x)),$$

and dividing by $s > 0$ we get $\varphi(s) \leq \varphi(t)$: the difference quotient grows with the step. A nondecreasing function of t has a limit as $t \rightarrow 0^+$, equal to its infimum over $t > 0$ (possibly $-\infty$). Finally, if x is interior, fix $t_0 > 0$ with $x - t_0d \in \text{dom } f$ and set $\psi(t) = (f(x - td) - f(x))/t$, which by what we just proved (with direction $-d$) is nondecreasing for $t > 0$. Convexity of f on the segment from $x - td$ to $x + td$ gives $\varphi(t) \geq -\psi(t)$, and monotonicity of ψ gives $-\psi(t) \geq -\psi(t_0)$ for $0 < t \leq t_0$. Hence $\varphi(t) \geq -\psi(t_0) > -\infty$ for all small t : the infimum is finite. $\square$

Intuition

Monotonicity of the difference quotient is *the* working property of convex functions in one variable: “chords get steeper as they get longer”. From it we get almost for free the existence of one sided derivatives, the monotonicity of the derivative, and, as we will now see, the link between directional derivative and subdifferential. When one sided derivatives of the value function appear in Milgrom–Segal [11], it is exactly this lemma that guarantees their existence.

The link with the subdifferential is an elegant identity: the directional derivative is the *support function* of the subdifferential (recall from ch. 5: the support function of a set S is $\sigma_S(d) = \sup\{\langle g, d \rangle : g \in S\}$).

Theorem 7.13 (Max formula). *Let f be convex and $x \in \text{int}(\text{dom } f)$. Then, for every $d \in \mathbb{R}^n$,*

$$f'(x; d) = \max\{\langle g, d \rangle : g \in \partial f(x)\},$$

and the maximum is attained.

Proof idea. (Inequality $\geq$, the easy one.) If $g \in \partial f(x)$, the subgradient inequality with $y = x + td$ gives $\frac{f(x+td) - f(x)}{t} \geq \langle g, d \rangle$ for all $t > 0$; taking the infimum in t (Lemma 7.12), $f'(x; d) \geq \langle g, d \rangle$. Hence $f'(x; d) \geq \sup_{g \in \partial f(x)} \langle g, d \rangle$.

(Inequality $\leq$, the substance.) One shows that the function $d \mapsto f'(x; d)$ is convex and positively homogeneous (it inherits both properties from Lemma 7.12 and from convexity of f). To such a function we can apply the existence theorem (Theorem 7.8) in the fixed direction d : there is a linear minorant $\langle g, \cdot \rangle$ of $f'(x; \cdot)$ that is exact at d , and one checks that such a g belongs to $\partial f(x)$ (because $f(y) - f(x) \geq f'(x; y - x) \geq \langle g, y - x \rangle$, where the first inequality is the monotonicity of the quotient with $t = 1$). That g attains the supremum, which is therefore a maximum. The full details are in [6], section 2.5, and in [3]. $\square$

In particular, $\partial f(x)$ (convex, closed and bounded in the interior) is *completely determined* by the directional derivatives: two descriptions of the same object, one analytic and one geometric. On $\mathbb{R}$, the formula simply says that $\partial f(x) = [f'(x; -1) \cdot (-1), f'(x; 1)] = [f'_-(x), f'_+(x)]$: the interval between the one sided derivatives, confirming the pattern of Examples 7.2 and 7.3.

7.5 The generalized Fermat rule [P3]

We now reach the reason why this whole apparatus exists. In classical calculus, the necessary condition for an interior minimum is $\nabla f(x^*) = 0$ (the Fermat rule), but for nondifferentiable convex functions that condition cannot even be written down. Its correct generalization is remarkably elegant: the proof fits in two lines because the definition of subgradient already contains all of global optimization.

Theorem 7.14 (Generalized Fermat rule). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper convex. Then $x^* \in \text{dom } f$ is a global minimizer of f if and only if*

$$0 \in \partial f(x^*).$$

Proof. By definition, $0 \in \partial f(x^*)$ means

$$f(y) \geq f(x^*) + \langle 0, y - x^* \rangle = f(x^*) \quad \text{for all } y \in \mathbb{R}^n,$$

which is *literally* the statement that x^* minimizes f over $\mathbb{R}^n$. $\square$

Stop for a moment at what just happened. In classical calculus, $\nabla f(x^*) = 0$ is only a *necessary* and *local* condition, and more work is needed (second order, convexity) to conclude anything global. Here the condition $0 \in \partial f(x^*)$ is *necessary and sufficient*, and global, with no extra hypotheses: convexity is already built into the definition of the subdifferential. Moreover the statement does not require differentiability anywhere. All the difficulty moves, of course, to *computing* ∂f ; that is the job of the next section.

Example 7.15 (Minimization with an absolute value). Let us minimize on $\mathbb{R}$ the function

$$f(x) = |x - 1| + \frac{x^2}{2},$$

convex because it is a sum of convex functions. Since $x^2/2$ is differentiable everywhere, the subdifferential of the sum is the sum of the gradient and the subdifferential (we justify this in Section 7.6; with one differentiable summand the rule holds with no extra conditions):

$$\partial f(x) = x + \partial |\cdot - 1|(x) = \begin{cases} \{x - 1\} & \text{if } x < 1, \\ [1 - 1, 1 + 1] = [0, 2] & \text{if } x = 1, \\ \{x + 1\} & \text{if } x > 1. \end{cases}$$

We look for x^* with $0 \in \partial f(x^*)$, checking the three cases:

- *Case $x < 1$:* we need $x - 1 = 0$, that is $x = 1$, which does *not* belong to the region $x < 1$. No candidates.
- *Case $x > 1$:* we need $x + 1 = 0$, that is $x = -1$, which does not belong to the region $x > 1$. No candidates.
- *Case $x = 1$:* we need $0 \in [0, 2]$, which is *true* (0 is the left endpoint of the interval).

By the generalized Fermat rule, $x^* = 1$ is the global minimizer, with value $f(1) = 1/2$. Note the typical phenomenon: the minimum occurs *exactly at the kink*, where no “piecewise derivative” vanishes, but the interval of subgradients captures zero. If we had naively looked for points with $f'(x) = 0$ region by region, we would have concluded (wrongly) that there is no minimum.

Exam strategy

Recipe for minimizing piecewise convex functions on $\mathbb{R}$ (very frequent in exams and in models with brackets):

1. Identify the kinks and the smooth regions between them.
2. In each smooth region, compute the classical derivative and look for zeros *inside the region* (always check membership: a zero outside its region does not count).
3. At each kink $\bar{x}$, compute $\partial f(\bar{x}) = [f'_-(\bar{x}), f'_+(\bar{x})]$: the interval between the left and the right derivative (for sums: add the smooth part to each endpoint).
4. The minimizer is the point (region or kink) where 0 falls in the subdifferential. By convexity, when the derivative goes from negative to positive, zero gets trapped there, and there is no need to compare values.

In $\mathbb{R}^n$ the logic is identical but step 3 uses the calculus rules of Section 7.6 (sum, affine composition, pointwise maximum).

7.6 Computing subdifferentials [P3]

For the Fermat rule to be usable we need a “calculus” of subdifferentials analogous to the rules of differentiation: sum, composition and maximum. The sum rule is the most famous and the most subtle result.

Theorem 7.16 (Moreau–Rockafellar: subdifferential of a sum). *Let $f_1, f_2 : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper convex, and let $x \in \text{dom } f_1 \cap \text{dom } f_2$.*

1. *The inclusion*

$$\partial f_1(x) + \partial f_2(x) \subseteq \partial(f_1 + f_2)(x)$$

always holds, where the sum of sets is $A + B = \{a + b : a \in A, b \in B\}$.

2. *If in addition the **qualification condition***

$$\text{ri}(\text{dom } f_1) \cap \text{ri}(\text{dom } f_2) \neq \emptyset$$

holds (it is enough, for example, that there is a point of $\text{dom } f_2$ at which f_1 is finite and continuous), then equality holds:

$$\partial(f_1 + f_2)(x) = \partial f_1(x) + \partial f_2(x).$$

Proof of the easy inclusion (1). Let $g_1 \in \partial f_1(x)$ and $g_2 \in \partial f_2(x)$. For every $y \in \mathbb{R}^n$ we have, by definition,

$$f_1(y) \geq f_1(x) + \langle g_1, y - x \rangle, \quad f_2(y) \geq f_2(x) + \langle g_2, y - x \rangle.$$

Adding the two inequalities:

$$(f_1 + f_2)(y) \geq (f_1 + f_2)(x) + \langle g_1 + g_2, y - x \rangle,$$

which is exactly the statement $g_1 + g_2 \in \partial(f_1 + f_2)(x)$. The proof of the reverse inclusion under the qualification is deeper (it needs a separation argument between suitable epigraphs); see [6], section 2.6, or [3]. $\square$

Why is the qualification condition needed? Because the reverse inclusion can *fail* when the domains only touch at the boundary, exactly where (recall the warning) subdifferentials can vanish.

Example 7.17 (Equality can fail without qualification). On $\mathbb{R}$, let

$$f_1(x) = \begin{cases} -\sqrt{x} & x \geq 0 \\ +\infty & x < 0 \end{cases}, \quad f_2(x) = \delta_{(-\infty, 0]}(x).$$

Both are convex and $\text{dom } f_1 \cap \text{dom } f_2 = \{0\}$: the domains $[0, +\infty)$ and $(-\infty, 0]$ touch only at their common boundary, so $\text{ri}(\text{dom } f_1) \cap \text{ri}(\text{dom } f_2) = (0, \infty) \cap (-\infty, 0) = \emptyset$ and the qualification fails. Now, $f_1 + f_2 = \delta_{\{0\}}$ (equal to 0 at $x = 0$ and $+\infty$ elsewhere), so, by Example 7.6,

$$\partial(f_1 + f_2)(0) = N_{\{0\}}(0) = \mathbb{R}.$$

But $\partial f_1(0) = \emptyset$ (vertical slope, as we saw in the warning), so, even though $\partial f_2(0) = N_{(-\infty, 0]}(0) = [0, +\infty)$, the sum of sets with one empty summand is empty: $\partial f_1(0) + \partial f_2(0) = \emptyset$. The easy inclusion $\emptyset \subseteq \mathbb{R}$ holds, but equality fails dramatically: the subdifferential of the sum is all of $\mathbb{R}$

and the sum of subdifferentials is empty. The Moreau–Rockafellar qualification is exactly what rules out this kind of boundary collapse.

Remark 7.18. A very useful special case: if f_2 is convex and *differentiable* on all of $\mathbb{R}^n$ (full domain), the qualification holds automatically and

$$\partial(f_1 + f_2)(x) = \partial f_1(x) + \{\nabla f_2(x)\} = \partial f_1(x) + \nabla f_2(x).$$

This is the rule we used in Example 7.15: adding the smooth part “shifts” the subdifferential of the nonsmooth part.

Proposition 7.19 (Affine composition). *Let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be convex and finite, A an $m \times n$ matrix and $b \in \mathbb{R}^m$. Then $h(x) = f(Ax + b)$ is convex and*

$$\partial h(x) = A^\top \partial f(Ax + b) = \{A^\top g : g \in \partial f(Ax + b)\}.$$

The inclusion $\supseteq$ is again direct from the definition: if $g \in \partial f(Ax + b)$, then for every y ,

$$h(y) = f(Ay + b) \geq f(Ax + b) + \langle g, Ay - Ax \rangle = h(x) + \langle A^\top g, y - x \rangle,$$

using the identity $\langle g, Az \rangle = \langle A^\top g, z \rangle$ (definition of the transpose). Equality requires, as always, a qualification condition, which here holds because f is finite on all of $\mathbb{R}^m$. This rule is what makes LAD regression tractable: $\beta \mapsto |y_i - x_i^\top \beta|$ is the composition of the absolute value with an affine function of β .

Theorem 7.20 (Pointwise maximum of smooth functions). *Let $f_1, \dots, f_m : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and differentiable, and $f(x) = \max\{f_1(x), \dots, f_m(x)\}$. Define the **active set** at x :*

$$I(x) = \{i \in \{1, \dots, m\} : f_i(x) = f(x)\}.$$

Then

$$\partial f(x) = \text{conv}\{\nabla f_i(x) : i \in I(x)\} :$$

the subdifferential is the convex hull of the gradients of the active functions (those attaining the maximum at x). In particular, if only one is active, f is differentiable at x with $\nabla f(x) = \nabla f_i(x)$.

The inclusion $\supseteq$ is proved just as in Example 7.3: each $\nabla f_i(x)$ with i active is a subgradient because $f \geq f_i$ with equality at x (so $f(y) \geq f_i(y) \geq f_i(x) + \langle \nabla f_i(x), y - x \rangle = f(x) + \langle \nabla f_i(x), y - x \rangle$), and $\partial f(x)$ is convex (Proposition 7.7), so it contains the convex hull. The inclusion $\subseteq$ uses the max formula (Theorem 7.13); see [6] or [2], where this formula is the basis of bundle methods and of the analysis of max functions.

Example 7.21 (Computing with the max formula). Let $f(x) = \max\{x^2, x\}$ on $\mathbb{R}$. The two pieces tie when $x^2 = x$, that is at $x = 0$ and $x = 1$; between them ($0 < x < 1$) x wins, and outside x^2 wins. Then:

- If $x < 0$ or $x > 1$: only x^2 is active, so $\partial f(x) = \{2x\}$.
- If $0 < x < 1$: only x is active, so $\partial f(x) = \{1\}$.

- At $x = 0$: both are active, with gradients $2 \cdot 0 = 0$ and 1 ; hence $\partial f(0) = \text{conv}\{0, 1\} = [0, 1]$.
- At $x = 1$: both are active, with gradients $2 \cdot 1 = 2$ and 1 ; hence $\partial f(1) = \text{conv}\{2, 1\} = [1, 2]$.

Along the way, the Fermat rule: $0 \in \partial f(x)$ happens at $x = 0$ (since $0 \in [0, 1]$), and nowhere else; the global minimizer is $x^* = 0$ with $f(0) = 0$.

7.7 Economic applications [P3]

Expenditure function and Hicksian demand: the general Shephard lemma

Recall the consumer expenditure minimization problem: given prices $p \in \mathbb{R}_{++}^n$ and a utility level u ,

$$e(p, u) = \min\{\langle p, x \rangle : u(x) \geq u, x \geq 0\},$$

with solution set $h(p, u)$ (**Hicksian** or compensated demand). A result from ch. 5 (a minimum of linear functions of p , indexed by feasible x) is that $p \mapsto e(p, u)$ is *concave* and positively homogeneous of degree 1.

Since $e(\cdot, u)$ is concave, the natural object is the *superdifferential*: g is a **supergradient** of a concave function ϕ at p if $\phi(q) \leq \phi(p) + \langle g, q - p \rangle$ for all q (the inequalities are reversed; equivalently, g is a supergradient of ϕ if and only if $-g$ is a subgradient of the convex function $-\phi$). The whole theory of this chapter translates unchanged, through the mirror.

Theorem 7.22 (Shephard lemma, general version). *Fix u and assume $e(\cdot, u)$ is finite in a neighborhood of $p \in \mathbb{R}_{++}^n$. Then Hicksian demand is exactly the superdifferential of the expenditure function:*

$$h(p, u) = \partial_p e(p, u) \quad (\text{superdifferential in } p).$$

In particular, $e(\cdot, u)$ is differentiable at p if and only if Hicksian demand is unique at p , and in that case $h(p, u) = \{\nabla_p e(p, u)\}$, which is the classical statement of the Shephard lemma [9, 10].

Proof idea (main inclusion). Let $x^* \in h(p, u)$: an optimal bundle at prices p , with $e(p, u) = \langle p, x^* \rangle$. For any other price vector q , the bundle x^* is still *feasible* (it reaches utility u ; feasibility does not depend on prices), although possibly no longer optimal. Therefore

$$e(q, u) \leq \langle q, x^* \rangle = \langle p, x^* \rangle + \langle x^*, q - p \rangle = e(p, u) + \langle x^*, q - p \rangle,$$

which is exactly the supergradient inequality: $x^* \in \partial_p e(p, u)$. The argument is pure “over-budgeting”: at new prices, buying the old bundle costs $\langle q, x^* \rangle$ and that bounds the minimum expenditure from above. The reverse inclusion (every supergradient is an optimal demand) uses homogeneity and the structure of the problem; see [9] (ch. 3) and, for the general envelope argument, [11]. $\square$

Example 7.23 (Perfect substitutes: kink and multiple demand). Let $u(x_1, x_2) = x_1 + x_2$ (perfect substitutes). The consumer buys only the cheaper good, so

$$e(p, u) = u \cdot \min\{p_1, p_2\},$$

concave in p (a minimum of two linear functions) and with a *kink* on the diagonal $p_1 = p_2$. Let us compute the superdifferential at a kink point $p_1 = p_2 = \bar{p}$ using (the concave version of) Theorem 7.20: the two linear pieces u_{p_1} and u_{p_2} are active, with gradients $(u, 0)$ and $(0, u)$, so

$$\partial_p e(\bar{p}, \bar{p}, u) = \text{conv}\{(u, 0), (0, u)\} = \{(\lambda u, (1 - \lambda)u) : \lambda \in [0, 1]\}.$$

And indeed, when $p_1 = p_2$ the consumer is indifferent among all bundles with $x_1 + x_2 = u$: Hicksian demand is exactly the segment $\{(\lambda u, (1 - \lambda)u) : \lambda \in [0, 1]\}$. Theorem 7.22 holds to the letter: *the set of demands is the superdifferential at the kink*. Off the diagonal, demand is unique (everything goes to the cheap good) and e is differentiable: $\nabla_p e = (u, 0)$ if $p_1 < p_2$, confirming the classical Shephard lemma.

The same happens, word for word, with the firm cost function $c(w, y) = \min\{\langle w, z \rangle : F(z) \geq y\}$ (w input prices): it is concave in w , conditional factor demands are its superdifferential, and when technologies have linearly substitutable inputs, kinks with multiple demands appear. And with the profit function $\pi(p)$ (convex in p): the general Hotelling lemma says that net supply is $\partial\pi(p)$.

Two more appearances you will see soon

Contracts and mechanisms. In a direct mechanism, the equilibrium utility of an agent of type θ is $U(\theta) = \max_{\hat{\theta}} [\text{allocation linear in } \theta]$: a maximum of affine functions of θ , hence convex and typically with kinks where the agent is indifferent between two reports. The Milgrom–Segal envelope theorem [11] characterizes U through its one sided derivatives, which exist by Lemma 7.12, and the monotonicity condition on the allocation is precisely monotonicity of the slopes (the subgradients) of U .

Econometrics: LAD and quantile regression. The least absolute deviations estimator solves

$$\min_{\beta \in \mathbb{R}^k} \sum_{i=1}^N |y_i - x_i^\top \beta|,$$

a nondifferentiable convex function (a sum of affine compositions of $|\cdot|$, Proposition 7.19). Its optimality conditions are $0 \in \partial(\cdot)$, not a system of equations: this is why the median (the case $k = 1$ with $x_i = 1$) can be an *interval* of solutions, and why quantile regression is studied with the tools of this chapter (see Exercise 7.28).

7.8 Summary and exercises [P3]

Chapter summary

- **Subgradient:** $g \in \partial f(x)$ if $f(y) \geq f(x) + \langle g, y - x \rangle$ for all y : the slope of an affine minorant that is exact at x ; geometrically, a nonvertical supporting hyperplane of the epigraph at $(x, f(x))$.
- **Basic examples:** $\partial|x|(0) = [-1, 1]$; $\partial\|\cdot\|(0) = \bar{B}(0, 1)$; $\partial\delta_C(x) = N_C(x)$ (normal cone); at a kink of a max of affine functions, the convex hull of the active slopes.

- **Properties:** $\partial f(x)$ is always convex and closed; it is nonempty (and compact) if $x \in \text{int}(\text{dom } f)$, via a supporting hyperplane to the epigraph; it can be empty on the boundary (vertical slope). If f is differentiable at x , then $\partial f(x) = \{\nabla f(x)\}$.
- **Directional derivative:** for convex functions it always exists ($\varphi(t)$ is monotone) and $f'(x; d) = \max\{\langle g, d \rangle : g \in \partial f(x)\}$: the support function of the subdifferential. On $\mathbb{R}$: $\partial f(x) = [f'_-(x), f'_+(x)]$.
- **Generalized Fermat:** x^* minimizes f (globally) if and only if $0 \in \partial f(x^*)$. Necessary, sufficient, and with no differentiability.
- **Calculus:** $\partial(f_1 + f_2) = \partial f_1 + \partial f_2$ under the qualification $\text{ri}(\text{dom } f_1) \cap \text{ri}(\text{dom } f_2) \neq \emptyset$ (Moreau–Rockafellar; the inclusion $\supseteq$ is always true and free); $\partial f(A \cdot + b)(x) = A^\top \partial f(Ax + b)$; $\partial \max_i f_i(x) = \text{conv}\{\nabla f_i(x) : i \text{ active}\}$.
- **Economics:** Hicksian demand = superdifferential of the expenditure function (general Shephard); supply = subdifferential of profit (general Hotelling); agent utilities in mechanisms are maxima of affine functions and are analyzed with one sided derivatives (Milgrom–Segal); LAD and quantile regression in econometrics.

References: [6] (secs. 2.5–2.6), [3], [1], [2]; for the applications, [9], [10], [11].

Exercise 7.24. Let $f(x) = |x| + |x - 1|$ on $\mathbb{R}$. Compute $\partial f(x)$ at every point and find *all* the global minimizers of f .

Solution. We write $f = f_1 + f_2$ with $f_1 = |\cdot|$ and $f_2 = |\cdot - 1|$; both are finite on all of $\mathbb{R}$, so Moreau–Rockafellar applies with equality (the qualification is automatic: $\text{ri}(\text{dom } f_1) \cap \text{ri}(\text{dom } f_2) = \mathbb{R}$). Using $\partial|\cdot|$ from Example 7.2 (shifted for f_2):

$$\partial f(x) = \begin{cases} \{-1\} + \{-1\} = \{-2\} & x < 0, \\ [-1, 1] + \{-1\} = [-2, 0] & x = 0, \\ \{1\} + \{-1\} = \{0\} & 0 < x < 1, \\ \{1\} + [-1, 1] = [0, 2] & x = 1, \\ \{1\} + \{1\} = \{2\} & x > 1. \end{cases}$$

We look for $0 \in \partial f(x)$: it fails for $x < 0$ and $x > 1$; it holds at $x = 0$ (since $0 \in [-2, 0]$), at every $x \in (0, 1)$ (since $\partial f(x) = \{0\}$) and at $x = 1$ (since $0 \in [0, 2]$). By the generalized Fermat rule, the set of minimizers is *the whole interval* $[0, 1]$, with optimal value $f(x) = 1$ on it. (Direct check: between 0 and 1, $|x| + |x - 1| = x + (1 - x) = 1$; outside, the sum grows.) This is the “median” phenomenon: with two points, any point between them minimizes the sum of distances. $\square$

Exercise 7.25. Let $f(x_1, x_2) = |x_1| + |x_2|$ (the norm $\|\cdot\|_1$ on $\mathbb{R}^2$). Prove from the definition that $\partial f(0, 0) = [-1, 1] \times [-1, 1]$, and compute $\partial f(3, 0)$.

Solution. $g = (g_1, g_2) \in \partial f(0, 0)$ if and only if

$$|y_1| + |y_2| \geq g_1 y_1 + g_2 y_2 \quad \forall (y_1, y_2) \in \mathbb{R}^2.$$

(*Necessity.*) Taking $y = (\pm 1, 0)$ we get $1 \geq \pm g_1$, that is $|g_1| \leq 1$; taking $y = (0, \pm 1)$, $|g_2| \leq 1$. Hence $g \in [-1, 1]^2$. (*Sufficiency.*) If $|g_1| \leq 1$ and $|g_2| \leq 1$, then $g_1 y_1 + g_2 y_2 \leq |g_1| |y_1| + |g_2| |y_2| \leq |y_1| + |y_2|$ for all y . We conclude $\partial f(0, 0) = [-1, 1]^2$ (the “unit square”, which is the unit ball of the dual norm $\|\cdot\|_\infty$: a general pattern for norms).

For $(3, 0)$: the first coordinate is in a smooth zone ($\partial|\cdot|(3) = \{1\}$) and the second is at the kink ($\partial|\cdot|(0) = [-1, 1]$). Since f is a separable sum of functions of each coordinate (finite on all of $\mathbb{R}$), Moreau–Rockafellar gives

$$\partial f(3, 0) = \{1\} \times [-1, 1],$$

a vertical segment: a single point in the smooth coordinate, an interval in the kinked one. $\square$

Exercise 7.26. Let $f(x) = e^{|x|}$ on $\mathbb{R}$. (a) Justify that f is convex. (b) Compute $\partial f(0)$. (c) Find the global minimizer.

Solution. (a) $f = \phi \circ |\cdot|$ where $\phi(t) = e^t$ is convex and *nondecreasing* on $[0, \infty)$ and $|\cdot|$ is convex; the composition of a nondecreasing convex function with a convex function is convex (composition rule of ch. 5).

(b) Since f is smooth for $x \neq 0$, on $\mathbb{R}$ the pattern $\partial f(0) = [f'_-(0), f'_+(0)]$ is enough (Theorem 7.13 in one variable). For $x > 0$, $f(x) = e^x$ and $f'(x) = e^x \rightarrow 1$ as $x \rightarrow 0^+$, so $f'_+(0) = 1$; for $x < 0$, $f(x) = e^{-x}$ and $f'(x) = -e^{-x} \rightarrow -1$, so $f'_-(0) = -1$. Therefore $\partial f(0) = [-1, 1]$.

Check from the definition (necessity): if $e^{|y|} \geq 1 + gy$ for all y , with $y = t > 0$ small, $g \leq (e^t - 1)/t \rightarrow 1$; with $y = -t$, $g \geq -(e^t - 1)/t \rightarrow -1$. (*Sufficiency*): if $|g| \leq 1$, then $1 + gy \leq 1 + |y| \leq e^{|y|}$, using the classical inequality $1 + u \leq e^u$ with $u = |y| \geq 0$.

(c) $0 \in \partial f(x)$: for $x \neq 0$ the unique subgradient is $\pm e^{|x|} \neq 0$; at $x = 0$, $0 \in [-1, 1]$. By the generalized Fermat rule, the global minimizer is $x^* = 0$, with $f(0) = 1$. $\square$

Exercise 7.27. Check in detail the claim in Example 7.17: for $f(x) = -\sqrt{x}$ if $x \geq 0$ (and $+\infty$ if $x < 0$), we have $\partial f(0) = \emptyset$, while $\partial f(x) = \{-1/(2\sqrt{x})\}$ for $x > 0$. Does this contradict Theorem 7.8?

Solution. At $x = 0$: $g \in \partial f(0)$ would require $-\sqrt{y} \geq 0 + g(y - 0)$ for all $y \geq 0$ (for $y < 0$ it is trivial). With $y > 0$, dividing by y : $g \leq -\sqrt{y}/y = -1/\sqrt{y}$. Letting $y \rightarrow 0^+$, the right hand side tends to $-\infty$, and no real number g satisfies $g \leq -\infty$. Hence $\partial f(0) = \emptyset$.

At $x > 0$: f is differentiable there with $f'(x) = -1/(2\sqrt{x})$, and f is convex on its domain (its second derivative $\frac{1}{4}x^{-3/2}$ is positive); by Proposition 7.10, $\partial f(x) = \{-1/(2\sqrt{x})\}$.

There is no contradiction with Theorem 7.8: it guarantees nonemptiness only on $\text{int}(\text{dom } f) = (0, \infty)$, and 0 is a *boundary* point of the domain. The example shows that the interiority hypothesis is indispensable: at 0 the only candidate supporting hyperplane of the epigraph is vertical, and vertical hyperplanes do not define subgradients. $\square$

Exercise 7.28. (The median as the LAD estimator.) Let $a_1 < a_2 < a_3$ be real numbers and

$$f(x) = |x - a_1| + |x - a_2| + |x - a_3|.$$

Using subdifferentials and the generalized Fermat rule, prove that the unique global minimizer is the median $x^* = a_2$.

Solution. Each summand is finite on all of $\mathbb{R}$, so $\partial f(x) = \sum_{i=1}^3 \partial|\cdot - a_i|(x)$ (Moreau–Rockafellar with automatic qualification), where each $\partial|\cdot - a_i|(x)$ equals $\{-1\}$ if $x < a_i$, $[-1, 1]$ if $x = a_i$ and $\{+1\}$ if $x > a_i$. Walking along the line:

$$\partial f(x) = \begin{cases} \{-3\} & x < a_1, \\ [-3, -1] & x = a_1, \\ \{-1\} & a_1 < x < a_2, \\ [-1, 1] & x = a_2, \\ \{1\} & a_2 < x < a_3, \\ [1, 3] & x = a_3, \\ \{3\} & x > a_3. \end{cases}$$

(For example, at $x = a_1$: $[-1, 1] + \{-1\} + \{-1\} = [-3, -1]$; on $a_1 < x < a_2$: $\{1\} + \{-1\} + \{-1\} = \{-1\}$; at $x = a_2$: $\{1\} + [-1, 1] + \{-1\} = [-1, 1]$.) The only point where $0 \in \partial f(x)$ is $x = a_2$, since $0 \in [-1, 1]$ and in no other case does the set contain zero. By the generalized Fermat rule, $x^* = a_2$ is the unique global minimizer.

Econometric reading: this is the simplest case of LAD regression (regression with an intercept only): the estimator is the sample median. With an even number of observations, the same computation produces an *interval* of minimizers (as in Exercise 7.24): the “median” of elementary statistics is really a subdifferential that traps zero along a whole stretch. Quantile regression replaces $|u|$ by the asymmetric loss function $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$, also convex and nondifferentiable, and is analyzed in the same way. $\square$

Exercise 7.29. (Shephard with a kink.) A consumer has utility $u(x_1, x_2) = x_1 + 2x_2$ and wants to reach $u = 6$. (a) Compute the expenditure function $e(p_1, p_2, 6)$. (b) Determine at which prices e is not differentiable. (c) At $p = (2, 4)$, compute the superdifferential $\partial_p e(p, 6)$ and check that it agrees with the set of Hicksian demands.

Solution. (a) With linear utility $x_1 + 2x_2 \geq 6$, the consumer buys a single good: buying only good 1 needs $x_1 = 6$ (cost $6p_1$); buying only good 2 needs $x_2 = 3$ (cost $3p_2$). The cheaper option wins one and no interior combination improves on both corners (the constraint and the objective are linear, the minimum is at a vertex):

$$e(p, 6) = \min\{6p_1, 3p_2\}.$$

(b) e is a minimum of two linear functions of p : concave, and not differentiable exactly where both pieces tie, $6p_1 = 3p_2$, that is on the ray $p_2 = 2p_1$. (This is the locus of prices where the price *per unit of utility* is the same across goods: $p_1/1 = p_2/2$.)

(c) At $p = (2, 4)$ we are exactly at the kink ($6 \cdot 2 = 12 = 3 \cdot 4$). The active pieces are $6p_1$ and

$3p_2$, with gradients $(6, 0)$ and $(0, 3)$. By the concave version of Theorem 7.20,

$$\partial_p e(p, 6) = \text{conv}\{(6, 0), (0, 3)\} = \{ (6\lambda, 3(1 - \lambda)) : \lambda \in [0, 1] \}.$$

Hicksian demands at $p = (2, 4)$: any bundle on the line $x_1 + 2x_2 = 6$ with $x_1 \in [0, 6]$ costs $2x_1 + 4x_2 = 2x_1 + 2(6 - x_1) = 12$: they all cost the same. The solution set is $\{(x_1, x_2) : x_1 + 2x_2 = 6, x_1, x_2 \geq 0\} = \{(6\lambda, 3(1 - \lambda)) : \lambda \in [0, 1]\}$, exactly the superdifferential we computed. The general Shephard lemma (Theorem 7.22) holds: at the kink of the expenditure function lives, as a set, the whole indeterminacy of demand. $\square$

Chapter 8

Unconstrained optimization: optimality conditions, projection and distance [P2]

Weeks 5–6 of the course. Bibliography: Nocedal and Wright [5]; Ruszczyński [6]; Aragón et al. [1].

What is this used for in Microeconomics?

This is probably the chapter you will use most often without noticing in Microeconomics. The *first and second order conditions* are the bread and butter of the course:

- **Theory of the firm (MWG ch. 5):** the profit maximization problem $\max_z pf(z) - w^\top z$ is solved by setting the gradient to zero and checking concavity. We will also see why with increasing returns to scale the maximum may *fail to exist* [9].
- **Games and auctions (MWG chs. 7–9 and 23):** the Cournot *best response* comes from a first order condition (FOC), and in auctions the optimal bid of each bidder solves a FOC; equilibrium is a system of FOCs [10].
- **Moral hazard and principal agent (MWG ch. 14):** the *first order approach* (Rogerson 1985) replaces the agent incentive compatibility constraint by the agent's FOC; its validity depends exactly on the conditions of this chapter (that the FOC characterizes the global optimum of the agent) [11].
- **The key bridge:** convexity (concavity, for maximization) turns *local* conditions into *global* ones. That is what justifies the sentence, repeated a thousand times in micro, “we solve the first order condition”.
- **Projection:** the projection onto a closed convex set appears in proofs of the second welfare theorem (via separation, as we saw in the chapter on separation) and in econometrics: the ordinary least squares (OLS) estimator *is* an orthogonal projection.

In the previous chapters we built the language (convex sets and functions, separation, sub-

differential). Now we use it to answer the central question of the whole course: given a problem

$$\min_{x \in \mathbb{R}^n} f(x),$$

how do we know it has a solution? How do we find the candidate solutions? And how do we certify that a candidate really is a minimum? We mainly follow [1] (chapter 3), [5] (chapter 2) and [7] (chapter 1).

8.1 Statement of the problem and existence of solutions [P2]

8.1.1 Local and global minima

Definition 8.1 (Global, local and strict minimum). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and let $x^* \in \mathbb{R}^n$.

- (a) x^* is a **global minimizer** of f if

$$f(x^*) \leq f(x) \quad \text{for all } x \in \mathbb{R}^n.$$

- (b) x^* is a **local minimizer** of f if there is $\varepsilon > 0$ such that

$$f(x^*) \leq f(x) \quad \text{for all } x \text{ with } \|x - x^*\| < \varepsilon.$$

That is, x^* wins the comparison only against its *nearby neighbors*.

- (c) The minimum (global or local) is **strict** if the inequality is strict, $f(x^*) < f(x)$, for every $x \neq x^*$ in the corresponding set.

The value $f(x^*)$ is called the *minimum value*, and the set of all global minimizers is denoted $\arg \min_x f(x)$.

Remark 8.2 (Maximizing is minimizing). Everything we say for minima translates to maxima by replacing f with $-f$: x^* maximizes f if and only if it minimizes $-f$, and $\max_x f(x) = -\min_x (-f(x))$. Also, f is concave if and only if $-f$ is convex. This is why in micro (where we maximize utilities and profits, concave functions) and in optimization (where by convention we minimize) *the same theorems* are proved. In this chapter we minimize; translate mentally as you go: “positive definite Hessian at a minimum” $\leftrightarrow$ “negative definite Hessian at a maximum”.

Example 8.3. On $\mathbb{R}$: (i) $f(x) = x^2$ has a unique strict global minimum at $x^* = 0$. (ii) $f(x) = e^x$ has no minimum: $\inf_x e^x = 0$ but no x attains that value (the infimum is not *attained*). (iii) $f(x) = x^3$ has neither a local nor a global minimum ($f(x) \rightarrow -\infty$). (iv) $f(x) = (x^2 - 1)^2$ has two global minima, at $x = 1$ and $x = -1$ (the global minimum is not unique, although each of the two is a strict local minimum), and a local maximum at $x = 0$.

Example (ii) shows that the first serious question is *existence*: before differentiating and setting to zero, you need to know that there is something to find.

8.1.2 Existence: Weierstrass and coercivity

Recall the classical theorem from analysis (we already used it in the preliminaries chapter):

Theorem 8.4 (Weierstrass). *If $K \subseteq \mathbb{R}^n$ is nonempty, closed and bounded (that is, compact) and $f : K \rightarrow \mathbb{R}$ is continuous, then f attains its minimum and its maximum on K : there are $x^*, x^{**} \in K$ such that $f(x^*) \leq f(x) \leq f(x^{**})$ for all $x \in K$.*

The problem is that in unconstrained optimization the domain is all of $\mathbb{R}^n$, which is not bounded, so Weierstrass does not apply directly. The way out is a condition on f that “recreates” compactness: requiring the function to grow to infinity in all directions, so that the search for the minimum can be confined to a bounded region.

Definition 8.5 (Coercive function). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **coercive** if

$$\lim_{\|x\| \rightarrow \infty} f(x) = +\infty,$$

that is: for every $M \in \mathbb{R}$ there is $R > 0$ such that $f(x) > M$ whenever $\|x\| > R$. Equivalently, for every sequence (x_k) with $\|x_k\| \rightarrow \infty$ we have $f(x_k) \rightarrow +\infty$.

Example 8.6. (i) $f(x) = \|x\|^2$ is coercive. (ii) $f(x) = e^x$ on $\mathbb{R}$ is *not* coercive: as $x \rightarrow -\infty$, $|x| \rightarrow \infty$ but $f(x) \rightarrow 0$. (iii) $f(x, y) = x^2$ on $\mathbb{R}^2$ is not coercive: along the sequence $(0, k)$ the norm blows up but f is always 0. Coercivity requires growth *in all directions*, not only in some. (iv) $f(x, y) = x^4 + y^4 - 3xy$ is coercive (we prove it in Exercise 8.37): the fourth degree terms dominate the cross term.

Theorem 8.7 (Existence under coercivity). *If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous and coercive, then f attains a global minimum on $\mathbb{R}^n$: the set $\arg \min_x f(x)$ is nonempty (and compact).*

Proof. The idea is to reduce the problem to a compact set and apply Weierstrass.

Step 1: confine the search. Fix any point $x_0 \in \mathbb{R}^n$ (for example $x_0 = 0$) and consider the sublevel set

$$S = \{x \in \mathbb{R}^n : f(x) \leq f(x_0)\}.$$

S is nonempty because $x_0 \in S$.

Step 2: S is closed. S is the preimage of the closed interval $(-\infty, f(x_0)]$ under the continuous function f , and the preimage of a closed set under a continuous function is closed.

Step 3: S is bounded. Suppose not. Then there would be a sequence $(x_k) \subseteq S$ with $\|x_k\| \rightarrow \infty$. By coercivity, $f(x_k) \rightarrow +\infty$. But each x_k lies in S , so $f(x_k) \leq f(x_0)$ for all k , which prevents $f(x_k)$ from diverging. Contradiction. Hence S is bounded.

Step 4: Weierstrass. S is nonempty, closed and bounded, that is, compact, and f is continuous on S . By Theorem 8.4, there is $x^* \in S$ with $f(x^*) \leq f(x)$ for all $x \in S$.

Step 5: the minimum over S is global. Let $x \in \mathbb{R}^n$ be arbitrary. If $x \in S$, we already know $f(x^*) \leq f(x)$. If $x \notin S$, then $f(x) > f(x_0) \geq f(x^*)$ (the last inequality because $x_0 \in S$). In both cases $f(x^*) \leq f(x)$, so x^* is a global minimizer. Finally, $\arg \min f = \{x : f(x) \leq f(x^*)\}$ is closed and bounded by the same arguments of steps 2 and 3, hence compact. $\square$

Intuition

Coercivity is a “wall at infinity”: since the function rises without bound far from the origin, no candidate minimum can escape outwards, and everything happens inside a

ball. Inside the ball, Weierstrass rules. Note the division of labor: *continuity* guarantees that the infimum is attained on compact sets; *coercivity* guarantees that we can work on a compact set. Without the first, the example of a function with a “hole” at the good point fails; without the second, e^x fails, since its infimum escapes toward the infinite edge of the domain (as $x \rightarrow -\infty$) and is never attained.

Example 8.8 (Why firm profit may fail to attain a maximum). Consider a competitive firm with one input, technology $f(z) = z^\alpha$ with $\alpha > 0$, output price $p > 0$ and input price $w > 0$. Its profit is

$$\pi(z) = pz^\alpha - wz, \quad z \geq 0.$$

Maximizing π is the same as minimizing $-\pi(z) = wz - pz^\alpha$.

- **Decreasing returns** ($\alpha < 1$): $-\pi$ is coercive on $[0, \infty)$, since wz grows linearly and pz^α only sublinearly, so $-\pi(z) \rightarrow +\infty$ as $z \rightarrow \infty$. By the argument of Theorem 8.7 (adapted to the closed set $[0, \infty)$), the maximum of π exists.
- **Increasing returns** ($\alpha > 1$): now $\pi(z) = pz^\alpha - wz \rightarrow +\infty$: doubling the scale more than doubles revenue, and the firm problem *has no solution*; the supremum of profit is $+\infty$. The FOC may have a solution (a critical point), but it is an interior *minimum* of profit, not a maximum.
- **Constant returns** ($\alpha = 1$): if $p > w$ profit grows without bound; if $p = w$ every z is optimal with profit 0; if $p < w$ the optimum is $z = 0$.

What “repairs” existence? Either assumptions on the technology that make $-\pi$ coercive (decreasing returns, convex costs growing faster than revenue), or *compactifying* the feasible set (finite installed capacity, limited inputs: $z \in [0, \bar{z}]$, and Weierstrass does the rest). This discussion is exactly the one in [9] about nonexistence of the optimum with nondecreasing returns.

Exam strategy

For any optimization problem, in an exam (in this course or in micro) always follow this order:

1. **Existence:** is the domain compact (Weierstrass)? Is the objective coercive (or is $-f$, if you are maximizing)? If you cannot guarantee existence, say so: that may be exactly what is being asked (increasing returns).
2. **Candidates:** solve $\nabla f(x) = 0$. Do not forget points of nondifferentiability if there are any (there the candidate condition is $0 \in \partial f$).
3. **Local classification:** Hessian at each candidate (positive definite $\Rightarrow$ strict local minimum; negative definite $\Rightarrow$ strict local maximum; indefinite $\Rightarrow$ saddle; semidefinite $\Rightarrow$ the test does not decide, analyze by hand).
4. **Globality:** is f convex (or concave)? Then the critical point is a *global* optimum and you are done. If not, compare values across candidates and study the behavior at infinity.

Writing these four steps, even when some of them are trivial, organizes the solution and usually earns points by itself.

8.2 First order necessary condition [P1]

In this section we assume that f is differentiable. Recall from the preliminaries chapter that the *gradient* $\nabla f(x)$ is the vector of partial derivatives and that the *directional derivative* of f at x in the direction $d \in \mathbb{R}^n$ is

$$f'(x; d) = \lim_{t \downarrow 0} \frac{f(x + td) - f(x)}{t} = \langle \nabla f(x), d \rangle,$$

the instantaneous rate of change of f when moving from x in the direction d .

Theorem 8.9 (First order necessary condition, Fermat). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at x^* . If x^* is a local minimizer of f , then*

$$\nabla f(x^*) = 0.$$

Proof. Fix an arbitrary direction $d \in \mathbb{R}^n$ and consider the one variable function

$$g(t) = f(x^* + td),$$

which “walks along” f on the line through x^* with direction d . Since x^* is a local minimum, there is $\varepsilon > 0$ such that $f(x^*) \leq f(x)$ for all x with $\|x - x^*\| < \varepsilon$. If $d \neq 0$ and $0 < t < \varepsilon / \|d\|$, the point $x^* + td$ lies in that neighborhood, so

$$f(x^* + td) - f(x^*) \geq 0 \implies \frac{f(x^* + td) - f(x^*)}{t} \geq 0,$$

where dividing by $t > 0$ preserves the direction of the inequality. Taking the limit as $t \downarrow 0$ (limits preserve weak inequalities):

$$\langle \nabla f(x^*), d \rangle = \lim_{t \downarrow 0} \frac{f(x^* + td) - f(x^*)}{t} \geq 0.$$

This holds for *every* direction d . In particular, applying it to the direction $d = -\nabla f(x^*)$:

$$\langle \nabla f(x^*), -\nabla f(x^*) \rangle = -\|\nabla f(x^*)\|^2 \geq 0,$$

which forces $\|\nabla f(x^*)\|^2 \leq 0$. Since a squared norm is always ≥ 0 , we conclude $\|\nabla f(x^*)\| = 0$, that is, $\nabla f(x^*) = 0$. $\square$

Intuition

The proof says two things. First: at a local minimum, moving in *any* direction cannot lower the function to first order ($f'(x^*; d) \geq 0$ for every d). Second: if the gradient were not zero, the direction $-\nabla f(x^*)$ (the direction of *steepest descent*, the one used by

gradient algorithms [5, 8]) would strictly lower the function, contradicting minimality. In one sentence: at the bottom of the valley, the ground is flat.

Definition 8.10 (Critical point). A point x with $\nabla f(x) = 0$ is called a **critical** or **stationary point** of f .

Careful (common mistake)

Theorem 8.9 is a **necessary** condition, not a sufficient one. The correct implication is:

$$x^* \text{ local minimum} \implies \nabla f(x^*) = 0, \quad \text{and not the other way round.}$$

Classic exam mistakes:

- Solving $\nabla f = 0$ and declaring “optimum” without checking second order or globality: the point may be a maximum, a saddle, or nothing at all (example: $x = 0$ for $f(x) = x^3$).
- Forgetting existence: the FOC may have no solution ($f(x) = e^x$: $f'(x) = e^x \neq 0$ always) or have one without the optimum existing (the firm with increasing returns of Example 8.8).
- Missing the “necessary under differentiability” clause: $f(x) = |x|$ has a minimum at 0 with no gradient there; the correct condition is $0 \in \partial f(0)$ (Section 8.4).

The only situation where “FOC $\Rightarrow$ global optimum” with no further work is when f is convex (Theorem 8.23); that is the deep reason why in micro concavity is almost always assumed.

Example 8.11 (Saddle point). Let $f(x, y) = x^2 - y^2$. Its gradient is $\nabla f(x, y) = (2x, -2y)$, which vanishes only at $(0, 0)$. Is it an optimum? Look at f along the axes:

$$f(x, 0) = x^2 \geq 0 = f(0, 0) \quad \text{and} \quad f(0, y) = -y^2 \leq 0 = f(0, 0).$$

Along the x axis, the origin looks like a minimum; along the y axis, like a maximum. In *every* neighborhood of $(0, 0)$ there are points where f is larger and points where it is smaller: $(0, 0)$ is neither a local minimum nor a local maximum. Such a critical point is called a **saddle point**, after the saddle shape of the graph. The Hessian is

$$\nabla^2 f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix},$$

with eigenvalues 2 and -2 : *indefinite*, the algebraic signature of a saddle, as we formalize next.

8.3 Second order conditions [P1]

The FOC filters candidates; second order classifies them. The tool is the second order Taylor expansion (preliminaries chapter): if f is twice differentiable at x^* , then as $h \rightarrow 0$,

$$f(x^* + h) = f(x^*) + \langle \nabla f(x^*), h \rangle + \frac{1}{2} h^\top \nabla^2 f(x^*) h + o(\|h\|^2), \quad (8.1)$$

where $\nabla^2 f(x^*)$ is the Hessian matrix (symmetric, by the Schwarz theorem when f is of class C^2) and $o(\|h\|^2)$ denotes a remainder with $o(\|h\|^2)/\|h\|^2 \rightarrow 0$ as $h \rightarrow 0$. Recall also from the preliminaries chapter: a symmetric matrix A is *positive semidefinite* (PSD) if $d^\top A d \geq 0$ for all d , and *positive definite* (PD) if $d^\top A d > 0$ for all $d \neq 0$.

Theorem 8.12 (Second order necessary condition). *Let f be twice differentiable at x^* . If x^* is a local minimizer of f , then $\nabla f(x^*) = 0$ and the Hessian $\nabla^2 f(x^*)$ is positive semidefinite.*

Proof. That $\nabla f(x^*) = 0$ is Theorem 8.9. Fix a direction $d \in \mathbb{R}^n$, $d \neq 0$, and take $h = td$ with $t > 0$ small in (8.1). Since the first order term vanishes:

$$f(x^* + td) - f(x^*) = \frac{t^2}{2} d^\top \nabla^2 f(x^*) d + o(t^2).$$

For t small enough, $x^* + td$ falls in the neighborhood where x^* is a minimum, so the left hand side is ≥ 0 . Dividing by $t^2 > 0$:

$$\frac{1}{2} d^\top \nabla^2 f(x^*) d + \frac{o(t^2)}{t^2} \geq 0.$$

Letting $t \downarrow 0$, the term $o(t^2)/t^2$ tends to 0 (that is the definition of $o(t^2)$), and we are left with $d^\top \nabla^2 f(x^*) d \geq 0$. Since d was arbitrary, $\nabla^2 f(x^*)$ is positive semidefinite. $\square$

Theorem 8.13 (Second order sufficient condition). *Let f be twice differentiable at x^* , with*

$$\nabla f(x^*) = 0 \quad \text{and} \quad \nabla^2 f(x^*) \text{ positive definite.}$$

Then x^ is a strict local minimizer of f .*

Proof. Write $H = \nabla^2 f(x^*)$ and let $\lambda_{\min} > 0$ be its smallest eigenvalue (it is positive precisely because H is positive definite).

Step 1: quadratic lower bound. Since H is symmetric, the spectral theorem (preliminaries chapter) gives an orthonormal basis of eigenvectors $v_1, \dots, v_n$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n = \lambda_{\min}$. Writing $h = \sum_i c_i v_i$ (with $\|h\|^2 = \sum_i c_i^2$ by orthonormality):

$$h^\top H h = \sum_{i=1}^n \lambda_i c_i^2 \geq \lambda_{\min} \sum_{i=1}^n c_i^2 = \lambda_{\min} \|h\|^2 \quad \text{for all } h \in \mathbb{R}^n.$$

That is: the quadratic form cannot be flatter than its smallest eigenvalue.

Step 2: Taylor. With $\nabla f(x^*) = 0$, expansion (8.1) and Step 1 give

$$f(x^* + h) - f(x^*) = \frac{1}{2} h^\top H h + o(\|h\|^2) \geq \frac{\lambda_{\min}}{2} \|h\|^2 + o(\|h\|^2).$$

Step 3: the remainder does not spoil the bound. By the definition of $o(\|h\|^2)$, there is $\delta > 0$ such that if $0 < \|h\| < \delta$ then

$$|o(\|h\|^2)| \leq \frac{\lambda_{\min}}{4} \|h\|^2.$$

(We are using the definition of limit with $\varepsilon = \lambda_{\min}/4 > 0$.) Then, for $0 < \|h\| < \delta$,

$$f(x^* + h) - f(x^*) \geq \frac{\lambda_{\min}}{2} \|h\|^2 - \frac{\lambda_{\min}}{4} \|h\|^2 = \frac{\lambda_{\min}}{4} \|h\|^2 > 0.$$

Hence $f(x^* + h) > f(x^*)$ for every $h \neq 0$ with $\|h\| < \delta$: x^* is a strict local minimizer. $\square$

Intuition

The argument is a race between the quadratic term (which pushes upward with strength at least $\lambda_{\min} \|h\|^2 / 2$) and the Taylor remainder (which is *negligible compared to* $\|h\|^2$ near x^*). Since the quadratic push has a fixed positive rate and the remainder has a rate going to zero, near x^* the quadratic term wins. It is essential here that $\lambda_{\min} > 0$ is a fixed number: if H were only positive semidefinite, some direction would have $\lambda = 0$, the quadratic term would push nothing in that direction, and the remainder (of unknown sign) would decide the outcome. This is why PSD is not enough as a sufficient condition.

Example 8.14 (The borderline case: when the test does not decide). On $\mathbb{R}$, consider $f(x) = x^4$ and $g(x) = x^3$. Both satisfy $f'(0) = g'(0) = 0$ and $f''(0) = g''(0) = 0$: in both cases the 1×1 “Hessian” is the zero matrix, which is positive semidefinite (and also negative semidefinite). However:

- $x = 0$ is a strict global minimum of $f(x) = x^4$ ($x^4 > 0$ for $x \neq 0$);
- $x = 0$ is *neither* a local minimum nor a local maximum of $g(x) = x^3$ (g changes sign: it is an inflection point).

Conclusions: (1) the necessary condition (PSD) is not sufficient, as x^3 shows; (2) the sufficient condition (PD) is not necessary, as x^4 shows (a strict minimum with a Hessian that is not PD). When the Hessian at a critical point is semidefinite but not definite, the second order test **concludes nothing** and you have to analyze the function directly (higher order terms, convexity, or inspection).

8.3.1 Worked classification examples

Example 8.15 (A quadratic). Classify the critical points of

$$f(x, y) = 2x^2 + y^2 + xy - 4x - 2y.$$

Step 1 (FOC). $\nabla f(x, y) = (4x + y - 4, 2y + x - 2) = (0, 0)$ gives the linear system

$$4x + y = 4, \quad x + 2y = 2.$$

From the first, $y = 4 - 4x$; substituting into the second: $x + 8 - 8x = 2 \Rightarrow -7x = -6 \Rightarrow x^* = 6/7$, and then $y^* = 4 - 24/7 = 4/7$. A single critical point: $(6/7, 4/7)$.

Step 2 (SOC). The Hessian is constant:

$$\nabla^2 f = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its leading principal minors are $4 > 0$ and $\det = 4 \cdot 2 - 1 \cdot 1 = 7 > 0$: by the Sylvester criterion (preliminaries chapter), it is positive definite. By Theorem 8.13, $(6/7, 4/7)$ is a strict local minimum.

Step 3 (globality). Since the Hessian is PD *everywhere*, f is (strictly) convex (chapter on convex functions), and by Theorem 8.23 below, the point is the *global* minimum, and it is unique. The minimum value is $f(6/7, 4/7) = -16/7$ (direct check).

Example 8.16 (Several critical points). Classify the critical points of

$$f(x, y) = x^3 - 3x + y^2.$$

FOC. $\nabla f = (3x^2 - 3, 2y) = (0, 0)$ gives $x^2 = 1$ and $y = 0$: two critical points, $(1, 0)$ and $(-1, 0)$.

$$\text{SOC. } \nabla^2 f(x, y) = \begin{pmatrix} 6x & 0 \\ 0 & 2 \end{pmatrix}.$$

- At $(1, 0)$: $\text{diag}(6, 2)$, eigenvalues $6, 2 > 0$, positive definite $\Rightarrow$ **strict local minimum**, with $f(1, 0) = -2$.
- At $(-1, 0)$: $\text{diag}(-6, 2)$, eigenvalues $-6 < 0 < 2$, **indefinite** $\Rightarrow$ **saddle point** (along the y axis the function goes up; along the x axis, down).

Globality. $(1, 0)$ is *not* a global minimum: $f(x, 0) = x^3 - 3x \rightarrow -\infty$ as $x \rightarrow -\infty$, so $\inf f = -\infty$ and there is no global minimum. Moral: local classification says nothing about global behavior without convexity or coercivity.

Example 8.17 (Monopolist with linear demand and quadratic cost). A monopolist faces inverse demand $p(q) = a - bq$ (with $a, b > 0$) and has cost $C(q) = cq + \frac{d}{2}q^2$ (with $0 \leq c < a$, $d \geq 0$): a marginal cost $c + dq$ increasing in scale. Its profit is

$$\pi(q) = p(q)q - C(q) = (a - bq)q - cq - \frac{d}{2}q^2 = (a - c)q - \left(b + \frac{d}{2}\right)q^2.$$

Existence. $-\pi(q) = (b + d/2)q^2 - (a - c)q$ is coercive (the quadratic term with positive coefficient dominates), so the maximum exists (Theorem 8.7).

FOC. $\pi'(q) = (a - c) - (2b + d)q = 0$, so

$$q^* = \frac{a - c}{2b + d} > 0.$$

Economically, the FOC is the equality *marginal revenue* = *marginal cost*: $a - 2bq = c + dq$.

SOC. $\pi''(q) = -(2b + d) < 0$ for all q : π is strictly concave (equivalently, $-\pi$ is strictly convex), so q^* is not only a strict local maximum but the **unique global maximum** (Theorem 8.23 applied to $-\pi$). The monopoly price is $p^* = a - bq^* = \frac{a(b + d) + bc}{2b + d}$, which is checked to be above marginal cost at q^* (market power).

Example 8.18 (Cournot best response). Two firms produce a homogeneous good with constant marginal cost $c \in (0, a)$; the market price is $p = a - b(q_1 + q_2)$. Given the quantity $q_2 \geq 0$ of the rival, firm 1 solves

$$\max_{q_1 \geq 0} \pi_1(q_1) = (a - b(q_1 + q_2))q_1 - cq_1.$$

FOC. $\pi'_1(q_1) = a - 2bq_1 - bq_2 - c = 0$, which gives the **best response function**:

$$q_1 = BR_1(q_2) = \frac{a - c - bq_2}{2b} \quad (\text{and } q_1 = 0 \text{ if } q_2 \geq (a - c)/b).$$

SOC. $\pi''_1(q_1) = -2b < 0$: π_1 is strictly concave in q_1 , so the FOC characterizes the global maximum. *This is what validates the standard procedure of game theory*: “differentiate, set to zero, solve for the best response”. In the symmetric Nash equilibrium, $q_1 = q_2 = q^*$ solves $q^* = (a - c - bq^*)/(2b)$, that is,

$$q^* = \frac{a - c}{3b}, \quad Q^* = \frac{2(a - c)}{3b}, \quad p^* = \frac{a + 2c}{3}.$$

The same scheme, FOC + concavity, validates the equilibrium bid in first price auctions and the first order approach in moral hazard [10, 11].

8.4 Minimizing convex functions [P1]

Here is the magic that makes this course special: for convex functions, all the pathologies above (local minima that are not global, critical points that are saddles) disappear. Recall from the chapter on convex functions: $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if for all x, y and $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y),$$

and **strictly convex** if the inequality is strict when $x \neq y$ and $\lambda \in (0, 1)$.

Theorem 8.19 (Every local minimum is global). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex. If x^* is a local minimizer of f , then x^* is a global minimizer.*

Proof. By contradiction: suppose x^* is a local minimum but not a global one. Then there is $y \in \mathbb{R}^n$ with $f(y) < f(x^*)$. Consider the points of the segment between x^* and y :

$$x_\lambda = (1 - \lambda)x^* + \lambda y, \quad \lambda \in (0, 1].$$

By convexity,

$$f(x_\lambda) \leq (1 - \lambda)f(x^*) + \lambda f(y) < (1 - \lambda)f(x^*) + \lambda f(x^*) = f(x^*),$$

where the strict inequality uses $f(y) < f(x^*)$ and $\lambda > 0$. That is, *every* point of the segment (other than x^*) has value strictly below $f(x^*)$. But $\|x_\lambda - x^*\| = \lambda\|y - x^*\| \rightarrow 0$ as $\lambda \downarrow 0$: there are points x_λ *arbitrarily close* to x^* with $f(x_\lambda) < f(x^*)$. This contradicts x^* being a local minimum (no neighborhood of x^* can satisfy $f(x^*) \leq f$). Hence no such y exists, and x^* is a global minimum. $\square$

Intuition

If there were a better point somewhere else, the “chord” joining the local valley with that better point would lie, by convexity, above the graph; walking along the segment toward the better point, the function goes down *from the very first step*. A local valley you can walk out of downhill is not a valley. In a convex function there are no false valleys.

Theorem 8.20 (The set of minimizers is convex). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex. Then $\arg \min_x f(x)$ is a convex set (possibly empty).*

Proof. If $\arg \min f$ is empty or has a single point, it is trivially convex. Let $x, y \in \arg \min f$, both with the minimum value $p^* = f(x) = f(y)$, and let $\lambda \in [0, 1]$. By convexity,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) = \lambda p^* + (1 - \lambda)p^* = p^*.$$

But p^* is the minimum value of f , so also $f(\lambda x + (1 - \lambda)y) \geq p^*$. Hence $f(\lambda x + (1 - \lambda)y) = p^*$ and the point $\lambda x + (1 - \lambda)y$ belongs to $\arg \min f$. (In fact, $\arg \min f = \{x : f(x) \leq p^*\}$ is a sublevel set of a convex function, and we already saw in the chapter on convex functions that these are convex.) $\square$

Theorem 8.21 (Uniqueness under strict convexity). *If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly convex, then f has at most one global minimizer.*

Proof. Suppose $x \neq y$ are both global minimizers, with value $p^* = f(x) = f(y)$. Take the midpoint $m = \frac{1}{2}x + \frac{1}{2}y$. By *strict* convexity (with $\lambda = 1/2 \in (0, 1)$ and $x \neq y$):

$$f(m) < \frac{1}{2}f(x) + \frac{1}{2}f(y) = p^*.$$

This contradicts p^* being the minimum value. Hence there cannot be two distinct minimizers. $\square$

Remark 8.22. Strict convexity gives *uniqueness* but not *existence*: $f(x) = e^x$ is strictly convex and has no minimizer. The full package “exists and is unique” requires, for example, strict convexity + coercivity. In micro, this is the standard argument for demand being a *function* (not a correspondence): strictly quasiconcave utility $\Rightarrow$ unique optimal choice [9, 10].

Theorem 8.23 (The FOC is sufficient under convexity). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and differentiable. Then x^* is a global minimizer of f **if and only if** $\nabla f(x^*) = 0$.*

Proof. ($\Rightarrow$) This is Theorem 8.9 (a global minimum is in particular a local one).

($\Leftarrow$) We use the *first order characterization of convexity* proved in the chapter on convex functions: a convex differentiable f satisfies, for all x, y ,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle,$$

that is, the tangent at any point stays below the graph (it is a *global minorant*). Applying it at $x = x^*$ with $\nabla f(x^*) = 0$:

$$f(y) \geq f(x^*) + \langle 0, y - x^* \rangle = f(x^*) \quad \text{for all } y \in \mathbb{R}^n,$$

which is exactly the definition of a global minimum. $\square$

Remark 8.24 (The nondifferentiable version: the subdifferential). If f is convex but not differentiable (think of $f(x) = |x|$ or of $\|\cdot\|_1$, ubiquitous in statistics), the correct optimality condition uses the subdifferential of the previous chapter:

$$x^* \in \arg \min_x f(x) \iff 0 \in \partial f(x^*).$$

The proof is immediate from the definition of subgradient: $0 \in \partial f(x^*)$ means $f(y) \geq f(x^*) + \langle 0, y - x^* \rangle = f(x^*)$ for all y , which is global optimality; and conversely, if x^* is a global minimum that same inequality says that 0 is a subgradient. The Fermat rule $\nabla f(x^*) = 0$ is the special case where $\partial f(x^*) = \{\nabla f(x^*)\}$. See [6] (chapter 3) and [3].

8.4.1 Star example: ordinary least squares

Example 8.25 (OLS). This is the most frequently solved optimization problem in the history of economics. We have m observations of a dependent variable, $y \in \mathbb{R}^m$, and a matrix of regressors $X \in \mathbb{R}^{m \times n}$ (each column is an explanatory variable; $m \geq n$). We look for the coefficient vector $\beta \in \mathbb{R}^n$ minimizing the sum of squared residuals:

$$\min_{\beta \in \mathbb{R}^n} f(\beta) = \|y - X\beta\|^2.$$

Step 1: expand. Using $\|v\|^2 = v^\top v$ and the transposition rules:

$$f(\beta) = (y - X\beta)^\top (y - X\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta.$$

(The cross term appears twice, $y^\top X\beta$ and $\beta^\top X^\top y$, and both are the same scalar.)

Step 2: gradient. With the matrix differentiation rules of the preliminaries chapter ($\nabla_\beta(b^\top \beta) = b$ and $\nabla_\beta(\beta^\top A\beta) = 2A\beta$ for symmetric A ; here $A = X^\top X$ is symmetric):

$$\nabla f(\beta) = -2X^\top y + 2X^\top X \beta.$$

The FOC $\nabla f(\beta) = 0$ gives the famous **normal equations**:

$$X^\top X \beta = X^\top y. \tag{8.2}$$

Step 3: convexity. The Hessian is $\nabla^2 f(\beta) = 2X^\top X$, constant. For any $d \in \mathbb{R}^n$,

$$d^\top (2X^\top X) d = 2(Xd)^\top (Xd) = 2\|Xd\|^2 \geq 0:$$

$X^\top X$ is always positive semidefinite, so f is **convex** and, by Theorem 8.23, *every* solution of (8.2) is a global minimum. Nothing else needs to be checked: the FOC is enough.

Step 4: uniqueness. If X has *full column rank* ($\text{rank } X = n$, that is, no perfect collinearity among regressors), then $Xd = 0$ implies $d = 0$, so $\|Xd\|^2 > 0$ for $d \neq 0$: $X^\top X$ is positive

definite, f is strictly convex, the solution is unique (Theorem 8.21) and the system is invertible:

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

A note for the economist. This is exactly the OLS estimator formula from econometrics. The “no perfect multicollinearity” assumption of econometrics courses is, mathematically, the strict convexity condition that guarantees uniqueness. And in Section 8.6 we will see that $\hat{\beta}$ also has a geometric reading: $X\hat{\beta}$ is the *projection* of y onto the space spanned by the columns of X .

8.5 Distance to a set and the projection map [P2]

We now move from a “free” question (minimizing f on all of $\mathbb{R}^n$) to the most geometric of all optimization questions: given a set C and a point x outside it, which point of C is closest to x ? This problem is itself an optimization problem (minimizing the function $y \mapsto \|x - y\|$ over C), and its solution, the *projection*, is one of the most used tools in convex analysis: in the chapter on separation we used it (stated without a full proof) to build separating hyperplanes; here we develop it with all the details.

Definition 8.26 (Distance to a set). Let $C \subseteq \mathbb{R}^n$ be nonempty. The **distance function** to C is

$$d_C(x) = \inf_{y \in C} \|x - y\|, \quad x \in \mathbb{R}^n.$$

We use $\inf$ (the greatest lower bound, which always exists because norms are ≥ 0) and not $\min$ because, in general, the closest point may fail to exist: if $C = (0, 1)$ in $\mathbb{R}$ and $x = 2$, then $d_C(x) = 1$ but no $y \in C$ satisfies $|2 - y| = 1$ (the candidate would be $y = 1 \notin C$). The problem is that C is not closed. Closedness is enough:

Proposition 8.27 (Existence of the closest point). *If $C \subseteq \mathbb{R}^n$ is nonempty and closed, then for every $x \in \mathbb{R}^n$ the infimum is attained: there is $\bar{y} \in C$ with $\|x - \bar{y}\| = d_C(x)$.*

Proof. Fix x and any point $y_0 \in C$, and let $r = \|x - y_0\|$. The points of C at distance greater than r from x cannot be the closest ones, so

$$\inf_{y \in C} \|x - y\| = \inf_{y \in K} \|x - y\|, \quad \text{where } K = C \cap \{y : \|x - y\| \leq r\}.$$

The set K is nonempty ($y_0 \in K$), closed (an intersection of two closed sets: C and a closed ball) and bounded (it lies inside the ball of radius r centered at x): it is compact. The function $y \mapsto \|x - y\|$ is continuous (a consequence of the triangle inequality). By Weierstrass (Theorem 8.4), the infimum over K is attained at some $\bar{y} \in K \subseteq C$. $\square$

Note the same move as in Theorem 8.7: trim the problem down to a compact set and let Weierstrass work. It is a pattern worth recognizing.

Proposition 8.28 (The distance to a convex set is convex). *If $C \subseteq \mathbb{R}^n$ is nonempty and convex, then $d_C : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function.*

Proof. Let $x_1, x_2 \in \mathbb{R}^n$, $\lambda \in [0, 1]$ and $x_\lambda = \lambda x_1 + (1 - \lambda)x_2$. Fix an arbitrary $\varepsilon > 0$. By definition of infimum, there are $y_1, y_2 \in C$ that are “almost optimal”:

$$\|x_1 - y_1\| \leq d_C(x_1) + \varepsilon, \quad \|x_2 - y_2\| \leq d_C(x_2) + \varepsilon.$$

Since C is convex, the point $y_\lambda = \lambda y_1 + (1 - \lambda)y_2$ belongs to C , and is therefore an admissible candidate in the infimum defining $d_C(x_\lambda)$:

$$\begin{aligned} d_C(x_\lambda) &\leq \|x_\lambda - y_\lambda\| = \|\lambda(x_1 - y_1) + (1 - \lambda)(x_2 - y_2)\| \\ &\leq \lambda \|x_1 - y_1\| + (1 - \lambda) \|x_2 - y_2\| \quad (\text{triangle inequality and homogeneity}) \\ &\leq \lambda d_C(x_1) + (1 - \lambda)d_C(x_2) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, letting $\varepsilon \downarrow 0$ we get $d_C(x_\lambda) \leq \lambda d_C(x_1) + (1 - \lambda)d_C(x_2)$: d_C is convex. (Note where each hypothesis was used: convexity of C so that $y_\lambda \in C$; the norm properties to split the distance.) $\square$

8.5.1 The projection theorem

Definition 8.29 (Projection onto a set). Let $C \subseteq \mathbb{R}^n$ be nonempty, closed and convex. The **projection** of $x \in \mathbb{R}^n$ onto C , written $\text{proj}_C(x)$, is the point of C closest to x :

$$\text{proj}_C(x) = \arg \min_{y \in C} \|x - y\|.$$

The next theorem justifies the definite article “the point”: the projection exists and is unique, so $\text{proj}_C : \mathbb{R}^n \rightarrow C$ is a well defined function.

Theorem 8.30 (Projection theorem). *Let $C \subseteq \mathbb{R}^n$ be nonempty, closed and convex, and let $x \in \mathbb{R}^n$. Then:*

- (a) **(Existence)** *There is at least one point of C closest to x .*
- (b) **(Uniqueness)** *That point is unique; we write it $\text{proj}_C(x)$.*
- (c) **(Variational characterization)** *A point $p \in C$ is the projection of x if and only if*

$$\langle x - p, y - p \rangle \leq 0 \quad \text{for all } y \in C. \quad (8.3)$$

Proof. **(a) Existence.** This is Proposition 8.27 (it only requires C closed and nonempty; convexity is not used yet).

(b) Uniqueness. Here convexity enters, through the *parallelogram identity*, valid for the Euclidean norm:

$$\|a + b\|^2 + \|a - b\|^2 = 2\|a\|^2 + 2\|b\|^2 \quad \text{for all } a, b \in \mathbb{R}^n$$

(check it by expanding $\|a \pm b\|^2 = \|a\|^2 \pm 2\langle a, b \rangle + \|b\|^2$ and adding). Suppose $p_1, p_2 \in C$ are both closest points: $\|x - p_1\| = \|x - p_2\| = d$, where $d = d_C(x)$. Apply the identity with $a = x - p_1$

and $b = x - p_2$:

$$\|2x - p_1 - p_2\|^2 + \|p_2 - p_1\|^2 = 2\|x - p_1\|^2 + 2\|x - p_2\|^2 = 4d^2.$$

The first term is $\|2x - p_1 - p_2\|^2 = 4\|x - m\|^2$ where $m = \frac{1}{2}(p_1 + p_2)$ is the midpoint. Solving:

$$\|x - m\|^2 = d^2 - \frac{1}{4}\|p_1 - p_2\|^2.$$

By convexity of C , $m \in C$, so $\|x - m\| \geq d$ (nobody in C is closer than d). Combining:

$$d^2 \leq \|x - m\|^2 = d^2 - \frac{1}{4}\|p_1 - p_2\|^2 \implies \|p_1 - p_2\|^2 \leq 0 \implies p_1 = p_2.$$

(Geometrically: if there were two distinct closest points, their midpoint, which also lies in C , would be *strictly* closer to x ; impossible.)

(c) Characterization. ($\Rightarrow$) Let $p = \text{proj}_C(x)$ and let $y \in C$ be arbitrary. For $t \in (0, 1]$, the point $p_t = p + t(y - p) = (1 - t)p + ty$ lies in C by convexity. Since p minimizes the distance (and therefore the squared distance):

$$\|x - p\|^2 \leq \|x - p_t\|^2 = \|(x - p) - t(y - p)\|^2 = \|x - p\|^2 - 2t\langle x - p, y - p \rangle + t^2\|y - p\|^2.$$

Cancelling $\|x - p\|^2$ and dividing by $t > 0$:

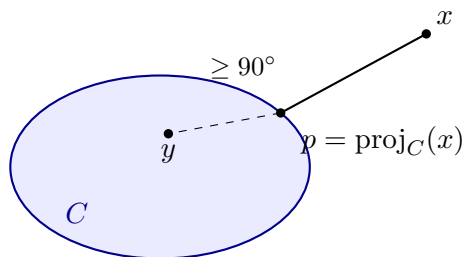
$$0 \leq -2\langle x - p, y - p \rangle + t\|y - p\|^2.$$

Letting $t \downarrow 0$ we get $\langle x - p, y - p \rangle \leq 0$. (Compare with the proof of Theorem 8.9: it is the same “directional derivative ≥ 0 ” argument, now with moves that stay inside C .)

($\Leftarrow$) Suppose $p \in C$ satisfies (8.3). For any $y \in C$, we expand:

$$\begin{aligned} \|x - y\|^2 &= \|(x - p) - (y - p)\|^2 \\ &= \|x - p\|^2 - 2\langle x - p, y - p \rangle + \|y - p\|^2 \\ &\geq \|x - p\|^2 + \|y - p\|^2 && \text{(by (8.3), } -2\langle \cdot, \cdot \rangle \geq 0) \\ &\geq \|x - p\|^2. \end{aligned}$$

Hence p is the point of C closest to x , that is, $p = \text{proj}_C(x)$. Note that the first inequality also gives the quantitative information $\|x - y\|^2 \geq \|x - p\|^2 + \|y - p\|^2$, a “Pythagorean inequality” saying not only that p is closest but by how much. $\square$



The characterization (8.3): the angle at p between $x - p$ and any $y - p$ with $y \in C$ is obtuse (cosine ≤ 0).

Intuition

The inequality $\langle x - p, y - p \rangle \leq 0$ says that, standing at the projection p and looking toward x , the whole set C is “behind us”: the angle between the direction to x and the direction to any other point of C is at least 90 degrees. Put differently, the hyperplane through p with normal $x - p$ leaves C on one side and x on the other. **This is exactly the construction we used in the chapter on separation:** to separate a point $x \notin C$ from a closed convex set C , the separating hyperplane is built from the projection ($a = x - \text{proj}_C(x)$ as normal vector). There the projection theorem was the hidden engine; here you have all its pieces proved.

Theorem 8.31 (Nonexpansiveness of the projection). *Let $C \subseteq \mathbb{R}^n$ be nonempty, closed and convex. Then the map proj_C is nonexpansive:*

$$\|\text{proj}_C(x) - \text{proj}_C(z)\| \leq \|x - z\| \quad \text{for all } x, z \in \mathbb{R}^n.$$

In particular, proj_C is continuous (in fact Lipschitz with constant 1).

Proof. Write $p = \text{proj}_C(x)$ and $q = \text{proj}_C(z)$. We apply the characterization (8.3) twice, crossing the points (note that $q \in C$ and $p \in C$, so each one is a valid “ y ” for the other):

$$\langle x - p, q - p \rangle \leq 0, \quad \langle z - q, p - q \rangle \leq 0.$$

In the second one, $\langle z - q, p - q \rangle = -\langle z - q, q - p \rangle$, so $\langle q - z, q - p \rangle \leq 0$. Adding this to the first:

$$\langle (x - p) + (q - z), q - p \rangle \leq 0 \iff \langle x - z, q - p \rangle + \langle q - p, q - p \rangle \leq 0,$$

where we regrouped $(x - p) + (q - z) = (x - z) + (q - p)$. Therefore,

$$\|q - p\|^2 \leq \langle x - z, p - q \rangle \leq \|x - z\| \|p - q\|,$$

using Cauchy–Schwarz in the last step. If $p = q$ there is nothing to prove; if $p \neq q$, we divide by $\|p - q\| > 0$ and obtain $\|p - q\| \leq \|x - z\|$. $\square$

Remark 8.32. Nonexpansiveness says that projecting *does not separate* points: two nearby points have nearby projections. This property is the basis of *projected gradient* algorithms for constrained optimization [5, 2]: alternate a gradient step with a projection onto the feasible set never amplifies errors.

8.5.2 Projections you can compute by hand

Example 8.33 (Projection onto a ball). Let $C = \{y \in \mathbb{R}^n : \|y\| \leq r\}$ be the closed ball of radius $r > 0$ centered at the origin. We claim that

$$\text{proj}_C(x) = \begin{cases} x, & \text{if } \|x\| \leq r, \\ \frac{r}{\|x\|} x, & \text{if } \|x\| > r, \end{cases}$$

that is: if you are inside you do not move; if you are outside, you are “shrunk” radially onto the surface. Check with the characterization (8.3) in the case $\|x\| > r$: let $p = (r/\|x\|)x$, so that $x - p = (1 - r/\|x\|)x = \alpha x$ with $\alpha = 1 - r/\|x\| > 0$. For any $y \in C$:

$$\langle x - p, y - p \rangle = \alpha \langle x, y \rangle - \alpha \langle x, p \rangle = \alpha (\langle x, y \rangle - r \|x\|) \leq \alpha (\|x\| \|y\| - r \|x\|) = \alpha \|x\| (\|y\| - r) \leq 0,$$

using $\langle x, p \rangle = (r/\|x\|) \|x\|^2 = r \|x\|$, Cauchy–Schwarz and $\|y\| \leq r$. By part (c) of Theorem 8.30, $p = \text{proj}_C(x)$. Numerical example: with $r = 2$ and $x = (3, 4)$, $\|x\| = 5$ and $\text{proj}_C(x) = \frac{2}{5}(3, 4) = (6/5, 8/5)$.

Example 8.34 (Projection onto a box). Let $C = [a_1, b_1] \times \cdots \times [a_n, b_n]$ be a product of intervals (a “box”). The problem $\min_{y \in C} \|x - y\|^2 = \sum_i (x_i - y_i)^2$ separates into n independent one dimensional problems: minimize $(x_i - y_i)^2$ with $y_i \in [a_i, b_i]$. The solution of each one is to clip x_i to the interval:

$$(\text{proj}_C(x))_i = \min\{\max\{x_i, a_i\}, b_i\} = \begin{cases} a_i, & x_i < a_i, \\ x_i, & a_i \leq x_i \leq b_i, \\ b_i, & x_i > b_i. \end{cases}$$

An important special case: the **nonnegative orthant** $\mathbb{R}_+^n = \{y : y_i \geq 0\}$ (a box with $a_i = 0$, $b_i = +\infty$):

$$(\text{proj}_{\mathbb{R}_+^n}(x))_i = \max\{x_i, 0\} =: x_i^+,$$

the componentwise *positive part*. This is the natural feasible set of economics (nonnegative quantities), and its projection is simply “set the negative entries to zero”.

Example 8.35 (Projection onto a hyperplane: formula step by step). Let $H = \{y \in \mathbb{R}^n : \langle a, y \rangle = b\}$ be a hyperplane, with $a \neq 0$. We derive the formula for $\text{proj}_H(x)$ from scratch.

Step 1: shape of the candidate. Geometrically, the shortest path from x to the hyperplane must be *perpendicular* to it, that is, parallel to the normal vector a . So we look for the projection in the form

$$p = x - t a \quad \text{for some scalar } t \in \mathbb{R}.$$

Step 2: impose feasibility. p must lie in H :

$$\langle a, p \rangle = \langle a, x \rangle - t \langle a, a \rangle = \langle a, x \rangle - t \|a\|^2 = b \quad \implies \quad t = \frac{\langle a, x \rangle - b}{\|a\|^2}.$$

Step 3: formula.

$$\text{proj}_H(x) = x - \frac{\langle a, x \rangle - b}{\|a\|^2} a. \tag{8.4}$$

Step 4: rigorous check. Our “geometric argument” in Step 1 was heuristic; let us certify it with the characterization (8.3). With p given by (8.4), $x - p = ta$. For any $y \in H$, since $\langle a, y \rangle = b = \langle a, p \rangle$, we have $\langle a, y - p \rangle = 0$, and therefore

$$\langle x - p, y - p \rangle = t \langle a, y - p \rangle = 0 \leq 0. \quad \checkmark$$

(On a hyperplane the inequality holds with *equality*: the projection is orthogonal in the classical sense.) Along the way, the distance to the hyperplane comes for free:

$$d_H(x) = \|x - p\| = |t| \|a\| = \frac{|\langle a, x \rangle - b|}{\|a\|},$$

the “point to plane distance” formula of analytic geometry. Numerical example: for $H = \{y \in \mathbb{R}^3 : y_1 + y_2 + y_3 = 3\}$ (that is, $a = (1, 1, 1)$, $b = 3$) and $x = 0$: $t = (0 - 3)/3 = -1$ and $\text{proj}_H(0) = 0 - (-1)(1, 1, 1) = (1, 1, 1)$, with $d_H(0) = \sqrt{3}$. Economic note: a hyperplane $\langle p, y \rangle = w$ is a *budget line* in $\mathbb{R}^n$; projecting onto it means finding the bundle that exhausts the budget and is closest to a given bundle.

8.6 Economic applications of the projection [P3]

8.6.1 OLS is an orthogonal projection

Let us return to Example 8.25 with fresh eyes. The problem $\min_{\beta} \|y - X\beta\|^2$ can be read in two different spaces:

- In the space of *parameters* $\mathbb{R}^n$: we look for the best coefficient vector β . That is how we solved it before.
- In the space of *observations* $\mathbb{R}^m$: the vectors of the form $X\beta$ sweep out the subspace

$$\text{col}(X) = \{X\beta : \beta \in \mathbb{R}^n\} \subseteq \mathbb{R}^m,$$

the *column space* of X (all linear combinations of the regressors). Then

$$\min_{\beta \in \mathbb{R}^n} \|y - X\beta\|^2 = \min_{v \in \text{col}(X)} \|y - v\|^2 :$$

we are computing the **projection of y onto $\text{col}(X)$** . A vector subspace is a closed convex set, so Theorem 8.30 guarantees that the vector of *fitted values* $\hat{y} = X\hat{\beta}$ exists and is unique (even if $\hat{\beta}$ is not unique under collinearity: the projection $\hat{y}$ always is).

The characterization (8.3) on a subspace V becomes *orthogonality*: if $v \in V$ then also $p \pm v \in V$, and applying the inequality to $y' = p + v$ and to $y' = p - v$ we get $\langle y - p, v \rangle \leq 0$ and $\langle y - p, v \rangle \geq 0$, that is,

$$\langle y - \hat{y}, v \rangle = 0 \quad \text{for all } v \in \text{col}(X).$$

Taking as v each column of X : $X^\top(y - X\hat{\beta}) = 0$, which are *exactly the normal equations* (8.2). In econometrics this is stated as “the residuals are orthogonal to the regressors” (and with a constant in the regression, they sum to zero). With full rank, $\hat{y} = X(X^\top X)^{-1}X^\top y = Py$, where $P = X(X^\top X)^{-1}X^\top$ is the *projection matrix* or “hat matrix”, symmetric and idempotent ($P^2 = P$: projecting twice is projecting once). The whole geometry of OLS (fitted values, residuals, R^2 as the squared cosine of the angle between centered y and its projection) is Theorem 8.30 in action [2].

8.6.2 Rationing: the feasible bundle closest to the desired one

A consumer wants the bundle $x^* \in \mathbb{R}^n$ (the optimum without extra constraints), but faces *rationing*: maximum quotas q_i for each good and nonnegativity, so that the feasible set is the box

$$C = [0, q_1] \times \cdots \times [0, q_n].$$

If the criterion for choosing is to minimize the quadratic “discomfort distance” $\|x - x^*\|^2$ (a second order approximation of the utility loss around the optimum, via Taylor), the choice is exactly the projection onto the box of Example 8.34:

$$x_i^{\text{chosen}} = \min\{\max\{x_i^*, 0\}, q_i\} :$$

the consumer takes what is wanted of the nonrationed goods and “sticks” to the quota on the rationed ones. Nonexpansiveness (Theorem 8.31) gives a *comparative statics* result for free: if the desire x^* changes little, the rationed choice changes by at most the same amount. And the same “closest feasible point” logic appears in the proofs of the **second welfare theorem** [9]: to support a Pareto optimum as an equilibrium one separates the (convex) set of preferred allocations from the feasible set, and the separating hyperplane (built from the projection, as we saw) delivers the supporting price vector.

Summary

Chapter summary

Existence. Weierstrass: continuous + compact $\Rightarrow$ the optimum is attained. Without compactness: f continuous and *coercive* ($f \rightarrow +\infty$ as $\|x\| \rightarrow \infty$) attains a global minimum (Theorem 8.7). In economics: increasing returns $\Rightarrow$ profit may have no maximum.

First order. x^* a local minimum and f differentiable $\Rightarrow \nabla f(x^*) = 0$ (Fermat, Theorem 8.9). Necessary, not sufficient: there are saddles ($x^2 - y^2$) and inflections (x^3).

Second order. Necessary: at a local minimum, $\nabla^2 f(x^*)$ is PSD (Theorem 8.12). Sufficient: $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ PD $\Rightarrow$ strict local minimum (Theorem 8.13, via Taylor + smallest eigenvalue). PSD without PD: the test does not decide (x^4 vs. x^3).

Convexity = a license for the FOC. If f is convex: every local minimum is global (Theorem 8.19); $\arg \min f$ is convex (Theorem 8.20); with strict convexity the minimizer is unique (Theorem 8.21); and $\nabla f(x^*) = 0 \Leftrightarrow x^*$ is a global minimum (Theorem 8.23). Without differentiability: $0 \in \partial f(x^*)$. Star example: OLS, normal equations $X^\top X\beta = X^\top y$, unique under full rank.

Distance and projection. $d_C(x) = \inf_{y \in C} \|x - y\|$; the infimum is attained if C is closed; d_C is convex if C is. On a closed convex set, the projection exists, is unique and is characterized by $\langle x - \text{proj}_C(x), y - \text{proj}_C(x) \rangle \leq 0$ for all $y \in C$ (Theorem 8.30); it is nonexpansive (Theorem 8.31). Formulas: ball (shrink radially), box and $\mathbb{R}_+^n$ (clip componentwise), hyperplane ($x - \frac{\langle a, x \rangle - b}{\|a\|^2} a$). OLS is the projection of y onto $\text{col}(X)$; the separation of the previous chapter and the second welfare theorem rest on this theorem.

Main references: [1] (ch. 3), [5] (ch. 2), [7] (ch. 1), [6] (ch. 3), [3].

Exam-style exercises

Exercise 8.36 (Full classification of critical points). Find and classify all the critical points of

$$f(x, y) = x^3 + y^3 - 3xy.$$

Does f have a global minimum?

Solution. *FOC.* $\nabla f = (3x^2 - 3y, 3y^2 - 3x) = (0, 0)$ gives the system $y = x^2, x = y^2$. Substituting: $x = (x^2)^2 = x^4$, that is, $x(x^3 - 1) = 0$, with real solutions $x = 0$ and $x = 1$. Critical points: $(0, 0)$ and $(1, 1)$.

SOC. The Hessian is $\nabla^2 f(x, y) = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}$.

- At $(0, 0)$: $\begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}$, with determinant $-9 < 0$. A symmetric 2×2 matrix with negative determinant has eigenvalues of opposite signs (the determinant is their product): *indefinite*. Therefore $(0, 0)$ is a **saddle point** (it cannot be a minimum by Theorem 8.12 nor a maximum by the analogous result).
- At $(1, 1)$: $\begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$, with leading minors $6 > 0$ and $36 - 9 = 27 > 0$: positive definite by Sylvester. By Theorem 8.13, $(1, 1)$ is a **strict local minimum**, with $f(1, 1) = 1 + 1 - 3 = -1$.

Globality. There is no global minimum: along the x axis, $f(x, 0) = x^3 \rightarrow -\infty$ as $x \rightarrow -\infty$, so $\inf f = -\infty$. (Note that f is neither coercive nor convex; the local minimum at $(1, 1)$ is genuinely only local.) $\square$

Exercise 8.37 (Coercivity and existence). Prove that $f(x, y) = x^4 + y^4 - 3xy$ is coercive and conclude that it attains a global minimum on $\mathbb{R}^2$. (Hint: bound $|xy|$ by $\frac{1}{2}(x^2 + y^2)$ and use $t^4 \geq 2t^2 - 1$.)

Solution. *Step 1: two auxiliary inequalities.* (i) From $(|x| - |y|)^2 \geq 0$ we get $2|x||y| \leq x^2 + y^2$, hence $|xy| \leq \frac{1}{2}(x^2 + y^2)$ and therefore $-3xy \geq -\frac{3}{2}(x^2 + y^2)$. (ii) From $(t^2 - 1)^2 \geq 0$ we get $t^4 - 2t^2 + 1 \geq 0$, that is, $t^4 \geq 2t^2 - 1$ for every $t \in \mathbb{R}$.

Step 2: coercive lower bound. Applying (ii) to x and to y , and then (i):

$$f(x, y) \geq (2x^2 - 1) + (2y^2 - 1) - \frac{3}{2}(x^2 + y^2) = \frac{1}{2}(x^2 + y^2) - 2 = \frac{1}{2}\|(x, y)\|^2 - 2.$$

As $\|(x, y)\| \rightarrow \infty$, the right hand side tends to $+\infty$, and by comparison $f(x, y) \rightarrow +\infty$: f is coercive.

Step 3: existence. f is a polynomial, hence continuous. By Theorem 8.7, f attains a global minimum on $\mathbb{R}^2$. (As a complement: the FOC $4x^3 = 3y, 4y^3 = 3x$ has solutions $(0, 0)$ and $\pm(\sqrt{3}/2, \sqrt{3}/2)$; comparing values, the global minima are the two symmetric nonzero points, and $(0, 0)$ is a saddle.) $\square$

Exercise 8.38 (Cournot with quadratic costs). Two firms compete à la Cournot with inverse demand $p = a - (q_1 + q_2)$ and costs $C_i(q_i) = \frac{c}{2}q_i^2$, with $a, c > 0$. (a) State the problem of firm 1 given q_2 , check that it has a solution and that the FOC characterizes it. (b) Find the best response $BR_1(q_2)$. (c) Compute the symmetric Nash equilibrium.

Solution. (a) Firm 1 maximizes $\pi_1(q_1) = (a - q_1 - q_2)q_1 - \frac{c}{2}q_1^2$ over $q_1 \geq 0$. It is a quadratic polynomial with leading coefficient $-(1 + \frac{c}{2}) < 0$, so $-\pi_1$ is coercive and strictly convex: the maximum exists and is unique (Theorems 8.7 and 8.21). Since $\pi_1''(q_1) = -(2+c) < 0$ everywhere, π_1 is strictly concave and, by Theorem 8.23 applied to $-\pi_1$, the FOC characterizes the global maximum (in the interior; at the corner $q_1 = 0$ you check the sign of $\pi_1'(0) = a - q_2$).

(b) FOC: $\pi_1'(q_1) = a - 2q_1 - q_2 - cq_1 = 0$, so

$$BR_1(q_2) = \frac{a - q_2}{2 + c} \quad \text{if } q_2 \leq a, \quad BR_1(q_2) = 0 \text{ if } q_2 > a.$$

(c) By symmetry, $q_1 = q_2 = q^*$ with $q^* = \frac{a - q^*}{2 + c}$, that is, $(2 + c)q^* = a - q^*$, hence

$$q^* = \frac{a}{3 + c}, \quad p^* = a - 2q^* = \frac{a(1 + c)}{3 + c}.$$

Sanity check: with $c = 0$ we recover classical Cournot $q^* = a/3$. The strict concavity in (a) is what validates the whole procedure: without it, “solving the FOC” would not identify the best response. $\square$

Exercise 8.39 (Regression with shrinkage (ridge)). Let $a \in \mathbb{R}^n$ be fixed and $\lambda > 0$. Consider

$$f(x) = \|x - a\|^2 + \lambda \|x\|^2.$$

(a) Prove that f is strictly convex and coercive. (b) Find its unique minimizer. (c) Interpret the result as λ grows.

Solution. (a) Expanding, $f(x) = (1 + \lambda) \|x\|^2 - 2 \langle a, x \rangle + \|a\|^2$. Its Hessian is $\nabla^2 f(x) = 2(1 + \lambda)I$, with single eigenvalue $2(1 + \lambda) > 0$: positive definite everywhere, so f is strictly convex (second order characterization from the chapter on convex functions). Coercivity: by Cauchy–Schwarz, $f(x) \geq (1 + \lambda) \|x\|^2 - 2 \|a\| \|x\| \rightarrow +\infty$ as $\|x\| \rightarrow \infty$ (the quadratic term dominates the linear one).

(b) Existence and uniqueness: Theorems 8.7 and 8.21. FOC (sufficient by convexity, Theorem 8.23):

$$\nabla f(x) = 2(x - a) + 2\lambda x = 0 \implies (1 + \lambda)x = a \implies x^* = \frac{a}{1 + \lambda}.$$

(c) The minimizer is the target a *shrunk* toward the origin by the factor $1/(1 + \lambda)$: as $\lambda \rightarrow 0$, $x^* \rightarrow a$ (with no penalty, stay at a); as $\lambda \rightarrow \infty$, $x^* \rightarrow 0$ (the penalty dominates). This is the logic of the *ridge* estimator in econometrics, $\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$: the penalty $\lambda \|\beta\|^2$ makes the problem strictly convex (useful exactly when $X^\top X$ is singular because of collinearity) and shrinks the coefficients toward zero. $\square$

Exercise 8.40 (Computing projections). Compute: (a) $\text{proj}_C(2, -1)$ for the box $C = [0, 1] \times [0, 2]$; (b) $\text{proj}_H(1, 2, 3)$ for the hyperplane $H = \{y \in \mathbb{R}^3 : y_1 + 2y_2 + 2y_3 = 3\}$, and check the characterization (8.3) in this case; (c) $\text{proj}_B(3, 4)$ for the ball $B = \{y \in \mathbb{R}^2 : \|y\| \leq 1\}$.

Solution. (a) Componentwise (Example 8.34): the first coordinate $2 > 1$ is clipped to 1; the second, $-1 < 0$, is clipped to 0. Hence $\text{proj}_C(2, -1) = (1, 0)$.

(b) With $a = (1, 2, 2)$, $b = 3$, $x = (1, 2, 3)$: $\|a\|^2 = 1 + 4 + 4 = 9$ and $\langle a, x \rangle = 1 + 4 + 6 = 11$. By (8.4), $t = (11 - 3)/9 = 8/9$ and

$$\text{proj}_H(x) = (1, 2, 3) - \frac{8}{9}(1, 2, 2) = \left(\frac{1}{9}, \frac{2}{9}, \frac{11}{9}\right).$$

Feasibility check: $\frac{1}{9} + \frac{4}{9} + \frac{22}{9} = \frac{27}{9} = 3$. ✓ Check of (8.3): here $x - p = \frac{8}{9}a$, and for any $y \in H$ we have $\langle a, y - p \rangle = \langle a, y \rangle - \langle a, p \rangle = 3 - 3 = 0$, hence $\langle x - p, y - p \rangle = \frac{8}{9} \langle a, y - p \rangle = 0 \leq 0$. ✓

(c) $\|(3, 4)\| = 5 > 1$, so (Example 8.33) $\text{proj}_B(3, 4) = \frac{1}{5}(3, 4) = (3/5, 4/5)$, and the distance from $(3, 4)$ to the ball is $5 - 1 = 4$. □

Exercise 8.41 (The distance is 1-Lipschitz). Let $C \subseteq \mathbb{R}^n$ be nonempty (it need not be convex or closed). Prove that the distance function satisfies

$$|d_C(x) - d_C(z)| \leq \|x - z\| \quad \text{for all } x, z \in \mathbb{R}^n.$$

Conclude that d_C is continuous on all of $\mathbb{R}^n$.

Solution. Let $x, z \in \mathbb{R}^n$ and $\varepsilon > 0$. By definition of infimum there is $y \in C$ with $\|z - y\| \leq d_C(z) + \varepsilon$. Since that same y is a candidate in the infimum defining $d_C(x)$, the triangle inequality gives

$$d_C(x) \leq \|x - y\| \leq \|x - z\| + \|z - y\| \leq \|x - z\| + d_C(z) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, $d_C(x) - d_C(z) \leq \|x - z\|$. Swapping the roles of x and z (the argument is symmetric) we get $d_C(z) - d_C(x) \leq \|z - x\| = \|x - z\|$. The two inequalities together give $|d_C(x) - d_C(z)| \leq \|x - z\|$.

Continuity: given $\varepsilon > 0$, it is enough to take $\delta = \varepsilon$ in the definition: if $\|x - z\| < \delta$ then $|d_C(x) - d_C(z)| < \varepsilon$. (A Lipschitz function is uniformly continuous.) Note: compare with Proposition 8.28 and Theorem 8.31: the *distance* is 1-Lipschitz for any C ; the *convexity* of d_C and the nonexpansiveness of proj_C do require C to be convex. □

Chapter 9

Constrained optimization: Lagrange and the Karush–Kuhn–Tucker conditions [P1]

Week 7 of the course. Bibliography: Ruszczyński [6]; Nocedal and Wright [5], Ch. 12; Aragón *et al.* [1].

What is this used for in Microeconomics?

This is, without exaggeration, **the most important chapter of this notebook for your Microeconomics course**: all of modern microeconomics is constrained optimization. In individual decision theory (MWG, chs. 1–5), the consumer solves the utility maximization problem (UMP), that is, maximize $u(x)$ subject to the budget constraint $p \cdot x \leq m$ and to $x \geq 0$, and its dual expenditure minimization problem (EMP); the firm maximizes profit or minimizes cost subject to its technological constraint (the production function). Every Marshallian or Hicksian demand that appears in [9] or [10] is the solution of a KKT system. In models of contracts and asymmetric information (MWG, chs. 13–14), the principal maximizes expected profit subject to the *participation constraint* (IR, individual rationality) and to the *incentive compatibility constraint* (IC); the analysis of which constraint “bites” at the optimum (does IR bind? which IC constraints matter?) is **exactly** the complementary slackness analysis of KKT carried out by Holmström in moral hazard or Baron–Myerson in regulation under asymmetric information. In mechanism design and auctions (MWG, ch. 23), the optimal Myerson mechanism solves a problem with a *continuum* of IC constraints (one for each pair of types), and the standard technique is to identify which ones bind (the “local” ones) and discard the rest: again, KKT case analysis. Finally, Lagrange multipliers are *shadow prices*: the marginal utility of income in the UMP, marginal cost in the firm’s problem, the cost to the principal of raising the agent’s reservation utility. Knowing how to (i) write the Lagrangian with the correct signs, (ii) list the full KKT conditions, (iii) organize the case analysis according to which constraints are active, and (iv) read the multipliers economically, is **the** operational skill of the micro course. Give this chapter as much time as it takes.

In Chapter 8 we learned to solve $\min_{x \in \mathbb{R}^n} f(x)$ without constraints: the first order condition is $\nabla f(x^*) = 0$. But almost no economic problem looks like that: the consumer cannot spend more than their income, the firm cannot produce without inputs, quantities cannot be negative. This chapter develops the theory of optimality when there are constraints: first the geometry (tangent and normal cones), then the classical Lagrange theorem for equality constraints, and finally the central theorem of the course: the **Karush–Kuhn–Tucker (KKT) conditions**, with their proof through the Farkas lemma from the separation chapter. We close with a step by step solution manual and a battery of economic examples solved in full detail. Main references: [1] (chs. 3–4), [6] (ch. 3), [2] (ch. 5), [5] (ch. 12) and appendix M of [9].

9.1 The general problem and sign conventions [P1]

9.1.1 The standard nonlinear problem

Definition 9.1 (Constrained nonlinear problem). Let $f, g_1, \dots, g_m, h_1, \dots, h_p : \mathbb{R}^n \rightarrow \mathbb{R}$ be functions (which we assume differentiable whenever needed). The *nonlinear programming problem* (NLP) in *standard form* is

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p. \end{aligned} \tag{9.1}$$

The function f is the *objective function*; the g_i are the *inequality constraints* and the h_j the *equality constraints*. The set

$$X = \{x \in \mathbb{R}^n : g_i(x) \leq 0 \ \forall i, \ h_j(x) = 0 \ \forall j\}$$

is the *feasible set*, and its elements are the *feasible points*. A point $x^* \in X$ is a *global minimum* if $f(x^*) \leq f(x)$ for every $x \in X$, and a *local minimum* if there is $\varepsilon > 0$ such that $f(x^*) \leq f(x)$ for every $x \in X$ with $\|x - x^*\| < \varepsilon$.

Remark 9.2. Before looking for optimality conditions it helps to recall *when an optimum exists*: by the Weierstrass theorem (Chapter 1), if f is continuous and X is compact (closed and bounded) and nonempty, the global minimum exists. In typical economic problems (a budget set with positive prices and finite income) the feasible set is compact, so existence is guaranteed and the real work is to *characterize* the optimum. The conditions in this chapter do precisely that.

9.1.2 Active and inactive constraints

Definition 9.3 (Active constraint). Let $x \in X$ be a feasible point of problem (9.1). We say that the inequality constraint g_i is *active* (or *binding*; in economic jargon, that it “bites”) at x if

$$g_i(x) = 0,$$

and that it is *inactive* (or *slack*) if $g_i(x) < 0$. The *active index set* at x is

$$A(x) = \{i \in \{1, \dots, m\} : g_i(x) = 0\}.$$

Equality constraints are, by definition, always active at every feasible point.

Intuition

The active/inactive distinction is **the** organizing idea of the whole chapter. If $g_i(x^*) < 0$ at the optimum, then near x^* constraint i does not notice anything we do: there is a small ball around x^* where g_i stays negative (by continuity), so *locally* it is as if that constraint did not exist. Only active constraints can “push” the optimum away from where it would be without them. Economic example: if the consumer has income to spare and does not spend it all, the budget constraint is inactive and its “price” (the multiplier, as we will see) is zero; if they spend down to the last coin, the constraint bites and extra income has positive marginal value. All of KKT analysis consists of finding out, case by case, *which subset of constraints is active at the optimum*.

9.1.3 Sign conventions: how to translate between books

Here is the number one source of mistakes when a student moves from an optimization book (which minimizes with ≤ 0) to a microeconomics book like [9] (which maximizes with $\geq$). Let us go slowly.

Problem (9.1) is a *convention*, not a restriction of the theory. Any optimization problem can be brought to that form with the following mechanical translations:

- (i) **Maximize** $\leftrightarrow$ **minimize**. Maximizing f is the same as minimizing $-f$, and the optimal points coincide:

$$\arg \max_{x \in X} f(x) = \arg \min_{x \in X} (-f(x)), \quad \max_{x \in X} f(x) = -\min_{x \in X} (-f(x)).$$

- (ii) **Inequalities** $\geq \leftrightarrow \leq$. The constraint $g(x) \geq 0$ is equivalent to $-g(x) \leq 0$. The constraint $g(x) \leq b$ is equivalent to $g(x) - b \leq 0$.
- (iii) **Equalities as pairs of inequalities**. $h(x) = 0$ is equivalent to the pair $h(x) \leq 0$ and $-h(x) \leq 0$. (This is useful in proofs; in practice equalities are handled directly with sign free multipliers, as we will see.)
- (iv) **Nonnegativity**. The constraint $x_k \geq 0$ is simply the inequality $g(x) = -x_k \leq 0$, with gradient $\nabla g = -e_k$ (minus the k -th canonical vector). It is not a “special” constraint: it enters the KKT system like any other.

For example, the consumer problem as it appears in [9],

$$\max_{x \in \mathbb{R}^n} u(x) \quad \text{s.t.} \quad p \cdot x \leq m, \quad x \geq 0,$$

is written in the standard form (9.1) as

$$\min_{x \in \mathbb{R}^n} -u(x) \quad \text{s.t.} \quad g_0(x) = p \cdot x - m \leq 0, \quad g_k(x) = -x_k \leq 0 \quad (k = 1, \dots, n).$$

Careful (common mistake)

The signs of the multipliers depend on the convention. This is the most frequent mistake in exams. Memorize the rule and where it comes from:

- In standard form (**min** with $g_i \leq 0$), the Lagrangian is $L = f + \sum_i \mu_i g_i$ and the inequality multipliers satisfy $\mu_i \geq 0$.
- In the “economist” form (**max** with $g_i(x) \leq 0$, for example $p \cdot x - m \leq 0$), the Lagrangian is written $L = f - \sum_i \lambda_i g_i$ (with a minus sign) and again $\lambda_i \geq 0$. Both recipes are the same: maximizing f is minimizing $-f$, and $(-f) + \sum \lambda_i g_i = -(f - \sum \lambda_i g_i)$.
- If you write the constraint as $\geq$ ($u(x) \geq \bar{u}$, typical in the EMP or in a participation constraint inside a *maximization* problem), convert it first to $\leq$ ($\bar{u} - u(x) \leq 0$) or, equivalently, add it with a $+$ sign: $L = f + \lambda(u(x) - \bar{u})$ with $\lambda \geq 0$.
- The multipliers of *equality* constraints have no sign restriction in any convention.

The safe way never to make a mistake: *convert everything to the standard form (9.1) before writing the Lagrangian*, apply KKT as it is, and at the end translate back. With practice you will work directly in the max form, but until then the mechanical conversion saves you. An infallible consistency check: at the optimum, the multiplier of a resource constraint ($p \cdot x \leq m$) must come out ≥ 0 and must be readable as the marginal value of the resource; if you get a negative number, you dropped a sign.

Example 9.4 (Full translation of conventions). The expenditure minimization problem (EMP) of [9] is

$$\min_{x \geq 0} p \cdot x \quad \text{s.t.} \quad u(x) \geq \bar{u}.$$

Let us write it in standard form: the objective is already a minimum, $f(x) = p \cdot x$. The constraints: $u(x) \geq \bar{u}$ becomes $g_0(x) = \bar{u} - u(x) \leq 0$, and $x_k \geq 0$ becomes $g_k(x) = -x_k \leq 0$. The standard Lagrangian is

$$L(x, \mu) = p \cdot x + \mu_0 (\bar{u} - u(x)) - \sum_{k=1}^n \mu_k x_k, \quad \mu_0, \mu_1, \dots, \mu_n \geq 0.$$

The stationarity condition in coordinate k is $p_k - \mu_0 \partial u / \partial x_k - \mu_k = 0$. At an interior solution ($x_k > 0$, hence $\mu_k = 0$ by complementary slackness, as we will see) this gives $p_k = \mu_0 \partial u / \partial x_k$: the price of the good equals its marginal utility valued at the shadow price μ_0 of the utility constraint, which is exactly the inverse of the marginal utility of income. Compare: in many micro texts you will see $L = p \cdot x - \lambda(u(x) - \bar{u})$; this is *the same* Lagrangian with $\lambda = \mu_0 \geq 0$.

9.2 Geometry of optimality: tangent and normal cones [P1]

Before any theorem, let us build the mental picture. All of KKT theory is the formalization of a 2D drawing that you should be able to reproduce with your eyes closed.

9.2.1 The picture in two dimensions

Consider $\min f(x_1, x_2)$ subject to a single constraint $g(x_1, x_2) \leq 0$, and suppose that at the candidate optimum x^* the constraint is *active*: $g(x^*) = 0$, that is, x^* lies on the boundary of the feasible region.

Recall two facts about gradients from Chapter 8:

- $-\nabla f(x^*)$ is the direction of *steepest descent* of f : moving a little in a direction d reduces f if and only if $\langle \nabla f(x^*), d \rangle < 0$ (to first order).
- $\nabla g(x^*)$ is perpendicular to the level curve $\{g = 0\}$ and points toward where g *increases*, that is, *outward* from the feasible region $\{g \leq 0\}$.

Now reason as follows: if x^* is a local minimum, there must be no direction that is at the same time *feasible* (does not take us out of the region) and *descent* (reduces f). The feasible directions at x^* are, to first order, those forming an obtuse or right angle with $\nabla g(x^*)$ (those that do not increase g): $\langle \nabla g(x^*), d \rangle \leq 0$. The descent directions are those forming an obtuse angle with $\nabla f(x^*)$. Having no intersection between the two sets of directions forces $-\nabla f(x^*)$ and $\nabla g(x^*)$ to point in *the same direction*:

$$-\nabla f(x^*) = \mu \nabla g(x^*) \quad \text{for some } \mu \geq 0, \quad \text{that is,} \quad \nabla f(x^*) + \mu \nabla g(x^*) = 0. \quad (9.2)$$

Geometrically: at the optimum, *the level curve of f is tangent to the boundary of the feasible region*, and the two gradients are collinear with the correct sign.

What if there are *two* active constraints, $g_1(x^*) = g_2(x^*) = 0$ (the optimum is at a “corner” of the feasible region)? Then exact tangency with either boundary is no longer needed: it is enough that $-\nabla f(x^*)$ be “trapped” *between* the two gradients, that is, in the cone they generate:

$$-\nabla f(x^*) = \mu_1 \nabla g_1(x^*) + \mu_2 \nabla g_2(x^*), \quad \mu_1, \mu_2 \geq 0. \quad (9.3)$$

If $-\nabla f(x^*)$ fell outside that cone, there would be a direction that enters the feasible region and at the same time decreases f , and x^* would not be optimal.

Equations (9.2) and (9.3) *already are* the KKT conditions; the rest of the chapter states them in general, proves them and exploits them. But to do this rigorously we need two objects from Chapter 3 adapted to this setting: the cone of directions that “do not leave” the feasible region (tangent cone) and its polar cone (normal cone).

9.2.2 Feasible and descent directions

Definition 9.5 (Descent direction). Let f be differentiable at x . A vector $d \in \mathbb{R}^n$ is a *descent direction* of f at x if $\langle \nabla f(x), d \rangle < 0$.

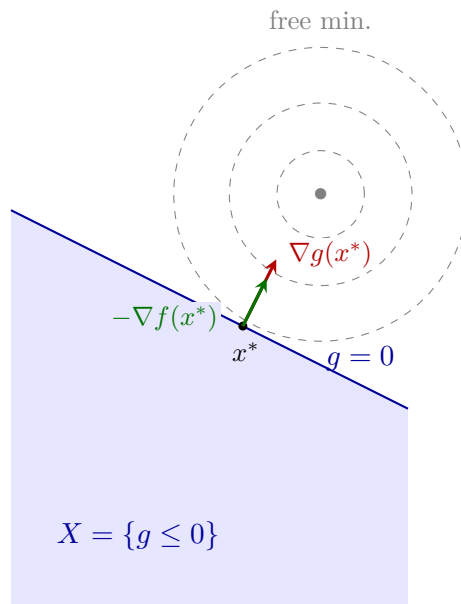


Figure 9.1: One active constraint: at the optimum x^* , the level curve of f (dashed lines) is tangent to the boundary $\{g = 0\}$ and $-\nabla f(x^*)$ is a nonnegative multiple of $\nabla g(x^*)$. The unconstrained minimum lies outside the feasible region.

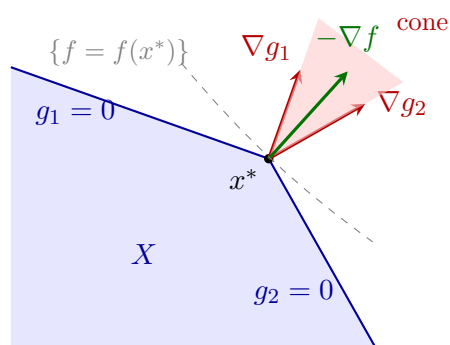


Figure 9.2: Two active constraints (corner solution): optimality requires that $-\nabla f(x^*)$ lie in the cone generated by $\nabla g_1(x^*)$ and $\nabla g_2(x^*)$ (shaded region). There is no longer tangency with either individual boundary.

By the first order Taylor expansion, if d is a descent direction then $f(x + td) = f(x) + t \langle \nabla f(x), d \rangle + o(t) < f(x)$ for every $t > 0$ small enough: moving a little in direction d strictly reduces the value of f .

Definition 9.6 (Cone of feasible directions). Let $X \subseteq \mathbb{R}^n$ and $x \in X$. A vector $d \in \mathbb{R}^n$ is a *feasible direction* of X at x if there is $\bar{t} > 0$ such that $x + td \in X$ for every $t \in [0, \bar{t}]$. The set of feasible directions at x , which we denote $D_X(x)$, is a cone (if d is feasible, so is αd for $\alpha > 0$).

With these two notions we can state the basic principle, which is an immediate generalization of the argument in Chapter 8:

Proposition 9.7 (No feasible descent directions at a local minimum). *If x^* is a local minimum of f over X and f is differentiable at x^* , then*

$$\langle \nabla f(x^*), d \rangle \geq 0 \quad \text{for every } d \in D_X(x^*).$$

Proof. Let $d \in D_X(x^*)$. Then $x^* + td \in X$ for $t \in [0, \bar{t}]$, and since x^* is a local minimum, $f(x^* + td) \geq f(x^*)$ for small $t > 0$. Therefore

$$\langle \nabla f(x^*), d \rangle = \lim_{t \downarrow 0} \frac{f(x^* + td) - f(x^*)}{t} \geq 0. \quad \square$$

The problem with $D_X(x)$ is that it can be too small when the boundary of X is *curved*. With the disk $X = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$ and $x = (1, 0)$ we are still fine: the disk is convex, so every segment pointing inward stays in X and $D_X(x)$ is a reasonable half-plane. But replace the disk by the *circle* $X = \{x_1^2 + x_2^2 = 1\}$ and the disaster is complete: no straight segment stays on the curve, so $D_X(x) = \{0\}$ and Proposition 9.7 says absolutely nothing. Since every equality constraint produces exactly that pathology, we need a more generous cone, one that allows approaches along *curves* and not only along straight segments: the tangent cone.

9.2.3 The tangent cone

Definition 9.8 (Tangent cone). Let $X \subseteq \mathbb{R}^n$ and $x \in X$. A vector $d \in \mathbb{R}^n$ is a *tangent direction* to X at x if there are a sequence of points $(x^k)_{k \in \mathbb{N}} \subseteq X$ and a sequence of scalars $t_k \downarrow 0$ (positive and converging to zero) such that

$$x^k \rightarrow x \quad \text{and} \quad \frac{x^k - x}{t_k} \rightarrow d.$$

The set of all tangent directions is the *tangent cone* (or Bouligand cone) of X at x , denoted $T_X(x)$.

The definition says: d is tangent if we can approach x *from inside* X along a (discrete) path whose limit direction is d . It does not require the segment $x + td$ to lie in X : it is enough that there are feasible points “asymptotically aligned” with d .

Proposition 9.9 (Basic properties of the tangent cone). *For every X and $x \in X$: (i) $T_X(x)$ is a closed cone containing 0; (ii) $D_X(x) \subseteq T_X(x)$; (iii) if $x \in \text{int } X$, then $T_X(x) = \mathbb{R}^n$; (iv) if X*

is convex, then $T_X(x) = \text{cl}\{\alpha(y - x) : y \in X, \alpha \geq 0\}$, the closure of the cone generated by the directions toward other points of X .

Proof (sketch). (i) $0 \in T_X(x)$ by taking $x^k = x$. If $d \in T_X(x)$ and $\alpha > 0$, the same sequences with $t'_k = t_k/\alpha$ give $\alpha d \in T_X(x)$; closedness follows from a standard diagonal argument. (ii) If $d \in D_X(x)$, take $t_k = \bar{t}/k$ and $x^k = x + t_k d \in X$; then $(x^k - x)/t_k = d$ exactly. (iii) If x is interior, for any d the points $x + t_k d$ are in X for small t_k . (iv) If X is convex and $y \in X$, the segment $x + t(y - x)$ is in X for $t \in [0, 1]$, so $y - x \in D_X(x) \subseteq T_X(x)$; with (i) we get “ $\supseteq$ ”. The reverse inclusion follows directly from the definition: if $(x^k - x)/t_k \rightarrow d$ with $x^k \in X$, each quotient is a positive multiple of $x^k - x$ with $x^k \in X$, so d lies in the closure of the described cone. Details in [6], sec. 3.1. $\square$

Example 9.10 (Tangent cones computed). Let us compute $T_X(x)$ from the definition in three cases that will come back all the time.

(a) *The nonnegative orthant at the origin.* $X = \mathbb{R}_+^2 = \{x : x_1 \geq 0, x_2 \geq 0\}$, $x = (0, 0)$. We claim that $T_X(0) = \mathbb{R}_+^2$. If $d \geq 0$, then $x^k = t_k d \in X$ for any $t_k \downarrow 0$ and $(x^k - 0)/t_k = d$, so $d \in T_X(0)$. Conversely, if $d \in T_X(0)$, then $d = \lim(x^k - 0)/t_k$ with $x^k \in \mathbb{R}_+^2$ and $t_k > 0$; each quotient has both coordinates ≥ 0 , and the limit of nonnegative numbers is nonnegative, so $d \geq 0$. Conclusion: at a corner of the orthant, the tangent directions are exactly those that do not make any coordinate negative.

(b) *The orthant at a point with one positive coordinate.* $X = \mathbb{R}_+^2$, $x = (a, 0)$ with $a > 0$. Now the constraint $x_1 \geq 0$ is *inactive* and does not get in the way: $T_X(x) = \mathbb{R} \times \mathbb{R}_+ = \{d : d_2 \geq 0\}$. The first coordinate of d can be anything because for small t_k , $a + t_k d_1$ is still positive. Note the pattern: *only active constraints restrict the tangent cone*.

(c) *A circle (equality constraint).* $X = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$, $x = (1, 0)$. Here $D_X(x) = \{0\}$ (no straight segment stays on the circle), but the tangent cone is the tangent line: $T_X(x) = \{d : d_1 = 0\} = \{0\} \times \mathbb{R}$. Let us check the inclusion “ $\supseteq$ ” from the definition: let $d = (0, \delta)$. Take $x^k = (\cos(\delta/k), \sin(\delta/k)) \in X$ and $t_k = 1/k$. Then $x^k \rightarrow (1, 0)$ and

$$\frac{x^k - x}{t_k} = \left(k(\cos(\delta/k) - 1), k \sin(\delta/k)\right) \rightarrow (0, \delta) = d,$$

using $\cos \varepsilon - 1 = O(\varepsilon^2)$ and $\sin \varepsilon \approx \varepsilon$. For “ $\subseteq$ ”: if $(x^k - x)/t_k \rightarrow d$, from $\|x^k\| = \|x\| = 1$ we get $0 = \|x^k\|^2 - \|x\|^2 = 2\langle x, x^k - x \rangle + \|x^k - x\|^2$; dividing by t_k and passing to the limit (the second term is $o(t_k)$ because $\|x^k - x\|/t_k \rightarrow \|d\|$ and $\|x^k - x\| \rightarrow 0$), we get $\langle x, d \rangle = d_1 = 0$. This computation shows why the sequence definition is the right one: it captures the directions along the *curve*, which is what equality constraints need.

With the tangent cone, Proposition 9.7 becomes stronger:

Theorem 9.11 (Geometric first order necessary condition). *If x^* is a local minimum of f over X and f is differentiable at x^* , then*

$$\langle \nabla f(x^*), d \rangle \geq 0 \quad \text{for every } d \in T_X(x^*). \quad (9.4)$$

Proof. Let $d \in T_X(x^*)$, with sequences $x^k \in X$, $x^k \rightarrow x^*$, $t_k \downarrow 0$, $(x^k - x^*)/t_k \rightarrow d$. Since x^* is a local minimum and $x^k \rightarrow x^*$ with $x^k \in X$, for large k we have $f(x^k) \geq f(x^*)$. By differentiability,

$$0 \leq f(x^k) - f(x^*) = \langle \nabla f(x^*), x^k - x^* \rangle + o(\|x^k - x^*\|).$$

We divide by $t_k > 0$:

$$0 \leq \left\langle \nabla f(x^*), \frac{x^k - x^*}{t_k} \right\rangle + \frac{o(\|x^k - x^*\|)}{\|x^k - x^*\|} \cdot \frac{\|x^k - x^*\|}{t_k}.$$

The first term tends to $\langle \nabla f(x^*), d \rangle$; in the second, the first factor tends to 0 and the second to $\|d\|$ (finite), so the whole second term tends to 0. (If $x^k = x^*$ for infinitely many k , then $d = 0$ and there is nothing to prove.) Taking the limit, $\langle \nabla f(x^*), d \rangle \geq 0$. $\square$

9.2.4 The normal cone and the condition $-\nabla f(x^*) \in N_X(x^*)$

Condition (9.4) says that $\nabla f(x^*)$ forms a non obtuse angle with *all* tangent directions. That is exactly saying that $-\nabla f(x^*)$ belongs to the *polar* cone of the tangent cone. This polar deserves its own name; you already met it in Chapter 7 for convex sets, and here we use it in that same version.

Definition 9.12 (Normal cone). Let $X \subseteq \mathbb{R}^n$ and $x \in X$. The *normal cone* of X at x is the polar of the tangent cone:

$$N_X(x) = (T_X(x))^\circ = \{v \in \mathbb{R}^n : \langle v, d \rangle \leq 0 \ \forall d \in T_X(x)\}.$$

When X is convex, this definition coincides (by Proposition 9.9(iv)) with the one in Chapter 7:

$$N_X(x) = \{v \in \mathbb{R}^n : \langle v, y - x \rangle \leq 0 \ \forall y \in X\}.$$

Example 9.13. From the computations in Example 9.10: (a) $N_{\mathbb{R}_+^2}(0) = -\mathbb{R}_+^2 = \mathbb{R}_-^2$ (the vectors with both coordinates ≤ 0); (b) at $x = (a, 0)$ with $a > 0$, $N_{\mathbb{R}_+^2}(x) = \{0\} \times \mathbb{R}_-$ (you can only “point outward” through the active constraint); (c) for the circle at $(1, 0)$, $N_X(x) = \mathbb{R} \times \{0\}$, *the whole* normal line (with both signs), reflecting that an equality constrains on both sides. At an interior point, $T_X(x) = \mathbb{R}^n$ and $N_X(x) = \{0\}$: the condition $-\nabla f(x^*) \in N_X(x^*)$ collapses to $\nabla f(x^*) = 0$, the unconstrained condition of Chapter 8. Everything fits.

Theorem 9.14 (Geometric condition; always necessary, sufficient in the convex case). *Let f be differentiable at $x^* \in X$.*

- (a) *If x^* is a local minimum of f over X , then $-\nabla f(x^*) \in N_X(x^*)$.*
- (b) *If in addition X is convex and f is convex, the condition is also sufficient: if $-\nabla f(x^*) \in N_X(x^*)$, then x^* is a global minimum of f over X .*

Proof. (a) This is a restatement of Theorem 9.11: for every $d \in T_X(x^*)$ we have $\langle \nabla f(x^*), d \rangle \geq 0$, that is, $\langle -\nabla f(x^*), d \rangle \leq 0$; by definition of polar, $-\nabla f(x^*) \in (T_X(x^*))^\circ = N_X(x^*)$.

(b) Let $y \in X$ be arbitrary. Since X is convex, $y - x^* \in T_X(x^*)$ (Proposition 9.9(iv)), and the hypothesis $-\nabla f(x^*) \in N_X(x^*)$ gives

$$\langle \nabla f(x^*), y - x^* \rangle \geq 0.$$

Since f is convex and differentiable, its graph lies above its tangent plane (gradient inequality, chapter on convex functions):

$$f(y) \geq f(x^*) + \langle \nabla f(x^*), y - x^* \rangle \geq f(x^*).$$

Since $y \in X$ was arbitrary, x^* is a global minimum. $\square$

Intuition

Theorem 9.14 is *all* of constrained optimization theory compressed into one line: **at the optimum, minus the gradient of the objective must be an “outward normal” to the feasible region.** If $-\nabla f(x^*)$ had a component pointing into X (or along X), we could move a little in that direction without leaving and reduce f . The force that “pushes” the objective down ($-\nabla f$) must be exactly balanced by the “walls” of the feasible region, and walls can only push in the direction of their normals. What Lagrange and KKT will do is give this geometric condition an *algebraic computable* form: express $N_X(x^*)$ in terms of the gradients of the active constraints. The whole business of constraint qualifications (Section 9.4) is precisely about when that algebraic expression of the normal cone is correct.

9.3 The Lagrange theorem: equality constraints [P1]

We start with the historically first and technically cleanest case: only equality constraints. This is the one you use when economic theory guarantees in advance that the constraints bind (for example, monotone preferences $\Rightarrow$ the budget constraint holds with equality).

Definition 9.15 (Lagrangian and regular point). For the problem $\min f(x)$ s.t. $h_j(x) = 0$, $j = 1, \dots, p$, with f, h_j of class $\mathcal{C}^1$, the *Lagrangian function* (or *Lagrangian*) is

$$L(x, \lambda) = f(x) + \sum_{j=1}^p \lambda_j h_j(x), \quad \lambda = (\lambda_1, \dots, \lambda_p) \in \mathbb{R}^p.$$

The scalars λ_j are the *Lagrange multipliers*. A feasible point x is *regular* if the constraint gradients are linearly independent at x :

$$\text{rank}(\nabla h_1(x), \dots, \nabla h_p(x)) = p.$$

Theorem 9.16 (Lagrange, 1788). Let x^* be a local minimum (or maximum) of f subject to $h_j(x) = 0$, $j = 1, \dots, p$, with $f, h_j \in \mathcal{C}^1$ in a neighborhood of x^* . If x^* is a regular point, then

there is a unique vector of multipliers $\lambda^* \in \mathbb{R}^p$ such that

$$\nabla_x L(x^*, \lambda^*) = \nabla f(x^*) + \sum_{j=1}^p \lambda_j^* \nabla h_j(x^*) = 0. \quad (9.5)$$

That is: $\nabla f(x^*)$ is a linear combination of the constraint gradients. The multipliers λ_j^* have no sign restriction.

Proof (detailed sketch through the implicit function theorem). The idea is to reduce the constrained problem to an *unconstrained* one over a local parametrization of the feasible surface, and apply there the condition $\nabla = 0$ of Chapter 8.

Step 1: parametrize the feasible surface. Let $M = \{x : h(x) = 0\}$ with $h = (h_1, \dots, h_p) : \mathbb{R}^n \rightarrow \mathbb{R}^p$. By regularity, the Jacobian matrix $Dh(x^*) \in \mathbb{R}^{p \times n}$ has rank p ; after reordering coordinates, we can write $x = (y, z)$ with $y \in \mathbb{R}^{n-p}$, $z \in \mathbb{R}^p$, so that the block $D_z h(x^*) \in \mathbb{R}^{p \times p}$ is invertible. The *implicit function theorem* (Chapter 1) then guarantees that, near x^* , the surface M is the graph of a C^1 function: there is $\varphi : U \subseteq \mathbb{R}^{n-p} \rightarrow \mathbb{R}^p$ with

$$h(y, \varphi(y)) = 0 \quad \forall y \in U, \quad x^* = (y^*, \varphi(y^*)).$$

Differentiating the identity $h(y, \varphi(y)) \equiv 0$ with the chain rule: $D_y h + D_z h \cdot D\varphi(y^*) = 0$, that is, $D\varphi(y^*) = -(D_z h)^{-1} D_y h$ (everything evaluated at x^*).

Step 2: optimize without constraints in the parameters. The function $F(y) = f(y, \varphi(y))$ has an *unconstrained* local minimum at y^* (because the points $(y, \varphi(y))$ run exactly through the nearby feasible ones). Hence $\nabla F(y^*) = 0$, and by the chain rule:

$$0 = \nabla_y f + (D\varphi(y^*))^\top \nabla_z f = \nabla_y f - (D_y h)^\top (D_z h)^{-\top} \nabla_z f.$$

Step 3: read off the multipliers. Define $\lambda^* = -(D_z h)^{-\top} \nabla_z f \in \mathbb{R}^p$. The equation in step 2 says $\nabla_y f + (D_y h)^\top \lambda^* = 0$, and the definition of λ^* says $\nabla_z f + (D_z h)^\top \lambda^* = 0$. Together, these two are exactly (9.5) written in blocks: $\nabla f(x^*) + Dh(x^*)^\top \lambda^* = 0$. The *uniqueness* of λ^* follows from the linear independence of the $\nabla h_j(x^*)$: if two multiplier vectors worked, their difference would give a nontrivial vanishing linear combination of the gradients. Full details in [1] or [5], ch. 12. $\square$

Remark 9.17 (Geometric reading). The set of tangent directions to $M = \{h = 0\}$ at a regular point is the kernel of the Jacobian, $T_M(x^*) = \{d : Dh(x^*)d = 0\}$ (the proof of this equality is essentially step 1 above: the implicit function theorem lets you build feasible curves with any initial velocity in the kernel). Condition (9.4) requires $\langle \nabla f(x^*), d \rangle \geq 0$ on a subspace, which being symmetric (d and $-d$ are both tangent) forces $\langle \nabla f(x^*), d \rangle = 0$: the gradient of the objective is *orthogonal to the tangent space*. And the orthogonal complement of the kernel of Dh is the range of $Dh^\top$, that is, the space spanned by the ∇h_j . That is (9.5). Note that here the normal “cone” is a whole subspace (which is why the λ_j have no sign), while with inequalities it will be a genuine cone and the signs $\mu_i \geq 0$ will appear.

9.3.1 The canonical example: Cobb–Douglas with a budget

Example 9.18 (Cobb–Douglas UMP by Lagrange, step by step). Let us solve the favorite problem of every micro course:

$$\max_{x,y>0} u(x,y) = x^\alpha y^\beta \quad \text{s.t.} \quad px + qy = m,$$

with $\alpha, \beta > 0$, prices $p, q > 0$ and income $m > 0$. (We work with the constraint as an equality: since u is strictly increasing in both goods, at the optimum all income is spent; in Section 9.7 we redo this problem with inequalities and full KKT. And since u vanishes on the axes while there are feasible bundles with positive utility, the optimum is interior, so ignoring $x, y \geq 0$ is legitimate.)

Step 1: form and Lagrangian. This is a maximization problem with one equality $h(x,y) = px + qy - m = 0$. We use the “economist” Lagrangian

$$L(x,y,\lambda) = x^\alpha y^\beta - \lambda(px + qy - m).$$

(In standard form we would minimize $-x^\alpha y^\beta$ with $+\lambda h$; the candidates are the same and λ only changes name.)

Step 2: check regularity. $\nabla h = (p, q) \neq (0, 0)$: every feasible point is regular. The Lagrange theorem applies with no fine print.

Step 3: first order conditions.

$$\frac{\partial L}{\partial x} = \alpha x^{\alpha-1} y^\beta - \lambda p = 0 \implies \alpha x^{\alpha-1} y^\beta = \lambda p, \quad (9.6)$$

$$\frac{\partial L}{\partial y} = \beta x^\alpha y^{\beta-1} - \lambda q = 0 \implies \beta x^\alpha y^{\beta-1} = \lambda q, \quad (9.7)$$

$$\frac{\partial L}{\partial \lambda} = -(px + qy - m) = 0 \implies px + qy = m. \quad (9.8)$$

Step 4: solve the system. Divide (9.6) by (9.7) (legitimate because at an interior candidate $x, y, \lambda > 0$):

$$\frac{\alpha x^{\alpha-1} y^\beta}{\beta x^\alpha y^{\beta-1}} = \frac{\alpha}{\beta} \cdot \frac{y}{x} = \frac{p}{q} \implies qy = \frac{\beta}{\alpha} px.$$

This is the famous tangency condition: *the marginal rate of substitution equals the price ratio*, $MRS = \frac{u_x}{u_y} = \frac{\alpha y}{\beta x} = \frac{p}{q}$. Substituting into the budget (9.8):

$$px + \frac{\beta}{\alpha} px = m \implies px \frac{\alpha + \beta}{\alpha} = m \implies \boxed{x^* = \frac{\alpha}{\alpha + \beta} \cdot \frac{m}{p}, \quad y^* = \frac{\beta}{\alpha + \beta} \cdot \frac{m}{q}.$$

The classical reading: the Cobb–Douglas consumer spends the fixed fraction $\alpha/(\alpha + \beta)$ of income on good x and $\beta/(\alpha + \beta)$ on good y , regardless of prices.

Step 5: the multiplier and its interpretation. From (9.6),

$$\lambda^* = \frac{\alpha(x^*)^{\alpha-1}(y^*)^\beta}{p} = \frac{u(x^*, y^*)(\alpha + \beta)}{m},$$

where the last equality comes from substituting the demands (do the computation: it is good algebra practice). The meaning: λ^* is the *marginal utility of income*. If you define the indirect utility function $v(p, q, m) = u(x^*(m), y^*(m))$ and differentiate with respect to m , you get exactly $\partial v / \partial m = \lambda^*$; we prove this in general in Chapter 11 (envelope theorem), and we check it numerically in Section 9.8. One extra unit of income relaxes the constraint and buys λ^* extra “utils”: that is why λ^* is the *shadow price* of income.

Careful (common mistake)

Two classic mistakes with Lagrange. **(1) Forgetting the rank condition.** The conclusion of Theorem 9.16 can fail at non regular points: the optimum may exist and satisfy (9.5) for no λ at all (we will see the analogous phenomenon with inequalities in Section 9.4). In an exam, check regularity (usually one line, as in step 2 above) before invoking the theorem. **(2) Believing that solving the first order system is enough.** The Lagrange conditions are *necessary*, not sufficient: their solutions are *candidates* (they can be maxima, minima or saddle points on the constraint). You need an extra argument: existence by Weierstrass + comparison of candidates, concavity/convexity (Section 9.6), or second order conditions (Section 9.9).

9.4 Constraint qualifications: why they are needed [P1]

Before stating KKT in general, we need to understand its fine print. The geometric condition $-\nabla f(x^*) \in N_X(x^*)$ (Theorem 9.14) is *always* necessary. The delicate step is *algebraic*: we want to replace the geometric normal cone by the cone generated by the gradients of the active constraints,

$$N_X(x^*) \stackrel{?}{=} \left\{ \sum_{i \in A(x^*)} \mu_i \nabla g_i(x^*) + \sum_{j=1}^p \lambda_j \nabla h_j(x^*) : \mu_i \geq 0, \lambda_j \in \mathbb{R} \right\}.$$

That equality can *fail*: the gradients may describe the local geometry of the feasible region badly. A *constraint qualification* (CQ) is any hypothesis guaranteeing that the algebraic description is faithful. Let us first see the failure live.

9.4.1 A complete counterexample: the cusp

Example 9.19 (KKT fails without a constraint qualification). Consider the problem in $\mathbb{R}^2$:

$$\min_{x \in \mathbb{R}^2} f(x) = x_1 \quad \text{s.t.} \quad g_1(x) = x_2 - x_1^3 \leq 0, \quad g_2(x) = -x_2 - x_1^3 \leq 0.$$

Step 1: understand the feasible region. The two constraints together say $x_2 \leq x_1^3$ and $-x_2 \leq x_1^3$, that is, $|x_2| \leq x_1^3$. For some x_2 to satisfy them we need $x_1^3 \geq 0$, that is, $x_1 \geq 0$. The feasible region is

$$X = \{x : x_1 \geq 0, |x_2| \leq x_1^3\},$$

a very thin “tongue” glued to the positive x_1 semiaxis, with a *cusp* (sharp point) at the origin:

near 0, the region has width $2x_1^3$, which flattens much faster than any cone.

Step 2: find the optimum by hand. We want the smallest possible x_1 , and every feasible point has $x_1 \geq 0$. The value $x_1 = 0$ is attained only at $x^* = (0, 0)$ (since $|x_2| \leq 0$). Conclusion: $x^* = (0, 0)$ is the *global minimum*, no discussion.

Step 3: try KKT at x^ .* Both constraints are active at the origin ($g_1(0) = g_2(0) = 0$). We compute gradients:

$$\nabla f(x) = (1, 0), \quad \nabla g_1(x) = (-3x_1^2, 1), \quad \nabla g_2(x) = (-3x_1^2, -1),$$

and at $x^* = (0, 0)$:

$$\nabla f(x^*) = (1, 0), \quad \nabla g_1(x^*) = (0, 1), \quad \nabla g_2(x^*) = (0, -1).$$

The KKT stationarity condition would require $\mu_1, \mu_2 \geq 0$ with

$$\nabla f(x^*) + \mu_1 \nabla g_1(x^*) + \mu_2 \nabla g_2(x^*) = 0 \iff (1, 0) + \mu_1(0, 1) + \mu_2(0, -1) = (0, 0).$$

The first coordinate demands $1 = 0$: **impossible**. No multipliers exist. KKT *fails at the global minimum*.

Step 4: geometric diagnosis. What went wrong? The active gradients generate the cone $\text{cone}\{(0, 1), (0, -1)\} = \{0\} \times \mathbb{R}$, the vertical axis: according to the algebra, the cone of allowed directions near 0 would be $\{d : d_2 \leq 0\} \cap \{d : -d_2 \leq 0\} = \mathbb{R} \times \{0\}$, the entire horizontal axis, including $d = (-1, 0)$. But the real geometry is different: the true tangent cone is

$$T_X(0) = \mathbb{R}_+ \times \{0\}$$

(you can only enter the tongue from the right: if $x^k \in X$ and $x^k/t_k \rightarrow d$, from $x_1^k \geq 0$ we get $d_1 \geq 0$, and from $|x_2^k| \leq (x_1^k)^3$ we get $|d_2| = \lim |x_2^k|/t_k \leq \lim (x_1^k)^3/t_k = \lim (x_1^k/t_k) (x_1^k)^2 = d_1 \cdot 0 = 0$). The “linearized” cone seen by the gradients is strictly larger than the true tangent cone, the true normal cone $N_X(0) = \mathbb{R}_- \times \mathbb{R}$ is larger than the one generated by the gradients, and $-\nabla f(0) = (-1, 0)$ lives in the first but not in the second. *The linearization lies* because the cusp is a cubic order object that the gradients (first order objects) cannot see. Constraint qualifications are exactly the conditions that forbid this kind of pathology.

Remark 9.20. An even shorter one dimensional version of the same phenomenon: $\min x$ s.t. $g(x) = x^2 \leq 0$. The only feasible point is $x^* = 0$ (hence it is the minimum), but $\nabla f(0) = 1$ and $\nabla g(0) = 0$, and the equation $1 + \mu \cdot 0 = 0$ has no solution. Here the gradient of the active constraint *vanishes* and carries no directional information: an extreme case of gradients describing the geometry badly.

9.4.2 The three qualifications you should know

Definition 9.21 (LICQ). The feasible point x satisfies the *linear independence constraint qualification* (LICQ) if the gradients of **all active constraints** at x (the g_i with $i \in A(x)$ and all

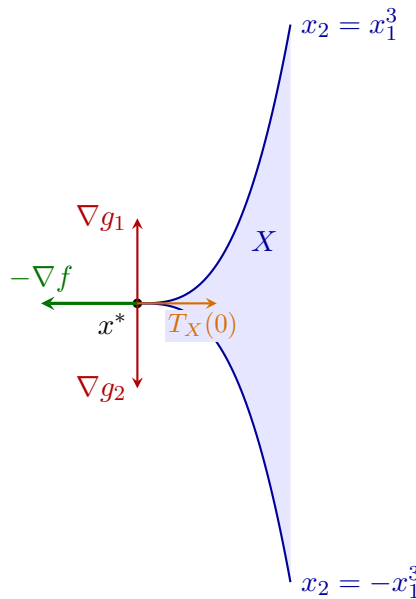


Figure 9.3: The cusp of Example 9.19. The true tangent cone at the origin is the positive horizontal semi-axis, but the active gradients (vertical) generate a cone that “does not see” the point: $-\nabla f$ cannot be written as a nonnegative combination of ∇g_1 and ∇g_2 , and KKT fails at the global minimum.

the h_j) are linearly independent:

$$\{\nabla g_i(x) : i \in A(x)\} \cup \{\nabla h_j(x) : j = 1, \dots, p\} \text{ is linearly independent.}$$

Definition 9.22 (Slater condition). A *convex* problem (f and the g_i convex, the h_j affine) satisfies the *Slater condition* if there is a strictly feasible point: some $\hat{x} \in \mathbb{R}^n$ with

$$g_i(\hat{x}) < 0 \quad \forall i, \quad h_j(\hat{x}) = 0 \quad \forall j.$$

That is, the feasible region has a point where *no* inequality bites. Unlike LICQ, this is a *global* condition on the problem (not on each point) and almost trivial to check.

Definition 9.23 (Mangasarian–Fromovitz). The feasible point x satisfies the *Mangasarian–Fromovitz constraint qualification* (MFCQ) if (i) the gradients $\{\nabla h_j(x)\}_{j=1}^p$ are linearly independent, and (ii) there is a direction $d \in \mathbb{R}^n$ with

$$\langle \nabla g_i(x), d \rangle < 0 \quad \forall i \in A(x), \quad \langle \nabla h_j(x), d \rangle = 0 \quad \forall j.$$

Intuitively: there is a direction that “enters strictly” the feasible region, moving away from all active boundaries at once.

Theorem 9.24 (Hierarchy of qualifications). *At a feasible point x :*

$$\text{LICQ} \implies \text{MFCQ} \implies \text{Abadie CQ} \implies (\text{the KKT theorem holds at } x),$$

where the Abadie qualification is the equality between the tangent cone and the linearized cone

that we will see in Definition 9.27 and in Lemma 9.28. In a convex problem, Slater provides the strictly entering direction required by MFCQ at every feasible point (and, with affine equalities whose gradients are linearly independent, full MFCQ); moreover, if all constraints are affine (linear), the KKT theorem holds with no additional hypothesis. Under LICQ the multipliers are unique; under MFCQ or Slater they exist but may not be unique.

Proof (only LICQ $\Rightarrow$ MFCQ). Let $v_1, \dots, v_k$ be the active gradients (inequalities) and $w_1, \dots, w_p$ those of the equalities, all linearly independent. The linear map

$$d \longmapsto (\langle v_1, d \rangle, \dots, \langle v_k, d \rangle, \langle w_1, d \rangle, \dots, \langle w_p, d \rangle)$$

from $\mathbb{R}^n$ to $\mathbb{R}^{k+p}$ is *surjective* (its matrix has linearly independent rows), so there is d with image $(-1, \dots, -1, 0, \dots, 0)$: exactly what MFCQ requires. The other implications are proved in [5], ch. 12, and [6], sec. 3.2; the Slater one is covered in Section 9.6. $\square$

Example 9.25 (Diagnosis of the counterexample). Let us check that the cusp of Example 9.19 violates all qualifications at $x^* = 0$. *LICQ*: $\nabla g_1(0) = (0, 1)$ and $\nabla g_2(0) = (0, -1)$ are collinear ($\nabla g_2 = -\nabla g_1$): it fails. *MFCQ*: we would need d with $\langle (0, 1), d \rangle = d_2 < 0$ and $\langle (0, -1), d \rangle = -d_2 < 0$ at the same time: impossible, it fails. *Slater*: here there is a strictly feasible point, for example $\hat{x} = (1, 0)$, where $g_1(\hat{x}) = g_2(\hat{x}) = -1 < 0$; and yet KKT fails. There is no contradiction: g_1 and g_2 are not convex (they contain $-x_1^3$), so the problem is not convex and the Slater condition simply does not apply. The moral is twofold: *Slater is a privilege of convex problems*, and with non convex g_i exhibiting an interior point guarantees nothing. By contrast, in the one dimensional example $\min x$ s.t. $x^2 \leq 0$, the constraint is convex but there is no $\hat{x}$ with $\hat{x}^2 < 0$: Slater fails, and with it KKT.

Exam strategy

Which qualification do I use in practice? Decision rule for the exam (and for life):

1. **Are all constraints linear or affine?** (budgets, non negativity, resources): KKT holds automatically, check nothing. This is the case of the standard UMP. Write one line: “the constraints are affine, so no additional qualification is required”.
2. **Is the problem convex with some nonlinear constraint?** (EMP with concave utility, that is, $\bar{u} - u(x)$ convex; cost minimization with a concave production requirement): check **Slater** by exhibiting ONE strictly feasible point. This is usually trivial: “for large $\hat{x}$, $u(\hat{x}) > \bar{u}$ strictly”.
3. **Non convex problem?** Check **LICQ** at the candidates: compute the active gradients and check linear independence (with one or two active constraints, this is just “no active gradient is zero” and “they are not collinear”).

MFCQ is the intermediate theoretical tool: you cite it when LICQ fails but there is still a strictly entering direction; in micro exams you will almost never need it, but you should be able to state it.

9.5 The Karush–Kuhn–Tucker theorem [P1]

We reach the heart of the chapter and of the course. Fix the standard problem (9.1) and define the full Lagrangian:

$$L(x, \mu, \lambda) = f(x) + \sum_{i=1}^m \mu_i g_i(x) + \sum_{j=1}^p \lambda_j h_j(x), \quad \mu \in \mathbb{R}_+^m, \lambda \in \mathbb{R}^p.$$

Theorem 9.26 (Karush–Kuhn–Tucker necessary conditions). *Let x^* be a local minimum of problem (9.1), with f, g_i, h_j of class $\mathcal{C}^1$ in a neighborhood of x^* . Suppose that at x^* some constraint qualification holds (for example LICQ, or MFCQ, or that the problem is convex with Slater, or that all constraints are affine). Then there exist multiplier vectors $\mu^* \in \mathbb{R}^m$ and $\lambda^* \in \mathbb{R}^p$ such that:*

$$(S) \text{ Stationarity:} \quad \nabla f(x^*) + \sum_{i=1}^m \mu_i^* \nabla g_i(x^*) + \sum_{j=1}^p \lambda_j^* \nabla h_j(x^*) = 0; \quad (9.9)$$

$$(PF) \text{ Primal feasibility:} \quad g_i(x^*) \leq 0, \quad h_j(x^*) = 0 \quad \forall i, j; \quad (9.10)$$

$$(DF) \text{ Dual feasibility:} \quad \mu_i^* \geq 0 \quad \forall i; \quad (9.11)$$

$$(CS) \text{ Complementary slackness:} \quad \mu_i^* g_i(x^*) = 0 \quad \forall i. \quad (9.12)$$

A point x^* for which such (μ^*, λ^*) exist is called a KKT point. The λ_j^* have no sign restriction.

Before the proof, let us make sure we read the four conditions correctly:

- **(S)** says that x^* is a critical point of the Lagrangian in x : $\nabla_x L(x^*, \mu^*, \lambda^*) = 0$. Geometrically, it is $-\nabla f(x^*) \in \text{cone}\{\nabla g_i(x^*) : i \in A(x^*)\} + \text{span}\{\nabla h_j(x^*)\}$: the algebraic version of $-\nabla f(x^*) \in N_X(x^*)$.
- **(PF)** only recalls that the candidate must be feasible. In exams it is forgotten with astonishing frequency: a “candidate” that violates a constraint is not a candidate.
- **(DF)** is the sign of the inequality multipliers, which comes from the geometry: walls only push inward (figure 9.1).
- **(CS)** is the star condition for the economist, so much so that we devote the whole Subsection 9.5.2 to it: for each i , either the constraint is active ($g_i(x^*) = 0$) or its multiplier is zero ($\mu_i^* = 0$), or both.

9.5.1 Proof through linearization and the Farkas lemma

The proof is a gem: it combines the geometric condition (Theorem 9.11) with the Farkas lemma we proved in the separation chapter. The structure is:

$$\begin{array}{ccc} \text{local optimum} & \xrightarrow{(1)} & \nabla f \text{ sees no descent in } T_X(x^*) \xrightarrow{(2), \text{CQ}} \text{nor in the linearized cone} \\ & & \xrightarrow{(3), \text{Farkas}} \text{multipliers.} \end{array}$$

Definition 9.27 (Linearized cone). The *linearized cone* (or cone of linearized feasible directions) of problem (9.1) at a feasible point x is

$$L_X(x) = \{ d \in \mathbb{R}^n : \langle \nabla g_i(x), d \rangle \leq 0 \quad \forall i \in A(x), \quad \langle \nabla h_j(x), d \rangle = 0 \quad \forall j \}.$$

It is the tangent cone of the problem “linearized” at x : it replaces each active constraint by its first order approximation.

Lemma 9.28. *At every feasible point, $T_X(x) \subseteq L_X(x)$. Under LICQ (or MFCQ) the equality $T_X(x) = L_X(x)$ holds.*

Proof (the inclusion always holds; equality sketched). Let $d \in T_X(x)$, with $x^k \in X$, $t_k \downarrow 0$, $(x^k - x)/t_k \rightarrow d$. For $i \in A(x)$, by differentiability and $g_i(x) = 0$:

$$0 \geq g_i(x^k) = \underbrace{g_i(x)}_{=0} + \langle \nabla g_i(x), x^k - x \rangle + o(\|x^k - x\|).$$

Dividing by t_k and taking the limit (same handling of the remainder as in Theorem 9.11): $\langle \nabla g_i(x), d \rangle \leq 0$. For the equalities, the same argument applied to $h_j \leq 0$ and $-h_j \leq 0$ gives $\langle \nabla h_j(x), d \rangle = 0$. Hence $d \in L_X(x)$.

The reverse inclusion under LICQ is the technical step (where the cusp of Example 9.19 is excluded): given $d \in L_X(x)$, one has to *construct* a feasible sequence with limit direction d . The tool is again the implicit function theorem: thanks to the linear independence of the active gradients, the system of active constraints can be solved locally and the straight path $x + td$ can be “projected” back into the feasible region with an error $o(t)$, giving feasible points $x(t) = x + td + o(t)$. The detailed construction is in [5], lemma 12.2, and [6], lemma 3.13. $\square$

Lemma 9.29 (Farkas, version with equalities). *Let $c, a_1, \dots, a_k, b_1, \dots, b_p \in \mathbb{R}^n$. Exactly one of the following alternatives holds:*

(I) *there is $d \in \mathbb{R}^n$ with $\langle a_i, d \rangle \leq 0 \quad \forall i$, $\langle b_j, d \rangle = 0 \quad \forall j$, and $\langle c, d \rangle > 0$;*

(II) *there are $\mu \in \mathbb{R}_+^k$ and $\lambda \in \mathbb{R}^p$ with $c = \sum_{i=1}^k \mu_i a_i + \sum_{j=1}^p \lambda_j b_j$.*

That is: either c belongs to the cone generated by the a_i plus the subspace generated by the b_j , or there is a “certificate” d exposing it.

Proof. First, the two alternatives are *incompatible*: if (I) and (II) held at once,

$$0 < \langle c, d \rangle = \sum_i \mu_i \underbrace{\langle a_i, d \rangle}_{\leq 0} + \sum_j \lambda_j \underbrace{\langle b_j, d \rangle}_{=0} \leq 0,$$

absurd (here we use $\mu_i \geq 0$; the λ_j may have any sign because they multiply zero).

That at least one holds is the Farkas lemma of the separation chapter (Theorem 4.23), in the row version: either c is a nonnegative combination of the rows of a matrix A , or there is d with $Ad \leq 0$ and $\langle c, d \rangle > 0$. Apply it to the matrix A whose rows are

$$a_1, \dots, a_k, \quad b_1, \dots, b_p, \quad -b_1, \dots, -b_p$$

(each equality is split into two inequalities). The first alternative gives weights $\mu_i \geq 0$, $\lambda_j^+ \geq 0$, $\lambda_j^- \geq 0$ with $c = \sum_i \mu_i a_i + \sum_j (\lambda_j^+ - \lambda_j^-) b_j$; defining $\lambda_j = \lambda_j^+ - \lambda_j^- \in \mathbb{R}$ (with no sign restriction) gives (II). The second alternative, $Ad \leq 0$ with $\langle c, d \rangle > 0$, says exactly $\langle a_i, d \rangle \leq 0$, $\langle b_j, d \rangle \leq 0$ and $\langle -b_j, d \rangle \leq 0$, that is, $\langle b_j, d \rangle = 0$: that is (I). Note how splitting the equalities explains at the root why the λ_j have no sign restriction: they are the difference of two nonnegative multipliers. $\square$

Proof of Theorem 9.26. Step 1 (geometry). By Theorem 9.11, since x^* is a local minimum, $\langle \nabla f(x^*), d \rangle \geq 0$ for every $d \in T_X(x^*)$.

Step 2 (qualification). The constraint qualification guarantees $T_X(x^*) = L_X(x^*)$ (Lemma 9.28; for Slater and affine constraints the argument is different but the relevant conclusion is the same, see [6], sec. 3.2). Therefore:

$$\text{there is no } d \text{ with } \langle \nabla g_i(x^*), d \rangle \leq 0 \ \forall i \in A(x^*), \quad \langle \nabla h_j(x^*), d \rangle = 0 \ \forall j, \quad \langle -\nabla f(x^*), d \rangle > 0.$$

(We write $\langle -\nabla f(x^*), d \rangle > 0$, that is, “ d is a descent direction”: if such a d existed, it would be in $L_X(x^*) = T_X(x^*)$ and would violate step 1.)

Step 3 (Farkas). The nonexistence of such a d is exactly the negation of alternative (I) of Lemma 9.29 with $c = -\nabla f(x^*)$, $a_i = \nabla g_i(x^*)$ for $i \in A(x^*)$, $b_j = \nabla h_j(x^*)$. Hence alternative (II) holds: there are $\mu_i^* \geq 0$ ($i \in A(x^*)$) and $\lambda_j^* \in \mathbb{R}$ with

$$-\nabla f(x^*) = \sum_{i \in A(x^*)} \mu_i^* \nabla g_i(x^*) + \sum_{j=1}^p \lambda_j^* \nabla h_j(x^*),$$

which is (9.9) restricted to the active indices.

Step 4 (complete and check CS). Define $\mu_i^* = 0$ for every $i \notin A(x^*)$. Then (9.9) holds with the full sums (the added terms are zero), (9.11) holds by construction, (9.10) holds because x^* is feasible, and complementary slackness (9.12) is checked index by index: if $i \in A(x^*)$, then $g_i(x^*) = 0$ and the product $\mu_i^* g_i(x^*)$ is zero; if $i \notin A(x^*)$, then $\mu_i^* = 0$ and the product is zero as well. *Complementary slackness is not an extra hypothesis: it is the bookkeeping that inactive constraints do not take part in the geometry of the optimum.* $\square$

Intuition

It is worth recapping the proof in one sentence per step, because its logic is that of economic analysis itself. (1) At the optimum there is no direction that is feasible and improves the objective (if there were, we would not be at the optimum). (2) The constraint qualification lets us check “feasible” with the derivatives of the constraints instead of with the true region. (3) The Farkas lemma turns “no such direction exists” (a statement about

infinitely many directions, impossible to check one by one) into “multipliers exist” (a *finite* system of equations and signs, which can be solved). That jump from the universal negative to the existential positive is the whole computational value of KKT: instead of proving that no deviation improves, you solve a system. Compare it with the trick of the first welfare theorem or with no arbitrage arguments in finance: “no improvement is possible” $\Leftrightarrow$ “there are prices that support the allocation”. The multipliers *are* those prices.

9.5.2 Complementary slackness: the economist’s condition

Condition (9.12), $\mu_i^* g_i(x^*) = 0$ for each i , deserves its own subsection because it organizes *all* the practical work and carries the most economic content.

What it says exactly. Since $\mu_i^* \geq 0$ and $g_i(x^*) \leq 0$, the product $\mu_i^* g_i(x^*)$ is always ≤ 0 ; asking it to be zero is the same as asking that *for each constraint, at least one of the two factors vanishes*:

$$\begin{aligned} \text{for each } i : \quad & \underbrace{g_i(x^*) < 0}_{\text{slack constraint}} \implies \mu_i^* = 0, \\ & \text{equivalently} \quad \mu_i^* > 0 \implies g_i(x^*) = 0. \end{aligned}$$

Careful with what it does *not* say: it is perfectly possible that $g_i(x^*) = 0$ and $\mu_i^* = 0$ at the same time (a constraint active “by chance”, touching the boundary without exerting pressure; this is called the *degenerate* case and appears in Exercise 9.51).

Why it is intuitive. Think of μ_i^* as the *shadow price* of constraint i : how much the optimal value would improve if we relaxed the constraint marginally (Section 9.8). If the constraint does not bite ($g_i(x^*) < 0$), relaxing it a little changes nothing (the optimum does not even touch it), so its shadow price must be zero: *a resource in excess is free*. Conversely, if the resource has a positive price ($\mu_i^* > 0$), it is because it is being used to the limit ($g_i(x^*) = 0$). This is the same market logic you already know: a good in excess supply has zero equilibrium price (Walras law with inequalities, the “free goods” condition). In contracts: if the agent’s participation constraint does not bind, the principal is paying excess rent and could reduce the payment; at the optimum, either IR binds or its multiplier is zero, and Holmström’s analysis consists of finding out which.

How it organizes the case analysis. Complementary slackness turns the KKT system (which mixes equations, inequalities and products) into a *finite disjunction of systems of equations*. For each inequality constraint there are two exhaustive possibilities (active or zero multiplier), so with m inequalities there are at most 2^m “regimes” or cases:

$$\text{for each } i, \text{ choose: } \begin{cases} \text{Case A}_i : & g_i(x) = 0 \quad (\mu_i \text{ free } \geq 0), \\ \text{Case B}_i : & \mu_i = 0 \quad (g_i(x) \text{ free } \leq 0). \end{cases}$$

In each regime, the remaining system consists of equations (stationarity + the chosen equalities) and is solved with algebra; then you *check* the pending inequalities ($\mu_i \geq 0$ in the A cases, $g_i \leq 0$ in the B cases) and discard the inconsistent cases. Section 9.7 turns this into a recipe, with economic theory as a case pruner: monotonicity $\Rightarrow$ the budget binds; Inada $\Rightarrow$ the nonnegativity constraints do not bind; and so on. Without those prunings, 2^m explodes fast; with them, exam problems reduce to two or three cases.

Remark 9.30 (KKT generalizes everything before it). If there are no constraints ($m = p = 0$), KKT reduces to $\nabla f(x^*) = 0$ (Chapter 8). If there are only equalities, it reduces to the Lagrange theorem 9.16. If no inequality is active at the optimum, complementary slackness kills all the μ_i^* and again we get $\nabla f(x^*) = 0$: an interior optimum does not notice the constraints. Equations (9.2) and (9.3) of the motivation are the cases of one and two active constraints. One single theory for everything.

9.6 KKT under convexity: necessity and sufficiency [P1]

So far, KKT is *necessary*: it filters candidates. The gem of the convex case is that it is also *sufficient*: every KKT point is a global optimum. For the economist this is gold: in the UMP with concave utility or in the EMP, finding ONE KKT point *finishes* the problem.

Definition 9.31 (Convex problem). Problem (9.1) is *convex* if f is convex, each g_i is convex and each h_j is affine ($h_j(x) = \langle a_j, x \rangle - b_j$). In that case the feasible set X is convex (an intersection of sublevel sets of convex functions, which are convex by the chapter on convex functions, and hyperplanes). In the “economist” version: maximizing a *concave* function with convex constraints $g_i(x) \leq 0$ is a convex problem after the change $\min -f$.

Theorem 9.32 (Sufficiency of KKT in convex problems). *Consider problem (9.1) convex with f, g_i differentiable. If (x^*, μ^*, λ^*) satisfies the four KKT conditions (9.9)–(9.12), then x^* is a global minimum. No constraint qualification is needed for this direction.*

Proof. The proof is short and elegant; each step uses exactly one hypothesis. Let $x \in X$ be any feasible point; we want $f(x) \geq f(x^*)$.

Step 1: convexity of f (gradient inequality).

$$f(x) \geq f(x^*) + \langle \nabla f(x^*), x - x^* \rangle.$$

Step 2: stationarity (9.9). We substitute $\nabla f(x^*) = -\sum_i \mu_i^* \nabla g_i(x^*) - \sum_j \lambda_j^* \nabla h_j(x^*)$:

$$f(x) \geq f(x^*) - \sum_{i=1}^m \mu_i^* \langle \nabla g_i(x^*), x - x^* \rangle - \sum_{j=1}^p \lambda_j^* \langle \nabla h_j(x^*), x - x^* \rangle.$$

Step 3: the affine equalities disappear. Since h_j is affine, $\langle \nabla h_j(x^*), x - x^* \rangle = h_j(x) - h_j(x^*) = 0 - 0 = 0$ (both points are feasible). The whole third sum vanishes.

Step 4: convexity of the g_i and dual feasibility. By the gradient inequality applied to g_i : $\langle \nabla g_i(x^*), x - x^* \rangle \leq g_i(x) - g_i(x^*)$. Multiplying by $\mu_i^* \geq 0$ (this is where the sign is used) and

summing:

$$\sum_i \mu_i^* \langle \nabla g_i(x^*), x - x^* \rangle \leq \sum_i \mu_i^* g_i(x) - \sum_i \mu_i^* g_i(x^*).$$

Step 5: complementary slackness and primal feasibility. By (9.12), $\sum_i \mu_i^* g_i(x^*) = 0$. And since x is feasible ($g_i(x) \leq 0$) and $\mu_i^* \geq 0$, we have $\sum_i \mu_i^* g_i(x) \leq 0$. Hence $\sum_i \mu_i^* \langle \nabla g_i(x^*), x - x^* \rangle \leq 0$, and going back to step 2:

$$f(x) \geq f(x^*) - \underbrace{\sum_i \mu_i^* \langle \nabla g_i(x^*), x - x^* \rangle}_{\leq 0} \geq f(x^*).$$

Since $x \in X$ was arbitrary, x^* is a global minimum. $\square$

Corollary 9.33 (Full characterization in the convex case). *In a differentiable convex problem satisfying the Slater condition, x^* is a global minimum **if and only if** there are (μ^*, λ^*) such that (x^*, μ^*, λ^*) satisfies KKT. (Necessity: Theorem 9.26 with Slater as CQ; sufficiency: Theorem 9.32.) In particular, solving the KKT system solves the problem: this is the standard mode of operation in all of microeconomics with concave preferences.*

9.6.1 Beyond concavity: Arrow–Enthoven

In micro it is common for utility to be only *quasiconcave* (convex indifference curves), not concave; recall from the quasiconcavity chapter that quasiconcavity does not get along with sums or with gradient inequalities. Even so, the sufficiency of KKT survives under surprisingly weak hypotheses:

Theorem 9.34 (Arrow–Enthoven, 1961; statement). *Consider $\max f(x)$ s.t. $g_i(x) \leq 0$, $x \geq 0$, where f is differentiable and quasiconcave, and each g_i is differentiable and quasiconvex. Let (x^*, μ^*) be a KKT point of the problem. If in addition one of the following conditions holds:*

- (a) *f is pseudoconcave at x^* ; in particular, it is enough that f be quasiconcave, twice differentiable in a neighborhood of x^* and with $\nabla f(x^*) \neq 0$ (the Arrow–Enthoven condition that holds in micro: some good is desirable at the margin); or*
- (b) *f is concave (the case already covered),*

then x^ is a global maximum. Recall from the quasiconcavity chapter: pseudoconcave means that $\langle \nabla f(x), y - x \rangle \leq 0$ implies $f(y) \leq f(x)$; every differentiable concave function is pseudoconcave, and every pseudoconcave function is quasiconcave. The practical moral: for monotone quasiconcave utilities (nonzero gradient), like almost all of those in micro, KKT is still sufficient. Full statement and proof in [1] and in appendix M of [9].*

Remark 9.35. Cobb–Douglas $u = x^\alpha y^\beta$ with $\alpha + \beta > 1$ is not concave, but it is quasiconcave with nonzero gradient in the positive orthant: Arrow–Enthoven guarantees that the KKT point of the UMP is the global maximum in that case too. Another practical way out: maximize $\ln u = \alpha \ln x + \beta \ln y$, which is concave and has the same maximizers since $\ln$ is strictly increasing (a monotone transformation does not change the argmax, although it can change concavity). This trick of “concavifying” with a monotone transformation is standard in micro and worth keeping at hand.

9.7 Manual for solving KKT problems [P1]

This section is a recipe, deliberately mechanical, for solving KKT problems in an exam, followed by four economic examples solved with *all* cases analyzed. The recipe first:

Exam strategy

KKT protocol in seven steps (for an exam problem):

- P1. Normalize.** Write the problem in a fixed form (I recommend: max with constraints $g_i(x) \leq 0$, Lagrangian $L = f - \sum_i \mu_i g_i$; or convert everything to the standard min form). Identify every constraint, including the nonnegativity ones. Write down how many inequalities there are: that is your universe of cases.
- P2. Justify the method.** One line on existence (Weierstrass if the feasible set is compact) and one line on qualification and sufficiency: are the constraints affine? Slater? is the objective concave or quasiconcave (Arrow–Enthoven)? This decides whether the KKT point you find is *the* optimum or only a candidate.
- P3. Write the Lagrangian and the full conditions:** stationarity (one equation per variable), primal feasibility, $\mu_i \geq 0$, and complementary slackness (one product equation per inequality). Write them ALL: half the exam points are usually here.
- P4. Prune cases with economics.** Before enumerating 2^m cases, use the structure: monotonicity $\Rightarrow$ the budget is active; Inada conditions ($u_x \rightarrow \infty$ as $x \rightarrow 0$) $\Rightarrow$ nonnegativity inactive; $\lambda > 0$ deduced from stationarity $\Rightarrow$ its constraint is active. Document each pruning with one sentence.
- P5. Enumerate and solve the remaining cases.** In each case, the choices “active” ($g_i = 0$) or “zero price” ($\mu_i = 0$) turn the system into pure equations. Solve it.
- P6. Check and discard.** In each solved case, check the conditions you did not use: signs of multipliers and feasibility of the constraints assumed inactive. A case is discarded by *showing* the violation (for example “ $\mu_2 = -3 < 0$, contradiction”). Never discard it on a hunch.
- P7. Conclude and interpret.** Report the optimum, the optimal value and the multipliers with their economic reading (shadow prices). If step P2 gave sufficiency, declare global optimality; if not, compare values among the surviving candidates.

9.7.1 Example 1: Cobb–Douglas UMP with nonnegativity (interior solution)

Example 9.36 (Full UMP with KKT).

$$\max_{x,y} u(x,y) = x^\alpha y^\beta \quad \text{s.t.} \quad px + qy \leq m, \quad x \geq 0, \quad y \geq 0,$$

with $\alpha, \beta \in (0, 1)$, $\alpha + \beta \leq 1$, and $p, q, m > 0$.

P1 (normalize). Three inequalities: $g_0 = px + qy - m \leq 0$, $g_1 = -x \leq 0$, $g_2 = -y \leq 0$. Universe: $2^3 = 8$ possible cases.

P2 (justify). The budget set is compact (closed, and bounded because $p, q > 0$) and u is continuous: the maximum exists (Weierstrass). The three constraints are *affine*: KKT is necessary with no further requirements. With $\alpha + \beta \leq 1$, u is concave on $\mathbb{R}_+^2$ (and in general it is quasiconcave with nonzero gradient in the interior, so Arrow–Enthoven covers the case $\alpha + \beta > 1$): KKT will also be sufficient.

P3 (Lagrangian and conditions). With multipliers $\lambda, \mu_1, \mu_2 \geq 0$:

$$L = x^\alpha y^\beta - \lambda(px + qy - m) + \mu_1 x + \mu_2 y.$$

KKT conditions:

$$\begin{aligned} (S_x) \quad & \alpha x^{\alpha-1} y^\beta - \lambda p + \mu_1 = 0, \\ (S_y) \quad & \beta x^\alpha y^{\beta-1} - \lambda q + \mu_2 = 0, \\ (PF) \quad & px + qy \leq m, \quad x \geq 0, \quad y \geq 0, \\ (DF) \quad & \lambda \geq 0, \quad \mu_1 \geq 0, \quad \mu_2 \geq 0, \\ (CS) \quad & \lambda(px + qy - m) = 0, \quad \mu_1 x = 0, \quad \mu_2 y = 0. \end{aligned}$$

P4 (prune). *Pruning 1: the corners on the axes are not optimal.* If $x = 0$ or $y = 0$, then $u = 0$, while the feasible point $(\frac{m}{2p}, \frac{m}{2q})$ gives positive utility. Hence the optimum has $x^* > 0$ and $y^* > 0$, and by complementary slackness $\mu_1 = \mu_2 = 0$. (Technical caution: at $x = 0$ with $\alpha < 1$ the derivative $\alpha x^{\alpha-1}$ is not even finite; the argument by direct comparison of values cleanly avoids the problem, and it is the one I recommend in an exam.) This eliminates at once the 6 cases with some corner. *Pruning 2: the budget binds.* Suppose $px + qy < m$ at the optimum; by (CS), $\lambda = 0$; with $\mu_1 = 0$, (S_x) would give $\alpha x^{\alpha-1} y^\beta = 0$, impossible with $x, y > 0$. Contradiction: therefore $px^* + qy^* = m$. (Economic reading: marginal utility is positive, so leaving income unspent cannot be optimal.) ONE case remains.

P5 (solve the interior case with active budget). With $\mu_1 = \mu_2 = 0$, (S_x) and (S_y) are exactly conditions (9.6)–(9.7) of Example 9.18; dividing them and using the budget:

$$x^* = \frac{\alpha}{\alpha + \beta} \cdot \frac{m}{p}, \quad y^* = \frac{\beta}{\alpha + \beta} \cdot \frac{m}{q}, \quad \lambda^* = \frac{\alpha(x^*)^{\alpha-1}(y^*)^\beta}{p} > 0.$$

P6 (check). $x^*, y^* > 0$ ✓; budget exact by construction ✓; $\lambda^* > 0$ ✓; $\mu_1 = \mu_2 = 0 \geq 0$ ✓. All KKT conditions hold and no other case remains.

P7 (conclude). By sufficiency (concave or quasiconcave problem with affine constraints), (x^*, y^*) is the global maximum. Note how KKT *rediscovers* the Lagrange solution of Example 9.18, but now with the budget binding and interiority *proved* instead of assumed: that is what the inequality machinery adds.

9.7.2 Example 2: corner solution with quasilinear utility

Example 9.37 (Quasilinear: complementary slackness detects the corner).

$$\max_{x,y} u(x, y) = \ln x + y \quad \text{s.t.} \quad px + y \leq m, \quad x \geq 0, \quad y \geq 0,$$

with $p > 0$, $m > 0$ (the price of y is normalized to 1). This is the perfect example to see KKT really at work: depending on the income level, the solution is interior or a corner, and it is *complementary slackness* that discovers this by itself.

P1–P2. Inequalities: $g_0 = px + y - m \leq 0$, $g_1 = -x \leq 0$, $g_2 = -y \leq 0$. Affine constraints (KKT necessary with no extra CQ); u is concave (a sum of concave functions: $\ln x$ strictly concave, y linear), so KKT is sufficient. Effective domain: $x > 0$ (since $\ln x \rightarrow -\infty$ as $x \downarrow 0$: the corner $x = 0$ is infinitely bad and is pruned immediately; formally, $u \rightarrow -\infty$ guarantees that the supremum is not attained at $x = 0$). Therefore $\mu_1 = 0$ always.

P3. Lagrangian and conditions ($\lambda, \mu_2 \geq 0$):

$$L = \ln x + y - \lambda(px + y - m) + \mu_2 y,$$

$$(S_x) \quad \frac{1}{x} - \lambda p = 0 \implies \lambda = \frac{1}{px} > 0,$$

$$(S_y) \quad 1 - \lambda + \mu_2 = 0 \implies \mu_2 = \lambda - 1,$$

$$(CS) \quad \lambda(px + y - m) = 0, \quad \mu_2 y = 0,$$

plus primal and dual feasibility.

P4 (prune). From (S_x) , $\lambda > 0$ automatically; by (CS) , *the budget always binds*: $px + y = m$. Only two cases remain, governed by the constraint $y \geq 0$:

P5, Case A: $y > 0$ (interior). By (CS) , $\mu_2 = 0$; by (S_y) , $\lambda = 1$; by (S_x) , $x = 1/(\lambda p) = 1/p$. The budget gives $y = m - px = m - 1$. *Check (P6)*: we need $y > 0$, that is, $\boxed{m > 1}$. If $m > 1$, everything fits:

$$x^* = \frac{1}{p}, \quad y^* = m - 1, \quad \lambda^* = 1, \quad \mu_2^* = 0.$$

Note the classical quasilinear property: the demand for x does not depend on income; all extra income goes to the linear good y , and the marginal utility of income is constant ($\lambda^* = 1$).

P5, Case B: $y = 0$ (corner). The binding budget gives $x = m/p$. From (S_x) : $\lambda = 1/(px) = 1/m$. From (S_y) : $\mu_2 = \lambda - 1 = \frac{1}{m} - 1 = \frac{1-m}{m}$. *Check (P6)*: we need $\mu_2 \geq 0$, that is, $\boxed{m \leq 1}$. If $m < 1$:

$$x^* = \frac{m}{p}, \quad y^* = 0, \quad \lambda^* = \frac{1}{m} > 1, \quad \mu_2^* = \frac{1-m}{m} > 0.$$

P7 (conclude and interpret). The two cases *split* the parameter space without overlapping (except at the boundary $m = 1$, where both give the same solution $x^* = 1/p$, $y^* = 0$ with $\mu_2^* = 0$: the degenerate case “active with zero price”). The economic reading of the corner: when income is low ($m < 1$), the consumer would like to “sell” good y to buy more x (the interior tangency would require $y = m - 1 < 0$), but cannot: the constraint $y \geq 0$ *bites*, and its multiplier $\mu_2^* = (1 - m)/m > 0$ measures how much utility the consumer would gain if allowed a marginally negative y (borrowing in the numeraire good). Moreover $\lambda^* = 1/m > 1$: income is more valuable when you are poor, exactly as concavity dictates. Note the logical flow: we *did not guess* the corner; complementary slackness organized the cases and the sign checks ($y > 0$ vs. $\mu_2 \geq 0$) said exactly for which parameters each regime holds. Contract problems are solved the same way: the Baron–Myerson regimes (“IR binds for the low type, IC for the high type”)

are KKT cases checked in exactly this manner.

9.7.3 Example 3: cost minimization by the firm (the dual of production)

Example 9.38 (Minimum cost with Cobb–Douglas technology). The firm wants to produce at least $q > 0$ units at the lowest cost:

$$\min_{K,L} wL + rK \quad \text{s.t.} \quad F(K, L) = K^{1/2}L^{1/2} \geq q, \quad K \geq 0, \quad L \geq 0,$$

with wage $w > 0$ and capital rental rate $r > 0$.

P1. Standard form: $f = wL + rK$; $g_0 = q - K^{1/2}L^{1/2} \leq 0$; $g_1 = -K \leq 0$; $g_2 = -L \leq 0$.

P2. The objective is linear (convex). F is concave on $\mathbb{R}_+^2$ (Cobb–Douglas with constant returns), so $g_0 = q - F$ is convex: *convex problem*. Slater: the point $\hat{K} = \hat{L} = 2q$ gives $F = 2q > q$ and $\hat{K}, \hat{L} > 0$: a strictly feasible point exists ✓. By Corollary 9.33, KKT characterizes the global optimum. (Existence: the feasible set is not compact, but we can restrict to $\{wL + rK \leq wL_0 + rK_0\}$ for any feasible point (K_0, L_0) and apply Weierstrass on the intersection, which is compact.)

P3. With $\mu, \mu_K, \mu_L \geq 0$ (and calling $\mathcal{L}$ the Lagrangian, because here the letter L is already labor):

$$\mathcal{L}(K, L, \mu, \mu_K, \mu_L) = wL + rK + \mu(q - K^{1/2}L^{1/2}) - \mu_K K - \mu_L L,$$

$$(S_K) \quad r - \frac{\mu}{2} K^{-1/2} L^{1/2} - \mu_K = 0,$$

$$(S_L) \quad w - \frac{\mu}{2} K^{1/2} L^{-1/2} - \mu_L = 0,$$

$$(CS) \quad \mu(q - K^{1/2}L^{1/2}) = 0, \quad \mu_K K = 0, \quad \mu_L L = 0.$$

P4 (prune). To produce $q > 0$ both inputs are needed ($F = 0$ if $K = 0$ or $L = 0$): at any feasible point $K, L > 0$, hence $\mu_K = \mu_L = 0$. Can $\mu = 0$? Then (S_K) would give $r = 0$, a contradiction. Hence $\mu > 0$ and, by complementary slackness, *the production constraint binds*: $K^{1/2}L^{1/2} = q$. (Economics: producing more costs money and adds nothing; the requirement bites and the multiplier, which will turn out to be marginal cost, is positive.) Again a single case survives.

P5 (solve). Dividing (S_K) by (S_L) :

$$\frac{r}{w} = \frac{K^{-1/2}L^{1/2}}{K^{1/2}L^{-1/2}} = \frac{L}{K} \quad \implies \quad L = \frac{r}{w}K.$$

This is the firm's tangency condition: the marginal rate of technical substitution ($MRTS = F_L/F_K = K/L$) equals the relative input price w/r . Substituting into $K^{1/2}L^{1/2} = q$:

$$K^{1/2}\left(\frac{r}{w}K\right)^{1/2} = q \implies K\sqrt{\frac{r}{w}} = q \implies \boxed{K^* = q\sqrt{\frac{w}{r}}, \quad L^* = q\sqrt{\frac{r}{w}}}.$$

The minimum cost (the *cost function*) is

$$C(w, r, q) = wL^* + rK^* = wq\sqrt{\frac{r}{w}} + rq\sqrt{\frac{w}{r}} = 2q\sqrt{wr},$$

and the multiplier comes from (S_L) , which with $\mu_L = 0$ says $w = \frac{\mu}{2} K^{1/2} L^{-1/2}$; solving for μ and evaluating at the optimum:

$$\mu^* = \frac{w}{\frac{1}{2}(K^*)^{1/2}(L^*)^{-1/2}} = 2w\sqrt{\frac{L^*}{K^*}} = 2w\sqrt{\frac{r}{w}} = 2\sqrt{wr}.$$

P6. $K^*, L^* > 0$ ✓, exact production ✓, $\mu^* > 0$ ✓.

P7. A revealing check: $\partial C / \partial q = 2\sqrt{wr} = \mu^*$. *The multiplier of the production constraint is marginal cost:* the shadow price of requiring the firm to deliver one more unit of output. This equality is no coincidence: it is the envelope theorem (Chapter 11) and it reappears throughout the theory of the firm (the competitive supply curve is $p = C'(q) = \mu^*$).

9.7.4 Example 4: mini principal agent problem with symmetric information

Example 9.39 (Optimal risk sharing: IR binds and the agent is fully insured). A risk neutral *principal* hires a risk averse *agent*. Output is random: it equals y_H with probability $\pi \in (0, 1)$ and $y_L < y_H$ with probability $1 - \pi$ (effort is not an issue here: symmetric information, no moral hazard). The contract specifies contingent wages $w_H, w_L \geq 0$. The agent has Bernoulli utility $v(w)$ with $v' > 0$, $v'' < 0$ (strictly concave) and reservation utility $\bar{v}$ (what they get in their best alternative). The principal solves:

$$\max_{w_H, w_L} \pi(y_H - w_H) + (1 - \pi)(y_L - w_L) \quad \text{s.t.} \quad \underbrace{\pi v(w_H) + (1 - \pi)v(w_L)}_{(\text{IR})} \geq \bar{v}.$$

(We assume parameters such that the optimal wages come out positive, and we omit $w \geq 0$ to focus on IR; Exercise 9.56 repeats the analysis with a concrete specification.)

P1–P2. A single inequality: $g(w_H, w_L) = \bar{v} - \pi v(w_H) - (1 - \pi)v(w_L) \leq 0$. Since v is concave, $-v$ is convex and g is convex; the objective (to be minimized: $-\pi(y_H - w_H) - (1 - \pi)(y_L - w_L)$, linear) is convex: *convex problem*. Slater: by paying huge wages, the agent's expected utility exceeds $\bar{v}$ strictly ✓. KKT characterizes the global optimum.

P3. With $\gamma \geq 0$ the multiplier of IR:

$$L = \pi(y_H - w_H) + (1 - \pi)(y_L - w_L) + \gamma [\pi v(w_H) + (1 - \pi)v(w_L) - \bar{v}],$$

$$(S_H) \quad -\pi + \gamma \pi v'(w_H) = 0 \implies \gamma v'(w_H) = 1, \quad (9.13)$$

$$(S_L) \quad -(1 - \pi) + \gamma (1 - \pi) v'(w_L) = 0 \implies \gamma v'(w_L) = 1, \quad (9.14)$$

$$(CS) \quad \gamma [\pi v(w_H) + (1 - \pi)v(w_L) - \bar{v}] = 0.$$

P4 (does IR bind?). Suppose not: $\gamma = 0$. Then (S_H) says $-\pi = 0$: a contradiction. Hence $\boxed{\gamma > 0}$ and, by complementary slackness, *IR binds*:

$$\pi v(w_H) + (1 - \pi)v(w_L) = \bar{v}.$$

Pure economics: if the agent had utility to spare, the principal could cut both wages a little,

still hire them, and earn more. At the optimum, the principal extracts all the surplus and leaves the agent exactly at their reservation utility. The multiplier $\gamma > 0$ is the shadow price of IR: how much profit the principal loses for each extra util that must be guaranteed to the agent.

P5 (solve: full insurance). From (9.13) and (9.14):

$$v'(w_H) = \frac{1}{\gamma} = v'(w_L).$$

Since $v'' < 0$, v' is strictly decreasing and therefore *injective*: $v'(w_H) = v'(w_L)$ implies

$$w_H = w_L = \bar{w}.$$

The optimal wage does not depend on the state. The risk neutral principal absorbs all the risk and *fully insures* the risk averse agent. This is Borch's optimal risk sharing condition and the *first best* of the principal agent model (MWG, ch. 14): with symmetric information, efficiency = full insurance + surplus extraction. The binding IR pins down the level: $v(\bar{w}) = \bar{v}$, that is, $\bar{w} = v^{-1}(\bar{v})$, and the multiplier is $\gamma = 1/v'(\bar{w}) > 0$.

P6–P7. Checks: $\gamma > 0$ ✓, IR with equality ✓; by convexity, this is the global optimum. The principal's profit: $\pi y_H + (1 - \pi)y_L - v^{-1}(\bar{v})$. *Numerical illustration*: $v(w) = \sqrt{w}$, $\bar{v} = 2$, $\pi = 1/2$, $y_H = 16$, $y_L = 8$. Then $\bar{w} = \bar{v}^2 = 4$, $\gamma = 1/v'(4) = 2\sqrt{4} = 4$, and profit is $12 - 4 = 8$. Envelope check: if $\bar{v}$ rises to 2.1, the new wage is 4.41 and profit 7.59; the drop $\approx 0.41 \approx \gamma \cdot 0.1 = 0.4$ ✓ (the difference is the second order term). When you add moral hazard, the IC constraint will break full insurance ($w_H > w_L$ to provide incentives); the KKT scaffolding (who binds, what sign each multiplier has) will be identical to this one.

Careful (common mistake)

The most common exam mistakes in problems like the above:

- **Dividing by something that may be zero.** Before dividing (S_x) by (S_y) , justify that the denominator does not vanish (interiority, monotonicity). In corner solutions that division is precisely what you *cannot* do.
- **Forgetting cases.** “Assume an interior solution” without checking afterwards the conditions of that case or examining the others: you lose exactly the points the problem meant to test (the corners). The P4–P6 protocol exists for that.
- **Concluding optimality without sufficiency.** A KKT point of a non convex problem can be a local maximum, a constrained saddle point or even the worst point of the region. Always state your reason: convexity + Slater, Arrow–Enthoven, or exhaustive comparison of candidates with existence by Weierstrass.
- **Multiplier with the wrong sign** from mixing min/max conventions: review Section 9.1.3 and do the shadow price check.

9.8 Economic interpretation of the multipliers: shadow prices [P1]

So far the multipliers have been an algebraic device: numbers that make a combination of gradients vanish. This section explains why they are, by far, the *economically* most interesting object in the whole apparatus. The sentence you should be able to say in an exam (and in a seminar) is:

the multiplier of a constraint is the shadow price of the resource that constraint rations.

9.8.1 The value function and the formal statement

We start from the problem of Definition 9.1, but now *indexed* by the resource level. For $b = (b_1, \dots, b_m) \in \mathbb{R}^m$ consider

$$P(b) : \quad \max_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad g_i(x) \leq b_i \quad (i = 1, \dots, m), \quad h_j(x) = 0 \quad (j = 1, \dots, p). \quad (9.15)$$

The original problem is $P(0)$. Increasing b_i *relaxes* constraint i : the feasible set grows, and therefore the optimal value cannot get worse.

Definition 9.40 (Value function). The *value function* of family (9.15) is

$$V(b) = \sup \{f(x) : g_i(x) \leq b_i \ \forall i, \ h_j(x) = 0 \ \forall j\},$$

with the convention $V(b) = -\infty$ if $P(b)$ is infeasible. In economics V gets its own names depending on the context: *indirect utility* $v(p, m)$ if the resource is income, *cost function* $C(w, q)$ if the resource is the required output level, *maximum profit* $\pi(p)$, *value of the principal's program* in contract theory.

Definition 9.41 (Shadow price). The *shadow price* of resource i at b is the marginal rate at which the optimal value responds to an extra endowment of that resource, $\partial V(b)/\partial b_i$. It is an *implicit* price: no market quotes it, it is generated by scarcity inside the problem.

Theorem 9.42 (The multiplier is the shadow price). Let $\bar{b} \in \mathbb{R}^m$ and suppose that:

- (H1) f, g_i, h_j are of class C^2 in a neighborhood of the solution;
- (H2) $P(\bar{b})$ has a unique solution $x^*(\bar{b})$;
- (H3) at $x^*(\bar{b})$ LICQ holds (Definition 9.21), so that the multipliers (μ^*, λ^*) of Theorem 9.26 exist and are unique (written in the maximization convention of Section 9.1.3: $L = f - \sum_i \mu_i (g_i - b_i)$ with $\mu_i \geq 0$);
- (H4) strict complementary slackness holds ($\mu_i^* > 0$ for every active constraint) together with the second order sufficient condition (Section 9.9).

Then there is a neighborhood of $\bar{b}$ in which $x^*(\cdot)$, $\mu^*(\cdot)$ and $\lambda^*(\cdot)$ are C^1 functions, V is differentiable, and

$$\frac{\partial V}{\partial b_i}(\bar{b}) = \mu_i^*(\bar{b}), \quad i = 1, \dots, m.$$

In particular $\mu_i^* \geq 0$ recovers the obvious monotonicity (relaxing cannot hurt) and $\mu_i^* = 0$ for inactive constraints: a resource in excess has zero shadow price, which is exactly the complementary slackness of Section 9.5.2 read backwards.

Careful (common mistake)

Signs: relaxing versus tightening. The statement is written for the *maximization* problem (9.15), where raising b_i *relaxes* the constraint: that is why $\partial V / \partial b_i = +\mu_i^* \geq 0$. If you work with the standard *minimization* form (9.1) and perturb in the same way ($g_i(x) \leq b_i$), relaxing *lowers* the optimal value and the same result reads

$$\frac{\partial V_{\min}}{\partial b_i}(\bar{b}) = -\mu_i^* \leq 0,$$

as you will check numerically in Exercise 9.51(c). The rule that never fails, in either convention: $\mu_i^* \geq 0$ always measures *the improvement of the objective per unit of relaxation*, or, what is the same, the cost of *tightening* the constraint by one unit. That is why in Example 9.38 the parameter that moves is the requirement q (tightening) and we get $\partial C / \partial q = +\mu^*$.

The proof (through the implicit function theorem to differentiate the KKT system, or through the envelope theorem in its constrained version) is given in Chapter 11; there you will also see the versions that do *not* require differentiability of V (with subgradients: $\mu^* \in \partial V(b)$ in the convex case, Chapter 10) and the Milgrom–Segal version for arbitrary choice sets, which is the one used in mechanism design. Here we care about the statement, the reading and the numerical check.

Careful (common mistake)

Hypotheses (H2)–(H4) are not decorative. If the solution is not unique, V can have a *kink* at $\bar{b}$ and only directional derivatives exist; if the multiplier is not unique (LICQ fails, even if MFCQ holds), what you get is

$$\liminf_{t \downarrow 0} \frac{V(\bar{b} + te_i) - V(\bar{b})}{t} \geq \min_{\mu \in \Lambda^*} \mu_i, \quad \limsup_{t \downarrow 0} \frac{V(\bar{b} + te_i) - V(\bar{b})}{t} \leq \max_{\mu \in \Lambda^*} \mu_i,$$

where Λ^* is the set (compact under MFCQ) of multipliers: it bounds the directional derivatives on both sides, but there is no single number. And the interpretation is *marginal*: it holds for small perturbations. If you double b , $V(2b) \neq V(b) + \mu^{*\top} b$ except in special cases (for example, if V is linear in b).

9.8.2 Reading it in the consumer and in the firm

Consumer: λ^* is the marginal utility of income. In the UMP of Example 9.36 the resource is income m and the value function is indirect utility $v(p, q, m)$. Theorem 9.42 says

$$\frac{\partial v}{\partial m}(p, q, m) = \lambda^*(p, q, m).$$

One extra unit of income buys λ^* “utils”. From this follow three consequences that appear all the time in Microeconomics:

- The tangency condition $u_x/u_y = p/q$ can be rewritten as $u_x/p = u_y/q = \lambda^*$: *at the optimum, the last unit of money spent on each good yields the same utility*. That is the law of equalized marginal utility, and the common value is the multiplier.
- λ^* is the conversion unit between utility and money. That is why in Roy’s identity $x_k^* = -(\partial v / \partial p_k) / (\partial v / \partial m)$ it appears in the denominator: it converts “utility lost by raising p_k ” into “money”.
- With quasilinear utility $\lambda^* = 1$ in the interior region (Example 9.37): utility is already measured in money and there is no income effect. This is exactly why quasilinearity makes surplus analysis legitimate and dominates auction and mechanism theory.

Firm: μ^* is marginal cost (or the marginal value of the input). In Example 9.38 the “resource” is the production requirement q and we obtained $\mu^* = 2\sqrt{wr} = \partial C / \partial q$: the shadow price of requiring one more unit of output is marginal cost, the basis of the competitive supply curve $p = C'(q)$. If instead the firm maximizes profit subject to a fixed endowment of an input (installed capacity $\bar{K}$, an import quota, a license, an emission permit), the multiplier of that constraint is *how much the firm would pay for one extra unit*: its willingness to pay, the price at which it would rent capacity or buy a permit. In general equilibrium with complete markets, the shadow prices of scarce resources coincide with market prices: that is the operational content of the first welfare theorem, and the reason why a planner solving an allocation problem “discovers” competitive prices as multipliers (see Section 9.6 and Chapter 10, where the multipliers are the dual variables).

9.8.3 A full numerical example: checking the shadow price

Example 9.43 (Numerical check of $\partial V / \partial b = \lambda^*$). Consider the logarithmic UMP with prices $p = 2$, $q = 1$ and income b :

$$V(b) = \max_{x, y > 0} \ln x + \ln y \quad \text{s.t.} \quad 2x + y \leq b.$$

Step 1: solve for generic b . The objective is concave, the constraint affine and Slater holds, so KKT characterizes the global optimum (Corollary 9.33). With $L = \ln x + \ln y - \lambda(2x + y - b)$:

$$(S_x) \quad \frac{1}{x} = 2\lambda, \quad (S_y) \quad \frac{1}{y} = \lambda, \quad (\text{CS}) \quad \lambda(2x + y - b) = 0.$$

Since $1/x > 0$, necessarily $\lambda > 0$ and the budget binds. From (S_x) and (S_y) : $x = 1/(2\lambda)$, $y = 1/\lambda$. Substituting into $2x + y = b$:

$$\frac{1}{\lambda} + \frac{1}{\lambda} = b \implies \lambda^*(b) = \frac{2}{b}, \quad x^*(b) = \frac{b}{4}, \quad y^*(b) = \frac{b}{2}.$$

The optimal value is

$$V(b) = \ln \frac{b}{4} + \ln \frac{b}{2} = 2 \ln b - \ln 8.$$

Step 2: two concrete resource levels. Take $b = 10$ and $b = 11$.

b	x^*	y^*	λ^*	$V(b) = 2 \ln b - \ln 8$
10	2.5	5	0.200000	2.525729
11	2.75	5.5	0.181818	2.716349

(Check of the first row: $V(10) = \ln 2.5 + \ln 5 = 0.916291 + 1.609438$.)

Step 3: compare. The exact change in the optimal value is

$$V(11) - V(10) = 2 \ln \frac{11}{10} = 0.190620,$$

while the shadow price prediction is $\lambda^*(10) \cdot \Delta b = 0.2 \cdot 1 = 0.2$. They agree up to an error of 0.0094, which is a pure second order effect: $\Delta b = 1$ is not “marginal” for an income of 10. Note also that $\lambda^*(11) = 0.1818 < 0.1906 < 0.2 = \lambda^*(10)$: the true change is trapped between the two shadow prices, because V is concave in b and λ^* is decreasing (decreasing marginal utility of income).

Step 4: make it really marginal. With $\Delta b = 0.1$:

$$V(10.1) - V(10) = 2 \ln(1.01) = 0.0199007 \quad \text{versus} \quad \lambda^*(10) \cdot 0.1 = 0.02,$$

an error of 10^{-4} . With $\Delta b = 0.01$ the error drops to 10^{-6} . This is what “derivative” means: the shadow price is exact in the limit and an excellent approximation for small perturbations. Direct check by calculus: $V'(b) = 2/b = \lambda^*(b)$ ✓, which is Theorem 9.42 in this case.

Exam strategy

The “shadow price check” is the best self correction tool you have in an exam. When you finish a problem: (i) look at the sign of the multiplier and ask yourself whether relaxing that constraint *should* improve the objective; if your μ^* has the opposite sign to what economics dictates, you mixed conventions (Section 9.1.3); (ii) if you have the optimal value in closed form, differentiate with respect to the parameter and compare it with the multiplier: if they do not agree, there is an algebra mistake somewhere. This check costs two lines and saves entire problems.

9.8.4 Reading it in contract theory: IR and IC

In a principal agent program there are two families of constraints and their multipliers have precise economic names. We write the principal's problem as

$$\begin{array}{ll} \max_{\text{contract}} & \Pi(\text{contract}) \quad \text{s.t.} \\ & \underbrace{U(\text{contract}) \geq \bar{u}}_{(\text{IR})} \\ & \underbrace{U(\text{contract} \mid \text{desired action}) \geq U(\text{contract} \mid \text{deviation})}_{(\text{IC})}. \end{array}$$

The IR multiplier: the cost of the reservation utility. Write (IR) in standard form, $g(\text{contract}) = \bar{u} - U \leq 0$, with multiplier $\gamma \geq 0$ and Lagrangian $L = \Pi - \gamma(\bar{u} - U)$. Here the perturbed “resource” is $-\bar{u}$, so Theorem 9.42 gives

$$\frac{\partial \Pi^*}{\partial \bar{u}} = -\gamma^* \leq 0 : \quad \text{raising } \bar{u} \text{ by } \varepsilon \text{ costs the principal } \gamma^* \varepsilon.$$

(The sign is pure convention: what matters is the magnitude and the economic direction, which is always “an agent with better outside options is more expensive”.) It is literally the *price of the agent*: how much profit must be given up for each extra util that the outside labor market guarantees them. If the agent has better alternatives (higher $\bar{u}$), the principal earns less, at a rate of γ^* per util. In Example 9.39 we checked this numerically: $\gamma = 4$ and raising $\bar{v}$ from 2 to 2.1 reduced profit by ≈ 0.4 .

That (IR) *binds* ($\gamma^* > 0$) is the rule in the standard case: if the agent had utility to spare, the principal would cut the payment. When (IR) does *not* bind for some agent type (and in adverse selection models with several types this is normal for the high types), that agent gets a strict surplus above their outside option: this is the *informational rent*.

The IC multiplier: the price of asymmetric information. If δ is the multiplier of (IC), δ^* measures how much it costs the principal that the incentive constraint exists. The key comparison is:

$$\underbrace{\Pi^{FB}}_{\text{without (IC): symmetric information}} - \underbrace{\Pi^{SB}}_{\text{with (IC): asymmetric information}} = \text{agency cost},$$

and δ^* is its marginal version: if you relax (IC) by one unit (for example, because a monitoring technology makes deviation harder, or because the required effort differential falls), the principal gains δ^* . From this follow two readings that reappear in every principal agent model (MWG, ch. 14):

- *Moral hazard*: $\delta^* > 0$ forces the wage to depend on the observable outcome; the extra cost relative to the *first best* is the risk premium that must be paid to a risk averse agent for exposing them to randomness. Compare with Example 9.39, where without (IC) the optimal contract was full insurance ($w_H = w_L$): (IC) is exactly what breaks that equality. Exercise 9.56 solves it with numbers.

- *Adverse selection:* δ^* (the multiplier of the IC of the efficient type, who wants to pass as the inefficient one) is the source of the *informational rent* and of the “no distortion at the top, distortion below” result. The chain is always the same: (IC) binds $\Rightarrow$ rent must be left $\Rightarrow$ it pays to distort the low type’s quantity to make that rent cheaper.

Which constraints bind is therefore the central economic result of a contract model, and determining it is an exercise in complementary slackness. The whole chapter boils down, for a theoretical economist, to that sentence.

Intuition

The multiplier is the price of a rationing coupon. Imagine that constraint $g_i(x) \leq b_i$ is a quota: you may only use b_i tons of steel. If the quota is slack, one extra ton is worth nothing: nobody would pay for a steel coupon. If the quota is tight, each extra ton is worth something very concrete (the extra profit it would allow) and that number is μ_i^* . It is the price at which a secondary market for coupons would exist if we let one appear. Three consequences that hold for the whole course:

1. **Complementary slackness is “you do not pay for what is in excess”:** a free resource $\Rightarrow$ zero price; a positive price $\Rightarrow$ the resource is exhausted.
2. **The Lagrangian is a market problem.** In the convex case, maximizing $f(x) - \sum_i \mu_i^* (g_i(x) - b_i)$ without constraints gives the same optimum: instead of forbidding excess use of the resource, you charge for it at its shadow price and the agent rations themselves. That is the whole idea of duality (Chapter 10), of Pigouvian taxes and of emission permit markets.
3. **In contracts, the “resource” is the agent’s utility and information.** The shadow price of IR is how much the agent costs; that of IC, how much it costs not to be able to observe them. When in a contract model you read “the participation constraint of the low type and the incentive constraint of the high type are the ones that bind”, you are reading the result of a KKT case analysis like the ones in this chapter.

9.9 Second order conditions with constraints [P2]

KKT is a *first order* condition: it identifies candidates. When the problem is convex (Section 9.6) that is enough and nothing else is needed. But if the objective is not concave, or the constraints do not define a convex set, a KKT point can be a local maximum, a local minimum or a constrained saddle point, and we must look at curvature. The right question is not “how does f curve?” but “how does the *Lagrangian* curve along the directions the constraint allows?”. Two ideas, both subtle:

1. **We examine the Hessian of L , not that of f .** Curved constraints “bend” the feasible set, and that bending enters the analysis through the $\nabla^2 g_i$, $\nabla^2 h_j$ appearing in $\nabla_{xx}^2 L$. If the constraint is affine, both objects coincide; otherwise they do not.

2. **We examine it only on the critical tangent subspace (or cone).** What happens in directions leaving the feasible set does not matter: those curvatures are irrelevant because the problem does not let you move there.

9.9.1 The critical cone and the statements

We work with the standard *minimization* problem (9.1) and the Lagrangian $L(x, \mu, \lambda) = f(x) + \sum_i \mu_i g_i(x) + \sum_j \lambda_j h_j(x)$ of Section 9.5. We write

$$\nabla_{xx}^2 L(x^*, \mu^*, \lambda^*) = \nabla^2 f(x^*) + \sum_{i=1}^m \mu_i^* \nabla^2 g_i(x^*) + \sum_{j=1}^p \lambda_j^* \nabla^2 h_j(x^*).$$

Definition 9.44 (Critical cone). Let (x^*, μ^*, λ^*) be a KKT point and $A(x^*)$ the set of active indices (Definition 9.3). The *critical cone* is

$$C(x^*, \mu^*) = \left\{ d \in \mathbb{R}^n : \begin{array}{ll} \nabla h_j(x^*)^\top d = 0 & \text{for every } j, \\ \nabla g_i(x^*)^\top d = 0 & \text{for } i \in A(x^*) \text{ with } \mu_i^* > 0, \\ \nabla g_i(x^*)^\top d \leq 0 & \text{for } i \in A(x^*) \text{ with } \mu_i^* = 0 \end{array} \right\}.$$

The logic of the three lines is transparent: equalities do not let you move off their level surface; an inequality that binds *with a positive price* ($\mu_i^* > 0$) is, locally, an equality (moving inward would change the value to first order, so those directions are already discarded by the first order conditions); and a *degenerate* active constraint ($\mu_i^* = 0$) leaves a half space open. Note that $C(x^*, \mu^*)$ is exactly the set of directions of the linearized cone (Definition 9.27) on which the directional derivative of the objective vanishes: they are the “neutral” directions to first order, the only ones where curvature decides.

Theorem 9.45 (Second order necessary condition). *Let x^* be a local minimum of (9.1) with f, g_i, h_j of class $\mathcal{C}^2$, and suppose LICQ holds at x^* (Definition 9.21), with multipliers (μ^*, λ^*) . Then*

$$d^\top \nabla_{xx}^2 L(x^*, \mu^*, \lambda^*) d \geq 0 \quad \text{for every } d \in C(x^*, \mu^*).$$

(Positive semidefinite on the critical cone.)

Theorem 9.46 (Second order sufficient condition). *Let (x^*, μ^*, λ^*) be a point satisfying the KKT conditions (9.9)–(9.12) with f, g_i, h_j of class $\mathcal{C}^2$. If*

$$d^\top \nabla_{xx}^2 L(x^*, \mu^*, \lambda^*) d > 0 \quad \text{for every } d \in C(x^*, \mu^*), \ d \neq 0,$$

then x^ is a strict local minimum of (9.1). For maxima, everything reverses: one requires $d^\top \nabla_{xx}^2 L d < 0$ on the critical cone (negative definite) and gets a strict local maximum.*

Comment on the proof. The idea is a second order Taylor expansion along feasible curves $x(t)$ with $x(0) = x^*$, $\dot{x}(0) = d$: feasibility forces $h_j(x(t)) = 0$, and differentiating that identity twice produces precisely the term $\lambda_j^* \nabla^2 h_j$ that turns the Hessian of f into that of the Lagrangian. The full argument (with the use of LICQ to build the curves in the necessary case) is in [5, ch.

12, theorems 12.5 and 12.6] and in [6, ch. 3]; the version in economic notation, with a bordered Hessian, is in [9, appendix M.D–M.E]. $\square$

Careful (common mistake)

Four classic mistakes with second order conditions:

- **Using $\nabla^2 f$ instead of $\nabla_{xx}^2 L$.** This is valid only if all constraints are affine. With curved constraints the result can change sign.
- **Evaluating on all of $\mathbb{R}^n$ instead of the critical cone.** This is a “conservative” mistake that discards legitimate optima: there are problems with $\nabla_{xx}^2 L$ indefinite on $\mathbb{R}^n$ and positive definite on the tangent subspace, precisely Exercise 9.53.
- **Confusing necessary with sufficient.** Necessary: ≥ 0 . Sufficient: > 0 for $d \neq 0$. In the borderline case (≥ 0 with some d where it equals 0) the test does not decide, just like x^3 versus x^4 in the unconstrained case (Chapter 8).
- **Forgetting that sufficiency gives *local* optimality.** For global optimality you need convexity (Theorem 9.32), Arrow–Enthoven (Theorem 9.34) or an exhaustive comparison of candidates with guaranteed existence.

9.9.2 The bordered Hessian criterion (equality case)

In practice (and in micro exams) the test is implemented with determinants. Consider the problem with *only* equality constraints

$$\max_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad h_j(x) = 0, \quad j = 1, \dots, m, \quad m < n,$$

(in this subsection, and only here, m counts the *equality* constraints, as is traditional in the bordered Hessian literature), let x^* be a Lagrange point (Theorem 9.16) with multipliers λ^* in the convention $L = f + \sum_j \lambda_j h_j$ (if you write the “economist” Lagrangian with a minus sign, replace λ_j by $-\lambda_j$ when forming H ; with affine constraints it makes no difference, since $\nabla^2 h_j = 0$), and let

$$A = [\nabla h_j(x^*)^\top]_{j=1, \dots, m} \in \mathbb{R}^{m \times n} \quad (\text{the Jacobian of the constraints}),$$

$$H = \nabla_{xx}^2 L(x^*, \lambda^*) \in \mathbb{R}^{n \times n}.$$

Definition 9.47 (Bordered Hessian). The *bordered Hessian* is the symmetric matrix of order $n + m$

$$\bar{H} = \begin{pmatrix} 0_{m \times m} & A \\ A^\top & H \end{pmatrix}.$$

We write $\bar{H}_k$ for its leading principal submatrix of order k (the first k rows and columns) and $D_k = \det \bar{H}_k$ for the corresponding leading principal minor.

Theorem 9.48 (Sign rule for the bordered Hessian). *Suppose $\text{rank } A = m$ (LICQ). Only the last $n - m$ leading principal minors matter, that is, $D_{2m+1}, D_{2m+2}, \dots, D_{n+m}$. Then:*

- **Strict local maximum** (H negative definite on $\{d : Ad = 0\}$) if those $n - m$ minors alternate in sign, starting with the sign of $(-1)^{m+1}$ in D_{2m+1} ; equivalently, if

$$\text{sign}(D_{2m+k}) = (-1)^{m+k}, \quad k = 1, \dots, n - m,$$

so that the last one, D_{n+m} , has the sign of $(-1)^n$.

- **Strict local minimum** (H positive definite on $\{d : Ad = 0\}$) if those $n - m$ minors all have the same sign, that of $(-1)^m$.
- If some minor has the “wrong” sign (and is not zero), the point is not a local optimum; if some minor vanishes, the criterion does not decide.

Remark 9.49 (Cases worth memorizing). • $n = 2, m = 1$ (the exam case par excellence: two goods, one budget). There is $n - m = 1$ relevant minor, the full determinant $D_3 = \det \bar{H}$, of order 3. Maximum: $D_3 > 0$ (sign of $(-1)^n = (-1)^2$). Minimum: $D_3 < 0$ (sign of $(-1)^m = -1$).

- $n = 3, m = 1$. Two minors, D_3 and D_4 . Maximum: $D_3 > 0, D_4 < 0$ (they alternate). Minimum: $D_3 < 0, D_4 < 0$ (both with the sign of $(-1)^1$).
- $m = 0$ (no constraints): the rule reduces to the familiar one of Chapter 8: all minors positive for a minimum, alternating starting from negative for a maximum.
- **With inequalities:** the criterion is applied treating the *active constraints with positive multiplier* as if they were equalities (and if there are degenerate active ones with $\mu_i^* = 0$, the bordered Hessian is no longer enough: you must go back to the critical cone of Definition 9.44).

Exam strategy

In an exam, for $n = 2, m = 1$ it is almost always faster *not* to use determinants: parametrize the tangent subspace. With the constraint $a_1 d_1 + a_2 d_2 = 0$ take $d = (a_2, -a_1)$ (a single vector spans the tangent line) and compute the scalar

$$Q = d^\top H d = a_2^2 H_{11} - 2a_1 a_2 H_{12} + a_1^2 H_{22}.$$

$Q < 0 \Rightarrow$ strict local maximum; $Q > 0 \Rightarrow$ strict local minimum. It is one line of arithmetic, there are no sign rules to remember, and in fact $\det \bar{H} = -Q$ in this case, so both routes are the same computation. Use the bordered Hessian when n is large or when the problem explicitly asks for it.

9.9.3 A solved 2×2 example

Example 9.50 (Checking that the tangency point is a maximum). Solve and verify with second order conditions:

$$\max_{x,y>0} u(x,y) = x^{1/2} y^{1/2} \quad \text{s.t.} \quad 2x + y = 12.$$

Step 1: Lagrange point. With $L = x^{1/2}y^{1/2} - \lambda(2x + y - 12)$:

$$(S_x) \quad \frac{1}{2}x^{-1/2}y^{1/2} = 2\lambda, \quad (S_y) \quad \frac{1}{2}x^{1/2}y^{-1/2} = \lambda.$$

Dividing (both quantities are positive in the positive orthant, so the division is legitimate):

$$\frac{y}{x} = 2 \implies y = 2x.$$

Substituting into the constraint: $2x + 2x = 12$, hence

$$x^* = 3, \quad y^* = 6, \quad \lambda^* = \frac{1}{2}\sqrt{\frac{x^*}{y^*}} = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{4} \approx 0.35355, \quad u(x^*, y^*) = \sqrt{18} = 3\sqrt{2}.$$

Step 2: the Hessian of the Lagrangian. The constraint is affine, so $\nabla^2 h = 0$ and $\nabla_{xx}^2 L = \nabla^2 u$. Differentiating:

$$u_{xx} = -\frac{1}{4}x^{-3/2}y^{1/2}, \quad u_{xy} = \frac{1}{4}x^{-1/2}y^{-1/2}, \quad u_{yy} = -\frac{1}{4}x^{1/2}y^{-3/2}.$$

At $(3, 6)$, writing $k = \sqrt{2}/48 \approx 0.0294628$ to avoid dragging radicals:

$$u_{xx} = -\frac{\sqrt{2}}{12} = -4k, \quad u_{xy} = \frac{\sqrt{2}}{24} = 2k, \quad u_{yy} = -\frac{\sqrt{2}}{48} = -k, \quad H = \begin{pmatrix} -4k & 2k \\ 2k & -k \end{pmatrix}.$$

Note that $\det H = 4k^2 - 4k^2 = 0$: *the Hessian is singular* (negative semidefinite, but not definite). The “unconstrained” test would conclude nothing. This is exactly the kind of situation where you must restrict to the tangent space.

Step 3a: through the tangent subspace (recommended). The tangent space to $2x + y = 12$ is $\{d : 2d_1 + d_2 = 0\}$, spanned by $d = (1, -2)$. Then

$$d^\top H d = (-4k)(1)^2 + 2(2k)(1)(-2) + (-k)(-2)^2 = -4k - 8k - 4k = -16k = -\frac{\sqrt{2}}{3} \approx -0.4714 < 0.$$

Since the critical cone is exactly this line (there are no inequalities) and the quadratic form is strictly negative on it, Theorem 9.46 gives: $(3, 6)$ is a **strict local maximum**.

Step 3b: through the bordered Hessian (check). Here $n = 2$, $m = 1$, $A = (2, 1)$ and

$$\bar{H} = \begin{pmatrix} 0 & 2 & 1 \\ 2 & -4k & 2k \\ 1 & 2k & -k \end{pmatrix},$$

so that

$$\begin{aligned} D_3 = \det \bar{H} &= -2[(2)(-k) - (2k)(1)] + [(2)(2k) - (-4k)(1)] \\ &= -2(-4k) + 8k = 16k > 0. \end{aligned}$$

There is only $n - m = 1$ relevant minor (D_3), and for a *maximum* it must have the sign of $(-1)^n = (-1)^2 = +1$: it does $\checkmark$. Note the announced relation: $D_3 = 16k = -d^\top H d$, the two methods are the same computation.

Step 4: from local to global. $u = x^{1/2}y^{1/2}$ is concave on $\mathbb{R}_{++}^2$ (Cobb–Douglas with

$\alpha + \beta = 1$) and the constraint is affine: the problem is convex and the local maximum is global (Theorem 9.32). In this example, then, the second order condition was *redundant*, and that is exactly the moral: **if you can invoke convexity, do it, and save yourself the determinants**. Second order conditions earn their keep when the problem is *not* convex; Exercise 9.53 shows such a case, with $\nabla^2 f$ indefinite and yet a strict minimum on the constraint.

Intuition

Think of a mountain (f) and a trail drawn on it (the constraint). The first order conditions say that at the optimum the trail is *flat*: it neither rises nor falls locally. The second order ones ask whether that flat spot is the top of a ridge or the bottom of a dip *along the trail*. That the mountain rises in directions off the trail is irrelevant (you cannot go there), and that is the reason for restricting to the critical cone. The term $\sum_j \lambda_j \nabla^2 h_j$ that turns $\nabla^2 f$ into $\nabla_{xx}^2 L$ is the curvature of the *trail* itself: if the path bends, your altitude changes even if the mountain does not, and that must be netted out.

Summary

Chapter summary

Standard problem. $\min f(x)$ s.t. $g_i(x) \leq 0$ ($i \leq m$), $h_j(x) = 0$ ($j \leq p$); Lagrangian $L = f + \sum_i \mu_i g_i + \sum_j \lambda_j h_j$. Every statement depends on the sign convention: in the max form with $g_i \leq 0$ one uses $L = f - \sum_i \mu_i g_i$ and the $\mu_i \geq 0$ keep their shadow price reading (Section 9.1.3). *Before writing anything, state your convention.* Constraint active at x^* : $g_i(x^*) = 0$.

1. Lagrange theorem (equalities only) (Theorem 9.16). *Exact hypotheses:* x^* a local optimum; $f, h_j \in \mathcal{C}^1$ in a neighborhood of x^* ; **the gradients** $\nabla h_1(x^*), \dots, \nabla h_p(x^*)$ **are linearly independent** (x^* is a regular point). *Conclusion:* there is a **unique** $\lambda^* \in \mathbb{R}^p$ with $\nabla f(x^*) + \sum_j \lambda_j^* \nabla h_j(x^*) = 0$. The λ_j^* have no sign restriction. Geometry: $\nabla f(x^*)$ belongs to the subspace spanned by the constraint gradients; equivalently, the level curves are tangent.

2. KKT, necessity (Theorem 9.26). *Exact hypotheses:* x^* a local minimum; $f, g_i, h_j \in \mathcal{C}^1$ near x^* ; **and some constraint qualification** (CQ). *Conclusion:* there are $\mu^* \in \mathbb{R}^m$, $\lambda^* \in \mathbb{R}^p$ with

- (i) **Stationarity:** $\nabla f(x^*) + \sum_i \mu_i^* \nabla g_i(x^*) + \sum_j \lambda_j^* \nabla h_j(x^*) = 0$;
- (ii) **Primal feasibility:** $g_i(x^*) \leq 0$, $h_j(x^*) = 0$;
- (iii) **Dual feasibility (nonnegativity):** $\mu_i^* \geq 0$ (the λ_j^* free);
- (iv) **Complementary slackness:** $\mu_i^* g_i(x^*) = 0$ for every i : either the constraint binds, or its price is zero, or both (degenerate case).

3. Constraint qualifications (Section 9.4). None of them is about the optimum: they are all about the *geometry of the feasible set* at x^* .

- **LICQ:** $\{\nabla g_i(x^*)\}_{i \in A(x^*)} \cup \{\nabla h_j(x^*)\}_j$ linearly independent. It is the strongest of the usual ones: it implies MFCQ and gives *unique* multipliers.
- **Slater** (convex problems only: f, g_i convex, h_j affine): there is $\hat{x}$ with $g_i(\hat{x}) < 0$ for every i (*strictly* feasible) and $h_j(\hat{x}) = 0$. It is a *global* condition and easy to check; it requires nothing about gradients.
- **MFCQ:** $\{\nabla h_j(x^*)\}$ linearly independent and there is d with $\nabla h_j(x^*)^\top d = 0$ and $\nabla g_i(x^*)^\top d < 0$ for every active constraint (there is a strictly entering direction). It is equivalent to the multiplier set being nonempty and *bounded*.
- **Affine constraints:** if all g_i and h_j are affine, KKT is necessary *with no further hypotheses* (Abadie CQ holds automatically). This covers budgets, linear resources and linear programming.
- Hierarchy: LICQ $\Rightarrow$ MFCQ $\Rightarrow$ Abadie $\Rightarrow$ KKT necessary (Theorem 9.24).

4. The cusp counterexample (Example 9.19). It serves to recall *why* CQs are not fine print: at a point where the feasible set has a cusp, the tangent cone is strictly smaller than the linearized cone, the active gradients are linearly dependent (LICQ fails and MFCQ fails) and the optimum **admits no multipliers**: KKT is false there. Exam moral: KKT necessary = “optimum + CQ”; without a CQ you can lose the optimum. And CQs do not depend on f : only on what the feasible set looks like.

5. KKT, sufficiency (Theorem 9.32 and Corollary 9.33). *Exact hypotheses (convex problem):* f **convex**, each g_i **convex**, each h_j **affine** (affine, not “convex”), all differentiable. *Conclusion:* if (x^*, μ^*, λ^*) satisfies the four KKT conditions, then x^* is a **global minimum**. No CQ is needed for sufficiency. With Slater you also get the full characterization: x^* optimal $\iff x^*$ is a KKT point. In max form: f concave, g_i convex (X convex), h_j affine. *Arrow–Enthoven variant* (Theorem 9.34): it is enough for f to be **quasiconcave** (instead of concave) and the g_i quasiconvex, plus one extra technical condition (for example $\nabla f(x^*) \neq 0$). This is what saves Cobb–Douglas with increasing returns ($\alpha + \beta > 1$), which is quasiconcave but not concave. Alternative trick: maximize $\ln u$, an increasing monotone transformation of u .

6. Second order conditions (Section 9.9). One examines $\nabla_{xx}^2 L$ (not $\nabla^2 f$) on the *critical cone* $C(x^*, \mu^*)$: $\nabla h_j^\top d = 0$; $\nabla g_i^\top d = 0$ if $\mu_i^* > 0$; $\nabla g_i^\top d \leq 0$ if i is active with $\mu_i^* = 0$. *Necessary* (with LICQ): $d^\top \nabla_{xx}^2 L d \geq 0$ on C , for a minimum. *Sufficient* (no CQ): $d^\top \nabla_{xx}^2 L d > 0$ for every $d \in C$, $d \neq 0 \Rightarrow$ *strict* local minimum. Maxima: the other way round. *Bordered Hessian* (n variables, m equalities, $m < n$): with $\bar{H} = \begin{pmatrix} 0 & A \\ A^\top & H \end{pmatrix}$ of order $n + m$, what matters are the last $n - m$ leading principal minors $D_{2m+1}, \dots, D_{n+m}$: for a **maximum** they must alternate with $\text{sign}(D_{2m+k}) = (-1)^{m+k}$ (the last with the sign of $(-1)^n$); for a **minimum** all with the sign of $(-1)^m$. Case $n = 2, m = 1$: maximum if $\det \bar{H} > 0$, minimum if $\det \bar{H} < 0$.

7. Shadow price (Theorem 9.42). With $V(b)$ the optimal value of $\max f$ s.t. $g_i(x) \leq b_i$, and under (i) $\mathcal{C}^2$, (ii) a unique solution, (iii) LICQ, (iv) strict complementary slackness +

the second order sufficient condition: $\partial V/\partial b_i = \mu_i^*$. Readings: λ^* of the UMP = marginal utility of income ($\partial v/\partial m$); μ^* of the production constraint = marginal cost; the multiplier of (IR) = cost to the principal of the agent's reservation utility; the multiplier of (IC) = marginal cost of asymmetric information (informational rent, risk premium). The general proof is in Chapter 11; the nondifferentiable version with subgradients, in Chapter 10.

8. Exam protocol (Section 9.7): normalize $\rightarrow$ justify existence and CQ or sufficiency $\rightarrow$ write the four full conditions $\rightarrow$ prune cases with economic arguments $\rightarrow$ enumerate and solve the remaining $2^{\# \text{inequalities}}$ cases $\rightarrow$ discard by *showing* the violation (primal feasibility or multiplier sign) $\rightarrow$ conclude and interpret the multipliers.

Main references: [1, chs. 3–4], [6, ch. 3], [2, ch. 5], [5, ch. 12], [9, appendix M], [10, ch. A2].

9.10 Solved exercises [P1]

The following six exercises cover, between them, the full repertoire of what can be asked: exhaustive case analysis, corner solutions with a threshold, second order conditions, failure of the constraint qualification, shadow prices checked numerically and a contract program. All solutions are complete: there is no “left to the reader”. Try each one on paper before reading the solution; the value is in the case analysis, not in the final answer.

Exercise 9.51 (Two inequalities: enumerating the four cases). Consider the problem of projecting the point $(3, 4)$ onto a feasible set defined by two inequalities:

$$\min_{x,y} f(x,y) = (x-3)^2 + (y-4)^2 \quad \text{s.t.} \quad x+y \leq 2, \quad x \geq 0.$$

(a) Write the full KKT conditions. (b) Enumerate the four activity cases, solve each one and discard by *showing* the corresponding violation. (c) Justify that the survivor is the global minimum. (d) Redo the problem changing the target point to $(0, 2)$ and explain what is special about the solution.

Solution. (a) **Setup.** Standard minimization form: $g_1(x,y) = x+y-2 \leq 0$ and $g_2(x,y) = -x \leq 0$, with constant gradients $\nabla g_1 = (1, 1)$, $\nabla g_2 = (-1, 0)$ and $\nabla f = (2(x-3), 2(y-4))$. With $L = f + \mu_1 g_1 + \mu_2 g_2$:

$$(S_x) \quad 2(x-3) + \mu_1 - \mu_2 = 0,$$

$$(S_y) \quad 2(y-4) + \mu_1 = 0,$$

$$(PF) \quad x+y \leq 2, \quad x \geq 0,$$

$$(DF) \quad \mu_1 \geq 0, \quad \mu_2 \geq 0,$$

$$(CS) \quad \mu_1(x+y-2) = 0, \quad \mu_2 x = 0.$$

Before computing: both constraints are *affine*, so KKT is necessary with no additional hypothesis, and f is convex (a sum of squares; its Hessian is $2I \succ 0$), so KKT will also be sufficient

(Theorem 9.32). With $m = 2$ inequalities there are $2^2 = 4$ cases.

(b) The four cases.

Case 1: none active ($\mu_1 = \mu_2 = 0$). The (S) equations give the unconstrained minimum $x = 3, y = 4$. Primal feasibility check: $x + y = 7 > 2$. **Discarded** for violating $g_1 \leq 0$.

Case 2: g_1 active, g_2 inactive ($x + y = 2, \mu_2 = 0, x > 0$). From (S_x) and (S_y) with $\mu_2 = 0$:

$$\mu_1 = -2(x - 3) = -2(y - 4) \implies x - 3 = y - 4 \implies y = x + 1.$$

With $x + y = 2$: $2x + 1 = 2$, that is,

$$x^* = \frac{1}{2}, \quad y^* = \frac{3}{2}, \quad \mu_1 = -2\left(\frac{1}{2} - 3\right) = 5.$$

Check: $\mu_1 = 5 \geq 0 \checkmark$; $\mu_2 = 0 \geq 0 \checkmark$; $x^* = 1/2 > 0$, so g_2 is indeed inactive and (CS) holds $\checkmark$; $x^* + y^* = 2 \checkmark$. **It survives.**

Case 3: g_1 inactive, g_2 active ($x = 0, \mu_1 = 0, x + y < 2$). From (S_y) with $\mu_1 = 0$: $y = 4$. But then $x + y = 4 > 2$. **Discarded** by primal feasibility (it also contradicts the assumption $x + y < 2$).

Case 4: both active ($x = 0$ and $x + y = 2$, hence $y = 2$). From (S_y) : $\mu_1 = -2(2 - 4) = 4$. From (S_x) : $2(0 - 3) + 4 - \mu_2 = 0$, that is,

$$\mu_2 = -6 + 4 = -2 < 0.$$

Discarded by dual feasibility ($\mu_2 \geq 0$ fails).

(c) Conclusion. The only KKT point is $(x^*, y^*) = (1/2, 3/2)$ with $(\mu_1^*, \mu_2^*) = (5, 0)$, and the optimal value is

$$f(x^*, y^*) = \left(\frac{1}{2} - 3\right)^2 + \left(\frac{3}{2} - 4\right)^2 = \frac{25}{4} + \frac{25}{4} = 12.5, \quad \text{distance} = \sqrt{12.5} \approx 3.5355.$$

Since the problem is convex with affine constraints, KKT is necessary and sufficient: $(1/2, 3/2)$ is the global minimum (and it is unique because f is strictly convex). Geometric check: the vector $(3, 4) - (1/2, 3/2) = (5/2, 5/2)$ is orthogonal to the line $x + y = 2$, that is, parallel to $\nabla g_1 = (1, 1)$, exactly the characterization of the projection onto a hyperplane from Chapter 8. Moreover $\mu_1^* = 5$ is the shadow price: if the resource went from 2 to $2 + \varepsilon$ (constraint $x + y \leq 2 + \varepsilon$), the *squared* distance would fall at a rate of 5 per unit. Check it: with resource b , the optimal point is $(\frac{b-1}{2}, \frac{b+1}{2})$ and $V(b) = 2\left(\frac{7-b}{2}\right)^2 = \frac{(7-b)^2}{2}$, hence $V'(2) = -(7-2) = -5 = -\mu_1^*$ (negative because we *minimize*: relaxing lowers the optimal value, and μ^* measures the improvement) $\checkmark$.

(d) The degenerate case. With target point $(0, 2)$, the problem is

$$\min x^2 + (y - 2)^2 \quad \text{s.t.} \quad x + y \leq 2, \quad x \geq 0.$$

The point $(0, 2)$ is *already feasible* ($0 + 2 = 2 \leq 2$ and $x = 0 \geq 0$), so the solution is the point itself, with value 0. Its multipliers: $\nabla f(0, 2) = (0, 0)$, so (S_x) and (S_y) are satisfied with

$$\mu_1^* = \mu_2^* = 0, \quad \text{but both constraints are active.}$$

This is the **degenerate case** announced in Section 9.5.2: complementary slackness holds ($\mu_i g_i = 0$) because *both* factors vanish. The constraints touch the optimum but exert no pressure: relaxing either of them improves nothing, because the unconstrained minimum sits right at the corner. Two fine and very examinable observations:

- Here no constraint qualification fails: $\nabla g_1 = (1, 1)$ and $\nabla g_2 = (-1, 0)$ are linearly independent, so LICQ holds and the multiplier is unique (the zero one). What fails is *strict complementary slackness*, hypothesis (H4) of Theorem 9.42.
- Consequence: the value function $V(b)$ (with resource b instead of 2) has a *kink* at $b = 2$: $V(b) = 0$ for $b \geq 2$ and $V(b) = (2 - b)^2$ for $b < 2$ (with $b < 2$ the minimum is attained at $x = 0, y = b$, because the projection onto the line $x + y = b$ would have $x < 0$). It happens to be differentiable here ($V'(2^-) = 0 = V'(2^+)$), but in general degeneracy is the warning sign that the derivative of the optimal value may not exist and that only directional derivatives are available.

□

Exercise 9.52 (Quasilinear: exact threshold between corner and interior). A consumer has utility $u(x, y) = 2\sqrt{x} + y$, price $p > 0$ for x and 1 for y , and income $m > 0$:

$$\max_{x, y} 2\sqrt{x} + y \quad \text{s.t.} \quad px + y \leq m, \quad x \geq 0, \quad y \geq 0.$$

(a) Solve by KKT distinguishing the interior and corner regimes, and find the **exact income threshold** separating them. (b) Write the Marshallian demands and the indirect utility in each regime, and check that $\partial v / \partial m = \lambda^*$. (c) Illustrate with $p = 1/4$ and incomes $m = 1$ and $m = 9$.

Solution. Preliminaries (P1–P2). Inequalities: $g_0 = px + y - m \leq 0$, $g_1 = -x \leq 0$, $g_2 = -y \leq 0$; all affine, so KKT is necessary with no additional CQ. u is concave ($2\sqrt{x}$ is concave, y is linear), so KKT is also sufficient. Existence: the budget set is compact and u is continuous.

Immediate pruning: $u_x = 1/\sqrt{x} \rightarrow +\infty$ as $x \downarrow 0$ (Inada condition), so the corner $x = 0$ is never optimal: starting from $x = 0$, moving ε of spending from y to x raises utility by $2\sqrt{\varepsilon/p} - \varepsilon$, which is positive for small ε . Hence $x^* > 0$ and $\mu_1 = 0$.

P3 (conditions). With $L = 2\sqrt{x} + y - \lambda(px + y - m) + \mu_2 y$:

$$\begin{aligned} (S_x) \quad \frac{1}{\sqrt{x}} - \lambda p &= 0 & \implies \quad \lambda &= \frac{1}{p\sqrt{x}} > 0, \\ (S_y) \quad 1 - \lambda + \mu_2 &= 0 & \implies \quad \mu_2 &= \lambda - 1, \\ (CS) \quad \lambda(px + y - m) &= 0, & \mu_2 y &= 0. \end{aligned}$$

Since (S_x) forces $\lambda > 0$, complementary slackness implies that **the budget always binds**: $px + y = m$. Two cases remain, according to $y \geq 0$.

Case A (interior, $y > 0$). By (CS), $\mu_2 = 0$; by (S_y) , $\lambda = 1$; by (S_x) ,

$$1 = \frac{1}{p\sqrt{x}} \implies \sqrt{x} = \frac{1}{p} \implies x^* = \frac{1}{p^2},$$

and the budget gives $y^* = m - px^* = m - \frac{1}{p}$. *Check:* the case requires $y^* > 0$, that is,

$$\boxed{m > \frac{1}{p}} \quad (\text{interior regime}).$$

All other signs hold: $\lambda^* = 1 > 0$, $\mu_2^* = 0$, $x^* > 0$ ✓.

Case B (corner, $y = 0$). The binding budget gives $x^* = m/p$. Then

$$\lambda^* = \frac{1}{p\sqrt{m/p}} = \frac{1}{\sqrt{pm}}, \quad \mu_2^* = \lambda^* - 1 = \frac{1}{\sqrt{pm}} - 1.$$

Check: the case requires $\mu_2^* \geq 0$, that is, $\sqrt{pm} \leq 1$, in other words,

$$\boxed{m \leq \frac{1}{p}} \quad (\text{corner regime}).$$

Threshold. The two regimes split the parameter space exactly and the threshold is $\hat{m} = 1/p$. At $m = \hat{m}$ both coincide: $x^* = 1/p^2 = m/p$, $y^* = 0$, $\lambda^* = 1$, $\mu_2^* = 0$ (degenerate case: active with zero price, as in Exercise 9.51(d)).

(b) Demands and indirect utility.

$$x^*(p, m) = \begin{cases} \frac{m}{p}, & m \leq 1/p, \\ \frac{1}{p^2}, & m > 1/p, \end{cases} \quad y^*(p, m) = \begin{cases} 0, & m \leq 1/p, \\ m - \frac{1}{p}, & m > 1/p, \end{cases}$$

$$v(p, m) = \begin{cases} 2\sqrt{m/p}, & m \leq 1/p, \\ m + \frac{1}{p}, & m > 1/p. \end{cases}$$

Shadow price check in each branch:

- Interior regime: $\partial v / \partial m = 1 = \lambda^*$ ✓. The marginal utility of income is *constant*: the trademark of quasilinearity, and the reason there are no income effects on x (all extra income goes to the numeraire y ; the Engel curve of x is vertical).
- Corner regime: $\frac{\partial}{\partial m} 2\sqrt{m/p} = \frac{1}{\sqrt{pm}} = \lambda^*$ ✓. Here there is an income effect, and $\lambda^* > 1$: when the consumer is too poor to consume the numeraire, income is worth more than one “util” per unit of money.

Note also that v is continuous and C^1 at the threshold ($v = 2/p$ and $\partial v / \partial m = 1$ on both sides): the value function joins smoothly even though the regime changes.

(c) Numbers with $p = 1/4$ (threshold $\hat{m} = 1/p = 4$).

m	regime	x^*	y^*	λ^*	μ_2^*
1	corner ($m < 4$)	$m/p = 4$	0	$1/\sqrt{pm} = 2$	1
9	interior ($m > 4$)	$1/p^2 = 16$	$9 - 4 = 5$	1	0

Checks: with $m = 1$, $u = 2\sqrt{4} + 0 = 4 = 2\sqrt{m/p} = 2\sqrt{4}$ ✓; with $m = 9$, $u = 2\sqrt{16} + 5 = 13 =$

$m + 1/p = 9 + 4 \checkmark$; and spending is $\frac{1}{4} \cdot 16 + 5 = 9 = m \checkmark$. Economic reading of $\mu_2^* = 1$ in the poor regime: the consumer would like a negative y (“borrowing” in the numeraire to buy more x); the constraint $y \geq 0$ prevents this and its shadow price is worth 1 util per unit of y they were allowed to sell. $\square$

Exercise 9.53 (Quadratic with one equality and one inequality, with second order conditions). Consider

$$\min_{x,y} f(x,y) = x^2 - 4xy + y^2 \quad \text{s.t.} \quad x + y = 2, \quad x \geq 0.$$

(a) Check that f is *not* convex. (b) Solve by KKT with an exhaustive case analysis. (c) Check the second order conditions, first on the tangent subspace and then with the bordered Hessian. (d) Is the point a global minimum of the constrained problem?

Solution. (a) f is **not convex**. Its Hessian is constant,

$$\nabla^2 f = \begin{pmatrix} 2 & -4 \\ -4 & 2 \end{pmatrix},$$

with eigenvalues $2 - 4 = -2$ and $2 + 4 = 6$ (for a matrix $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ the eigenvalues are $a \pm b$, with eigenvectors $(1, 1)$ and $(1, -1)$). There is a negative eigenvalue: $\nabla^2 f$ is *indefinite* and f is not convex (in fact the origin is a saddle point of f). Therefore we *cannot* invoke Theorem 9.32: KKT will give candidates and we will have to check with second order conditions.

(b) **KKT.** Standard form: $h(x, y) = x + y - 2 = 0$ and $g(x, y) = -x \leq 0$, with $\nabla h = (1, 1)$, $\nabla g = (-1, 0)$ and $\nabla f = (2x - 4y, -4x + 2y)$. Since both constraints are affine, KKT is necessary with no additional CQ (and, in passing: LICQ holds at every feasible point, because $(1, 1)$ and $(-1, 0)$ are linearly independent). With $L = f + \lambda h + \mu g$:

$$(S_x) \quad 2x - 4y + \lambda - \mu = 0,$$

$$(S_y) \quad -4x + 2y + \lambda = 0,$$

$$(PF) \quad x + y = 2, \quad x \geq 0,$$

$$(DF) \quad \mu \geq 0, \quad \lambda \in \mathbb{R} \text{ free},$$

$$(CS) \quad \mu x = 0.$$

A single inequality, hence two cases.

Case 1: g inactive ($x > 0, \mu = 0$). Subtracting (S_y) from (S_x) :

$$(2x - 4y) - (-4x + 2y) = 6x - 6y = 0 \implies x = y.$$

With $x + y = 2$: $x = y = 1$. Then $\lambda = 4x - 2y = 4 - 2 = 2$. *Check:* $x = 1 > 0 \checkmark$ (consistent with $\mu = 0$ and with (CS)), $\mu = 0 \geq 0 \checkmark$, equality constraint satisfied $\checkmark$. **It survives:** $(x^*, y^*) = (1, 1)$, $\lambda^* = 2$, $\mu^* = 0$, with $f(1, 1) = 1 - 4 + 1 = -2$.

Case 2: g active ($x = 0$). The equality gives $y = 2$. From (S_y) : $\lambda = 4x - 2y = -4$. From (S_x) : $\mu = 2x - 4y + \lambda = -8 - 4 = -12 < 0$. **Discarded** by dual feasibility.

(c) **Second order.** Since all constraints are affine, $\nabla_{xx}^2 L = \nabla^2 f$ (the terms $\lambda \nabla^2 h$ and $\mu \nabla^2 g$ vanish). The *critical cone* at $(1, 1)$: the equality imposes $\nabla h^\top d = d_1 + d_2 = 0$; the inequality is

inactive, so it imposes nothing. Hence

$$C = \{d : d_1 + d_2 = 0\} = \text{span}\{(1, -1)\},$$

a line. We evaluate the quadratic form at $d = (1, -1)$:

$$d^\top \nabla^2 f d = 2(1)^2 + 2(-4)(1)(-1) + 2(-1)^2 = 2 + 8 + 2 = 12 > 0.$$

Strictly positive on $C \setminus \{0\}$: by Theorem 9.46, $(1, 1)$ is a **strict local minimum**. Note the essential point: on $\mathbb{R}^2$ the Hessian is indefinite (in the direction $(1, 1)$ it gives $2 + (-8) + 2 = -4 < 0$), but that direction *leaves* the feasible set, so it is irrelevant. Restricted to the feasible line, the curvature is positive.

With the bordered Hessian. Here $n = 2$, $m = 1$ (only the equality borders; the inactive inequality does not take part), $A = (1, 1)$ and

$$\bar{H} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 2 & -4 \\ 1 & -4 & 2 \end{pmatrix}, \quad D_3 = \det \bar{H} = -1[(1)(2) - (-4)(1)] + [(1)(-4) - (2)(1)] = -6 - 6 = -12.$$

There is $n - m = 1$ relevant minor and, for a *minimum*, it must have the sign of $(-1)^m = -1$: $D_3 = -12 < 0$ ✓. This agrees with the previous computation, as Section 9.9 anticipated: $D_3 = -d^\top \nabla^2 f d = -12$.

(d) Global. Yes. Parametrize the constraint with $y = 2 - x$:

$$\varphi(x) = f(x, 2 - x) = x^2 - 4x(2 - x) + (2 - x)^2 = 6x^2 - 12x + 4,$$

a *convex* parabola ($\varphi'' = 12 > 0$) with vertex at $\varphi'(x) = 12x - 12 = 0$, that is, $x = 1$, and value $\varphi(1) = -2$. Since the feasible set in this parametrization is $\{x \geq 0\}$ and the vertex falls inside, $(1, 1)$ is the *global* minimum of the constrained problem, with $f = -2$. Moral: a non convex function can be convex *on the feasible set*; that is exactly what constrained second order conditions measure, and that is why they are the right tool when the convex sufficiency theorem does not apply. $\square$

Exercise 9.54 (When KKT fails: violation of the constraint qualification). Consider

$$\max_{x,y} x \quad \text{s.t.} \quad y \leq (1 - x)^3, \quad x \geq 0, \quad y \geq 0.$$

(a) Find the optimum by direct reasoning. (b) Show that at that optimum **no KKT multipliers exist**. (c) Diagnose which qualification fails and compare the tangent cone with the linearized cone. (d) Show how a rewriting of the same feasible set restores KKT, and draw the moral.

Solution. **(a) The optimum.** Every feasible point satisfies $0 \leq y \leq (1 - x)^3$, which forces $(1 - x)^3 \geq 0$, that is, $x \leq 1$. Hence the highest possible value of the objective is $x = 1$, and it is attained: with $x = 1$ the chain $0 \leq y \leq 0$ forces $y = 0$, and the point $(1, 0)$ is feasible. Therefore

$$(x^*, y^*) = (1, 0), \quad \text{optimal value} = 1.$$

(Existence is guaranteed: the feasible set is compact (closed and contained in $[0, 1] \times [0, 1]$) and the objective is continuous.)

(b) There are no multipliers. Let us pass to the standard minimization form, $\min -x$, with

$$g_1(x, y) = y - (1 - x)^3 \leq 0, \quad g_2(x, y) = -x \leq 0, \quad g_3(x, y) = -y \leq 0.$$

Activity at $(1, 0)$: $g_1 = 0 - 0 = 0$ **active**; $g_2 = -1 < 0$ inactive (hence $\mu_2 = 0$ by complementary slackness); $g_3 = 0$ **active**. Gradients:

$$\nabla f = (-1, 0), \quad \nabla g_1 = (3(1 - x)^2, 1)|_{(1,0)} = (0, 1), \quad \nabla g_3 = (0, -1).$$

The stationarity condition would require $\mu_1, \mu_3 \geq 0$ with

$$(-1, 0) + \mu_1(0, 1) + \mu_3(0, -1) = (0, 0) \iff \begin{cases} -1 + 0 = 0 & \text{(first component),} \\ \mu_1 - \mu_3 = 0 & \text{(second component).} \end{cases}$$

The first equation is $-1 = 0$: **impossible**. No pair (μ_1, μ_3) exists, not even allowing arbitrary signs. The global optimum of the problem is *not* a KKT point. The intuitive reason: all active gradients are vertical, so no combination of them can cancel the horizontal component of ∇f , and the horizontal component is precisely the direction in which we would like to move.

(c) Diagnosis.

- **LICQ fails:** $\nabla g_1 = (0, 1)$ and $\nabla g_3 = (0, -1)$ are linearly dependent ($\nabla g_1 + \nabla g_3 = 0$).
- **MFCQ fails:** we would have to find d with $\nabla g_1^\top d = d_2 < 0$ and $\nabla g_3^\top d = -d_2 < 0$ simultaneously, that is, $d_2 < 0$ and $d_2 > 0$. Impossible: there is no direction entering strictly into the feasible set.
- **Slater does not apply:** the problem is not convex. g_1 is not convex (in x , $-(1 - x)^3$ has second derivative $-6(1 - x) < 0$ for $x < 1$: it is concave), and the feasible set is not convex either: $(0, 1)$ and $(1, 0)$ are feasible, but their midpoint $(\frac{1}{2}, \frac{1}{2})$ would require $\frac{1}{2} \leq (\frac{1}{2})^3 = \frac{1}{8}$, which is false.
- **Cones.** The tangent cone at $(1, 0)$ is $T = \{d : d_1 \leq 0, d_2 = 0\}$: you can only leave to the left, and you cannot go up, because along any feasible curve $y(t) \leq (1 - x(t))^3 = O(t^3)$, so $y(t)/t \rightarrow 0$. The linearized cone (Definition 9.27), by contrast, is

$$L_X(x^*) = \{d : \nabla g_1^\top d \leq 0, \nabla g_3^\top d \leq 0\} = \{d : d_2 \leq 0, -d_2 \leq 0\} = \{d : d_2 = 0\},$$

which does *not* restrict d_1 . So $T \subsetneq L_X(x^*)$ strictly (Lemma 9.28 holds, but with proper inclusion), and the direction $d = (1, 0) \in L_X(x^*) \setminus T$ is a descent direction for the objective ($\nabla f^\top d = -1 < 0$). That is exactly the gap that breaks the proof of Theorem 9.26: the Farkas lemma (Lemma 9.29) is applied to the linearized cone, and here the linearized cone “sees” an improvement that the true feasible set does not allow.

Compare with Example 9.19: it is the same phenomenon (a boundary with higher order contact, a cusp or a degenerate tangency) in another disguise.

(d) A rewriting that restores KKT. The feasible set does not change if we add the *redundant* constraint $g_4(x, y) = x - 1 \leq 0$ (we already saw that every feasible solution satisfies $x \leq 1$). Now at $(1, 0)$ there are three active constraints and $\nabla g_4 = (1, 0)$, so stationarity

$$(-1, 0) + \mu_1(0, 1) + \mu_3(0, -1) + \mu_4(1, 0) = (0, 0)$$

is solved with $\mu_4 = 1$ and $\mu_1 = \mu_3$ (for example $\mu_1 = \mu_3 = 0$): all ≥ 0 ✓. With this description, the linearized cone becomes $\{d : d_2 = 0, d_1 \leq 0\} = T$, that is, the Abadie qualification holds and KKT is necessary again (even though LICQ and MFCQ still fail, which is why the multiplier is not unique).

Exam moral. Constraint qualifications are properties of the *algebraic description* of the feasible set, not only of the set. Two systems of inequalities defining exactly the same region can differ in whether KKT is valid. Practical consequences: (i) never write “by KKT, the optimum satisfies...” without checking a CQ; (ii) if a constraint can be written in affine form, write it that way; (iii) when the KKT system turns out to be inconsistent, the first suspect should be a CQ failure, not an algebra mistake. $\square$

Exercise 9.55 (Allocating a resource and the shadow price). A planner splits a scarce resource $R > 0$ between two uses, with benefits

$$\max_{x_1, x_2} f(x_1, x_2) = 10x_1 - x_1^2 + 8x_2 - x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \leq R, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

(a) Solve by KKT for $R = 6$. (b) Interpret the multiplier as a shadow price and check it numerically by perturbing the resource to $R = 6.1$ and to $R = 7$. (c) Obtain the full map of regimes: for which values of R does the resource constraint bind and for which does $x_2 \geq 0$ bind? Write $V(R)$ and check that $V'(R) = \lambda^*(R)$ on each branch.

Solution. **(a) Solution with $R = 6$.** The objective is concave (Hessian $-2I \prec 0$) and the constraints are affine: KKT is necessary and sufficient, and the maximum exists and is unique. With $L = f - \lambda(x_1 + x_2 - R) + \mu_1 x_1 + \mu_2 x_2$:

$$(S_1) \quad 10 - 2x_1 - \lambda + \mu_1 = 0,$$

$$(S_2) \quad 8 - 2x_2 - \lambda + \mu_2 = 0,$$

$$(CS) \quad \lambda(x_1 + x_2 - R) = 0, \quad \mu_1 x_1 = 0, \quad \mu_2 x_2 = 0,$$

$$(DF) \quad \lambda, \mu_1, \mu_2 \geq 0.$$

Pruning. With $R = 6$, if the resource were slack ($\lambda = 0$) we would have, with $\mu_i = 0$, $x_1 = 5$ and $x_2 = 4$, whose sum $9 > 6$ is infeasible: a contradiction. Hence $\lambda > 0$ and the resource is exhausted: $x_1 + x_2 = 6$. Suppose also $x_1, x_2 > 0$ (we will check it), so that $\mu_1 = \mu_2 = 0$. Then (S_1) and (S_2) equate the *marginal benefits*:

$$10 - 2x_1 = \lambda = 8 - 2x_2 \implies x_1 - x_2 = 1.$$

Together with $x_1 + x_2 = 6$:

$$x_1^* = 3.5, \quad x_2^* = 2.5, \quad \lambda^* = 10 - 2(3.5) = 3.$$

Check: $x_1^*, x_2^* > 0$ ✓ (consistent with $\mu_1 = \mu_2 = 0$), $\lambda^* = 3 > 0$ ✓, resource exhausted ✓.

Optimal value:

$$V(6) = 10(3.5) - 3.5^2 + 8(2.5) - 2.5^2 = 35 - 12.25 + 20 - 6.25 = 36.5.$$

(b) Shadow price and numerical check. The reading of $\lambda^* = 3$: one extra unit of the resource raises total benefit by about 3. It is also the *common marginal benefit* of the two uses: at the optimum the planner equates $\partial f / \partial x_1 = \partial f / \partial x_2 = \lambda^*$, which is the efficiency condition in the allocation (if one use paid more at the margin, it would be worth shifting resource to it). In a decentralized economy, λ^* would be the *market price* of the resource.

Solving for generic R in the same regime: $x_1 = \frac{R+1}{2}$, $x_2 = \frac{R-1}{2}$, $\lambda^*(R) = 10 - (R+1) = 9 - R$, and

$$V(R) = 10 \frac{R+1}{2} + 8 \frac{R-1}{2} - \frac{(R+1)^2 + (R-1)^2}{4} = (9R + 1) - \frac{R^2 + 1}{2} = -\frac{R^2}{2} + 9R + \frac{1}{2}.$$

R	x_1^*	x_2^*	λ^*	$V(R)$	ΔV versus $\lambda^*(6) \Delta R$
6	3.5	2.5	3	36.5	.
6.1	3.55	2.55	2.9	36.795	0.295 vs. $3 \cdot 0.1 = 0.3$
7	4	3	2	39	2.5 vs. $3 \cdot 1 = 3$

Checks: $V(6.1) = -18.605 + 54.9 + 0.5 = 36.795$; $V(7) = -24.5 + 63 + 0.5 = 39$. The error of the linear approximation is exactly the second order term: since $V'' = -1$,

$$V(R + \Delta) - V(R) = \lambda^* \Delta - \frac{1}{2} \Delta^2,$$

which gives errors of $\frac{1}{2}(0.1)^2 = 0.005$ and $\frac{1}{2}(1)^2 = 0.5$: they match the table to the cent ✓. And indeed $V'(R) = -R + 9 = \lambda^*(R)$ ✓, which is Theorem 9.42. Note the concavity of V : the shadow price *falls* as the resource becomes abundant ($\lambda^*(6) = 3 > \lambda^*(7) = 2$), which is the law of decreasing marginal returns seen from the value function.

(c) Full map of regimes. The previous regime requires $x_2 \geq 0$ and $\lambda \geq 0$, that is, $R \geq 1$ and $R \leq 9$.

- **Abundant resource**, $R \geq 9$. The resource constraint does not bind: $\lambda^* = 0$, $\mu_1 = \mu_2 = 0$, and the optimum is the unconstrained one $x_1^* = 5$, $x_2^* = 4$ (which is feasible because $9 \leq R$), with $V(R) = 25 + 16 = 41$ constant. Zero shadow price: *what is in excess is free*, complementary slackness in its purest form. Note the continuity at $R = 9$: $-40.5 + 81 + 0.5 = 41$ ✓, and $V'(9) = 0 = \lambda^*(9)$ ✓.
- **Interior regime**, $1 \leq R \leq 9$. The one in part (a).
- **Very scarce resource**, $R \leq 1$. Here the formula would give $x_2 < 0$: the nonnegativity of the second use binds. With $x_2 = 0$ and the resource exhausted, $x_1 = R$; then (S₁) gives $\lambda = 10 - 2R$ and (S₂) gives $\mu_2 = \lambda - 8 = 2 - 2R \geq 0$ if and only if $R \leq 1$ ✓ (consistency of the regime). $V(R) = 10R - R^2$, and $V'(R) = 10 - 2R = \lambda^*(R)$ ✓. Economics: when

the resource is so scarce that its shadow price exceeds the maximum marginal benefit of the second use (8), all of it goes to the first.

At $R = 1$ and $R = 9$ the regimes join with continuity of V and of λ^* ($\lambda^*(1) = 8$ on both sides; $\lambda^*(9) = 0$ on both sides): the value function is C^1 and concave, and its derivative is the shadow price on every branch. $\square$

Exercise 9.56 (Principal agent with moral hazard: IR, IC and their multipliers). A risk neutral principal hires a risk averse agent with utility $v(w) = \sqrt{w}$ and reservation utility $\bar{u} = 1$. The agent chooses effort $e \in \{A, B\}$ (high or low). Output is worth 10 if there is success and 0 if there is failure, with success probability

$$p = \Pr(\text{success} \mid A) = 0.8, \quad q = \Pr(\text{success} \mid B) = 0.4.$$

High effort costs the agent $c = 0.5$ in utility units; low effort costs 0. The agent's utility is $\sqrt{w} - c(e)$. The principal observes the outcome but *not* the effort, and wants to implement $e = A$ at minimum expected cost with contingent wages w_S (success) and w_F (failure), $w_S, w_F \geq 0$. (a) Write the program with the (IR) and (IC) constraints and turn it into a convex problem. (b) Write the KKT conditions. (c) Determine which ones bind, solve and compute the multipliers. (d) Interpret γ (multiplier of IR) and δ (multiplier of IC), checking them numerically, and compare with the *first best*. (e) Check that the principal prefers to implement A .

Solution. (a) **The program and the key change of variable.** The principal minimizes the expected wage subject to the agent (i) accepting and (ii) preferring to exert effort:

$$\min_{w_S, w_F \geq 0} p w_S + (1 - p) w_F$$

$$\text{(IR)} \quad p\sqrt{w_S} + (1 - p)\sqrt{w_F} - c \geq \bar{u}, \quad \text{(IC)} \quad p\sqrt{w_S} + (1 - p)\sqrt{w_F} - c \geq q\sqrt{w_S} + (1 - q)\sqrt{w_F}.$$

As written, the problem is ugly: the objective is linear but the constraints are concave in w and the feasible set is not convex in those variables. The standard (and examinable) trick is to change variables to *utilities* instead of wages:

$$u_S = \sqrt{w_S}, \quad u_F = \sqrt{w_F} \quad \Longleftrightarrow \quad w_S = u_S^2, \quad w_F = u_F^2, \quad u_S, u_F \geq 0.$$

Then the objective becomes $p u_S^2 + (1 - p) u_F^2$, which is *convex*, and both constraints become **linear**. Simplifying (IC): $p u_S + (1 - p) u_F - c \geq q u_S + (1 - q) u_F$ is equivalent to $(p - q)(u_S - u_F) \geq c$. With the numbers ($p = 0.8, q = 0.4, c = 0.5, \bar{u} = 1$):

$$\text{(IR)} \quad 0.8 u_S + 0.2 u_F \geq 1.5, \quad \text{(IC)} \quad 0.4(u_S - u_F) \geq 0.5 \Longleftrightarrow u_S - u_F \geq 1.25.$$

A convex problem with affine constraints: KKT is necessary and sufficient (Corollary 9.33); Slater holds trivially (take u_S large and $u_F = 0$).

(b) **KKT conditions.** In standard form, $g_1 = \bar{u} + c - p u_S - (1 - p) u_F \leq 0$ (IR) and $g_2 = c - (p - q)(u_S - u_F) \leq 0$ (IC), with multipliers $\gamma, \delta \geq 0$. We ignore $u_S, u_F \geq 0$ for now and

will check at the end that they hold. With $L = pu_S^2 + (1-p)u_F^2 + \gamma g_1 + \delta g_2$:

$$(S_S) \quad 2pu_S - \gamma p - \delta(p-q) = 0 \quad \implies \quad 2u_S = \gamma + \delta\left(1 - \frac{q}{p}\right),$$

$$(S_F) \quad 2(1-p)u_F - \gamma(1-p) + \delta(p-q) = 0 \quad \implies \quad 2u_F = \gamma - \delta\frac{p-q}{1-p},$$

$$(CS) \quad \gamma g_1 = 0, \quad \delta g_2 = 0, \quad (DF) \quad \gamma \geq 0, \quad \delta \geq 0.$$

These two equations are the *Holmstrom condition*: since $1/v'(w) = 2\sqrt{w}$, they say

$$\frac{1}{v'(w_s)} = \gamma + \delta \left(1 - \frac{\Pr(s|B)}{\Pr(s|A)}\right),$$

that is, *the optimal payment adjusts according to how much each outcome tells you about effort* (the likelihood ratio). With $p = 0.8$, $q = 0.4$: $1 - q/p = 0.5$ for success and $(p-q)/(1-p) = 2$ for failure.

(c) Which ones bind? Four cases, and three fall immediately.

Case $\delta = 0$ (IC slack). Then (S_S) and (S_F) give $2u_S = \gamma = 2u_F$, that is, full insurance $u_S = u_F$. But then (IC) requires $0.4 \cdot 0 = 0 \geq 0.5$: **false**. A contradiction, so $\boxed{\delta > 0}$ and (IC) binds. *Economics*: without incentives the agent would not exert effort; under asymmetric information the contract *must* expose the agent to risk. Compare with Example 9.39, where without (IC) the optimum was exactly full insurance.

Case $\gamma = 0$ (IR slack). Then (S_F) would give $2u_F = -\delta(p-q)/(1-p) = -2\delta < 0$, impossible because $u_F = \sqrt{w_F} \geq 0$. A contradiction, so $\boxed{\gamma > 0}$ and (IR) binds. (Equivalent economic argument: if the agent had utility to spare, the principal would lower both wages by the same amount of utility, which does not affect (IC), since that depends only on the *difference*, and reduces cost.)

Case both slack: discarded by either of the two previous arguments. A single case remains: **both bind**, and the system is linear:

$$\begin{cases} u_S - u_F = 1.25, \\ 0.8u_S + 0.2u_F = 1.5. \end{cases}$$

Substituting $u_F = u_S - 1.25$: $0.8u_S + 0.2u_S - 0.25 = 1.5$, hence

$$\boxed{u_S = 1.75, \quad u_F = 0.5, \quad w_S = u_S^2 = 3.0625, \quad w_F = u_F^2 = 0.25.}$$

Both ≥ 0 ✓ (the nonnegativity constraint we had ignored holds with slack). Expected cost of the contract:

$$C^{SB} = 0.8(3.0625) + 0.2(0.25) = 2.45 + 0.05 = 2.50.$$

Multipliers: from (S_S) and (S_F) ,

$$2(1.75) = 3.5 = \gamma + 0.5\delta, \quad 2(0.5) = 1 = \gamma - 2\delta.$$

Subtracting: $2.5 = 2.5\delta$, hence $\delta = 1$ and $\gamma = 1 + 2 = 3$. Both positive ✓, consistent with both

constraints binding.

(d) Interpretation and numerical check.

$\gamma = 3$: *the price of the reservation utility.* It is the marginal cost (in expected wage) of guaranteeing the agent one extra util. Check: raise $\bar{u}$ from 1 to 1.1. The two binding constraints give $u_S - u_F = 1.25$ and $0.8u_S + 0.2u_F = 1.6$, hence $u_S = 1.85$, $u_F = 0.6$ and

$$C = 0.8(1.85)^2 + 0.2(0.6)^2 = 2.738 + 0.072 = 2.81, \quad \Delta C = 0.31 \approx \gamma \cdot 0.1 = 0.3 \checkmark$$

(the 0.01 difference is the second order term). If the labor market improves for the agent, the principal pays: at a rate of 3 expected units of money per util.

$\delta = 1$: *the price of asymmetric information.* It is the marginal cost of the incentive constraint existing. Check: tighten (IC) by requiring $(p-q)(u_S - u_F) \geq c+t$ with $t = 0.1$ (imagine deviation becomes more tempting). Then $u_S - u_F = 0.6/0.4 = 1.5$ and, with (IR) unchanged, $u_S = 1.8$, $u_F = 0.3$:

$$C = 0.8(1.8)^2 + 0.2(0.3)^2 = 2.592 + 0.018 = 2.61, \quad \Delta C = 0.11 \approx \delta \cdot 0.1 = 0.1 \checkmark$$

Comparison with the first best. If effort were observable (symmetric information), (IC) would disappear and only (IR) would remain: by the argument of the case $\delta = 0$, the optimal contract would be **full insurance** $u_S = u_F = u$, with (IR) binding: $u = \bar{u} + c = 1.5$, that is, $w_S = w_F = 2.25$ and

$$C^{FB} = 2.25 \quad \text{versus} \quad C^{SB} = 2.50.$$

The **agency cost** is $C^{SB} - C^{FB} = 0.25$: it is the *risk premium* that must be paid to a risk averse agent for exposing them to the wage lottery $(3.0625, 0.25)$, whose expected utility is the same as that of the safe wage 2.25, but whose expected cost is higher by Jensen inequality. Note the conceptual difference, which is worth having very clear when studying the principal agent problem: here (IR) binds and the agent gets *no* rent at all; the cost of asymmetric information takes the form of inefficient risk sharing. In *adverse selection* models (private types instead of hidden action), by contrast, the (IC) of the efficient type usually forces its (IR) to be slack, and then the cost of asymmetric information appears as **informational rent** paid to the efficient type. In both cases, the diagnosis “which constraints bind” is obtained with the same KKT case analysis.

(e) Is it worth implementing high effort? With high effort, expected output is $0.8 \cdot 10 = 8$ and the principal's profit is $8 - C^{SB} = 8 - 2.50 = 5.50$. If instead the principal implemented low effort, no (IC) would be needed (low effort is what the agent chooses by default), so the optimal contract would be a fixed wage w with $\sqrt{w} \geq \bar{u} = 1$, that is, $w = 1$; expected output would be $0.4 \cdot 10 = 4$ and profit $4 - 1 = 3.00 < 5.50$. Therefore the principal implements high effort $\checkmark$. (In the *first best*, profit with high effort would have been $8 - 2.25 = 5.75$: moral hazard costs the principal exactly the 0.25 computed above.) $\square$

Chapter 10

Saddle point and Lagrangian duality

[P2]

Week 8 of the course. Bibliography: Boyd and Vandenberghe [2]; Brinkhuis [3]; Ruszczyński [6].

What is this used for in Microeconomics?

Duality is one of the ideas in this course with the most direct echo in Microeconomics:

- **Individual choice (consumer, MWG ch. 3):** the UMP/EMP pair (maximize utility subject to a budget, Marshallian demand, versus minimize expenditure subject to utility, Hicksian demand) mirrors Lagrangian duality: they are “inverse” problems (objective and constraint swap roles) and their multipliers are reciprocals. The Roy and Shephard identities that connect them live in the envelope chapter, but the underlying structure is the one in this chapter [9, 10].
- **General equilibrium (welfare, MWG chs. 16–18):** the multipliers of the dual problem are *shadow prices* that support efficient allocations; the second welfare theorem, which says that every efficient allocation is an equilibrium with transfers, is at bottom a duality result: the prices that “decentralize” the allocation are the dual variables of the planner’s problem [9].
- **Game theory (MWG ch. 8):** von Neumann’s minimax theorem for zero sum games, the historical basis of all game theory, is exactly a *saddle point* theorem: the value of the game exists because each player’s problem is the linear dual of the other’s problem.
- **Market failures (contracts and information, MWG chs. 13–14):** in the principal’s problem with incentive compatibility (IC) and participation (IR) constraints, the Lagrangian is analyzed through duality: the multipliers of the IC constraints identify *which types “bind”* (which constraints are active) in Baron–Myerson and Mussa–Rosen, and those of the IR constraints measure the cost of leaving rent to the agent [11].

In the previous chapter we defined the Lagrangian and the KKT conditions: given a candidate x^* , the multipliers certify that it is optimal. In this chapter we change the point of view: instead of treating the multipliers (λ, μ) as auxiliary witnesses, we turn them into *decision*

variables of a new optimization problem, the **dual problem**. We will see that:

- (i) the dual is always a *concave* problem (even if the primal is not convex), and it always gives *lower bounds* on the primal value (weak duality);
- (ii) under convexity and a regularity condition (Slater), the bound is exact (strong duality), and the joint primal-dual solution is a *saddle point* of the Lagrangian;
- (iii) the dual variables have a precise economic reading: they are the *prices* at which a market would value the scarce resources of the problem.

Our main reference is [2] (chapter 5); we also follow [6] (chapter 4), [3] and [1].

10.1 The Lagrange dual function [P2]

10.1.1 The primal problem and its Lagrangian

We work with the same problem as in the KKT chapter, which we now call the **primal problem**:

$$\begin{aligned} \min_{x \in X} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p, \end{aligned} \tag{10.1}$$

where $f, g_1, \dots, g_m, h_1, \dots, h_p : \mathbb{R}^n \rightarrow \mathbb{R}$ and $X \subseteq \mathbb{R}^n$ is a nonempty set (usually $X = \mathbb{R}^n$, although allowing a general X , for example $X = \mathbb{R}_+^n$ or an interval, will be useful). We write p^* for the **primal optimal value**:

$$p^* := \inf \{ f(x) : x \in X, g_i(x) \leq 0, h_j(x) = 0 \},$$

with the convention $p^* = +\infty$ if the problem is infeasible.

The **Lagrangian** is, as before,

$$L(x, \lambda, \mu) := f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \mu_j h_j(x), \quad x \in X, \lambda \in \mathbb{R}_+^m, \mu \in \mathbb{R}^p.$$

The idea of duality comes from a simple question: instead of imposing the constraints rigidly, what happens if we *charge* for them? The term $\lambda_i g_i(x)$ is a linear *fine* for violating constraint i : if $g_i(x) > 0$ (violation) we pay $\lambda_i g_i(x) > 0$; if $g_i(x) < 0$ (slack) we get a reward. The Lagrangian is the original cost plus the fines, and minimizing it *without constraints* is easier than solving the primal. The function that records the result of that minimization is the central object of the chapter.

Definition 10.1 (Lagrange dual function). The **Lagrange dual function** associated with problem (10.1) is the function $q : \mathbb{R}_+^m \times \mathbb{R}^p \rightarrow \mathbb{R} \cup \{-\infty\}$ defined by

$$q(\lambda, \mu) := \inf_{x \in X} L(x, \lambda, \mu) = \inf_{x \in X} \left[f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \mu_j h_j(x) \right].$$

Its **effective domain** is $\text{dom}(q) := \{(\lambda, \mu) : q(\lambda, \mu) > -\infty\}$. The **dual problem** is

$$\max_{\lambda \geq 0, \mu \in \mathbb{R}^p} q(\lambda, \mu), \quad (10.2)$$

and its optimal value is written d^* . A pair (λ, μ) with $\lambda \geq 0$ is called **dual feasible** (and it is *useful* only if it also belongs to $\text{dom}(q)$).

Two immediate remarks. First, the infimum is taken over *all* of X , ignoring the constraints: this is why q can often be computed in closed form. Second, q may equal $-\infty$ for some (λ, μ) (the Lagrangian is unbounded below); those points carry no information, and this is why in practice the dual is restricted to $\text{dom}(q)$.

10.1.2 Concavity of the dual: free and always

Theorem 10.2 (The dual function is always concave). *For any problem (10.1), convex or not, the dual function q is concave in (λ, μ) .*

Proof. Fix $x \in X$ and look at the Lagrangian as a function of the multipliers:

$$(\lambda, \mu) \mapsto L(x, \lambda, \mu) = \underbrace{f(x)}_{\text{constant}} + \sum_{i=1}^m \underbrace{g_i(x)}_{\text{constant}} \lambda_i + \sum_{j=1}^p \underbrace{h_j(x)}_{\text{constant}} \mu_j.$$

With x fixed, the values $f(x), g_i(x), h_j(x)$ are numbers: the function is *affine* (linear plus a constant) in (λ, μ) . Now recall, from the chapter on convex functions, an operation that preserves convexity: the pointwise supremum of a family of convex functions is convex; by symmetry, *the pointwise infimum of a family of concave functions is concave*. Since every affine function is concave, and

$$q(\lambda, \mu) = \inf_{x \in X} L(x, \lambda, \mu)$$

is the pointwise infimum of the family $\{L(x, \cdot, \cdot)\}_{x \in X}$ of affine functions, q is concave. (Direct check, in case the abstract argument is not convincing: for $(\lambda^1, \mu^1), (\lambda^2, \mu^2)$ and $t \in [0, 1]$, affinity in (λ, μ) gives, for each x , $L(x, t\lambda^1 + (1-t)\lambda^2, t\mu^1 + (1-t)\mu^2) = tL(x, \lambda^1, \mu^1) + (1-t)L(x, \lambda^2, \mu^2) \geq tq(\lambda^1, \mu^1) + (1-t)q(\lambda^2, \mu^2)$; taking the infimum over x on the left gives the concavity inequality.) $\square$

Intuition

The dual is concave *by construction*, not by assumption: q is an infimum of lines (in dimension 1) or of hyperplanes, and the pointwise minimum of a family of flat functions always curves downward. This has a huge practical consequence: even if the primal is horrible (not convex, not differentiable, even combinatorial), *the dual is always a concave maximization problem*, that is, a “good” problem where local maxima are global. Duality is a machine that builds convex problems out of arbitrary problems; the price you pay is that in general the dual only gives a *bound* on the primal, as we will see.

10.1.3 Two fully worked examples

Example 10.3 (Dual of a one variable QP). Consider the quadratic program

$$\min_{x \in \mathbb{R}} x^2 \quad \text{s.t.} \quad x \geq 1,$$

which we write in standard form with $g(x) = 1 - x \leq 0$. By inspection: the solution is $x^* = 1$ with $p^* = 1$. Let us build the dual step by step.

Step 1: Lagrangian. $L(x, \lambda) = x^2 + \lambda(1 - x)$, with $\lambda \geq 0$.

Step 2: minimize in x . For fixed λ , L is a convex parabola in x ; its minimum is found with the first order condition $\partial L / \partial x = 2x - \lambda = 0$, that is $x(\lambda) = \lambda/2$. Substituting:

$$q(\lambda) = \left(\frac{\lambda}{2}\right)^2 + \lambda \left(1 - \frac{\lambda}{2}\right) = \frac{\lambda^2}{4} + \lambda - \frac{\lambda^2}{2} = \lambda - \frac{\lambda^2}{4}.$$

As Theorem 10.2 announces, q is concave (an inverted parabola).

Step 3: dual problem. $\max_{\lambda \geq 0} \lambda - \lambda^2/4$. The first order condition gives $1 - \lambda/2 = 0$, that is $\lambda^* = 2 \geq 0$, with value

$$d^* = q(2) = 2 - 1 = 1 = p^*.$$

There is no gap: the best lower bound of the dual *attains* the primal value. Moreover $x(\lambda^*) = \lambda^*/2 = 1 = x^*$: minimizing the Lagrangian with the optimal multiplier recovers the primal solution. Note also that $\lambda^* = 2 > 0$ and the constraint is active ($g(x^*) = 0$): complementary slackness, as in KKT.

Example 10.4 (Dual of a linear program). Take the linear program in standard form

$$\min_{x \in \mathbb{R}^n} c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0,$$

with $A \in \mathbb{R}^{p \times n}$, $b \in \mathbb{R}^p$, $c \in \mathbb{R}^n$. We write $x \geq 0$ as $-x_k \leq 0$ with multipliers $\lambda \in \mathbb{R}_+^n$, and $Ax = b$ with multipliers $\mu \in \mathbb{R}^p$ (we use the form $b - Ax = 0$ so that the signs come out conveniently):

$$L(x, \lambda, \mu) = c^\top x - \lambda^\top x + \mu^\top (b - Ax) = b^\top \mu + (c - A^\top \mu - \lambda)^\top x.$$

The Lagrangian is *affine in x* : its infimum over $x \in \mathbb{R}^n$ is $-\infty$ unless the coefficient vector is exactly zero. Therefore,

$$q(\lambda, \mu) = \begin{cases} b^\top \mu & \text{if } c - A^\top \mu - \lambda = 0, \\ -\infty & \text{otherwise.} \end{cases}$$

The dual maximizes q over $\lambda \geq 0$, with μ free. Since λ only appears in the condition $\lambda = c - A^\top \mu$, requiring $\lambda \geq 0$ is the same as $A^\top \mu \leq c$, and λ disappears as a slack variable. The dual is then the linear program

$$\max_{\mu \in \mathbb{R}^p} b^\top \mu \quad \text{s.t.} \quad A^\top \mu \leq c,$$

which is exactly classical *linear programming duality*. Note the symmetry: the primal minimizes with equality constraints and nonnegative variables; the dual maximizes with free variables and inequality constraints. The roles of b and c swap.

Exam strategy

Recipe for computing duals in the exam (and in life):

- (1) Write the problem in standard *minimization* form with constraints $g_i \leq 0$ and $h_j = 0$, deciding which constraints go into the Lagrangian and which stay in X (simple bounds such as $x \geq 0$ or $x \in [a, b]$ are sometimes better left in X).
- (2) Build $L(x, \lambda, \mu)$ and *group by powers of x* : collect all terms linear in x , the quadratic ones, and so on.
- (3) Minimize in x ignoring the constraints: if L is quadratic and convex in x , use the first order condition; if it is affine in x , the infimum is $-\infty$ unless the coefficient of x vanishes, and that vanishing becomes a *constraint of the dual*; if X is an interval and L is concave in x , the minimum is at an endpoint.
- (4) Write $\max q(\lambda, \mu)$ s.t. $\lambda \geq 0$ and the constraints that appeared in the previous step. Check that q came out concave: if not, there is a computational error.

10.2 Weak duality and the duality gap [P2]

The first duality theorem is so short that it fits in two lines, and yet it is the basis of everything else: *any* value of the dual function is a lower bound on the primal optimal value.

Theorem 10.5 (Weak duality). *For every $\lambda \in \mathbb{R}_+^m$ and every $\mu \in \mathbb{R}^p$,*

$$q(\lambda, \mu) \leq p^*.$$

Consequently, $d^ \leq p^*$.*

Proof. Let $\bar{x}$ be any feasible point of the primal: $\bar{x} \in X$, $g_i(\bar{x}) \leq 0$, $h_j(\bar{x}) = 0$. Then

$$q(\lambda, \mu) = \inf_{x \in X} L(x, \lambda, \mu) \leq L(\bar{x}, \lambda, \mu) = f(\bar{x}) + \sum_{i=1}^m \lambda_i g_i(\bar{x}) + \sum_{j=1}^p \mu_j h_j(\bar{x}) \leq f(\bar{x}).$$

Let us explain each inequality. The first is the definition of infimum: the infimum over all $x \in X$ does not exceed the value at the particular point $\bar{x}$. The second uses feasibility and the sign of the multipliers: each term $\lambda_i g_i(\bar{x})$ is the product of a number ≥ 0 and a number ≤ 0 , hence it is ≤ 0 ; and each $\mu_j h_j(\bar{x}) = \mu_j \cdot 0 = 0$. So $q(\lambda, \mu) \leq f(\bar{x})$ for *every* feasible $\bar{x}$; taking the infimum over the feasible points, $q(\lambda, \mu) \leq p^*$. Finally, taking the supremum over (λ, μ) with $\lambda \geq 0$ gives $d^* \leq p^*$. $\square$

Definition 10.6 (Duality gap). The quantity $p^* - d^* \geq 0$ is called the **duality gap**. If $p^* = d^*$ we say that there is **strong duality**, or zero gap.

Corollary 10.7 (Optimality certificates). *If $\hat{x}$ is primal feasible, $(\hat{\lambda}, \hat{\mu})$ is dual feasible and*

$$f(\hat{x}) = q(\hat{\lambda}, \hat{\mu}),$$

then $\hat{x}$ is primal optimal, $(\hat{\lambda}, \hat{\mu})$ is dual optimal and the gap is zero.

Proof. By weak duality, $q(\hat{\lambda}, \hat{\mu}) \leq d^* \leq p^* \leq f(\hat{x})$. If the two ends coincide, the whole chain consists of equalities: $f(\hat{x}) = p^*$ ($\hat{x}$ primal optimal) and $q(\hat{\lambda}, \hat{\mu}) = d^*$ ($(\hat{\lambda}, \hat{\mu})$ dual optimal). $\square$

Intuition

The dual is a *factory of certifiable lower bounds*. Suppose you solved a minimization problem numerically and got a feasible point $\hat{x}$ with $f(\hat{x}) = 10.02$. Is that good? You do not know... unless someone shows you a dual feasible $(\hat{\lambda}, \hat{\mu})$ with $q(\hat{\lambda}, \hat{\mu}) = 10.00$: then *nobody* can do better than 10.00, and your solution is at most 0.02 away from the optimum, with a guarantee. The dual pair is a certificate that anyone can check with a single function evaluation, without repeating the optimization. This logic is the heart of modern solvers (which report the primal-dual gap as a stopping criterion, see [2, 5]) and also of the theory: proving optimality *is* exhibiting a dual feasible point with no gap.

Remark 10.8. Weak duality holds with the usual conventions about infinities: if the primal is infeasible ($p^* = +\infty$), the bound is trivial; if the dual is unbounded ($d^* = +\infty$), then the primal is infeasible; and if the primal is unbounded ($p^* = -\infty$), then $q \equiv -\infty$ (the dual is infeasible in the sense that $\text{dom}(q) = \emptyset$).

10.3 Economic reading of the dual [P2]

This section deserves a calm reading: duality is not an algebraic trick, it is a *theory of prices*. The idea, in one sentence: **the dual problem is the problem of a market that puts prices on the scarce resources of the primal**, and strong duality says that this market, with the right prices, reproduces the optimal allocation exactly.

10.3.1 Multipliers as resource prices

Let us write the primal as the problem of a firm that *allocates resources*. The firm chooses activity levels $x \in X \subseteq \mathbb{R}_+^n$ (how much to produce of each good) to maximize a profit $R(x)$ (concave), subject to the condition that its resource use does not exceed the available endowments:

$$\max_{x \in X} R(x) \quad \text{s.t.} \quad G_i(x) \leq b_i, \quad i = 1, \dots, m, \quad (10.3)$$

where $G_i(x)$ is the amount of resource i (labor hours, machine hours, raw material) used by the plan x , and b_i is the endowment. To fit our minimization format, the primal is $\min_{x \in X} -R(x)$ s.t. $G_i(x) - b_i \leq 0$, and the Lagrangian (with the sign flipped so we can read it in terms of profit) is

$$L(x, \lambda) = R(x) + \sum_{i=1}^m \lambda_i (b_i - G_i(x)) = R(x) - \sum_{i=1}^m \lambda_i G_i(x) + \sum_{i=1}^m \lambda_i b_i.$$

Read it as an income statement: if resource i has price λ_i , then L is the production profit $R(x)$, minus the *cost of the resources used* $\sum_i \lambda_i G_i(x)$, plus the *market value of the endowment* $\sum_i \lambda_i b_i$.

(the firm could sell its resources instead of using them). The dual function

$$q(\lambda) = \sum_{i=1}^m \lambda_i b_i + \max_{x \in X} \left[R(x) - \sum_{i=1}^m \lambda_i G_i(x) \right]$$

is the largest “total value” the firm can get when resources trade at prices λ : value of the endowment plus the net profit from operating and buying resources at those prices. Weak duality (in the maximization version: $q(\lambda) \geq$ the optimal primal profit, for every $\lambda \geq 0$) says that *putting prices on resources never undervalues the firm*: with resource markets, the firm can always do at least as well as being self sufficient, because it keeps the option of replicating its original plan. The dual problem,

$$\min_{\lambda \geq 0} q(\lambda),$$

looks for the *most demanding* prices: the ones that push that valuation to the minimum. They are the prices that give nothing away, and at the optimum (with strong duality) the market valuation coincides exactly with the productive profit: the dual prices λ^* are the **shadow prices** of the resources.

10.3.2 The linear case and the no arbitrage condition

With $R(x) = c^\top x$ and $G(x) = Gx$ (linear technology, $X = \mathbb{R}_+^n$), the inner maximization is

$$\max_{x \geq 0} (c - G^\top \lambda)^\top x = \begin{cases} 0 & \text{if } G^\top \lambda \geq c, \\ +\infty & \text{if } (G^\top \lambda)_k < c_k \text{ for some } k. \end{cases}$$

The economic reading is exact: $(G^\top \lambda)_k = \sum_i \lambda_i G_{ik}$ is the *resource cost* of one unit of activity k at prices λ , and c_k is its unit revenue. If some activity had a strictly positive unit profit ($c_k > (G^\top \lambda)_k$), the firm would scale it without limit: there is *arbitrage* and the valuation blows up. The dual becomes

$$\min_{\lambda \geq 0} b^\top \lambda \quad \text{s.t.} \quad G^\top \lambda \geq c,$$

that is: **find the cheapest resource prices (the minimum valuation of the endowment) such that no activity is strictly profitable**. This is the *no arbitrage condition*: at equilibrium, each activity that is used covers exactly its resource cost (complementary slackness: $x_k^* > 0 \Rightarrow c_k = (G^\top \lambda^*)_k$) and each resource with a positive price is exhausted ($\lambda_i^* > 0 \Rightarrow (Gx^*)_i = b_i$: a resource left over is a free good, with price zero).

Example 10.9 (Resource allocation in a firm). A firm produces two goods with unit profits $c = (3, 2)$. It has 4 hours of labor and 5 units of input; good 1 uses 1 hour and 2 inputs per unit, good 2 uses 1 hour and 1 input. The primal:

$$\max_{x \geq 0} 3x_1 + 2x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 4 \quad (\text{labor}), \quad 2x_1 + x_2 \leq 5 \quad (\text{input}).$$

Primal solution. With two constraints and positive profits, the natural candidate is the vertex where both are active: $x_1 + x_2 = 4$ and $2x_1 + x_2 = 5$ give $x^* = (1, 3)$, with profit $p^* = 3 + 6 = 9$.

Dual. With prices λ_1 (labor) and λ_2 (input):

$$\min_{\lambda \geq 0} 4\lambda_1 + 5\lambda_2 \quad \text{s.t.} \quad \lambda_1 + 2\lambda_2 \geq 3, \quad \lambda_1 + \lambda_2 \geq 2.$$

The first constraint says: the resources used by one unit of good 1 (one hour, two inputs) must be worth at least its profit 3; the same for the second. Since $x_1^*, x_2^* > 0$, complementary slackness forces both to hold with equality: $\lambda_1 + 2\lambda_2 = 3$, $\lambda_1 + \lambda_2 = 2$, so $\lambda^* = (1, 1)$: one hour of labor is worth 1 and one unit of input is worth 1. The dual value is $d^* = 4(1) + 5(1) = 9 = p^*$: the valuation of the endowment at shadow prices exhausts the profit exactly (the “exhaustion of the product” result). Marginal reading: one extra hour of labor would raise profit by $\lambda_1^* = 1$; that is the highest price the firm would pay for it.

Intuition

Think of the dual as an *adversarial appraiser*. The primal is the engineer: it assigns resources to activities. The dual is the accountant and market: it does not look at activities, it only puts prices on resources, with two rules: it cannot leave arbitrage (no activity may be strictly profitable at its prices) and it wants the lowest possible total valuation. Weak duality says that any appraisal without arbitrage is an upper bound on the attainable profit; strong duality says that the best appraisal is *exact*. This is the same mechanism as the second welfare theorem in general equilibrium: the efficient allocation (the planner’s primal) can be *decentralized* with prices (dual variables) that make no agent want to deviate [9].

Remark 10.10 (A bit of history). This reading is not a metaphor added later: it was born with the theory. Leonid Kantorovich (1939), studying the planning of Soviet production, introduced the multipliers as “objectively determined valuations” of resources, shadow prices in an economy without markets, and Tjalling Koopmans developed activity analysis in parallel in the West; they shared the Nobel prize in economics in 1975 for the theory of optimal resource allocation. Formal linear duality took shape with von Neumann, Gale, Kuhn and Tucker around 1947–1951. For an economist, the moral is that *linear programming and price theory are the same mathematics* [3].

10.4 Saddle point of the Lagrangian [P2]

10.4.1 Definition and first reading

So far the primal and the dual are two separate problems joined by an inequality. The concept that welds them into one is the saddle point: a point that is *a minimum in the primal variables and a maximum in the dual variables, at the same time*.

Definition 10.11 (Saddle point). A pair $(x^*, (\lambda^*, \mu^*))$ with $x^* \in X$, $\lambda^* \in \mathbb{R}_+^m$, $\mu^* \in \mathbb{R}^p$ is a **saddle point** of the Lagrangian L if

$$L(x^*, \lambda, \mu) \leq L(x^*, \lambda^*, \mu^*) \leq L(x, \lambda^*, \mu^*) \quad \text{for all } x \in X, \lambda \in \mathbb{R}_+^m, \mu \in \mathbb{R}^p.$$

That is: fixing $x = x^*$, the multipliers (λ^*, μ^*) maximize L over $\lambda \geq 0$, μ free; and fixing $(\lambda, \mu) = (\lambda^*, \mu^*)$, the point x^* minimizes L over X .

The name comes from the geometry of a horse saddle: the surface $(x, \lambda) \mapsto L(x, \lambda)$ goes up in the direction of x (minimum) and down in the direction of λ (maximum), like the function $x^2 - \lambda^2$ at the origin. The game reading is immediate and will come back in the minimax theorem: one player chooses x to minimize, another chooses (λ, μ) to maximize, and the saddle point is a pair of strategies from which neither wants to deviate unilaterally: an equilibrium.

10.4.2 The central theorem

Theorem 10.12 (Characterization of saddle points). *The pair $(x^*, (\lambda^*, \mu^*))$ with $\lambda^* \geq 0$ is a saddle point of the Lagrangian if and only if the following three conditions hold:*

- (a) x^* is an optimal solution of the primal (10.1);
- (b) (λ^*, μ^*) is an optimal solution of the dual (10.2);
- (c) the duality gap is zero: $p^* = d^*$.

Proof. ($\Rightarrow$) **Suppose $(x^*, (\lambda^*, \mu^*))$ is a saddle point.**

Step 1: x^ is feasible.* The first saddle inequality says that the function

$$(\lambda, \mu) \mapsto L(x^*, \lambda, \mu) = f(x^*) + \sum_i \lambda_i g_i(x^*) + \sum_j \mu_j h_j(x^*)$$

attains its supremum over $\lambda \geq 0$, $\mu \in \mathbb{R}^p$ at (λ^*, μ^*) , and in particular that supremum is *finite*. But it is an affine function of (λ, μ) : if some $g_i(x^*) > 0$, sending $\lambda_i \rightarrow +\infty$ makes the value blow up to $+\infty$; if some $h_j(x^*) \neq 0$, choosing $\mu_j \rightarrow \pm\infty$ with the right sign also makes it blow up. For the supremum to be finite we need $g_i(x^*) \leq 0$ for all i and $h_j(x^*) = 0$ for all j : x^* is feasible.

Step 2: complementary slackness and the saddle value. Taking $\lambda = 0$, $\mu = \mu^*$ in the first inequality (and using $h_j(x^*) = 0$):

$$f(x^*) = L(x^*, 0, \mu^*) \leq L(x^*, \lambda^*, \mu^*) = f(x^*) + \sum_i \lambda_i^* g_i(x^*) \leq f(x^*),$$

where the last inequality uses $\lambda_i^* \geq 0$, $g_i(x^*) \leq 0$. The whole chain is an equality, hence $\sum_i \lambda_i^* g_i(x^*) = 0$ (and since each summand is ≤ 0 , each one is zero: $\lambda_i^* g_i(x^*) = 0$, the KKT complementary slackness) and

$$L(x^*, \lambda^*, \mu^*) = f(x^*).$$

Step 3: close with weak duality. The second saddle inequality says that x^* minimizes $L(\cdot, \lambda^*, \mu^*)$ over X , that is,

$$q(\lambda^*, \mu^*) = \inf_{x \in X} L(x, \lambda^*, \mu^*) = L(x^*, \lambda^*, \mu^*) = f(x^*).$$

Since x^* is feasible, $p^* \leq f(x^*)$; and by weak duality (Theorem 10.5), $q(\lambda^*, \mu^*) \leq d^* \leq p^*$. Putting these together:

$$p^* \leq f(x^*) = q(\lambda^*, \mu^*) \leq d^* \leq p^*,$$

so all of them are equalities: $f(x^*) = p^*$ (x^* primal optimal), $q(\lambda^*, \mu^*) = d^*$ ((λ^*, μ^*) dual optimal) and $p^* = d^*$ (zero gap).

($\Leftarrow$) **Suppose (a), (b), (c) hold.** Then

$$f(x^*) = p^* \stackrel{(c)}{=} d^* = q(\lambda^*, \mu^*) = \inf_{x \in X} L(x, \lambda^*, \mu^*) \leq L(x^*, \lambda^*, \mu^*) = f(x^*) + \sum_i \lambda_i^* g_i(x^*),$$

using the feasibility of x^* ($h_j(x^*) = 0$) in the last equality. Since $\lambda_i^* \geq 0$ and $g_i(x^*) \leq 0$, the last term is $\leq f(x^*)$. Hence the whole chain is an equality and we get two things:

- $\inf_{x \in X} L(x, \lambda^*, \mu^*) = L(x^*, \lambda^*, \mu^*)$: the infimum is attained at x^* , which is exactly the *second* saddle inequality;
- $\sum_i \lambda_i^* g_i(x^*) = 0$ and $L(x^*, \lambda^*, \mu^*) = f(x^*)$.

For the *first* saddle inequality, let $\lambda \geq 0$ and μ be arbitrary; by feasibility of x^* ,

$$L(x^*, \lambda, \mu) = f(x^*) + \sum_i \lambda_i g_i(x^*) + \sum_j \mu_j \cdot 0 \leq f(x^*) = L(x^*, \lambda^*, \mu^*).$$

Both inequalities hold: $(x^*, (\lambda^*, \mu^*))$ is a saddle point. □

Remark 10.13. The theorem does not require convexity: it is pure bookkeeping of inequalities. Convexity enters in the *existence* of the saddle point (strong duality, section 10.5): without convexity it may happen that $p^* > d^*$, and then, by the theorem, *no saddle point exists*, no matter how well behaved the primal solution is.

10.4.3 Saddle point and KKT

Theorem 10.14 (Under convexity, KKT is equivalent to a saddle point). *Suppose that in (10.1) the functions $f, g_1, \dots, g_m$ are convex and differentiable, the h_j are affine, and $X = \mathbb{R}^n$. Then $(x^*, (\lambda^*, \mu^*))$ with $\lambda^* \geq 0$ is a saddle point of the Lagrangian if and only if it satisfies the KKT conditions:*

- (i) (stationarity) $\nabla_x L(x^*, \lambda^*, \mu^*) = \nabla f(x^*) + \sum_i \lambda_i^* \nabla g_i(x^*) + \sum_j \mu_j^* \nabla h_j(x^*) = 0$;
- (ii) (primal feasibility) $g_i(x^*) \leq 0$, $h_j(x^*) = 0$;
- (iii) (dual feasibility) $\lambda^* \geq 0$;
- (iv) (complementary slackness) $\lambda_i^* g_i(x^*) = 0$ for every i .

Proof. (Saddle $\Rightarrow$ KKT.) Steps 1 and 2 of the proof of Theorem 10.12 give (ii), (iii) and (iv) directly (we did not use convexity there). For (i): the second saddle inequality says that x^* minimizes over *all* of $\mathbb{R}^n$ the differentiable function $x \mapsto L(x, \lambda^*, \mu^*)$; by the necessary first order condition of the chapter on unconstrained optimization, its gradient at x^* vanishes, which is (i).

(KKT $\Rightarrow$ saddle.) Here we do use convexity. With $\lambda^* \geq 0$ fixed, the function $x \mapsto L(x, \lambda^*, \mu^*)$ is convex: it is the sum of f (convex), of the $\lambda_i^* g_i$ (convex because $\lambda_i^* \geq 0$) and of the $\mu_j^* h_j$ (affine, convex for any sign). For a differentiable convex function, a zero gradient at

x^* implies a *global* minimum: condition (i) then gives

$$L(x^*, \lambda^*, \mu^*) \leq L(x, \lambda^*, \mu^*) \quad \text{for all } x \in \mathbb{R}^n,$$

the second saddle inequality. For the first one, we use (ii) and (iv): for any $\lambda \geq 0, \mu$,

$$L(x^*, \lambda, \mu) = f(x^*) + \sum_i \lambda_i \underbrace{g_i(x^*)}_{\leq 0} + \sum_j \mu_j \underbrace{h_j(x^*)}_{=0} \leq f(x^*) = f(x^*) + \underbrace{\sum_i \lambda_i^* g_i(x^*)}_{=0 \text{ by (iv)}} = L(x^*, \lambda^*, \mu^*).$$

□

Careful (common mistake)

The equivalence “KKT $\Leftrightarrow$ saddle point” **depends crucially on convexity** in the direction KKT $\Rightarrow$ saddle. Without convexity, a KKT point need not even be a local minimum, and even if x^* is the global optimum of the primal there may be no saddle point (there will be a duality gap, see Example 10.17). The typical exam mistake is to invoke “KKT, hence optimal” or “optimal, hence there is λ^* making a saddle” without checking: (1) convexity of f and the g_i , affinity of the h_j , and (2) a constraint qualification such as Slater. Always check both before using strong duality.

10.5 Strong duality and the Slater condition [P2]

10.5.1 The statement

When is the duality gap zero? The classical answer combines convexity with a regularity condition that asks for a *strictly* feasible point.

Definition 10.15 (Slater condition). Problem (10.1), with f, g_i convex, h_j affine (we write the equalities as $Ax = b$) and X convex, satisfies the **Slater condition** if there is a point $\hat{x} \in \text{ri}(X)$ such that

$$g_i(\hat{x}) < 0 \quad \text{for all } i = 1, \dots, m, \quad \text{and} \quad A\hat{x} = b.$$

That is: there is a strictly feasible point (the inequalities hold with slack). If some g_i are affine, it is enough to ask $g_i(\hat{x}) \leq 0$ for them (refined version).

Theorem 10.16 (Strong duality). Suppose problem (10.1) is convex (f, g_i convex, h_j affine, X convex) and satisfies the Slater condition. Then

$$p^* = d^*,$$

and if $p^* > -\infty$ the dual value is attained: there is (λ^*, μ^*) with $\lambda^* \geq 0$ and $q(\lambda^*, \mu^*) = d^* = p^*$. If in addition the primal has a solution x^* , the pair $(x^*, (\lambda^*, \mu^*))$ is a saddle point of the Lagrangian [2, 6].

Note the fine asymmetry: Slater guarantees that the *dual* is attained (there are optimal multipliers), but not that the primal is attained; and the primal does not need to be attained for $p^* = d^*$ to hold.

10.5.2 The geometry: the set of attainable values

The full proof is in [2], section 5.3.2; here we give the schematic argument, which is a direct application of the separation theorem of the separation chapter, and which is worth understanding because *the picture explains all of duality*. To lighten the notation, suppose there is a single inequality constraint ($m = 1$, no equalities).

The set $\mathcal{A}$. We define the set of pairs (violation, cost) that are *attainable or worse*:

$$\mathcal{A} := \{(u, t) \in \mathbb{R}^2 : \exists x \in X \text{ with } g(x) \leq u, f(x) \leq t\}.$$

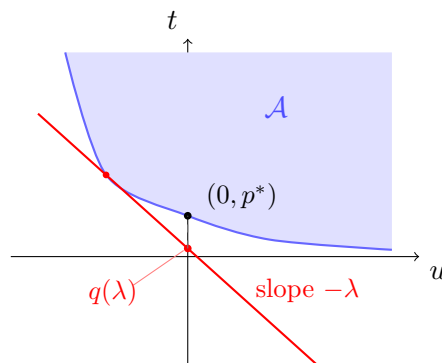
A point $(u, t) \in \mathcal{A}$ means: “there is a decision that violates the constraint by at most u and costs at most t ”. The set $\mathcal{A}$ is closed upward and to the right (if a pair is attainable, every worse pair is attainable too). Two key facts:

- (1) **The primal value is read off the vertical axis:** $p^* = \inf\{t : (0, t) \in \mathcal{A}\}$, because $(0, t) \in \mathcal{A}$ means that there is a feasible x ($g(x) \leq 0$) with $f(x) \leq t$.
- (2) **The dual function is read with supporting lines:** for $\lambda \geq 0$,

$$q(\lambda) = \inf_{x \in X} [f(x) + \lambda g(x)] = \inf \{t + \lambda u : (u, t) \in \mathcal{A}\}.$$

Geometrically: among all lines with slope $-\lambda$ (of the form $t = -\lambda u + \beta$), $q(\lambda)$ is the largest intercept β such that the line stays *below* $\mathcal{A}$. That is, each $\lambda \geq 0$ defines a line that supports $\mathcal{A}$ from below, and $q(\lambda)$ is its intersection with the vertical axis $u = 0$.

Weak duality is now *visible*: the supporting line passes below $\mathcal{A}$, in particular below the point $(0, p^*)$, so its intercept satisfies $q(\lambda) \leq p^*$. And the dual problem consists of *tilting the line* (choosing λ) to push that intercept as high as possible.



The line with slope $-\lambda$ that supports $\mathcal{A}$ from below cuts the axis $u = 0$ at $q(\lambda) \leq p^*$; the dashed distance is the loss of that particular λ . The dual tilts the line looking for the largest intercept, which under convexity and Slater reaches exactly $(0, p^*)$.

Where convexity and Slater enter. If the problem is convex, $\mathcal{A}$ is a *convex* set (an instructive exercise: combine two points of $\mathcal{A}$ using the convexity of f , g and X). The point $(0, p^*)$ is on the boundary of $\mathcal{A}$, so by the supporting hyperplane theorem (separation chapter) there is $(\nu_1, \nu_2) \neq (0, 0)$ with

$$\nu_1 u + \nu_2 t \geq \nu_2 p^* \quad \text{for all } (u, t) \in \mathcal{A}.$$

Since $\mathcal{A}$ is unbounded upward and to the right, we get $\nu_1 \geq 0$ and $\nu_2 \geq 0$ (if some coefficient were negative, the left side would go to $-\infty$). Two cases remain:

- If $\nu_2 > 0$, we normalize $\lambda := \nu_1/\nu_2$ and the inequality becomes $t + \lambda u \geq p^*$ on $\mathcal{A}$, that is $q(\lambda) \geq p^*$; together with weak duality, $q(\lambda) = p^* = d^*$: strong duality, with optimal multiplier λ .
- If $\nu_2 = 0$ (a *vertical* hyperplane), the inequality says $\nu_1 u \geq 0$ on $\mathcal{A}$ with $\nu_1 > 0$, that is $u \geq 0$ for every attainable point. **This is where Slater saves the argument:** the strictly feasible point $\hat{x}$ produces $(g(\hat{x}), f(\hat{x})) \in \mathcal{A}$ with $u = g(\hat{x}) < 0$, a contradiction. Slater rules out the vertical hyperplane.

Without convexity, $\mathcal{A}$ may have a nonconvex “dent” above the vertical axis: no supporting line reaches the point $(0, p^*)$, and the best intercept d^* stays strictly below. That vertical distance *is* the duality gap.

10.5.3 Positive gap without convexity: a worked counterexample

Example 10.17 (Positive duality gap). Consider the *nonconvex* problem (the objective is concave):

$$\min_{x \in X} -x^2 \quad \text{s.t.} \quad x - 1 \leq 0, \quad X = [0, 2].$$

Primal. The feasible points are $x \in [0, 1]$; on them, $-x^2$ is minimized by pushing x as high as possible: $x^* = 1$, $p^* = -1$.

Dual. $L(x, \lambda) = -x^2 + \lambda(x - 1)$. For fixed λ , $L(\cdot, \lambda)$ is *concave* in x , so its minimum over the interval $[0, 2]$ is attained at an endpoint (a concave function on an interval takes its minimum on the boundary):

$$L(0, \lambda) = -\lambda, \quad L(2, \lambda) = -4 + \lambda, \quad \implies \quad q(\lambda) = \min\{-\lambda, \lambda - 4\}.$$

It is the minimum of two lines: concave and piecewise linear (Theorem 10.2 at work again). Its maximum over $\lambda \geq 0$ is where the lines cross: $-\lambda = \lambda - 4$ gives $\lambda^* = 2$ and

$$d^* = q(2) = -2 < -1 = p^*.$$

The gap is $p^* - d^* = 1 > 0$. By Theorem 10.12, this Lagrangian *has no saddle point*, even though the primal has a perfectly well defined solution. Geometrically, the set $\mathcal{A} = \{(u, t) : g(x) \leq u, f(x) \leq t, x \in [0, 2]\}$ has as its lower boundary the curve $t = -(1 + u)^2$ for $u \in [-1, 1]$ (parametrized by $u = x - 1$, $t = -x^2$), which bends *downward*: this is the nonconvex “dent”. The best supporting line from below touches the curve at the endpoints and passes below the point $(0, p^*) = (0, -1)$, cutting the axis at -2 .

Remark 10.18. In economic practice, Slater is almost always harmless: in the consumer’s problem it is enough to have an interior consumption plan that does not exhaust the budget ($p^\top \hat{x} < w$, available if $w > 0$); in the planner’s problem, an allocation with slack in all resources. This is why in micro strong duality is applied without mentioning it, but Example 10.17 shows that *convexity* is not negotiable: with nonconcave utilities or nonconvex production sets (increasing

returns), the “prices that decentralize” may fail to exist, which is exactly the known failure of the second welfare theorem without convexity [9].

10.6 Three worked examples: consumer, QP and minimax [P2]

10.6.1 The dual of the consumer problem and Marshall–Hicks duality

Example 10.19 (Dual of the UMP). The consumer’s utility maximization problem (UMP) with utility u strictly increasing and concave, prices $p \gg 0$ and wealth $w > 0$ is

$$\max_{x \in \mathbb{R}_+^n} u(x) \quad \text{s.t.} \quad p^\top x \leq w.$$

In minimization form: $\min_{x \in X} -u(x)$ s.t. $p^\top x - w \leq 0$, with $X = \mathbb{R}_+^n$. The Lagrangian and the dual function:

$$L(x, \lambda) = -u(x) + \lambda(p^\top x - w), \quad q(\lambda) = -\lambda w - \sup_{x \geq 0} [u(x) - \lambda p^\top x].$$

Note the structure of the inner supremum: it is a “profit problem” in which the consumer *sells utility at price 1 and buys goods at prices λp* . Let us make it concrete with the log Cobb–Douglas utility, $u(x_1, x_2) = \alpha \ln x_1 + (1 - \alpha) \ln x_2$ with $\alpha \in (0, 1)$ (we take $X = \mathbb{R}_{++}^2$; the constraint $x \geq 0$ is never active because the logarithm has infinite slope at zero).

Step 1: the inner supremum. It is separable. For good 1: $\max_{x_1 > 0} \alpha \ln x_1 - \lambda p_1 x_1$; the first order condition $\alpha/x_1 = \lambda p_1$ gives $x_1(\lambda) = \alpha/(\lambda p_1)$, with value $\alpha \ln \frac{\alpha}{\lambda p_1} - \alpha$. Similarly for good 2. Then

$$q(\lambda) = -\lambda w - \alpha \ln \frac{\alpha}{\lambda p_1} - (1 - \alpha) \ln \frac{1 - \alpha}{\lambda p_2} + 1.$$

Step 2: maximize the dual. Differentiating with respect to λ (use that $-\alpha \ln \frac{\alpha}{\lambda p_1}$ contributes α/λ , and similarly for the other term):

$$q'(\lambda) = -w + \frac{\alpha}{\lambda} + \frac{1 - \alpha}{\lambda} = \frac{1}{\lambda} - w = 0 \quad \implies \quad \lambda^* = \frac{1}{w}.$$

The optimal multiplier is the *marginal utility of wealth* (with logarithmic u , exactly $1/w$).

Step 3: recover the primal. Minimizing $L(\cdot, \lambda^*)$ (that is, evaluating the inner supremum at λ^*) gives

$$x_1^* = \frac{\alpha}{\lambda^* p_1} = \frac{\alpha w}{p_1}, \quad x_2^* = \frac{(1 - \alpha)w}{p_2},$$

the Cobb–Douglas Marshallian demands: spend the fraction α of wealth on good 1. One checks $p^\top x^* = w$ (active constraint, consistent with $\lambda^* > 0$) and, substituting, $-q(\lambda^*)$ is exactly the indirect utility $v(p, w) = \alpha \ln \frac{\alpha w}{p_1} + (1 - \alpha) \ln \frac{(1 - \alpha)w}{p_2}$: zero gap, as Slater guarantees (the plan $\hat{x} = (w/(3p_1), w/(3p_2))$ is strictly feasible).

Remark 10.20 (Marshall–Hicks duality, stated precisely). In consumer theory (MWG ch. 3) the name “duality” is used for the relation between the UMP and the expenditure minimization problem (EMP):

$$\min_{x \geq 0} p^\top x \quad \text{s.t.} \quad u(x) \geq \bar{u},$$

whose solutions are the Hicksian demands. Let us be precise about what kind of duality this is, because it is *not* literally Lagrangian duality: the Lagrange dual of the UMP is a problem in the scalar variable λ (the one we just solved), not the EMP. The correct relation is the following:

- UMP and EMP are **inverse problems**: they swap objective and constraint. The UMP maximizes u with expenditure bounded by w ; the EMP minimizes expenditure with u bounded below by $\bar{u}$. If $\bar{u} = v(p, w)$, both have the same solution x^* (with u increasing and continuous): $x^m(p, w) = x^h(p, v(p, w))$ and $e(p, v(p, w)) = w$ [9, 10].
- **Lagrangian duality is what links their multipliers**. In the UMP, KKT gives $\nabla u(x^*) = \lambda^* p$; in the EMP (with multiplier ρ for $\bar{u} - u(x) \leq 0$), KKT gives $p = \rho^* \nabla u(x^*)$. Comparing: $\rho^* = 1/\lambda^*$. The UMP multiplier is the marginal utility of wealth ($\partial v/\partial w = \lambda^*$, envelope chapter); the EMP one is the marginal cost of utility ($\partial e/\partial \bar{u} = \rho^*$). They are reciprocals because measuring “utility per sol” and “soles per util” is the same rate seen from the two sides.
- In both problems, the pair (solution, multiplier) is a *saddle point* of the respective Lagrangian (concave utility, linear or convex constraints, easy Slater), and the Roy and Shephard identities that you will use in demand theory are consequences of reading those multipliers as derivatives of the value functions [11]. In Example 10.19: $\lambda^* = 1/w = \partial v/\partial w$, exactly.

Moral: the Marshall–Hicks pair is a *manifestation* of the primal-dual structure (constraint and objective swapped, reciprocal shadow prices, same saddle point), and this is why textbooks call it duality, even though the “formal dual” of each problem is the Lagrange one.

10.6.2 Dual of a QP with linear constraints

Example 10.21 (QP: full computation). Let us solve by duality the problem

$$\min_{x \in \mathbb{R}^2} \frac{1}{2}(x_1^2 + x_2^2) \quad \text{s.t.} \quad x_1 + x_2 \geq 2,$$

that is $g(x) = 2 - x_1 - x_2 \leq 0$ (geometrically: the point of the half plane $x_1 + x_2 \geq 2$ closest to the origin, a projection, as in the optimality chapter).

Dual function. $L(x, \lambda) = \frac{1}{2}(x_1^2 + x_2^2) + \lambda(2 - x_1 - x_2)$, convex and differentiable in x ; the first order conditions $x_1 - \lambda = 0$, $x_2 - \lambda = 0$ give $x(\lambda) = (\lambda, \lambda)$ and

$$q(\lambda) = \frac{1}{2}(\lambda^2 + \lambda^2) + \lambda(2 - 2\lambda) = 2\lambda - \lambda^2.$$

Dual. $\max_{\lambda \geq 0} 2\lambda - \lambda^2$: the first order condition $2 - 2\lambda = 0$ gives $\lambda^* = 1$, $d^* = 1$.

Recovery and check. $x^* = x(\lambda^*) = (1, 1)$, feasible ($1 + 1 = 2$), with $p^* = \frac{1}{2}(1 + 1) = 1 = d^*$: zero gap (the problem is convex and Slater holds with $\hat{x} = (2, 2)$). The pair $((1, 1), 1)$ is a saddle point, and $\lambda^* = 1$ is the shadow price: relaxing the constraint to $x_1 + x_2 \geq 2 - \varepsilon$ would reduce the optimal cost at a (marginal) rate of 1 per unit of ε .

The general case, for your toolbox. For

$$\min_x \frac{1}{2}x^\top Qx + c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad Q \text{ symmetric positive definite,}$$

the same computation (solve $\nabla_x L = Qx + c + A^\top \lambda = 0$, that is $x(\lambda) = -Q^{-1}(c + A^\top \lambda)$, and substitute) gives the concave quadratic dual function

$$q(\lambda) = -\frac{1}{2}(c + A^\top \lambda)^\top Q^{-1}(c + A^\top \lambda) - b^\top \lambda,$$

so the dual of a QP is another QP (concave), but with the simple constraint $\lambda \geq 0$ instead of $Ax \leq b$: this is why many algorithms solve the dual [5, 2].

10.6.3 Von Neumann's minimax theorem

We close with the historical gem: zero sum games, where duality *is* the equilibrium. Consider a matrix game: the row player chooses a row $i \in \{1, \dots, m\}$, the column player chooses a column $k \in \{1, \dots, n\}$, and the column pays a_{ik} to the row (zero sum: what one wins, the other loses). With *mixed strategies* $s \in \Delta_m := \{s \geq 0 : \sum_i s_i = 1\}$ for the row and $y \in \Delta_n$ for the column, the expected payoff is

$$s^\top Ay = \sum_{i,k} s_i a_{ik} y_k.$$

The row wants to maximize it; the column, to minimize it. If the row moves “first” it guarantees $\max_s \min_y s^\top Ay$; if it moves “second” it can aim at $\min_y \max_s s^\top Ay$. It always holds (this is weak duality; see exercise 10.28) that

$$\max_{s \in \Delta_m} \min_{y \in \Delta_n} s^\top Ay \leq \min_{y \in \Delta_n} \max_{s \in \Delta_m} s^\top Ay.$$

Theorem 10.22 (von Neumann minimax, 1928). *For every matrix $A \in \mathbb{R}^{m \times n}$,*

$$\max_{s \in \Delta_m} \min_{y \in \Delta_n} s^\top Ay = \min_{y \in \Delta_n} \max_{s \in \Delta_m} s^\top Ay =: v,$$

the value of the game. Equivalently, there is a saddle point (s^, y^*) of the bilinear function $(s, y) \mapsto s^\top Ay$ over $\Delta_m \times \Delta_n$:*

$$s^\top Ay^* \leq (s^*)^\top Ay^* \leq (s^*)^\top Ay \quad \text{for all } s \in \Delta_m, y \in \Delta_n.$$

Proof idea (through linear duality). Against a fixed s , the column minimizes a linear function of y over the simplex, and the minimum is attained at a vertex (a pure column): $\min_{y \in \Delta_n} s^\top Ay = \min_k (A^\top s)_k$. So the row's problem is

$$\max_{s \in \Delta_m} \min_k (A^\top s)_k \iff \max_{s,v} v \quad \text{s.t.} \quad A^\top s \geq v\mathbf{1}, \quad s \in \Delta_m,$$

a *linear* problem (the epigraph trick: raise a bound v below all the components). The analogous problem for the column, $\min_{y,w} w$ s.t. $Ay \leq w\mathbf{1}$, $y \in \Delta_n$, turns out to be *exactly the linear programming dual* of the row's problem (a direct computation with Example 10.4). Both linear programs are feasible and bounded, and strong duality for linear programming (which does not need Slater: for affine constraints strong duality holds whenever p^* is finite, see [2]) gives equality of the values. The equivalence between “equal values” and “there is a saddle point” is Theorem 10.12 in minimax form. $\square$

Example 10.23 (2×2 game solved with the primal-dual pair). Take the zero sum game with payoff matrix (to the row)

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}.$$

There is no saddle point in pure strategies: the best pure guarantee of the row is $\max_i \min_k a_{ik} = \max\{1, 0\} = 1$, and that of the column is $\min_k \max_i a_{ik} = \min\{3, 2\} = 2$; since $1 < 2$, mixing is needed.

Row problem (primal). With $s = (\sigma, 1 - \sigma)$, the payoff against each pure column is

$$\text{col. 1: } 3\sigma + 0(1 - \sigma) = 3\sigma, \quad \text{col. 2: } \sigma + 2(1 - \sigma) = 2 - \sigma.$$

The row guarantees $\min\{3\sigma, 2 - \sigma\}$, a piecewise concave function of σ that is maximized where the two lines cross (if the crossing falls in $[0, 1]$): $3\sigma = 2 - \sigma$ gives $\sigma^* = 1/2$, with guaranteed value $v = 3/2$. Optimal strategy of the row: $s^* = (1/2, 1/2)$.

Column problem (dual). With $y = (\tau, 1 - \tau)$, the payoff against each pure row is

$$\text{row 1: } 3\tau + (1 - \tau) = 1 + 2\tau, \quad \text{row 2: } 2(1 - \tau) = 2 - 2\tau.$$

The column makes sure it pays at most $\max\{1 + 2\tau, 2 - 2\tau\}$, piecewise convex, minimized at the crossing: $1 + 2\tau = 2 - 2\tau$ gives $\tau^* = 1/4$, with value $1 + 2(1/4) = 3/2$. Optimal strategy: $y^* = (1/4, 3/4)$.

Saddle point check. Both values coincide: $v = 3/2$ (zero gap, as the theorem requires). And $(s^*)^\top A y^* = \frac{1}{2}(1 + 2 \cdot \frac{1}{4}) + \frac{1}{2}(2 - 2 \cdot \frac{1}{4}) = \frac{1}{2} \cdot \frac{3}{2} + \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{2}$. Against y^* , *any* row mixture yields exactly $3/2$ (both rows yield $3/2$ against y^*): the *indifference* property that you will use constantly in game theory to compute mixed equilibria is, here, the complementary slackness of the primal-dual pair: the pure strategies played with positive probability must be optimal against the rival's mixture.

Remark 10.24 (Note for game theory). The minimax theorem is the “zero sum” case of Nash equilibrium, and it was proved by von Neumann (1928) almost two decades before Nash's theorem (1950): historically, *game theory started as a duality theorem*. In MWG ch. 8 you will see that in nonzero sum games the value stops existing and Kakutani's fixed point theorem is needed; but for strictly competitive games, all you need is this chapter: value of the game = common primal-dual value, optimal strategies = primal and dual solutions, indifference = complementary slackness.

10.7 Perturbation: the deep geometry of duality [P2]

We finish with the point of view that unifies everything above and that prepares the envelope chapter: instead of solving *one* problem, we solve a whole *family* of problems indexed by how much we relax the constraints, and we study how the value changes.

Definition 10.25 (Perturbation function or value function). For problem (10.1) (without equal-

ities, for simplicity), the **perturbation function** or **value function** is

$$p(u) := \inf \{f(x) : x \in X, g_i(x) \leq u_i, i = 1, \dots, m\}, \quad u \in \mathbb{R}^m,$$

with $p(u) = +\infty$ if there are no feasible points for that u . Note that $p(0) = p^*$ is the value of the original problem; $u_i > 0$ *relaxes* constraint i and $u_i < 0$ *tightens* it. By construction, p is nonincreasing in each u_i (relaxing never hurts).

Proposition 10.26 (Convexity of the value function). *If X is convex and $f, g_1, \dots, g_m$ are convex, then p is a convex function of u .*

Proof idea. The region “above” p is essentially the set $\mathcal{A} = \{(u, t) : \exists x \in X, g(x) \leq u, f(x) \leq t\}$ of section 10.5 (indeed, $p(u) = \inf\{t : (u, t) \in \mathcal{A}\}$, so $\text{epi}(p)$ and $\mathcal{A}$ agree except for boundary points). One checks that $\mathcal{A}$ is convex: if (u^1, t^1) and (u^2, t^2) are in $\mathcal{A}$ with witnesses x^1, x^2 , then for $x_t = tx^1 + (1-t)x^2 \in X$ the convexity of g_i and of f gives $g_i(x_t) \leq tu_i^1 + (1-t)u_i^2$ and $f(x_t) \leq tt^1 + (1-t)t^2$. A function whose epigraph is convex is convex (chapter on convex functions). $\square$

Theorem 10.27 (Multipliers as subgradients of the value function). *Suppose the problem is convex with strong duality and optimal dual multiplier $\lambda^* \geq 0$. Then, for every $u \in \mathbb{R}^m$,*

$$p(u) \geq p(0) - \langle \lambda^*, u \rangle,$$

that is, $-\lambda^ \in \partial p(0)$: the vector $-\lambda^*$ is a subgradient of the value function at $u = 0$. If in addition p is differentiable at 0, then*

$$\lambda_i^* = -\frac{\partial p}{\partial u_i}(0), \quad i = 1, \dots, m :$$

multiplier i is exactly the rate at which the optimal cost falls when constraint i is relaxed: the shadow price.

Proof. Fix u and let x be any point with $x \in X, g(x) \leq u$. Then

$$f(x) + \langle \lambda^*, u \rangle \geq f(x) + \langle \lambda^*, g(x) \rangle \geq \inf_{z \in X} L(z, \lambda^*) = q(\lambda^*) = p(0),$$

where the first inequality uses $\lambda^* \geq 0$ and $g(x) \leq u$ componentwise, and the last equality is strong duality. Taking the infimum over the x admissible for u : $p(u) + \langle \lambda^*, u \rangle \geq p(0)$, which is the subgradient inequality. The differentiable case is immediate: if p is differentiable at 0, its only subgradient is the gradient (subdifferential chapter), hence $\nabla p(0) = -\lambda^*$. $\square$

Intuition

This is the best way to *remember* all of duality at once. The value function $p(u)$ is a convex, decreasing surface; at $u = 0$ sits the real problem. The dual function with multiplier λ is the best affine lower bound with slope $-\lambda$ below that surface, and in fact $q(\lambda) = \inf_u [p(u) + \langle \lambda, u \rangle]$: the dual “feels” the value function with hyperplanes. Strong duality = at $u = 0$ the surface admits a nonvertical supporting hyperplane, and its slope is

$-\lambda^*$. This is why the multiplier *is* a price: at the margin it measures how much one more unit of the resource (relaxation) is worth. The envelope chapter will turn this observation into the envelope theorem (the derivatives of value functions such as $v(p, w)$ or $e(p, u)$ with respect to their parameters), which in micro produces Roy, Shephard and Hotelling [11]. And the connection with Example 10.17: there the value function is $p(u) = -(1+u)^2$ for $u \in [-1, 1]$, concave instead of convex; at $u = 0$ there is no affine minorant touching it, and this is why there is a gap.

10.8 Summary and exercises [P2]

Chapter summary

- **Dual function:** $q(\lambda, \mu) = \inf_{x \in X} L(x, \lambda, \mu)$. It is *always concave* (pointwise infimum of functions affine in (λ, μ)), whether the primal is convex or not. The dual problem is $\max_{\lambda \geq 0, \mu} q(\lambda, \mu)$, with value d^* .
- **Weak duality:** $q(\lambda, \mu) \leq p^*$ for every $\lambda \geq 0$ (proof: infimum $\leq$ value at a feasible point, and at a feasible point the fines are ≤ 0). Hence $d^* \leq p^*$; the difference $p^* - d^* \geq 0$ is the duality gap. A feasible primal-dual pair with $f(\hat{x}) = q(\hat{\lambda}, \hat{\mu})$ certifies that both are optimal.
- **Economic reading:** the multipliers are shadow prices of the resources; the dual is the problem of a market that looks for the minimum valuation of the endowments without leaving arbitrage (no activity strictly profitable). Kantorovich–Koopmans, Nobel 1975.
- **Saddle point:** $L(x^*, \lambda, \mu) \leq L(x^*, \lambda^*, \mu^*) \leq L(x, \lambda^*, \mu^*)$. Central theorem: there is a saddle point $\iff x^*$ primal optimal + (λ^*, μ^*) dual optimal + zero gap (with no convexity assumption). Under convexity and differentiability, saddle point $\iff$ KKT.
- **Strong duality:** convex problem + Slater (a strictly feasible point) $\Rightarrow p^* = d^*$ and the dual is attained. Proof through a supporting hyperplane to the set $\mathcal{A}$ of pairs (violation, cost) that are attainable; Slater rules out the vertical hyperplane [2]. Without convexity the gap can be positive (Example 10.17: $p^* = -1 > d^* = -2$).
- **Worked cases:** dual of a linear program = transposed linear program (no arbitrage); dual of a positive definite QP = concave QP in $\lambda \geq 0$; UMP: λ^* = marginal utility of wealth, and Marshall–Hicks duality is the primal-dual structure with reciprocal multipliers; von Neumann minimax = linear programming duality between the two players' problems, value of the game = common value.
- **Perturbation:** $p(u) = \inf\{f : g \leq u\}$ is convex if the problem is; $-\lambda^* \in \partial p(0)$, and with differentiability $\lambda_i^* = -\partial p / \partial u_i(0)$: a direct bridge to the envelope theorem.

Exercise 10.28. (General max-min inequality.) Let $F : S \times Y \rightarrow \mathbb{R}$ be any function and let

S, Y be nonempty sets. Prove that

$$\sup_{s \in S} \inf_{y \in Y} F(s, y) \leq \inf_{y \in Y} \sup_{s \in S} F(s, y),$$

and explain why weak duality (Theorem 10.5) is a particular case.

Solution. Fix arbitrary $s \in S$ and $y \in Y$. By definition of infimum, $\inf_{y'} F(s, y') \leq F(s, y)$; by definition of supremum, $F(s, y) \leq \sup_{s'} F(s', y)$. Chaining these,

$$\inf_{y'} F(s, y') \leq \sup_{s'} F(s', y) \quad \text{for all } s, y.$$

The left side does not depend on y and the right side does not depend on s : taking the supremum over s on the left (the right side is a fixed upper bound) and then the infimum over y on the right, we get the required inequality. (The lemma in words: *whoever moves second has the advantage*.)

For weak duality, take $F((\lambda, \mu), x) = L(x, \lambda, \mu)$ with $S = \mathbb{R}_+^m \times \mathbb{R}^p$ and $Y = X$. The left side is $\sup_{\lambda \geq 0, \mu} \inf_x L = d^*$. For the right side, compute the inner supremum: if x is feasible, $\sup_{\lambda \geq 0, \mu} L(x, \lambda, \mu) = f(x)$ (the optimal fines are zero); if x violates some constraint, that supremum is $+\infty$ (infinite fine). Therefore $\inf_x \sup_{\lambda, \mu} L = \inf\{f(x) : x \text{ feasible}\} = p^*$, and the max-min inequality is exactly $d^* \leq p^*$. (In passing: this shows that the primal is the “inf sup” problem and the dual the “sup inf” problem of the same Lagrangian, and that strong duality means being able to swap the order.) $\square$

Exercise 10.29. (Dual of a linear program in inequality form.) Derive the dual problem of

$$\min_{x \in \mathbb{R}^n} c^\top x \quad \text{s.t.} \quad Ax \geq b, \quad x \geq 0,$$

with $A \in \mathbb{R}^{m \times n}$, and give the no arbitrage reading of its constraints.

Solution. We write the constraints as $b - Ax \leq 0$ (multipliers $\lambda \in \mathbb{R}_+^m$) and $-x \leq 0$ (multipliers $s \in \mathbb{R}_+^n$):

$$L(x, \lambda, s) = c^\top x + \lambda^\top (b - Ax) - s^\top x = b^\top \lambda + (c - A^\top \lambda - s)^\top x.$$

Affine in x : the infimum over $x \in \mathbb{R}^n$ is $-\infty$ unless $c - A^\top \lambda - s = 0$, in which case $q(\lambda, s) = b^\top \lambda$. Since $s \geq 0$ acts as slack, the condition is the same as $A^\top \lambda \leq c$. The dual is

$$\max_{\lambda \geq 0} b^\top \lambda \quad \text{s.t.} \quad A^\top \lambda \leq c.$$

Reading (think of x as levels of activities that must cover requirements b , with unit costs c): λ_i is the shadow price of requirement i ; the dual constraint $(A^\top \lambda)_k \leq c_k$ says that the value imputed to what one unit of activity k produces cannot exceed its cost, that is, no activity can generate value strictly greater than its cost (no arbitrage), and the objective looks for the maximum valuation of the requirements compatible with that. At the optimum, complementary slackness: activities that are used have exact imputation, requirements with slack have price zero. $\square$

Exercise 10.30. (QP with an equality.) Solve by duality the problem

$$\min_{x \in \mathbb{R}^2} x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + 2x_2 = 5 :$$

compute the dual function $q(\mu)$, solve the dual, recover x^* , check that there is no gap, and verify the reading of μ^* as a shadow price by recomputing the optimal value with the constraint $x_1 + 2x_2 = 5 + \varepsilon$.

Solution. *Dual function.* With multiplier $\mu \in \mathbb{R}$ (free: the constraint is an equality), $L(x, \mu) = x_1^2 + x_2^2 + \mu(5 - x_1 - 2x_2)$, convex in x . The first order conditions $2x_1 - \mu = 0$ and $2x_2 - 2\mu = 0$ give $x(\mu) = (\mu/2, \mu)$, and

$$q(\mu) = \frac{\mu^2}{4} + \mu^2 + \mu \left(5 - \frac{\mu}{2} - 2\mu \right) = \frac{5\mu^2}{4} + 5\mu - \frac{5\mu^2}{2} = 5\mu - \frac{5\mu^2}{4},$$

concave, as it must be.

Dual and recovery. $q'(\mu) = 5 - \frac{5}{2}\mu = 0$ gives $\mu^* = 2$ and $d^* = 10 - 5 = 5$. Then $x^* = x(\mu^*) = (1, 2)$, which is feasible ($1 + 4 = 5$) with value $p^* = 1 + 4 = 5 = d^*$: zero gap (by Corollary 10.7 this certifies that x^* and μ^* are optimal; for affine constraints strong duality was guaranteed). Geometrically, x^* is the projection of the origin onto the line.

Shadow price. With the constraint $x_1 + 2x_2 = 5 + \varepsilon$, the same computation (or the projection formula: the squared distance from the origin to the line $a^\top x = \beta$ is $\beta^2 / \|a\|^2$ with $a = (1, 2)$, $\|a\|^2 = 5$) gives the value function

$$p(\varepsilon) = \frac{(5 + \varepsilon)^2}{5}, \quad p'(0) = \frac{2 \cdot 5}{5} = 2 = \mu^*.$$

(With our sign convention for equalities, μ^* measures the *increase* in cost when the right hand side increases; the sign depends on how h is written, so in an exam always state it.) The derivative of the value function at the point of interest coincides with the multiplier: Theorem 10.27. $\square$

Exercise 10.31. (Positive gap.) Consider $\min_{x \in [0, 1]} -x^2$ s.t. $x \leq \frac{1}{2}$ (that is, $g(x) = x - \frac{1}{2} \leq 0$, $X = [0, 1]$). Compute p^* , the dual function $q(\lambda)$, the dual value d^* and the gap. Is there a saddle point of the Lagrangian? Justify with the relevant theorem.

Solution. *Primal.* Feasible points: $x \in [0, \frac{1}{2}]$. Minimizing $-x^2$ pushes to $x^* = \frac{1}{2}$: $p^* = -\frac{1}{4}$.

Dual. $L(x, \lambda) = -x^2 + \lambda(x - \frac{1}{2})$ is concave in x , so its minimum over the interval $[0, 1]$ is at an endpoint:

$$L(0, \lambda) = -\frac{\lambda}{2}, \quad L(1, \lambda) = -1 + \frac{\lambda}{2} \quad \implies \quad q(\lambda) = \min \left\{ -\frac{\lambda}{2}, \frac{\lambda}{2} - 1 \right\}.$$

The first branch decreases and the second increases; the maximum is at the crossing $-\lambda/2 = \lambda/2 - 1$, that is $\lambda^* = 1$, with $d^* = q(1) = -\frac{1}{2}$.

Gap and saddle point. $p^* - d^* = -\frac{1}{4} + \frac{1}{2} = \frac{1}{4} > 0$. By Theorem 10.12, a saddle point exists if and only if there are primal and dual optima *with zero gap*; since the gap is positive, **there is no saddle point**, even though the primal has a solution. The culprit is the nonconvexity of

the objective: the value function $p(u) = -(\frac{1}{2} + u)^2$ (for $u \in [-\frac{1}{2}, \frac{1}{2}]$) is concave, and at $u = 0$ it admits no affine minorant touching it, which is exactly what the multiplier would need to be. $\square$

Exercise 10.32. (Log-linear consumer.) For $u(x_1, x_2) = \frac{1}{2} \ln x_1 + \frac{1}{2} \ln x_2$, prices $p_1, p_2 > 0$ and wealth $w > 0$:

- (a) set up the UMP as a minimization problem and compute its dual function $q(\lambda)$;
- (b) solve the dual and show that $\lambda^* = 1/w$;
- (c) recover the Marshallian demands by minimizing $L(\cdot, \lambda^*)$ and check that there is no gap;
- (d) set up the EMP with $\bar{u} = v(p, w)$ and check that its multiplier is $\rho^* = 1/\lambda^* = w$.

Solution. (a) The UMP is $\min_{x>0} \left[-\frac{1}{2} \ln x_1 - \frac{1}{2} \ln x_2 \right]$ s.t. $p^\top x - w \leq 0$. The Lagrangian $L(x, \lambda) = -\frac{1}{2} \ln x_1 - \frac{1}{2} \ln x_2 + \lambda(p^\top x - w)$ is convex in x ; the first order conditions $\frac{1}{2x_k} = \lambda p_k$ give $x_k(\lambda) = \frac{1}{2\lambda p_k}$. Substituting,

$$q(\lambda) = \frac{1}{2} \ln(2\lambda p_1) + \frac{1}{2} \ln(2\lambda p_2) + \lambda \left(\frac{1}{2\lambda} + \frac{1}{2\lambda} \right) - \lambda w = \ln(2\lambda) + \frac{1}{2} \ln(p_1 p_2) + 1 - \lambda w.$$

(b) $q'(\lambda) = 1/\lambda - w = 0$ gives $\lambda^* = 1/w$ (> 0), and q is concave for $\lambda > 0$ (second derivative $-1/\lambda^2 < 0$), so this is the maximum of the dual.

(c) $x_k^* = x_k(\lambda^*) = \frac{w}{2p_k}$ for $k = 1, 2$: the Cobb–Douglas demands with shares $\frac{1}{2}$. They are feasible with equality: $p^\top x^* = \frac{w}{2} + \frac{w}{2} = w$. The primal value is $-u(x^*) = -v(p, w)$ with $v(p, w) = \frac{1}{2} \ln \frac{w}{2p_1} + \frac{1}{2} \ln \frac{w}{2p_2} = \ln \frac{w}{2} - \frac{1}{2} \ln(p_1 p_2)$; and $q(\lambda^*) = \ln \frac{2}{w} + \frac{1}{2} \ln(p_1 p_2) + 1 - 1 = -v(p, w)$. They are equal: zero gap (consistent with Slater: an $\hat{x}$ with $p^\top \hat{x} < w$ exists because $w > 0$).

(d) EMP: $\min_{x>0} p^\top x$ s.t. $\bar{u} - u(x) \leq 0$ with $\bar{u} = v(p, w)$. KKT: $p_k = \rho^* \cdot \frac{1}{2x_k^*}$, and the Hicksian solution coincides with x^* from part (c) (same indifference curve, same tangency). From $p_k = \rho^*/(2x_k^*)$ with $x_k^* = w/(2p_k)$ we get $\rho^* = 2p_k x_k^* = w$. Indeed $\rho^* = w = 1/\lambda^*$: reciprocal multipliers, as Remark 10.20 states: “extra utility per sol” ($1/w$) and “soles per extra util” (w) are the same rate inverted. $\square$

Exercise 10.33. (Zero sum game.) The row and the column play the game with payoff matrix to the row

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}.$$

Show that there is no saddle point in pure strategies, find the optimal mixed strategies s^*, y^* and the value v of the game by solving the primal-dual pair, and check the indifference property.

Solution. *Pure strategies.* Row guarantee: $\max_i \min_k a_{ik} = \max\{0, 1\} = 1$ (row 2). Column guarantee: $\min_k \max_i a_{ik} = \min\{2, 3\} = 2$ (column 1). Since $1 < 2$, there is no pure saddle point.

Row (primal). With $s = (\sigma, 1 - \sigma)$: against column 1 it gets $2\sigma + (1 - \sigma) = 1 + \sigma$; against column 2 it gets $3(1 - \sigma) = 3 - 3\sigma$. It maximizes $\min\{1 + \sigma, 3 - 3\sigma\}$: the crossing $1 + \sigma = 3 - 3\sigma$ gives $\sigma^* = \frac{1}{2}$, value $v = \frac{3}{2}$. Hence $s^* = (\frac{1}{2}, \frac{1}{2})$.

Column (dual). With $y = (\tau, 1 - \tau)$: row 1 gets 2τ ; row 2 gets $\tau + 3(1 - \tau) = 3 - 2\tau$. The column minimizes $\max\{2\tau, 3 - 2\tau\}$: crossing $2\tau = 3 - 2\tau$, that is $\tau^* = \frac{3}{4}$, value $2 \cdot \frac{3}{4} = \frac{3}{2}$. Hence $y^* = (\frac{3}{4}, \frac{1}{4})$.

Value and indifference. Both problems have value $v = \frac{3}{2}$: zero gap, and by Theorem 10.22 (s^*, y^*) is a saddle point of $s^\top Ay$. Indifference: against $y^* = (\frac{3}{4}, \frac{1}{4})$, row 1 yields $2 \cdot \frac{3}{4} = \frac{3}{2}$ and row 2 yields $\frac{3}{4} + 3 \cdot \frac{1}{4} = \frac{3}{2}$, equal, which is why the row can mix; against $s^* = (\frac{1}{2}, \frac{1}{2})$, column 1 costs $\frac{1}{2}(2 + 1) = \frac{3}{2}$ and column 2 costs $\frac{1}{2}(0 + 3) = \frac{3}{2}$, equal as well. Every pure strategy used with positive probability is a best response: complementary slackness of the primal-dual pair, and the standard method in game theory for computing mixed equilibria. $\square$

Chapter 11

Envelope theorems and sensitivity analysis [P1]

Between Lagrangian duality and Fenchel conjugation. Bibliography: Aragón et al. [1]; Milgrom and Segal [11].

What is this used for in Microeconomics?

The envelope theorem is possibly **the** most used theorem in all of microeconomics. You will use it constantly and in every part of the course. In **individual choice** (consumer and firm, MWG chs. 3 and 5) it is the machine that produces the three central results of duality: *Shephard's lemma* (differentiating the expenditure function gives Hicksian demand), *Hotelling's lemma* (differentiating the profit function gives supply and input demands) and *Roy's identity* (a ratio of derivatives of indirect utility gives Marshallian demand); combined with the curvature properties of value functions, it also delivers the law of supply and the Slutsky matrix, symmetric and negative semidefinite. In **welfare and incentives** (mechanism design and auctions, MWG ch. 23), the integral version of Milgrom and Segal (2002) is the one that generates the *payment formula* and Myerson's *revenue equivalence theorem*: the equilibrium utility of a bidder is an integral of the probability of winning, and therefore the expected revenue of the seller is pinned down by the allocation rule. In **market failures** (contracts, moral hazard and adverse selection, MWG chs. 13 and 14), the agent's "information rent" is characterized by applying the envelope to the incentive compatibility (IC) constraint: the utility of type θ is the integral of the partial derivatives with respect to the type. Finally, *sensitivity analysis* (the Lagrange multiplier as a *shadow price*) is the economic reading of the KKT conditions of the previous chapter: λ^* measures how much, in units of the objective, one extra unit of a resource is worth. Master this chapter and you will have mastered the grammar in which modern micro is written.

This chapter answers a question that looks modest:

How does the optimal value of an optimization problem change when a parameter changes?

The answer, surprisingly simple, is that to first order *only the direct effect of the parameter*

matters: the indirect effect, through the readjustment of the optimal solution, is second order and disappears. That observation, once formalized, is the envelope theorem. The whole chapter is a systematic exploitation of that idea.

11.1 Value functions [P1]

11.1.1 Definition and examples you already know

Definition 11.1 (Value function). Let $f : \mathbb{R}^n \times \Theta \rightarrow \mathbb{R}$ be an objective function that depends on a decision variable $x \in \mathbb{R}^n$ and on a *parameter* $\theta \in \Theta \subseteq \mathbb{R}^k$, and let $\Gamma(\theta) \subseteq \mathbb{R}^n$ be the feasible set (which may depend on the parameter). The *value function* of the problem

$$\max_x f(x, \theta) \quad \text{subject to} \quad x \in \Gamma(\theta)$$

is the function that assigns to each parameter the best attainable value:

$$V(\theta) := \sup_{x \in \Gamma(\theta)} f(x, \theta).$$

The set of optimal solutions is written $X^*(\theta) := \arg \max_{x \in \Gamma(\theta)} f(x, \theta)$ and is called the *argmax correspondence*; when it is a single point for each θ we write $x(\theta)$ and speak of the *policy function*.

Two comments on the definition. First, we use $\sup$ and not $\max$ because in general the maximum may fail to be attained; when $\Gamma(\theta)$ is compact and $f(\cdot, \theta)$ is continuous, the Weierstrass theorem (preliminaries chapter) guarantees that the supremum is attained and we can write $\max$ with a clear conscience. Second, the value function does *not* depend on x : the decision variable has already been “maximized out”. V only sees the parameters.

The reason this object is central in economics is that **you already know several value functions**, even if you never called them that:

Example 11.2 (The four value functions of microeconomics). Let $p \in \mathbb{R}_{++}^n$ be a vector of goods prices, $w \in \mathbb{R}_{++}^m$ a vector of input prices, $m > 0$ the consumer’s income, $u(\cdot)$ a utility function and Y a production set (or F a production function).

- (a) **Indirect utility**: the value of the consumer’s problem,

$$v(p, m) := \max_{x \geq 0} u(x) \quad \text{s.t.} \quad \langle p, x \rangle \leq m.$$

Here the decision is the bundle x and the parameters are (p, m) .

- (b) **Expenditure function**: the value of the dual problem of minimizing expenditure to reach utility u ,

$$e(p, u) := \min_{x \geq 0} \langle p, x \rangle \quad \text{s.t.} \quad u(x) \geq u.$$

It is a *minimization* problem, with parameters (p, u) . Its optimal solution $h(p, u)$ is the *Hicksian or compensated demand*.

- (c) **Profit function**: for a firm with production set $Y \subseteq \mathbb{R}^n$ (with the sign convention: inputs

negative, outputs positive),

$$\pi(p) := \sup_{y \in Y} \langle p, y \rangle.$$

In the version with a single output production function: $\pi(p, w) = \max_{x \geq 0} p F(x) - \langle w, x \rangle$. Note that $\pi(p) = \sup_{y \in Y} \langle p, y \rangle$ is exactly the *support function* of Y that we studied in the chapter on convex functions: the theory of the competitive firm is convex analysis applied.

(d) **Cost function:** the value of minimizing the cost of producing the quantity y ,

$$c(w, y) := \min_{x \geq 0} \langle w, x \rangle \quad \text{s.t.} \quad F(x) \geq y.$$

Its solution $x(w, y)$ is the *conditional factor demand*: conditional because it takes the output level y as given.

All of the “duality” of MWG chs. 3 and 5 [9] consists of studying these four value functions and extracting from them, by differentiation, the demand and supply functions. The envelope theorem is the tool that performs that extraction.

Remark 11.3. Everything we say for maximization problems holds for minimization with the obvious sign changes: if $V(\theta) = \min_x f(x, \theta)$, then $V(\theta) = -\max_x [-f(x, \theta)]$, so every theorem has its “min” version. In particular, a value function that is a *maximum* of affine functions of the parameter will be *convex*, and one that is a *minimum* of affine functions will be *concave*; this will explain in Section 11.5 why π is convex and e is concave.

11.1.2 Why “envelope”? The geometric picture

The name comes from a picture worth memorizing. Fix each possible decision x and look at the function *of the parameter* that results from never reoptimizing:

$$g_x(\theta) := f(x, \theta), \quad \theta \in \Theta.$$

We get a *family of curves* $\{g_x\}_x$, one for each frozen decision. By definition of supremum,

$$V(\theta) = \sup_x g_x(\theta) :$$

at each θ , the value function takes the value of the highest curve in the family. We then say that V is the *upper envelope* of the family: the graph of V “wraps” all the graphs g_x from above, touching at each point the curve that is optimal there.

Two immediate geometric consequences, which are the heart of the whole chapter:

- (i) **Dominance:** $V(\theta) \geq g_x(\theta)$ for every x and every θ , with equality exactly when x is optimal at θ . That is, each curve of the family lies *below* the envelope and *touches* it where that decision is the best one.
- (ii) **Tangency:** if V and g_x are differentiable and g_x touches V at θ_0 (that is, $x \in X^*(\theta_0)$), then the difference $V - g_x$ attains a minimum (equal to zero) at θ_0 ; by the first order condition of that interior minimum, $V'(\theta_0) = g'_x(\theta_0)$. *The slopes coincide at the contact point:* the envelope is tangent to each member of the family where it touches it.

Consequence (ii) *already is* the envelope theorem: the derivative of the value function is the partial derivative of the objective with respect to the parameter, *holding the decision fixed at its optimal level*. We will formalize it in Section 11.2, but first a fully worked example.

Example 11.4 (An envelope computed by hand). Let $f(x, \theta) = \theta x - \frac{1}{2}x^2$ with $x, \theta \in \mathbb{R}$ (think of a firm that chooses output x with marginal revenue θ and quadratic cost). For each θ , the first order condition $\partial f / \partial x = \theta - x = 0$ gives $x(\theta) = \theta$ (it is a global maximum because f is strictly concave in x). The value function is

$$V(\theta) = f(x(\theta), \theta) = \theta \cdot \theta - \frac{1}{2}\theta^2 = \frac{1}{2}\theta^2.$$

Now look at the family of curves with a frozen decision: for each fixed x ,

$$g_x(\theta) = \theta x - \frac{1}{2}x^2$$

is a *line* in θ , with slope x and intercept $-\frac{1}{2}x^2$. The parabola $V(\theta) = \frac{1}{2}\theta^2$ is the upper envelope of this family of lines:

- *Dominance*: $\frac{1}{2}\theta^2 - (\theta x - \frac{1}{2}x^2) = \frac{1}{2}(\theta - x)^2 \geq 0$, with equality if and only if $x = \theta$. Each line lies below the parabola and touches it exactly at $\theta = x$.
- *Tangency*: at the contact point $\theta_0 = x$, the slope of the line is x and the slope of the parabola is $V'(\theta_0) = \theta_0 = x$. They coincide.
- *Envelope*: $V'(\theta) = \theta = x(\theta) = \frac{\partial f}{\partial \theta}(x(\theta), \theta)$. The derivative of the optimal value is the partial derivative of the objective with respect to θ with the decision frozen at its optimal value. (In passing: since V is the supremum of affine functions of θ , it is convex; we will come back to this.)

Figure 11.1 shows the parabola and five lines of the family.

Intuition

The picture to keep: **the value function is the “ceiling” of a family of curves, and at each point it is tangent to the curve that holds it up.** This is why, to compute the slope of the ceiling at θ_0 , you do not need to know how the optimal decision *moves* when θ changes: the slope of the individual curve that touches the ceiling at θ_0 is enough, that is, the partial derivative $\partial f / \partial \theta$ with x frozen at $x(\theta_0)$. The optimal readjustment of x contributes nothing *to first order* precisely because $x(\theta_0)$ was already optimal: moving away from an optimum has zero marginal effect on the objective. That is the “envelope logic” and you will use it hundreds of times.

11.2 The envelope theorem without constraints [P1]

Let us formalize the discussion above in the simplest case: unconstrained optimization (or, which is the same for our purposes, with a solution in the interior of the feasible set and constraints that do not depend on the parameter).

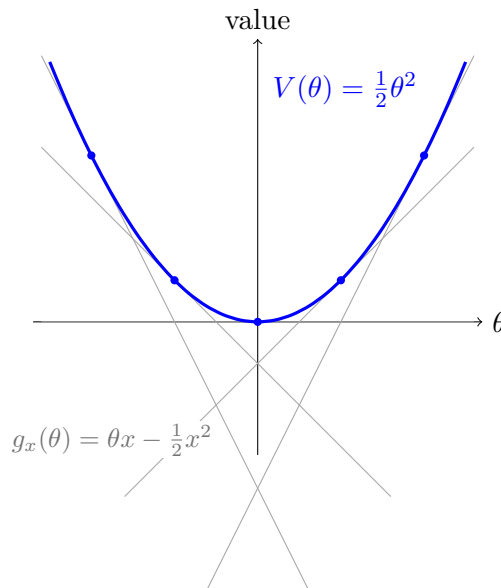


Figure 11.1: The value function $V(\theta) = \frac{1}{2}\theta^2$ (thick curve) as the upper envelope of the family of lines $g_x(\theta) = \theta x - \frac{1}{2}x^2$ (one line for each frozen decision $x \in \{-2, -1, 0, 1, 2\}$). Each line stays below V and touches it tangentially at $\theta = x$, where that decision is the optimal one.

Theorem 11.5 (Envelope, unconstrained case). *Let $X \subseteq \mathbb{R}^n$ be open, $\Theta \subseteq \mathbb{R}$ an open interval and $f : X \times \Theta \rightarrow \mathbb{R}$ of class C^1 . Suppose that:*

(H1) *for each $\theta \in \Theta$ the problem $\max_{x \in X} f(x, \theta)$ has a solution $x(\theta) \in X$ (an interior solution, since X is open);*

(H2) *the selection $\theta \mapsto x(\theta)$ is differentiable at $\theta_0 \in \Theta$.*

Then the value function $V(\theta) = f(x(\theta), \theta)$ is differentiable at θ_0 and

$$V'(\theta_0) = \frac{\partial f}{\partial \theta}(x(\theta_0), \theta_0).$$

That is: the derivative of the optimal value is the partial derivative of the objective with respect to the parameter, evaluated at the optimal solution, with no readjustment term.

Proof. We give every step. Since $x(\cdot)$ is differentiable at θ_0 and f is C^1 , the composition $V(\theta) = f(x(\theta), \theta)$ is differentiable at θ_0 and the chain rule (for the composition $\theta \mapsto (x(\theta), \theta) \mapsto f$) gives

$$V'(\theta_0) = \underbrace{\langle \nabla_x f(x(\theta_0), \theta_0), x'(\theta_0) \rangle}_{\text{indirect effect: readjustment of } x} + \underbrace{\frac{\partial f}{\partial \theta}(x(\theta_0), \theta_0)}_{\text{direct effect}}. \quad (*)$$

Now we use optimality. Since $x(\theta_0)$ maximizes $f(\cdot, \theta_0)$ over the open set X and $f(\cdot, \theta_0)$ is differentiable, the necessary first order condition for an interior maximum (optimality chapter) requires

$$\nabla_x f(x(\theta_0), \theta_0) = 0.$$

Substituting into $(*)$, the first summand vanishes *whatever* the value of $x'(\theta_0)$, and we are left with $V'(\theta_0) = \partial f / \partial \theta(x(\theta_0), \theta_0)$, as claimed. $\square$

It is worth also giving the “geometric” proof that does not use the chain rule on $x(\cdot)$, because it is the one that generalizes to Milgrom and Segal:

Proposition 11.6 (Version by comparing tangents). *With the notation above, suppose only that V is differentiable at θ_0 , that $x_0 \in X^*(\theta_0)$ and that $\theta \mapsto f(x_0, \theta)$ is differentiable at θ_0 . Then $V'(\theta_0) = \frac{\partial f}{\partial \theta}(x_0, \theta_0)$.*

Proof. Define $\varphi(\theta) := V(\theta) - f(x_0, \theta)$. By dominance (V is the supremum over x), $\varphi(\theta) \geq 0$ for every θ , and by optimality of x_0 at θ_0 , $\varphi(\theta_0) = 0$. Hence φ attains a *global minimum* at the interior point θ_0 . Since φ is differentiable at θ_0 (a difference of two functions differentiable there), the first order condition gives $\varphi'(\theta_0) = 0$, that is $V'(\theta_0) = \partial f / \partial \theta(x_0, \theta_0)$. $\square$

11.2.1 Discussion of the hypotheses

No hypothesis of Theorem 11.5 is decorative; it is worth knowing what fails if each one is dropped.

- (a) **Interior solution.** If the optimum is on the boundary of the feasible set, the condition $\nabla_x f = 0$ no longer holds and the indirect effect in (*) does not vanish by itself. The constrained version (Section 11.3) fixes this: the boundary terms are captured by the multipliers.
- (b) **Differentiability of the selection $x(\cdot)$.** This is the most uncomfortable hypothesis, because it is about an *endogenous* object: it is not assumed on the data of the problem but on its solution. In practice it is guaranteed by the implicit function theorem applied to the FOC $\nabla_x f(x, \theta) = 0$, which requires $f \in C^2$ and a nonsingular Hessian $\nabla_{xx}^2 f(x(\theta_0), \theta_0)$ (for example, negative definite: the strict second order sufficient condition). Proposition 11.6 shows that it can be replaced by “ V differentiable”, and the theorems of Danskin and Milgrom and Segal (Sections 11.7 and 11.6) remove it almost completely.
- (c) **Uniqueness (or a good selection).** If $X^*(\theta_0)$ has several points, different selections can give different “candidate derivatives” $\partial f / \partial \theta(x, \theta_0)$, $x \in X^*(\theta_0)$, and then V typically is *not* differentiable at θ_0 : it has a kink, with one sided derivatives given by the largest and the smallest of the candidates. We will see this precisely (and with subdifferentials) in Section 11.7.
- (d) **Existence of the maximum.** Without it V is only a supremum; the theory still stands in an “inequalities” version but the exact formulas require someone to reach the ceiling.

Example 11.7 (Full numerical example: profit of a firm). A firm produces $y = 2\sqrt{x}$ with one input $x \geq 0$, sells at price $p > 0$ and buys the input at price $w > 0$. Its problem is

$$\pi(p, w) = \max_{x \geq 0} f(x; p, w), \quad f(x; p, w) = 2p\sqrt{x} - wx.$$

Step 1: solve. FOC: $\partial f / \partial x = p/\sqrt{x} - w = 0$, hence $\sqrt{x} = p/w$ and $x(p, w) = p^2/w^2$. It is a global maximum because f is strictly concave in x ($\partial^2 f / \partial x^2 = -\frac{p}{2}x^{-3/2} < 0$), and it is interior ($x > 0$). Supply is $y(p, w) = 2\sqrt{x(p, w)} = 2p/w$.

Step 2: *value function*.

$$\pi(p, w) = 2p \cdot \frac{p}{w} - w \cdot \frac{p^2}{w^2} = \frac{2p^2}{w} - \frac{p^2}{w} = \frac{p^2}{w}.$$

Step 3: *check the envelope with respect to p*. Direct derivative of the value function: $\partial\pi/\partial p = 2p/w$. Partial derivative of the objective with x frozen: $\partial f/\partial p = 2\sqrt{x}$, which evaluated at $x(p, w) = p^2/w^2$ gives $2p/w$. They coincide, and moreover both equal the *supply* $y(p, w)$: we have just discovered Hotelling's lemma in a concrete case.

Step 4: *check with respect to w*. Direct: $\partial\pi/\partial w = -p^2/w^2$. Frozen: $\partial f/\partial w = -x$, evaluated at the optimum: $-p^2/w^2$. They coincide, and both equal *minus the input demand*.

Step 5: *see what was ignored*. Had we differentiated “with everything”, by the chain rule: $\frac{d\pi}{dp} = (\frac{p}{\sqrt{x}} - w) \frac{\partial x}{\partial p} + 2\sqrt{x}$. The bracket is exactly the FOC, which equals zero at the optimum: the readjustment term $\partial x/\partial p = 2p/w^2 \neq 0$ exists and is not small, but it is multiplied by a zero. That is the envelope in action.

Intuition

Why does the readjustment “cost nothing”? Because at the optimum the firm is *locally indifferent* between its choice and the neighboring ones: the marginal benefit of moving x is zero (that is what the FOC says). When p rises a little, the firm does readjust x , but that readjustment moves it among options that were worth *the same at the margin*; the first order gain comes only from the fact that each unit it was already producing is now worth more. In accounting terms: given dp , profit rises by (units sold) $\times dp$ plus a readjustment effect of order $(dp)^2$. Economists summarize this as: “*to first order, evaluate parameter changes at constant behavior*”.

11.3 The envelope theorem with constraints [P1]

In economics almost every problem has constraints that *depend on the parameters* (the budget constraint depends on p and m ; the technological constraint, on y). The previous theorem does not apply directly because the solution is not interior in the required sense. The general version says something very elegant: **the derivative of the value function is the partial derivative of the Lagrangian with respect to the parameter, evaluated at the optimal point (with its optimal multipliers)**.

Consider the parametrized problem

$$V(\theta) = \max_{x \in \mathbb{R}^n} f(x, \theta) \quad \text{s.t.} \quad g_j(x, \theta) \leq 0, \quad j = 1, \dots, r, \quad (\text{P}_\theta)$$

with Lagrangian

$$L(x, \lambda, \theta) = f(x, \theta) - \sum_{j=1}^r \lambda_j g_j(x, \theta).$$

(Equality constraints are handled the same way, without the sign requirement on their multipliers; to fix ideas we write inequalities, which is the KKT case of the previous chapter.)

Theorem 11.8 (Constrained envelope, Lagrangian version). *Let $f, g_1, \dots, g_r$ be of class C^1 in (x, θ) . Suppose that for each θ in a neighborhood of θ_0 :*

(H1) *problem (P_θ) has a solution $x(\theta)$ with KKT multipliers $\lambda(\theta) \in \mathbb{R}_+^r$ (a constraint qualification holds, for example Slater or linear independence of the active gradients, as in the KKT chapter);*

(H2) *the selections $x(\cdot)$ and $\lambda(\cdot)$ are differentiable at θ_0 ;*

(H3) *strict complementary slackness: for each j , either $g_j(x(\theta_0), \theta_0) < 0$ (inactive), or $\lambda_j(\theta_0) > 0$ (active with a positive multiplier), so that the set of active constraints is locally constant.*

Then V is differentiable at θ_0 and

$$V'(\theta_0) = \frac{\partial L}{\partial \theta}(x(\theta_0), \lambda(\theta_0), \theta_0) = \frac{\partial f}{\partial \theta}(x(\theta_0), \theta_0) - \sum_{j=1}^r \lambda_j(\theta_0) \frac{\partial g_j}{\partial \theta}(x(\theta_0), \theta_0).$$

Proof. Write $x_0 = x(\theta_0)$, $\lambda_0 = \lambda(\theta_0)$ and omit the evaluation points when they are clear. The proof has four steps.

Step 1 (the value coincides with the Lagrangian). By complementary slackness (a KKT condition from the previous chapter), $\lambda_j(\theta) g_j(x(\theta), \theta) = 0$ for every j and every θ near θ_0 : if the constraint is inactive its multiplier is zero, and if the multiplier is positive the constraint holds with equality. Summing,

$$V(\theta) = f(x(\theta), \theta) = f(x(\theta), \theta) - \sum_j \lambda_j(\theta) g_j(x(\theta), \theta) = L(x(\theta), \lambda(\theta), \theta).$$

Step 2 (differentiate with the chain rule). All the pieces are differentiable at θ_0 by assumption, so

$$V'(\theta_0) = \langle \nabla_x L, x'(\theta_0) \rangle + \sum_j \frac{\partial L}{\partial \lambda_j} \lambda'_j(\theta_0) + \frac{\partial L}{\partial \theta},$$

everything evaluated at $(x_0, \lambda_0, \theta_0)$.

Step 3 (the term in x vanishes). The KKT stationarity condition says exactly $\nabla_x L(x_0, \lambda_0, \theta_0) = \nabla_x f - \sum_j \lambda_{0j} \nabla_x g_j = 0$. First term: zero, whatever $x'(\theta_0)$ is.

Step 4 (the term in λ vanishes). We have $\partial L / \partial \lambda_j = -g_j(x_0, \theta_0)$. Split by strict complementary slackness. If $g_j(x_0, \theta_0) < 0$ (inactive), then by continuity it stays inactive near θ_0 , so $\lambda_j(\theta) = 0$ in a neighborhood and $\lambda'_j(\theta_0) = 0$: the summand $-g_j \cdot \lambda'_j$ is zero. If instead $\lambda_{0j} > 0$ (active), then $g_j(x_0, \theta_0) = 0$ and the factor $-g_j$ of the summand is zero. In both cases the j -th summand dies. We are left with

$$V'(\theta_0) = \frac{\partial L}{\partial \theta}(x_0, \lambda_0, \theta_0) = \frac{\partial f}{\partial \theta}(x_0, \theta_0) - \sum_j \lambda_{0j} \frac{\partial g_j}{\partial \theta}(x_0, \theta_0),$$

which is the announced formula. □

Remark 11.9. The practical rule to memorize: *to differentiate a value function with respect to a parameter, write the Lagrangian, differentiate it partially with respect to the parameter treating x and λ as constants, and evaluate at the optimum.* No differentiating the policies $x(\theta)$: all those

terms cancel by stationarity and complementary slackness, as we just saw. This rule contains Theorem 11.5 as a particular case (no constraints, $L = f$).

Careful (common mistake)

Three classic exam mistakes with this formula. **(1)** Forgetting the constraint term: if the parameter appears in g_j , the derivative of V is *not* just $\partial f/\partial\theta$; it carries the term $-\sum_j \lambda_j \partial g_j/\partial\theta$. Example: in the consumer problem, p_i does not appear in $u(x)$ but it does appear in the budget constraint; all the action is in the multiplier (this is how Roy's identity comes out). **(2)** The sign depends on how you wrote the constraint. With $L = f - \lambda(g - b)$ and constraint $g(x) \leq b$ you get $\partial V/\partial b = +\lambda$; if you write the constraint as $b - g(x) \geq 0$ and change the Lagrangian convention, the signs move. Before differentiating, fix your convention and be consistent. **(3)** Applying the formula at a point where the solution is not unique or where a constraint is just “about to become active” ($g_j = 0$ with $\lambda_j = 0$): there V may have a kink and only one sided derivatives exist (Section 11.7).

11.3.1 The multiplier as a shadow price

The most important special case of Theorem 11.8 is the one where the parameter is the *endowment of a resource* and enters only on the right hand side of a constraint.

Corollary 11.10 (The multiplier is the shadow price). *Consider $V(b) = \max_x f(x)$ s.t. $g(x) \leq b$, with the hypotheses of Theorem 11.8 (here $\theta = b$). Then*

$$V'(b) = \lambda^*(b),$$

where $\lambda^*(b) \geq 0$ is the KKT multiplier of the constraint at the optimum.

Proof. We write the constraint in standard form $\tilde{g}(x, b) := g(x) - b \leq 0$, with Lagrangian $L(x, \lambda, b) = f(x) - \lambda(g(x) - b)$. The objective f does not depend on b and $\partial \tilde{g}/\partial b = -1$. Theorem 11.8 gives

$$V'(b) = \frac{\partial L}{\partial b} = \underbrace{\frac{\partial f}{\partial b}}_{=0} - \lambda^* \cdot \underbrace{\frac{\partial \tilde{g}}{\partial b}}_{=-1} = \lambda^*. \quad \square$$

Economic reading (read it twice). $V(b)$ is the most that can be achieved with b units of the resource; $V'(b) = \lambda^*$ says that *one extra marginal unit of the resource is worth exactly λ^* units of the objective*. This is why λ^* is called the *shadow price* of the resource: it is the highest price the optimizer would be willing to pay (measured in units of the objective) to relax the constraint by one marginal unit. Consequences you will see again and again:

- In the consumer problem, the multiplier of the budget constraint is $\lambda^* = \partial v/\partial m$: the *marginal utility of income*. One extra sol is worth λ^* “utils”.
- If a constraint is inactive at the optimum, its multiplier is zero (complementary slackness) and indeed relaxing it is worth nothing: the leftover resource has zero shadow price. KKT and the envelope tell the same economic story.

- In duality (previous chapter): the dual problem “puts prices” on the constraints; the envelope confirms that the optimal dual prices are the sensitivities of the primal value. It is the same coin seen from its two sides.
- In planning and in finance: the shadow prices of capacity constraints are the basis for valuing productive assets and for the internal transfer prices of an organization [2, 3].

Example 11.11 (Numerical check of the shadow price). Problem: $V(b) = \max_{x_1, x_2 > 0} x_1 x_2$ s.t. $x_1 + x_2 \leq b$, with $b > 0$.

Solve. Since the objective is increasing in each variable, the constraint binds: $x_1 + x_2 = b$ and $\lambda > 0$. Lagrangian $L = x_1 x_2 - \lambda(x_1 + x_2 - b)$. FOC: $x_2 = \lambda$ and $x_1 = \lambda$, hence $x_1 = x_2$ and, with the constraint, $x_1 = x_2 = b/2$, $\lambda^* = b/2$. (It is a maximum: on the budget line the objective is $x_1(b - x_1)$, an inverted parabola with vertex at $b/2$.)

Explicit value function. $V(b) = \frac{b}{2} \cdot \frac{b}{2} = \frac{b^2}{4}$, so $V'(b) = \frac{b}{2} = \lambda^*(b)$. The formula of Corollary 11.10 is verified exactly.

Check with two values of b . Take $b = 10$ and $b = 10.1$:

$$V(10) = \frac{100}{4} = 25, \quad V(10.1) = \frac{(10.1)^2}{4} = \frac{102.01}{4} = 25.5025.$$

The actual increment is $\Delta V = 0.5025$. The shadow price prediction is $\lambda^*(10) \cdot \Delta b = 5 \times 0.1 = 0.5$. The error, 0.0025, is exactly the second order term $\frac{1}{2} V''(10)(\Delta b)^2 = \frac{1}{2} \cdot \frac{1}{2} \cdot (0.1)^2 = 0.0025$: the first order approximation with the multiplier is as good as calculus promises.

Intuition

Why can the planner ignore *how* the extra unit of the resource will be allocated? Again the envelope logic: at the optimum, the resource is already allocated so that its marginal product is *the same in all uses* (otherwise it would pay to reallocate). That common marginal productivity is λ^* . One extra unit can enter through any use (they all yield the same at the margin) and this is why the value of the extra unit is simply λ^* , with no need to solve the allocation problem again. The multiplier compresses all the complexity of the optimal allocation into a single number.

11.4 The four gems of microeconomics [P1]

We now apply the machinery to the four value functions of Example 11.2. Each result in this section is a theorem with a name of its own that you will use continuously in Microeconomics; each one is a two line corollary of the envelope theorem, and that is precisely the moral: *the duality of the consumer and of the firm is not a collection of tricks, it is one theorem applied four times* [9, 10].

Throughout the section, for the Cobb–Douglas checks we use the utility and the technology

$$u(x_1, x_2) = x_1^{1/2} x_2^{1/2}, \quad F(x_1, x_2) = x_1^{1/2} x_2^{1/2},$$

with goods prices $p = (p_1, p_2) \gg 0$, input prices $w = (w_1, w_2) \gg 0$ and income $m > 0$.

11.4.1 Gem 1: Hotelling's lemma

Theorem 11.12 (Hotelling's lemma). *Let $\pi(p, w) = \max_{x \geq 0} pF(x) - \langle w, x \rangle$ be the profit function of a competitive firm with one output, and suppose that at (p, w) the solution $x(p, w)$ is unique, interior and differentiable, with F of class C^1 . Then π is differentiable at (p, w) and*

$$\begin{aligned} \frac{\partial \pi}{\partial p}(p, w) &= F(x(p, w)) = y(p, w) \quad (\text{supply}), \\ \frac{\partial \pi}{\partial w_i}(p, w) &= -x_i(p, w) \quad (\text{minus the demand for input } i). \end{aligned}$$

In the production set formulation, $\pi(p) = \sup_{y \in Y} \langle p, y \rangle$ and, if the optimal plan $y(p)$ is unique, $\nabla \pi(p) = y(p)$, the net supply.

Proof. This is Theorem 11.5 with objective $f(x; p, w) = pF(x) - \langle w, x \rangle$, an unconstrained problem (interior solution) and parameters $\theta = (p, w)$, applied coordinate by coordinate. For p : freezing x at its optimal value,

$$\frac{\partial \pi}{\partial p} = \frac{\partial f}{\partial p} \Big|_{x=x(p, w)} = F(x(p, w)) = y(p, w).$$

For w_i : similarly,

$$\frac{\partial \pi}{\partial w_i} = \frac{\partial f}{\partial w_i} \Big|_{x=x(p, w)} = -x_i(p, w).$$

In the production set version, $f(y; p) = \langle p, y \rangle$ is linear in p with $\partial f / \partial p_k = y_k$, and the same envelope gives $\partial \pi / \partial p_k = y_k(p)$. $\square$

Reading. If the output price rises by dp , each of the y units the firm was already selling is worth dp more; the production readjustment is second order. This is why the marginal profit of p is exactly the quantity supplied. Symmetrically, if w_i rises, the firm pays dw_i more for each of the x_i units of the input it was already using: hence the negative sign. In empirical practice this is gold: if you know (or estimate) the profit function, *differentiating it* recovers the whole behavior of the firm.

Example 11.13 (Hotelling in Cobb–Douglas, all the computations). With $F(x_1, x_2) = x_1^{1/2} x_2^{1/2}$ the profit function is not well defined (constant returns: profit 0 or $+\infty$), so we use its cousin with decreasing returns $F(x) = x^{1/2}$, one input, as in Example 11.7 but without the factor 2: $\pi(p, w) = \max_{x \geq 0} p x^{1/2} - wx$. FOC: $\frac{p}{2} x^{-1/2} = w$, hence $x(p, w) = \frac{p^2}{4w^2}$ and $y(p, w) = x^{1/2} = \frac{p}{2w}$. The value function:

$$\pi(p, w) = p \cdot \frac{p}{2w} - w \cdot \frac{p^2}{4w^2} = \frac{p^2}{2w} - \frac{p^2}{4w} = \frac{p^2}{4w}.$$

Hotelling check:

$$\frac{\partial \pi}{\partial p} = \frac{2p}{4w} = \frac{p}{2w} = y(p, w) \quad \checkmark, \quad \frac{\partial \pi}{\partial w} = -\frac{p^2}{4w^2} = -x(p, w) \quad \checkmark.$$

Additional note: $\pi(p, w) = p^2/(4w)$ is convex in p ($\partial^2 \pi / \partial p^2 = \frac{1}{2w} > 0$) and homogeneous of

degree 1 in (p, w) ; both properties are general (Section 11.5).

11.4.2 Gem 2: Shephard's lemma

Theorem 11.14 (Shephard's lemma). *Let $e(p, u) = \min_{x \geq 0} \langle p, x \rangle$ s.t. $u(x) \geq u$, with $u(\cdot)$ continuous and locally nonsatiated, and suppose that at (p, u) the Hicksian demand $h(p, u)$ is unique and the hypotheses of Theorem 11.8 hold. Then*

$$\frac{\partial e}{\partial p_i}(p, u) = h_i(p, u) \quad \text{for } i = 1, \dots, n.$$

Differentiating the expenditure function with respect to a price gives the Hicksian demand for that good.

Proof. This is a *minimization* problem, so we apply the “min” version of Theorem 11.8 (Remark 11.3). The minimization Lagrangian is

$$L(x, \mu; p, u) = \langle p, x \rangle - \mu(u(x) - u), \quad \mu \geq 0.$$

The parameter p_i appears *only* in the objective $\langle p, x \rangle = \sum_k p_k x_k$, not in the constraint $u(x) \geq u$. By the envelope,

$$\frac{\partial e}{\partial p_i} = \frac{\partial L}{\partial p_i} \Big|_{x=h(p,u), \mu=\mu^*} = x_i \Big|_{x=h(p,u)} = h_i(p, u).$$

There is no multiplier term because $\partial/\partial p_i$ of the constraint is zero. (In passing, differentiating with respect to the parameter u : $\partial e/\partial u = \mu^* > 0$, the shadow price of requiring more utility.) $\square$

Reading. If p_i rises by dp_i , keeping the same standard of living costs $h_i dp_i$ more: to first order, the compensated consumer does not substitute (the readjustment towards other goods is second order), and simply pays more for what was already being bought. This lemma is the basis of *cost of living indices* and of the measurement of welfare changes (compensating and equivalent variation, consumer theory, MWG ch. 3).

Example 11.15 (Shephard in Cobb–Douglas, all the computations). With $u(x_1, x_2) = x_1^{1/2} x_2^{1/2}$, let us first compute e and h from scratch. Minimize $p_1 x_1 + p_2 x_2$ s.t. $x_1^{1/2} x_2^{1/2} = u$ (the constraint binds by monotonicity). FOC of the Lagrangian $L = p_1 x_1 + p_2 x_2 - \mu(x_1^{1/2} x_2^{1/2} - u)$:

$$p_1 = \frac{\mu}{2} x_1^{-1/2} x_2^{1/2}, \quad p_2 = \frac{\mu}{2} x_1^{1/2} x_2^{-1/2}.$$

Dividing: $\frac{p_1}{p_2} = \frac{x_2}{x_1}$, that is $x_2 = \frac{p_1}{p_2} x_1$. Substituting into the constraint: $x_1^{1/2} \left(\frac{p_1}{p_2} x_1\right)^{1/2} = x_1 \left(\frac{p_1}{p_2}\right)^{1/2} = u$, hence

$$h_1(p, u) = u \left(\frac{p_2}{p_1}\right)^{1/2}, \quad h_2(p, u) = u \left(\frac{p_1}{p_2}\right)^{1/2}.$$

The expenditure function:

$$e(p, u) = p_1 h_1 + p_2 h_2 = u p_1^{1/2} p_2^{1/2} + u p_1^{1/2} p_2^{1/2} = 2u \sqrt{p_1 p_2}.$$

Shephard check:

$$\frac{\partial e}{\partial p_1} = 2u \cdot \frac{1}{2} p_1^{-1/2} p_2^{1/2} = u \left(\frac{p_2}{p_1} \right)^{1/2} = h_1(p, u) \checkmark,$$

and symmetrically $\partial e / \partial p_2 = u(p_1/p_2)^{1/2} = h_2$. Note that e is concave in p : for example with p_2 fixed, $\partial^2 e / \partial p_1^2 = -\frac{u}{2} p_1^{-3/2} p_2^{1/2} < 0$; full concavity is proved in general in Section 11.5.

11.4.3 Gem 3: Roy's identity

For indirect utility the parameter p_i enters *in the constraint*, not in the objective, so here the Lagrangian version shows its full power.

Theorem 11.16 (Roy's identity). *Let $v(p, m) = \max_{x \geq 0} u(x)$ s.t. $\langle p, x \rangle \leq m$, with u of class C^1 , locally nonsatiated, unique and interior Marshallian solution $x(p, m)$, and the hypotheses of Theorem 11.8. If $\partial v / \partial m > 0$, then*

$$x_i(p, m) = - \frac{\partial v / \partial p_i}{\partial v / \partial m} \quad \text{for } i = 1, \dots, n.$$

Proof. Lagrangian: $L(x, \lambda; p, m) = u(x) - \lambda(\langle p, x \rangle - m)$. We apply the envelope twice, at the same optimal point $(x(p, m), \lambda^*(p, m))$.

(1) *With respect to p_i .* The parameter p_i appears only in the constraint, through $\langle p, x \rangle = \sum_k p_k x_k$:

$$\frac{\partial v}{\partial p_i} = \frac{\partial L}{\partial p_i} \Big|_{\text{opt}} = -\lambda^* x_i(p, m). \quad (\text{R1})$$

(2) *With respect to m .* It also appears only in the constraint:

$$\frac{\partial v}{\partial m} = \frac{\partial L}{\partial m} \Big|_{\text{opt}} = \lambda^*. \quad (\text{R2})$$

(R2) is Corollary 11.10: the multiplier is the marginal utility of income. Since $\partial v / \partial m = \lambda^* > 0$ by assumption, we can divide (R1) by (R2):

$$\frac{\partial v / \partial p_i}{\partial v / \partial m} = \frac{-\lambda^* x_i}{\lambda^*} = -x_i(p, m),$$

which rearranged is Roy's identity. □

Reading. (R1) says: if p_i rises by dp_i , the consumer loses purchasing power worth $x_i dp_i$ soles (paying more for what was already being bought), and each sol lost is worth λ^* utils; the utility damage is $\lambda^* x_i dp_i$. Dividing by λ^* converts that loss from “utils” into soles, leaving the quantity x_i . Roy is the bridge that lets you go from an unobservable object (v , measured in utility) to an observable one (demand): this is why it is the standard tool to *recover demands from functional forms of indirect utility* in applied work.

Example 11.17 (Roy in Cobb–Douglas, all the computations). With $u = x_1^{1/2} x_2^{1/2}$, the Marshallian demand is the familiar $x_i(p, m) = \frac{m}{2p_i}$ (each good takes half of expenditure; it follows

from the FOC $\frac{\partial u/\partial x_1}{\partial u/\partial x_2} = \frac{x_2}{x_1} = \frac{p_1}{p_2}$ plus the constraint). Indirect utility:

$$v(p, m) = \left(\frac{m}{2p_1}\right)^{1/2} \left(\frac{m}{2p_2}\right)^{1/2} = \frac{m}{2\sqrt{p_1 p_2}}.$$

We compute the three derivatives we need:

$$\frac{\partial v}{\partial p_1} = \frac{m}{2} \cdot \frac{\partial}{\partial p_1} (p_1^{-1/2} p_2^{-1/2}) = -\frac{m}{4} p_1^{-3/2} p_2^{-1/2}, \quad \frac{\partial v}{\partial m} = \frac{1}{2\sqrt{p_1 p_2}} = \frac{1}{2} p_1^{-1/2} p_2^{-1/2}.$$

Applying Roy:

$$-\frac{\partial v/\partial p_1}{\partial v/\partial m} = \frac{\frac{m}{4} p_1^{-3/2} p_2^{-1/2}}{\frac{1}{2} p_1^{-1/2} p_2^{-1/2}} = \frac{m}{4} \cdot \frac{2}{1} \cdot p_1^{-3/2+1/2} = \frac{m}{2p_1} = x_1(p, m) \checkmark.$$

By symmetry, $-(\partial v/\partial p_2)/(\partial v/\partial m) = m/(2p_2) = x_2$. You can also check (R2): the λ^* of the Cobb–Douglas consumer problem turns out to be $\lambda^* = \frac{1}{2\sqrt{p_1 p_2}} = \partial v/\partial m$.

11.4.4 Gem 4: the derivative of the cost function

Theorem 11.18 (Shephard for costs). *Let $c(w, y) = \min_{x \geq 0} \langle w, x \rangle$ s.t. $F(x) \geq y$, under hypotheses analogous to those of Theorem 11.14. Then*

$$\frac{\partial c}{\partial w_i}(w, y) = x_i(w, y),$$

the conditional demand for input i . Moreover $\partial c/\partial y = \mu^$, the multiplier of the technological constraint, which is marginal cost.*

Proof. Identical to the proof of Shephard’s lemma, changing names: minimization Lagrangian $L = \langle w, x \rangle - \mu(F(x) - y)$; the parameter w_i only touches the objective, with $\partial L/\partial w_i = x_i$, and the envelope evaluates at the optimum: $\partial c/\partial w_i = x_i(w, y)$. For y : it appears only in the constraint, with $\partial L/\partial y = +\mu$, hence $\partial c/\partial y = \mu^* \geq 0$: the shadow price of requiring one more unit of output is, by economic definition, marginal cost. $\square$

Example 11.19 (Costs in Cobb–Douglas, all the computations). With $F(x_1, x_2) = x_1^{1/2} x_2^{1/2}$: minimize $w_1 x_1 + w_2 x_2$ s.t. $x_1^{1/2} x_2^{1/2} = y$. It is the same problem as Example 11.15 with (p, u) renamed (w, y) , so

$$x_1(w, y) = y \left(\frac{w_2}{w_1}\right)^{1/2}, \quad x_2(w, y) = y \left(\frac{w_1}{w_2}\right)^{1/2}, \quad c(w, y) = 2y\sqrt{w_1 w_2}.$$

Check: $\frac{\partial c}{\partial w_1} = y \left(\frac{w_2}{w_1}\right)^{1/2} = x_1(w, y) \checkmark$. And $\partial c/\partial y = 2\sqrt{w_1 w_2}$ is marginal cost (constant in y : constant returns to scale). The conditional input demand is decreasing in its own price ($\partial x_1/\partial w_1 = -\frac{y}{2} w_1^{-3/2} w_2^{1/2} < 0$), which is the “production” version of the law of compensated demand; in the next section we will see that this is not an accident of the example but a consequence of the concavity of c in w .

Exam strategy

Exam recipe for “derive demand or supply from the value function” (it always shows up, in this course and in Microeconomics):

1. Identify which problem the given object is the value function of (π : max profit; e : min expenditure; c : min cost; v : max utility).
2. Write the Lagrangian *with your sign convention fixed* and locate where the parameter of interest appears: objective, constraint or both?
3. Apply the rule of Remark 11.9: differentiate L partially with respect to the parameter, freeze (x, λ) at the optimum. If the parameter is only in the objective: direct derivative (Hotelling, Shephard). If it is only in the constraint: the multiplier appears (Roy, shadow price).
4. If the result carries an unwanted λ^* , look for a second envelope equation that also carries it and divide to eliminate it (this is how Roy is closed).
5. Check signs with economics: $\partial\pi/\partial p > 0$, $\partial e/\partial p_i \geq 0$, $\partial v/\partial p_i \leq 0$, $\partial v/\partial m > 0$. If a sign does not fit, review the Lagrangian convention.
6. Free checks: homogeneity (degree 1 for e, c, π in prices; degree 0 for the demands) and curvature (Section 11.5). In an exam, checking your answer with Cobb–Douglas takes two minutes and catches almost every error.

11.5 Curvature of value functions: second order sensitivity [P1]

The envelope theorem is a *first order* result: it gives the derivatives of V . *Second order* sensitivity analysis asks about the curvature of V , and the answer comes for free from the “envelope of affine functions” structure: recall from the chapter on convex functions that *the pointwise supremum of convex functions (in particular, affine ones) is convex, and the pointwise infimum of concave functions (in particular, affine ones) is concave*. Combined with the gems of the previous section, curvature produces the comparative behavior laws of micro: law of supply, law of compensated demand and the Slutsky matrix.

Theorem 11.20 (Convexity of the profit function). *For any production set $Y \subseteq \mathbb{R}^n$ (with no convexity assumption on Y), the function $\pi(p) = \sup_{y \in Y} \langle p, y \rangle$ is convex and homogeneous of degree 1 in p .*

Proof. Let p, p' be such that π is finite, $t \in [0, 1]$ and $p_t = tp + (1 - t)p'$. For any $y \in Y$,

$$\langle p_t, y \rangle = t \langle p, y \rangle + (1 - t) \langle p', y \rangle \leq t \pi(p) + (1 - t) \pi(p'),$$

where the inequality uses the definition of supremum twice: $\langle p, y \rangle \leq \pi(p)$ and $\langle p', y \rangle \leq \pi(p')$. Taking the supremum over $y \in Y$ on the left, $\pi(p_t) \leq t \pi(p) + (1 - t) \pi(p')$: convexity. Homogeneity is immediate: $\pi(\alpha p) = \sup_y \langle \alpha p, y \rangle = \alpha \sup_y \langle p, y \rangle = \alpha \pi(p)$ for $\alpha > 0$. (As anticipated, π is the support function of Y , and support functions are always convex and homogeneous of degree 1: this is the “two line proof through the supremum of linear functions”.) $\square$

Corollary 11.21 (Law of supply). *If π is twice differentiable at p , then the substitution matrix of the firm*

$$D_p y(p) = \nabla^2 \pi(p)$$

is symmetric and positive semidefinite. In particular, $\frac{\partial y_k}{\partial p_k} = \frac{\partial^2 \pi}{\partial p_k^2} \geq 0$ for each k : supply curves have nonnegative slope and (unconditional) input demands have nonpositive slope, with no assumption on preferences or income effects.

Proof. By Hotelling's lemma (Theorem 11.12), $y(p) = \nabla \pi(p)$; differentiating again, $D_p y(p) = \nabla^2 \pi(p)$. This matrix is symmetric by Schwarz's theorem (equality of cross derivatives of a C^2 function) and positive semidefinite because π is convex (second order characterization of convexity, chapter on convex functions). The diagonal entries of a positive semidefinite matrix are nonnegative (take $z = e_k$ in $\langle z, \nabla^2 \pi z \rangle \geq 0$). For an input, $y_k = -x_k < 0$ is the net supply, so $\partial(-x_k)/\partial p_k \geq 0$ means that the input demand falls with its own price. $\square$

Theorem 11.22 (Concavity of the expenditure function). *For any utility function u (it is enough that the upper contour set $\{x \geq 0 : u(x) \geq u\}$ be nonempty), the function $e(p, u) = \inf\{\langle p, x \rangle : u(x) \geq u, x \geq 0\}$ is concave and homogeneous of degree 1 in p . The same holds, with an identical proof, for the cost function $c(w, y)$ in w .*

Proof. It is the mirror argument: an infimum of linear functions of p . Let $p, p' \gg 0$, $t \in [0, 1]$, $p_t = tp + (1-t)p'$, and let x_t be (any) solution of the expenditure problem at prices p_t (if the infimum is not attained, take x_t almost optimal and pass to the limit). Then x_t is *feasible* for the problems at prices p and p' (the constraint $u(x) \geq u$ does not involve prices), hence by the definition of infimum

$$\langle p, x_t \rangle \geq e(p, u) \quad \text{and} \quad \langle p', x_t \rangle \geq e(p', u).$$

Multiplying by $t \geq 0$ and $(1-t) \geq 0$ and adding:

$$e(p_t, u) = \langle p_t, x_t \rangle = t \langle p, x_t \rangle + (1-t) \langle p', x_t \rangle \geq t e(p, u) + (1-t) e(p', u) :$$

concavity. Homogeneity: $e(\alpha p, u) = \inf \langle \alpha p, x \rangle = \alpha \inf \langle p, x \rangle = \alpha e(p, u)$ for $\alpha > 0$. $\square$

Intuition

The concavity of e has a beautiful economic reading. Imagine that p_1 rises and the consumer *could not substitute*: the expenditure needed to keep u would grow *linearly* in p_1 (paying for the old bundle at the new prices: the line $\langle p, h(p^0, u) \rangle$). Since substitution *is* possible, actual expenditure stays *below* that line: $e(\cdot, u)$ lies always below its tangent lines, which is exactly concavity. The curvature of e measures how much substitution capacity the consumer has. The same reasoning with signs reversed explains the convexity of π : a firm that can readjust its plan always stays *above* the “frozen plan” line, and the curvature of π measures its technological flexibility.

Theorem 11.23 (Slutsky matrix: symmetry and negative semidefiniteness). *If $e(\cdot, u)$ is twice differentiable, the Slutsky substitution matrix*

$$S(p, u) := D_p h(p, u) = \nabla_{pp}^2 e(p, u)$$

is symmetric and negative semidefinite. In particular $\partial h_i / \partial p_i \leq 0$ (law of compensated demand) and $\partial h_i / \partial p_j = \partial h_j / \partial p_i$ (symmetry of the compensated cross effects).

Proof sketch. By Shephard's lemma, $h(p, u) = \nabla_p e(p, u)$; differentiating, $D_p h = \nabla_{pp}^2 e$. Symmetry: Schwarz. Negative semidefiniteness: second order characterization of the concavity of e in p (Theorem 11.22). The regularity details (when e is twice differentiable) are in [9], ch. 3 and appendix M. It is exactly the same argument as Corollary 11.21 with the signs flipped: *Shephard + concavity = Slutsky NSD*, just as *Hotelling + convexity = law of supply*. $\square$

Proposition 11.24 (Slutsky equation). *Under the hypotheses above, if in addition consumer duality gives $h_i(p, u) = x_i(p, e(p, u))$ for every p (the Hicksian-Marshallian identity), then*

$$\underbrace{\frac{\partial x_i}{\partial p_j}}_{\text{total effect}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{substitution effect}} - \underbrace{x_j \frac{\partial x_i}{\partial m}}_{\text{income effect}}.$$

Proof. We differentiate the identity $h_i(p, u) = x_i(p, e(p, u))$ with respect to p_j using the chain rule (the second argument of x_i depends on p_j through e):

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial m} \cdot \frac{\partial e}{\partial p_j}.$$

By Shephard's lemma, $\partial e / \partial p_j = h_j(p, u)$, and evaluating at $u = v(p, m)$ (so that $e(p, u) = m$ and $h_j = x_j(p, m)$ by duality):

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial x_i}{\partial p_j} + x_j \frac{\partial x_i}{\partial m}.$$

Solving for $\partial x_i / \partial p_j$ gives the Slutsky equation. Note the role of the envelope: it is Shephard who turns the abstract term $\partial e / \partial p_j$ into the observable object x_j . $\square$

Theorem 11.25 (Quasiconvexity of indirect utility). *Indirect utility $v(p, m) = \max\{u(x) : \langle p, x \rangle \leq m, x \geq 0\}$ is quasiconvex in (p, m) : for every $\bar{v} \in \mathbb{R}$, the lower contour set $N(\bar{v}) = \{(p, m) : v(p, m) \leq \bar{v}\}$ is convex.*

Proof. Let (p, m) and (p', m') be in $N(\bar{v})$, $t \in [0, 1]$, and $(p'', m'') = t(p, m) + (1 - t)(p', m')$. We must prove $v(p'', m'') \leq \bar{v}$. Take any $x \geq 0$ feasible in the problem with data (p'', m'') , that is $\langle p'', x \rangle \leq m''$. We claim that x is feasible at (p, m) or at (p', m') . If it were feasible at neither, we would have $\langle p, x \rangle > m$ and $\langle p', x \rangle > m'$; multiplying the first by t , the second by $(1 - t)$ and adding:

$$\langle p'', x \rangle = t \langle p, x \rangle + (1 - t) \langle p', x \rangle > tm + (1 - t)m' = m'',$$

contradicting feasibility at (p'', m'') . Hence x is feasible in at least one of the two original budgets, and therefore $u(x) \leq \max\{v(p, m), v(p', m')\} \leq \bar{v}$. Taking the supremum over the x feasible at (p'', m'') : $v(p'', m'') \leq \bar{v}$. So $N(\bar{v})$ is convex and v is quasiconvex. (Geometrically:

the “average” budget does not contain any bundle that was not already in one of the extreme budgets; averaging prices and incomes can never be strictly better than the better of the two original situations.) $\square$

Remark 11.26. Mental table for Microeconomics: $\pi(p)$ *convex* and homogeneous of degree 1 (sup of linear functions); $e(p, u)$ and $c(w, y)$ *concave* and homogeneous of degree 1 in prices (inf of linear functions); $v(p, m)$ *quasiconvex*, homogeneous of degree 0, decreasing in p , increasing in m . These properties, together with the derivatives of the four gems, are *all* the testable restrictions that theory imposes on empirically estimated functions: they are the basis of “integrability” tests (could this estimated demand have come from a rational consumer?) [9, 10].

11.6 The Milgrom and Segal envelope theorem and its applications to mechanisms [P1]

11.6.1 Motivation: when you cannot differentiate in x

All the previous theorems assume differentiable structure in the decision variable: x lives in $\mathbb{R}^n$, f is C^1 in x , there are first order conditions. In mechanism design (MWG ch. 23) this collapses: an agent’s “decision” can be a *report* in an abstract set of messages, a *bid* in a set of strategies, or a choice in a discrete menu of contracts. There is no $\nabla_x f$, no FOC, and the solution $x(\theta)$ can jump discontinuously with the type. Milgrom and Segal [11] showed that none of that matters: all the envelope needs is differentiability *in the parameter* θ , which in mechanisms is the agent’s *type*, where smoothness is a natural assumption.

Theorem 11.27 (Milgrom and Segal, integral version). *Let X be an arbitrary set (no topology, no linear structure) and $f : X \times [0, 1] \rightarrow \mathbb{R}$. Define $V(\theta) = \sup_{x \in X} f(x, \theta)$ and suppose:*

- (MS1) *for each $x \in X$, the function $\theta \mapsto f(x, \theta)$ is differentiable, and there is an integrable function $b : [0, 1] \rightarrow \mathbb{R}_+$ such that $|\partial f(x, \theta)/\partial \theta| \leq b(\theta)$ for every x and almost every θ ;*
- (MS2) *the supremum is attained: there is a selection $x(\theta) \in X^*(\theta)$ for each θ .*

Then V is absolutely continuous (in particular, differentiable almost everywhere) and admits the integral representation

$$V(\theta) = V(0) + \int_0^\theta \frac{\partial f}{\partial \theta}(x(s), s) ds \quad \text{for every } \theta \in [0, 1],$$

valid for any measurable selection $x(\cdot)$ of the argmax. Moreover, at every θ where V is differentiable and $x(\theta) \in X^(\theta)$, $V'(\theta) = \partial f(x(\theta), \theta)/\partial \theta$.*

Proof idea (honest, following [11]). Step 1: V is absolutely continuous. For $\theta' > \theta$ and any $\varepsilon > 0$, take x that is ε -optimal at θ' . Then

$$V(\theta') - V(\theta) \leq f(x, \theta') + \varepsilon - f(x, \theta) = \varepsilon + \int_\theta^{\theta'} \frac{\partial f}{\partial \theta}(x, s) ds \leq \varepsilon + \int_\theta^{\theta'} b(s) ds,$$

using the fundamental theorem of calculus in the variable θ (legitimate because $f(x, \cdot)$ is differentiable with derivative dominated by b). With the symmetric argument, $|V(\theta') - V(\theta)| \leq$

$\int_{\theta}^{\theta'} b(s) ds$. Since b is integrable, this gives the absolute continuity of V , and by the fundamental theorem of calculus (Lebesgue version), $V(\theta) = V(0) + \int_0^{\theta} V'(s) ds$ with V' defined almost everywhere.

Step 2: identify V' at the good points. Let s be a point where V is differentiable and take $x(s) \in X^*(s)$. This is exactly the argument of Proposition 11.6, which used no structure on X : the function $\varphi(\theta) = V(\theta) - f(x(s), \theta)$ is nonnegative, vanishes at $\theta = s$ and is differentiable there, hence $\varphi'(s) = 0$, that is $V'(s) = \partial f(x(s), s)/\partial \theta$. Since this holds almost everywhere, we can substitute $V'(s)$ inside the integral of Step 1, and the formula is proved. $\square$

Remark 11.28. Look carefully at what was *not* required: nothing about the continuity or differentiability of the selection $x(\cdot)$, nothing about convexity or even structure on X , no first order conditions in x . The optimal selection can jump from one point of X to a completely different one when θ crosses a threshold; at those thresholds V simply has a nondifferentiable point (of measure zero), and the *integral* representation does not even notice: an integral does not see sets of measure zero. This is the key difference with Theorem 11.5, whose conclusion was pointwise (“ $V'(\theta_0) = \dots$ ”) and which therefore needed strong pointwise hypotheses. Milgrom and Segal trade a pointwise conclusion for an integral one and in exchange drop almost all the hypotheses: a perfect trade for mechanism theory.

11.6.2 Application: auctions and revenue equivalence

Consider an auction (or, more generally, a mechanism) with n bidders. The bidder we care about has a *private type* $\theta \in [0, 1]$ (the valuation of the object), distributed with continuous cdf F . In the mechanism, once the equilibrium strategies of the rivals are fixed, by choosing an action or report $x \in X$ (a bid, a message) the bidder gets a probability of winning $q(x) \in [0, 1]$ and makes an expected payment $t(x) \in \mathbb{R}$. The expected utility, if the type is θ , is *quasilinear*:

$$f(x, \theta) = \theta q(x) - t(x).$$

The equilibrium payoff is the value function

$$U(\theta) = \sup_{x \in X} [\theta q(x) - t(x)],$$

because in equilibrium each type chooses its action optimally given the behavior of the others (this is exactly the incentive compatibility constraint IC of the micro course).

Proposition 11.29 (Formula for the equilibrium payoff). *In the framework above, let $x(\theta)$ be a selection of equilibrium actions and write $Q(\theta) := q(x(\theta))$ for the probability that type θ wins in equilibrium. Then U is convex, absolutely continuous, and*

$$U(\theta) = U(0) + \int_0^{\theta} Q(s) ds.$$

Consequently, the expected payment of type θ is completely determined by the allocation rule and by $U(0)$:

$$T(\theta) := t(x(\theta)) = \theta Q(\theta) - U(0) - \int_0^{\theta} Q(s) ds.$$

Proof. We check the Milgrom and Segal hypotheses. For each fixed x , $f(x, \theta) = \theta q(x) - t(x)$ is *affine* in θ , hence differentiable with $\partial f / \partial \theta = q(x) \in [0, 1]$: bound (MS1) holds with $b \equiv 1$. Assuming the optimum is attained (MS2, true in equilibrium), Theorem 11.27 gives

$$U(\theta) = U(0) + \int_0^\theta \frac{\partial f}{\partial \theta}(x(s), s) ds = U(0) + \int_0^\theta q(x(s)) ds = U(0) + \int_0^\theta Q(s) ds.$$

(In passing, U is a supremum of affine functions of θ , hence convex; and its derivative almost everywhere is $Q(\theta)$, which by convexity must be nondecreasing: the *monotonicity of the allocation* is a necessary condition of IC, another famous result that comes for free.) The payment formula follows by solving: since $U(\theta) = \theta Q(\theta) - T(\theta)$ in equilibrium, we get $T(\theta) = \theta Q(\theta) - U(\theta) = \theta Q(\theta) - U(0) - \int_0^\theta Q(s) ds$. $\square$

Theorem 11.30 (Revenue equivalence, Myerson). *Consider any two auction mechanisms with bidders whose private types are independent, such that in their respective equilibria: (i) the allocation rule is the same (for example, both assign the object to the bidder with the highest type), and (ii) the lowest type gets the same utility (for example, $U(0) = 0$ in both). Then the expected payment of each type of each bidder is the same in both mechanisms and therefore the expected revenue of the seller is identical.*

Schematic (but honest) derivation. By Proposition 11.29, in any mechanism the expected payment of type θ is $T(\theta) = \theta Q(\theta) - U(0) - \int_0^\theta Q(s) ds$, an expression that only involves the equilibrium allocation function $Q(\cdot)$ and the constant $U(0)$. If two mechanisms share (i) the same Q (in the symmetric case with efficient allocation, $Q(\theta) = F(\theta)^{n-1}$, the probability that the other $n-1$ types are lower) and (ii) the same $U(0)$, then they share $T(\theta)$ for every θ . The expected revenue of the seller is the sum over bidders of $\mathbb{E}_\theta[T(\theta)]$, hence it coincides. This explains the result, surprising at first sight, that the first price auction, the second price auction, the English and the Dutch ones (with symmetric, risk neutral bidders and independent types) raise *exactly the same* on average: their payment rules look different, but the envelope shows that expected payments are tied down by the allocation. The fine details (existence of equilibrium, measurability of the selection) are in [11] and in the mechanism design material (MWG ch. 23). $\square$

Intuition

Why is the marginal utility of the type *the probability of winning* and nothing else? It is the envelope logic in its purest form. A bidder of type $\theta + d\theta$ could imitate the type θ perfectly (bid the same): it would win with the same probability $Q(\theta)$ and pay the same. The only difference is that, on winning, the object is worth $d\theta$ more: gain $Q(\theta) d\theta$. And the fact that the bidder will actually bid differently (reoptimize)? Second order effect, because the starting point is an optimal strategy. This is why $U'(\theta) = Q(\theta)$: *the slope of the agent's rent is what "carries" the private information*, and the principal cannot tax it without breaking the incentives. Exactly the same argument appears in screening (Mussa–Rosen, Baron–Myerson, MWG ch. 14): the utility of the agent with type θ in any IC contract satisfies $U(\theta) = U(\underline{\theta}) + \int_{\underline{\theta}}^\theta \frac{\partial u}{\partial \theta}(q(s), s) ds$; that integral is the *information rent* that the principal must give up to the high types so that they do not pretend to be low,

and it appears subtracted in the principal's objective function, giving rise to Myerson's "virtual utility".

11.7 General sensitivity: Berge, Danskin and multiple solutions [P2]

We close with the general picture: what can be said about V *without* assuming uniqueness or differentiable selections? The answer comes in three levels: continuity (always, under compactness), differentiability (under uniqueness) and subdifferentiability (in general).

Theorem 11.31 (Berge's maximum theorem). *Let $f : \mathbb{R}^n \times \Theta \rightarrow \mathbb{R}$ be continuous and let $\Gamma : \Theta \rightrightarrows \mathbb{R}^n$ be a continuous correspondence (upper and lower hemicontinuous) with nonempty compact values. Then:*

- (i) *the value function $V(\theta) = \max_{x \in \Gamma(\theta)} f(x, \theta)$ is continuous on Θ ;*
- (ii) *the solution correspondence $X^*(\theta)$ has nonempty compact values and is upper hemicontinuous; in particular, if $X^*(\theta)$ is a single point $x(\theta)$ for each θ , then $x(\cdot)$ is a continuous function.*

We do not prove Berge here (it is an analysis exercise with sequences and compactness; you can see [1] and appendix M of [9]); what matters is knowing *what it is used for*: it guarantees that $v(p, m)$, $e(p, u)$, $c(w, y)$ are continuous and that demands move "without exploding" when prices move, that the value functions of dynamic programming are continuous, and that the limit steps in equilibrium existence theorems are legitimate. Berge is the continuity insurance behind almost all comparative statics.

Theorem 11.32 (Danskin). *Let $K \subseteq \mathbb{R}^n$ be compact, $f : K \times \Theta \rightarrow \mathbb{R}$ continuous, with partial derivative $\partial f / \partial \theta$ continuous in (x, θ) , and $V(\theta) = \max_{x \in K} f(x, \theta)$ with $\Theta \subseteq \mathbb{R}$ open. Then V has directional derivatives at every point, given by*

$$V'(\theta; +1) = \max_{x \in X^*(\theta)} \frac{\partial f}{\partial \theta}(x, \theta), \quad V'(\theta; -1) = - \min_{x \in X^*(\theta)} \frac{\partial f}{\partial \theta}(x, \theta).$$

In particular, if $X^(\theta) = \{x(\theta)\}$ is a single point, V is differentiable at θ with $V'(\theta) = \partial f(x(\theta), \theta) / \partial \theta$ (the envelope formula) without needing to assume that $x(\cdot)$ is differentiable or to use first order conditions.*

Danskin (see [1, 6]; in optimization it is used constantly for max functions and for minimax methods [5, 8]) is the perfect middle ground between Theorem 11.5 (strong hypotheses, pointwise conclusion) and Milgrom and Segal (weak hypotheses, integral conclusion): with compactness and continuity, uniqueness of the maximizer is enough to differentiate V .

11.7.1 Multiple solutions: kinks and the subdifferential

What if $X^*(\theta_0)$ has several elements? Danskin already says it: the one sided derivatives of V exist but may differ, and then V has a *kink* at θ_0 . The minimal example says it all.

Example 11.33 (The simplest kink in the world). Let $X = \{-1, +1\}$ (two points, with no differentiable structure in x) and $f(x, \theta) = \theta x$. Then

$$V(\theta) = \max\{\theta, -\theta\} = |\theta|.$$

For $\theta > 0$ the only optimum is $x = +1$ and $V'(\theta) = +1 = \partial f / \partial \theta$; for $\theta < 0$ the only optimum is $x = -1$ and $V'(\theta) = -1$. At $\theta_0 = 0$ both decisions are optimal, the candidate derivatives are $\{-1, +1\}$, and $V = |\cdot|$ is not differentiable: right derivative $+1$ (the maximum of the candidates, as Danskin says), left derivative -1 . Since V is convex (a maximum of linear functions), the subdifferential chapter gives us the exact language for the bad point:

$$\partial V(0) = [-1, 1] = \text{conv} \left\{ \frac{\partial f}{\partial \theta}(x, 0) : x \in X^*(0) \right\}.$$

This is the general pattern for convex V (for example $\pi(p)$, or the $U(\theta)$ of auctions): the subdifferential of the value function is the convex hull of the partial derivatives over the set of optima; there is a derivative exactly where the optimum is unique. For the firm: $\partial \pi(p) = \text{conv } Y^*(p)$, the set of optimal plans; if there are several optimal plans, “supply” is a correspondence and π has an edge. Figure 11.2 illustrates the kink.

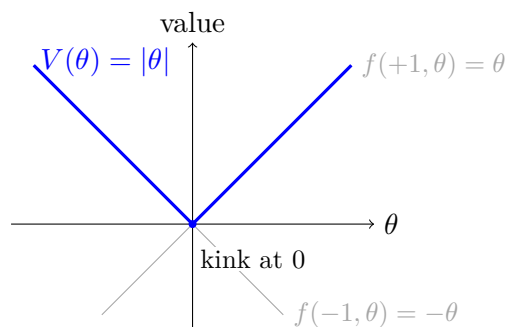


Figure 11.2: With two possible decisions, the value function is the maximum of two lines. Where the optimum changes hands ($\theta = 0$, both decisions tie) the envelope has a kink: there is no derivative, but there is a subdifferential $\partial V(0) = [-1, 1]$, the interval generated by the slopes of the two lines.

Chapter summary

The essentials of the chapter.

- **Value function:** $V(\theta) = \max_{x \in \Gamma(\theta)} f(x, \theta)$. It is the upper envelope of the family $\{f(x, \cdot)\}_x$: it dominates each member and is tangent to the one that touches it. Examples: $v(p, m)$, $e(p, u)$, $\pi(p, w)$, $c(w, y)$.
- **Unconstrained envelope:** $V'(\theta) = \frac{\partial f}{\partial \theta}(x(\theta), \theta)$. The readjustment of x does not count to first order because $\nabla_x f = 0$ at the optimum.
- **Constrained envelope:** $V'(\theta) = \frac{\partial L}{\partial \theta}$ evaluated at the optimum: differentiate the Lagrangian with respect to the parameter with (x, λ) frozen. Key case: if the

parameter is the resource b in $g(x) \leq b$, then $V'(b) = \lambda^*$: the multiplier is the shadow price.

- **The four gems:** Hotelling ($\nabla_p \pi = y$, $\partial \pi / \partial w_i = -x_i$), Shephard ($\partial e / \partial p_i = h_i$), Roy ($x_i = -\frac{\partial v / \partial p_i}{\partial v / \partial m}$, through $\partial v / \partial m = \lambda^*$), cost ($\partial c / \partial w_i = x_i$ conditional, $\partial c / \partial y =$ marginal cost).
- **Curvature:** π convex (sup of linear functions) $\Rightarrow$ law of supply; e, c concave (inf of linear functions) $\Rightarrow$ Slutsky $= \nabla_{pp}^2 e$ symmetric and NSD, law of compensated demand; v quasiconvex. Slutsky equation = Shephard + duality.
- **Milgrom and Segal:** with arbitrary X , $V(\theta) = V(0) + \int_0^\theta f_\theta(x(s), s) ds$; it does not require $x(\cdot)$ to be differentiable. In auctions: $U'(\theta) = Q(\theta)$ (probability of winning) $\Rightarrow$ payment formula $\Rightarrow$ revenue equivalence. In screening: information rent = envelope integral.
- **General picture:** Berge gives continuity of V and upper hemicontinuity of the argmax; Danskin gives directional derivatives $V'(\theta; +1) = \max_{x \in X^*} f_\theta$ and differentiability under uniqueness; with multiple optima, V has kinks and $\partial V(\theta) = \text{conv}\{f_\theta(x, \theta) : x \in X^*(\theta)\}$ in the convex case.

11.8 Exam-style exercises [P1]

Exercise 11.34 (Shadow price, full check). Consider $V(b) = \max_{x_1, x_2 > 0} \ln x_1 + \ln x_2$ s.t. $2x_1 + x_2 \leq b$, with $b > 0$.

- Solve the problem and find $x(b)$ and $\lambda^*(b)$.
- Compute $V(b)$ explicitly and check that $V'(b) = \lambda^*(b)$.
- Compare $V(4.1) - V(4)$ with the prediction $\lambda^*(4) \cdot (0.1)$ and explain the discrepancy.

Solution. (a) The objective is increasing, so the constraint binds and $\lambda > 0$. Lagrangian $L = \ln x_1 + \ln x_2 - \lambda(2x_1 + x_2 - b)$. FOC: $1/x_1 = 2\lambda$ and $1/x_2 = \lambda$, that is $x_1 = \frac{1}{2\lambda}$, $x_2 = \frac{1}{\lambda}$. In the constraint: $2 \cdot \frac{1}{2\lambda} + \frac{1}{\lambda} = \frac{2}{\lambda} = b$, hence $\lambda^*(b) = \frac{2}{b}$, $x_1(b) = \frac{b}{4}$, $x_2(b) = \frac{b}{2}$. (It is a maximum: the objective is strictly concave and the constraint is linear, so KKT is sufficient, KKT chapter.)

(b) $V(b) = \ln \frac{b}{4} + \ln \frac{b}{2} = 2 \ln b - \ln 8$. Differentiating directly: $V'(b) = \frac{2}{b} = \lambda^*(b) \checkmark$, exactly as Corollary 11.10 predicts.

(c) $V(4.1) - V(4) = 2 \ln \frac{4.1}{4} = 2 \ln(1.025) \approx 2 \times 0.024693 = 0.049385$. The linear prediction is $\lambda^*(4) \cdot 0.1 = \frac{2}{4} \cdot 0.1 = 0.05$. The difference (≈ -0.0006) is the second order term $\frac{1}{2} V''(4)(0.1)^2 = \frac{1}{2} \cdot \left(-\frac{2}{16}\right) \cdot 0.01 = -0.000625$: the shadow price slightly overestimates the gain because V is concave in b (decreasing marginal returns to the resource). $\square$

Exercise 11.35 (Shephard's lemma in a Leontief case). A consumer has Leontief preferences $u(x_1, x_2) = \min\{x_1, x_2\}$.

- Find the Hicksian demand $h(p, u)$ and the expenditure function $e(p, u)$ directly (without differential calculus: think which bundle is the cheapest one that guarantees u).

- (b) Check Shephard's lemma.
- (c) Check that e is concave and homogeneous of degree 1 in p , and compute the Slutsky matrix $S = \nabla_{pp}^2 e$. Interpret your result: how much substitution is there?

Solution. (a) To reach $\min\{x_1, x_2\} \geq u$ you need $x_1 \geq u$ and $x_2 \geq u$; since the prices are positive, the cheapest is exactly $x_1 = x_2 = u$. So $h_1(p, u) = h_2(p, u) = u$ (it does not depend on prices) and

$$e(p, u) = p_1 u + p_2 u = u(p_1 + p_2).$$

(b) $\frac{\partial e}{\partial p_1} = u = h_1(p, u)$ ✓ and $\frac{\partial e}{\partial p_2} = u = h_2(p, u)$ ✓. Note that here the objective is not differentiable in x and the optimum is at a “corner” of the indifference curve; even so Shephard works, because what the envelope needs is smoothness *in the parameter* p (expenditure $\langle p, x \rangle$ is linear in p), not in the decision: the spirit of Milgrom and Segal (Remark 11.28).

(c) e is linear in p , hence concave (and convex at the same time: a boundary case) and clearly homogeneous of degree 1: $e(\alpha p, u) = \alpha u(p_1 + p_2)$. The Slutsky matrix is $S = \nabla_{pp}^2 e = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$: symmetric and negative semidefinite (on the boundary). Reading: Leontief is the case of *zero substitution*; the compensated consumer does not react at all to relative prices, and the concavity of e degenerates into linearity. (Compare with Example 11.15, where $e = 2u\sqrt{p_1 p_2}$ is strictly concave: there is substitution there.) □

Exercise 11.36 (Roy's identity with quasilinear utility). Let $u(x_1, x_2) = \ln x_1 + x_2$, with prices (p_1, p_2) and income m large enough for the solution to be interior (assume $m > p_2$).

- (a) Solve the consumer problem and find $x_1(p, m)$, $x_2(p, m)$ and $v(p, m)$.
- (b) Recover x_1 through Roy's identity and check that they agree.
- (c) Compute λ^* and confirm that $\partial v / \partial m = \lambda^*$. What is special about the marginal utility of income in this case and why do economists love quasilinearity?

Solution. (a) Lagrangian $L = \ln x_1 + x_2 - \lambda(p_1 x_1 + p_2 x_2 - m)$. FOC: $\frac{1}{x_1} = \lambda p_1$ and $1 = \lambda p_2$. From the second, $\lambda^* = \frac{1}{p_2}$; substituting into the first, $x_1(p, m) = \frac{p_2}{p_1}$. From the constraint (which binds, since u is increasing in x_2): $x_2(p, m) = \frac{m - p_1 x_1}{p_2} = \frac{m}{p_2} - 1$ (interior because $m > p_2$). Indirect utility:

$$v(p, m) = \ln \frac{p_2}{p_1} + \frac{m}{p_2} - 1.$$

(b) We compute the derivatives: $\frac{\partial v}{\partial p_1} = \frac{\partial}{\partial p_1} (\ln p_2 - \ln p_1) = -\frac{1}{p_1}$ and $\frac{\partial v}{\partial m} = \frac{1}{p_2}$. Roy:

$$x_1 = -\frac{\partial v / \partial p_1}{\partial v / \partial m} = -\frac{-1/p_1}{1/p_2} = \frac{p_2}{p_1} \quad \checkmark,$$

which agrees with (a). (Extra mental exercise: Roy for good 2 gives $x_2 = -(\frac{1}{p_2} - \frac{m}{p_2^2})p_2 = \frac{m}{p_2} - 1$ ✓.)

(c) $\lambda^* = \frac{1}{p_2} = \partial v / \partial m$ ✓ (Corollary 11.10 and equation (R2)). What is special: λ^* *does not depend on* m : the marginal utility of income is constant. This is why with quasilinearity there are no income effects on good 1 ($\partial x_1 / \partial m = 0$), the Marshallian and Hicksian demands for good

1 coincide, and consumer surplus is an exact measure of welfare. It is the favorite assumption in industrial organization and in mechanisms (including Section 11.6.2) precisely because it makes “utils” and “soles” interchangeable at a fixed rate. $\square$

Exercise 11.37 (Inverse Hotelling: from the profit function to the firm). You are told that a competitive firm with one output and two inputs has profit function

$$\pi(p, w_1, w_2) = \frac{p^2}{4} \left(\frac{1}{w_1} + \frac{1}{w_2} \right).$$

- (a) Recover supply $y(p, w)$ and the input demands $x_1(p, w)$, $x_2(p, w)$ through Hotelling’s lemma.
- (b) Check that π is homogeneous of degree 1 and convex in (p, w) “in its own directions” (check at least that $\partial^2 \pi / \partial p^2 \geq 0$ and $\partial^2 \pi / \partial w_i^2 \geq 0$), and confirm the law of supply.
- (c) Propose a technology that generates exactly this π .

Solution. (a) By Hotelling (Theorem 11.12):

$$y = \frac{\partial \pi}{\partial p} = \frac{p}{2} \left(\frac{1}{w_1} + \frac{1}{w_2} \right), \quad x_1 = -\frac{\partial \pi}{\partial w_1} = \frac{p^2}{4w_1^2}, \quad x_2 = -\frac{\partial \pi}{\partial w_2} = \frac{p^2}{4w_2^2}.$$

- (b) Homogeneity: $\pi(tp, tw) = \frac{t^2 p^2}{4} \left(\frac{1}{tw_1} + \frac{1}{tw_2} \right) = t \pi(p, w)$ ✓ (degree 1, like every support function). Own curvatures: $\partial^2 \pi / \partial p^2 = \frac{1}{2} \left(\frac{1}{w_1} + \frac{1}{w_2} \right) > 0$ and $\partial^2 \pi / \partial w_i^2 = \frac{p^2}{2w_i^3} > 0$ ✓. Law of supply: $\partial y / \partial p = \partial^2 \pi / \partial p^2 > 0$ (supply increasing in price) and $\partial x_i / \partial w_i = -\partial^2 \pi / \partial w_i^2 = -\frac{p^2}{2w_i^3} < 0$ (input demand decreasing in its price), exactly as Corollary 11.21 requires.

(c) The additive form suggests two independent “plants”, each like the one in Example 11.13: $F(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}$. Quick check: $\max_x p(\sqrt{x_1} + \sqrt{x_2}) - w_1 x_1 - w_2 x_2$ separates into two problems, each with value $p^2 / (4w_i)$; adding them gives exactly the given π ✓. $\square$

Exercise 11.38 (Milgrom and Segal in a first price auction). Two risk neutral bidders have independent private valuations θ_1, θ_2 uniform on $[0, 1]$. Consider the *first price* auction (the highest bid wins and the winner pays that bid) in its symmetric increasing equilibrium $\beta(\theta)$, and assume $U(0) = 0$.

- (a) Use Proposition 11.29 to compute the equilibrium utility $U(\theta)$ of a bidder of type θ .
- (b) Deduce the equilibrium bid $\beta(\theta)$.
- (c) Check revenue equivalence by comparing with the second price auction.

Solution. (a) In the symmetric equilibrium with increasing β the highest type wins, so the probability that type θ wins is $Q(\theta) = \Pr(\theta_2 < \theta) = F(\theta) = \theta$ (uniform, $n - 1 = 1$ rival). By the Milgrom and Segal envelope formula (Proposition 11.29):

$$U(\theta) = U(0) + \int_0^\theta Q(s) ds = 0 + \int_0^\theta s ds = \frac{\theta^2}{2}.$$

Note that we did not need to know β for this: the envelope computes the equilibrium utility *before* knowing the strategy.

(b) In a first price auction, the utility of type θ , who wins with probability $Q(\theta) = \theta$ paying $\beta(\theta)$ on winning, is $U(\theta) = (\theta - \beta(\theta))\theta$. Setting this equal to (a):

$$(\theta - \beta(\theta))\theta = \frac{\theta^2}{2} \implies \beta(\theta) = \theta - \frac{\theta}{2} = \frac{\theta}{2}.$$

Each bidder bids half of their valuation (in general, with n uniform bidders, $\beta(\theta) = \frac{n-1}{n}\theta$: the expectation of the highest rival conditional on winning).

(c) Expected (unconditional) payment of type θ in the first price auction: $T^I(\theta) = Q(\theta)\beta(\theta) = \theta \cdot \frac{\theta}{2} = \frac{\theta^2}{2}$. In the second price auction, bidding your valuation is dominant and the winner pays the rival's bid: $T^{II}(\theta) = \Pr(\theta_2 < \theta) \cdot \mathbb{E}[\theta_2 \mid \theta_2 < \theta] = \theta \cdot \frac{\theta}{2} = \frac{\theta^2}{2}$. Equal type by type, as Theorem 11.30 requires (same allocation, same $U(\theta)$). Expected revenue of the seller in both: $\mathbb{E}[T(\theta_1) + T(\theta_2)] = 2 \int_0^1 \frac{\theta^2}{2} d\theta = \frac{1}{3}$, which is $\mathbb{E}[\min\{\theta_1, \theta_2\}]$: on average, the seller collects the second highest valuation. Revenue equivalence verified $\checkmark$. $\square$

Exercise 11.39 (Kinks, Danskin and the subdifferential). Let $V(\theta) = \max_{x \in [0,1]} \theta x - x^2$, with $\theta \in \mathbb{R}$.

- (a) Find $x(\theta)$ and $V(\theta)$ explicitly for every θ (careful with the corners of the interval $[0, 1]$).
- (b) At which points is V differentiable? Compute V' where it exists using the envelope and check against the explicit formula.
- (c) Modify the problem to $\tilde{V}(\theta) = \max_{x \in \{0,1\}} \theta x - x^2$ (only two decisions). Find $\tilde{V}$, locate its kink and compute $\partial\tilde{V}$ at that point.

Solution. (a) The objective $x \mapsto \theta x - x^2$ is strictly concave with unconstrained maximum at $x = \theta/2$. Projecting onto $[0, 1]$ (optimality chapter: in one dimensional concave problems, the constrained solution is the projection of the unconstrained one):

$$x(\theta) = \begin{cases} 0, & \theta \leq 0, \\ \theta/2, & 0 \leq \theta \leq 2, \\ 1, & \theta \geq 2, \end{cases} \quad V(\theta) = \begin{cases} 0, & \theta \leq 0, \\ \theta^2/4, & 0 \leq \theta \leq 2, \\ \theta - 1, & \theta \geq 2. \end{cases}$$

(b) The maximizer is *unique for every* θ , so by Danskin (Theorem 11.32, with $K = [0, 1]$ compact) V is differentiable at every θ , with $V'(\theta) = \partial(\theta x - x^2)/\partial\theta|_{x=x(\theta)} = x(\theta)$. Check with the explicit formula: for $\theta < 0$, $V' = 0 = x(\theta)$; for $0 < \theta < 2$, $V' = \theta/2 = x(\theta)$; for $\theta > 2$, $V' = 1 = x(\theta)$; and at the junction points $\theta = 0, 2$ the one sided derivatives agree (0 and 1 respectively) $\checkmark$. Moral: even though the *policy* $x(\theta)$ has corners at $\theta = 0, 2$ (it is not differentiable there), the *value function* is C^1 : uniqueness of the optimum protects the smoothness of V , not that of x .

(c) Now $\tilde{V}(\theta) = \max\{0, \theta - 1\}$: the decision $x = 0$ gives 0 and the decision $x = 1$ gives $\theta - 1$. For $\theta < 1$ the only optimum is $x = 0$ and $\tilde{V} = 0$, $\tilde{V}' = 0$; for $\theta > 1$ the only optimum is $x = 1$ and $\tilde{V}' = 1$. At $\theta = 1$ both decisions are optimal ($X^*(1) = \{0, 1\}$) and $\tilde{V}$ has a kink: one sided derivatives 0 (left) and 1 (right), which are the minimum and the maximum of

$\{f_\theta(x, 1) : x \in X^*(1)\} = \{0, 1\}$, as Danskin dictates. Since $\tilde{V}$ is convex (a maximum of affine functions),

$$\partial\tilde{V}(1) = \text{conv}\{0, 1\} = [0, 1].$$

This is exactly the pattern of Example 11.33: discretizing the decision, the multiplicity of optima at the threshold produces the kink, and the subdifferential (subdifferential chapter) collects all the “candidate slopes”. $\square$

Chapter 12

Fenchel conjugation [P3]

Week 9 of the course. Bibliography: Brinkhuis [3]; Ruszczyński [6]; Boyd and Vandenberghe [2].

What is this used for in Microeconomics?

Fenchel conjugation is the exact mathematical language of the *duality of production theory* that you will see in Microeconomics, inside the individual decision part ([9], Ch. 5). Three concrete links:

- **Profit and cost.** The profit function of a firm is, literally, the *Fenchel conjugate* of its cost function: $\pi(p) = \sup_y \{py - c(y)\} = c^*(p)$. The whole dual theory of the firm (Hotelling’s lemma, Shephard’s lemma, recovery of the technology) is a particular case of the theorems in this chapter.
- **Recoverability.** The Fenchel–Moreau theorem ($f^{**} = \text{clconv } f$) explains *exactly* what can be recovered about the technology from the profit or the cost function: the closed convex hull. This is why duality is said to “convexify”: two technologies with the same convex envelope are indistinguishable from market data. The same happens between indirect and direct utility, and between the expenditure function and utility: they are conjugate pairs (up to signs and monotonicity).
- **Mechanisms and taxation (MWG Ch. 23).** In *screening* and in optimal taxation à la Mirrlees, the *virtual surplus* is “ironed” (*ironing*, Mussa–Rosen, Myerson) by replacing a function with its concave or convex envelope: that is taking a biconjugate. Understanding f^{**} is understanding why ironing works.

Priority 3: it is not the first thing to be examined, but it lights up all the dual material of Micro and comes back in mechanism design.

In earlier chapters we learned that a closed convex set is completely described by its supporting hyperplanes (separation chapter) and that this description is encoded in the *support function* (convex functions chapter). In this chapter we do the same with *functions*: we will describe a convex function by its supporting lines (or hyperplanes), and the bookkeeping of those lines will be a new function, the *Fenchel conjugate*. The economic surprise is that this construction, apparently abstract, is something you already know: the step from the cost function to the profit function of a firm. The central reference of the course for this topic is [3]; also useful are

[6] (sec. 2.6) and [2] (sec. 3.3).

12.1 Motivation: describing a function by its supporting lines [P3]

Let us start in dimension 1 and go slowly. A non vertical line in the plane is determined by two numbers: its *slope* p and its *vertical intercept* b (the value where it crosses the vertical axis):

$$\ell_{p,b}(x) = px - b.$$

(We write $-b$ instead of $+b$ for pure convenience: the signs will come out cleaner; with this convention the line crosses the vertical axis at $-b$.)

Let $f: \mathbb{R} \rightarrow \mathbb{R} \cup \{+\infty\}$ be a function. We say that the line $\ell_{p,b}$ is an **affine minorant** of f if it stays below the graph at every point:

$$px - b \leq f(x) \quad \text{for all } x \in \mathbb{R}.$$

If it also touches the graph at some point, we say that it is a **supporting line** of f .

Now fix the slope p and ask the key question:

among all lines of slope p that stay below f , which one is the highest?

The line $x \mapsto px - b$ stays below f if and only if $b \geq px - f(x)$ for all x , that is, if and only if

$$b \geq \sup_x \{px - f(x)\}.$$

The smallest admissible b , the one giving the highest line, is exactly that supremum. That number, which depends only on the slope p , deserves a name: it will be $f^*(p)$, the conjugate of f evaluated at p .

Geometrically, $px - f(x)$ is the *signed vertical distance* between the line $y = px$ (which passes through the origin) and the graph of f at the abscissa x : positive where the line is above f , negative where it is below. The supremum $f^*(p)$ is the *largest vertical advantage* that the line of slope p gets over f . Figure 12.1 shows this for $f(x) = x^2/2$ and $p = 1$.

Intuition

Think of f^* as a *change of coordinates to describe the same curve*. The usual description of f answers the question “at the point x , how high is the graph?”. The conjugate description answers “for the slope p , where does the supporting line of that slope cross the vertical axis?” (the answer is $-f^*(p)$). For closed convex functions both descriptions carry the same information: knowing all the supporting lines you reconstruct the function as their supremum. This is the functional analogue of describing a closed convex set by its support function.

The economic reading that will guide the whole chapter. Let $c(y)$ be the *cost* function of a firm that produces the quantity $y \geq 0$ of a good whose price is p (the firm is a price taker).

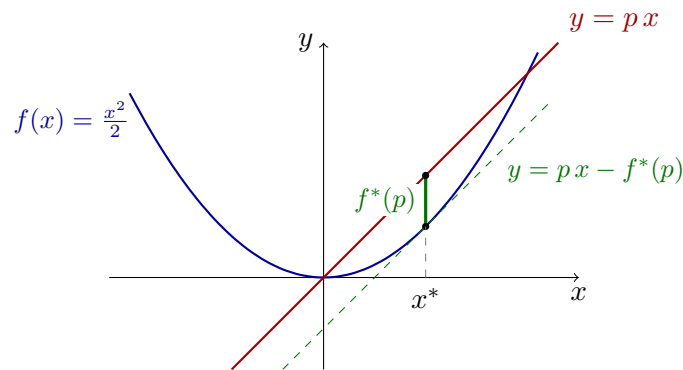


Figure 12.1: The conjugate $f^*(p)$ is the largest vertical gap between the line $y = px$ and the graph of f ; it is attained at the point x^* where the slope of f equals p . Lowering the line $y = px$ by exactly $f^*(p)$ units gives the supporting line of slope p (dashed line).

Its profit when producing y is revenue minus cost, $py - c(y)$, and the firm chooses y to maximize it. The maximum attainable profit is

$$\pi(p) = \sup_y \{ py - c(y) \},$$

which is *exactly* the previous construction with $f = c$:

The profit function is the conjugate of the cost function: $\pi = c^*$.

Each geometric concept then has an economic translation: the slope p is the selling price; the vertical gap $py - c(y)$ is the profit of producing y ; the supremum is the firm's optimization; the point x^* where it is attained is the optimal supply $y(p)$; and the supporting line $y \mapsto py - \pi(p)$ is the maximum isoprofit line that “leans on” the cost curve. We will keep this dictionary active throughout the chapter.

12.2 The Fenchel conjugate: definition and first remarks [P3]

Let us formalize in $\mathbb{R}^n$. Recall from the convex functions chapter that we work with *extended* functions $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$, that f is **proper** if it is not identically $+\infty$, that its **effective domain** is $\text{dom } f = \{x : f(x) < +\infty\}$, and that f is **closed** if its epigraph $\text{epi } f = \{(x, t) : t \geq f(x)\}$ is a closed set (equivalently, f is lower semicontinuous).

Definition 12.1 (Fenchel conjugate). Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper. Its **Fenchel conjugate**, also called the **Legendre–Fenchel transform**, is the function $f^*: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ defined by

$$f^*(p) = \sup_{x \in \mathbb{R}^n} \{ \langle p, x \rangle - f(x) \}.$$

The **biconjugate** is $f^{**} = (f^*)^*$, that is, $f^{**}(x) = \sup_p \{ \langle p, x \rangle - f^*(p) \}$.

Three immediate remarks to read the definition well.

Remark 12.2. (i) The supremum is in fact taken over $x \in \text{dom } f$: outside the domain, $f(x) = +\infty$ and the term $\langle p, x \rangle - f(x) = -\infty$ does not compete. This is the great advantage of extended functions: the constraints are already built in.

- (ii) $f^*(p)$ can be $+\infty$ for some p (if no line of slope p stays below f), but if f is proper and has at least one affine minorant, then f^* never takes the value $-\infty$ and f^* is proper.
- (iii) The variable p lives in the *dual space*: if x are quantities, p are prices; if x are positions, p are slopes. The product $\langle p, x \rangle$ must have “units of value”.

Careful (common mistake)

Three frequent mistakes when computing conjugates:

- (a) **Forgetting the case analysis.** The supremum $\sup_x \{\langle p, x \rangle - f(x)\}$ is *not* always attained at an interior critical point. You have to ask, for *each* value of p : is the supremum finite? is it attained? is it attained at the boundary of the domain or at infinity? A computation that only sets the derivative to zero and solves usually gives a formula valid only for *some* p , and the conjugate is left undefined elsewhere (compare examples 12.6 and 12.7).
- (b) **Confusing the argument.** f^* is a function of p , not of x . After optimizing in x , the variable x must have disappeared from the final formula.
- (c) **Expecting $f^{**} = f$ always.** This is true only for f convex, closed and proper (Fenchel–Moreau theorem, section 12.6). In general $f^{**} \leq f$ and the inequality can be strict.

Exam strategy

Recipe to compute $f^*(p) = \sup_x \{\langle p, x \rangle - f(x)\}$ in an exam:

1. Write $h_p(x) = \langle p, x \rangle - f(x)$ and fix p as a parameter.
2. If f is differentiable in the interior of the domain, look for critical points: $\nabla f(x) = p$. Note that h_p is *concave* when f is convex, so an interior critical point is a global maximum.
3. Determine for which values of p the equation $\nabla f(x) = p$ has a solution in the domain. For those p , substitute and simplify.
4. For the other p , study the behavior of h_p on the boundary of $\text{dom } f$ and when $\|x\| \rightarrow \infty$: the supremum will be a boundary value or $+\infty$.
5. Check: f^* must be convex, and Fenchel–Young ($f(x) + f^*(p) \geq \langle p, x \rangle$) must hold at some easy test point, for example $x = 0$ or $p = 0$. With $p = 0$: $f^*(0) = -\inf_x f(x)$, a very fast check.

12.3 Examples computed in detail [P3]

Example 12.3 (The parabola is self conjugate). Let $f(x) = \frac{x^2}{2}$ on $\mathbb{R}$. Let us compute f^* step by step. Fix $p \in \mathbb{R}$ and study

$$h_p(x) = px - \frac{x^2}{2}.$$

This is an inverted parabola in x (negative quadratic coefficient), so its supremum is attained at the unique critical point:

$$h'_p(x) = p - x = 0 \iff x^* = p.$$

There are no cases here: for *every* $p \in \mathbb{R}$ the equation has a solution. Substituting,

$$f^*(p) = h_p(p) = p \cdot p - \frac{p^2}{2} = \frac{p^2}{2}.$$

Conclusion: $f^*(p) = p^2/2$. The function $x \mapsto x^2/2$ is **self conjugate**: $f^* = f$. One can prove that it is the only function with this property.

Example 12.4 (Quadratic cost and quadratic profit). Let us translate the previous example into economics. A firm has cost $c(y) = y^2/2$ (increasing marginal cost $c'(y) = y$: the first unit is almost free, each extra unit costs more). Its maximum profit at price $p > 0$ is

$$\pi(p) = \sup_y \{p y - c(y)\} = c^*(p) = \frac{p^2}{2},$$

attained at the *supply* $y(p) = p$ (produce until marginal cost equals price: $c'(y) = p$). Note two things that we will generalize: (i) π is convex in p even if c had not been convex in y ; (ii) $\pi'(p) = p = y(p)$: *the derivative of profit with respect to price is supply*. This is Hotelling's lemma, and in section 12.5 we will see that it is a general theorem about conjugates, not an accident of the example.

Example 12.5 (Powers and Hölder dual exponents). Let $q > 1$ and $f(x) = \frac{1}{q}|x|^q$ on $\mathbb{R}$. We claim that

$$f^*(p) = \frac{1}{q'} |p|^{q'}, \quad \text{where } \frac{1}{q} + \frac{1}{q'} = 1 \quad (\text{that is, } q' = \frac{q}{q-1}).$$

The exponents q and q' are the *Hölder conjugates*; Fenchel conjugation explains why they appear together in so many inequalities.

Full computation. First a useful symmetry: f is even ($f(-x) = f(x)$), and for even functions $f^*(-p) = f^*(p)$ (replace x by $-x$ inside the supremum). So it is enough to compute for $p \geq 0$.

Case $p = 0$: $f^*(0) = \sup_x \{-f(x)\} = -\inf_x f(x) = 0$, since $f \geq 0$ and $f(0) = 0$.

Case $p > 0$: since $p x - f(x) \leq p x \leq 0$ for $x \leq 0$ and $h_p(0) = 0$, the supremum is searched for on $x \geq 0$, where $h_p(x) = p x - \frac{1}{q} x^q$. Differentiating,

$$h'_p(x) = p - x^{q-1} = 0 \iff x^* = p^{\frac{1}{q-1}} > 0.$$

Since h_p is concave on $[0, \infty)$ (its second derivative is $-(q-1)x^{q-2} \leq 0$), x^* is a global maximum. We substitute, using that $\frac{1}{q-1} + 1 = \frac{q}{q-1} = q'$:

$$f^*(p) = p \cdot p^{\frac{1}{q-1}} - \frac{1}{q} p^{\frac{q}{q-1}} = p^{q'} - \frac{1}{q} p^{q'} = \left(1 - \frac{1}{q}\right) p^{q'} = \frac{1}{q'} p^{q'}.$$

By symmetry, $f^*(p) = \frac{1}{q'} |p|^{q'}$ for all $p \in \mathbb{R}$.

Check with $q = 2$: then $q' = 2$ and we recover example 12.3. The parabola $x^2/2$ is the fixed point of the duality $q \leftrightarrow q'$.

Byproduct: the Fenchel–Young inequality (theorem 12.10) applied to this pair gives *Young’s inequality*

$$p x \leq \frac{|x|^q}{q} + \frac{|p|^{q'}}{q'},$$

the basis of the Hölder inequality that you used in the preliminaries chapter.

Example 12.6 (Exponential and entropy). Let $f(x) = e^x$ on $\mathbb{R}$. Here the case analysis is mandatory, because the critical point equation does not always have a solution. Fix p and study $h_p(x) = p x - e^x$.

Case $p < 0$: when $x \rightarrow -\infty$, the term $p x \rightarrow +\infty$ (negative times negative) while $e^x \rightarrow 0$. Hence $h_p(x) \rightarrow +\infty$ and $f^*(p) = +\infty$: no line of negative slope stays below e^x , because e^x flattens against the horizontal axis on the left and any decreasing line ends up above it and in fact escapes upwards.

Case $p = 0$: $h_0(x) = -e^x < 0$ for all x , but $h_0(x) \rightarrow 0$ when $x \rightarrow -\infty$. The supremum is 0 and it is *not attained*. So $f^*(0) = 0$: the supremum in the definition can be finite without being attained.

Case $p > 0$: h_p is concave ($h_p'' = -e^x < 0$) and

$$h_p'(x) = p - e^x = 0 \iff x^* = \ln p,$$

which exists precisely because $p > 0$. Substituting,

$$f^*(p) = p \ln p - e^{\ln p} = p \ln p - p.$$

In summary,

$$f^*(p) = \begin{cases} p \ln p - p, & p > 0, \\ 0, & p = 0, \\ +\infty, & p < 0. \end{cases}$$

Note that $p \ln p - p \rightarrow 0$ when $p \downarrow 0$, so f^* is continuous at $p = 0$ within its domain $[0, \infty)$: it is a closed function, as it must be (theorem 12.9). The function $p \ln p - p$ (and its cousin $p \ln p$) is the **negative entropy**; the pair $(e^x, p \ln p - p)$ is central in information theory, in large deviations and in the entropic regularization methods that you may meet in optimal transport.

Example 12.7 (Affine functions: the conjugate is an indicator). Let $f(x) = \langle a, x \rangle + b$ with $a \in \mathbb{R}^n$, $b \in \mathbb{R}$. Then

$$f^*(p) = \sup_x \{ \langle p, x \rangle - \langle a, x \rangle - b \} = \sup_x \langle p - a, x \rangle - b.$$

If $p \neq a$, choose $x = t(p - a)$ with $t \rightarrow +\infty$ and $\langle p - a, x \rangle = t \|p - a\|^2 \rightarrow +\infty$: the supremum

is $+\infty$. If $p = a$, all the terms $\langle p - a, x \rangle$ are 0 and the supremum is $-b$. Therefore

$$f^*(p) = \begin{cases} -b, & p = a, \\ +\infty, & p \neq a, \end{cases} \quad \text{that is,} \quad f^* = \delta_{\{a\}} - b,$$

where δ_C is the indicator function from the convex functions chapter. This makes perfect geometric sense: a line (hyperplane) only admits supporting lines of *its own* slope a , and the best (the only) one has vertical intercept b , that is $-f^*(a) = b$. Moral: the “flatter” f is (the fewer distinct slopes it shows), the more “concentrated” f^* is, and conversely. An affine function (a single possible gradient) has a conjugate whose domain is a single point; at the other extreme, a function with a kink (many subgradients at one point) has a conjugate with a flat piece (exercise 12.22).

Example 12.8 (The conjugate of the indicator is the support function). Let $C \subseteq \mathbb{R}^n$ be non empty and δ_C its indicator ($\delta_C(x) = 0$ if $x \in C$, $+\infty$ otherwise). Then

$$\delta_C^*(p) = \sup_x \{\langle p, x \rangle - \delta_C(x)\} = \sup_{x \in C} \langle p, x \rangle = \sigma_C(p),$$

the **support function** of C from the convex functions chapter. Conjugation thus unifies two theories: the duality convex sets $\leftrightarrow$ support functions is the particular case of Fenchel duality in which the function only distinguishes “inside/outside”. In production theory this is the profit function of a firm with production set Y : $\pi(p) = \sup_{y \in Y} \langle p, y \rangle = \sigma_Y(p) = \delta_Y^*(p)$, with the convention that input quantities enter with a negative sign in y ([9], Ch. 5).

12.4 Basic properties of the conjugate [P3]

The first property is the most important and the cheapest: the conjugate is *always* convex and closed, without asking anything of f .

Theorem 12.9 (The conjugate is always convex and closed). *Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper. Then f^* is convex and closed (lower semicontinuous), even if f is neither convex nor closed.*

Proof. For each fixed $x \in \text{dom } f$, the function

$$p \mapsto \langle p, x \rangle - f(x)$$

is *affine in p* (linear plus constant: here x and $f(x)$ are constants and the variable is p). By definition,

$$f^* = \sup_{x \in \text{dom } f} [p \mapsto \langle p, x \rangle - f(x)],$$

a pointwise supremum of affine functions. In the convex functions chapter we proved that the pointwise supremum of any family of convex functions is convex (the epigraphs intersect, and the intersection of convex sets is convex) and that the supremum of continuous functions is lower semicontinuous (an intersection of closed epigraphs is closed). Affine functions are convex and continuous, hence f^* is convex and closed. $\square$

Intuition

This theorem is the heart of the phrase “duality convexifies”. The process of conjugating only looks at the *affine minorants* of f , and that kind of information is intrinsically convex: no matter how wild f is, what the supporting lines see of it is a convex object. In economics: the profit function $\pi = c^*$ is convex in prices *always*, even if the underlying technology is horrible (non convex, discontinuous). This is why in the data (which record responses to prices) the convexity of π is not an assumption about the technology: it is a tautology of maximization.

The second property is a one line inequality with its own name.

Theorem 12.10 (Fenchel–Young inequality). *For every proper f , every $x \in \mathbb{R}^n$ and every $p \in \mathbb{R}^n$,*

$$f(x) + f^*(p) \geq \langle p, x \rangle.$$

Proof. By definition of supremum, $f^*(p) \geq \langle p, x \rangle - f(x)$ for the given x (the supremum dominates each term). If $f(x) = +\infty$ the inequality is trivial; otherwise, add $f(x)$ to both sides. $\square$

In economic terms with $f = c$ and $f^* = \pi$: for any production plan y and any price p ,

$$\pi(p) \geq p y - c(y),$$

the *maximum* profit dominates the profit of any concrete plan. Equality holds exactly when y is an optimal response to the price p ; that equality case is so important that we devote section 12.5 to it.

Proposition 12.11 (Conjugation reverses the order). *If $f \leq g$ pointwise (both proper), then $f^* \geq g^*$.*

Proof. For each x : $\langle p, x \rangle - f(x) \geq \langle p, x \rangle - g(x)$, because subtracting something smaller leaves something larger. Taking the supremum in x on both sides (the supremum preserves pointwise inequalities), $f^*(p) \geq g^*(p)$. $\square$

Economically: if the technology of firm A is cheaper than that of B ($c_A \leq c_B$), then A gets more profit at any price ($\pi_A \geq \pi_B$). Conjugation is a “mirror”: the small becomes large.

Proposition 12.12 (Calculus rules: translations and scalings). *Let f be proper, $x_0, a \in \mathbb{R}^n$, $\beta \in \mathbb{R}$ and $\lambda > 0$. Then:*

<i>Transformation of f</i>	<i>Resulting conjugate</i>
$g(x) = f(x) + \beta$	$g^*(p) = f^*(p) - \beta$
$g(x) = f(x - x_0)$ (horizontal translation)	$g^*(p) = f^*(p) + \langle p, x_0 \rangle$
$g(x) = f(x) + \langle a, x \rangle$ (affine tilt)	$g^*(p) = f^*(p - a)$
$g(x) = \lambda f(x)$ (vertical scaling)	$g^*(p) = \lambda f^*(p/\lambda)$
$g(x) = f(\lambda x)$ (horizontal scaling)	$g^*(p) = f^*(p/\lambda)$

Proof. All of them are direct substitutions in the definition; we prove two as a model and leave the rest as a mental exercise. *Horizontal translation:* with the change of variable $z = x - x_0$ (a bijection of $\mathbb{R}^n$, so the supremum over x is the supremum over z),

$$g^*(p) = \sup_x \{ \langle p, x \rangle - f(x - x_0) \} = \sup_z \{ \langle p, z + x_0 \rangle - f(z) \} = \langle p, x_0 \rangle + f^*(p).$$

Vertical scaling: with $\lambda > 0$,

$$g^*(p) = \sup_x \{ \langle p, x \rangle - \lambda f(x) \} = \lambda \sup_x \left\{ \left\langle \frac{p}{\lambda}, x \right\rangle - f(x) \right\} = \lambda f^*\left(\frac{p}{\lambda}\right). \quad \square$$

Example 12.13 (Using the rules). Let us compute the conjugate of $g(x) = \frac{1}{2}(x - 3)^2 + 5x$ without optimizing anything. We write $g(x) = f(x - 3) + 5x$ with $f(x) = x^2/2$, whose conjugate we know: $f^* = f$. First the affine tilt: if $g_1(x) = f_1(x) + 5x$ then $g_1^*(p) = f_1^*(p - 5)$, with $f_1(x) = f(x - 3)$. By the translation rule, $f_1^*(p) = f^*(p) + 3p = \frac{p^2}{2} + 3p$. Composing,

$$g^*(p) = f_1^*(p - 5) = \frac{(p - 5)^2}{2} + 3(p - 5).$$

Check by direct computation: g is convex and differentiable, so we solve $g'(x) = x - 3 + 5 = p$, that is $x^* = p - 2$, and substitute:

$$g^*(p) = p(p - 2) - g(p - 2) = (p - 2)(p - 5) - \frac{1}{2}(p - 5)^2 = (p - 5)\left(\frac{p+1}{2}\right),$$

which is exactly $\frac{(p-5)^2}{2} + 3(p - 5)$ expanded (both expressions factor as $(p - 5)(p + 1)/2$). The rules and the direct computation agree, as they must.

12.5 Equality in Fenchel–Young and subdifferentials [P3]

The Fenchel–Young inequality becomes deep theory when we ask *when it is an equality*. The answer connects the conjugate with the subdifferential of the corresponding chapter and, in economics, it is Hotelling’s lemma.

Recall the definition: $p \in \partial f(x)$ (“ p is a subgradient of f at x ”) if

$$f(z) \geq f(x) + \langle p, z - x \rangle \quad \text{for all } z \in \mathbb{R}^n,$$

that is, if the affine function $z \mapsto f(x) + \langle p, z - x \rangle$ is a minorant of f that touches f at x : a *supporting line at x with slope p* .

Theorem 12.14 (Characterization of equality in Fenchel–Young). *Let f be proper, $x \in \text{dom } f$ and $p \in \mathbb{R}^n$. The following statements are equivalent:*

- (i) $p \in \partial f(x)$;
- (ii) $f(x) + f^*(p) = \langle p, x \rangle$ (Fenchel–Young with equality);
- (iii) $x \in \arg \max_z \{ \langle p, z \rangle - f(z) \}$ (the supremum defining $f^*(p)$ is attained at x).

Any of them implies

(iv) $x \in \partial f^*(p)$;

and if f is convex, closed and proper, then (iv) is also equivalent to the previous ones: in that case

$$p \in \partial f(x) \iff x \in \partial f^*(p).$$

Proof. (i) $\Leftrightarrow$ (iii). The condition $p \in \partial f(x)$ says $f(z) \geq f(x) + \langle p, z - x \rangle$ for all z . Rearranging (we move $f(z)$ and $\langle p, x \rangle$ to the other side, a legitimate operation because $f(x)$ is finite):

$$\langle p, x \rangle - f(x) \geq \langle p, z \rangle - f(z) \quad \text{for all } z,$$

which says exactly that $z = x$ maximizes $z \mapsto \langle p, z \rangle - f(z)$. Each step is reversible, so the two conditions are equivalent.

(iii) $\Leftrightarrow$ (ii). If the supremum is attained at x , then $f^*(p) = \langle p, x \rangle - f(x)$, which is (ii). Conversely, if $f^*(p) = \langle p, x \rangle - f(x)$, the value of the supremum is achieved at the point x , which is (iii).

(ii) $\Rightarrow$ (iv). We apply what is already proved, but with f^* playing the role of f . We want to see that $x \in \partial f^*(p)$, that is, by the equivalence (i) $\Leftrightarrow$ (ii) applied to f^* , that

$$f^*(p) + f^{**}(x) = \langle p, x \rangle.$$

We know (ii): $f^*(p) + f(x) = \langle p, x \rangle$. Moreover, by Fenchel–Young applied to the pair (f^*, x) : $f^{**}(x) \geq \langle p, x \rangle - f^*(p) = f(x)$; and always $f^{**} \leq f$ (each term $\langle p, x \rangle - f^*(p)$ is the value at x of an affine minorant of f , and the supremum of minorants does not exceed f). Hence $f^{**}(x) = f(x)$ and substituting we get the desired equality.

(iv) $\Rightarrow$ (ii) when f is convex closed proper. In that case the Fenchel–Moreau theorem (theorem 12.15, which we will prove without using this implication) gives $f^{**} = f$. Condition (iv), via the equivalence (i) $\Leftrightarrow$ (ii) applied to f^* , says $f^*(p) + f^{**}(x) = \langle p, x \rangle$; substituting $f^{**}(x) = f(x)$ we obtain (ii). $\square$

Intuition

The symmetry $p \in \partial f(x) \Leftrightarrow x \in \partial f^*(p)$ says that *the subdifferentials of f and of f^* are the same relation read in a mirror*: the “graph” $\{(x, p) : p \in \partial f(x)\}$ of a closed convex function and that of its conjugate are inverse to each other. When f is differentiable and strictly convex this reads as: ∇f and ∇f^* are inverse functions. In example 12.3, $\nabla f(x) = x$ is its own inverse: this is why $x^2/2$ is self conjugate.

Hotelling’s lemma is Fenchel–Young with equality. Let us apply the theorem to the cost–profit pair, $f = c$ and $c^* = \pi$. The optimal supply at price p is, by definition,

$$y(p) \in \arg \max_y \{py - c(y)\},$$

which is condition (iii). The theorem makes it equivalent to:

- (i) $p \in \partial c(y(p))$: at the optimum, *price is a marginal (sub)cost*. If c is differentiable, $p = c'(y(p))$: the classical rule “price equals marginal cost”. The version with subdifferential

covers costs with kinks (capacity steps, fixed costs in steps), where “the” marginal cost does not exist but the interval of subgradients does.

- (iv) $y(p) \in \partial\pi(p)$: *supply is a subgradient of the profit function*. If π is differentiable at p ,

$$y(p) = \pi'(p),$$

which is **Hotelling’s lemma**. In the envelope chapter we obtained it as an envelope theorem (with the hypotheses of [11]): $\pi(p) = \max_y \{py - c(y)\}$ and the derivative of the optimal value with respect to the parameter p is the partial derivative of the objective evaluated at the optimum, that is $y(p)$. Now we rediscover it without differentiating anything: it is pure conjugation. The Fenchel version is more general (it needs neither uniqueness of the optimum nor differentiability: if there are several optimal supplies, *all* of them are subgradients of π and $\partial\pi(p)$ is exactly the set of optimal supplies, after taking the convex closure).

This is a pattern worth fixing: *envelope theorem in problems linear in the parameter = subdifferential of a conjugate*. The expenditure function and Shephard’s lemma admit the same treatment with the signs adapted (expenditure is a “concave” conjugate of the indicator of the set of bundles at least as good as u).

12.6 Biconjugate and the Fenchel–Moreau theorem [P3]

If conjugating once translates “heights” into “supporting lines”, conjugating twice should give us back the original function, provided that the original function was reconstructible from its supporting lines. Let us see what that means precisely.

Let us expand f^{**} :

$$f^{**}(x) = \sup_p \{ \langle p, x \rangle - f^*(p) \}.$$

For each p with $f^*(p) < +\infty$, the function $x \mapsto \langle p, x \rangle - f^*(p)$ is precisely *the highest possible affine minorant of f with slope p* (that is how we defined $f^*(p)$ in section 12.1). Therefore:

*f^{**} is the pointwise supremum of all the affine minorants of f .*

Two immediate consequences: $f^{**} \leq f$ (a supremum of minorants of f cannot exceed f), and f^{**} is convex and closed (a supremum of affine functions, theorem 12.9). The question is whether the supremum *fills in* everything that can be filled in.

Theorem 12.15 (Fenchel–Moreau). *Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper with at least one affine minorant. Then:*

- (i) *f^{**} is the largest closed convex function that minorizes f ; equivalently, $\text{epi } f^{**} = \text{cl conv}(\text{epi } f)$. We say that f^{**} is the **closed convex envelope** of f , and we write $f^{**} = \text{cl conv } f$.*

- (ii) *$f^{**} = f$ if and only if f is convex and closed.*

Proof (sketch, pointing at where separation enters). Let $g = \text{cl conv } f$ be the largest closed convex minorant of f (it exists: it is the supremum of all closed convex minorants, which is convex

and closed, and the hypothesis of an affine minorant guarantees that it is proper). We want $f^{**} = g$.

Step 1: $f^{**} \leq g$. We already saw that f^{**} is convex, closed and $\leq f$; since g is the *largest* with those properties, $f^{**} \leq g$.

Step 2: $g \leq f^{**}$, the step where separation enters. It is enough to prove that every proper closed convex function is the supremum of its affine minorants; applied to g (whose affine minorants are the same as those of f , since an affine function stays below f if and only if it stays below g , by maximality of g), this gives $g = \sup\{\text{affine minorants of } f\} = f^{**}$.

So let g be proper closed convex, $x_0 \in \mathbb{R}^n$ and $t_0 < g(x_0)$ any height strictly below the graph. The point $(x_0, t_0) \in \mathbb{R}^n \times \mathbb{R}$ does *not* belong to the set $\text{epi } g$, which is convex (because g is convex) and closed (because g is closed). **Here we invoke the strict separation theorem** from the separation chapter: there is a hyperplane in $\mathbb{R}^{n+1}$ that strictly separates the compact point $\{(x_0, t_0)\}$ from the closed convex set $\text{epi } g$: there are $(q, s) \in \mathbb{R}^n \times \mathbb{R}$, $(q, s) \neq 0$, and $\alpha \in \mathbb{R}$ with

$$\langle q, x \rangle + st > \alpha > \langle q, x_0 \rangle + st_0 \quad \text{for all } (x, t) \in \text{epi } g.$$

Since $\text{epi } g$ contains vertical upward rays ($t \rightarrow +\infty$ with x fixed), necessarily $s \geq 0$ (if $s < 0$, the left hand side would go to $-\infty$ and violate the inequality). Two cases remain:

- *Non vertical case* ($s > 0$). Dividing by s and rearranging, the inequality over the points $(x, g(x)) \in \text{epi } g$ gives an affine function $a(x) = -\frac{1}{s}\langle q, x \rangle + \frac{\alpha}{s}$ with $a \leq g$ and $a(x_0) > t_0$: an affine minorant that exceeds the height t_0 at x_0 .
- *Vertical case* ($s = 0$). The inequality separates x_0 from $\text{dom } g$ in $\mathbb{R}^n$, which can only happen if $x_0 \notin \text{cl}(\text{dom } g)$. Then take any affine minorant a_0 of g (it exists because g is proper, convex and closed, again by separation applied to a point below the graph) and add to it a large multiple of the vertical separation term: $a_\mu(x) = a_0(x) + \mu(\alpha - \langle q, x \rangle)$ with the sign arranged so that $\alpha - \langle q, x \rangle \leq 0$ on $\text{dom } g$. Each a_μ is still an affine minorant of g , and at x_0 the added term is strictly positive, so with μ large enough $a_\mu(x_0) > t_0$.

In both cases there is an affine minorant of g whose value at x_0 exceeds t_0 . Since $t_0 < g(x_0)$ was arbitrary, the supremum of the affine minorants at x_0 is at least $g(x_0)$; the other inequality is obvious, and Step 2 is proved.

Part (ii). If f is convex and closed (and proper), then $\text{cl conv } f = f$ and (i) gives $f^{**} = f$. Conversely, if $f^{**} = f$, then f is convex and closed because it is a biconjugate (theorem 12.9). $\square$

Intuition

f^{**} is what an observer who can only lay straight rulers below the graph “sees” of f . Anything that sticks inwards into a local concavity is invisible: the ruler bridges it. In figure 12.2, no line can touch the central crest of the W without crossing the graph; the conjugate twice process replaces the crest by the bridge. Conjugating is a summary *with loss* for non convex functions, and *without loss* exactly for the closed convex ones.

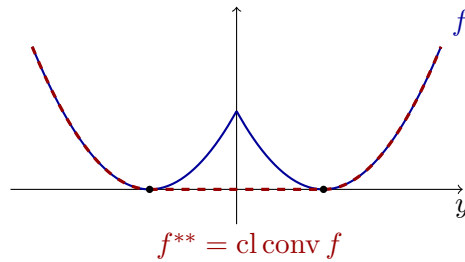


Figure 12.2: A non convex function f (solid line) and its biconjugate f^{**} (dashed line): they agree outside the central “hump”, but f^{**} replaces it by the segment that bridges it. No supporting line can see the hump: this is why conjugation erases it.

12.6.1 Economic application: recovering the technology from profit or cost

This subsection develops the star application for Microeconomics ([9], Ch. 5; [10], Ch. 3).

The problem. The econometrician does not observe the technology of the firm (its cost function c , or its production set Y). What is observed is *behavior in response to prices*: profits $\pi(p)$ and supplies $y(p)$. Can the technology be reconstructed? The answer given by conjugation is exact:

Proposition 12.16 (Cost–profit duality). *Let $c: \mathbb{R} \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper cost function with some affine minorant, and $\pi = c^*$ the profit function. Then the “recovered technology” produced by the dual procedure*

$$\hat{c}(y) = \pi^*(y) = \sup_p \{py - \pi(p)\}$$

*satisfies $\hat{c} = c^{**} = \text{cl conv } c$. As a consequence:*

- (i) *if c is convex and closed, the recovery is perfect: $\hat{c} = c$;*
- (ii) *if not, what is recovered is the closed convex envelope, and any other cost function $\tilde{c}$ with $\text{cl conv } \tilde{c} = \text{cl conv } c$ generates exactly the same profit function and is indistinguishable from c using market data.*

Proof. This is the Fenchel–Moreau theorem applied to c : $\pi^* = (c^*)^* = c^{**} = \text{cl conv } c$. For (ii), if $\text{cl conv } \tilde{c} = \text{cl conv } c$ then $\tilde{c}^* = (\text{cl conv } \tilde{c})^* = (\text{cl conv } c)^* = c^*$, where we use that conjugating a function and conjugating its closed convex envelope give the same result (the affine minorants are the same). $\square$

In the version with production sets it is even more geometric: $\pi = \sigma_Y = \delta_Y^*$ (example 12.8), and the set recovered by the dual procedure

$$\hat{Y} = \{y \in \mathbb{R}^n : \langle p, y \rangle \leq \pi(p) \text{ for all } p\}$$

is $\hat{Y} = \text{cl conv } Y$: the biconjugate of the indicator is the indicator of the closed convex hull, which is the theorem “set = intersection of supporting halfspaces” from the separation chapter, now derived as a particular case of Fenchel–Moreau.

Remark 12.17 (Why this matters in practice). Suppose the technology has a *non convexity*: a fixed cost, a region of increasing returns, indivisibilities. Then $c \neq \text{cl conv } c$ and the proposition says two things. First, the maximizing firm *never chooses* to produce in the region where $c > \text{cl conv } c$ (the “hump” of figure 12.2): at any price, it prefers to jump between the ends of the bridge (for example, alternating between not producing and producing at a large scale) rather than operating in the intermediate region. Second, precisely for that reason the data never reveal the hump: market observation identifies the technology *only up to its convex hull*. This is not a limitation of the econometrician but of the information contained in optimizing behavior. The same logic governs the pair expenditure–utility: from the expenditure function you recover the preference only up to convexifying the indifference curves.

Ironing in mechanism design. The other star appearance of the biconjugate in economic theory: in *screening* (Mussa–Rosen) and in optimal auctions (Myerson), the designer maximizes a *virtual surplus* that may fail to be monotone or concave where needed; the feasibility constraint (monotonicity of the allocation) forces one to replace the problematic function by its concave envelope (or its concave biconjugate, looking at the cumulative integral), a procedure called *ironing*: the humps are flattened just as in figure 12.2, and on the ironed pieces the optimal mechanism groups different types into the same allocation (*bunching*). The mathematical machinery is exactly $f \mapsto f^{**}$ (with concave signs). The same happens in optimal taxation à la Mirrlees when the “relaxed” solution violates monotonicity.

12.7 Fenchel duality [P4]

Conjugation generates its own duality theory for optimization problems, parallel to the Lagrangian duality of the corresponding chapter. The standard format considers an objective split into two convex pieces:

$$(P) \quad v_P = \inf_{x \in \mathbb{R}^n} \{ f(x) + g(x) \},$$

with $f, g: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ proper and convex. The format is less restrictive than it looks: a constraint $x \in C$ is encoded by taking $g = \delta_C$ (indicator functions working for free again), and many problems with the structure “smooth objective + non smooth term” (regularization, penalties) already come split this way.

The **Fenchel dual problem** is

$$(D) \quad v_D = \sup_{p \in \mathbb{R}^n} \{ -f^*(p) - g^*(-p) \}.$$

Proposition 12.18 (Weak Fenchel duality). *Always $v_P \geq v_D$.*

Proof. Fix any x and p . Fenchel–Young twice:

$$f(x) + f^*(p) \geq \langle p, x \rangle, \quad g(x) + g^*(-p) \geq \langle -p, x \rangle = -\langle p, x \rangle.$$

Adding, the inner products cancel: $f(x) + g(x) \geq -f^*(p) - g^*(-p)$. Taking the infimum in x on the left and the supremum in p on the right, $v_P \geq v_D$. $\square$

Theorem 12.19 (Strong Fenchel duality). *Let f, g be proper and convex, and assume the constraint qualification*

$$\text{ri}(\text{dom } f) \cap \text{ri}(\text{dom } g) \neq \emptyset$$

(it is enough, for example, that there is a point where both are finite and one of them is continuous). Then $v_P = v_D$ and, if v_P is finite, the dual supremum is attained: there is an optimal p^ for the dual.*

Proof idea. One defines the *perturbation function* $\varphi(u) = \inf_x \{f(x) + g(x + u)\}$: how much the optimal value improves if g is allowed to be evaluated at a shifted point u . A direct computation shows that φ is convex (it is an infimal convolution, a mental exercise with joint convexity) and that $v_P = \varphi(0)$, while the dual (D) amounts to evaluating the biconjugate: $v_D = \varphi^{**}(0)$. The duality gap is then the difference between $\varphi(0)$ and its closed convexification at 0. The constraint qualification guarantees that $0 \in \text{ri}(\text{dom } \varphi)$, and a theorem from the convex functions chapter says that a proper convex function agrees with its closure on the relative interior of its domain and is subdifferentiable there; the subgradient $p^* \in \partial\varphi(0)$ produced by separation (again the separation theorem, applied to $\text{epi } \varphi$ at the point $(0, \varphi(0))$) is exactly an optimal solution of the dual. The details are in [6], sec. 2.6, and [3]. $\square$

Remark 12.20 (Fenchel and Lagrange: the same theory in different packaging). The Lagrangian duality of the previous chapter and Fenchel duality are interchangeable:

- *From Lagrange to Fenchel.* The Lagrangian dual of $\inf\{f_0(x) : Ax = b\}$ is obtained by writing the problem as $\inf_x \{f_0(x) + \delta_{\{b\}}(Ax)\}$ and applying the Fenchel version with linear composition. With the convention of the previous chapter ($L(x, \lambda) = f_0(x) + \langle \lambda, b - Ax \rangle$) the Lagrangian dual function turns out to be

$$q(\lambda) = \inf_x \{f_0(x) - \langle A^\top \lambda, x \rangle\} + \langle b, \lambda \rangle = \langle b, \lambda \rangle - f_0^*(A^\top \lambda).$$

That is: *the conjugate of the objective is the block from which the Lagrangian dual is built*. If you already computed f_0^* , the dual comes for free.

- *From Fenchel to Lagrange.* (P) is rewritten with a duplicated variable: $\inf\{f(x) + g(z) : x - z = 0\}$. The Lagrangian is $f(x) + g(z) + \langle p, z - x \rangle$, and minimizing separately in x and in z gives

$$q(p) = \inf_x \{f(x) - \langle p, x \rangle\} + \inf_z \{g(z) + \langle p, z \rangle\} = -f^*(p) - g^*(-p),$$

exactly the Fenchel dual objective. The multiplier of the artificial constraint $x = z$ is the dual variable p .

Moral: there are not two dualities, there is one, and it pays to master the dictionary because optimization books use Fenchel ([6, 3]) and economics books use Lagrange or direct support arguments ([9]).

Example 12.21 (A problem solved from both sides). Let us solve

$$v_P = \inf_{x \in \mathbb{R}} \left\{ \frac{x^2}{2} + |x - 1| \right\}, \quad f(x) = \frac{x^2}{2}, \quad g(x) = |x - 1|.$$

Primal side. For $x \leq 1$: the objective is $\frac{x^2}{2} + 1 - x$, with derivative $x - 1 \leq 0$: it decreases up to $x = 1$. For $x \geq 1$: it is $\frac{x^2}{2} + x - 1$, with derivative $x + 1 > 0$: it increases. The minimum is $x^* = 1$ with $v_P = \frac{1}{2} + 0 = \frac{1}{2}$.

Dual side. We need the conjugates. We already know $f^*(p) = p^2/2$. For $g(x) = |x - 1|$ we use the rules (proposition 12.12): $g(x) = h(x - 1)$ with $h(x) = |x|$, and the conjugate of the absolute value is the indicator of $[-1, 1]$:

$$h^*(p) = \sup_x \{px - |x|\} = \begin{cases} 0, & |p| \leq 1 \\ +\infty, & |p| > 1 \end{cases} = \delta_{[-1,1]}(p),$$

(for $|p| \leq 1$ each term $px - |x| \leq 0$ with equality at $x = 0$; for $|p| > 1$ take $x = tp$, $t \rightarrow +\infty$, and $px - |x| = t(p^2 - |p|) \rightarrow +\infty$). By the translation rule, $g^*(p) = h^*(p) + p \cdot 1 = p + \delta_{[-1,1]}(p)$. The dual is

$$v_D = \sup_p \{-f^*(p) - g^*(-p)\} = \sup_{|p| \leq 1} \left\{-\frac{p^2}{2} + p\right\},$$

where we use $g^*(-p) = -p + \delta_{[-1,1]}(-p)$ and that $\delta_{[-1,1]}(-p) = \delta_{[-1,1]}(p)$. The function $p - \frac{p^2}{2}$ is concave with derivative $1 - p$, zero at $p = 1$, which lies just at the edge of the admissible interval: the maximum is $p^* = 1$ with $v_D = 1 - \frac{1}{2} = \frac{1}{2}$.

Checking the theory. $v_P = v_D = \frac{1}{2}$: strong duality, as guaranteed by theorem 12.19 ($\text{dom } f = \text{dom } g = \mathbb{R}$, the qualification is automatic). Moreover the Fenchel–Young type optimality conditions hold: $p^* = 1 \in \partial f(x^*) = \{f'(1)\} = \{1\}$ and $-p^* = -1 \in \partial g(x^*) = \partial|\cdot - 1|(1) = [-1, 1]$. The pair $(x^*, p^*) = (1, 1)$ certifies optimality on both sides without needing to compare values.

12.8 Summary and exercises [P3]

Chapter summary

- **Conjugate:** $f^*(p) = \sup_x \{\langle p, x \rangle - f(x)\}$. Geometric reading: $-f^*(p)$ is the vertical intercept of the supporting line of slope p . Economic reading: if $f = c$ is a cost, $f^* = \pi$ is profit; profit is the conjugate of cost.
- **Always convex and closed:** f^* is a supremum of affine functions, hence convex and closed even if f is not. This is why profit functions are convex in prices without assumptions on the technology.
- **Fenchel–Young:** $f(x) + f^*(p) \geq \langle p, x \rangle$, with equality if and only if $p \in \partial f(x)$, if and only if x solves the supremum, and (for f convex closed) if and only if $x \in \partial f^*(p)$. Economic case: Hotelling’s lemma, $y(p) \in \partial \pi(p)$.
- **Pairs to memorize:** $(\frac{x^2}{2}, \frac{p^2}{2})$ (self conjugate); $(\frac{|x|^q}{q}, \frac{|p|^{q'}}{q'})$ with $\frac{1}{q} + \frac{1}{q'} = 1$; $(e^x, p \ln p - p)$ on $p \geq 0$; affine $\leftrightarrow$ indicator of a point; $\delta_C \leftrightarrow \sigma_C$; $|x| \leftrightarrow \delta_{[-1,1]}$.
- **Fenchel–Moreau:** $f^{**} = \text{cl conv } f$ (closed convex envelope); $f^{**} = f$ if and only if f is proper closed convex. The proof separates a point below the epigraph with the separation theorem. Economically: from π (or from the expenditure function) you

recover the technology (or the preference) only up to its convex hull; the *ironing* of mechanism design is a biconjugation.

- **Fenchel duality:** $\inf\{f + g\} \geq \sup_p\{-f^*(p) - g^*(-p)\}$, with equality and the dual attained if $\text{ri}(\text{dom } f) \cap \text{ri}(\text{dom } g) \neq \emptyset$. It is the Lagrangian duality of the problem $\inf\{f(x) + g(z) : x = z\}$.

Exercise 12.22. Let $a > 0$ and $f(x) = a|x|$ on $\mathbb{R}$. Compute f^* with the full case analysis and give its geometric interpretation.

Solution. Fix p and study $h_p(x) = px - a|x|$.

Case $|p| \leq a$: for all x , by the inequality $px \leq |p||x|$ we have $h_p(x) \leq |p||x| - a|x| = (|p| - a)|x| \leq 0$, and at $x = 0$ it equals $h_p(0) = 0$. The supremum is 0, attained at $x = 0$.

Case $|p| > a$: take $x = tp$ with $t > 0$ and let $t \rightarrow +\infty$:

$$h_p(tp) = tp^2 - at|p| = t|p|(|p| - a) \rightarrow +\infty,$$

since $|p| - a > 0$. The supremum is $+\infty$.

Conclusion:

$$f^*(p) = \begin{cases} 0, & |p| \leq a, \\ +\infty, & |p| > a, \end{cases} \quad \text{that is,} \quad f^* = \delta_{[-a, a]}.$$

Geometry: the only slopes that admit a supporting line below the “V” with slopes $\pm a$ are those in the interval $[-a, a]$, and the best line of each slope passes through the origin (vertical intercept 0). Note the duality of shapes: a function with a “kink” (not differentiable at 0, with $\partial f(0) = [-a, a]$) has a conjugate with a flat “floor” exactly on $[-a, a]$, and conversely. $\square$

Exercise 12.23. Compute the conjugate of the logarithmic barrier

$$f(x) = \begin{cases} -\ln x, & x > 0, \\ +\infty, & x \leq 0. \end{cases}$$

Solution. By remark 12.2, the supremum is taken over the domain: $f^*(p) = \sup_{x>0}\{px + \ln x\}$.

Case $p \geq 0$: when $x \rightarrow +\infty$ we have $px \geq 0$ and $\ln x \rightarrow +\infty$, hence $px + \ln x \rightarrow +\infty$ and $f^*(p) = +\infty$. Geometrically: since $-\ln x \rightarrow -\infty$ when $x \rightarrow \infty$, the graph goes down without bound and no line of slope $p \geq 0$ (which does not go down) can stay below it on the whole domain.

Case $p < 0$: the function $x \mapsto px + \ln x$ is concave on $(0, \infty)$ (second derivative $-1/x^2 < 0$) with derivative $p + \frac{1}{x}$, which vanishes at $x^* = -\frac{1}{p} > 0$. Substituting,

$$f^*(p) = p\left(-\frac{1}{p}\right) + \ln\left(-\frac{1}{p}\right) = -1 - \ln(-p).$$

Conclusion:

$$f^*(p) = \begin{cases} -1 - \ln(-p), & p < 0, \\ +\infty, & p \geq 0. \end{cases}$$

Up to constants and signs, the logarithmic barrier is (almost) self conjugate: $f^*(p) = f(-p) - 1$. This symmetry is the reason for its central role in interior point methods ([5], [2]). $\square$

Exercise 12.24. A firm has cost function

$$c(y) = \begin{cases} \frac{y^2}{2}, & y \geq 0, \\ +\infty, & y < 0 \end{cases}$$

(it cannot produce negative quantities). Compute the profit function $\pi = c^*$ for all $p \in \mathbb{R}$ (including negative prices), determine the supply $y(p)$ and check Hotelling's lemma $y(p) = \pi'(p)$, including at $p = 0$.

Solution. $\pi(p) = \sup_{y \geq 0} \{py - \frac{y^2}{2}\}$.

Case $p \leq 0$: for $y \geq 0$, both terms are ≤ 0 : $py \leq 0$ and $-y^2/2 \leq 0$. The supremum is 0, attained at $y = 0$ (with $p < 0$ it is the unique maximizer; with $p = 0$ it is also unique because $-y^2/2 < 0$ if $y > 0$). The firm does not produce: $y(p) = 0$.

Case $p > 0$: the interior critical point $y = p > 0$ is admissible and the function is concave, so

$$\pi(p) = p \cdot p - \frac{p^2}{2} = \frac{p^2}{2}, \quad y(p) = p.$$

In summary,

$$\pi(p) = \frac{(\max\{p, 0\})^2}{2}, \quad y(p) = \max\{p, 0\}.$$

Checking Hotelling: for $p > 0$, $\pi'(p) = p = y(p)$; for $p < 0$, $\pi'(p) = 0 = y(p)$. At $p = 0$, the one sided derivatives are $\lim_{h \uparrow 0} \frac{\pi(h) - 0}{h} = 0$ and $\lim_{h \downarrow 0} \frac{h^2/2}{h} = 0$: π is differentiable there too, with $\pi'(0) = 0 = y(0)$. The function π is convex, non decreasing and differentiable, even though the constraint $y \geq 0$ “becomes active” at $p \leq 0$: conjugation smooths the corner because the maximizer changes continuously. (If the cost had a linear piece, π would have a corner and supply would be a correspondence: exactly $\partial\pi(p)$.) $\square$

Exercise 12.25. For $f(x) = e^x$ (example 12.6): (a) check the Fenchel–Young inequality numerically at the point $x = 0$, $p = 2$; (b) for $p = 1$, find the x that gives equality and check the three equivalent conditions of theorem 12.14.

Solution. (a) $f(0) = e^0 = 1$ and $f^*(2) = 2 \ln 2 - 2 \approx -0.614$. Then

$$f(0) + f^*(2) \approx 1 - 0.614 = 0.386 \geq \langle 2, 0 \rangle = 0. \checkmark$$

The slack 0.386 measures how far $x = 0$ is from being an optimal response to the “slope” $p = 2$ (the optimal response would be $x = \ln 2$).

(b) With $p = 1$: equality requires $p = f'(x) = e^x$, that is $x = 0$. Let us check the three conditions:

- (ii) Equality: $f(0) + f^*(1) = 1 + (1 \cdot \ln 1 - 1) = 1 - 1 = 0 = \langle 1, 0 \rangle$. $\checkmark$
- (iii) Supremum attained: $h_1(x) = x - e^x$ has derivative $1 - e^x$, positive if $x < 0$ and negative if $x > 0$: global maximum at $x = 0$. $\checkmark$

- (i) Subgradient: f is differentiable and convex, so $\partial f(0) = \{f'(0)\} = \{1\} \ni 1$. ✓

Moreover, (iv): $x = 0$ must belong to $\partial f^*(1)$. Since $(f^*)'(p) = \ln p$ for $p > 0$, indeed $(f^*)'(1) = \ln 1 = 0$. The maps $x \mapsto e^x$ and $p \mapsto \ln p$ are inverses: the gradients of a conjugate pair invert each other. $\square$

Exercise 12.26. Let $f(x) = \min\{(x+1)^2, (x-1)^2\} = (|x|-1)^2$, the “W” function of figure 12.2 (two efficient scales of operation, at $x = -1$ and $x = 1$, with an inefficient intermediate zone). Prove that

$$f^{**}(x) = \begin{cases} 0, & |x| \leq 1, \\ (|x| - 1)^2, & |x| > 1, \end{cases}$$

and give the economic interpretation.

Solution. Call h the candidate function. By the Fenchel–Moreau theorem it is enough to prove that $h = \text{cl conv } f$, that is: (1) h is convex and closed, (2) $h \leq f$, and (3) every closed convex minorant of f stays below h (that is, h is the *largest* one).

(1) h is continuous (the three pieces match with value 0 at $x = \pm 1$), and it is convex: on each piece it is, and at the junctions the derivative is increasing: $h'(x) = 2(x+1) \leq 0$ on $x < -1$, $h' = 0$ on $(-1, 1)$, $h'(x) = 2(x-1) \geq 0$ on $x > 1$; h' is non decreasing on all of $\mathbb{R}$, which characterizes convexity for continuous piecewise differentiable functions.

(2) $h \leq f$: outside $[-1, 1]$ they are equal; inside, $h = 0 \leq (|x| - 1)^2 = f$.

(3) Let g be convex with $g \leq f$. At the minimizers of f : $g(-1) \leq f(-1) = 0$ and $g(1) \leq f(1) = 0$. Every $x \in [-1, 1]$ is written as a convex combination $x = \lambda(-1) + (1 - \lambda)(1)$ with $\lambda = \frac{1-x}{2} \in [0, 1]$, and the convexity of g gives

$$g(x) \leq \lambda g(-1) + (1 - \lambda) g(1) \leq 0 = h(x).$$

Outside $[-1, 1]$, $g \leq f = h$ directly. Hence $g \leq h$ at every point.

By (1) to (3), h is the largest closed convex minorant of f : $f^{**} = \text{cl conv } f = h$.

Interpretation. If f is (a stylized version of) a cost with two efficient scales, the firm never operates in the intermediate zone: at any price it prefers to mix the scales -1 and 1 over time (or across plants), obtaining the cost of the segment, which is 0. The market only reveals f^{**} ; the central hump of f is economically invisible. This is also the mechanism of *ironing*: replace the function by its envelope and accept *bunching* on the flattened piece. $\square$

Exercise 12.27. Using only the known pairs and proposition 12.12, compute the conjugate of the general quadratic cost function

$$c(y) = \frac{\alpha}{2} y^2 + \beta y, \quad \alpha > 0, \beta \geq 0,$$

and interpret each piece of the result (with β as the minimum marginal cost).

Solution. We write $c(y) = \alpha f(y) + \beta y$ with $f(y) = y^2/2$, $f^* = f$. First the vertical scaling: $(\alpha f)^*(p) = \alpha f^*(p/\alpha) = \alpha \cdot \frac{(p/\alpha)^2}{2} = \frac{p^2}{2\alpha}$. Then the affine tilt ($g = \tilde{f} + \langle a, \cdot \rangle$ with $a = \beta$ gives $g^*(p) = \tilde{f}^*(p - \beta)$):

$$\pi(p) = c^*(p) = \frac{(p - \beta)^2}{2\alpha}.$$

(Direct check: $c'(y) = \alpha y + \beta = p$ gives the supply $y(p) = \frac{p-\beta}{\alpha}$ and $\pi(p) = py(p) - c(y(p)) = \frac{(p-\beta)^2}{2\alpha}$, after simplifying.)

Interpretation: β is the marginal cost of the first unit; profit only starts to grow when p exceeds β (for $p < \beta$ this formula gives the branch with $y < 0$; if $y \geq 0$ is imposed as in exercise 12.24, profit is 0 for $p \leq \beta$). And α , the curvature of the cost, appears *inverted* in the profit: very convex costs (large α , rigid supply) give profits that are not very sensitive to price, and conversely. Conjugation exchanges curvatures: it is the same inversion $\nabla c \leftrightarrow \nabla \pi$ of theorem 12.14, and the seed of why in Micro the substitution matrices come from second derivatives of value functions ([9], Ch. 5; [10]). $\square$

Chapter 13

Linear programming and applications [P3]

Week 10 of the course. Bibliography: Nocedal and Wright [5], Ch. 13; Aragón et al. [1]; Li [4].

What is this used for in Microeconomics?

Linear programming (LP) is the *cleanest laboratory* for understanding two ideas that reappear in all modern microeconomic analysis: the **shadow prices** and the **complementary slackness**. In LP these notions are computed by hand, without regularity technicalities, so it pays to master them here before meeting them again in more abstract settings:

- **Market failures: contracts and screening (MWG Ch. 14):** the complementary slackness conditions (“an inactive constraint has a zero multiplier”) are exactly the argument used to determine *which incentive constraints bind* in an adverse selection problem with finitely many types; in fact, once the allocation is fixed, the problem in the transfers is a linear program (Section 13.7.3).
- **Game theory (MWG Ch. 8):** zero sum matrix games are solved as a *primal–dual pair* of linear programs; von Neumann’s minimax theorem is, literally, strong LP duality (Section 13.7.1).
- **General equilibrium (MWG Chs. 15–18):** Koopmans’ activity analysis and linear production models describe the technology as a polyhedron; the equilibrium prices that “value” the resources are the dual variables of the production program [9].
- **Welfare and incentives: mechanism design and auctions (MWG Ch. 23):** the assignment (matching) problem between buyers and objects is a transportation LP; its dual delivers the prices that support the efficient allocation.

Priority 3: it is not the technical core of the Microeconomics course, but it is the best training ground for KKT and duality, which are.

13.1 Motivation: the linear case [P3]

In earlier chapters we studied general optimization problems: KKT conditions, Lagrangian duality, constraint qualifications such as the Slater condition. This chapter specializes to the case where *everything is linear*: the objective function is linear and the constraints are linear equalities or inequalities. This type of problem is called a **linear program** (LP).

Why devote a chapter to a special case? For three reasons.

1. **Everything can be computed.** In LP the Lagrangian dual function $q(\lambda)$ is computed in closed form, strong duality holds without extra hypotheses (feasibility of the problem is enough), and complementary slackness lets you recover the dual solution from the primal with elementary algebra. It is the only setting where *all* the machinery of the previous chapters can be seen working from start to finish without technical assumptions.
2. **The geometry is visible.** The feasible region is a polyhedron (an intersection of halfspaces), and the optimum, when it exists, is attained at a *vertex*. This connects directly with the chapter on cones and hulls: the vertices are the extreme points of the polyhedron, and we already proved that a linear function attains its optimum at an extreme point.
3. **Economics uses it everywhere.** Production with linear technology, the diet problem, transportation and assignment, zero sum games, contracts with finitely many types: all of them are linear programs or primal–dual pairs of linear programs.

A concrete example that will accompany us through the whole chapter:

Example 13.1 (Production problem). A firm makes two goods. Good 1 sells at 3 soles per unit and good 2 at 5 soles per unit. Producing requires three scarce resources:

- hours in plant A: each unit of good 1 uses 1 hour; 4 hours available;
- hours in plant B: each unit of good 2 uses 2 hours; 12 hours available;
- hours in plant C: each unit of good 1 uses 3 hours and each unit of good 2 uses 2 hours; 18 hours available.

If $x_1, x_2 \geq 0$ denote the quantities produced, the problem of maximizing revenue is

$$\begin{aligned}
 \max_{x_1, x_2} \quad & 3x_1 + 5x_2 \\
 \text{s.t.} \quad & x_1 \leq 4, \\
 & 2x_2 \leq 12, \\
 & 3x_1 + 2x_2 \leq 18, \\
 & x_1 \geq 0, \ x_2 \geq 0.
 \end{aligned} \tag{13.1}$$

We will solve it graphically in Section 13.3, build its dual in Section 13.4 and interpret it economically in Section 13.5.

13.2 Standard and canonical forms [P3]

A linear program can be written in many ways: minimize or maximize, with $\leq$ or $\geq$ inequalities, with equalities, with variables that may be negative. For theory and algorithms it pays to fix *reference formats* and to know how to convert any LP into them.

Definition 13.2 (Standard form). An LP is in **standard form** if it is written as

$$\min_{x \in \mathbb{R}^n} \langle c, x \rangle \quad \text{s.t.} \quad Ax = b, \quad x \geq 0, \quad (13.2)$$

where $c \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, A is an $m \times n$ matrix, and $x \geq 0$ means $x_j \geq 0$ for each component $j = 1, \dots, n$. That is: *minimization, equality constraints* and all the variables *non negative*.

Definition 13.3 (Canonical form). An LP is in **canonical form** if it is written as

$$\max_{x \in \mathbb{R}^n} \langle c, x \rangle \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0, \quad (13.3)$$

that is: *maximization, $\leq$ inequality constraints* and non negative variables. (The vector inequality $Ax \leq b$ is read row by row: $\langle a_i, x \rangle \leq b_i$ for $i = 1, \dots, m$, where $a_i^\top$ is row i of A .)

Standard form is the one used by algorithms (simplex, interior points); the canonical one is the most convenient for duality and for economic applications. The production problem (13.1) is already in canonical form with

$$c = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix}.$$

Converting between forms: the four operations

Any LP is converted to standard (or canonical) form with four mechanical tricks. It pays to memorize them because they are used constantly.

(T1) From max to min (and back). Maximizing $\langle c, x \rangle$ is equivalent to minimizing $\langle -c, x \rangle$; the optima x^* are the same and the optimal values differ by a sign: $\max \langle c, x \rangle = -\min \langle -c, x \rangle$.

(T2) From inequality to equality: slack variables. The constraint $\langle a_i, x \rangle \leq b_i$ is equivalent to

$$\langle a_i, x \rangle + s_i = b_i, \quad s_i \geq 0,$$

where s_i is a new variable called a **slack variable**: it measures how much of resource i is “left over”. For $\langle a_i, x \rangle \geq b_i$ one subtracts a **surplus variable**: $\langle a_i, x \rangle - e_i = b_i$ with $e_i \geq 0$.

(T3) From equality to inequalities. The equality $\langle a_i, x \rangle = b_i$ is equivalent to the pair $\langle a_i, x \rangle \leq b_i$ and $-\langle a_i, x \rangle \leq -b_i$.

(T4) Free variables as a difference of non negative ones. If x_j has no sign constraint (a *free variable*), it is replaced by

$$x_j = x_j^+ - x_j^-, \quad x_j^+ \geq 0, \quad x_j^- \geq 0.$$

Every real number is a difference of two non negative ones (for example, $-7 = 0 - 7$), so no solution is lost.

Example 13.4 (Conversion to standard form, step by step). Let us convert to standard form the problem

$$\max_{x_1, x_2} 2x_1 - x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 5, \quad x_1 - 3x_2 \geq -2, \quad x_1 \geq 0,$$

where x_2 is free (it may be negative).

Step 1 (max $\rightarrow$ min, T1). The problem is equivalent to $\min(-2x_1 + x_2)$ with the same constraints.

Step 2 (free variable, T4). We write $x_2 = u - v$ with $u, v \geq 0$. The objective becomes $-2x_1 + u - v$ and the constraints

$$x_1 + u - v \leq 5, \quad x_1 - 3u + 3v \geq -2, \quad x_1, u, v \geq 0.$$

Step 3 (inequalities $\rightarrow$ equalities, T2). We introduce the slack $s_1 \geq 0$ in the first constraint and the surplus $e_2 \geq 0$ in the second:

$$x_1 + u - v + s_1 = 5, \quad x_1 - 3u + 3v - e_2 = -2.$$

Step 4 (assemble). With the variable vector $z = (x_1, u, v, s_1, e_2)^\top \in \mathbb{R}^5$, the problem in standard form (13.2) has

$$c = (-2, 1, -1, 0, 0)^\top, \quad A = \begin{pmatrix} 1 & 1 & -1 & 1 & 0 \\ 1 & -3 & 3 & 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ -2 \end{pmatrix}.$$

Note that the auxiliary variables have zero cost: they do not change the optimal value, they only “reformat” the problem. If the optimal value of the problem in standard form is p^* , that of the original (maximization) problem is $-p^*$ by (T1).

Remark 13.5. The conversions (T1) to (T4) preserve the solution set in the following sense: there is a correspondence between feasible points of one format and of the other that preserves the value of the objective (up to the sign in T1). This is why “solving the LP in standard form” solves the original problem. The price is an increase in dimension: Example 13.4 went from 2 variables to 5.

Careful (common mistake)

Three frequent mistakes when converting:

- **Forgetting the sign of the optimal value with (T1).** The $\arg \max$ and the

arg min agree, but the *value* changes sign. If the exam asks for the optimal value of the original problem, remember to undo the change.

- **Putting the slack with the wrong sign.** In $\langle a, x \rangle \leq b$ the slack is *added* ($\langle a, x \rangle + s = b$); in $\langle a, x \rangle \geq b$ it is *subtracted*. A checking trick: the auxiliary variable must always be ≥ 0 at a feasible point. If with your sign it would come out negative, you wrote it backwards.
- **Treating a free variable as if it were non negative.** If the statement does not say $x_j \geq 0$, the variable is free and you must apply (T4); ignoring this can remove precisely the optimum.

13.3 Geometry: polyhedra and vertices [P3]

13.3.1 The feasible region is a polyhedron

Definition 13.6 (Polyhedron). A **polyhedron** is a set of the form

$$P = \{x \in \mathbb{R}^n : Ax \leq b\},$$

that is, a finite intersection of closed halfspaces $\{x : \langle a_i, x \rangle \leq b_i\}$. A bounded polyhedron is called a **polytope**.

Since each halfspace is convex and closed, and the intersection of closed convex sets is convex and closed (convex sets chapter), every polyhedron is *convex and closed*. The feasible region of any LP is a polyhedron: the equalities $\langle a_i, x \rangle = b_i$ are written as two inequalities, and the conditions $x_j \geq 0$ are linear inequalities.

Definition 13.7 (Vertex or extreme point). A point $v \in P$ is a **vertex** or **extreme point** of P if it cannot be written as a strict convex combination of two other points of P : there are no $y, z \in P$ with $y \neq z$ and $\lambda \in (0, 1)$ such that $v = \lambda y + (1 - \lambda)z$. (This is the same definition of extreme point from the cones and hulls chapter, applied to the convex set P .)

13.3.2 Basic solutions: the algebra of vertices

Vertices have a purely algebraic description which is the basis of the simplex method. Let us work with the standard form: let

$$P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\},$$

where A is $m \times n$ with $\text{rank}(A) = m$ (if there were redundant rows, they are removed) and $m \leq n$.

Definition 13.8 (Basis, basic solution, basic feasible solution). Let $A_j \in \mathbb{R}^m$ denote column j of A .

- A **basis** is a set of indices $B \subseteq \{1, \dots, n\}$ with $|B| = m$ such that the columns $\{A_j : j \in B\}$ are linearly independent (equivalently, the submatrix A_B formed by those columns is invertible).

- The **basic solution** associated with B is the unique $x \in \mathbb{R}^n$ satisfying $Ax = b$ with $x_j = 0$ for all $j \notin B$. Concretely: $x_B = A_B^{-1}b$ and $x_N = 0$, where N is the complement of B . The variables x_j with $j \in B$ are called **basic**; the others, **non basic**.
- If in addition $x_B \geq 0$ (that is, $x \in P$), we say that x is a **basic feasible solution** (BFS).

Example 13.9 (Full enumeration of basic solutions). Consider the region in $\mathbb{R}^2$ given by

$$x_1 + x_2 \leq 4, \quad x_1 + 3x_2 \leq 6, \quad x_1, x_2 \geq 0.$$

In standard form, with slacks $s_1, s_2 \geq 0$:

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 6 \end{pmatrix}, \quad x = (x_1, x_2, s_1, s_2)^\top \geq 0.$$

Here $m = 2$, $n = 4$: each basis picks 2 of the 4 columns. There are at most $\binom{4}{2} = 6$ possible bases. Let us go one by one (for each basis we set the non basic variables to zero and solve the 2×2 system):

- $B = \{3, 4\}$ (the two slacks): $s_1 = 4$, $s_2 = 6$. Solution $(0, 0, 4, 6)$: **BFS**, it corresponds to the origin $(x_1, x_2) = (0, 0)$.
- $B = \{1, 4\}$: $x_1 = 4$ (from the first equation) and $s_2 = 6 - 4 = 2$. Solution $(4, 0, 0, 2)$: **BFS**, point $(4, 0)$.
- $B = \{1, 2\}$: system $x_1 + x_2 = 4$, $x_1 + 3x_2 = 6$. Subtracting: $2x_2 = 2$, hence $x_2 = 1$, $x_1 = 3$. Solution $(3, 1, 0, 0)$: **BFS**, point $(3, 1)$.
- $B = \{2, 3\}$: from the second equation $3x_2 = 6$, that is $x_2 = 2$; from the first, $s_1 = 4 - 2 = 2$. Solution $(0, 2, 2, 0)$: **BFS**, point $(0, 2)$.
- $B = \{2, 4\}$: from the first equation $x_2 = 4$; then $s_2 = 6 - 12 = -6 < 0$. Basic solution $(0, 4, 0, -6)$: *not feasible* (the point $(0, 4)$ violates $x_1 + 3x_2 \leq 6$).
- $B = \{1, 3\}$: from the second equation $x_1 = 6$; then $s_1 = 4 - 6 = -2 < 0$. Basic solution $(6, 0, -2, 0)$: *not feasible* (the point $(6, 0)$ violates $x_1 + x_2 \leq 4$).

The four BFS correspond exactly to the four vertices of the feasible polygon in the plane: $(0, 0)$, $(4, 0)$, $(3, 1)$ and $(0, 2)$. The other two basic solutions are intersections of constraint lines that fall *outside* the region. This pattern is general, as the following theorem says.

Theorem 13.10 (Vertices = basic feasible solutions). *Let $P = \{x : Ax = b, x \geq 0\}$ with $\text{rank}(A) = m$. A point $x \in P$ is a vertex of P if and only if the columns $\{A_j : x_j > 0\}$ are linearly independent, that is, if and only if x is a basic feasible solution for some basis B .*

Proof. ($\Leftarrow$) Let x be a BFS with support $S = \{j : x_j > 0\}$ and columns $\{A_j\}_{j \in S}$ linearly independent. Suppose $x = \lambda y + (1 - \lambda)z$ with $y, z \in P$, $\lambda \in (0, 1)$. For $j \notin S$ we have $x_j = 0$; since $y_j, z_j \geq 0$ and $\lambda y_j + (1 - \lambda)z_j = 0$ with $\lambda \in (0, 1)$, necessarily $y_j = z_j = 0$. Then y and z have support contained in S , and from $Ay = Az = b$ it follows that $A(y - z) = 0$, that is

$\sum_{j \in S} (y_j - z_j) A_j = 0$. By linear independence, $y_j = z_j$ for all $j \in S$, hence $y = z$. So x is not a strict convex combination of distinct points: it is a vertex.

($\Rightarrow$) By contraposition: suppose the columns $\{A_j\}_{j \in S}$ are linearly *dependent*. Then there is $d \neq 0$ with support in S such that $Ad = 0$. For small $\varepsilon > 0$, the points $y = x + \varepsilon d$ and $z = x - \varepsilon d$ satisfy $Ay = Az = b$, and since $x_j > 0$ for $j \in S$ (and $d_j = 0$ outside S), also $y, z \geq 0$ if ε is small enough. But $x = \frac{1}{2}y + \frac{1}{2}z$ with $y \neq z$, so x is not a vertex. (If $|S| < m$, the independent columns of S are completed to a basis with other columns of A , which shows that every vertex is a BFS of some basis.) $\square$

Intuition

A vertex is a point where “the degrees of freedom run out”: there are as many linearly independent active constraints as dimensions. In standard form, each non basic variable at zero is an active constraint $x_j \geq 0$; together with the m equations $Ax = b$, the point is completely determined. This is why a BFS has at most m positive components: the system “closes” and no direction is left in which to move without leaving P . The proof of the theorem is exactly that idea: if there is linear dependence, one degree of freedom is left over (d with $Ad = 0$) and the point can “slide” in two directions, so it is not extreme.

Remark 13.11 (Degeneracy). Different bases can produce the *same* BFS: this happens when some basic variable equals zero (more than n active constraints pass through the same point, like three concurrent lines in the plane). Such a BFS is called **degenerate**. Degeneracy does not affect the theory but it complicates the simplex method (Section 13.6).

13.3.3 The fundamental theorem of linear programming

Theorem 13.12 (Fundamental theorem of LP). *Consider the LP in standard form (13.2) with feasible region $P = \{x : Ax = b, x \geq 0\} \neq \emptyset$.*

(i) *P has at least one vertex.*

(ii) *If the optimal value is finite, then there is a vertex of P that is an optimal solution.*

As a consequence, to solve an LP it is enough to examine the finite set of basic feasible solutions.

Proof. (i) P is contained in the orthant $x \geq 0$, so it *contains no full lines* (a line has points with some coordinate as negative as you like). By the result of the cones and hulls chapter, every non empty closed convex set containing no lines has at least one extreme point; in a polyhedron, the extreme points are its vertices.

(ii) By the polyhedron representation theorem (Minkowski–Weyl, seen in the cones and hulls chapter), P decomposes as

$$P = \text{conv}\{v^1, \dots, v^k\} + \text{cone}\{r^1, \dots, r^p\},$$

where $v^1, \dots, v^k$ are the vertices of P and $r^1, \dots, r^p$ generate its recession cone. Every $x \in P$ is then written as

$$x = \sum_{i=1}^k \lambda_i v^i + \sum_{\ell=1}^p \mu_\ell r^\ell, \quad \lambda_i \geq 0, \sum_i \lambda_i = 1, \mu_\ell \geq 0.$$

If the optimal value $p^* = \inf_{x \in P} \langle c, x \rangle$ is finite, then $\langle c, r^\ell \rangle \geq 0$ for every ray ℓ (if some $\langle c, r^\ell \rangle < 0$, moving along that ray would make $\langle c, x \rangle \rightarrow -\infty$, contradicting that p^* is finite). Hence, for every $x \in P$,

$$\langle c, x \rangle = \sum_i \lambda_i \langle c, v^i \rangle + \sum_\ell \mu_\ell \langle c, r^\ell \rangle \geq \sum_i \lambda_i \langle c, v^i \rangle \geq \min_{1 \leq i \leq k} \langle c, v^i \rangle,$$

because a weighted average is never smaller than the minimum of the terms. Taking the infimum over $x \in P$ gives $p^* \geq \min_i \langle c, v^i \rangle$; and since each v^i is feasible, also $p^* \leq \min_i \langle c, v^i \rangle$. Therefore the minimum is attained at the vertex v^{i^*} that minimizes $\langle c, v^i \rangle$. $\square$

Intuition

A linear function has no “curvature”: on a segment, its value is an average of the values at the endpoints, so it is never worth staying in the interior. Always pushing in the direction where the objective improves, you end at the boundary; inside the boundary (a face of the polyhedron), the same argument pushes towards a face of lower dimension, and so on until reaching a face of dimension zero: a vertex. This is the same principle that in Micro makes optimal production “specialize” in extreme activities with linear technology, instead of mixing a little of everything.

13.3.4 Graphical solution of an LP in two variables

For LPs with two variables, Theorem 13.12 gives a complete graphical method:

Exam strategy

Recipe to solve a 2D LP graphically (and how to present it in an exam):

1. Draw each constraint line $\langle a_i, x \rangle = b_i$ and shade the feasible halfplane (evaluate at a test point, typically the origin).
2. Identify the feasible region (intersection of the halfplanes) and compute *all* its vertices by solving the 2×2 systems of the lines that cross (check that the crossing point is feasible).
3. Draw a level curve of the objective $\langle c, x \rangle = k$ and determine the direction of increase: it is the gradient vector c , perpendicular to the level curves.
4. Slide the level curve in the direction of improvement until the last contact point with the region: that contact is the optimum (a vertex, or a whole edge if the level curve is parallel to it).
5. *Check*: evaluate the objective at all the vertices; the best value must agree with the one from step 4. In an exam, this table of vertices is the safest way to justify the answer.

If the region is unbounded, check in addition whether the objective improves indefinitely along some ray of the region: in that case the LP is *unbounded* and there is no optimum.

Example 13.13 (Graphical solution of the production problem). Let us solve (13.1): $\max 3x_1 + 5x_2$ subject to $x_1 \leq 4$, $2x_2 \leq 12$, $3x_1 + 2x_2 \leq 18$, $x \geq 0$.

Steps 1 and 2: region and vertices. The relevant lines are $x_1 = 4$, $x_2 = 6$ and $3x_1 + 2x_2 = 18$. Intersecting them in pairs (and discarding infeasible crossings) we get the vertices:

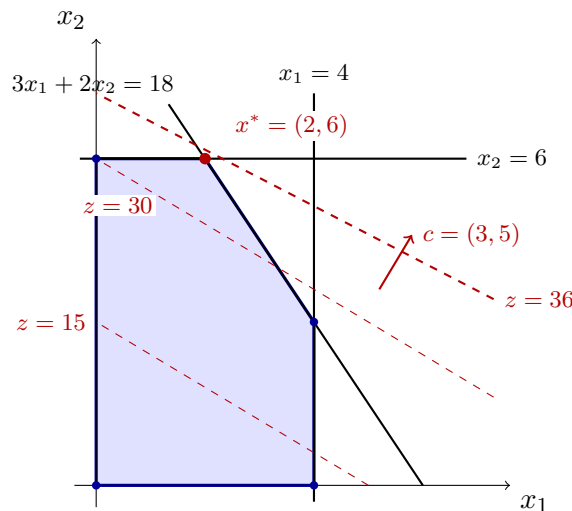
$$(0, 0), \quad (4, 0), \quad (4, 3), \quad (2, 6), \quad (0, 6).$$

For example, $(4, 3)$ solves $x_1 = 4$ and $3x_1 + 2x_2 = 18$ (since $12 + 2x_2 = 18$ gives $x_2 = 3$), and $(2, 6)$ solves $x_2 = 6$ and $3x_1 + 2x_2 = 18$ (since $3x_1 + 12 = 18$ gives $x_1 = 2$). The crossing of $x_1 = 4$ with $x_2 = 6$ is $(4, 6)$, which violates $3(4) + 2(6) = 24 > 18$: it is not a vertex of the region.

Steps 3 to 5: level curves and vertex table. The level curves $3x_1 + 5x_2 = k$ are lines of slope $-3/5$; the objective increases in the direction $c = (3, 5)$. We evaluate:

vertex	$z = 3x_1 + 5x_2$
$(0, 0)$	0
$(4, 0)$	12
$(4, 3)$	27
$(2, 6)$	36
$(0, 6)$	30

The maximum is $z^* = 36$ at $x^* = (2, 6)$: produce 2 units of good 1 and 6 of good 2. At the optimum, the constraints of plants B and C are *active* ($2 \cdot 6 = 12$ and $3 \cdot 2 + 2 \cdot 6 = 18$), while plant A has slack: $2 < 4$ hours are used. Let us keep this fact: it will be key for duality.



The figure shows the feasible region (shaded), three level curves of the objective (dashed) and the optimum: the level curve $z = 36$ touches the region for the last time exactly at the vertex $(2, 6)$.

13.4 Duality in linear programming [P2]

13.4.1 The dual from Lagrangian duality

In the Lagrangian duality chapter we defined, for a general minimization problem, the dual function $q(\lambda) = \inf_x L(x, \lambda)$ and the dual problem $\max_{\lambda} q(\lambda)$. Let us apply that construction, *with no shortcuts*, to the LP in standard form

$$p^* = \min_x \langle c, x \rangle \quad \text{s.t.} \quad Ax = b, \quad x \geq 0.$$

We treat the equalities $Ax = b$ with a free multiplier $\lambda \in \mathbb{R}^m$ (one per equation, with no sign constraint, as corresponds to equalities) and keep the simple constraint $x \geq 0$ *inside* the infimum. The Lagrangian is

$$L(x, \lambda) = \langle c, x \rangle + \langle \lambda, b - Ax \rangle = \langle b, \lambda \rangle + \langle c - A^\top \lambda, x \rangle,$$

where we use $\langle \lambda, Ax \rangle = \langle A^\top \lambda, x \rangle$ (definition of transpose). The dual function is

$$q(\lambda) = \inf_{x \geq 0} L(x, \lambda) = \langle b, \lambda \rangle + \inf_{x \geq 0} \langle c - A^\top \lambda, x \rangle.$$

Let us compute that infimum component by component. The term is $\sum_{j=1}^n (c_j - (A^\top \lambda)_j) x_j$ with each $x_j \geq 0$ independent:

- if $c_j - (A^\top \lambda)_j < 0$ for some j , taking $x_j \rightarrow +\infty$ the term goes to $-\infty$: the infimum is $-\infty$;
- if $c_j - (A^\top \lambda)_j \geq 0$ for every j , the best choice is $x = 0$ and the infimum equals 0.

Therefore

$$q(\lambda) = \begin{cases} \langle b, \lambda \rangle & \text{if } A^\top \lambda \leq c, \\ -\infty & \text{otherwise.} \end{cases} \quad (13.4)$$

The dual problem $\max_{\lambda} q(\lambda)$ automatically discards the λ with $q(\lambda) = -\infty$ (they are never optimal), so what is left is

$$d^* = \max_{\lambda \in \mathbb{R}^m} \langle b, \lambda \rangle \quad \text{s.t.} \quad A^\top \lambda \leq c. \quad (13.5)$$

The dual of an LP is another LP: linear objective $\langle b, \lambda \rangle$, linear constraints $A^\top \lambda \leq c$, free variables. The data swap roles: the resource vector b moves to the objective, the cost vector c moves to the constraints, and the matrix is transposed.

The same computation applied to the canonical form (useful for the economic applications) gives the symmetric pair that we will use from now on:

$$(P) \quad \max \langle c, x \rangle \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0, \quad (D) \quad \min \langle b, y \rangle \quad \text{s.t.} \quad A^\top y \geq c, \quad y \geq 0, \quad (13.6)$$

with one dual variable $y_i \geq 0$ for each primal constraint $\langle a_i, x \rangle \leq b_i$ and one dual constraint for each primal variable. (The sign $y_i \geq 0$ comes from the fact that in a maximization problem with a $\leq$ constraint, relaxing the constraint can only help; it is the same sign convention as in

KKT.) A direct computation (or applying the Lagrangian construction twice) shows that *the dual of the dual is the primal*: duality in LP is a perfect involution.

13.4.2 Rules for building the dual

In practice LPs come mixed: some equality constraints, others inequalities, free and non negative variables. The general rules (all of them derived with the same Lagrangian computation as above) are summarized in the following table. Read it by rows: each object of the primal generates an object of the dual.

Primal: $\max \langle c, x \rangle$	Dual: $\min \langle b, y \rangle$
constraint i : $\langle a_i, x \rangle \leq b_i$	variable $y_i \geq 0$
constraint i : $\langle a_i, x \rangle \geq b_i$	variable $y_i \leq 0$
constraint i : $\langle a_i, x \rangle = b_i$	variable y_i free
variable $x_j \geq 0$	constraint j : $\langle A_j, y \rangle \geq c_j$
variable $x_j \leq 0$	constraint j : $\langle A_j, y \rangle \leq c_j$
variable x_j free	constraint j : $\langle A_j, y \rangle = c_j$

Here $a_i^\top$ is row i of A and A_j is column j . Mnemonic rule: in the direction $\max \rightarrow \min$, “natural” constraints ($\leq$ in the max) generate “natural” variables ($y_i \geq 0$), the equalities generate free variables, and the other way around for the variables. If the primal is a minimization, the rows are read from right to left.

Example 13.14 (Dual of the production problem). Problem (13.1) is a max with three $\leq$ constraints and two variables ≥ 0 . Its dual has three variables $y_1, y_2, y_3 \geq 0$ (one per plant) and two constraints (one per good):

$$\begin{aligned}
 \min_y \quad & 4y_1 + 12y_2 + 18y_3 \\
 \text{s.t.} \quad & y_1 + 3y_3 \geq 3, & (\text{column of good 1}) \\
 & 2y_2 + 2y_3 \geq 5, & (\text{column of good 2}) \\
 & y_1, y_2, y_3 \geq 0.
 \end{aligned} \tag{13.7}$$

Check how it was built: the column of good 1 in A is $(1, 0, 3)^\top$ and its cost is 3, hence $1 \cdot y_1 + 0 \cdot y_2 + 3 \cdot y_3 \geq 3$. We will solve this dual with no extra computation in Example 13.19, using complementary slackness.

13.4.3 Weak and strong duality

Theorem 13.15 (Weak duality). *Let x be feasible for the primal (P) of (13.6) and y feasible for the dual (D). Then*

$$\langle c, x \rangle \leq \langle b, y \rangle.$$

As a consequence: (i) the optimal primal value never exceeds the dual one ($p^* \leq d^*$); (ii) if $\langle c, x \rangle = \langle b, y \rangle$ for a feasible pair, then x is primal optimal and y is dual optimal; (iii) if the primal is unbounded, the dual is infeasible, and if the dual is unbounded (towards $-\infty$), the primal is infeasible.

Proof. Everything follows from chaining two inequalities of scalar products with non negative vectors. Since $y \geq 0$ and $Ax \leq b$ (componentwise),

$$\langle y, Ax \rangle \leq \langle y, b \rangle = \langle b, y \rangle.$$

Since $x \geq 0$ and $A^\top y \geq c$,

$$\langle c, x \rangle \leq \langle A^\top y, x \rangle = \langle y, Ax \rangle.$$

Putting them together: $\langle c, x \rangle \leq \langle y, Ax \rangle \leq \langle b, y \rangle$, which is the claim. The consequences (i) to (iii) are immediate: (ii) because no primal feasible point can exceed $\langle b, y \rangle$ and x already attains it (and symmetrically for y); (iii) because if there were a feasible point of the other problem, it would bound the first one. $\square$

Theorem 13.16 (Strong duality in LP). *If the primal (P) has an optimal solution x^* , then the dual (D) has an optimal solution y^* and the values agree:*

$$\langle c, x^* \rangle = \langle b, y^* \rangle.$$

Moreover, if both problems are feasible, both have an optimal solution and $p^* = d^*$. The duality gap in LP can only be non zero when one of the two problems is infeasible.

We do not give the complete proof (it can be done with Farkas' lemma from the separation chapter, or constructively with the simplex method: on termination, the simplex delivers the optimal multipliers as a byproduct; see [5], Ch. 13, and [1]). What matters is the contrast with the non linear case:

Remark 13.17 (In LP you do not need Slater). In general convex programming, strong duality requires a *constraint qualification* such as the Slater condition (existence of a strictly feasible point). In LP it does **not**: it is enough that the problem is feasible, even if the whole region has empty interior (for example, if there are equalities). The deep reason is that linear constraints cannot “curve” and generate the boundary phenomena that break duality in the non linear case; technically, the separation of polyhedra can always be done *properly* (Farkas' lemma), with no need for interior points. This is a fact worth quoting precisely in an exam: “*constraint qualifications are automatically satisfied when the constraints are affine*”.

13.4.4 Complementary slackness

Theorem 13.18 (Complementary slackness). *Let x^* be feasible for (P) and y^* feasible for (D) in (13.6). Then x^* and y^* are optimal for their respective problems if and only if*

$$y_i^* \cdot (b_i - \langle a_i, x^* \rangle) = 0 \quad \text{for all } i = 1, \dots, m, \quad (13.8)$$

$$x_j^* \cdot (\langle A_j, y^* \rangle - c_j) = 0 \quad \text{for all } j = 1, \dots, n. \quad (13.9)$$

In words: (i) if a primal constraint has slack, its dual price is zero; if a dual price is positive, its constraint is active. (ii) if a primal variable is positive, its dual constraint is active; if a dual constraint has slack, the corresponding primal variable is zero.

Proof. Let us go through the weak duality chain with a magnifying glass:

$$\langle c, x^* \rangle \stackrel{(\alpha)}{\leq} \langle A^\top y^*, x^* \rangle = \langle y^*, Ax^* \rangle \stackrel{(\beta)}{\leq} \langle b, y^* \rangle.$$

The total gap is the sum of two non negative terms:

$$\langle b, y^* \rangle - \langle c, x^* \rangle = \underbrace{\langle y^*, b - Ax^* \rangle}_{\text{gap } (\beta)} + \underbrace{\langle A^\top y^* - c, x^* \rangle}_{\text{gap } (\alpha)} = \sum_{i=1}^m y_i^* (b_i - \langle a_i, x^* \rangle) + \sum_{j=1}^n x_j^* (\langle A_j, y^* \rangle - c_j).$$

Each summand is a product of two non negative numbers (by feasibility of x^* and y^*), hence it is ≥ 0 .

($\Rightarrow$) If x^* and y^* are optimal, strong duality (Theorem 13.16) gives $\langle b, y^* \rangle - \langle c, x^* \rangle = 0$: a sum of non negative terms equal to zero forces *each* term to be zero. That is exactly (13.8) and (13.9).

($\Leftarrow$) If (13.8) and (13.9) hold, all the summands are zero, hence $\langle c, x^* \rangle = \langle b, y^* \rangle$, and by part (ii) of weak duality both points are optimal. $\square$

Intuition

Think of y_i^* as the *shadow price* of resource i . Complementary slackness says two very reasonable things:

- **Leftover resource, zero price.** If at the optimum resource i is left over ($\langle a_i, x^* \rangle < b_i$), having a bit more of that resource would not increase revenue: its marginal value is $y_i^* = 0$. This is the logic of free goods in general equilibrium: what is abundant has no price.
- **Activity used, fair cost.** If activity j is carried out ($x_j^* > 0$), the cost of the resources it consumes, valued at shadow prices, is exactly equal to its unit revenue ($\langle A_j, y^* \rangle = c_j$): the activity leaves neither accounting profit nor loss. If instead the resources it would consume are worth more than what it generates ($\langle A_j, y^* \rangle > c_j$), the activity is not used ($x_j^* = 0$).

This is the same complementary slackness condition of KKT ($\mu_i g_i(x^*) = 0$), but in LP it can be *exploited algebraically* to compute one solution from the other, as the following example shows.

Example 13.19 (Recovering the dual solution from the primal). We already know (Example 13.13) that the primal optimum of the production problem is $x^* = (2, 6)$ with value 36. Let us use complementary slackness to find $y^* = (y_1^*, y_2^*, y_3^*)$ of the dual (13.7) *without solving the dual directly*.

Step 1: which primal constraints have slack? At $x^* = (2, 6)$:

$$x_1^* = 2 < 4 \quad (\text{slack } 2), \quad 2x_2^* = 12 = 12 \quad (\text{active}), \quad 3x_1^* + 2x_2^* = 18 = 18 \quad (\text{active}).$$

By (13.8), the first constraint has positive slack, so its price is zero:

$$y_1^* = 0.$$

Step 2: which primal variables are positive? Both: $x_1^* = 2 > 0$ and $x_2^* = 6 > 0$. By (13.9), both dual constraints must be active (hold with equality):

$$y_1^* + 3y_3^* = 3, \quad 2y_2^* + 2y_3^* = 5.$$

Step 3: solve the system. With $y_1^* = 0$, the first equation gives $3y_3^* = 3$, that is $y_3^* = 1$. Substituting in the second: $2y_2^* + 2 = 5$, that is $y_2^* = 3/2$. So

$$y^* = \left(0, \frac{3}{2}, 1\right).$$

Step 4: full check. (a) Dual feasibility: $y^* \geq 0$; $0 + 3(1) = 3 \geq 3$ and $2(3/2) + 2(1) = 5 \geq 5$.
(b) Dual value:

$$4(0) + 12\left(\frac{3}{2}\right) + 18(1) = 0 + 18 + 18 = 36 = \langle c, x^* \rangle.$$

The values agree, so by weak duality (part (ii) of Theorem 13.15) both points are optimal. We have solved *two* linear programs for the price of one.

Careful (common mistake)

When using complementary slackness to recover the dual solution:

- The implications go only one way: “positive slack $\Rightarrow$ zero price”, but *not* “active constraint $\Rightarrow$ positive price”. It can happen that $y_i^* = 0$ with constraint i active (dual degeneracy). This is why the system built from complementary slackness may have several solutions: any of them that is also *dual feasible* works, and you must always check feasibility and equality of values at the end.
- Do not mix the roles: (13.8) relates *primal* constraints with *dual* variables; (13.9), *primal* variables with *dual* constraints. A typical exam mistake is to impose “dual constraint with slack $\Rightarrow$ zero dual price”, which means nothing.

13.5 Economic interpretation of duality [P2]

13.5.1 The dual values the resources: shadow prices

Let us go back to the production problem (13.1) and its dual (13.7), with optima $x^* = (2, 6)$ and $y^* = (0, 3/2, 1)$. The economic reading of the dual is the following.

- **The dual variables are prices of the resources.** y_i^* is the **shadow price** of resource i : the marginal value, in units of the objective, of one extra unit of that resource. Here: an

extra hour in plant B is worth $3/2$ soles of additional revenue; an extra hour in plant C is worth 1 sol; an extra hour in plant A is worth nothing ($y_1^* = 0$), because hours are already left over. Formally, if $V(b)$ denotes the optimal value as a function of the resource vector, then $y_i^* = \partial V / \partial b_i$ wherever the derivative exists; this is the envelope theorem in its cleanest version [11].

- **The dual objective is the accounting value of the endowment.** $\langle b, y \rangle = 4y_1 + 12y_2 + 18y_3$ is what *all* the resources of the firm are worth at prices y . The dual looks for the prices that *minimize* the imputed value of the endowment, subject to no product being undervalued.
- **The dual constraints are no arbitrage conditions.** The constraint $\langle A_j, y \rangle \geq c_j$ says: *the imputed cost of the resources consumed by one unit of activity j must cover its unit revenue*. If there were a product with $\langle A_j, y \rangle < c_j$, the prices y would undervalue the resources: someone could “buy” the inputs at those prices, produce good j and obtain pure profit. The optimal shadow prices eliminate every arbitrage opportunity, and by complementary slackness the activities actually used ($x_j^* > 0$) have exactly zero accounting profit: $\langle A_j, y^* \rangle = c_j$.
- **Strong duality = product exhaustion.** The equality $\langle c, x^* \rangle = \langle b, y^* \rangle$ says that the maximum revenue of the firm is exactly the value of its resource endowment at shadow prices: all the revenue is “imputed” to the resources, with no residual. It is the linear analogue of the product exhaustion theorem with constant returns to scale, and the entry door to Koopmans’ activity analysis in general equilibrium [9].

Intuition

One way to fix the story: imagine that an *external buyer* offers to rent all the resources of the firm, paying y_i per unit of resource i . For the firm to prefer accepting rather than producing, each activity it could carry out must give it, by selling its inputs, at least what it would give by producing: $\langle A_j, y \rangle \geq c_j$ for every j . The buyer, in turn, wants to pay the least: $\min \langle b, y \rangle$. Strong duality says that at the optimum the buyer pays *exactly* the maximum revenue of the firm: shadow prices transfer all the value, no more and no less. The dual is not an algebraic trick: it is the problem of *valuing the firm through its resources*.

13.5.2 The diet problem

The classical dual example, proposed by Stigler in the forties, reverses the roles of costs and requirements.

Example 13.20 (Diet problem). There are n foods with prices $c_1, \dots, c_n$ and m nutrients. Food j provides N_{ij} units of nutrient i per unit consumed, and a healthy diet requires at least r_i units of nutrient i . The consumer looks for the cheapest diet:

$$\min_{x \geq 0} \langle c, x \rangle \quad \text{s.t.} \quad Nx \geq r.$$

By the rules of the table (read in the direction $\min \rightarrow \max$), the dual is

$$\max_{y \geq 0} \langle r, y \rangle \quad \text{s.t.} \quad N^\top y \leq c.$$

Interpretation: y_i is the *price per unit of nutrient i*. Imagine a seller of pure nutrient pills who wants to replace food: the seller chooses prices $y_i \geq 0$ to maximize the revenue from selling the required basket r , but with the competitive constraint that “rebuilding” any food j with pills must not cost more than the food itself ($\sum_i N_{ij}y_i \leq c_j$); if it cost more, nobody would buy. Strong duality says that the minimum spending on food equals the maximum revenue of the pill seller: nutrients, not foods, are what is really being bought, and the dual reveals their implicit prices. Complementary slackness completes the picture: the foods that do *not* enter the optimal diet are exactly those which, at nutrient prices, are overvalued ($\sum_i N_{ij}y_i^* < c_j$), and the nutrients consumed in excess of the requirement have price zero.

13.5.3 Sensitivity analysis: validity of shadow prices

The shadow price y_i^* measures the effect of one *marginal* unit of resource, but only as long as the change in b_i does not alter *which* constraints are active at the optimum (that is, as long as the optimal basis does not change). **Sensitivity analysis** computes the range $[b_i - \Delta^-, b_i + \Delta^+]$ in which the shadow price is still valid; outside that range, the value function $V(b)$, which is concave and piecewise linear in b , changes slope and the shadow price jumps to a smaller value.

Example 13.21 (Sensitivity in the production problem). In (13.1), let us increase the hours of plant B: $12 + \Delta$. As long as the active constraints are still those of B and C, the optimum moves to the crossing of $2x_2 = 12 + \Delta$ with $3x_1 + 2x_2 = 18$:

$$x_2 = 6 + \frac{\Delta}{2}, \quad x_1 = \frac{18 - 12 - \Delta}{3} = 2 - \frac{\Delta}{3},$$

with value $z(\Delta) = 3(2 - \Delta/3) + 5(6 + \Delta/2) = 36 + \frac{3}{2}\Delta$: each extra hour is indeed worth $y_2^* = 3/2$. But this requires $x_1 \geq 0$ (that is $\Delta \leq 6$) and $x_1 \leq 4$ (that is $\Delta \geq -6$): the shadow price $3/2$ holds in the range $\Delta \in [-6, 6]$, that is, for $b_2 \in [6, 18]$. With $\Delta > 6$ the production of good 1 would reach zero, the optimal basis changes and additional hours of B become worth less.

13.6 Algorithms: simplex and interior points [P4]

This section is conceptual: what matters is *the idea* of each algorithm and its connection with the theory, not the mechanics of tableaus. The detailed reference is [5], Ch. 13; see also [4] for worked examples and [8] for the general context.

13.6.1 The simplex method: walking along vertices

Theorem 13.12 reduces the LP to a finite problem: examining the basic feasible solutions. But enumerating all of them is unfeasible ($\binom{n}{m}$ grows explosively). The **simplex method** (Dantzig, 1947) organizes the search:

1. **Start** at a BFS (a vertex). If none is known, an auxiliary *phase I* finds one by solving another LP whose initial vertex is obvious.
2. **Move to a neighboring vertex that improves the objective.** Two vertices are neighbors if their bases differ in a single column: the step consists in letting a non basic variable whose *reduced cost* indicates improvement enter the basis, and removing the basic variable that first reaches zero as you move along the corresponding edge.
3. **Stop** when no neighbor improves. By convexity of the problem, that vertex is a global optimum: in an LP, a local optimum over the vertices is global (the optimality check of the simplex amounts to exhibiting a feasible dual solution with the same value, that is, a duality certificate).

Why does it terminate? At each step the objective improves strictly (if there is no degeneracy), and there are finitely many vertices, so none is repeated and the process ends at an optimum or detects that the problem is unbounded (an infinite edge along which the objective improves without bound). With **degeneracy** (Remark 13.11) there can be steps that change the basis without moving from the vertex or improving the objective, and in theory the method could *cycle* visiting the same bases; careful selection rules (such as Bland's rule, which always picks the smallest eligible index) guarantee termination. In practice the simplex is remarkably fast, although there are purpose built examples (the Klee–Minty cube) where it visits exponentially many vertices.

A valuable byproduct: on termination, the simplex delivers *for free* the optimal dual solution $y^* = (A_B^{-1})^\top c_B$ (the “simplex multipliers” of the final basis), which gives a constructive proof of strong duality.

13.6.2 Interior points: crossing the polyhedron

The modern alternative (Karmarkar, 1984) gives up the vertices and travels through the *interior* of the feasible region. The connection with the chapters on penalties and barriers is direct: the constraint $x \geq 0$ is replaced by a **log barrier** that penalizes getting close to the boundary:

$$\min_x \langle c, x \rangle - \mu \sum_{j=1}^n \ln x_j \quad \text{s.t.} \quad Ax = b,$$

with parameter $\mu > 0$. For each μ the problem is *strictly convex* in the interior and has a unique minimizer $x(\mu)$; the curve $\{x(\mu) : \mu > 0\}$ is called the **central path**. When $\mu \downarrow 0$, the barrier weighs less and less and $x(\mu)$ converges to an optimal solution of the LP. The algorithm follows the path numerically: for the current μ it takes one or two Newton steps towards $x(\mu)$, reduces μ , and repeats. The optimality conditions of the barrier problem are exactly the KKT conditions of the LP with *perturbed* complementary slackness: $x_j s_j = \mu$ instead of $x_j s_j = 0$ (where $s = c - A^\top \lambda$ is the dual slack), which shows that the method solves the primal and the dual simultaneously, closing the duality gap as $\mu \rightarrow 0$. Unlike the simplex, interior point methods have guaranteed polynomial complexity and are the most efficient on very large and sparse problems [5, 2].

Remark 13.22. For an economist, the algorithmic moral is twofold. The simplex formalizes the intuition of “comparing corners”: with linear technology, the optimal solutions are specialized and are found by exchanging one activity for another at the margin. Interior points formalize the intuition of “prices that adjust gradually”: the central path is a family of approximately optimal allocations whose multipliers converge to the shadow prices. Both readings reappear in the algorithms for computing equilibria.

13.7 Applications [P3]

13.7.1 Zero sum matrix games

A **two person zero sum matrix game** is described by an $m \times n$ matrix A : if the row player chooses row i and the column player chooses column j , the row receives a_{ij} and the column pays a_{ij} . A **mixed strategy** of the row is a probability vector $x \in \Delta_m = \{x \in \mathbb{R}^m : x \geq 0, \sum_i x_i = 1\}$; similarly $y \in \Delta_n$ for the column. The expected payoff is $x^\top A y = \sum_{i,j} x_i a_{ij} y_j$.

The row player's problem. The row wants to guarantee a high payoff *against the worst the column can do*. If the row plays x , the column will respond with the column j that minimizes the expected payoff $\sum_i x_i a_{ij} = (A^\top x)_j$. The row then solves

$$\max_{x \in \Delta_m} \min_{1 \leq j \leq n} (A^\top x)_j.$$

The standard trick turns this max–min into an LP: one introduces a variable v (the guaranteed payoff) and requires that *all* the columns give at least v :

$$\begin{aligned} \max_{x,v} \quad & v \\ \text{s.t.} \quad & \sum_{i=1}^m a_{ij} x_i \geq v, \quad j = 1, \dots, n, \\ & \sum_{i=1}^m x_i = 1, \quad x \geq 0, \quad v \text{ free.} \end{aligned} \tag{13.10}$$

Proposition 13.23 (The dual is the column player's problem). *The dual of the LP (13.10) is*

$$\begin{aligned} \min_{y,w} \quad & w \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} y_j \leq w, \quad i = 1, \dots, m, \\ & \sum_{j=1}^n y_j = 1, \quad y \geq 0, \quad w \text{ free,} \end{aligned} \tag{13.11}$$

which is exactly the column player's problem: $\min_{y \in \Delta_n} \max_i (Ay)_i$, guaranteeing a small payment against the best response of the row.

Proof. We write (13.10) with the constraints in $\leq$ format: $v - \sum_i a_{ij} x_i \leq 0$ for each j , plus the equality $\sum_i x_i = 1$. We apply the table of rules: each inequality j generates a dual variable $y_j \geq 0$; the equality generates a free variable w . The dual objective is $\langle b, (y, w) \rangle = 0 \cdot y + 1 \cdot w = w$ (the right hand sides of the inequalities are zero and that of the equality is one). The dual constraints, one per primal variable: the variable $x_i \geq 0$ (coefficient 0 in the objective, $-a_{ij}$ in inequality j , 1 in the equality) generates $\sum_j (-a_{ij}) y_j + w \geq 0$, that is $\sum_j a_{ij} y_j \leq w$; the free variable v (coefficient 1 in the objective and 1 in each inequality) generates the equality

$\sum_j y_j = 1$. That is (13.11). $\square$

Theorem 13.24 (Von Neumann minimax theorem). *Every matrix game has a **value** v^* and optimal mixed strategies $x^* \in \Delta_m$, $y^* \in \Delta_n$:*

$$\max_{x \in \Delta_m} \min_{y \in \Delta_n} x^\top A y = \min_{y \in \Delta_n} \max_{x \in \Delta_m} x^\top A y = v^*.$$

Proof. The LP (13.10) is feasible (any $x \in \Delta_m$ with v negative enough) and bounded (with $\sum_i x_i = 1$, no column can give more than $\max_{ij} a_{ij}$, so $v \leq \max_{ij} a_{ij}$). By Theorem 13.16 (strong duality), the primal (13.10) and the dual (13.11) have optimal solutions with the same value $v^* = w^*$. Moreover, the minimum of $x^\top A y$ over $y \in \Delta_n$ is always attained at a pure column (a linear function on the simplex is minimized at a vertex, again Theorem 13.12 in miniature), and similarly the maximum over $x \in \Delta_m$ is attained at a pure row; this is why the optimal value of the primal is the max–min and that of the dual is the min–max. The equality of both is the theorem. $\square$

Example 13.25 (A 2×2 game solved completely). Consider the game with payoff matrix (to the row)

$$A = \begin{pmatrix} 3 & -1 \\ -2 & 2 \end{pmatrix}.$$

There is no equilibrium in pure strategies. The best guaranteed pure payoff is

$$\max\{\min(3, -1), \min(-2, 2)\} = -1 \quad (\text{row}), \quad \min\{\max(3, -2), \max(-1, 2)\} = 2 \quad (\text{column});$$

since $-1 < 2$, mixing is needed.

Row player. With $x = (p, 1 - p)$, the expected payoffs against each pure column are

$$\text{col. 1: } 3p - 2(1 - p) = 5p - 2, \quad \text{col. 2: } -p + 2(1 - p) = 2 - 3p.$$

The LP (13.10) maximizes $v = \min\{5p - 2, 2 - 3p\}$. Since one line increases in p and the other decreases, the maximum of the minimum is where they cross (if the crossing falls in $[0, 1]$, which must be checked):

$$5p - 2 = 2 - 3p \implies 8p = 4 \implies p = \frac{1}{2}, \quad v^* = 5 \cdot \frac{1}{2} - 2 = \frac{1}{2}.$$

Optimal strategy $x^* = (1/2, 1/2)$, value of the game $v^* = 1/2$.

Column player (through the dual). With $y = (q, 1 - q)$, the expected payoffs of each pure row are

$$\text{row 1: } 3q - (1 - q) = 4q - 1, \quad \text{row 2: } -2q + 2(1 - q) = 2 - 4q.$$

The dual (13.11) minimizes $w = \max\{4q - 1, 2 - 4q\}$:

$$4q - 1 = 2 - 4q \implies q = \frac{3}{8}, \quad w^* = 4 \cdot \frac{3}{8} - 1 = \frac{1}{2}.$$

Optimal strategy $y^* = (3/8, 5/8)$ and $w^* = 1/2 = v^*$: the primal and dual values agree, confirming strong duality.

Reading through complementary slackness. Both variables $x_i^* > 0$, so both dual constraints are active: against y^* , the two rows give exactly $1/2$ (check it: $4 \cdot 3/8 - 1 = 1/2$ and $2 - 4 \cdot 3/8 = 1/2$). And both $y_j^* > 0$, so against x^* both columns give exactly $1/2$. This is the *indifference condition* of mixed equilibria in game theory: each player mixes only between actions that give the same expected payoff, and that condition *is* complementary slackness of the primal–dual pair.

13.7.2 Transportation and assignment

Example 13.26 (Transportation problem). There are m origins (plants) with supplies $s_1, \dots, s_m$ and n destinations (markets) with demands $d_1, \dots, d_n$, with $\sum_i s_i = \sum_j d_j$ (balanced market). Shipping one unit from i to j costs c_{ij} . If $x_{ij} \geq 0$ is the quantity shipped from i to j , the planner solves

$$\min_{x \geq 0} \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \quad \text{s.t.} \quad \sum_{j=1}^n x_{ij} = s_i \quad \forall i, \quad \sum_{i=1}^m x_{ij} = d_j \quad \forall j.$$

It is an LP with mn variables and $m + n$ equality constraints. Its dual, with free variables u_i (one per origin) and v_j (one per destination), is

$$\max_{u, v} \sum_{i=1}^m s_i u_i + \sum_{j=1}^n d_j v_j \quad \text{s.t.} \quad u_i + v_j \leq c_{ij} \quad \forall i, j.$$

Economic interpretation: $-u_i$ is the price of the good at origin i and v_j its price at destination j (the absolute levels are undetermined: adding a constant to all the u and subtracting it from all the v changes nothing; only the *differentials* matter). The dual constraint $u_i + v_j \leq c_{ij}$ is again *spatial no arbitrage*: the margin earned by buying at i and selling at j cannot exceed the transport cost; and by complementary slackness, the routes actually used ($x_{ij}^* > 0$) satisfy $u_i^* + v_j^* = c_{ij}$: on the active routes, the price differential exactly covers the freight. It is the basic model of market integration and the finite version of the optimal transport problem.

Remark 13.27 (Assignment and matching). The particular case $m = n$, $s_i = d_j = 1$ is the **assignment problem**: assign n workers to n tasks (or buyers to objects) minimizing the total cost, with $x_{ij} \in \{0, 1\}$. At first sight it is an *integer* problem, but the linear relaxation ($0 \leq x_{ij}$, with the supply and demand equalities) has a remarkable property: its vertices are exactly the permutation matrices (Birkhoff–von Neumann theorem: the polytope of doubly stochastic matrices is the convex hull of the permutations). By Theorem 13.12, solving the LP already delivers an integer assignment: linearity “gives away” integrality. The dual variables (u_i, v_j) are the wages or prices that *support* the efficient matching: each assigned pair satisfies $u_i^* + v_j^* = c_{ij}$ and no unassigned pair could block. This structure is the basis of the model of matching with transfers (Shapley–Shubik, Becker) and of the auctions for multiple objects in mechanism design [9].

13.7.3 Screening with finitely many types as a linear program

In adverse selection (*screening*) models with a finite number of types (the monopolist designing a menu of contracts, the regulator facing a firm with unknown cost [10, 9]), the principal chooses, for each type θ_k ($k = 1, \dots, K$), an allocation q_k and a transfer t_k , subject to *participation* (IR) and *incentive compatibility* (IC) constraints:

$$\theta_k v(q_k) - t_k \geq 0 \quad \forall k, \quad \theta_k v(q_k) - t_k \geq \theta_k v(q_\ell) - t_\ell \quad \forall k, \ell.$$

Note the structure: *once the allocations* $(q_1, \dots, q_K)$ are fixed, both the principal's objective (expected transfers minus costs) and all the IR and IC constraints are *linear in the transfers* $(t_1, \dots, t_K)$. The problem of extracting the maximum payment given the menu of quantities is therefore a *linear program*, and all the technology of this chapter applies:

- the solution is at a vertex of the polyhedron of feasible transfers, that is, at a point where “enough” IR and IC constraints are active (Theorem 13.10);
- the complementary slackness of the LP identifies *which* of them bind: with the usual ordering conditions on the types, the ones that turn out to be active are the IR of the lowest type and the adjacent “downward” IC constraints, exactly the pattern taught in contract theory; the remaining constraints have zero multiplier and can be ignored;
- the dual multipliers of the IC constraints measure the marginal cost (in informational rent) of each incentive constraint, and feed the “virtual” distortion formula of the optimum.

In advanced courses this approach is exploited computationally: the mechanism design problems with finitely many types are solved literally with an LP solver.

13.8 Summary and exercises [P3]

Chapter summary

- **Forms.** Every LP is taken to standard form ($\min \langle c, x \rangle, Ax = b, x \geq 0$) or canonical form ($\max \langle c, x \rangle, Ax \leq b, x \geq 0$) with four tricks: change the sign of the objective, slacks and surpluses, split equalities, and free variables as $x_j^+ - x_j^-$.
- **Geometry.** The feasible region is a polyhedron; its vertices are the basic feasible solutions ($x_B = A_B^{-1}b \geq 0, x_N = 0$). Fundamental theorem: if the optimum is finite, it is attained at a vertex; this is why 2D LPs are solved by evaluating the objective at the vertices.
- **Duality.** The Lagrangian dual function of the standard LP is $q(\lambda) = \langle b, \lambda \rangle$ if $A^\top \lambda \leq c$ and $-\infty$ if not: the dual of an LP is another LP. Weak duality: $\langle c, x \rangle \leq \langle b, y \rangle$ for any feasible pair. Strong duality: in LP feasibility is enough (Slater is not needed). Complementary slackness: $y_i^*(b_i - \langle a_i, x^* \rangle) = 0$ and $x_j^*(\langle A_j, y^* \rangle - c_j) = 0$; it allows you to recover the dual solution from the primal by solving a linear system.
- **Economics.** The dual variables are shadow prices of the resources ($y_i^* = \partial V / \partial b_i$).

within a range); the dual constraints are no arbitrage conditions (the imputed cost covers the revenue); strong duality = product exhaustion.

- **Algorithms.** Simplex: walk along neighboring vertices improving; it terminates by finiteness (with anticycling rules if there is degeneracy). Interior points: follow the central path of the log barrier; perturbed complementary slackness $x_j s_j = \mu$.
- **Applications.** Zero sum games = primal–dual pair (minimax = strong duality; indifference = complementary slackness); transportation and assignment (dual prices that support the matching); screening with finitely many types = LP in the transfers.

References: [1], [5] Ch. 13, [2] Ch. 4, [4], [3].

Exercise 13.28 (Conversion to standard form). Take to standard form the problem

$$\min_{x_1, x_2, x_3} x_1 - 2x_2 + x_3 \quad \text{s.t.} \quad x_1 + x_2 + x_3 = 10, \quad x_1 - x_2 \geq 3, \quad x_1 \geq 0, \quad x_3 \geq 0,$$

where x_2 is free. State the final dimensions of A , b and c .

Solution. It is already a minimization problem, so (T1) is not needed.

Free variable (T4): $x_2 = u - v$, with $u, v \geq 0$. The objective becomes $x_1 - 2u + 2v + x_3$; the constraints:

$$x_1 + u - v + x_3 = 10, \quad x_1 - u + v \geq 3.$$

Inequality to equality (T2): in the second constraint ($\geq$) a surplus $e \geq 0$ is subtracted:

$$x_1 - u + v - e = 3.$$

Assemble: with $z = (x_1, u, v, x_3, e)^\top \in \mathbb{R}^5$,

$$c = (1, -2, 2, 1, 0)^\top, \quad A = \begin{pmatrix} 1 & 1 & -1 & 1 & 0 \\ 1 & -1 & 1 & 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 10 \\ 3 \end{pmatrix},$$

and the problem is $\min \langle c, z \rangle$ s.t. $Az = b$, $z \geq 0$: A is 2×5 , $b \in \mathbb{R}^2$, $c \in \mathbb{R}^5$. The optimal value agrees with the original one, and from a solution z^* you recover $x^* = (z_1^*, z_2^* - z_3^*, z_4^*)$. $\square$

Exercise 13.29 (Graphical 2D LP + dual + complementary slackness). Consider the problem

$$\max_{x_1, x_2} 2x_1 + 3x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 6, \quad x_1 + 2x_2 \leq 8, \quad x_1, x_2 \geq 0.$$

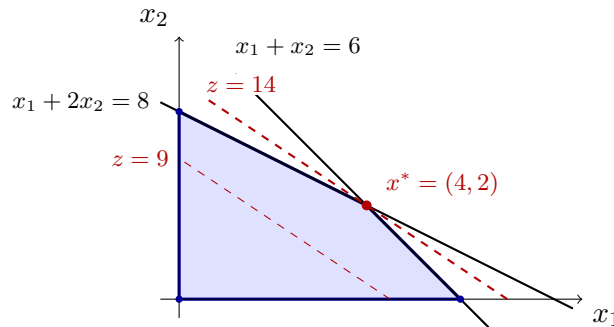
(a) Solve it graphically, computing all the vertices. (b) Write its dual. (c) Recover the optimal dual solution through complementary slackness and check strong duality.

Solution. (a) *Vertices.* The lines are $x_1 + x_2 = 6$ and $x_1 + 2x_2 = 8$, together with the axes. Feasible crossings: $(0, 0)$; $(6, 0)$ (axis 1 with the first line; the crossing $(8, 0)$ of the second violates the first); $(0, 4)$ (axis 2 with the second line; the crossing $(0, 6)$ violates the second); and the

intersection of both lines: subtracting, $x_2 = 2$, hence $x_1 = 4$: the point $(4, 2)$. Vertex table:

vertex	$z = 2x_1 + 3x_2$
$(0, 0)$	0
$(6, 0)$	12
$(4, 2)$	14
$(0, 4)$	12

Optimum: $x^* = (4, 2)$ with $z^* = 14$.



(b) *Dual*. A max with two $\leq$ constraints and variables ≥ 0 : the dual has $y_1, y_2 \geq 0$ and one constraint per primal variable:

$$\min_{y \geq 0} 6y_1 + 8y_2 \quad \text{s.t.} \quad y_1 + y_2 \geq 2, \quad y_1 + 2y_2 \geq 3.$$

(c) *Complementary slackness*. At $x^* = (4, 2)$ both primal constraints are active ($4 + 2 = 6$, $4 + 4 = 8$), so (13.8) does not force any $y_i^* = 0$. Both primal variables are positive, so by (13.9) both dual constraints are active:

$$y_1^* + y_2^* = 2, \quad y_1^* + 2y_2^* = 3.$$

Subtracting: $y_2^* = 1$; hence $y_1^* = 1$. Both are ≥ 0 : dual feasible. Dual value: $6(1) + 8(1) = 14 = z^*$. Strong duality checked; the shadow prices are $y^* = (1, 1)$: one more unit of either of the two resources would increase the objective by one unit. $\square$

Exercise 13.30 (Basic solutions). For the region $x_1 + 2x_2 \leq 6$, $2x_1 + x_2 \leq 6$, $x_1, x_2 \geq 0$ in standard form (with slacks s_1, s_2), enumerate *all* the basic solutions, state which ones are feasible and give the vertex of $\mathbb{R}^2$ to which each BFS corresponds.

Solution. The standard form has

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 6 \\ 6 \end{pmatrix}, \quad x = (x_1, x_2, s_1, s_2)^\top.$$

There are $\binom{4}{2} = 6$ choices of basis; in each one, the variables outside the basis are set to zero:

- $B = \{s_1, s_2\}$: $(0, 0, 6, 6)$. **Feasible**: vertex $(0, 0)$.

- $B = \{x_1, s_1\}$: from row 2, $2x_1 = 6$, $x_1 = 3$; from row 1, $s_1 = 6 - 3 = 3$. $(3, 0, 3, 0)$. **Feasible**: vertex $(3, 0)$.
- $B = \{x_1, s_2\}$: from row 1, $x_1 = 6$; from row 2, $s_2 = 6 - 12 = -6 < 0$. *Not feasible* (the point $(6, 0)$ violates the second constraint).
- $B = \{x_2, s_1\}$: from row 2, $x_2 = 6$; from row 1, $s_1 = 6 - 12 = -6 < 0$. *Not feasible*.
- $B = \{x_2, s_2\}$: from row 1, $2x_2 = 6$, $x_2 = 3$; from row 2, $s_2 = 6 - 3 = 3$. $(0, 3, 0, 3)$. **Feasible**: vertex $(0, 3)$.
- $B = \{x_1, x_2\}$: system $x_1 + 2x_2 = 6$, $2x_1 + x_2 = 6$. Multiplying the first by 2 and subtracting: $3x_2 = 6$, $x_2 = 2$, $x_1 = 2$. $(2, 2, 0, 0)$. **Feasible**: vertex $(2, 2)$.

Four BFS (the vertices $(0, 0)$, $(3, 0)$, $(0, 3)$, $(2, 2)$ of the polygon) and two infeasible basic solutions, which are crossings of lines outside the region. No BFS is degenerate: all of them have exactly $m = 2$ positive components. $\square$

Exercise 13.31 (Weak duality and unboundedness). Using only weak duality (Theorem 13.15), prove: if the primal (P) of (13.6) is unbounded ($\sup \langle c, x \rangle = +\infty$), then the dual (D) is infeasible. Give an example in $\mathbb{R}$ (one variable, one constraint) where *both* problems are infeasible (this shows that the converse is false).

Solution. Suppose, for contradiction, that the dual has a feasible point y^0 . By weak duality, *every* primal feasible x satisfies $\langle c, x \rangle \leq \langle b, y^0 \rangle < +\infty$: the fixed number $\langle b, y^0 \rangle$ is an upper bound of the primal objective, contradicting that $\sup \langle c, x \rangle = +\infty$. Hence the dual has no feasible points.

Example with both infeasible. Take $n = m = 1$, $c = 1$, $A = (0)$, $b = -1$:

$$(P) \max x \text{ s.t. } 0 \cdot x \leq -1, \ x \geq 0; \quad (D) \min (-1)y \text{ s.t. } 0 \cdot y \geq 1, \ y \geq 0.$$

The primal requires $0 \leq -1$: infeasible. The dual requires $0 \geq 1$: infeasible. So “dual infeasible” does not imply “primal unbounded”: the third possibility is that both are infeasible at the same time. $\square$

Exercise 13.32 (2×2 zero sum game). Solve the matrix game with payoffs to the row

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} :$$

set up the LP of the row player, find the optimal mixed strategies of both players and the value of the game, and check that the primal and dual values agree.

Solution. There is no solution in pure strategies: the pure max–min is $\max\{\min(2, -1), \min(-1, 1)\} = -1$ and the pure min–max is $\min\{\max(2, -1), \max(-1, 1)\} = 1$.

Row. With $x = (p, 1-p)$, the LP (13.10) is to maximize $v = \min\{2p - (1-p), -p + (1-p)\} = \min\{3p - 1, 1 - 2p\}$. One line increases and the other decreases; the crossing:

$$3p - 1 = 1 - 2p \implies 5p = 2 \implies p = \frac{2}{5}, \quad v^* = 3 \cdot \frac{2}{5} - 1 = \frac{1}{5}.$$

Since $p = 2/5 \in [0, 1]$, it is admissible: $x^* = (2/5, 3/5)$ and value $v^* = 1/5$.

Column (dual). With $y = (q, 1 - q)$, minimize $w = \max\{2q - (1 - q), -q + (1 - q)\} = \max\{3q - 1, 1 - 2q\}$:

$$3q - 1 = 1 - 2q \implies q = \frac{2}{5}, \quad w^* = \frac{1}{5}.$$

So $y^* = (2/5, 3/5)$ and $w^* = 1/5 = v^*$: strong duality confirmed, the value of the game is $1/5$ (the game favors the row). Indifference check (complementary slackness): against y^* , row 1 gives $2(2/5) - 3/5 = 1/5$ and row 2 gives $-2/5 + 3/5 = 1/5$; against x^* , both columns also give $1/5$. $\square$

Exercise 13.33 (Shadow prices and an investment decision). In the production problem (13.1), the firm can hire one extra hour in *only one* of the three plants, at a cost of 1.2 soles per hour. (a) In which plant is it worth hiring it, and what is the net gain? (b) Would the answer change if the cost were 1.8 soles? (c) Why is the answer to (a) valid only “locally”?

Solution. (a) The shadow prices (Example 13.19) are $y^* = (0, 3/2, 1)$: an extra hour is worth 0 in plant A, 1.5 soles in B and 1 sol in C. At a cost of 1.2, only plant B gives a positive net gain: $1.5 - 1.2 = 0.3$ soles. It is worth hiring in B.

(b) With a cost of $1.8 > 1.5$, no plant gives more than the hour costs (1.5 is the largest shadow price): it is not worth hiring anywhere.

(c) Because the shadow price is a derivative of the value function, valid only while the optimal basis does not change. By Example 13.21, the price $3/2$ of plant B holds for $b_2 \in [6, 18]$; if the firm hired many hours (more than 6 extra), the constraint $x_1 \geq 0$ would become the bottleneck, the basis would change and the additional hours of B would be worth less than $3/2$. For one hour, the computation in (a) is exact because $13 \in [6, 18]$. $\square$

Chapter 14

Support vector machines [P5]

Week 11 of the course. Bibliography: Wright and Recht [8]; Li [4].

What is this used for in Microeconomics?

Let us be honest: support vector machines *do not appear* in the Microeconomics syllabus. There will be no exam question about SVM. So why devote a chapter to them? For three reasons:

- (i) **It is the perfect integrating review.** The SVM is a single problem that uses *everything* we have built: separation of convex sets (separation chapter), KKT conditions with complementary slackness (KKT chapter) and Lagrangian duality with strong duality through Slater (duality chapter). Solving an SVM by hand is the best possible audit of whether you master those three chapters.
- (ii) **Econometrics and machine learning applied to economics.** Classification methods are today standard tools in applied economics: prediction of credit default, classification of poor and non poor households with survey data or satellite images, and text as data analysis in monetary policy documents. The SVM is one of the classical classifiers worth understanding from the optimization side, not as a black box.
- (iii) **It is an excellent topic for a final course presentation.** It combines clean theory (convexity, KKT, duality) with a computational implementation that is accessible in Python or Julia. At the end of the chapter we give a concrete project suggestion.

14.1 The binary classification problem [P5]

14.1.1 Data, labels and linear classifiers

So far we have optimized functions coming from economic theory (utility, profit, cost). Let us change the setting: now the data are in charge.

Definition 14.1 (Binary classification problem). We have a *training sample*

$$\{(x_i, y_i)\}_{i=1}^m, \quad x_i \in \mathbb{R}^n, \quad y_i \in \{-1, +1\},$$

where x_i is a vector of *features* and y_i is the *label* indicating which of two classes observation i belongs to. The goal is to build a function $f: \mathbb{R}^n \rightarrow \{-1, +1\}$, called a *classifier*, that correctly predicts the label of new observations.

Example 14.2 (Credit scoring). A bank observes, for each past client i , a vector $x_i = (\text{income}_i, \text{debt}_i, \text{job tenure}_i, \dots) \in \mathbb{R}^n$ and a label $y_i = +1$ if the client repaid the loan and $y_i = -1$ if the client defaulted. The bank wants a rule that, for a new applicant with features x , predicts whether the loan will be repaid. That is exactly a binary classification problem.

The simplest family of classifiers is that of *linear classifiers*: rules that draw a hyperplane in $\mathbb{R}^n$ and assign one class to each side.

Definition 14.3 (Linear classifier). Given $w \in \mathbb{R}^n$, $w \neq 0$, and $b \in \mathbb{R}$, the *linear classifier* associated with the hyperplane $H = \{x \in \mathbb{R}^n : \langle w, x \rangle = b\}$ is

$$f(x) = \text{sgn}(\langle w, x \rangle - b) = \begin{cases} +1 & \text{if } \langle w, x \rangle - b > 0, \\ -1 & \text{if } \langle w, x \rangle - b < 0. \end{cases}$$

The vector w is the *normal vector* to the hyperplane and b is the *bias* or *intercept*.

Note a compact way of saying “the classifier is right at point i ”: since $y_i \in \{-1, +1\}$, the point x_i lies on the correct side of the hyperplane if and only if

$$y_i(\langle w, x_i \rangle - b) > 0. \tag{14.1}$$

Indeed: if $y_i = +1$ we need $\langle w, x_i \rangle - b > 0$, and if $y_i = -1$ we need $\langle w, x_i \rangle - b < 0$; multiplying by y_i unifies both cases in a single inequality. This small algebraic trick will be used constantly throughout the chapter.

14.1.2 Linear separability and the separation theorem

Definition 14.4 (Linear separability). The sample $\{(x_i, y_i)\}_{i=1}^m$ is *linearly separable* if there are $w \in \mathbb{R}^n$, $w \neq 0$, and $b \in \mathbb{R}$ such that

$$y_i(\langle w, x_i \rangle - b) > 0 \quad \text{for all } i = 1, \dots, m.$$

When does such a hyperplane exist? Here the separation theorem for convex sets that we studied in the separation chapter takes the leading role. Let us define the sets of points of each class,

$$P = \{x_i : y_i = +1\}, \quad N = \{x_i : y_i = -1\},$$

and their convex hulls $\text{conv}(P)$ and $\text{conv}(N)$ (the smallest convex set containing each class; recall that the convex hull of finitely many points is the set of all their convex combinations, a compact polytope).

Theorem 14.5 (Separability = separation of convex hulls). *The sample is linearly separable if and only if $\text{conv}(P) \cap \text{conv}(N) = \emptyset$.*

Proof. ($\Leftarrow$) Suppose $\text{conv}(P) \cap \text{conv}(N) = \emptyset$. Both sets are convex hulls of finitely many points, hence convex and compact (closed and bounded). By the *strict* separation theorem for a compact convex set and a disjoint closed convex set (seen in the separation chapter; see also [2, sec. 2.5]), there are $w \neq 0$ and $b \in \mathbb{R}$ such that

$$\langle w, u \rangle > b \quad \forall u \in \text{conv}(P), \quad \langle w, v \rangle < b \quad \forall v \in \text{conv}(N).$$

In particular, for each x_i with $y_i = +1$ we have $\langle w, x_i \rangle - b > 0$, and for each x_i with $y_i = -1$ we have $\langle w, x_i \rangle - b < 0$. By (14.1), the hyperplane $\{x : \langle w, x \rangle = b\}$ separates the sample.

($\Rightarrow$) Suppose there is (w, b) with $y_i(\langle w, x_i \rangle - b) > 0$ for all i . Let $u \in \text{conv}(P)$ be arbitrary: $u = \sum_{i: y_i = +1} \lambda_i x_i$ with $\lambda_i \geq 0$, $\sum \lambda_i = 1$. Since the function $x \mapsto \langle w, x \rangle - b$ is affine,

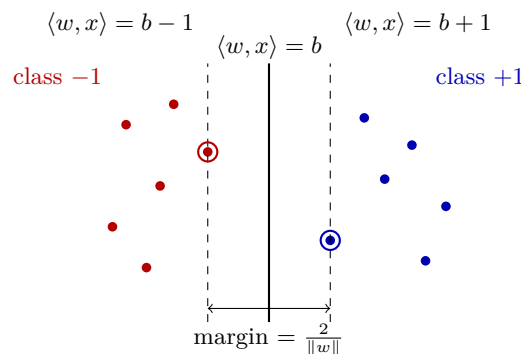
$$\langle w, u \rangle - b = \sum_{i: y_i = +1} \lambda_i (\langle w, x_i \rangle - b) > 0,$$

because it is a weighted average of strictly positive quantities (and some $\lambda_i > 0$). Similarly, $\langle w, v \rangle - b < 0$ for every $v \in \text{conv}(N)$. If there were $z \in \text{conv}(P) \cap \text{conv}(N)$, we would have simultaneously $\langle w, z \rangle - b > 0$ and $\langle w, z \rangle - b < 0$, which is absurd. Hence the hulls are disjoint. $\square$

Intuition

Theorem 14.5 is the conceptual bridge of the chapter: *classifying with a line is separating two convex sets*. All the geometric machinery that we developed (supporting hyperplanes, strict separation of compact sets) stops being abstract and becomes a prediction algorithm. It also anticipates the geometry of the solution: the best hyperplane will pass “through the middle” of the corridor left between the two convex hulls, and only the points on the boundary of that corridor will matter. Those points will be the *support vectors*.

If the sample is separable, there are *infinitely many* separating hyperplanes: any line inside the “corridor” between the clouds works. Which one to choose? The SVM answer: the one that leaves the *widest safety strip* on both sides. Making that idea precise requires measuring distances from points to hyperplanes, which is the topic of the next section.



Two separable classes, the maximum margin hyperplane (solid line), the canonical lines (dashed) and, circled, the support vectors.

14.2 Margin and the primal problem [P5]

14.2.1 Distance from a point to a hyperplane

We need a formula for the distance from a point to a hyperplane. We derive it from scratch.

Lemma 14.6 (Point to hyperplane distance). *Let $H = \{x \in \mathbb{R}^n : \langle a, x \rangle = \beta\}$ with $a \neq 0$, and let $x_0 \in \mathbb{R}^n$. Then*

$$\text{dist}(x_0, H) = \min_{x \in H} \|x_0 - x\| = \frac{|\langle a, x_0 \rangle - \beta|}{\|a\|}.$$

Proof. Step 1 (geometry). The point of H closest to x_0 is the orthogonal projection of x_0 onto H , and the segment joining them is parallel to the normal vector a . So let us look for the point of the form

$$x^* = x_0 - t a, \quad t \in \mathbb{R},$$

that belongs to H : we move from x_0 in the direction of $-a$ until we touch the hyperplane.

Step 2 (compute t). We impose $\langle a, x^* \rangle = \beta$:

$$\langle a, x_0 - t a \rangle = \langle a, x_0 \rangle - t \langle a, a \rangle = \langle a, x_0 \rangle - t \|a\|^2 = \beta \implies t = \frac{\langle a, x_0 \rangle - \beta}{\|a\|^2}.$$

Step 3 (compute the distance). Since $x_0 - x^* = t a$,

$$\|x_0 - x^*\| = |t| \cdot \|a\| = \frac{|\langle a, x_0 \rangle - \beta|}{\|a\|^2} \cdot \|a\| = \frac{|\langle a, x_0 \rangle - \beta|}{\|a\|}.$$

Step 4 (check that it is the minimum). For any other $x \in H$, by the Cauchy-Schwarz inequality,

$$\|a\| \cdot \|x_0 - x\| \geq |\langle a, x_0 - x \rangle| = |\langle a, x_0 \rangle - \underbrace{\langle a, x \rangle}_{=\beta}| = |\langle a, x_0 \rangle - \beta|,$$

hence $\|x_0 - x\| \geq |\langle a, x_0 \rangle - \beta| / \|a\|$ for every $x \in H$, with equality at x^* . This proves the formula. $\square$

14.2.2 Functional margin, geometric margin and canonical normalization

Definition 14.7 (Margins). Let (w, b) be a separating hyperplane of the sample $\{(x_i, y_i)\}_{i=1}^m$. For each point i :

- the *functional margin* is $\hat{\gamma}_i = y_i(\langle w, x_i \rangle - b)$;
- the *geometric margin* is the distance from the point to the hyperplane,

$$\gamma_i = \frac{y_i(\langle w, x_i \rangle - b)}{\|w\|} = \frac{\hat{\gamma}_i}{\|w\|}.$$

The (*geometric*) *margin of the sample* is $\gamma = \min_{i=1, \dots, m} \gamma_i$: the distance from the closest point to the hyperplane.

Note that in γ_i we use Lemma 14.6 with $a = w$, $\beta = b$: since the hyperplane separates, the sign of $\langle w, x_i \rangle - b$ agrees with y_i , so $y_i(\langle w, x_i \rangle - b) = |\langle w, x_i \rangle - b|$ and γ_i is exactly the (positive) distance from x_i to the hyperplane.

Careful (common mistake)

The functional margin $\hat{\gamma}_i$ is **illusory**: if you multiply (w, b) by 10, the geometric hyperplane $\{x : \langle w, x \rangle = b\}$ does not move a millimeter, but $\hat{\gamma}_i$ is multiplied by 10. “Enlarging the functional margin” by rescaling improves nothing. This is why the object with geometric meaning is $\gamma_i = \hat{\gamma}_i / \|w\|$, which is *invariant* under rescalings $(w, b) \mapsto (tw, tb)$ with $t > 0$. That invariance is also what will let us, in a moment, fix a convenient normalization without loss of generality. Do not confuse the two margins: it is the most common conceptual mistake when studying SVM for the first time.

The hard margin SVM chooses, among all separating hyperplanes, the one that *maximizes the geometric margin of the sample*:

$$\max_{w \neq 0, b} \min_{i=1, \dots, m} \frac{y_i(\langle w, x_i \rangle - b)}{\|w\|}. \quad (14.2)$$

As written, (14.2) is ugly: a max–min, not differentiable, not convex (the quotient with $\|w\|$ spoils everything). The magic of the SVM is that a clever normalization turns it into a flawless convex quadratic problem.

14.2.3 Derivation: from maximum margin to a quadratic program

Proposition 14.8 (Primal formulation of the hard margin SVM). *If the sample is linearly separable, problem (14.2) is equivalent to the quadratic program (QP)*

$$\begin{aligned} \min_{w \in \mathbb{R}^n, b \in \mathbb{R}} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y_i(\langle w, x_i \rangle - b) \geq 1, \quad i = 1, \dots, m, \end{aligned} \quad (14.3)$$

in the sense that both have the same optimal hyperplane, and the maximum margin equals $\gamma^ = 1/\|w^*\|$, where w^* solves (14.3).*

Step by step derivation. Step 1: introduce the margin as a variable. Problem (14.2) is equivalent to maximizing a lower bound γ of all the geometric margins:

$$\max_{w \neq 0, b, \gamma > 0} \gamma \quad \text{s.t.} \quad \frac{y_i(\langle w, x_i \rangle - b)}{\|w\|} \geq \gamma, \quad i = 1, \dots, m.$$

(Maximizing the minimum of some quantities is the same as maximizing a common lower bound: at the optimum, γ rises until it hits the minimum.)

Step 2: remove the scale ambiguity. We already saw that if (w, b) is feasible, so is (tw, tb) for every $t > 0$, with the *same* geometric margin: there is a whole family of representations of the same hyperplane. Let us break that redundancy by choosing the representative satisfying

the *canonical normalization*

$$\min_{i=1,\dots,m} y_i(\langle w, x_i \rangle - b) = 1,$$

that is, we scale (w, b) so that the minimum *functional* margin equals exactly 1. This is always possible: if the minimum functional margin of (w, b) is $\hat{\gamma} > 0$ (positive by separability), it is enough to replace (w, b) by $(w/\hat{\gamma}, b/\hat{\gamma})$.

Step 3: rewrite the margin. With the canonical normalization, the geometric margin of the sample is

$$\gamma = \min_i \frac{y_i(\langle w, x_i \rangle - b)}{\|w\|} = \frac{1}{\|w\|} \cdot \underbrace{\min_i y_i(\langle w, x_i \rangle - b)}_{=1} = \frac{1}{\|w\|}.$$

So maximizing γ is maximizing $1/\|w\|$, which in turn is minimizing $\|w\|$.

Step 4: relax the normalization to an inequality. The constraint “ $\min_i y_i(\langle w, x_i \rangle - b) = 1$ ” is awkward (it involves a minimum). Let us replace it by the m inequalities

$$y_i(\langle w, x_i \rangle - b) \geq 1, \quad i = 1, \dots, m,$$

and minimize $\|w\|$. Do we lose anything? No: at the optimum, *some* of the inequalities must be active (hold with equality). If all of them were strict, that is $\min_i y_i(\langle w, x_i \rangle - b) = c > 1$, we could rescale $(w, b) \mapsto (w/c, b/c)$: it would still be feasible and $\|w\|$ would drop, contradicting optimality. Hence the minimum with inequalities agrees with the minimum under the exact normalization.

Step 5: final cosmetics. Minimizing $\|w\|$ and minimizing $\frac{1}{2}\|w\|^2$ have the same solutions, because $t \mapsto \frac{1}{2}t^2$ is strictly increasing on $t \geq 0$. We prefer $\frac{1}{2}\|w\|^2 = \frac{1}{2}\langle w, w \rangle$ because it is differentiable with a linear gradient ($\nabla_w = w$): the $\frac{1}{2}$ is there only to keep the derivative clean. We obtain exactly (14.3), and the optimal margin is $1/\|w^*\|$. $\square$

Intuition

Read problem (14.3) with the eyes of an economist: it is a problem of “minimum cost subject to minimum service levels”. Each data point i requires being classified with functional margin at least 1 (its “participation constraint”), and the planner chooses the hyperplane of *minimum norm* that satisfies all of them. Since the geometric margin is $1/\|w\|$, small norm = wide corridor. And as in every economy with constraints, each constraint will have its “shadow price” $\alpha_i \geq 0$: the data points with positive shadow price are the ones that really constrain the planner, and those will be the support vectors.

14.3 The hard margin SVM: convexity and KKT [P5]

14.3.1 It is a convex QP with Slater

Proposition 14.9 (Convexity of the primal). *Problem (14.3) is a convex optimization problem: convex objective and affine inequality constraints. If the sample is linearly separable, the Slater condition also holds and the problem has a solution; the component w^* is unique.*

Proof. Objective. $f(w, b) = \frac{1}{2}\|w\|^2 = \frac{1}{2}w^\top w$ is a quadratic form with Hessian matrix $\nabla^2 f =$

$\begin{pmatrix} I_n & 0 \\ 0 & 0 \end{pmatrix} \succeq 0$ (positive semidefinite: identity block in w , zero in b), hence f is convex (and strictly convex in w).

Constraints. Each constraint $g_i(w, b) = 1 - y_i(\langle w, x_i \rangle - b) \leq 0$ is *affine* in (w, b) : it is a constant plus a linear function of the variables (x_i and y_i are data, not variables). Affine constraints define a polyhedron, which is convex.

Slater. For affine constraints, the Slater condition only asks for feasibility; but here we have more: if the sample is separable, by Theorem 14.5 there is (w_0, b_0) with $y_i(\langle w_0, x_i \rangle - b_0) > 0$ for all i , and rescaling by a large factor we obtain a point that satisfies all the constraints of (14.3) *strictly*. So Slater holds with room to spare.

Existence and uniqueness of w^ .* The feasible set is closed, non empty, and we can restrict the search to the compact set where $\|w\| \leq \|w_0\|$ (outside it, the objective already exceeds the value at w_0); a standard Weierstrass argument gives existence. If $w_1 \neq w_2$ were two optima with the same value, the midpoint would be feasible (convex set) and, by *strict* convexity of $w \mapsto \frac{1}{2} \|w\|^2$, would have a strictly smaller objective: a contradiction. Hence w^* is unique. $\square$

Operational conclusion: by the KKT chapter, in a convex problem with Slater the KKT conditions are *necessary and sufficient* for global optimality, and by the duality chapter there is *strong duality* (the dual attains the same optimal value). We have a green light to solve the SVM entirely through KKT.

14.3.2 Lagrangian and KKT conditions, step by step

We attach a multiplier $\alpha_i \geq 0$ to each constraint $1 - y_i(\langle w, x_i \rangle - b) \leq 0$. The *Lagrangian* is

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 + \sum_{i=1}^m \alpha_i (1 - y_i(\langle w, x_i \rangle - b)), \quad \alpha = (\alpha_1, \dots, \alpha_m) \geq 0. \quad (14.4)$$

Theorem 14.10 (KKT for the hard margin SVM). *(w^*, b^*) solves (14.3) if and only if there is $\alpha^* \in \mathbb{R}^m$ such that:*

$$(a) \text{ **Stationarity in } w\text{: } \nabla_w L = w^* - \sum_{i=1}^m \alpha_i^* y_i x_i = 0 \iff w^* = \sum_{i=1}^m \alpha_i^* y_i x_i.**$$

$$(b) \text{ **Stationarity in } b\text{: } \frac{\partial L}{\partial b} = \sum_{i=1}^m \alpha_i^* y_i = 0.**$$

$$(c) \text{ **Primal feasibility:** } y_i(\langle w^*, x_i \rangle - b^*) \geq 1 \text{ for all } i.$$

$$(d) \text{ **Dual feasibility:** } \alpha_i^* \geq 0 \text{ for all } i.$$

$$(e) \text{ **Complementary slackness:** } \alpha_i^* (1 - y_i(\langle w^*, x_i \rangle - b^*)) = 0 \text{ for all } i.$$

Checking the computations. The equivalence KKT $\Leftrightarrow$ optimum is the general theorem of the KKT chapter (convexity plus Slater, Proposition 14.9). Here we only check the derivatives. Writing $L = \frac{1}{2} \langle w, w \rangle + \sum_i \alpha_i - \sum_i \alpha_i y_i \langle w, x_i \rangle + b \sum_i \alpha_i y_i$: the gradient in w of the first term is w (rule $\nabla_w \frac{1}{2} \langle w, w \rangle = w$); that of the term $-\sum_i \alpha_i y_i \langle w, x_i \rangle$ is $-\sum_i \alpha_i y_i x_i$ (gradient of a linear function); the others do not depend on w . The derivative in b only sees the last term and equals $\sum_i \alpha_i y_i$. Setting them to zero gives (a) and (b). $\square$

14.3.3 Support vectors

Condition (e), complementary slackness, is the one that gives the method its name.

Definition 14.11 (Support vector). A point x_i of the sample is a *support vector* if its optimal multiplier is positive: $\alpha_i^* > 0$. By complementary slackness, every support vector satisfies its constraint with equality,

$$y_i(\langle w^*, x_i \rangle - b^*) = 1,$$

that is, it lies exactly on one of the two canonical lines $\langle w^*, x \rangle - b^* = \pm 1$, at distance exactly $1/\|w^*\|$ from the hyperplane: on the edge of the corridor.

Careful (common mistake)

The implication goes only one way: $\alpha_i^* > 0 \Rightarrow$ active constraint. The converse is false in general: there may be points *on* the edge of the corridor with $\alpha_i^* = 0$ (active constraint but zero multiplier, a degenerate case). What is always true, by reading complementary slackness backwards: if the constraint is *inactive* ($y_i(\langle w^*, x_i \rangle - b^*) > 1$, a point strictly inside its region), then $\alpha_i^* = 0$.

Theorem 14.12 (Representation by support vectors). *Let $S = \{i : \alpha_i^* > 0\}$ be the set of indices of support vectors. Then:*

- (i) $w^* = \sum_{i \in S} \alpha_i^* y_i x_i$: the optimal normal vector is a linear combination of the support vectors only.
- (ii) $S \neq \emptyset$, and there are support vectors of both classes.
- (iii) For any $j \in S$: $b^* = \langle w^*, x_j \rangle - y_j$.
- (iv) The classifier is evaluated using only inner products with the support vectors: $f(x) = \text{sgn}\left(\sum_{i \in S} \alpha_i^* y_i \langle x_i, x \rangle - b^*\right)$.

Proof. (i) It is condition (a) of Theorem 14.10, where the terms with $\alpha_i^* = 0$ vanish.

(ii) If $S = \emptyset$ then $w^* = 0$ by (i); but $w = 0$ makes the constraint $y_i(\langle 0, x_i \rangle - b) = -y_i b \geq 1$ impossible for points of both classes at once (it would require $-b \geq 1$ and $b \geq 1$). Hence $S \neq \emptyset$. Moreover, if all the support vectors belonged to class +1, condition (b) would give $0 = \sum_{i \in S} \alpha_i^* y_i = \sum_{i \in S} \alpha_i^* > 0$, absurd; the same with class -1. There is support in both classes.

(iii) If $j \in S$, by complementary slackness $y_j(\langle w^*, x_j \rangle - b^*) = 1$. Multiplying both sides by y_j and using $y_j^2 = 1$: $\langle w^*, x_j \rangle - b^* = y_j$, hence $b^* = \langle w^*, x_j \rangle - y_j$. (In numerical practice this formula is averaged over all $j \in S$ to reduce rounding error.)

(iv) Substitute (i) into $f(x) = \text{sgn}(\langle w^*, x \rangle - b^*)$ and use linearity of the inner product. $\square$

Intuition

Only the boundary matters. If you delete from the sample all the points that are not support vectors and train again, you obtain *exactly the same* hyperplane: those points had zero multiplier, they were not “pushing” the solution. It is the exact analogue of

shadow prices in economics: an inactive constraint (a resource that is not exhausted) has zero shadow price and removing it does not change the optimum. The SVM compresses the whole sample into a few boundary points, just as the equilibrium of a market is determined by the marginal agents, not by the inframarginal ones.

14.4 The dual problem and the kernel trick [P5]

14.4.1 Full derivation of the dual function

Following the duality chapter, the *dual function* is

$$q(\alpha) = \inf_{w \in \mathbb{R}^n, b \in \mathbb{R}} L(w, b, \alpha), \quad \alpha \geq 0,$$

and the dual problem is to maximize q . Let us compute the infimum *explicitly*, eliminating w and b .

Step 1: minimize in b . In (14.4), b appears only in the linear term $b \sum_i \alpha_i y_i$. A non constant linear function is not bounded below, so:

$$\inf_b L = \begin{cases} (\text{continue with the computation in } w) & \text{if } \sum_i \alpha_i y_i = 0, \\ -\infty & \text{if } \sum_i \alpha_i y_i \neq 0. \end{cases}$$

The α with $\sum_i \alpha_i y_i \neq 0$ give $q(\alpha) = -\infty$ and are discarded from the dual: the condition $\sum_i \alpha_i y_i = 0$ appears as a *dual constraint*.

Step 2: minimize in w . Assuming $\sum_i \alpha_i y_i = 0$, the Lagrangian as a function of w is

$$w \mapsto \frac{1}{2} \|w\|^2 - \left\langle \sum_i \alpha_i y_i x_i, w \right\rangle + \sum_i \alpha_i,$$

a strictly convex quadratic. Its minimum is attained where the gradient vanishes:

$$w - \sum_i \alpha_i y_i x_i = 0 \quad \implies \quad w(\alpha) = \sum_{i=1}^m \alpha_i y_i x_i.$$

(The same formula as KKT condition (a): this is no coincidence, the infimum of the Lagrangian is attained at the stationary point.)

Step 3: substitute. Write $v = \sum_i \alpha_i y_i x_i$, so that $w(\alpha) = v$. We evaluate term by term:

$$\frac{1}{2} \|w(\alpha)\|^2 = \frac{1}{2} \|v\|^2, \quad \left\langle \sum_i \alpha_i y_i x_i, w(\alpha) \right\rangle = \langle v, v \rangle = \|v\|^2,$$

and therefore

$$q(\alpha) = \frac{1}{2} \|v\|^2 - \|v\|^2 + \sum_i \alpha_i = \sum_{i=1}^m \alpha_i - \frac{1}{2} \|v\|^2.$$

Step 4: expand $\|v\|^2$ into inner products. By bilinearity of the inner product,

$$\|v\|^2 = \left\langle \sum_i \alpha_i y_i x_i, \sum_j \alpha_j y_j x_j \right\rangle = \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle.$$

We have proved:

Theorem 14.13 (Dual of the hard margin SVM). *The dual problem of (14.3) is*

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^m} \quad & \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle \\ \text{s.t.} \quad & \sum_{i=1}^m \alpha_i y_i = 0, \quad \alpha_i \geq 0, \quad i = 1, \dots, m. \end{aligned} \tag{14.5}$$

If the sample is separable, strong duality holds: the optimal values of (14.3) and (14.5) agree, the dual attains its maximum at some α^* , and the solutions are connected by

$$w^* = \sum_{i=1}^m \alpha_i^* y_i x_i, \quad b^* = \langle w^*, x_j \rangle - y_j \text{ for any } j \text{ with } \alpha_j^* > 0.$$

Proof. The derivation of q is above. Strong duality is the theorem of the duality chapter: a convex problem plus the Slater condition (Proposition 14.9) imply a zero duality gap and existence of optimal multipliers. The recovery of (w^*, b^*) is Theorem 14.12. $\square$

Remark 14.14 (The dual is another QP, and often more manageable). Defining the matrix $Q \in \mathbb{R}^{m \times m}$ with entries $Q_{ij} = y_i y_j \langle x_i, x_j \rangle$ (the labeled *Gram matrix*), the dual is $\max_{\alpha} \mathbf{1}^\top \alpha - \frac{1}{2} \alpha^\top Q \alpha$ subject to simple linear constraints. Q is positive semidefinite ($\alpha^\top Q \alpha = \|\sum_i \alpha_i y_i x_i\|^2 \geq 0$), so the dual is a *concave* QP in maximization form: convex again. The dual has m variables (one per data point) while the primal has $n + 1$ (one per feature); which one is computationally convenient depends on whether there are more data points or more features [8, 5].

14.4.2 The kernel trick

Look carefully at (14.5) and at the prediction formula of Theorem 14.12(iv). There is something remarkable:

Remark 14.15 (The data enter only through inner products). Neither the dual problem nor the final classifier needs the vectors x_i individually: they *only* need the inner products $\langle x_i, x_j \rangle$ (to train) and $\langle x_i, x \rangle$ (to predict).

This observation opens a huge door. Suppose the data are not linearly separable in $\mathbb{R}^n$, but they would be after transforming them with a non linear map $\phi: \mathbb{R}^n \rightarrow \mathbb{R}^N$ into a space of higher dimension (even much higher). To train the SVM in the transformed space we would only need the numbers $\langle \phi(x_i), \phi(x_j) \rangle$. The *kernel trick* consists in choosing ϕ so that this inner product has a cheap closed formula

$$K(x, z) = \langle \phi(x), \phi(z) \rangle,$$

and then *never computing ϕ explicitly*: in (14.5) each $\langle x_i, x_j \rangle$ is replaced by $K(x_i, x_j)$ and everything else stays the same. The function K is called a *kernel*.

Example 14.16 (Polynomial kernel). In $\mathbb{R}^2$, let $K(x, z) = \langle x, z \rangle^2 = (x_1 z_1 + x_2 z_2)^2$. Expanding the square:

$$K(x, z) = x_1^2 z_1^2 + 2x_1 z_1 x_2 z_2 + x_2^2 z_2^2 = \langle (x_1^2, \sqrt{2} x_1 x_2, x_2^2), (z_1^2, \sqrt{2} z_1 z_2, z_2^2) \rangle,$$

that is, $K(x, z) = \langle \phi(x), \phi(z) \rangle$ with $\phi(x) = (x_1^2, \sqrt{2} x_1 x_2, x_2^2) \in \mathbb{R}^3$. An SVM with this kernel looks for a hyperplane in $\mathbb{R}^3$ of quadratic coordinates, which seen in $\mathbb{R}^2$ is a *conic*: the classifier is no longer a line. The general polynomial kernel is $K(x, z) = (1 + \langle x, z \rangle)^d$, whose target space has dimension combinatorial in n and d , but evaluating K costs the same as an ordinary inner product.

Example 14.17 (Gaussian or RBF kernel). $K(x, z) = \exp(-\|x - z\|^2 / (2\sigma^2))$. It can be shown that it corresponds to a ϕ with values in a space of *infinite* dimension. The intuition: $K(x, z)$ measures *similarity* (it equals 1 if $x = z$ and decays to 0 if they are far apart), and the resulting classifier $f(x) = \text{sgn}(\sum_{i \in S} \alpha_i^* y_i K(x_i, x) - b^*)$ is a “weighted voter”: each support vector votes for its class with weight proportional to its closeness to the new point. The parameter σ controls the reach of the vote. The general theory of which functions K are valid kernels (reproducing kernel Hilbert spaces, RKHS) is beyond this notebook; for us the operational idea is enough. See [8] for the precise statement and [4] for the associated optimization.

Exam strategy

Review script with the SVM (an ideal integrating review before any evaluation of the course):

1. *Separation*: draw two clouds of points; ask yourself whether $\text{conv}(P) \cap \text{conv}(N) = \emptyset$ and which separation theorem applies (Theorem 14.5).
2. *Modeling*: derive (14.3) from the margin without looking at the book; the key step is the canonical normalization (Step 2 of Proposition 14.8).
3. *KKT*: write the Lagrangian and the five conditions from memory; identify what each one means geometrically.
4. *Duality*: eliminate w and b to obtain (14.5); check Slater and state strong duality.
5. *Numerical closing*: solve Example 14.22 by hand and check that primal and dual give the same value.

If you can do the five steps without help, you master the separation, KKT and duality chapters.

14.5 Soft margin: non separable data [P5]

14.5.1 Slack variables and the parameter C

In real data there is almost never perfect separability: there is noise, measurement errors, classes that overlap. If $\text{conv}(P) \cap \text{conv}(N) \neq \emptyset$, problem (14.3) is *infeasible* and the hard margin SVM

simply does not exist. The solution: allow some points to violate their constraint, but charge for it.

Definition 14.18 (Soft margin SVM). Given a parameter $C > 0$, the *soft margin SVM* is the problem

$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \xi_i \\ \text{s.t.} \quad & y_i(\langle w, x_i \rangle - b) \geq 1 - \xi_i, \quad i = 1, \dots, m, \\ & \xi_i \geq 0, \quad i = 1, \dots, m, \end{aligned} \tag{14.6}$$

where ξ_i is the *slack variable* of point i : it measures how much the point falls short of reaching the functional margin 1. If $0 < \xi_i \leq 1$, the point is inside the corridor but well classified; if $\xi_i > 1$, it is misclassified.

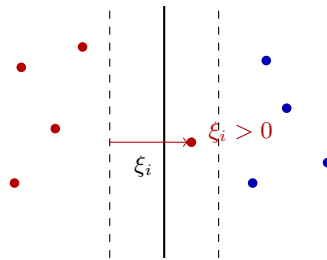
Problem (14.6) is still a convex QP (convex objective: a sum of a convex quadratic and a term linear in ξ ; affine constraints) and it is *always feasible*: given any (w, b) , it is enough to take $\xi_i = \max\{0, 1 - y_i(\langle w, x_i \rangle - b)\}$. Slater holds trivially (taking even larger ξ_i gives strict slack), so KKT and strong duality apply just as before.

Intuition

The parameter C is a *price*: each unit of margin violation costs C . It is the classical bias–variance tradeoff written as a planning problem:

- large C : violating is very expensive; the optimum imitates the hard margin, it fits the sample closely (risk of *overfitting* to noisy data).
- small C : violating is cheap; the optimum prefers a small $\|w\|$ (wide corridor) even if several points end up inside or misclassified (more *regularization*, a more stable boundary).

When $C \rightarrow \infty$ and the data are separable, the hard margin is recovered. In practice, C is chosen by cross validation.



Soft margin: a point of class -1 invades the corridor; its slack ξ_i measures the depth of the invasion, penalized at price C .

14.5.2 Equivalence with the hinge loss

Proposition 14.19 (Hinge loss formulation). *Problem (14.6) is equivalent to the unconstrained problem*

$$\min_{w, b} \quad \frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \max\{0, 1 - y_i(\langle w, x_i \rangle - b)\}, \tag{14.7}$$

where $t \mapsto \max\{0, 1 - t\}$ is the hinge loss. Equivalent means: the same optimal (w^*, b^*) and the same optimal value.

Proof. Fix (w, b) and ask for the best feasible ξ in (14.6) for that (w, b) . The constraints on ξ_i are

$$\xi_i \geq 1 - y_i(\langle w, x_i \rangle - b) \quad \text{and} \quad \xi_i \geq 0,$$

that is $\xi_i \geq \max\{0, 1 - y_i(\langle w, x_i \rangle - b)\}$. Since the objective is *increasing* in each ξ_i (coefficient $C > 0$), the best choice is to take each ξ_i at its lower bound:

$$\xi_i^*(w, b) = \max\{0, 1 - y_i(\langle w, x_i \rangle - b)\}.$$

Substituting this $\xi^*(w, b)$ into the objective of (14.6) we obtain exactly the objective of (14.7). Minimizing first in ξ (with (w, b) fixed) and then in (w, b) gives the same result as minimizing jointly, so both problems share optima and optimal value. $\square$

Remark 14.20 (Statistical reading). Form (14.7) reveals the SVM as a method of *regularized empirical risk minimization*: hinge loss (a convex approximation of the 0–1 loss “was I wrong or not?”) plus a penalty $\frac{1}{2} \|w\|^2$ (Tikhonov or ridge regularization). The hinge loss is not differentiable at the kink $t = 1$, but it is convex: the subgradient methods of the corresponding chapter apply directly, and this is how it is trained at large scale [8].

Theorem 14.21 (Soft margin dual, statement). *The Lagrangian dual of (14.6) is*

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^m} \quad & \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle \\ \text{s.t.} \quad & \sum_{i=1}^m \alpha_i y_i = 0, \quad 0 \leq \alpha_i \leq C, \quad i = 1, \dots, m, \end{aligned} \tag{14.8}$$

and strong duality holds. It is identical to the hard margin dual (14.5) except for the upper bounds $\alpha_i \leq C$ (box constraints).

The derivation repeats that of Section 14.4 with an extra multiplier $\mu_i \geq 0$ for each $\xi_i \geq 0$; stationarity in ξ_i gives $C - \alpha_i - \mu_i = 0$, and eliminating $\mu_i = C - \alpha_i \geq 0$ produces the bound $\alpha_i \leq C$. The details are left as a guided exercise (Exercise 14.26). The interpretation of the bounds is elegant: every point with $\xi_i^* > 0$ (those that pay a fine) has $\alpha_i^* = C$, since the complementary slackness of $\mu_i \xi_i = 0$ forces $\mu_i^* = 0$ and therefore $\alpha_i^* = C$ (the converse can fail in degenerate cases, just as with support vectors); the shadow price of each data point is *capped* by the price C of the violation, because there is always the option of “buying slack”.

14.6 An SVM solved by hand, from start to finish [P5]

Example 14.22 (SVM in $\mathbb{R}^2$ with four points). Consider the sample in $\mathbb{R}^2$:

i	1	2	3	4
x_i	(2, 0)	(4, 0)	(0, 0)	(−1, 2)
y_i	+1	+1	−1	−1

Find the maximum margin hyperplane, the support vectors, the multipliers and the margin; check with the dual.

Solution. Step 0: geometric guess. Draw the points: the positive ones are on the right ($x_{(1)} \geq 2$), the negative ones on the left ($x_{(1)} \leq 0$), where $x_{(1)}$ denotes the first coordinate. The closest pair of points across classes is $x_1 = (2, 0)$ and $x_3 = (0, 0)$, at distance 2. Guess: the optimal hyperplane is the vertical perpendicular bisector $x_{(1)} = 1$, that is w^* proportional to $(1, 0)$, with margin $\gamma^* = 1$. Now we *prove* it with KKT (remember: in a convex problem with Slater, finding a KKT point certifies the global optimum).

Step 1: primal candidate. If the margin is 1, then $\|w^*\| = 1/\gamma^* = 1$, and with direction $(1, 0)$: $w^* = (1, 0)$. The hyperplane $\langle w^*, x \rangle = b^*$ must be $x_{(1)} = 1$, so $b^* = 1$. Let us check primal feasibility (condition (c)), computing $y_i(\langle w^*, x_i \rangle - b^*)$:

$$\begin{aligned} i = 1: & \quad (+1)(2 - 1) = 1 \quad \geq 1 \quad (\text{active}) \\ i = 2: & \quad (+1)(4 - 1) = 3 \quad \geq 1 \quad (\text{inactive, slack } 2) \\ i = 3: & \quad (-1)(0 - 1) = 1 \quad \geq 1 \quad (\text{active}) \\ i = 4: & \quad (-1)(-1 - 1) = 2 \quad \geq 1 \quad (\text{inactive, slack } 1) \end{aligned}$$

Feasible, with active constraints exactly at $i \in \{1, 3\}$.

Step 2: complementary slackness. By condition (e), the inactive points must have zero multiplier: $\alpha_2^* = \alpha_4^* = 0$. The candidates for support vectors are x_1 and x_3 .

Step 3: stationarity. Condition (a) requires

$$w^* = \sum_i \alpha_i^* y_i x_i = \alpha_1^*(2, 0) - \alpha_3^*(0, 0) = (2\alpha_1^*, 0).$$

Setting this equal to $w^* = (1, 0)$: $2\alpha_1^* = 1$, hence $\alpha_1^* = \frac{1}{2}$. Condition (b) requires $\sum_i \alpha_i^* y_i = \alpha_1^* - \alpha_3^* = 0$, hence $\alpha_3^* = \frac{1}{2}$. (Note that $x_3 = (0, 0)$ contributes the zero vector to the sum in (a), but its multiplier is essential to balance condition (b): it is a support vector too.)

Step 4: dual feasibility. $\alpha^* = (\frac{1}{2}, 0, \frac{1}{2}, 0) \geq 0$. The five KKT conditions hold; since the problem is convex with Slater, $(w^*, b^*) = ((1, 0), 1)$ is the global optimum, with support vectors $S = \{1, 3\}$, margin $\gamma^* = 1/\|w^*\| = 1$ and primal value $\frac{1}{2}\|w^*\|^2 = \frac{1}{2}$.

Step 5: check through the dual. Let us solve (14.5) directly. Guided by the primal, look for the dual optimum with $\alpha_2 = \alpha_4 = 0$ and parametrize $\alpha_1 = \alpha_3 = a \geq 0$ (this imposes the constraint $\alpha_1 y_1 + \alpha_3 y_3 = a - a = 0$, feasible). The relevant inner products are

$$\langle x_1, x_1 \rangle = 4, \quad \langle x_1, x_3 \rangle = 0, \quad \langle x_3, x_3 \rangle = 0.$$

The dual objective becomes

$$q(a) = (a + a) - \frac{1}{2}[a^2 y_1 y_1(4) + 2a^2 y_1 y_3(0) + a^2 y_3 y_3(0)] = 2a - 2a^2.$$

It is a concave parabola; its maximum is at $q'(a) = 2 - 4a = 0$, that is $a^* = \frac{1}{2}$, with value $q(a^*) = 1 - \frac{1}{2} = \frac{1}{2}$. The dual value agrees with the primal one ($\frac{1}{2} = \frac{1}{2}$): *zero duality gap*, as strong duality predicts. We also recover $w^* = \frac{1}{2}(2, 0) - \frac{1}{2}(0, 0) = (1, 0)$ and, using the support vector $j = 1$: $b^* = \langle w^*, x_1 \rangle - y_1 = 2 - 1 = 1$. Everything fits.

Moral. The points $x_2 = (4, 0)$ and $x_4 = (-1, 2)$ can be deleted from the sample without altering the solution: they are not support vectors. The final classifier is $f(x) = \text{sgn}(x_{(1)} - 1)$. $\square$

14.7 Applied note: SVM in economics and a suggestion for the final presentation [P5]

SVM in applied economics. Three families of applications where the SVM (or its kernelized variants) appears in the empirical literature:

- *Credit risk*: classifying applicants into repay or default from history, income and sociodemographic variables (Example 14.2). Central banks and supervisors use classifiers of this kind for early warning of bank failures and crises.
- *Poverty targeting*: predicting whether a household is poor or not by combining surveys (ENAH in the Peruvian case) with cheap proxy variables, including features extracted from night time satellite images or from housing roofs; the classifier makes it possible to target social programs where there is no recent census.
- *Text as data*: classifying monetary policy statements into restrictive or expansionary (hawkish or dovish), news into positive or negative for uncertainty indices, or occupational descriptions into categories; each document is turned into a vector of word frequencies (n huge, which is why the dual and the kernels matter).

A concrete suggestion for the final course presentation. A well rounded project, feasible in two weeks:

1. *Theory* (10 minutes): present the chain separation $\rightarrow$ margin $\rightarrow$ primal QP $\rightarrow$ KKT $\rightarrow$ dual $\rightarrow$ kernel, with Example 14.22 solved on the board.
2. *Implementation* (Python with `cvxpy` and `scikit-learn`, or Julia with `JuMP`): solve the primal (14.6) and the dual (14.8) as explicit QPs on a public economic dataset (for example, a credit risk dataset, or household poverty built from a survey); check numerically (i) that primal and dual give the same value (strong duality), (ii) complementary slackness, and (iii) identify the support vectors and count how many there are ($|S| \ll m$ typically).
3. *Extension* (5 minutes): repeat with a Gaussian kernel and comment on the effect of C and σ on the boundary.

As project references: the SVM chapter of [8] (primal and dual formulations, large scale training and hinge loss), [4] for the duality and KKT framework in finite dimension with the rigor of this notebook, and [2, sec. 8.6] for the family of classification problems as convex optimization (including robust and logistic variants).

Chapter summary

- **Classifying is separating**: a labeled sample is linearly separable if and only if the convex hulls of the two classes are disjoint (Theorem 14.5); the linear classifier is a separating hyperplane.

- **Margin:** the distance from x_0 to the hyperplane $\langle a, x \rangle = \beta$ is $|\langle a, x_0 \rangle - \beta| / \|a\|$ (Lemma 14.6). Maximizing the geometric margin, after the canonical normalization $\min_i y_i (\langle w, x_i \rangle - b) = 1$, is equivalent to the convex QP $\min \frac{1}{2} \|w\|^2$ s.t. $y_i (\langle w, x_i \rangle - b) \geq 1$; the optimal margin is $1 / \|w^*\|$.
- **KKT:** stationarity gives $w^* = \sum_i \alpha_i^* y_i x_i$ and $\sum_i \alpha_i^* y_i = 0$; complementary slackness defines the *support vectors* ($\alpha_i^* > 0 \Rightarrow$ active constraint). Only they determine w^* and $b^* = \langle w^*, x_j \rangle - y_j$, $j \in S$.
- **Dual:** $\max_\alpha \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$ s.t. $\sum_i \alpha_i y_i = 0$, $\alpha \geq 0$; strong duality by Slater. The dual only sees inner products $\Rightarrow$ *kernel trick*: replacing $\langle x_i, x_j \rangle$ by $K(x_i, x_j)$ (polynomial, Gaussian) classifies non linearly without computing the map ϕ .
- **Soft margin:** with slacks ξ_i and price C , the problem $\min \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$ is always feasible; it is equivalent to minimizing a regularized hinge loss (Proposition 14.19); its dual only adds the bounds $0 \leq \alpha_i \leq C$.
- **Working method:** guess the hyperplane geometrically, check the five KKT conditions and confirm with the dual that the duality gap is zero (Example 14.22).

14.8 Exercises [P5]

Exercise 14.23 (Distances and margins). Consider the hyperplane $H = \{x \in \mathbb{R}^2 : 3x_{(1)} + 4x_{(2)} = 10\}$ and the sample $x_1 = (2, 2)$ with $y_1 = +1$ and $x_2 = (0, 0)$ with $y_2 = -1$. (a) Compute the distance from each point to H . (b) Compute the functional margin and the geometric margin of each point with respect to $(w, b) = ((3, 4), 10)$. (c) Rescale (w, b) to the canonical normalization for this sample and check that the geometric margins do not change.

Solution. (a) With $a = (3, 4)$, $\|a\| = \sqrt{9 + 16} = 5$ and $\beta = 10$ (Lemma 14.6):

$$\text{dist}(x_1, H) = \frac{|3(2) + 4(2) - 10|}{5} = \frac{|14 - 10|}{5} = \frac{4}{5}, \quad \text{dist}(x_2, H) = \frac{|0 - 10|}{5} = 2.$$

(b) Functional margins: $\hat{\gamma}_1 = (+1)(14 - 10) = 4$ and $\hat{\gamma}_2 = (-1)(0 - 10) = 10$; both positive, so the hyperplane classifies both points correctly. Geometric margins: $\gamma_1 = \hat{\gamma}_1 / \|w\| = 4/5$ and $\gamma_2 = 10/5 = 2$, which agree with the distances in (a), as they must. (c) The minimum functional margin is $\hat{\gamma} = \min\{4, 10\} = 4$; the canonical representative is $(w, b)/4 = ((\frac{3}{4}, 1), \frac{10}{4})$. With it: the functional margin of x_1 equals $4/4 = 1$ (the minimum, as the normalization requires) and that of x_2 equals $10/4$; and $\|w/4\| = 5/4$, so the geometric margins are $1/(5/4) = 4/5$ and $(10/4)/(5/4) = 2$: identical to (b). The geometric margin is invariant under rescaling; the functional one is not. $\square$

Exercise 14.24 (Tiny SVM solved by KKT). Solve by hand, through KKT, the hard margin SVM for the three points of $\mathbb{R}^2$:

$$x_1 = (1, 1), y_1 = +1; \quad x_2 = (-1, 1), y_2 = -1; \quad x_3 = (-2, -1), y_3 = -1.$$

Give w^* , b^* , the support vectors, the margin and the optimal value, and check all the KKT conditions.

Solution. *Guess.* The positive point is $x_1 = (1, 1)$; the closest negative point is $x_2 = (-1, 1)$, at distance 2 (same height). Perpendicular bisector: the vertical line $x_{(1)} = 0$. Candidate: $w^* = (1, 0)$, $b^* = 0$, margin 1.

Primal feasibility. $y_1(\langle w^*, x_1 \rangle - 0) = 1 \geq 1$ (active); $y_2(\langle w^*, x_2 \rangle - 0) = (-1)(-1) = 1 \geq 1$ (active); $y_3(\langle w^*, x_3 \rangle - 0) = (-1)(-2) = 2 \geq 1$ (inactive).

Complementary slackness. x_3 inactive $\Rightarrow \alpha_3^* = 0$.

Stationarity in w . $w^* = \alpha_1^*(1, 1) - \alpha_2^*(-1, 1) = (\alpha_1^* + \alpha_2^*, \alpha_1^* - \alpha_2^*)$. Setting this equal to $(1, 0)$: $\alpha_1^* + \alpha_2^* = 1$ and $\alpha_1^* - \alpha_2^* = 0$, hence $\alpha_1^* = \alpha_2^* = \frac{1}{2}$.

Stationarity in b . $\sum_i \alpha_i^* y_i = \frac{1}{2} - \frac{1}{2} - 0 = 0$. It holds.

Dual feasibility. $\alpha^* = (\frac{1}{2}, \frac{1}{2}, 0) \geq 0$.

The five KKT conditions hold and the problem is convex with Slater: the global optimum is certified. Support vectors $S = \{1, 2\}$; margin $1/\|w^*\| = 1$; optimal value $\frac{1}{2}\|w^*\|^2 = \frac{1}{2}$. Quick dual check: with $\alpha_1 = \alpha_2 = a$ and $\alpha_3 = 0$, the inner products are $\langle x_1, x_1 \rangle = 2$, $\langle x_1, x_2 \rangle = -1 + 1 = 0$ and $\langle x_2, x_2 \rangle = 2$, so that

$$\sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle = a^2(2) - 2a^2(0) + a^2(2) = 4a^2, \quad q(a) = 2a - 2a^2,$$

maximized at $a = \frac{1}{2}$ with value $q(\frac{1}{2}) = \frac{1}{2}$: zero duality gap, in agreement with the primal value. $\square$

Exercise 14.25 (SVM without intercept). Consider the variant of the hard margin SVM in which the hyperplane must pass through the origin (no intercept, $b = 0$): $\min_w \frac{1}{2}\|w\|^2$ s.t. $y_i \langle w, x_i \rangle \geq 1$, $i = 1, \dots, m$. Derive its full dual. Which constraint of the dual (14.5) disappears and why?

Solution. Lagrangian: $L(w, \alpha) = \frac{1}{2}\|w\|^2 + \sum_i \alpha_i(1 - y_i \langle w, x_i \rangle)$ with $\alpha \geq 0$. Now there is no variable b , so the only minimization step is in w : it is a strictly convex quadratic whose minimum satisfies $w(\alpha) = \sum_i \alpha_i y_i x_i$. Substituting, exactly as in Steps 3 and 4 of Section 14.4:

$$q(\alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle,$$

and the dual is $\max_{\alpha \geq 0} q(\alpha)$, *without* the constraint $\sum_i \alpha_i y_i = 0$. The reason: that constraint came from requiring the Lagrangian not to be $-\infty$ when minimizing over b (the coefficient of b had to vanish). With no b in the problem, there is no linear term to tame and the equality constraint disappears. Moral: each dual constraint is the footprint of a primal variable (and the other way around). $\square$

Exercise 14.26 (Soft margin dual). Derive the dual (14.8) of the soft margin problem (14.6). Hint: assign multipliers $\alpha_i \geq 0$ to the margin constraints and $\mu_i \geq 0$ to the constraints $\xi_i \geq 0$, and minimize the Lagrangian in (w, b, ξ) .

Solution. The Lagrangian is

$$L(w, b, \xi, \alpha, \mu) = \frac{1}{2} \|w\|^2 + C \sum_i \xi_i + \sum_i \alpha_i (1 - \xi_i - y_i (\langle w, x_i \rangle - b)) - \sum_i \mu_i \xi_i,$$

with $\alpha, \mu \geq 0$. We minimize block by block:

In w : as before, $w(\alpha) = \sum_i \alpha_i y_i x_i$.

In b : coefficient $\sum_i \alpha_i y_i$; to avoid falling to $-\infty$, the dual constraint $\sum_i \alpha_i y_i = 0$.

In ξ_i : the coefficient of ξ_i in L is $C - \alpha_i - \mu_i$. Since ξ_i ranges over all of $\mathbb{R}$ in the minimization of the Lagrangian (the constraints are already incorporated through μ), if $C - \alpha_i - \mu_i \neq 0$ the infimum is $-\infty$; hence the dual requires

$$C - \alpha_i - \mu_i = 0 \quad \Longleftrightarrow \quad \mu_i = C - \alpha_i.$$

With this condition, all the terms in ξ cancel and the rest of the substitution is identical to the hard case: $q(\alpha, \mu) = \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$. Finally, we eliminate μ : the dual feasibility $\mu_i \geq 0$ becomes $C - \alpha_i \geq 0$, that is $\alpha_i \leq C$. Together with $\alpha_i \geq 0$ we obtain the box bounds $0 \leq \alpha_i \leq C$ and problem (14.8). Since (14.6) is convex and satisfies Slater (take a large ξ), there is strong duality. $\square$

Exercise 14.27 (An explicit kernel). Let $K(x, z) = (1 + \langle x, z \rangle)^2$ for $x, z \in \mathbb{R}^2$. Find explicitly a map $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^6$ such that $K(x, z) = \langle \phi(x), \phi(z) \rangle$. What kind of decision boundaries can an SVM with this kernel produce in $\mathbb{R}^2$?

Solution. We expand with $\langle x, z \rangle = x_1 z_1 + x_2 z_2$:

$$K(x, z) = (1 + x_1 z_1 + x_2 z_2)^2 = 1 + x_1^2 z_1^2 + x_2^2 z_2^2 + 2x_1 z_1 + 2x_2 z_2 + 2x_1 z_1 x_2 z_2.$$

Grouping each summand as a product of a factor in x times the same factor in z :

$$\phi(x) = (1, x_1^2, x_2^2, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_1x_2) \in \mathbb{R}^6,$$

and one checks directly that $\langle \phi(x), \phi(z) \rangle$ reproduces the six summands of the expansion (the $\sqrt{2}$ generate the factors 2 when multiplied by themselves). A hyperplane in $\mathbb{R}^6$ over these coordinates is, seen in $\mathbb{R}^2$, a general second degree equation

$$c_0 + c_1 x_1^2 + c_2 x_2^2 + c_3 x_1 + c_4 x_2 + c_5 x_1 x_2 = 0,$$

that is, a *conic*: ellipses, parabolas, hyperbolas or pairs of lines. An SVM with the quadratic kernel can, for example, enclose a class inside an ellipse, something impossible for a linear classifier in $\mathbb{R}^2$; and all of it by solving the same dual QP (14.5) with K in place of the inner product. $\square$

Chapter 15

First-order optimization methods

[P4]

Block of numerical methods, in the final part of the course. Bibliography: Nocedal and Wright [5], Chs. 3 and 5; Wright and Recht [8], Chs. 3–4; Snyman and Wilke [7], Chs. 1–2.

What is this used for in Microeconomics?

Let us be honest: in the Microeconomics exam *nobody* will ask you to run a gradient descent. The theorems tested there are those of the previous chapters (concavity, KKT, envelope, duality). So why study this chapter? For three concrete reasons. (i) These methods are the *engine* of computational economics and structural econometrics: estimating a model *is* minimizing an objective function (the negative log likelihood in MLE, the quadratic criterion in GMM, the distance between simulated and observed moments when calibrating a general equilibrium model), and in practice that minimization is done by an iterative algorithm like the ones in this chapter. (ii) The Walrasian price adjustment (*tatonnement*) that appears in [9], ch. 17, is literally a gradient dynamic on excess demand: prices rise where there is excess demand and fall where there is excess supply. Understanding gradient descent is understanding why (and when) that process converges to equilibrium. (iii) They are the ideal topic for the computational assignments and the final presentation of the course: you can implement them in twenty lines of Python or Julia, they visualize very well, and the convergence proofs are elementary but elegant. References: [5] chs. 3 and 5, [8] chs. 3–4, [7] chs. 1–2, [6] ch. 5, [2] ch. 9.

15.1 Overview: why iterative methods [P4]

15.1.1 There are no closed formulas

In the previous chapters we learned to *characterize* the solutions of an optimization problem: if f is convex and differentiable, x^* minimizes f on $\mathbb{R}^n$ if and only if $\nabla f(x^*) = 0$; with constraints, the KKT conditions play that role. But characterizing is not the same as *computing*. The system $\nabla f(x) = 0$ is a system of n equations (nonlinear in general) with n unknowns, and except in very special cases (quadratic functions, problems with a lot of structure) *there is no closed formula*

for its solution. Think of the log likelihood of a probit model with 20 regressors: nobody can solve for the estimator by hand.

The way out is to build a **sequence of approximations** $x_0, x_1, x_2, \dots$ that converges to the solution. Almost every method in this chapter has the same general structure:

$$x_{k+1} = x_k + t_k d_k, \quad (15.1)$$

where:

- $d_k \in \mathbb{R}^n$ is the **search direction**: where we move to;
- $t_k > 0$ is the **step size** (or, by tradition in machine learning, the *learning rate*): how far we move.

A method is specified once you say how it chooses d_k and how it chooses t_k . They are called *first-order methods* because they only use values of f and of its gradient ∇f ; *second-order methods* (such as Newton) also use the Hessian, which in large dimensions can be very expensive to compute and store.

15.1.2 Descent directions

We want each step to *improve* the objective value. That motivates the next definition.

Definition 15.1 (Descent direction). Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and $x \in \mathbb{R}^n$. A vector $d \in \mathbb{R}^n$ is a **descent direction** of f at x if there exists $\delta > 0$ such that

$$f(x + td) < f(x) \quad \text{for all } t \in (0, \delta).$$

That is: moving from x in the direction d , with a small enough step, *strictly* reduces the value of f .

How do you recognize a descent direction without evaluating f at infinitely many points? With the gradient: it is enough that d makes an obtuse angle with $\nabla f(x)$.

Proposition 15.2 (Characterization by the gradient). Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at x . If

$$\langle \nabla f(x), d \rangle < 0,$$

then d is a descent direction of f at x . In particular, if $\nabla f(x) \neq 0$, the choice $d = -\nabla f(x)$ is always a descent direction, since $\langle \nabla f(x), -\nabla f(x) \rangle = -\|\nabla f(x)\|^2 < 0$.

Proof. Define the one variable function $\varphi(t) = f(x + td)$, which “slices” f along the line through x with direction d . By the chain rule (seen in the preliminaries chapter), φ is differentiable at $t = 0$ and

$$\varphi'(0) = \langle \nabla f(x), d \rangle < 0.$$

Now use the definition of the derivative as a limit:

$$\lim_{t \rightarrow 0^+} \frac{\varphi(t) - \varphi(0)}{t} = \varphi'(0) < 0.$$

Since the limit is a strictly negative number, there exists $\delta > 0$ such that the difference quotient is negative for all $t \in (0, \delta)$ (if not, there would be a sequence $t_j \rightarrow 0^+$ with quotient ≥ 0 , and the limit would be ≥ 0 , a contradiction). For those t , since $t > 0$,

$$\frac{\varphi(t) - \varphi(0)}{t} < 0 \implies \varphi(t) < \varphi(0),$$

that is, $f(x + td) < f(x)$. This is exactly Definition 15.1. $\square$

Intuition

The gradient $\nabla f(x)$ points in the direction of local *maximum growth* of f (we saw this with directional derivatives: $f'(x; d) = \langle \nabla f(x), d \rangle$ is maximized over $\|d\| = 1$ when d is parallel to $\nabla f(x)$). So $-\nabla f(x)$ points in the direction of local maximum *descent*: the direction where the terrain goes down fastest. The condition $\langle \nabla f(x), d \rangle < 0$ says that d has a positive component along $-\nabla f(x)$: you do not need to go down the steepest slope, it is enough that the path points downhill. A short sighted skier who only sees the slope under their feet can pick any descending direction; gradient descent is the skier who always picks the steepest one.

15.1.3 Stopping criterion

An iterative algorithm needs to know when to stop. Since an interior minimizer satisfies $\nabla f(x^*) = 0$, the natural criterion is to stop when the gradient is almost zero:

$$\|\nabla f(x_k)\| \leq \varepsilon,$$

for a tolerance $\varepsilon > 0$ fixed in advance (for example $\varepsilon = 10^{-6}$). Other criteria used in practice: relative progress $|f(x_{k+1}) - f(x_k)| \leq \varepsilon(1 + |f(x_k)|)$, the displacement $\|x_{k+1} - x_k\| \leq \varepsilon$, or a maximum number of iterations as a safeguard. For convex functions we will see something better: you can give *a priori guarantees* of the form “after k iterations, the error $f(x_k) - f^*$ is at most this much”. These bounds are the theoretical core of the chapter.

15.2 Smoothness and strong convexity [P4]

To *prove* that a method converges, and how fast, we need quantitative assumptions on f . The two star assumptions are: an *upper* bound on curvature (smoothness) and a *lower* bound (strong convexity).

15.2.1 Lipschitz gradient and the descent lemma

Definition 15.3 (L -smooth function). Let $L > 0$. A differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth, or has an L -Lipschitz gradient, if

$$\|\nabla f(y) - \nabla f(x)\| \leq L \|y - x\| \quad \text{for all } x, y \in \mathbb{R}^n.$$

The condition says that the gradient cannot change arbitrarily fast: if I move a small distance, the slope changes little. If f is twice differentiable, being L -smooth is equivalent to all eigenvalues of the Hessian $\nabla^2 f(x)$ being bounded by L in absolute value (in the convex case: $0 \preceq \nabla^2 f(x) \preceq LI$ for all x). For example, for a quadratic $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with Q symmetric positive semidefinite, $\nabla f(x) = Qx - b$ and

$$\|\nabla f(y) - \nabla f(x)\| = \|Q(y - x)\| \leq \lambda_{\max}(Q) \|y - x\|,$$

so $L = \lambda_{\max}(Q)$, the largest eigenvalue of Q (and it is the best possible constant: take $y - x$ equal to the associated eigenvector).

The key consequence of L -smoothness is that f is trapped *below* a parabola. This is the so called *descent lemma*, the most used tool in the whole chapter.

Lemma 15.4 (Descent lemma). *If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth, then for all $x, y \in \mathbb{R}^n$:*

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2. \quad (15.2)$$

Proof. The idea: walk along the segment from x to y , write the change in f as the integral of its derivative along the segment, and control how far that derivative can drift from its initial value using the Lipschitz condition.

Step 1 (parametrize the segment). Define $g(s) = f(x + s(y - x))$ for $s \in [0, 1]$, so $g(0) = f(x)$ and $g(1) = f(y)$. By the chain rule,

$$g'(s) = \langle \nabla f(x + s(y - x)), y - x \rangle.$$

Step 2 (fundamental theorem of calculus). Since g is differentiable with continuous derivative,

$$f(y) - f(x) = g(1) - g(0) = \int_0^1 g'(s) ds = \int_0^1 \langle \nabla f(x + s(y - x)), y - x \rangle ds.$$

Step 3 (add and subtract the linear term). We want to compare with $\langle \nabla f(x), y - x \rangle$, which is constant in s and therefore $\int_0^1 \langle \nabla f(x), y - x \rangle ds = \langle \nabla f(x), y - x \rangle$. Subtracting:

$$f(y) - f(x) - \langle \nabla f(x), y - x \rangle = \int_0^1 \langle \nabla f(x + s(y - x)) - \nabla f(x), y - x \rangle ds.$$

Step 4 (bound the integrand). For each fixed s , apply first the Cauchy–Schwarz inequality and then L -smoothness to the pair of points $x + s(y - x)$ and x , whose distance is $s\|y - x\|$:

$$\langle \nabla f(x + s(y - x)) - \nabla f(x), y - x \rangle \leq \|\nabla f(x + s(y - x)) - \nabla f(x)\| \cdot \|y - x\| \leq L s \|y - x\|^2.$$

Step 5 (integrate the bound). Since $\int_0^1 s ds = \frac{1}{2}$,

$$f(y) - f(x) - \langle \nabla f(x), y - x \rangle \leq L \|y - x\|^2 \int_0^1 s ds = \frac{L}{2} \|y - x\|^2,$$

which is exactly (15.2). $\square$

Intuition

Compare (15.2) with the gradient characterization of convexity: $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ says that the graph of f lies *above* its tangent; the descent lemma says that it lies *below* the tangent plus a quadratic correction $\frac{L}{2} \|y - x\|^2$. In other words, at each point x we can put a parabola of curvature L as a ceiling over f , touching it at x . This is gold for designing algorithms: if I cannot minimize f , I minimize the ceiling parabola (that one does have a formula!) and I am guaranteed to go down. We will see that minimizing the ceiling in y gives exactly the gradient step $y = x - \frac{1}{L} \nabla f(x)$.

15.2.2 Strong convexity and the condition number

Definition 15.5 (Strong convexity). Let $m > 0$. A differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is **strongly convex** with modulus m (or *m -strongly convex*) if for all $x, y \in \mathbb{R}^n$:

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2} \|y - x\|^2. \quad (15.3)$$

Equivalently: the function $x \mapsto f(x) - \frac{m}{2} \|x\|^2$ is convex. If f is twice differentiable, this is equivalent to $\nabla^2 f(x) \succeq mI$ for all x (all eigenvalues of the Hessian are $\geq m$).

While L -smoothness puts a parabolic *ceiling*, strong convexity puts a parabolic *floor*: the function has minimum curvature m in every direction, its level sets are oval and bounded, and the minimizer exists and is unique (the parabolic floor grows to infinity, so f is coercive; uniqueness comes from strict convexity). For a quadratic with matrix $Q \succ 0$, the optimal modulus is $m = \lambda_{\min}(Q)$.

When f is both m -strongly convex and L -smooth (necessarily $m \leq L$), the ratio

$$\kappa = \frac{L}{m} \geq 1 \quad (15.4)$$

is called the **condition number** of f . For a quadratic, $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q)$.

Intuition

Geometrically, κ measures how elongated the level curves are. For $f(x, y) = x^2 + 10y^2$ the level curves are ellipses whose semiaxis in x is $\sqrt{10} \approx 3.16$ times the semiaxis in y : a long narrow valley. Here $Q = \text{diag}(2, 20)$, so $m = 2$, $L = 20$ and $\kappa = 10$. If $\kappa = 1$ ($L = m$), the level curves are *circles* and the negative gradient points exactly at the minimizer: gradient descent arrives in one step. The larger κ is, the more misaligned the negative gradient (perpendicular to the level curve) and the direction to the optimum (along the valley) become, and the slower and more erratic the method gets, as we will see precisely. In econometrics, a huge κ shows up when two parameters are almost collinear (weak identification): the likelihood has a flat valley and optimizers struggle.

15.3 Gradient descent [P4]

15.3.1 The algorithm

Gradient descent is method (15.1) with the direction of local steepest descent, $d_k = -\nabla f(x_k)$:

$$x_{k+1} = x_k - t_k \nabla f(x_k). \quad (15.5)$$

Example 15.6 (The algorithm, step by step). Full description of the method, as a recipe:

1. **Input:** initial point $x_0 \in \mathbb{R}^n$, tolerance $\varepsilon > 0$, rule for choosing the step t_k .
2. **For** $k = 0, 1, 2, \dots$:
3. Compute $g_k = \nabla f(x_k)$.
4. **If** $\|g_k\| \leq \varepsilon$: **stop** and return x_k (the gradient is almost zero, we are near a stationary point; if f is convex, near a global minimizer).
5. Choose the step size $t_k > 0$ (see below).
6. Update $x_{k+1} = x_k - t_k g_k$.

Each iteration costs *one* gradient evaluation and $O(n)$ arithmetic operations: this is why the method scales to problems with millions of variables.

15.3.2 Choosing the step size

All the art is in t_k . The three classical rules:

(a) **Fixed step** $t_k = 1/L$. If f is L -smooth, the descent lemma tells us exactly how much we go down. Substituting $y = x_k - t g_k$ (with $g_k = \nabla f(x_k)$) into (15.2), and using $\langle g_k, -t g_k \rangle = -t \|g_k\|^2$:

$$f(x_k - t g_k) \leq f(x_k) - t \|g_k\|^2 + \frac{Lt^2}{2} \|g_k\|^2 = f(x_k) - t \left(1 - \frac{Lt}{2}\right) \|g_k\|^2.$$

The right hand side is a parabola in t that is minimized at $t = 1/L$; with that choice,

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L} \|\nabla f(x_k)\|^2. \quad (15.6)$$

Conclusion: with step $1/L$ the method *always* descends, and it descends more the larger the gradient is. Inequality (15.6) is called *sufficient decrease* and is the building block of all the proofs that follow. Note that any $t \in (0, 2/L)$ guarantees strict descent (the factor $1 - Lt/2$ is positive if $t < 2/L$); with $t > 2/L$ the method can *diverge* (Exercise 15.27).

(b) **Exact line search.** Choose the best possible step in the current direction:

$$t_k \in \arg \min_{t \geq 0} f(x_k + t d_k).$$

This is a one variable problem, but in general it has no closed formula, except for quadratics, where we derive it in full in Section 15.4.

(c) **Backtracking with the Armijo condition.** In practice we do not know L , and the exact search is expensive. The standard compromise: try an optimistic step and shrink it until the decrease obtained is a reasonable fraction of the decrease promised by the linear approximation. Fix parameters $c_1 \in (0, 1)$ (typically $c_1 = 10^{-4}$ or 0.3) and $\beta \in (0, 1)$ (typically $\beta = 1/2$):

1. Start with $t = t_{\text{ini}}$ (for example $t_{\text{ini}} = 1$).
2. **While** the *Armijo condition* fails,

$$f(x_k + t d_k) \leq f(x_k) + c_1 t \langle \nabla f(x_k), d_k \rangle, \quad (15.7)$$

replace $t \leftarrow \beta t$ (“backtrack”).

3. Take $t_k = t$.

What does (15.7) say? The first order Taylor approximation predicts a change $t \langle \nabla f(x_k), d_k \rangle < 0$; Armijo demands that at least the fraction c_1 of that promise be realized. Why does the loop *terminate*? Because, since d_k is a descent direction,

$$\lim_{t \rightarrow 0^+} \frac{f(x_k + t d_k) - f(x_k)}{t} = \langle \nabla f(x_k), d_k \rangle < c_1 \langle \nabla f(x_k), d_k \rangle,$$

where the last inequality uses that multiplying a *negative* number by $c_1 < 1$ makes it larger. By the definition of limit, there exists $\bar{t} > 0$ such that for all $t \in (0, \bar{t})$ the quotient is smaller than $c_1 \langle \nabla f(x_k), d_k \rangle$; multiplying by $t > 0$ gives (15.7). Since backtracking multiplies t by $\beta < 1$ each time, after finitely many reductions it falls inside $(0, \bar{t})$ and the loop stops. One can also prove (see [5], ch. 3) that the accepted step is not too small: if f is L -smooth, backtracking with $d_k = -g_k$ accepts some $t_k \geq \min\{t_{\text{ini}}, \beta 2(1 - c_1)/L\}$, so all the guarantees of the $1/L$ step are preserved (with different constants).

Careful (common mistake)

The step size is not an implementation detail: it is the difference between converging and blowing up. With t too large, the step jumps to the other side of the valley, farther and farther each time, and $f(x_k) \rightarrow +\infty$ (in the one dimensional quadratic $f(x) = \frac{L}{2}x^2$, the map is $x_{k+1} = (1 - tL)x_k$, which diverges if $t > 2/L$; see Exercise 15.27). With t too small, the method converges at a snail’s pace. And beware of a beginner mistake: comparing gradients of variables with different *units* (prices in soles, quantities in tons) without rescaling. That artificially inflates κ and ruins the method. Rescaling the variables so they have comparable magnitudes is, in practice, improvement number one.

15.3.3 Convergence in the convex case: rate $O(1/k)$

Theorem 15.7 (Gradient descent, convex L -smooth case). *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and L -smooth, with minimizer x^* and optimal value $f^* = f(x^*)$. Gradient descent with fixed step*

$t_k = 1/L$ satisfies, for all $k \geq 1$:

$$f(x_k) - f^* \leq \frac{L \|x_0 - x^*\|^2}{2k}.$$

In particular $f(x_k) \rightarrow f^*$, with sublinear rate $O(1/k)$: to guarantee an error $\leq \varepsilon$ it is enough to run $k \approx L \|x_0 - x^*\|^2 / (2\varepsilon)$ iterations.

Proof. The proof is long but completely mechanical: each step uses a single tool. We do it slowly. Write $g_k = \nabla f(x_k)$ and $t = 1/L$.

Step 1: how much we go down in one iteration. This is (15.6), a direct consequence of the descent lemma:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L} \|g_k\|^2.$$

Step 2: compare $f(x_k)$ with f^* using convexity. The first order characterization of convexity, applied at the point x_k and evaluated at $y = x^*$, gives $f(x^*) \geq f(x_k) + \langle g_k, x^* - x_k \rangle$, that is

$$f(x_k) \leq f^* + \langle g_k, x_k - x^* \rangle.$$

(The value at x_k does not exceed the optimal value plus what the tangent plane “recovers” when going from x^* to x_k .)

Step 3: combine. Substituting Step 2 into Step 1:

$$f(x_{k+1}) - f^* \leq \langle g_k, x_k - x^* \rangle - \frac{1}{2L} \|g_k\|^2.$$

Step 4: the completing the square identity. We claim that the right hand side is exactly

$$\frac{L}{2} \left(\|x_k - x^*\|^2 - \|x_k - x^* - \frac{1}{L} g_k\|^2 \right) = \frac{L}{2} \left(\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2 \right),$$

where in the last equality we used $x_{k+1} = x_k - \frac{1}{L} g_k$. Let us verify it by expanding the square (with $a = x_k - x^*$):

$$\left\| a - \frac{1}{L} g_k \right\|^2 = \|a\|^2 - \frac{2}{L} \langle a, g_k \rangle + \frac{1}{L^2} \|g_k\|^2,$$

hence

$$\frac{L}{2} \left(\|a\|^2 - \left\| a - \frac{1}{L} g_k \right\|^2 \right) = \frac{L}{2} \left(\frac{2}{L} \langle a, g_k \rangle - \frac{1}{L^2} \|g_k\|^2 \right) = \langle g_k, a \rangle - \frac{1}{2L} \|g_k\|^2. \checkmark$$

Therefore:

$$f(x_{k+1}) - f^* \leq \frac{L}{2} \left(\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2 \right). \quad (15.8)$$

Notice what we achieved: the error in *value* is controlled by the drop in the *distance to the optimum*. And distances form a telescoping sum.

Step 5: telescope. Sum (15.8) for $i = 0, 1, \dots, k-1$:

$$\sum_{i=0}^{k-1} (f(x_{i+1}) - f^*) \leq \frac{L}{2} \sum_{i=0}^{k-1} \left(\|x_i - x^*\|^2 - \|x_{i+1} - x^*\|^2 \right) = \frac{L}{2} \left(\|x_0 - x^*\|^2 - \|x_k - x^*\|^2 \right),$$

because in the sum on the right each term appears once with sign $+$ and once with sign $-$,

except the first and the last. Dropping the negative term:

$$\sum_{i=0}^{k-1} (f(x_{i+1}) - f^*) \leq \frac{L}{2} \|x_0 - x^*\|^2.$$

Step 6: the last iterate is the best. By Step 1 the sequence $f(x_i)$ is nonincreasing, so each summand is $\geq f(x_k) - f^*$ and the sum (which has k terms) satisfies

$$k(f(x_k) - f^*) \leq \sum_{i=0}^{k-1} (f(x_{i+1}) - f^*) \leq \frac{L}{2} \|x_0 - x^*\|^2.$$

Dividing by k gives the stated bound. $\square$

15.3.4 Convergence under strong convexity: linear rate

With minimum curvature $m > 0$ the picture improves dramatically: the error is multiplied by a fixed factor smaller than 1 at *every* iteration (*linear* or *geometric* convergence). We prove it in full detail for quadratics, where the eigenvalues make the mechanism transparent, and we state the general case.

Theorem 15.8 (Linear rate on quadratics). *Let $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with Q symmetric and $mI \preceq Q \preceq LI$, $m > 0$; let $x^* = Q^{-1}b$ be its minimizer. Gradient descent with fixed step $t \in (0, 2/L)$ satisfies*

$$\|x_k - x^*\| \leq \rho(t)^k \|x_0 - x^*\|, \quad \rho(t) = \max\{|1 - tm|, |1 - tL|\} < 1.$$

In particular, with $t = 1/L$: $\rho = 1 - m/L = 1 - 1/\kappa$; and the optimal step $t = 2/(m + L)$ gives $\rho = (\kappa - 1)/(\kappa + 1)$.

Proof. Since $\nabla f(x) = Qx - b$ and $b = Qx^*$, the iteration is written as

$$x_{k+1} - x^* = x_k - t(Qx_k - b) - x^* = (I - tQ)(x_k - x^*).$$

The error obeys a *linear* recursion with matrix $I - tQ$. Let $Q = \sum_{i=1}^n \lambda_i v_i v_i^\top$ be the spectral decomposition of Q (eigenvalues $\lambda_i \in [m, L]$, orthonormal eigenvectors v_i ; see in the preliminaries). Then $I - tQ$ has the same eigenvectors with eigenvalues $1 - t\lambda_i$. Expanding the initial error in the orthonormal basis, $x_0 - x^* = \sum_i c_i v_i$, we get

$$x_k - x^* = (I - tQ)^k (x_0 - x^*) = \sum_{i=1}^n c_i (1 - t\lambda_i)^k v_i,$$

and by the Pythagorean theorem (the v_i are orthonormal):

$$\|x_k - x^*\|^2 = \sum_{i=1}^n c_i^2 (1 - t\lambda_i)^{2k} \leq \left(\max_i |1 - t\lambda_i| \right)^{2k} \sum_i c_i^2 = \rho(t)^{2k} \|x_0 - x^*\|^2,$$

where $\rho(t) = \max_i |1 - t\lambda_i|$. Since the function $\lambda \mapsto |1 - t\lambda|$ is convex in λ , its maximum over $[m, L]$ is attained at an endpoint: $\rho(t) = \max\{|1 - tm|, |1 - tL|\}$. If $t \in (0, 2/L)$ then $|1 - t\lambda_i| < 1$

for all i , so $\rho(t) < 1$. With $t = 1/L$: $|1 - m/L| = 1 - 1/\kappa$ and $|1 - 1| = 0$, so $\rho = 1 - 1/\kappa$. The optimal step balances the two endpoints: choose t such that $1 - tm = -(1 - tL)$, that is $t = 2/(m + L)$, and then $\rho = 1 - \frac{2m}{m+L} = \frac{L-m}{L+m} = \frac{\kappa-1}{\kappa+1}$. $\square$

Theorem 15.9 (General strongly convex case; statement). *Let f be m -strongly convex and L -smooth. Gradient descent with step $t = 1/L$ satisfies*

$$\|x_k - x^*\|^2 \leq \left(1 - \frac{m}{L}\right)^k \|x_0 - x^*\|^2 \quad \text{and} \quad f(x_k) - f^* \leq \left(1 - \frac{m}{L}\right)^k (f(x_0) - f^*).$$

To push the error below ε it is enough to run $k \approx \kappa \log(1/\varepsilon)$ iterations: the cost grows with the logarithm of the precision (compare with the $1/\varepsilon$ of the merely convex case), but it is proportional to the condition number κ . The general proof combines the descent lemma with (15.3); see [8], ch. 3, or [2], sec. 9.3.

15.3.5 Worked example: the effect of conditioning

Example 15.10 (Iterations by hand with step $1/L$). Let $f(x, y) = x^2 + 10y^2$, with $\nabla f(x, y) = (2x, 20y)$, Hessian $\text{diag}(2, 20)$, and therefore $m = 2$, $L = 20$, $\kappa = 10$, $x^* = (0, 0)$, $f^* = 0$. Take $x_0 = (1, 1)$ and fixed step $t = 1/L = 0.05$.

Iteration 1. $g_0 = \nabla f(1, 1) = (2, 20)$.

$$x_1 = (1, 1) - 0.05 (2, 20) = (1 - 0.1, 1 - 1) = (0.9, 0).$$

Iteration 2. $g_1 = \nabla f(0.9, 0) = (1.8, 0)$.

$$x_2 = (0.9, 0) - 0.05 (1.8, 0) = (0.81, 0).$$

Iteration 3. $g_2 = (1.62, 0)$, and $x_3 = (0.729, 0)$.

The objective values: $f(x_0) = 11$, $f(x_1) = 0.81$, $f(x_2) = 0.6561$, $f(x_3) \approx 0.5314$. Notice the pattern: the *stiff* coordinate y (curvature $20 = L$) is annihilated in a single step, because $1 - tL = 0$; but the *soft* coordinate x (curvature $2 = m$) only contracts by the factor $1 - tm = 0.9$ per iteration, exactly the rate $1 - 1/\kappa$ of Theorem 15.8. To halve the error in x you need $\log(2)/\log(1/0.9) \approx 6.6$ iterations. The step is *dictated* by the most curved direction (if it were larger, we would diverge in y), but the *progress* is limited by the flattest one: that tension is conditioning. With exact line search the same problem shows the famous zigzag, which we analyze next.

15.4 Steepest descent on quadratics [P4]

Steepest descent is gradient descent with *exact line search*. On a quadratic, the exact step has a closed formula, and that lets us examine the behavior of the method under a magnifying glass.

Proposition 15.11 (Exact step on a quadratic). *Let $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q \succ 0$, and let $g = \nabla f(x) = Qx - b \neq 0$. The one dimensional problem $\min_{t \geq 0} f(x - tg)$ has the unique*

solution

$$t^{ex} = \frac{g^\top g}{g^\top Q g} = \frac{\|g\|^2}{\langle g, Qg \rangle}. \quad (15.9)$$

Proof. Define $\varphi(t) = f(x - tg)$ and differentiate with the chain rule:

$$\varphi'(t) = \langle \nabla f(x - tg), -g \rangle.$$

Compute the gradient at the shifted point using the linearity of $\nabla f(\cdot) = Q(\cdot) - b$:

$$\nabla f(x - tg) = Q(x - tg) - b = (Qx - b) - tQg = g - tQg.$$

Then

$$\varphi'(t) = \langle g - tQg, -g \rangle = -\|g\|^2 + t \langle g, Qg \rangle.$$

Since $Q \succ 0$ and $g \neq 0$, we have $\langle g, Qg \rangle > 0$: φ is a strictly convex parabola in t , and its minimum is obtained by setting $\varphi'(t) = 0$, which gives (15.9) (which is positive, as it should be). $\square$

An immediate structural consequence explains the zigzag: at the minimum of the line search, $0 = \varphi'(t_k) = \langle \nabla f(x_{k+1}), -g_k \rangle$, that is,

$$\langle \nabla f(x_{k+1}), \nabla f(x_k) \rangle = 0 : \quad (15.10)$$

each gradient is orthogonal to the previous one. The method advances in ninety degree zigzags; in an elongated valley this means crossing the valley again and again while advancing very little toward the exit (this orthogonality holds for *any* f with exact search, not only quadratics; you are asked to prove it in Exercise 15.26).

Example 15.12 (Zigzag computed by hand). Again $f(x, y) = x^2 + 10y^2$, that is $Q = \text{diag}(2, 20)$, $b = 0$, $x^* = (0, 0)$; now $x_0 = (10, 1)$ and exact search.

Iteration 1. $g_0 = (2 \cdot 10, 20 \cdot 1) = (20, 20)$. Then $\|g_0\|^2 = 400 + 400 = 800$ and $\langle g_0, Qg_0 \rangle = 2 \cdot 20^2 + 20 \cdot 20^2 = 800 + 8000 = 8800$, so $t_0 = 800/8800 = 1/11$. Therefore

$$x_1 = (10, 1) - \frac{1}{11}(20, 20) = \left(\frac{110-20}{11}, \frac{11-20}{11}\right) = \left(\frac{90}{11}, -\frac{9}{11}\right) = \frac{9}{11}(10, -1).$$

Iteration 2. Because of the symmetric form of x_1 , the computation repeats at a smaller scale: $g_1 = \frac{9}{11}(20, -20)$, and since step (15.9) is invariant to rescaling g , again $t_1 = 1/11$, so

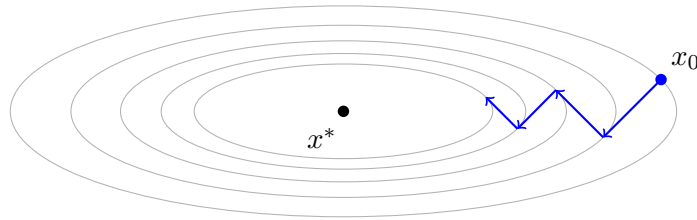
$$x_2 = \frac{9}{11} \left[(10, -1) - \frac{1}{11}(20, -20) \right] = \left(\frac{9}{11}\right)^2 (10, 1).$$

In general: $x_k = \left(\frac{9}{11}\right)^k (10, (-1)^k)$: the method bounces between the rays through $(10, 1)$ and $(10, -1)$, approaching the origin with exact ratio $9/11$ per iteration, and

$$f(x_k) = \left(\frac{9}{11}\right)^{2k} f(x_0) = \left(\frac{81}{121}\right)^k \cdot 110 \approx (0.669)^k \cdot 110.$$

Numerically: $x_1 \approx (8.18, -0.82)$, $x_2 \approx (6.69, 0.67)$, $x_3 \approx (5.48, -0.55)$; f falls $110 \rightarrow 73.6 \rightarrow$

49.3 $\rightarrow$ 33.0. Not bad, but note that with $\kappa = 10$ the ratio is $9/11 = (\kappa - 1)/(\kappa + 1)$: this starting point realizes the *worst case* of the next theorem.



Zigzag of steepest descent on the elliptical level curves of $f(x, y) = x^2 + 10y^2$: each step is orthogonal to the previous one.

Theorem 15.13 (Convergence bound; statement). *For $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q \succ 0$ and $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, steepest descent with exact search satisfies*

$$f(x_{k+1}) - f^* \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^2 (f(x_k) - f^*).$$

The proof uses the Kantorovich inequality; you can see it in [5], theorem 3.3, or [7], ch. 2.

Intuition

Read the bound like this: if $\kappa = 1$ (circular level sets), the factor is 0 and the method lands on x^* in one step, because the negative gradient points at the center. If $\kappa = 10$, the factor is $(9/11)^2 \approx 0.67$: you lose a third of the error per iteration, which is acceptable. If $\kappa = 1000$ (not unusual in econometrics with correlated regressors), the factor is $(999/1001)^2 \approx 0.9960$: you would need about 173 iterations to halve the error, because $\log(1/2)/\log(0.9960) \approx 173$. The frustrating part is that exact search (the best possible step in the chosen direction) cures nothing: the problem is not the step but the *direction*, which forgets everything learned in previous iterations and therefore repeats the same transversal mistake again and again. The cure is to give the method memory: that is the conjugate gradient.

15.5 The conjugate gradient method [P4]

15.5.1 Motivation: fixing the zigzag with memory

In the zigzag, each new direction undoes part of the work of the previous one: after minimizing along d_0 , the next step reintroduces error in the direction d_0 . The idea of the **conjugate gradient method** (CG) of Hestenes and Stiefel is to choose directions that *do not interfere with each other*: once minimized along d_0 , no future step spoils that direction again. The right notion of “no interference” is not the usual orthogonality, but orthogonality measured with the geometry of the quadratic itself.

In this section we work with

$$\min_{x \in \mathbb{R}^n} f(x) = \frac{1}{2} x^\top Qx - b^\top x, \quad Q \succ 0 \text{ symmetric,}$$

whose unique minimizer x^* solves $Qx^* = b$. We write the *residual* $r_k = b - Qx_k = -\nabla f(x_k)$ (the negative gradient).

Definition 15.14 (Q -conjugate directions). The nonzero vectors $d_0, d_1, \dots, d_m$ are Q -conjugate if

$$d_i^\top Q d_j = 0 \quad \text{for all } i \neq j.$$

This is orthogonality with respect to the inner product $\langle u, v \rangle_Q = u^\top Q v$ (which is a legitimate inner product because $Q \succ 0$).

Lemma 15.15 (Conjugate $\implies$ linearly independent). *If $d_0, \dots, d_m$ are nonzero and Q -conjugate with $Q \succ 0$, then they are linearly independent. In particular, in $\mathbb{R}^n$ there are at most n Q -conjugate directions.*

Proof. Suppose $\sum_{i=0}^m c_i d_i = 0$. Fix an index j and multiply the equality on the left by $d_j^\top Q$:

$$0 = d_j^\top Q \left(\sum_i c_i d_i \right) = \sum_i c_i d_j^\top Q d_i = c_j d_j^\top Q d_j,$$

because conjugacy kills all terms with $i \neq j$. Since $Q \succ 0$ and $d_j \neq 0$, we have $d_j^\top Q d_j > 0$, so $c_j = 0$. Since j was arbitrary, all coefficients are zero. $\square$

Theorem 15.16 (n conjugate directions solve in n steps). *Let $d_0, \dots, d_{n-1}$ be nonzero Q -conjugate directions in $\mathbb{R}^n$. From any x_0 , the conjugate direction method*

$$x_{k+1} = x_k + \alpha_k d_k, \quad \alpha_k = \frac{d_k^\top r_k}{d_k^\top Q d_k} \quad (\text{the exact line search step along } d_k),$$

reaches $x_n = x^$: the exact solution in at most n steps.*

Proof. Step 1: expand the total displacement to the optimum. By Lemma 15.15, $\{d_0, \dots, d_{n-1}\}$ is a basis of $\mathbb{R}^n$, so there is a unique expansion

$$x^* - x_0 = \sum_{i=0}^{n-1} \sigma_i d_i.$$

To solve for σ_j we use the trick from the lemma: multiply by $d_j^\top Q$ and conjugacy kills all terms except the j -th:

$$d_j^\top Q (x^* - x_0) = \sigma_j d_j^\top Q d_j \implies \sigma_j = \frac{d_j^\top Q (x^* - x_0)}{d_j^\top Q d_j}. \quad (15.11)$$

Step 2: the algorithm computes exactly those coefficients. After k steps, $x_k - x_0 = \sum_{i=0}^{k-1} \alpha_i d_i$ is a combination of the *previous* directions, so, by conjugacy,

$$d_k^\top Q (x_k - x_0) = \sum_{i=0}^{k-1} \alpha_i d_k^\top Q d_i = 0.$$

Therefore, in the numerator of (15.11) we can replace x_0 by x_k without changing anything:

$$d_k^\top Q(x^* - x_0) = d_k^\top Q(x^* - x_k) + \underbrace{d_k^\top Q(x_k - x_0)}_{=0} = d_k^\top (\underbrace{Qx^* - Qx_k}_{=b}) = d_k^\top r_k.$$

That is, $\sigma_k = \frac{d_k^\top r_k}{d_k^\top Q d_k} = \alpha_k$: the exact search step coincides with the expansion coefficient (that α_k is the exact step is checked as in Proposition 15.11, differentiating $\varphi(\alpha) = f(x_k + \alpha d_k)$ and setting it to zero).

Step 3: conclusion. After n steps,

$$x_n = x_0 + \sum_{k=0}^{n-1} \alpha_k d_k = x_0 + \sum_{k=0}^{n-1} \sigma_k d_k = x_0 + (x^* - x_0) = x^*. \quad \square$$

Intuition

In the geometry of the inner product $\langle \cdot, \cdot \rangle_Q$, the level ellipses of f become *spheres*, and Q -conjugate directions are simply perpendicular directions in that world. The theorem then says something obvious: to reach a point of $\mathbb{R}^n$ it is enough to adjust once each of n perpendicular coordinates, like someone who first walks all the way east and then all the way north, never retracing their steps. The zigzag of steepest descent happens precisely because its directions (the successive gradients) are orthogonal in the *wrong* world: the Euclidean one, not the one of Q .

15.5.2 The CG algorithm

We still need to know *where to get* the conjugate directions without diagonalizing Q (that would cost more than solving the system). The clever part of CG: build them on the fly, taking at each step the current negative gradient and “correcting” it with a multiple of the previous direction to restore conjugacy. A very short recursion is enough.

Example 15.17 (CG algorithm for quadratics). Data: $Q \succ 0$, b , initial point x_0 .

1. Initialize $r_0 = b - Qx_0$ (negative gradient) and $d_0 = r_0$ (the first step is pure gradient descent).
2. **For** $k = 0, 1, 2, \dots$ while $r_k \neq 0$:
3. Exact step along d_k : $\alpha_k = \frac{r_k^\top r_k}{d_k^\top Q d_k}$.
4. Advance: $x_{k+1} = x_k + \alpha_k d_k$.
5. Update the residual: $r_{k+1} = r_k - \alpha_k Q d_k$ (equivalent to $r_{k+1} = b - Qx_{k+1}$, but it reuses the product $Q d_k$ already computed).
6. Correction coefficient: $\beta_{k+1} = \frac{r_{k+1}^\top r_{k+1}}{r_k^\top r_k}$.
7. New direction: $d_{k+1} = r_{k+1} + \beta_{k+1} d_k$.

Comments on the formulas. (i) The expression for α_k is the exact search step $d_k^\top r_k / d_k^\top Q d_k$ of Theorem 15.16: you can prove by induction that $d_k^\top r_k = r_k^\top r_k$ (the residuals are orthogonal to all past directions). (ii) The coefficient β_{k+1} is exactly the one that makes $d_{k+1}^\top Q d_k = 0$; the remarkable part (and not obvious, you can see it in [5], theorem 5.3) is that with that *single* correction the new direction turns out to be conjugate with *all* the previous ones too. (iii) Each iteration costs a single matrix vector product $Q d_k$: CG never forms or factorizes Q , which is why it is the standard method for huge sparse linear systems.

Theorem 15.18 (Finite termination; statement). *Applied to a quadratic with $Q \succ 0$ in $\mathbb{R}^n$, CG generates Q -conjugate directions and mutually orthogonal residuals, and therefore reaches the exact solution $x^* = Q^{-1}b$ in at most n iterations. Even more, if Q has only $s < n$ distinct eigenvalues, it terminates in at most s iterations, and well before terminating it already satisfies the bound*

$$f(x_k) - f^* \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^{2k} (f(x_0) - f^*) :$$

the dependence on conditioning improves from κ to $\sqrt{\kappa}$ (see [5], ch. 5).

Example 15.19 (CG by hand in $\mathbb{R}^2$). Let us solve $\min f(x) = \frac{1}{2}x^\top Q x - b^\top x$ with

$$Q = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad x_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The exact solution of $Qx = b$ is $x^* = \frac{1}{11}(1, 7)^\top$ (check it: $4 \cdot \frac{1}{11} + \frac{7}{11} = 1$, $\frac{1}{11} + 3 \cdot \frac{7}{11} = 2$). Let us see that CG finds it in exactly 2 steps.

Initialization. $r_0 = b - Qx_0 = (1, 2)^\top = d_0$.

Iteration 1. $Qd_0 = (4 + 2, 1 + 6)^\top = (6, 7)^\top$; $r_0^\top r_0 = 1 + 4 = 5$; $d_0^\top Qd_0 = 1 \cdot 6 + 2 \cdot 7 = 20$. So $\alpha_0 = 5/20 = 1/4$ and

$$x_1 = (0, 0)^\top + \frac{1}{4}(1, 2)^\top = \left(\frac{1}{4}, \frac{1}{2}\right)^\top, \quad r_1 = r_0 - \frac{1}{4}Qd_0 = (1, 2)^\top - \left(\frac{6}{4}, \frac{7}{4}\right)^\top = \left(-\frac{1}{2}, \frac{1}{4}\right)^\top.$$

Iteration 2. $r_1^\top r_1 = \frac{1}{4} + \frac{1}{16} = \frac{5}{16}$, so $\beta_1 = \frac{5/16}{5} = \frac{1}{16}$ and

$$d_1 = r_1 + \frac{1}{16}d_0 = \left(-\frac{1}{2} + \frac{1}{16}, \frac{1}{4} + \frac{2}{16}\right)^\top = \frac{1}{16}(-7, 6)^\top.$$

(Conjugacy check: $d_0^\top Qd_1 = (6, 7) \cdot \frac{1}{16}(-7, 6)^\top = \frac{1}{16}(-42 + 42) = 0$. ✓) Now $Qd_1 = \frac{1}{16}(-28 + 6, -7 + 18)^\top = \frac{1}{16}(-22, 11)^\top$, $d_1^\top Qd_1 = \frac{1}{256}(154 + 66) = \frac{220}{256} = \frac{55}{64}$, and

$$\alpha_1 = \frac{5/16}{55/64} = \frac{5}{16} \cdot \frac{64}{55} = \frac{4}{11}, \quad x_2 = \left(\frac{1}{4}, \frac{1}{2}\right)^\top + \frac{4}{11} \cdot \frac{1}{16}(-7, 6)^\top = \left(\frac{11-7}{44}, \frac{22+6}{44}\right)^\top = \left(\frac{1}{11}, \frac{7}{11}\right)^\top = x^*.$$

Two variables, two steps, *exact* solution: finite termination in action. Gradient descent with a fixed step, by contrast, only *approaches* x^* geometrically and never arrives in finite time.

15.5.3 Nonlinear CG

To minimize a general f (not quadratic), the **Fletcher–Reeves** method imitates the recursion by replacing the residual with the negative gradient: $d_{k+1} = -\nabla f(x_{k+1}) + \beta_{k+1}^{FR} d_k$ with

$$\beta_{k+1}^{FR} = \frac{\|\nabla f(x_{k+1})\|^2}{\|\nabla f(x_k)\|^2},$$

and computes α_k with a line search (Armijo strengthened with the Wolfe conditions). Variants such as Polak–Ribière usually work better in practice. Near the minimum, where f looks like its quadratic Taylor model, nonlinear CG inherits the speed of the quadratic case; the details are in [5], ch. 5.

15.6 The projected gradient method [P4]

15.6.1 Motivation and algorithm

So far we minimized over all of $\mathbb{R}^n$. But economic problems almost always come with constraints: nonnegative quantities, budgets, probabilities that add up to one. Consider

$$\min_{x \in C} f(x), \quad C \subseteq \mathbb{R}^n \text{ convex, closed and nonempty, } f \text{ convex and differentiable.}$$

If we take a gradient step from $x_k \in C$, the point $x_k - t\nabla f(x_k)$ may fall out of C . The natural repair: bring it back to C with the *projection* proj_C studied in the chapter on projection onto convex sets (recall: $\text{proj}_C(z)$ is the point of C closest to z , and it exists and is unique because C is closed and convex). This gives the **projected gradient method**:

$$x_{k+1} = \text{proj}_C(x_k - t_k \nabla f(x_k)). \quad (15.12)$$

The method is practical exactly when C is *simple*: a box, a ball, a hyperplane or halfspace, the simplex, sets for which we have formulas (or fast algorithms) for the projection. If projecting onto C is as hard as the original problem, the method does not help.

Example 15.20 (The algorithm, step by step). 1. **Input:** $x_0 \in C$, step $t > 0$ (if f is L -smooth, $t = 1/L$), tolerance $\varepsilon > 0$.

2. **For** $k = 0, 1, 2, \dots$:

3. Gradient step: $y_k = x_k - t \nabla f(x_k)$.

4. Repair: $x_{k+1} = \text{proj}_C(y_k)$.

5. **If** $\|x_{k+1} - x_k\| \leq \varepsilon$: **stop** (we are almost a fixed point; by Theorem 15.21, almost optimal).

Note that the stopping criterion can no longer be $\|\nabla f(x_k)\| \approx 0$: at a boundary optimum the gradient does *not* vanish (KKT taught us that); what vanishes is the displacement of the algorithm.

15.6.2 Why it works

Two properties support the method. The first we proved in the projection chapter: the projection *does not expand* distances,

$$\|\text{proj}_C(u) - \text{proj}_C(v)\| \leq \|u - v\| \quad \text{for all } u, v \in \mathbb{R}^n.$$

Thanks to it, projecting never spoils the progress of the gradient step: all the descent and contraction estimates of the previous sections survive the projection. The second is that the *fixed points* of the iteration are exactly the solutions of the problem; we prove this now.

Theorem 15.21 (Fixed points = solutions). *Let f be convex and differentiable, C convex closed nonempty and $t > 0$ arbitrary. For $x^* \in C$ the following are equivalent:*

- (i) $x^* = \text{proj}_C(x^* - t \nabla f(x^*))$ (x^* is a fixed point of the iteration);
- (ii) $\langle \nabla f(x^*), y - x^* \rangle \geq 0$ for all $y \in C$ (variational inequality of optimality);
- (iii) $-\nabla f(x^*)$ belongs to the normal cone of C at x^* , $N_C(x^*)$;
- (iv) x^* minimizes f over C .

Proof. Recall the variational characterization of the projection (chapter on projection onto convex sets): for $z \in \mathbb{R}^n$ and $\bar{z} \in C$,

$$\bar{z} = \text{proj}_C(z) \iff \langle z - \bar{z}, y - \bar{z} \rangle \leq 0 \quad \text{for all } y \in C. \quad (15.13)$$

(“The vector going from the projection to the projected point makes an obtuse angle with any direction that enters C .”)

(i) $\Leftrightarrow$ (ii). Apply (15.13) with $z = x^* - t \nabla f(x^*)$ and projection candidate $\bar{z} = x^*$ (which is in C). Since $z - \bar{z} = -t \nabla f(x^*)$, the condition on the right reads: for all $y \in C$,

$$\langle -t \nabla f(x^*), y - x^* \rangle \leq 0 \iff \langle \nabla f(x^*), y - x^* \rangle \geq 0,$$

where we divided by $t > 0$ and changed the sign. This is exactly (ii), and equivalence (15.13) gives (i) $\Leftrightarrow$ (ii).

(ii) $\Leftrightarrow$ (iii). By definition, the normal cone is $N_C(x^*) = \{v : \langle v, y - x^* \rangle \leq 0 \ \forall y \in C\}$; condition (ii) says $\langle -\nabla f(x^*), y - x^* \rangle \leq 0$ for all $y \in C$, that is $-\nabla f(x^*) \in N_C(x^*)$. They are the same statement with different notation.

(ii) $\Rightarrow$ (iv). By convexity of f , for any $y \in C$: $f(y) \geq f(x^*) + \langle \nabla f(x^*), y - x^* \rangle \geq f(x^*)$, using (ii) in the last inequality. So x^* is a global minimizer on C .

(iv) $\Rightarrow$ (ii). Let $y \in C$. By convexity of C , the point $x^* + s(y - x^*)$ is in C for $s \in [0, 1]$, and by optimality $\varphi(s) = f(x^* + s(y - x^*))$ has a minimum at $s = 0$; its one sided derivative must be nonnegative: $0 \leq \varphi'(0^+) = \langle \nabla f(x^*), y - x^* \rangle$. $\square$

The theorem is the algorithmic version of the geometric optimality condition we already knew ($-\nabla f(x^*) \in N_C(x^*)$, the negative gradient “pushes outward” from C): iteration (15.12) stays put exactly at the optima. Convergence inherits the rates of the unconstrained case:

Theorem 15.22 (Convergence of projected gradient; statement). *If f is convex and L -smooth on C , and x^* minimizes f on C , method (15.12) with fixed step $t = 1/L$ satisfies*

$$f(x_k) - f(x^*) \leq \frac{L \|x_0 - x^*\|^2}{2k},$$

and if in addition f is m -strongly convex, the convergence is linear with ratio $1 - m/L$. The proofs copy those of Section 15.3, inserting the nonexpansiveness of proj_C where needed; see [6], ch. 5, and [8], ch. 4.

15.6.3 Full example on a box

Example 15.23 (Projected gradient on a box). Consider

$$\min_{(x,y) \in C} f(x,y) = (x-2)^2 + (y-1)^2, \quad C = [0,1] \times [0,1].$$

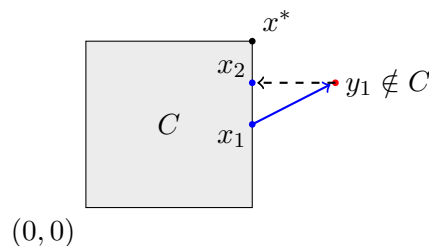
The unconstrained minimum is $(2,1) \notin C$. Recall from the projection chapter that projecting onto a box means *clipping* coordinate by coordinate: $\text{proj}_{[0,1]}(z) = \min\{\max\{z, 0\}, 1\}$. Here $\nabla f(x,y) = (2(x-2), 2(y-1))$ and the Hessian is $2I$, so $L = 2$; take, in order to see several iterations, $t = 1/4$ (with $t = 1/L = 1/2$ the example ends in a single step; check it). The gradient step is then, coordinate by coordinate, $x - \frac{1}{4} \cdot 2(x-2) = \frac{x}{2} + 1$ and $y - \frac{1}{4} \cdot 2(y-1) = \frac{y}{2} + \frac{1}{2}$. We start from $x_0 = (0,0)$.

Iteration 1. $y_0 = (\frac{0}{2} + 1, \frac{0}{2} + \frac{1}{2}) = (1, 0.5) \in C$: the projection does nothing, $x_1 = (1, 0.5)$.

Iteration 2. $y_1 = (\frac{1}{2} + 1, \frac{0.5}{2} + \frac{1}{2}) = (1.5, 0.75) \notin C$. We clip: the first coordinate $1.5 \mapsto 1$; the second stays. $x_2 = (1, 0.75)$.

Iteration 3. $y_2 = (1.5, 0.875)$, and again $x_3 = (1, 0.875)$.

The first coordinate is already stuck to the wall $x = 1$ (active constraint) and the second halves its distance to 1 at each step $(0, 0.5, 0.75, 0.875, \dots)$: $x_k \rightarrow x^* = (1, 1)$. And $(1, 1)$ had to be the optimum, because f is the squared distance to the point $(2, 1)$, so minimizing it over C is exactly projecting: $x^* = \text{proj}_C(2, 1) = (1, 1)$. Let us verify optimality with Theorem 15.21(iii): $\nabla f(1, 1) = (-2, 0)$, and the normal cone of the box at the vertex $(1, 1)$ is $N_C(1, 1) = \{(u, v) : u \geq 0, v \geq 0\}$ (the vectors pointing outward through the two faces that meet there). Since $-\nabla f(1, 1) = (2, 0) \in N_C(1, 1)$, indeed $(1, 1)$ is the optimum.



Projected gradient on the box: the gradient step (blue arrow) leaves C ; the projection (dashed arrow) brings it back to the face $x = 1$.

15.6.4 Full example on the simplex

The **probability simplex**

$$\Delta_n = \left\{ x \in \mathbb{R}^n : x_i \geq 0, \sum_{i=1}^n x_i = 1 \right\}$$

appears everywhere in economics: lotteries and mixed strategies, portfolios without short sales, market shares. Projecting onto Δ_n is an interesting problem *in itself*: there is no one line formula, but there is an almost formula. One can prove (with KKT, a good review exercise!) that

$$\text{proj}_{\Delta_n}(z) = ((z_1 - \tau)_+, \dots, (z_n - \tau)_+), \quad (u)_+ = \max\{u, 0\},$$

where the threshold $\tau \in \mathbb{R}$ is the unique number such that $\sum_i (z_i - \tau)_+ = 1$: you subtract the same amount from everyone and clip the negatives, choosing the discount so that the total is 1 (like a lump sum tax with limited liability). To find τ you sort the coordinates and test which subset stays positive: an $O(n \log n)$ algorithm.

Example 15.24 (Projected gradient on the simplex). Let us minimize the linear function

$$f(x) = 3x_1 + 2x_2 + x_3 \quad \text{over } \Delta_3,$$

whose obvious optimum is the vertex $e_3 = (0, 0, 1)$ (all the weight on the cheapest coefficient). Although f is linear (not strongly convex), the method works; take $t = \frac{1}{2}$, $x_0 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $\nabla f = c = (3, 2, 1)$ constant.

Iteration 1. Gradient step:

$$y_0 = x_0 - \frac{1}{2}(3, 2, 1) = \left(\frac{1}{3} - \frac{3}{2}, \frac{1}{3} - 1, \frac{1}{3} - \frac{1}{2}\right) = \left(-\frac{7}{6}, -\frac{2}{3}, -\frac{1}{6}\right).$$

We project: look for τ with $\sum_i (y_{0,i} - \tau)_+ = 1$. Let us try the guess that coordinates 2 and 3 stay positive and coordinate 1 is clipped: then $(-\frac{2}{3} - \tau) + (-\frac{1}{6} - \tau) = 1$ gives $\tau = -\frac{11}{12}$. Check: $y_{0,2} - \tau = -\frac{8}{12} + \frac{11}{12} = \frac{3}{12} > 0$, $y_{0,3} - \tau = -\frac{2}{12} + \frac{11}{12} = \frac{9}{12} > 0$, and $y_{0,1} - \tau = -\frac{14}{12} + \frac{11}{12} = -\frac{3}{12} < 0$: consistent. So

$$x_1 = \text{proj}_{\Delta_3}(y_0) = \left(0, \frac{1}{4}, \frac{3}{4}\right).$$

Iteration 2.

$$y_1 = \left(0 - \frac{3}{2}, \frac{1}{4} - 1, \frac{3}{4} - \frac{1}{2}\right) = \left(-\frac{3}{2}, -\frac{3}{4}, \frac{1}{4}\right).$$

Try support $\{3\}$: τ solves $\frac{1}{4} - \tau = 1$, that is $\tau = -\frac{3}{4}$; then $y_{1,2} - \tau = 0$ and $y_{1,1} - \tau = -\frac{3}{4} < 0$: only the third coordinate stays positive, and

$$x_2 = (0, 0, 1) = e_3.$$

We reached the optimum in two steps. Fixed point check (Theorem 15.21(ii)): for all $y \in \Delta_3$, $\langle c, y - e_3 \rangle = 3y_1 + 2y_2 + y_3 - 1 \geq y_1 + y_2 + y_3 - 1 = 0$, as the variational inequality requires. (In general the method does not terminate in finite time; here it does because the optimum is a vertex and the steps are large.)

Exam strategy

Facing a constrained optimization problem that you want to solve *numerically*, follow this order of questions. (1) Is C simple to project onto (box, ball, hyperplane, halfspace, simplex)? If so, projected gradient; recycle the formulas from the projection chapter. (2) Do you know L (for example $\lambda_{\max}$ of the Hessian, or a bound)? Use step $1/L$; if not, backtracking–Armijo, which requires knowing nothing. (3) Is the problem unconstrained, quadratic or nearly quadratic, with n large? Conjugate gradient. (4) Ill conditioned? Rescale the variables (or “precondition”) *before* blaming the algorithm. (5) To check your implementation, test it first on a small quadratic whose solution you know exactly, and verify the theoretical rates: if the error does not fall like $(1 - 1/\kappa)^k$, there is a bug. This protocol is, as it stands, a good script for the computational assignment of the course.

15.7 Applied note: computational economics [P4]**15.7.1 Walrasian tatonnement as gradient dynamics**

In the general equilibrium model, the Walrasian auctioneer adjusts the price vector $p \in \mathbb{R}_{++}^n$ according to the law of the market: the price of goods in excess demand goes up and the price of goods in excess supply goes down. If $z(p) \in \mathbb{R}^n$ is the *aggregate excess demand* function (demand minus supply at prices p), discrete time *tatonnement* is

$$p_{k+1} = p_k + t z(p_k),$$

formally identical to a gradient *ascent* (15.1) with direction $d_k = z(p_k)$. A Walrasian equilibrium p^* satisfies $z(p^*) = 0$: it is exactly a stationary point of the dynamics. When excess demand comes from a potential (for example, in economies with aggregable demands where $z(p) = \nabla W(p)$ for some concave function W), tatonnement *is* a gradient ascent on W and the theorems of this chapter give its convergence and its speed. Without a potential, the classical argument uses a Lyapunov function: under *gross substitutes* the distance $\|p_k - p^*\|^2$ decreases along the adjustment (the same proof scheme as Theorem 15.7, with the distance to the optimum as the monotone quantity) and tatonnement converges globally ([9], ch. 17). The algorithmic lesson also travels in the reverse direction: without substitutability type assumptions (the analogue of convexity), the adjustment can cycle or diverge, as in Scarf’s examples.

15.7.2 Structural estimation and machine learning

In structural econometrics, the estimator is the minimizer of a criterion:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} Q_n(\theta),$$

with Q_n the negative log likelihood (MLE), the quadratic form of moments $g_n(\theta)^\top W g_n(\theta)$ (GMM), or a distance between simulated and observed moments (calibration or SMM). Everything we learned applies directly: the Hessian of Q_n near $\hat{\theta}$ (the information matrix, in MLE)

determines κ and therefore the numerical difficulty; weakly identified parameters = flat valley = huge κ ; and the natural constraints (θ in a box, nonnegative variances, probabilities in the simplex) are handled with projected gradient. In machine learning and in models with massive data, $Q_n(\theta) = \frac{1}{n} \sum_{i=1}^n \ell_i(\theta)$ is an average of millions of terms, and evaluating the full gradient is prohibitive: *stochastic gradient descent* (SGD) replaces ∇Q_n by the gradient of a few randomly chosen summands (an unbiased and cheap estimator of the true gradient) and still converges (in expectation, with decreasing steps). SGD is today the algorithm that trains everything from giant logistic regressions to neural networks; see [8], ch. 5.

15.7.3 Suggestions for the computational project

Concrete ideas, calibrated for a final presentation of the course:

1. Implement gradient descent (fixed step, Armijo) and CG, and compare them on quadratics with $\kappa \in \{10, 100, 1000\}$: plot $\log(f(x_k) - f^*)$ against k and check the theoretical rates (lines with slopes $\log(1 - 1/\kappa)$ and much steeper ones for CG).
2. Reproduce the zigzag of Example 15.12 and the effect of preconditioning (change of variables $\tilde{x} = Q^{1/2}x$).
3. Estimate a logit or probit by maximum likelihood with your own gradient descent, comparing against the library optimizer.
4. Program the projection onto the simplex and use projected gradient for a minimum variance portfolio problem without short sales: $\min_{x \in \Delta_n} x^\top \Sigma x$.
5. Simulate a tatonnement in a CES exchange economy with two or three goods, and exhibit a convergent case (gross substitutes) and the Scarf cycle.

15.8 Summary and exercises [P4]

Chapter summary

General scheme: $x_{k+1} = x_k + t_k d_k$; d is a descent direction if $\langle \nabla f(x), d \rangle < 0$ (Proposition 15.2).

Smoothness: f is L -smooth if ∇f is L -Lipschitz; then the descent lemma holds, $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2$ (parabolic ceiling). **Strong convexity** (modulus m): parabolic floor. **Condition number:** $\kappa = L/m$, measures how elongated the level curves are; on quadratics, $\lambda_{\max}/\lambda_{\min}$.

Gradient descent $x_{k+1} = x_k - t_k \nabla f(x_k)$: with $t = 1/L$ it guarantees $f(x_{k+1}) \leq f(x_k) - \frac{1}{2L} \|\nabla f(x_k)\|^2$; without knowing L , backtracking–Armijo. Rates: convex L -smooth, $f(x_k) - f^* \leq L \|x_0 - x^*\|^2 / (2k)$ (sublinear, $O(1/k)$); strongly convex, linear with ratio $1 - 1/\kappa$ ($t = 1/L$).

Steepest descent (exact search): on quadratics $t^{\text{ex}} = \|g\|^2 / \langle g, Qg \rangle$; successive gradients orthogonal $\Rightarrow$ zigzag; worst case ratio $(\frac{\kappa-1}{\kappa+1})^2$ in f .

Conjugate gradient: directions with $d_i^\top Q d_j = 0$ ($i \neq j$) do not interfere; n of them

solve the quadratic in n exact steps (Theorem 15.16); CG builds them on the fly with $\alpha_k = \frac{r_k^\top r_k}{d_k^\top Q d_k}$, $\beta_{k+1} = \frac{r_{k+1}^\top r_{k+1}}{r_k^\top r_k}$, $d_{k+1} = r_{k+1} + \beta_{k+1} d_k$; dependence $\sqrt{\kappa}$ on conditioning; nonlinear version: Fletcher–Reeves.

Projected gradient: $x_{k+1} = \text{proj}_C(x_k - t \nabla f(x_k))$ for convex C that is simple to project onto; it works because proj_C is nonexpansive and because fixed point $\Leftrightarrow \langle \nabla f(x^*), y - x^* \rangle \geq 0 \forall y \in C \Leftrightarrow -\nabla f(x^*) \in N_C(x^*) \Leftrightarrow$ optimal (Theorem 15.21); same rates as without constraints.

For economists: tatonnement = gradient dynamics on excess demand; estimating (MLE, GMM, calibration) = minimizing; SGD for massive data; weak identification = bad conditioning.

Exercise 15.25. Let $f(x, y) = \frac{1}{2}(x^2 + 9y^2)$. (a) Compute ∇f , the Hessian, m , L and κ . (b) From $x_0 = (9, 1)$, run *by hand* three iterations of gradient descent with fixed step $t = 1/L$, reporting x_k and $f(x_k)$. (c) Identify the observed contraction ratio and compare it with the theoretical one of Theorem 15.8.

Solution. (a) $\nabla f(x, y) = (x, 9y)$ and $\nabla^2 f = \text{diag}(1, 9)$, constant. The eigenvalues are 1 and 9, so $m = 1$, $L = 9$ and $\kappa = 9$. The step is $t = 1/9$.

(b) The iteration is $x_{k+1} = x_k - \frac{1}{9}(x_k, 9y_k) = (\frac{8}{9}x_k, 0 \cdot y_k)$; let us make it explicit.

Iteration 1: $g_0 = (9, 9)$, so $x_1 = (9, 1) - \frac{1}{9}(9, 9) = (9 - 1, 1 - 1) = (8, 0)$.

Iteration 2: $g_1 = (8, 0)$, so $x_2 = (8, 0) - \frac{1}{9}(8, 0) = (\frac{64}{9}, 0) \approx (7.111, 0)$.

Iteration 3: $g_2 = (\frac{64}{9}, 0)$, so $x_3 = \frac{8}{9} \cdot \frac{64}{9}(1, 0) = (\frac{512}{81}, 0) \approx (6.321, 0)$.

Values: $f(x_0) = \frac{1}{2}(81 + 9) = 45$; $f(x_1) = \frac{1}{2} \cdot 64 = 32$; $f(x_2) = \frac{1}{2}(\frac{64}{9})^2 = \frac{2048}{81} \approx 25.28$; $f(x_3) = \frac{1}{2}(\frac{512}{81})^2 \approx 19.98$.

(c) The stiff coordinate y (curvature $L = 9$) dies in the first step because $1 - tL = 0$; from there on the error lies only in x and contracts exactly by $1 - tm = 8/9 \approx 0.889$ per iteration ($8 \rightarrow 64/9 \rightarrow 512/81$). This is the theoretical ratio $1 - m/L = 1 - 1/\kappa$ of Theorem 15.8: the worst eigenvalue rules. $\square$

Exercise 15.26. Prove that if t_k is chosen by *exact* line search along a descent direction d_k (and the minimum is attained at some $t_k \in (0, \infty)$), then $\langle \nabla f(x_{k+1}), d_k \rangle = 0$. Conclude that in steepest descent the successive gradients are orthogonal, and explain in one sentence why this produces the zigzag.

Solution. Let $\varphi(t) = f(x_k + t d_k)$, which is differentiable with $\varphi'(t) = \langle \nabla f(x_k + t d_k), d_k \rangle$ by the chain rule. By the definition of exact search, t_k minimizes φ over $[0, \infty)$, and since t_k is interior ($t_k > 0$), the first order condition of an interior minimum gives $\varphi'(t_k) = 0$, that is

$$\langle \nabla f(x_k + t_k d_k), d_k \rangle = \langle \nabla f(x_{k+1}), d_k \rangle = 0.$$

In steepest descent $d_k = -\nabla f(x_k)$, so $\langle \nabla f(x_{k+1}), \nabla f(x_k) \rangle = 0$: each new direction is perpendicular to the previous one. Since the method can only turn at right angles, in an elongated valley it advances by crossing the valley from side to side (large transversal component, small longitudinal component): that is the zigzag of Example 15.12. $\square$

Exercise 15.27. Consider $f(x) = \frac{L}{2}x^2$ on $\mathbb{R}$ (a single variable), which is L -smooth with exact constant L . (a) Write the gradient descent iteration with fixed step t as $x_{k+1} = cx_k$ and determine c . (b) For which values of t does the method converge to $x^* = 0$ from any x_0 ? For which does it diverge? (c) What happens exactly at $t = 2/L$?

Solution. (a) $f'(x) = Lx$, so $x_{k+1} = x_k - tLx_k = (1 - tL)x_k$: a geometric recursion with $c = 1 - tL$, so $x_k = (1 - tL)^k x_0$.

(b) $|x_k| = |1 - tL|^k |x_0| \rightarrow 0$ if and only if $|1 - tL| < 1$, that is $0 < t < 2/L$: it converges for every step in $(0, 2/L)$ (monotonically if $t \leq 1/L$, oscillating around 0 if $1/L < t < 2/L$). If $t > 2/L$ then $|1 - tL| > 1$ and $|x_k| \rightarrow \infty$ for every $x_0 \neq 0$: the method *diverges geometrically*, jumping to alternating sides of the minimum, farther each time.

(c) With $t = 2/L$, $c = -1$: the iteration bounces forever between x_0 and $-x_0$, neither converging nor diverging. This shows that the threshold $2/L$ of the theory is exact, not an artifact of the proof. $\square$

Exercise 15.28. Let $f(x) = \frac{1}{2}x^\top Qx$ with $Q = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. (a) Compute the eigenvalues of Q , the constants m and L , and the condition number κ . (b) Give the “safe” fixed step $1/L$ and the convergence ratio guaranteed by Theorem 15.8. (c) Give the optimal step $2/(m + L)$ and its ratio. (d) For which initial directions x_0 does the method converge in a single iteration with $t = 1/L$?

Solution. (a) The characteristic polynomial is $\det(Q - \lambda I) = (3 - \lambda)^2 - 1 = 0$, so $3 - \lambda = \pm 1$: $\lambda_1 = 2$ and $\lambda_2 = 4$, with eigenvectors $v_1 = \frac{1}{\sqrt{2}}(1, -1)^\top$ and $v_2 = \frac{1}{\sqrt{2}}(1, 1)^\top$. So $m = 2$, $L = 4$, $\kappa = 2$.

(b) $t = 1/L = 1/4$, with ratio $\rho = 1 - m/L = 1 - 1/\kappa = 1/2$: the error in norm is reduced at least by half per iteration.

(c) $t = 2/(m + L) = 2/6 = 1/3$, with ratio $\rho = (\kappa - 1)/(\kappa + 1) = 1/3$: somewhat better than $1/2$.

(d) The error evolves as $x_{k+1} = (I - \frac{1}{4}Q)x_k$, and $I - \frac{1}{4}Q$ has eigenvalues $1 - \frac{2}{4} = \frac{1}{2}$ (direction v_1) and $1 - \frac{4}{4} = 0$ (direction v_2). Therefore, if x_0 is proportional to $v_2 = (1, 1)^\top$ (the direction of the largest eigenvalue), a single step annihilates it: $x_1 = 0 = x^*$. In any other direction the component along v_1 survives and convergence is geometric with ratio $1/2$. $\square$

Exercise 15.29. Apply the conjugate gradient method to $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q = \text{diag}(1, 4)$, $b = (1, 2)^\top$, from $x_0 = (0, 0)^\top$. Report α_0 , x_1 , r_1 , β_1 , d_1 , α_1 and x_2 , and verify that x_2 solves $Qx = b$ and that $d_0^\top Qd_1 = 0$.

Solution. Initialization: $r_0 = b - Qx_0 = (1, 2)^\top = d_0$.

Iteration 1: $Qd_0 = (1, 8)^\top$; $r_0^\top r_0 = 1 + 4 = 5$; $d_0^\top Qd_0 = 1 + 16 = 17$. Then $\alpha_0 = 5/17$ and

$$x_1 = \frac{5}{17}(1, 2)^\top = \left(\frac{5}{17}, \frac{10}{17}\right)^\top, \quad r_1 = r_0 - \frac{5}{17}(1, 8)^\top = \left(\frac{12}{17}, -\frac{6}{17}\right)^\top.$$

Iteration 2: $r_1^\top r_1 = \frac{144+36}{289} = \frac{180}{289}$, so $\beta_1 = \frac{180/289}{5} = \frac{36}{289}$ and

$$d_1 = r_1 + \frac{36}{289}(1, 2)^\top = \left(\frac{204+36}{289}, \frac{-102+72}{289}\right)^\top = \frac{30}{289}(8, -1)^\top.$$

Conjugacy: $d_0^\top Q d_1 = (1, 8) \cdot \frac{30}{289}(8, -1)^\top = \frac{30}{289}(8 - 8) = 0$. ✓ Now $Q d_1 = \frac{30}{289}(8, -4)^\top$ and $d_1^\top Q d_1 = (\frac{30}{289})^2(64 + 4) = \frac{900 \cdot 68}{289^2}$, so

$$\alpha_1 = \frac{180/289}{900 \cdot 68/289^2} = \frac{180 \cdot 289}{900 \cdot 68} = \frac{17}{20}, \quad x_2 = x_1 + \frac{17}{20} \cdot \frac{30}{289}(8, -1)^\top = x_1 + \frac{3}{34}(8, -1)^\top.$$

Since $\frac{3}{34}(8, -1)^\top = (\frac{12}{17}, -\frac{3}{34})^\top$,

$$x_2 = (\frac{5}{17} + \frac{12}{17}, \frac{10}{17} - \frac{3}{34})^\top = (1, \frac{20-3}{34})^\top = (1, \frac{1}{2})^\top.$$

Check: $Q x_2 = (1 \cdot 1, 4 \cdot \frac{1}{2})^\top = (1, 2)^\top = b$. ✓ Termination in $n = 2$ steps, as Theorem 15.18 promises. $\square$

Exercise 15.30. Consider $\min_{(x,y) \in C} f(x, y) = (x - 3)^2 + (y + 1)^2$ with $C = [0, 2] \times [0, 2]$. (a) Run one iteration of projected gradient from $x_0 = (0, 0)$ with step $t = 1/2$. (b) Check that the point obtained is a fixed point of the iteration. (c) Certify its optimality using the normal cone condition of Theorem 15.21.

Solution. (a) $\nabla f(x, y) = (2(x - 3), 2(y + 1))$, so $\nabla f(0, 0) = (-6, 2)$ and the gradient step gives

$$y_0 = (0, 0) - \frac{1}{2}(-6, 2) = (3, -1),$$

which is precisely the unconstrained minimizer, outside C (with $t = 1/L = 1/2$ and f quadratic with Hessian $2I$, the gradient step always lands on the unconstrained minimizer). We project by clipping coordinates to $[0, 2]$: $3 \mapsto 2$ and $-1 \mapsto 0$, so $x_1 = (2, 0)$.

(b) $\nabla f(2, 0) = (-2, 2)$, so the tentative point is $(2, 0) - \frac{1}{2}(-2, 2) = (3, -1)$, whose projection is again $(2, 0)$: indeed $x_2 = x_1$, a fixed point.

(c) At $x^* = (2, 0)$ the upper bound on x ($x = 2$) and the lower bound on y ($y = 0$) are active; the normal cone of the box there is

$$N_C(2, 0) = \{(u, v) : u \geq 0, v \leq 0\}$$

(outward: to the right through the face $x = 2$, downward through the face $y = 0$). Since $-\nabla f(2, 0) = (2, -2)$ has $u = 2 \geq 0$ and $v = -2 \leq 0$, we get $-\nabla f(x^*) \in N_C(x^*)$ and, by Theorem 15.21, $x^* = (2, 0)$ minimizes f over C . (It is worth visualizing: the negative gradient points toward the unconstrained minimizer $(3, -1)$, which lies outside, crossing exactly through the corner $(2, 0)$.) $\square$

Chapter 16

Penalty methods and the augmented Lagrangian [P5]

Week 14 of the course. Bibliography: Boyd and Vandenberghe [2]; Nocedal and Wright [5], Ch. 17; Ruszczyński [6], Chs. 6–7.

What is this used for in Microeconomics?

Let us be honest: this chapter is *not* asked directly in Microeconomics. But the economic idea behind the augmented Lagrangian is too beautiful to skip: the algorithm coordinates decentralized decisions by *updating prices* (the Lagrange multiplier) until the “market constraint” clears, which is exactly the spirit of the Walrasian auctioneer in **general equilibrium** (MWG chs. 15–18): if there is excess demand, raise the price; if there is excess supply, lower it. The ADMM version we see at the end is, literally, a *computational second welfare theorem*: agents who solve private problems and a coordinator who only touches prices together reach the social optimum. Besides, the augmented Lagrangian is *the* computational tool used in practice to solve computable general equilibrium models and portfolio problems (Markowitz, which we develop in full detail), and it is a strong topic for the **final presentation of the course**: syllabi often call it the “augmented gradient method” and place it among numerical methods for constrained optimization.

In the previous chapters we learned to *characterize* solutions of constrained problems (KKT conditions, Lagrangian duality, saddle points) and to *compute* solutions of unconstrained problems or problems with very simple constraints (gradient method and projected gradient). The most important case in applied economics remains: how do you compute, with a computer, the solution of a problem with general equality and inequality constraints?

The unifying idea of this chapter is simple to state: *turn constraints into costs*. Instead of forbidding infeasibility, we *charge* for it: whoever violates a constraint pays a fine that grows with the violation. The constrained problem then becomes a sequence of *unconstrained* problems, which we already know how to solve with the methods of the previous chapter. We will see three ways of charging that fine, from the most naive to the most sophisticated:

- (i) the **quadratic penalty**, which fines the violation but needs infinite fines to be exact;

- (ii) the **logarithmic barrier**, which charges a growing toll for *approaching* the boundary from inside;
- (iii) the **augmented Lagrangian**, which combines the quadratic fine with a *price* (the Lagrange multiplier) that is updated iteratively, and achieves exactness with finite fines.

The main references are [5] (ch. 17), [6] (chs. 6–7), [7] and, for ADMM and its distributed applications, [2] and [8]. From this notebook we use the KKT conditions, the dual function $q(\lambda) = \min_x L(x, \lambda)$ and the notion of gradient ascent.

16.1 From constraints to costs: the quadratic penalty [P5]

16.1.1 Motivation: fining infeasibility

Consider the equality constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad h_j(x) = 0, \quad j = 1, \dots, m, \quad (16.1)$$

where $f, h_1, \dots, h_m : \mathbb{R}^n \rightarrow \mathbb{R}$ are differentiable functions. We write $h(x) = (h_1(x), \dots, h_m(x)) \in \mathbb{R}^m$ for the constraint vector, so feasibility means $h(x) = 0$.

The computational difficulty of (16.1) is that the feasible set $\{x : h(x) = 0\}$ is usually a “thin” surface in $\mathbb{R}^n$ (of dimension $n - m$): a gradient method taking free steps leaves it immediately. The first idea is to remove the constraint and replace it by a *cost*: if being at x with $h(x) \neq 0$ is expensive, the unconstrained minimizer will move close to the feasible surface. The most natural fine is proportional to the *square* of the violation (quadratic so that it is differentiable, and so that small fines near feasibility do not distort too much).

Definition 16.1 (Quadratic penalty function). For problem (16.1) and a penalty parameter $\rho > 0$, the *quadratic penalty function* is

$$Q(x; \rho) = f(x) + \frac{\rho}{2} \sum_{j=1}^m h_j(x)^2 = f(x) + \frac{\rho}{2} \|h(x)\|^2.$$

If the problem also has inequality constraints $g_i(x) \leq 0$, $i = 1, \dots, p$, we add the terms $\frac{\rho}{2} \sum_{i=1}^p (\max\{0, g_i(x)\})^2$: only the *violated* part of each inequality is fined (if $g_i(x) \leq 0$ the term is zero).

The *quadratic penalty method* consists of:

1. Choose an initial value $\rho_0 > 0$ (for example $\rho_0 = 1$), an initial point x^0 and a tolerance $\varepsilon > 0$.
2. At iteration k : minimize $Q(x; \rho_k)$ *without constraints* (with gradient, Newton, and so on), starting from the previous point x^{k-1} as a seed; call x^k the (approximate) minimizer obtained.
3. If $\|h(x^k)\| \leq \varepsilon$, stop and return x^k .
4. If not, increase the penalty, $\rho_{k+1} = \tau \rho_k$ with $\tau > 1$ (typically $\tau = 10$), and go back to step 2.

Intuition

Think of ρ as the size of the *fine for polluting*. A firm (the minimizer of Q) compares the private benefit of violating the rule (f goes down if it leaves the constraint) with the fine $\frac{\rho}{2} \|h\|^2$. With small fines it pollutes a lot; as the regulator raises the fine, the firm moves toward the required behavior, but, and this is the key point we now quantify, *as long as the fine is finite, it always pays to violate the rule a little*, because near $h = 0$ the marginal fine ρh is small compared with the marginal benefit of deviating. Exact feasibility is reached only in the limit $\rho \rightarrow \infty$.

16.1.2 A fully solved one dimensional example

Example 16.2 (The penalty is never exactly feasible). Consider the one variable problem

$$\min_{x \in \mathbb{R}} \frac{1}{2}x^2 \quad \text{s.t.} \quad h(x) = x - 1 = 0.$$

The solution is obvious: the only feasible point is $x^* = 1$, with value $f(x^*) = \frac{1}{2}$. Its Lagrange multiplier comes from the first order condition of the Lagrangian $L(x, \lambda) = \frac{1}{2}x^2 + \lambda(x - 1)$:

$$\frac{\partial L}{\partial x} = x + \lambda = 0 \quad \text{at } x^* = 1 \quad \implies \quad \lambda^* = -1.$$

Now apply the quadratic penalty. The penalized function is

$$Q(x; \rho) = \frac{1}{2}x^2 + \frac{\rho}{2}(x - 1)^2.$$

It is a sum of two convex parabolas, so it is strictly convex and its minimizer is the unique critical point. Differentiate and set to zero:

$$Q'(x; \rho) = x + \rho(x - 1) = (1 + \rho)x - \rho = 0 \quad \implies \quad x(\rho) = \frac{\rho}{1 + \rho}.$$

Let us analyze this formula carefully; it contains the whole theory of the method.

(a) **Convergence to the constrained optimum.** When $\rho \rightarrow \infty$, $x(\rho) = \rho/(1 + \rho) \rightarrow 1 = x^*$. With $\rho = 1$ we get $x = 1/2$; with $\rho = 10$, $x = 10/11 \approx 0.909$; with $\rho = 100$, $x = 100/101 \approx 0.990$.

(b) **But it is never feasible with finite ρ .** The error (the constraint violation) is exactly

$$h(x(\rho)) = x(\rho) - 1 = \frac{\rho}{1 + \rho} - \frac{1 + \rho}{1 + \rho} = -\frac{1}{1 + \rho} \neq 0 \quad \text{for every } \rho < \infty.$$

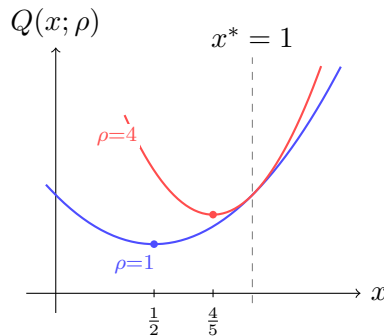
The violation decays like $1/\rho$: to halve the error you must *double* the fine; to gain one decimal digit of feasibility you must multiply it by ten.

(c) **The violation “remembers” the multiplier.** Multiply the violation by ρ :

$$\rho h(x(\rho)) = \rho \cdot \left(-\frac{1}{1 + \rho}\right) = -\frac{\rho}{1 + \rho} \xrightarrow{\rho \rightarrow \infty} -1 = \lambda^*.$$

The product $\rho h(x(\rho))$ converges to the Lagrange multiplier. This is no coincidence: the first order condition of Q is $f'(x(\rho)) + \rho h(x(\rho)) h'(x(\rho)) = 0$, which is exactly the KKT condition $f'(x) + \lambda h'(x) = 0$ with the “implicit multiplier” $\lambda = \rho h(x(\rho))$. The quadratic penalty estimates the multiplier *without meaning to*. This observation is the seed of the augmented Lagrangian.

The next figure shows the parabolas $Q(\cdot; \rho)$ for increasing ρ : the minimum (thick dot) approaches the feasible line $x = 1$ but never reaches it.



The phenomenon in the example is completely general; we state it as a theorem.

Theorem 16.3 (Convergence of the quadratic penalty). *Suppose f and h are continuous, that each $Q(\cdot; \rho_k)$ attains its global minimum at x^k , and that $\rho_k \rightarrow \infty$. Then every accumulation point of the sequence $\{x^k\}$ is a global minimizer of the constrained problem (16.1). If in addition f, h are differentiable, the x^k are approximate minimizers with increasing accuracy and the sequence converges to a regular point x^* (the Jacobian $\nabla h(x^*)$ has rank m), then*

$$\lambda^k = \rho_k h(x^k) \longrightarrow \lambda^*,$$

where λ^* is the vector of Lagrange multipliers of x^* .

Proof sketch. For the first part: let $\bar{x}$ be any feasible point. Since x^k minimizes $Q(\cdot; \rho_k)$,

$$f(x^k) + \frac{\rho_k}{2} \|h(x^k)\|^2 \leq f(\bar{x}) + \frac{\rho_k}{2} \|h(\bar{x})\|^2 = f(\bar{x}),$$

because $h(\bar{x}) = 0$. Hence $\frac{\rho_k}{2} \|h(x^k)\|^2 \leq f(\bar{x}) - f(x^k)$ is bounded, and since $\rho_k \rightarrow \infty$ we get $\|h(x^k)\| \rightarrow 0$: the limits are feasible. The same inequality gives $f(x^k) \leq f(\bar{x})$, so the limit is optimal. For the second part, the first order condition $\nabla f(x^k) + \nabla h(x^k)^\top [\rho_k h(x^k)] = 0$ identifies $\rho_k h(x^k)$ as an estimator of the multiplier; regularity of x^* allows solving for it and passing to the limit. The details are in [5] (Theorems 17.1 and 17.2) and [6] (ch. 6). $\square$

16.1.3 The numerical price: ill conditioning

If the penalty converges, why not always use it with a huge ρ and be done? Because the subproblem $\min_x Q(x; \rho)$ becomes *numerically intractable* as ρ grows. Let us see why. The

Hessian of Q (with h having components h_j) is

$$\nabla^2 Q(x; \rho) = \nabla^2 f(x) + \rho \nabla h(x)^\top \nabla h(x) + \rho \sum_{j=1}^m h_j(x) \nabla^2 h_j(x).$$

Near the solution, $h_j(x) \approx 0$ and the last term is negligible; the term $\rho \nabla h(x)^\top \nabla h(x)$ dominates, a symmetric positive semidefinite matrix of rank m (one per constraint). Consequently:

- in the m directions *normal* to the feasible surface (the rows of ∇h), the curvature of Q is of order ρ : eigenvalues $\approx \rho \cdot (\text{constant})$, huge;
- in the $n-m$ directions *tangent* to the feasible surface, the curvature is that of f : eigenvalues of order 1.

The *condition number* (ratio of the largest to the smallest eigenvalue) of the Hessian then grows like ρ : the “valley” of Q is a very long canyon with almost vertical walls. We know from the previous chapter that, with κ the condition number, gradient descent with step $1/L$ contracts the error at rate $1 - 1/\kappa$ and steepest descent at rate $(\frac{\kappa-1}{\kappa+1})^2$ in the objective value (Theorem 15.13): with $\rho = 10^6$, each iteration barely advances, and moreover the rounding errors in floating point arithmetic are amplified when solving Newton’s linear systems.

Careful (common mistake)

The temptation to “set $\rho = 10^9$ and minimize once” is a classic mistake. With a huge ρ : (i) the Hessian of Q has condition number of order ρ and first order methods crawl; (ii) the term $\frac{\rho}{2} \|h\|^2$ numerically crushes f (in double precision, adding 1 and 10^{16} loses the 1); (iii) even in exact arithmetic, the minimizer is still infeasible ($h \neq 0$). This is why the method increases ρ *gradually*, using each solution as a seed for the next; and why, ultimately, the augmented Lagrangian was invented, which does not need $\rho \rightarrow \infty$.

16.2 Barrier methods and the central path [P4]

16.2.1 The idea: a toll for approaching the boundary

The penalty fines infeasibility *from outside*. Barrier methods do the opposite: they always work *strictly inside* the feasible set and charge a toll that tends to infinity as the point approaches the boundary, so the iterate never crosses it. They apply to problems with inequalities,

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0, \quad i = 1, \dots, p,$$

whose feasible interior $\{x : g_i(x) < 0 \ \forall i\}$ is assumed nonempty (Slater’s condition, which you already know from the duality chapter).

Definition 16.4 (Logarithmic barrier and central path). The *logarithmic barrier function* with parameter $\mu > 0$ is

$$B(x; \mu) = f(x) - \mu \sum_{i=1}^p \log(-g_i(x)),$$

defined only where $g_i(x) < 0$ for all i . Its minimizer $x(\mu)$ (when it exists and is unique, for example if the problem is convex with Slater) describes, as $\mu > 0$ varies, a curve called the *central path*. The barrier method follows that curve: you minimize $B(\cdot; \mu_k)$ starting from the previous solution and reduce $\mu_{k+1} = \mu_k/\tau$, $\tau > 1$, until μ is small.

Note the logic of the sign: if $g_i(x) < 0$ (positive slack), then $-g_i(x) > 0$ and the logarithm is defined; when x approaches the boundary $g_i = 0$, $-\log(-g_i(x)) \rightarrow +\infty$: the toll explodes. With a large μ , the toll dominates and $x(\mu)$ stays near the “center” of the feasible set; with $\mu \rightarrow 0^+$, the toll switches off and $x(\mu)$ converges to the constrained optimum, sliding inside the set. This is the basis of the celebrated *interior point methods* that solve linear and convex programs in polynomial time [2].

Example 16.5 (Barrier in one dimension, computed). Let us solve

$$\min_x x \quad \text{s.t.} \quad -x \leq 0 \quad (\text{that is, } x \geq 0).$$

The solution is $x^* = 0$. The KKT multiplier comes from $1 - \lambda = 0$ (gradient of $x + \lambda(-x)$), that is $\lambda^* = 1$, with complementarity $\lambda^*g(x^*) = 0$ satisfied because $g(x^*) = 0$. The barrier function is

$$B(x; \mu) = x - \mu \log x, \quad x > 0,$$

which is strictly convex on $(0, \infty)$ (its second derivative is $\mu/x^2 > 0$). Its critical point:

$$B'(x; \mu) = 1 - \frac{\mu}{x} = 0 \quad \implies \quad x(\mu) = \mu.$$

The central path is, literally, $x(\mu) = \mu$: with $\mu = 1$, $x = 1$; with $\mu = 0.1$, $x = 0.1$; and $x(\mu) \rightarrow 0 = x^*$ as $\mu \rightarrow 0^+$, always from the interior $x > 0$. Moreover, the *implicit barrier multiplier*

$$\lambda(\mu) = \frac{\mu}{-g(x(\mu))} = \frac{\mu}{x(\mu)} = 1 = \lambda^*$$

hits the exact multiplier (in general it only converges to λ^* ; here it is exact for every μ because the problem is linear). And the gap $f(x(\mu)) - f(x^*) = \mu - 0 = \mu$ is exactly the value of the parameter: a general rule, as the next theorem states.

Theorem 16.6 (The central path gives approximate primal–dual pairs). *Consider the convex problem (f and all g_i convex and differentiable) with Slater’s condition, and let $x(\mu)$ be a point of the central path. Define*

$$\lambda_i(\mu) = \frac{\mu}{-g_i(x(\mu))} > 0, \quad i = 1, \dots, p.$$

Then the pair $(x(\mu), \lambda(\mu))$ satisfies: (i) $x(\mu)$ is strictly feasible; (ii) $\lambda(\mu) \geq 0$ and $x(\mu)$ minimizes the ordinary Lagrangian $L(x, \lambda(\mu)) = f(x) + \sum_i \lambda_i(\mu)g_i(x)$, so $\lambda(\mu)$ is feasible for the dual problem with dual value $q(\lambda(\mu)) = f(x(\mu)) - p\mu$; and therefore (iii) the duality gap is

$$f(x(\mu)) - q(\lambda(\mu)) = p\mu,$$

so $f(x(\mu)) - f(x^) \leq p\mu$: it is enough to bring μ down to ε/p to guarantee accuracy ε .*

Proof sketch. The first order condition of the barrier is

$$\nabla f(x(\mu)) + \sum_i \frac{\mu}{-g_i(x(\mu))} \nabla g_i(x(\mu)) = 0,$$

which, with the definition of $\lambda_i(\mu)$, reads $\nabla_x L(x(\mu), \lambda(\mu)) = 0$: since $L(\cdot, \lambda(\mu))$ is convex, $x(\mu)$ minimizes it globally and then

$$q(\lambda(\mu)) = L(x(\mu), \lambda(\mu)) = f(x(\mu)) + \sum_i \lambda_i(\mu) g_i(x(\mu)) = f(x(\mu)) - p\mu,$$

because each summand equals $\lambda_i(\mu) g_i(x(\mu)) = -\mu$. The final bound is weak duality. See [2] (section 11.2). $\square$

Intuition

The central path is a *relaxed complementarity*: instead of requiring $\lambda_i(-g_i) = 0$ (exact KKT), it requires $\lambda_i(-g_i) = \mu$ for every i . It is like a market where the product “shadow price $\times$ slack” is not zero but a uniform margin μ that the regulator squeezes toward zero: each point of the curve is a quasi equilibrium, and the limit $\mu \rightarrow 0$ is the exact KKT equilibrium. Note the duality of temperaments: the penalty *pushes from outside* toward feasibility; the barrier *holds you inside* and releases gradually. Both share the same defect: exactness only in the limit ($\rho \rightarrow \infty$ or $\mu \rightarrow 0$), with Hessians that get worse and worse conditioned. The augmented Lagrangian breaks that fate.

16.3 The augmented Lagrangian and the method of multipliers [P5]

16.3.1 Two defects, one cure

Let us recap the two findings of section 16.1. First, the pure penalty is *always infeasible* with finite ρ : in Example 16.2 the error was $-1/(1 + \rho)$. Second, the quantity $\rho h(x(\rho))$ estimates the multiplier λ^* . The clever idea of Hestenes and Powell (1969) was to combine both: if the penalty gives us an estimate of the price λ , why not *include that price explicitly* in the objective function, as in the Lagrangian, and keep the quadratic fine only as reinforcement? The price does the fine tuning of centering the minimum in the exact place; the fine does the rough work of providing curvature.

Definition 16.7 (Augmented Lagrangian). For problem (16.1), the *augmented Lagrangian* with parameter $\rho > 0$ is

$$L_\rho(x, \lambda) = f(x) + \sum_{j=1}^m \lambda_j h_j(x) + \frac{\rho}{2} \|h(x)\|^2 = f(x) + \langle \lambda, h(x) \rangle + \frac{\rho}{2} \|h(x)\|^2.$$

It is the ordinary Lagrangian $L(x, \lambda)$ plus the quadratic fine; or, if you prefer, the penalized function $Q(x; \rho)$ plus the price term $\langle \lambda, h(x) \rangle$. With $\rho = 0$ we recover the Lagrangian; with $\lambda = 0$, the pure penalty.

The motivation is twofold and it is worth keeping both faces in mind:

- (i) **It fixes infeasibility.** The linear term $\langle \lambda, h(x) \rangle$ “tilts” the penalized function. Recall Example 16.2: the minimum of Q fell short ($x(\rho) < 1$) because lowering f pushed toward 0. If we add $\lambda(x - 1)$ with λ *negative* (the correct sign, $\lambda^* = -1$), we reward moving toward $x > 1$ and compensate exactly for that bias. We will see that with $\lambda = \lambda^*$ the compensation is perfect: the minimum of $L_\rho(\cdot, \lambda^*)$ is *exactly* x^* , with finite ρ .
- (ii) **It convexifies locally.** The Hessian in x is

$$\nabla_{xx}^2 L_\rho = \nabla_{xx}^2 L(x, \lambda) + \rho \nabla h(x)^\top \nabla h(x) + \rho \sum_j h_j(x) \nabla^2 h_j(x).$$

In constrained problems, second order theory only guarantees that $\nabla_{xx}^2 L(x^*, \lambda^*)$ is positive definite *on the tangent subspace* $\{d : \nabla h(x^*)d = 0\}$; in the normal directions there may be negative curvature and the ordinary Lagrangian may have no minimum in x (only a saddle point). The term $\rho \nabla h^\top \nabla h$ adds curvature $\rho \|\nabla h(x^*)d\|^2 > 0$ precisely in those normal directions, without touching the tangent ones: for ρ large enough (but *finite*) the full Hessian becomes positive definite and the saddle point turns into a genuine local minimum, which can be minimized with the methods of the previous chapter.

16.3.2 The Hestenes–Powell algorithm

The *method of multipliers*, or augmented Lagrangian method (also called the “augmented gradient method”) is:

1. Choose $\lambda^0 \in \mathbb{R}^m$ (for example $\lambda^0 = 0$), $\rho > 0$, an initial point x^{-1} and a tolerance $\varepsilon > 0$.
2. At iteration $k = 0, 1, 2, \dots$: solve the *unconstrained* subproblem

$$x^k \in \arg \min_x L_\rho(x, \lambda^k),$$

using x^{k-1} as a seed.

3. Update the multiplier with the Hestenes–Powell rule:

$$\lambda^{k+1} = \lambda^k + \rho h(x^k).$$

4. If $\|h(x^k)\| \leq \varepsilon$ and $\|\lambda^{k+1} - \lambda^k\| \leq \rho \varepsilon$, stop; if not, go back to step 2. (Optional: if feasibility does not improve enough between iterations, increase ρ moderately.)

Where does the rule in step 3 come from? It is not a trick: it is *gradient ascent on the dual function*, and the derivation connects everything we know about duality. Let us do it slowly.

The update of λ is dual ascent. Define the *augmented dual function*

$$q_\rho(\lambda) = \min_x L_\rho(x, \lambda),$$

which is the ordinary dual function of the *equivalent* problem

$$\min_x f(x) + \frac{\rho}{2} \|h(x)\|^2 \quad \text{s.t.} \quad h(x) = 0,$$

because adding $\frac{\rho}{2} \|h\|^2$ changes neither the feasible set nor the value of f on it (there it is zero): same problem, same optimum, same multipliers. From the duality chapter we know two things about any dual function: it is *concave* in λ (a pointwise minimum of functions affine in λ), and its gradient, when the inner minimizer $x(\lambda)$ is unique, is obtained by the envelope theorem ([11]): you differentiate with respect to λ *without* differentiating $x(\lambda)$, because the term $\nabla_x L_\rho \cdot \partial x / \partial \lambda$ vanishes at the minimizer. So,

$$\nabla q_\rho(\lambda) = \frac{\partial}{\partial \lambda} \left[f(x) + \langle \lambda, h(x) \rangle + \frac{\rho}{2} \|h(x)\|^2 \right]_{x=x(\lambda)} = h(x(\lambda)).$$

The gradient of the dual is simply *the constraint violation at the current minimizer*. Maximizing the dual by gradient ascent with step size α gives $\lambda^{k+1} = \lambda^k + \alpha h(x^k)$; the Hestenes–Powell rule is exactly this with $\alpha = \rho$.

Why the step $\alpha = \rho$ and not another? Two reasons, one algebraic and one about convergence.

- (a) *The new multiplier makes the current point “KKT compatible”.* At the minimizer x^k of the subproblem, the first order condition is

$$0 = \nabla_x L_\rho(x^k, \lambda^k) = \nabla f(x^k) + \nabla h(x^k)^\top \underbrace{[\lambda^k + \rho h(x^k)]}_{= \lambda^{k+1}}.$$

Read it carefully: grouping the price term and the fine term, the point x^k satisfies *exactly* the KKT condition $\nabla f + \nabla h^\top \lambda = 0$ with the multiplier λ^{k+1} . That is, the update rule is the only one that turns each iterate into a stationary point of the ordinary Lagrangian with the new price; all that is missing is for feasibility $h(x^k) = 0$ to be achieved in the limit. (Compare with section 16.1: there the estimator was ρh ; here it is $\lambda^k + \rho h$, which accumulates the information of all iterations instead of starting from scratch.)

- (b) *It is the “safe” ascent step.* One can prove that q_ρ is the *Moreau regularization* of the ordinary dual q : smoothing it with the fine makes ∇q_ρ Lipschitz with constant $1/\rho$. From the chapter on gradient methods we know that for a function with $1/\rho$ -Lipschitz gradient the natural step is the reciprocal of the constant, that is ρ : it guarantees monotone ascent of the dual without overshooting. The larger the fine, the smoother the dual and the more aggressive the price step can be [6, 8].

Intuition

Economic interpretation: λ is a vector of *prices* and $h(x)$ a vector of *excess demands* (violations of the resource constraints). The subproblem in step 2 is the optimal response of the “economy” to the current prices; step 3 is a *Walrasian auctioneer*: where there is

excess demand ($h_j(x^k) > 0$), the price λ_j goes up; where there is excess supply, it goes down; and the sensitivity of the adjustment is ρ . The tatonnement process stops when the market clears ($h = 0$): at that point prices no longer move and we have a KKT pair, that is, a saddle point of the Lagrangian, the “equilibrium”. The quadratic fine is the modern ingredient: it makes the response of the economy to prices unique and stable (it convexifies), something the classical tatonnement did not guarantee.

16.3.3 Exactness with a finite penalty: the central theorem

The superiority of the augmented Lagrangian over the pure penalty is summed up in one statement: *with the right price you do not need an infinite fine.*

Theorem 16.8 (Exactness of the augmented Lagrangian, quadratic-linear case). *Consider the quadratic problem with linear constraints*

$$\min_x \frac{1}{2}x^\top Qx + c^\top x \quad s.t. \quad Ax = b,$$

with $Q \in \mathbb{R}^{n \times n}$ symmetric, $A \in \mathbb{R}^{m \times n}$ of rank m , and assume the second order sufficient condition: $d^\top Qd > 0$ for all $d \neq 0$ with $Ad = 0$. Let (x^*, λ^*) be the KKT pair, that is,

$$Qx^* + c + A^\top \lambda^* = 0, \quad Ax^* = b.$$

Then there exists $\bar{\rho} \geq 0$ such that for every $\rho > \bar{\rho}$, the function $x \mapsto L_\rho(x, \lambda^*)$ is strictly convex and its unique global minimizer is exactly x^* . No infinite fine, no residual error: with the true multiplier, the constrained problem is solved in a single unconstrained shot.

Proof. Here $h(x) = Ax - b$, so

$$L_\rho(x, \lambda^*) = \frac{1}{2}x^\top Qx + c^\top x + (\lambda^*)^\top (Ax - b) + \frac{\rho}{2} \|Ax - b\|^2,$$

a quadratic function in x . We check the two conditions for a global minimum of a quadratic: zero gradient and positive definite Hessian.

Step 1: x^* is a stationary point for every $\rho \geq 0$. We compute the gradient (using $\nabla_x \frac{1}{2} \|Ax - b\|^2 = A^\top (Ax - b)$) and evaluate it at x^* :

$$\nabla_x L_\rho(x^*, \lambda^*) = \underbrace{Qx^* + c + A^\top \lambda^*}_{=0 \text{ (KKT)}} + \rho A^\top \underbrace{(Ax^* - b)}_{=0 \text{ (feasibility)}} = 0.$$

Note the division of labor: KKT stationarity kills the Lagrangian block and feasibility kills the fine block. This holds for *any* ρ , even $\rho = 0$.

Step 2: the Hessian $H_\rho = Q + \rho A^\top A$ is positive definite for large ρ . Suppose, for contradiction, that no such $\bar{\rho}$ exists: then there is a sequence $\rho_k \rightarrow \infty$ and unit vectors d^k ($\|d^k\| = 1$) with

$$(d^k)^\top Qd^k + \rho_k \|Ad^k\|^2 \leq 0. \quad (*)$$

Since the unit sphere of $\mathbb{R}^n$ is compact, a subsequence satisfies $d^k \rightarrow d$ with $\|d\| = 1$. From (*), $\|Ad^k\|^2 \leq -(d^k)^\top Qd^k/\rho_k \rightarrow 0$ (the numerator is bounded by continuity), so $Ad = 0$: the limit direction is tangent. But also from (*), $(d^k)^\top Qd^k \leq 0$ for all k , and passing to the limit $d^\top Qd \leq 0$, contradicting the second order hypothesis $d^\top Qd > 0$ in nonzero tangent directions. Therefore there exists a finite $\bar{\rho}$ with $H_\rho \succ 0$ for $\rho > \bar{\rho}$.

Conclusion. For $\rho > \bar{\rho}$, $L_\rho(\cdot, \lambda^*)$ is a strictly convex quadratic with stationary point x^* ; that point is its unique global minimizer. $\square$

Remark 16.9. If Q is positive semidefinite on all of $\mathbb{R}^n$ (convex problem), any $\rho > 0$ works, that is $\bar{\rho} = 0$: the quadratic $H_\rho = Q + \rho A^\top A$ is positive definite as soon as Q is positive definite on $\ker A$ and $\rho > 0$. The theorem has a general nonlinear version: if (x^*, λ^*) satisfies the second order sufficient conditions of problem (16.1), then for ρ larger than a finite threshold, x^* is a strict local minimizer of $L_\rho(\cdot, \lambda^*)$, and the method of multipliers converges locally with a linear rate whose ratio is proportional to $1/\rho$ if λ^0 is close to λ^* ; see [5] (Theorem 17.5 and section 17.3) and [6] (ch. 6). In practice we do not know λ^* , but the algorithm *learns* it: each update brings λ^k closer to the true price, and the exactness of the theorem explains why the method finishes with a moderate ρ , without inheriting the ill conditioning of the pure penalty.

16.3.4 Full numerical example: three iterations by hand

Example 16.10 (Method of multipliers step by step). Let us solve with the method of multipliers the problem

$$\min_{x \in \mathbb{R}^2} x_1^2 + x_2^2 \quad \text{s.t.} \quad h(x) = x_1 + x_2 - 1 = 0.$$

Exact solution (for comparison). The Lagrangian is $L = x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1)$; the KKT conditions:

$$2x_1 + \lambda = 0, \quad 2x_2 + \lambda = 0, \quad x_1 + x_2 = 1.$$

The first two give $x_1 = x_2 = -\lambda/2$; substituting into the third, $-\lambda = 1$, that is $\lambda^* = -1$ and $x^* = (\frac{1}{2}, \frac{1}{2})$, with optimal value $\frac{1}{2}$. (Geometrically: the point of the line $x_1 + x_2 = 1$ closest to the origin.)

The subproblem in closed form. The augmented Lagrangian is

$$L_\rho(x, \lambda) = x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1) + \frac{\rho}{2}(x_1 + x_2 - 1)^2.$$

First order condition in x (differentiating with respect to each coordinate):

$$2x_1 + \lambda + \rho(x_1 + x_2 - 1) = 0, \quad 2x_2 + \lambda + \rho(x_1 + x_2 - 1) = 0.$$

Subtracting the two equations: $2x_1 - 2x_2 = 0$, so $x_1 = x_2 = t$. Substituting: $2t + \lambda + \rho(2t - 1) = 0$, hence

$$t(\lambda) = \frac{\rho - \lambda}{2(1 + \rho)}, \quad h(x(\lambda)) = 2t(\lambda) - 1 = \frac{\rho - \lambda}{1 + \rho} - 1 = \frac{-1 - \lambda}{1 + \rho}.$$

(The Hessian is $2I + \rho \mathbf{1}\mathbf{1}^\top \succ 0$, so this is a global minimum, as Remark 16.9 guarantees.) Take $\rho = 1$ and $\lambda^0 = 0$, and run three exact iterations.

Iteration $k = 0$ ($\lambda^0 = 0$):

$$t_0 = \frac{1-0}{2 \cdot 2} = \frac{1}{4}, \quad x^0 = \left(\frac{1}{4}, \frac{1}{4}\right), \quad h(x^0) = \frac{1}{2} - 1 = -\frac{1}{2}.$$

The point falls short (the fine alone is not enough, as in the pure penalty). We update the price:

$$\lambda^1 = \lambda^0 + \rho h(x^0) = 0 + 1 \cdot \left(-\frac{1}{2}\right) = -\frac{1}{2}.$$

Iteration $k = 1$ ($\lambda^1 = -\frac{1}{2}$):

$$t_1 = \frac{1 - (-\frac{1}{2})}{4} = \frac{3/2}{4} = \frac{3}{8}, \quad x^1 = \left(\frac{3}{8}, \frac{3}{8}\right), \quad h(x^1) = \frac{3}{4} - 1 = -\frac{1}{4},$$

$$\lambda^2 = -\frac{1}{2} + 1 \cdot \left(-\frac{1}{4}\right) = -\frac{3}{4}.$$

Iteration $k = 2$ ($\lambda^2 = -\frac{3}{4}$):

$$t_2 = \frac{1 - (-\frac{3}{4})}{4} = \frac{7/4}{4} = \frac{7}{16}, \quad x^2 = \left(\frac{7}{16}, \frac{7}{16}\right), \quad h(x^2) = \frac{7}{8} - 1 = -\frac{1}{8},$$

$$\lambda^3 = -\frac{3}{4} + 1 \cdot \left(-\frac{1}{8}\right) = -\frac{7}{8}.$$

Let us summarize the numerical evidence:

k	x^k	$h(x^k)$	λ^{k+1}
0	(0.25, 0.25)	-0.5	-0.5
1	(0.375, 0.375)	-0.25	-0.75
2	(0.4375, 0.4375)	-0.125	-0.875

Everything converges to the exact solution: $x^k \rightarrow (0.5, 0.5)$, $\lambda^k \rightarrow -1$, and *each error is divided exactly by 2 at each iteration*. This is no accident: from the closed formulas,

$$\lambda^{k+1} = \lambda^k + \rho \cdot \frac{-1 - \lambda^k}{1 + \rho} = \frac{\lambda^k - \rho}{1 + \rho} \quad \implies \quad \lambda^{k+1} - \lambda^* = \lambda^{k+1} + 1 = \frac{\lambda^k + 1}{1 + \rho},$$

that is, *linear* convergence with ratio $\frac{1}{1+\rho}$ (here $\frac{1}{2}$). Raise the fine to $\rho = 9$ and the ratio drops to $\frac{1}{10}$: one decimal digit per iteration. The parameter ρ no longer needs to go to infinity; it only tunes the *speed* of price learning. Compare with the pure penalty of Example 16.2, where the final accuracy was tied to the value of ρ .

Exam strategy

To run the method of multipliers by hand (in an exam or in the final presentation), always follow this script: (1) first solve the exact problem by KKT to know (x^*, λ^*) , which gives you a control; (2) write L_ρ and obtain the minimizer of the subproblem *as a function of λ* (in quadratic-linear problems this is a linear system; exploit symmetries as we did when subtracting the equations); (3) iterate the recursion $\lambda^{k+1} = \lambda^k + \rho h(x^k)$ in a table with columns $k, x^k, h(x^k), \lambda^{k+1}$; (4) close by identifying the convergence ratio $\frac{1}{1+\rho}$ (or its

analogue) by subtracting λ^* : showing the error recursion is more impressive than three rows of numbers. Typical mistakes to avoid: forgetting the factor ρ in the update, getting the sign of h wrong (always define $h = \text{“left hand side minus right hand side”}$), and updating λ with the x of the previous iteration instead of the newly computed one.

16.4 Inequalities and ADMM: decentralization through prices [P4]

16.4.1 Augmented Lagrangian with inequalities

What if the problem has inequalities $g_i(x) \leq 0$, $i = 1, \dots, p$? The standard recipe turns them into equalities with slack variables ($g_i(x) + s_i = 0$, $s_i \geq 0$), applies the equality augmented Lagrangian and minimizes over the slacks *analytically* (you will do it yourself in Exercise 16.16). The result is the following formula.

Proposition 16.11 (Augmented Lagrangian for inequalities). *For the problem $\min f(x)$ s.t. $g_i(x) \leq 0$, $i = 1, \dots, p$, the augmented Lagrangian is*

$$L_\rho(x, \lambda) = f(x) + \frac{1}{2\rho} \sum_{i=1}^p \left[(\max\{0, \lambda_i + \rho g_i(x)\})^2 - \lambda_i^2 \right],$$

and the multiplier update of the method is

$$\lambda_i^{k+1} = \max\{0, \lambda_i^k + \rho g_i(x^k)\}, \quad i = 1, \dots, p.$$

The formula looks complicated but its logic is transparent. Fix a constraint i and distinguish two regimes:

- **Active regime** ($\lambda_i + \rho g_i(x) > 0$: the constraint binds or the price is high). Term i equals $\lambda_i g_i(x) + \frac{\rho}{2} g_i(x)^2$ (expand the square and check that the λ_i^2 cancel): exactly the treatment of an *equality*, price plus fine.
- **Inactive regime** ($\lambda_i + \rho g_i(x) \leq 0$: ample slack and low price). The maximum is 0 and the term reduces to the constant $-\lambda_i^2/(2\rho)$: the constraint *disappears* from the optimization in x , as KKT complementarity requires ($\lambda_i^* = 0$ if $g_i(x^*) < 0$).

The update with $\max\{0, \cdot\}$ is the same auctioneer as before, but one that respects the floor $\lambda_i \geq 0$: shadow prices of inequalities cannot be negative, and if the Walrasian adjustment asks to go below zero, the price stays at zero (free good, inactive constraint). Note the consistency with the projected gradient of the previous chapter: the update is a *projected ascent step* on the orthant $\lambda \geq 0$, which is where the dual variable lives.

16.4.2 ADMM: the alternating direction method of multipliers

Many problems in economics and statistics have *separable* structure: the objective is a sum of pieces belonging to different agents, coupled only by a common linear constraint. The *alternating*

direction method of multipliers (ADMM) exploits exactly that structure [2]. The model problem is

$$\min_{x,z} f(x) + g(z) \quad \text{s.t.} \quad Ax + Bz = c,$$

with f, g convex. The augmented Lagrangian is

$$L_\rho(x, z, \lambda) = f(x) + g(z) + \langle \lambda, Ax + Bz - c \rangle + \frac{\rho}{2} \|Ax + Bz - c\|^2.$$

The pure method of multipliers would require minimizing in (x, z) *jointly*; the quadratic term mixes x with z and ruins separability. ADMM gives up joint minimization and alternates, Gauss–Seidel style:

1. Initialize z^0, λ^0 and $\rho > 0$.
2. Step in x : $x^{k+1} \in \arg \min_x L_\rho(x, z^k, \lambda^k)$ (with z and λ frozen).
3. Step in z : $z^{k+1} \in \arg \min_z L_\rho(x^{k+1}, z, \lambda^k)$ (already using the new x).
4. Dual step: $\lambda^{k+1} = \lambda^k + \rho(Ax^{k+1} + Bz^{k+1} - c)$.
5. Repeat 2–4 until the primal residual $\|Ax^{k+1} + Bz^{k+1} - c\|$ and the dual residual $\rho\|B(z^{k+1} - z^k)\|$ are small.

Under convexity of f and g and existence of a saddle point, ADMM converges for *any* $\rho > 0$ (objective value to the optimum, primal residual to zero, λ^k to the optimal multiplier); the proof, via monotonicity of the saddle point operator, is in [2] (section 3.2) and in the literature reviewed by [8] (ch. 10).

16.4.3 Economic reading: decentralized planning through prices

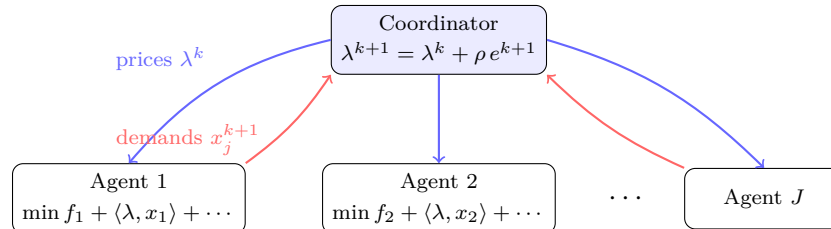
Here is the conceptual gem of the chapter. Consider an economy with J agents that share a resource: agent j chooses $x_j \in \mathbb{R}^n$ (their consumption or production plan) with private cost $f_j(x_j)$, and the market balance requires

$$\min_{x_1, \dots, x_J} \sum_{j=1}^J f_j(x_j) \quad \text{s.t.} \quad \sum_{j=1}^J x_j = d$$

(aggregate demand equal to the endowment d ; this is the *exchange problem* of [2], section 7.3). This is the *social planner's* problem. Solving it centrally would require someone to know *all* the functions f_j : exactly the classic informational critique of central planning. Apply instead the augmented Lagrangian with alternating updates (a variant of ADMM with averages; we sketch the simplest version, the method of multipliers with approximately separable subproblems):

1. The coordinator announces a price $\lambda^k \in \mathbb{R}^n$ (and the current market imbalance).
2. *Each agent j , in parallel and in private*, solves their subproblem: minimize their cost $f_j(x_j)$ plus the expenditure at announced prices $\langle \lambda^k, x_j \rangle$ plus a quadratic term that penalizes deviating from the split compatible with the balance (the ρ term). Nobody reveals their f_j : they only communicate their response x_j^{k+1} .

3. The coordinator observes the *aggregate excess demand* $e^{k+1} = \sum_j x_j^{k+1} - d$ and adjusts prices: $\lambda^{k+1} = \lambda^k + \rho e^{k+1}$. Goods in excess demand become more expensive; goods in excess supply become cheaper.
4. Repeat until the market clears ($e \approx 0$).



Intuition

This scheme is a *computational version of the second welfare theorem*. The theorem (general equilibrium, [9] ch. 16) says that every Pareto optimum can be *decentralized* with a suitable price vector: there exists λ^* such that the private optimal plans facing λ^* add up exactly to the social optimum. But the theorem is existential: it does not say *how to find* those prices. The augmented Lagrangian and ADMM are the missing algorithm: the tatonnement $\lambda^{k+1} = \lambda^k + \rho e^{k+1}$ is gradient ascent on the dual function, whose maximizer is precisely the supporting price λ^* , and it converges under convexity. Convexity of preferences and technologies, the central hypothesis of the second welfare theorem, reappears here as the convergence condition of the algorithm: this is not an accident, it is the same saddle point theorem seen from its dynamic side. And the quadratic term in ρ , absent from classical tatonnement, is what stabilizes the process when individual responses to prices are very elastic (functions f_j not strictly convex), the case where the pure Walrasian auctioneer can oscillate without converging.

ADMM also has a second modern life in statistics and machine learning (lasso, distributed regression on massive data, consensus across processors), always with the same grammar: private subproblems plus coordination through prices; see [2] and [8].

16.5 Worked application: the Markowitz portfolio [P4]

16.5.1 Formulation as a quadratic program

Markowitz's mean–variance portfolio selection problem is the canonical example of a quadratic program (QP) in finance [4]. There are n assets; asset i has expected return μ_i , and $\Sigma \in \mathbb{R}^{n \times n}$ is the covariance matrix of returns (symmetric and positive definite). A portfolio is a vector of weights $x \in \mathbb{R}^n$, where x_i is the fraction of wealth invested in asset i . Its expected return is $\mu^\top x$ and its variance is $x^\top \Sigma x$. The investor looks for the minimum variance compatible with a target return r and with investing all the wealth:

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top \Sigma x \quad \text{s.t.} \quad \mu^\top x = r, \quad \mathbf{1}^\top x = 1, \quad (16.2)$$

where $\mathbf{1} = (1, \dots, 1)$. (The factor $\frac{1}{2}$ is cosmetic; we allow short sales by not requiring $x \geq 0$.) It is a QP with a strictly convex objective and two linear equality constraints: exactly the class covered by Theorem 16.8.

16.5.2 Analytical solution by Lagrangian with three assets

Example 16.12 (Markowitz with three assets, all computations). Take $n = 3$ uncorrelated assets with

$$\Sigma = \text{diag}(1, 2, 4), \quad \mu = (1, 2, 3), \quad r = 2.$$

(Units: returns in percentage points; asset 3 yields more but is four times more volatile than asset 1.) The Lagrangian, with multipliers λ for the return and γ for the budget, is

$$L(x, \lambda, \gamma) = \frac{1}{2}x^\top \Sigma x - \lambda(\mu^\top x - r) - \gamma(\mathbf{1}^\top x - 1)$$

(the signs of λ, γ are a convention; with equalities there is no sign restriction). The first order condition in x :

$$\Sigma x = \lambda \mu + \gamma \mathbf{1} \quad \implies \quad x = \lambda \Sigma^{-1} \mu + \gamma \Sigma^{-1} \mathbf{1}.$$

With $\Sigma^{-1} = \text{diag}(1, \frac{1}{2}, \frac{1}{4})$ we compute the three classical scalars:

$$A := \mu^\top \Sigma^{-1} \mu = 1 \cdot 1 + 2 \cdot \frac{2}{2} + 3 \cdot \frac{3}{4} = 1 + 2 + \frac{9}{4} = \frac{21}{4},$$

$$B := \mu^\top \Sigma^{-1} \mathbf{1} = 1 + 1 + \frac{3}{4} = \frac{11}{4}, \quad C := \mathbf{1}^\top \Sigma^{-1} \mathbf{1} = 1 + \frac{1}{2} + \frac{1}{4} = \frac{7}{4}.$$

Imposing the two constraints on $x = \lambda \Sigma^{-1} \mu + \gamma \Sigma^{-1} \mathbf{1}$:

$$\mu^\top x = \lambda A + \gamma B = r = 2, \quad \mathbf{1}^\top x = \lambda B + \gamma C = 1,$$

that is, multiplying by 4, the 2×2 linear system

$$21\lambda + 11\gamma = 8, \quad 11\lambda + 7\gamma = 4.$$

Its determinant is $21 \cdot 7 - 11 \cdot 11 = 147 - 121 = 26$, and by Cramer's rule:

$$\lambda = \frac{8 \cdot 7 - 11 \cdot 4}{26} = \frac{56 - 44}{26} = \frac{12}{26} = \frac{6}{13}, \quad \gamma = \frac{21 \cdot 4 - 11 \cdot 8}{26} = \frac{84 - 88}{26} = -\frac{4}{26} = -\frac{2}{13}.$$

We substitute into the formula for x , coordinate by coordinate:

$$x_1 = \frac{6}{13} \cdot 1 - \frac{2}{13} \cdot 1 = \frac{4}{13}, \quad x_2 = \frac{6}{13} \cdot 1 - \frac{2}{13} \cdot \frac{1}{2} = \frac{6}{13} - \frac{1}{13} = \frac{5}{13},$$

$$x_3 = \frac{6}{13} \cdot \frac{3}{4} - \frac{2}{13} \cdot \frac{1}{4} = \frac{18-2}{52} = \frac{16}{52} = \frac{4}{13}.$$

Check (never skip it): budget $\frac{4+5+4}{13} = 1$; return $1 \cdot \frac{4}{13} + 2 \cdot \frac{5}{13} + 3 \cdot \frac{4}{13} = \frac{4+10+12}{13} = \frac{26}{13} = 2$. The optimal portfolio is

$$x^* = \left(\frac{4}{13}, \frac{5}{13}, \frac{4}{13} \right) \approx (0.308, 0.385, 0.308),$$

with minimum variance

$$(x^*)^\top \Sigma x^* = 1 \cdot \frac{16}{169} + 2 \cdot \frac{25}{169} + 4 \cdot \frac{16}{169} = \frac{16+50+64}{169} = \frac{130}{169} = \frac{10}{13} \approx 0.769.$$

Interpretation of the multipliers, by the envelope theorem [11]: if $V(r)$ is the optimal value of (16.2), then $V'(r) = \lambda = \frac{6}{13}$: requiring one hundredth more of target return costs $\frac{6}{13} \cdot 0.01$ units of the objective $\frac{1}{2}x^\top \Sigma x$ (half the variance): it is the *shadow price of return*, the (rescaled) slope of the efficient frontier. And $\gamma = -\frac{2}{13}$ is the marginal value of relaxing the budget.

16.5.3 How the augmented Lagrangian would solve it

In the previous example we inverted Σ by hand because it was diagonal and $n = 3$. With $n = 500$ assets, additional constraints (no short sales: $x \geq 0$; sector caps), the analytical route dies and the augmented Lagrangian steps in. For (16.2), with $h(x) = (\mu^\top x - r, \mathbf{1}^\top x - 1)$ and multipliers $\lambda = (\lambda_1, \lambda_2)$:

$$L_\rho(x, \lambda) = \frac{1}{2}x^\top \Sigma x + \lambda_1(\mu^\top x - r) + \lambda_2(\mathbf{1}^\top x - 1) + \frac{\rho}{2}[(\mu^\top x - r)^2 + (\mathbf{1}^\top x - 1)^2].$$

Each subproblem in x is a strictly convex quadratic: its solution is the linear system

$$[\Sigma + \rho(\mu\mu^\top + \mathbf{1}\mathbf{1}^\top)]x = \rho(r\mu + \mathbf{1}) - \lambda_1\mu - \lambda_2\mathbf{1},$$

solvable by gradient (previous chapter) or by factorization; then the dual step $\lambda_1^{k+1} = \lambda_1^k + \rho(\mu^\top x^k - r)$, $\lambda_2^{k+1} = \lambda_2^k + \rho(\mathbf{1}^\top x^k - 1)$. By Theorem 16.8 (quadratic-linear case with $\Sigma \succ 0$), the multipliers converge linearly to $(-\lambda, -\gamma) = (-\frac{6}{13}, \frac{2}{13})$ of the example (opposite signs because of the sign convention of the Lagrangian), and with the exact multiplier the subproblem delivers x^* *exactly* with finite ρ . If $x \geq 0$ is added, you use the max formula of Proposition 16.11 or an ADMM scheme in which the step in z is the projection onto the orthant (projected gradient again). On dynamic extensions of the mean-variance model, see [4].

With the same grammar one computes *computable general equilibrium* (CGE) models: by the Negishi approach, a Walrasian equilibrium is obtained by maximizing a weighted sum of utilities subject to the resource constraints, and the balance constraints of each market are handled with multipliers that are, literally, the equilibrium prices; the augmented Lagrangian and its distributed variants (section 16.4) are the standard numerical machinery for that computation.

16.6 Practical guide: which method do I use? [P3]

The chapter closes with a mental decision table:

- **Quadratic penalty:** useful as a first prototype because of its simplicity, or when only approximate feasibility is needed. Avoid it if you need high accuracy: it requires $\rho \rightarrow \infty$ and inherits ill conditioning (section 16.1).
- **Barrier / interior point:** the dominant option for *convex* problems with inequalities of medium size (LP, QP, conic programs): high accuracy in a few dozen Newton iterations.

It needs a strictly feasible starting point and does not make good use of a good previous seed (*warm start*).

- **Augmented Lagrangian:** the robust choice for nonlinear equality constraints and large problems; exact with finite ρ , tolerant of seeds and reoptimizations, the basis of professional nonlinear optimization programs (LANCELOT, ALGENCAN).
- **ADMM:** when the problem is separable or distributed (agents, data blocks, consensus): modest convergence in accuracy but very cheap and parallel iterations; ideal in large scale statistics [2, 8].

On software: in Python, `scipy.optimize.minimize` (methods SLSQP and `trust-constr`) and the modeling library `cvxpy` (which calls interior point solvers and OSQP, an ADMM for QP); in Julia, JuMP with Ipopt (interior point barrier) or Percival (augmented Lagrangian). For the final project of the course, `scipy` plus a twenty line hand written augmented Lagrangian is enough, like the one in Example 16.10.

Chapter summary

- **Quadratic penalty:** $Q(x; \rho) = f(x) + \frac{\rho}{2} \|h(x)\|^2$ (plus $\frac{\rho}{2} \sum_i \max\{0, g_i\}^2$ with inequalities). The minimizer $x(\rho)$ converges to the constrained optimum as $\rho \rightarrow \infty$, but with finite ρ it is *always infeasible* (error of order $1/\rho$) and the Hessian becomes ill conditioned (eigenvalues of order ρ in normal directions, of order 1 in tangent ones). Byproduct: $\rho h(x(\rho)) \rightarrow \lambda^*$.
- **Logarithmic barrier:** $B(x; \mu) = f(x) - \mu \sum_i \log(-g_i(x))$; its minimizers form the central path, which delivers primal–dual pairs with duality gap exactly $p\mu$ (the basis of interior point methods).
- **Augmented Lagrangian:** $L_\rho(x, \lambda) = f(x) + \langle \lambda, h(x) \rangle + \frac{\rho}{2} \|h(x)\|^2$. Method of multipliers (Hestenes–Powell): minimize in x , then $\lambda^{k+1} = \lambda^k + \rho h(x^k)$, which is *gradient ascent on the dual function* with the natural step ρ , and makes each x^k satisfy KKT stationarity with the new price.
- **Exactness:** with $\lambda = \lambda^*$, the minimum of $L_\rho(\cdot, \lambda^*)$ is exactly x^* for finite ρ (Theorem 16.8); the multipliers converge linearly with ratio $\frac{1}{1+\rho}$ in the model case.
- **Inequalities:** projected update $\lambda_i^{k+1} = \max\{0, \lambda_i^k + \rho g_i(x^k)\}$.
- **ADMM:** for separable objectives $f(x) + g(z)$ with linear coupling; it alternates block minimizations and a dual step. Economic reading: agents who optimize in private and a coordinator who adjusts prices according to excess demand: the second welfare theorem turned into an algorithm.
- **Applications:** Markowitz is a QP that the augmented Lagrangian solves with finite exactness; the multipliers are shadow prices of return and of the budget (envelope). CGE through Negishi: the multipliers of the balances are the equilibrium prices.
- References: [5] ch. 17, [6] chs. 6–7, [7], [2], [4].

Exercises

Exercise 16.13. Apply the quadratic penalty to the problem $\min_x \frac{1}{2}x^2$ s.t. $x = 2$. Compute $x(\rho)$, the violation $h(x(\rho))$, and verify that $\rho h(x(\rho))$ converges to the Lagrange multiplier of the problem. Which value of ρ guarantees $|h(x(\rho))| \leq 10^{-3}$?

Solution. The only feasible point is $x^* = 2$; from the Lagrangian $\frac{1}{2}x^2 + \lambda(x - 2)$, stationarity $x + \lambda = 0$ at $x^* = 2$ gives $\lambda^* = -2$. The penalized function is $Q(x; \rho) = \frac{1}{2}x^2 + \frac{\rho}{2}(x - 2)^2$, strictly convex; its critical point:

$$Q' = x + \rho(x - 2) = (1 + \rho)x - 2\rho = 0 \implies x(\rho) = \frac{2\rho}{1 + \rho}.$$

Violation: $h(x(\rho)) = x(\rho) - 2 = \frac{2\rho - 2(1 + \rho)}{1 + \rho} = -\frac{2}{1 + \rho} \neq 0$ for every finite ρ , and $x(\rho) \rightarrow 2$. Implicit multiplier: $\rho h(x(\rho)) = -\frac{2\rho}{1 + \rho} \rightarrow -2 = \lambda^*$. For the required accuracy: $\frac{2}{1 + \rho} \leq 10^{-3} \iff 1 + \rho \geq 2000 \iff \rho \geq 1999$: three digits of feasibility cost a fine of order 10^3 (and a condition number $Q'' = 1 + \rho \approx 2 \cdot 10^3$), showing why the pure penalty scales badly. $\square$

Exercise 16.14. Solve with the logarithmic barrier the problem $\min_x (-x)$ s.t. $x \leq 1$. Write $B(x; \mu)$, find the central path $x(\mu)$, the implicit multiplier $\lambda(\mu)$, and check Theorem 16.6 by computing the duality gap.

Solution. Here $f(x) = -x$ and $g(x) = x - 1 \leq 0$; the solution is $x^* = 1$ and, from KKT $-1 + \lambda = 0$, $\lambda^* = 1$. The barrier:

$$B(x; \mu) = -x - \mu \log(1 - x), \quad x < 1.$$

It is strictly convex on $(-\infty, 1)$ since $B'' = \mu/(1 - x)^2 > 0$. Critical point:

$$B' = -1 + \frac{\mu}{1 - x} = 0 \implies 1 - x = \mu \implies x(\mu) = 1 - \mu,$$

which tends to $x^* = 1$ from the interior as $\mu \rightarrow 0^+$. Implicit multiplier: $\lambda(\mu) = \frac{\mu}{-g(x(\mu))} = \frac{\mu}{1 - x(\mu)} = \frac{\mu}{\mu} = 1 = \lambda^*$ for every μ . Primal and dual values: $f(x(\mu)) = \mu - 1$; the dual is $q(\lambda) = \min_x [-x + \lambda(x - 1)]$, which equals $-\lambda$ if $\lambda = 1$ (if $\lambda \neq 1$ the infimum is unbounded and $q = -\infty$), so $q(\lambda(\mu)) = -1$. Gap: $f(x(\mu)) - q(\lambda(\mu)) = (\mu - 1) - (-1) = \mu = p\mu$ with $p = 1$ constraint, exactly as the theorem states. $\square$

Exercise 16.15. Run three iterations of the method of multipliers, with $\rho = 1$ and $\lambda^0 = 0$, for

$$\min_x \frac{1}{2}(x - 2)^2 \quad \text{s.t.} \quad x = 1.$$

Check that λ^k converges to the true multiplier and find the exact convergence ratio of the error $\lambda^k - \lambda^*$.

Solution. Exact solution: $x^* = 1$; from $(x - 2) + \lambda = 0$ at $x = 1$, $\lambda^* = 1$. The augmented Lagrangian is $L_\rho = \frac{1}{2}(x - 2)^2 + \lambda(x - 1) + \frac{\rho}{2}(x - 1)^2$, with first order condition

$$(x - 2) + \lambda + \rho(x - 1) = 0 \implies x(\lambda) = \frac{2 - \lambda + \rho}{1 + \rho}, \quad h(x(\lambda)) = x(\lambda) - 1 = \frac{1 - \lambda}{1 + \rho}.$$

With $\rho = 1$: $x(\lambda) = \frac{3-\lambda}{2}$ and $h = \frac{1-\lambda}{2}$. *Iteration 0* ($\lambda^0 = 0$): $x^0 = \frac{3}{2}$, $h = \frac{1}{2}$, $\lambda^1 = 0 + 1 \cdot \frac{1}{2} = \frac{1}{2}$. *Iteration 1*: $x^1 = \frac{3-1/2}{2} = \frac{5}{4}$, $h = \frac{1}{4}$, $\lambda^2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$. *Iteration 2*: $x^2 = \frac{3-3/4}{2} = \frac{9}{8}$, $h = \frac{1}{8}$, $\lambda^3 = \frac{3}{4} + \frac{1}{8} = \frac{7}{8}$. So $\lambda^k = 0, \frac{1}{2}, \frac{3}{4}, \frac{7}{8} \rightarrow 1 = \lambda^*$ and $x^k \rightarrow 1$. For the ratio, use the general recursion:

$$\lambda^{k+1} = \lambda^k + \rho \cdot \frac{1 - \lambda^k}{1 + \rho} \implies \lambda^{k+1} - 1 = \lambda^k - 1 - \rho \frac{\lambda^k - 1}{1 + \rho} = \frac{\lambda^k - 1}{1 + \rho} :$$

linear convergence with exact ratio $\frac{1}{1+\rho}$ (here $\frac{1}{2}$, visible in the table: errors $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$). With $\rho = 99$ the ratio would be $\frac{1}{100}$. $\square$

Exercise 16.16. Derive the formula of Proposition 16.11 for a single inequality $g(x) \leq 0$: introduce a slack $s \geq 0$ with $g(x) + s = 0$, write the augmented Lagrangian of the equality,

$$\ell(x, s) = f(x) + \lambda(g(x) + s) + \frac{\rho}{2}(g(x) + s)^2,$$

and minimize over $s \geq 0$ in closed form.

Solution. Fix x and write $u = g(x)$. As a function of s , $\varphi(s) = \lambda(u + s) + \frac{\rho}{2}(u + s)^2$ is a convex parabola in the variable $v = u + s$; completing the square, $\varphi = \frac{\rho}{2}(v + \frac{\lambda}{\rho})^2 - \frac{\lambda^2}{2\rho}$. Its unconstrained minimum is at $v = -\lambda/\rho$, that is $s = -u - \lambda/\rho$. Two cases according to the constraint $s \geq 0$: (i) if $-u - \lambda/\rho \geq 0$, that is $\lambda + \rho u \leq 0$, the unconstrained minimum is admissible and $\varphi_{\min} = -\frac{\lambda^2}{2\rho}$; (ii) if $\lambda + \rho u > 0$, the minimum with $s \geq 0$ is attained at the boundary $s = 0$ (the parabola increases to the right of its vertex, which lies at $s < 0$), with value $\varphi(0) = \lambda u + \frac{\rho}{2}u^2 = \frac{1}{2\rho}[(\lambda + \rho u)^2 - \lambda^2]$ (expand the square to check the identity). Both cases are written at once as

$$\min_{s \geq 0} \varphi(s) = \frac{1}{2\rho} [(\max\{0, \lambda + \rho g(x)\})^2 - \lambda^2],$$

which added to $f(x)$ gives the formula of the proposition. The update $\lambda^{k+1} = \lambda^k + \rho(g(x^k) + s^k) = \max\{0, \lambda^k + \rho g(x^k)\}$ comes from substituting the optimal slack s^k : in case (ii), $s^k = 0$ and we are left with $\lambda^k + \rho g(x^k) > 0$; in case (i), $g(x^k) + s^k = -\lambda^k/\rho$ and we are left with exactly 0. It is dual ascent projected onto $\lambda \geq 0$. $\square$

Exercise 16.17. Solve the Markowitz problem (16.2) with $\Sigma = I$ (uncorrelated assets with variance 1), $\mu = (1, 2, 3)$ and $r = 2$. Compute A, B, C , the multipliers and the optimal portfolio, and interpret economically the value you get for the return multiplier.

Solution. With $\Sigma^{-1} = I$: $A = \mu^\top \mu = 1 + 4 + 9 = 14$, $B = \mu^\top \mathbf{1} = 6$, $C = \mathbf{1}^\top \mathbf{1} = 3$. The system of section 16.5 is

$$14\lambda + 6\gamma = 2, \quad 6\lambda + 3\gamma = 1.$$

Multiplying the second by 2: $12\lambda + 6\gamma = 2$; subtracting from the first: $2\lambda = 0$, so $\lambda = 0$ and, from the second, $\gamma = \frac{1}{3}$. The portfolio:

$$x^* = \lambda\mu + \gamma\mathbf{1} = \frac{1}{3}\mathbf{1} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right),$$

with return $\mu^\top x^* = \frac{1+2+3}{3} = 2 = r$ (feasible) and variance $\|x^*\|^2 = \frac{1}{3}$. Interpretation: $\lambda = 0$ says that the return constraint has zero shadow price, and not by accident: the *global minimum variance* portfolio (ignoring the target return) is here the equally weighted one $\frac{1}{3}\mathbf{1}$, whose return is already exactly 2. The constraint $\mu^\top x = r$ holds without “costing” anything: $V'(r) = \lambda = 0$, the point where the mean–variance frontier has zero slope (its vertex). If the target were $r = 2.5$, the same system would give $\lambda = \frac{1}{4} > 0$ and $\gamma = -\frac{1}{6}$: requiring return above the vertex does make variance more expensive, and the multiplier again measures by how much. (Quick check of the case $r = 2.5$: $14\lambda + 6\gamma = 2.5$ and $6\lambda + 3\gamma = 1$ give $2\lambda = 0.5$, $\lambda = \frac{1}{4}$, $\gamma = \frac{1-6/4}{3} = -\frac{1}{6}$, and $x^* = \frac{1}{4}\mu - \frac{1}{6}\mathbf{1} = (\frac{1}{12}, \frac{1}{3}, \frac{7}{12})$, which loads more on the higher return asset.) $\square$

Appendix A

Study guide, exam and glossary [P2]

Final review. It gathers the exact definitions, the theorems with their hypotheses, the recurring techniques and the study plan for taking the exam.

This appendix introduces no new mathematics: it organizes everything above so you can *perform*. It contains the dependency map of the course, a week by week study plan, the “legal cheat sheet” of theorems with their exact hypotheses, the catalogue of classic exam mistakes, writing guides for proofs, assignments and the final presentation, the detailed bridge to Microeconomics and an English–Spanish glossary so you can read the bibliography ([2], [6], [9]) in either language without friction.

A.1 Course map and dependencies [P2]

A.1.1 The dependency graph

The notebook is not a linear list: it is a graph. Before studying a chapter it helps to know what must be solid. The essential dependencies are:

- **Ch. 1 (Preliminaries)** → everything. Topology (compactness, closure, interior), Weierstrass, gradient and Hessian are universal prerequisites.
- **Ch. 2 (Convex sets)** → 3, 4, 5, 8. The definition of a convex set and its stable operations is the geometric basis of the course.
- **Ch. 3 (Cones and hulls)** → 4, 9, 13. Farkas’ lemma lives here in its conic version and reappears in the proof of KKT.
- **Ch. 4 (Separation)** → 7, 10, 12, 13. It is the mother theorem of convex analysis: support, subgradients, duality and finite dimensional Hahn–Banach are corollaries of it.
- **Ch. 5 (Concave and convex functions)** → 6, 7, 8, 9, 15. Without the first and second order characterizations there are no optimality conditions.
- **Ch. 6 (Quasiconcavity)** → 9 (sufficiency with pseudoconvexity) and, above all, to consumer theory in Microeconomics.
- **Ch. 7 (Subdifferential)** → 8, 12, 15. It generalizes the gradient; Fermat’s rule $0 \in \partial f(x^*)$ unifies everything nondifferentiable.

- **Ch. 8 (Optimality and projection)** → 9, 14, 15. The projection onto a closed convex set is also the technical tool behind strict separation.
- **Ch. 9 (KKT)** → 10, 11, 14, 16. The operational heart of the course and of all applied microeconomics.
- **Ch. 10 (Saddle point and duality)** → 11, 12, 13, 14, 16. It gives meaning to multipliers as shadow prices.
- **Ch. 11 (Envelope)** → comparative statics, Shephard's lemma, Roy's identity, Milgrom–Segal [11] in auctions (mechanism design, [9] ch. 23).
- **Ch. 12 (Fenchel)** → 13, 16. The conjugate is the functional version of duality and the language of cost and profit functions.
- **Chs. 13, 14, 15, 16** are applications and algorithms: they consume everything above and are the natural quarry of topics for the final presentation.

A.1.2 The three backbones

Chapter summary

The course rests on three logical chains you should be able to recite:

1. **Geometric backbone:** separation (ch. 4) $\Rightarrow$ supporting hyperplane $\Rightarrow$ existence of subgradients (ch. 7) $\Rightarrow$ Lagrangian duality (ch. 10) $\Rightarrow$ Fenchel duality (ch. 12). Every duality theorem is, deep down, a well placed hyperplane.
2. **Analytic backbone:** convexity of functions (ch. 5) $\Rightarrow$ optimality conditions (ch. 8) $\Rightarrow$ KKT (ch. 9) $\Rightarrow$ envelope theorem (ch. 11). This is the chain that turns an economic problem into a system of equations and then into comparative statics.
3. **Economic backbone:** the two previous ones meet in economics. Separation proves the second welfare theorem; KKT solves the consumer, the firm and the principal; the envelope produces Shephard, Roy, Hotelling and revenue equivalence; duality turns multipliers into prices. The common thread: *every price in microeconomics is a separating hyperplane or a Lagrange multiplier, and almost always both at once.*

A.2 Weekly study plan [P2]

A.2.1 Week by week (W1–W15)

The plan follows the official schedule. “Redo” means solving the exercise again from a blank page, without looking at the solution, and comparing afterwards.

W1 (spaces and convex sets).

Read chs. 1–2. Bibliography: [1] chs. 1–2, [3] ch. 1. Redo: convexity proofs for intersections, Minkowski sums and budget sets.

W2 (cones and hulls).

Read ch. 3. Bibliography: [1] ch. 2, [6] ch. 2. Redo: Carathéodory with a numerical example in $\mathbb{R}^2$ and computation of polar cones of orthants.

W3 (separation).

Read ch. 4 *twice*: it is P1. Bibliography: [1] ch. 3, [2] sec. 2.5. Redo: strict separation via projection, counterexample without compactness, supporting hyperplane of a convex set with smooth boundary.

W4 (concave and quasiconcave functions).

Read chs. 5–6. Bibliography: [2] ch. 3, [4] on generalized concavity. Redo: convexity check by Hessian, quasiconcave Cobb–Douglas via bordered Hessian, the concave–pseudo–quasi hierarchy with counterexamples.

W5 (subdifferential).

Read ch. 7. Bibliography: [1] ch. 4, [6] ch. 2. Redo: $\partial|x|$, subdifferential of a maximum of affine functions, generalized Fermat rule.

W6 (optimality and projection).

Read ch. 8. Bibliography: [3] chs. 2–3, [7] ch. 1. Redo: variational characterization of the projection and uniqueness of minimizers under strict convexity.

W7 (KKT).

Read ch. 9 and solve *all* its exercises. Bibliography: [5] ch. 12, [2] ch. 5, [10] appendix A2. Redo: consumer with a corner solution, a problem where CQ fails, complete verification of a KKT point.

W8 (saddle point and duality).

Read ch. 10. Bibliography: [2] ch. 5, [6] ch. 4. Redo: construction of the dual of a QP, verification of Slater, duality gap in a nonconvex problem. **The pre midterm plan below starts here.**

W9 (Fenchel).

Read ch. 12. Bibliography: [1] ch. 5, [6] sec. 2.6. Redo: conjugates of e^x , $\frac{1}{2}\|x\|^2$, indicator functions; Fenchel–Young inequality with the equality case.

W10 (linear programming).

Read ch. 13. Bibliography: [3] ch. 4, [5] ch. 13. Redo: dual of an LP in standard form and verification by complementary slackness.

W11 (SVM).

Read ch. 14. Bibliography: [2] sec. 8.6, [8]. Redo: derivation of the dual of the hard margin SVM and identification of support vectors.

W12–W13 (first-order methods).

Read ch. 15. Bibliography: [8] chs. 3–4, [7] ch. 2. Redo: the $O(1/k)$ rate of gradient descent for L -smooth functions, choice of step $1/L$, projected gradient.

W14 (penalty and augmented Lagrangian).

Read ch. 16. Bibliography: [5] ch. 17, [8]. Redo: quadratic penalty versus augmented

Lagrangian on a problem with one equality.

W15 (applications).

Cross cutting review and full rehearsal of the final presentation (see sec. A.7); have the computational implementation ready.

A.2.2 Intensive pre midterm plan (the midterm falls around W8–W9)

The midterm is worth 30% and covers, with very high probability, W1–W8. The absolute priority is P1: separation, convex functions, quasiconcavity, KKT and duality.

Exam strategy

Plan for the 7 days before the midterm (about 3 hours a day):

1. **Day 1 (geometry).** Reread ch. 4: exact statements of the three versions of separation and of the support theorem, with their counterexamples. Write from memory the proof of strict separation via projection.
2. **Day 2 (functions).** Ch. 5: first and second order characterizations, operations that preserve convexity, Jensen. Redo 5 convexity checks by Hessian.
3. **Day 3 (quasiconcavity).** Ch. 6: the full hierarchy with memorized counterexamples and the bordered Hessian with its signs. Redo Cobb–Douglas and CES.
4. **Day 4 (KKT).** Ch. 9: write the full KKT system from memory (with the sign convention), list the CQs and the convex sufficiency theorem. Solve 3 problems with corner cases, discussing every complementary slackness case.
5. **Day 5 (duality).** Ch. 10: Lagrangian dual, weak duality (a 4 line proof, learn it), Slater, saddle point. Construct the dual of two different problems.
6. **Day 6 (mock exam).** A timed 3 hour exam with exercises from chs. 4, 5, 6, 9, 10 that you have not redone before.
7. **Day 7 (light review).** Only the cheat sheet of sec. A.3 and the mistakes of sec. A.4. Sleep well: a statement with correct hypotheses is worth more than an extra hour of late night study.

A.3 The star theorems with exact hypotheses [P2]

Format of each entry: **Hypotheses** (all of them, unabridged), **Conclusion** (one line) and **Typical mistake** when citing it badly. In an exam, citing a theorem with one hypothesis missing is citing a *false* statement.

A.3.1 Existence and projection

Weierstrass (ch. 1).

Hypotheses: $f: K \rightarrow \mathbb{R}$ continuous, $K \subseteq \mathbb{R}^n$ nonempty and compact (closed *and* bounded).

Conclusion: f attains a global maximum and a global minimum on K . **Typical mistake:** applying it with K closed but not bounded (the budget set with a zero price) or with f only semicontinuous without saying in which sense (for minima, lower semicontinuity is enough).

Projection theorem (ch. 8).

Hypotheses: $C \subseteq \mathbb{R}^n$ nonempty, closed and convex; $x \in \mathbb{R}^n$. **Conclusion:** there is a unique $\text{proj}_C(x)$, characterized by $\langle x - \text{proj}_C(x), y - \text{proj}_C(x) \rangle \leq 0$ for all $y \in C$. **Typical mistake:** forgetting that closedness gives existence and convexity gives uniqueness and the variational inequality; without convexity the projection can be multivalued.

A.3.2 Separation, support and Carathéodory

Point-set separation, strict (ch. 4).

Hypotheses: $C \subseteq \mathbb{R}^n$ nonempty, closed and convex; $x_0 \notin C$. **Conclusion:** there exist $p \neq 0$ and $\alpha \in \mathbb{R}$ with $\langle p, x_0 \rangle > \alpha > \langle p, y \rangle$ for all $y \in C$. **Typical mistake:** omitting “closed”: a point on the boundary of an open convex set cannot be strictly separated.

Separation of two convex sets (Minkowski, ch. 4).

Hypotheses: $C, D \subseteq \mathbb{R}^n$ nonempty, convex and disjoint ($C \cap D = \emptyset$). **Conclusion:** there exists $p \neq 0$ with $\sup_{x \in C} \langle p, x \rangle \leq \inf_{y \in D} \langle p, y \rangle$ (separation not necessarily strict). **Typical mistake:** promising strict inequalities; in finite dimension disjointness + convexity only guarantee weak separation.

Strict separation of sets (ch. 4).

Hypotheses: C, D nonempty, convex, disjoint, C closed and D compact. **Conclusion:** there exist $p \neq 0$, α with $\langle p, x \rangle < \alpha < \langle p, y \rangle$ for all $x \in C$, $y \in D$. **Typical mistake:** asking only for “both closed”: the x axis and the epigraph of $1/x$ on $x > 0$ are closed, convex, disjoint and cannot be strictly separated.

Proper separation (ch. 4).

Hypotheses: C, D nonempty convex with $\text{ri}(C) \cap \text{ri}(D) = \emptyset$. **Conclusion:** there is a hyperplane that separates them properly (it does not contain both sets at once); the condition is also necessary. **Typical mistake:** using ordinary interiors: if C has empty interior (a segment in $\mathbb{R}^2$) the correct criterion uses the *relative* interior.

Supporting hyperplane (ch. 4).

Hypotheses: $C \subseteq \mathbb{R}^n$ nonempty convex, $x_0 \in \text{bd}(C)$ (boundary). **Conclusion:** there exists $p \neq 0$ with $\langle p, x \rangle \leq \langle p, x_0 \rangle$ for all $x \in C$: a hyperplane through x_0 that leaves C in one halfspace. **Typical mistake:** invoking it at an interior point (it does not exist) or on nonconvex sets (it fails: the boundary of an annulus).

Carathéodory (ch. 3).

Hypotheses: $S \subseteq \mathbb{R}^n$, $x \in \text{conv}(S)$. **Conclusion:** x is a convex combination of at most $n + 1$ points of S (for $\text{cone}(S)$: at most n). **Typical mistake:** saying “ n points” for the convex hull; the triangle in $\mathbb{R}^2$ needs $3 = n + 1$.

A.3.3 Convexity of functions and the generalized hierarchy

First order characterization (ch. 5).

Hypotheses: f differentiable on an open set containing the convex set C . **Conclusion:** f is convex on C if and only if $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all $x, y \in C$ (the graph lies above each tangent); strict with $>$ for $y \neq x$. **Typical mistake:** forgetting that C must be convex: the pointwise condition does not chain if the segment leaves the domain.

Second order characterization (ch. 5).

Hypotheses: f of class C^2 on an open convex set C . **Conclusion:** f convex $\iff \nabla^2 f(x)$ positive semidefinite for all $x \in C$; if $\nabla^2 f(x)$ is positive definite on all of C , f is strictly convex. **Typical mistake:** believing the converse of the strict part: $f(x) = x^4$ is strictly convex with $f''(0) = 0$.

Concave–pseudo–quasi hierarchy (ch. 6).

Hypotheses: f defined on a convex set C ; for pseudoconcavity, f differentiable. **Conclusion:** concave $\Rightarrow$ pseudoconcave (if differentiable) $\Rightarrow$ quasiconcave; and strictly concave $\Rightarrow$ strictly quasiconcave. No implication reverses: x^3 is quasiconcave and not concave on $\mathbb{R}$. **Typical mistake:** using the chain backwards (“quasiconcave, therefore concave”) or applying pseudoconcavity without differentiability.

Quasiconcavity and level sets (ch. 6).

Hypotheses: $f: C \rightarrow \mathbb{R}$ with C convex. **Conclusion:** f is quasiconcave $\iff$ every upper contour set $\{x \in C : f(x) \geq \alpha\}$ is convex. **Typical mistake:** checking convexity of the lower level sets (that characterizes quasiconvexity) or of the level curves (not enough).

A.3.4 Optimality conditions

Generalized Fermat (chs. 7, 8).

Hypotheses: $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ convex and proper. **Conclusion:** x^* is a *global* minimum of f if and only if $0 \in \partial f(x^*)$; in the differentiable case, $\nabla f(x^*) = 0$ is necessary at interior points and, with convexity, also sufficient. **Typical mistake:** using $\nabla f(x^*) = 0$ as sufficient without convexity (saddle points, $f(x) = x^3$) or looking for *maxima* of a convex function with this rule (the maxima of a convex function on a convex compact set are at extreme points).

Lagrange multipliers, equalities only (ch. 9).

Hypotheses: x^* a local minimum of f subject to $h_j(x) = 0$, $j = 1, \dots, m$; f, h_j of class C^1 ; the gradients $\nabla h_1(x^*), \dots, \nabla h_m(x^*)$ linearly independent. **Conclusion:** there exists $\mu^* \in \mathbb{R}^m$ (unsigned) with $\nabla f(x^*) + \sum_j \mu_j^* \nabla h_j(x^*) = 0$. **Typical mistake:** forgetting linear independence (min x s.t. $x^2 = 0$ has no multiplier) or imposing $\mu^* \geq 0$: with equalities the multiplier has no sign.

KKT, necessity (ch. 9).

Hypotheses: x^* a local minimum of f s.t. $g_i(x) \leq 0$, $h_j(x) = 0$, everything C^1 , and a *constraint qualification* (CQ) at x^* : LICQ (gradients of active constraints linearly independent), or Mangasarian–Fromovitz, or affine constraints, or Slater in convex problems. **Conclusion:**

there exist $\lambda^* \geq 0, \mu^*$ with stationarity $\nabla f + \sum_i \lambda_i^* \nabla g_i + \sum_j \mu_j^* \nabla h_j = 0$ at x^* , feasibility and complementary slackness $\lambda_i^* g_i(x^*) = 0$. **Typical mistake:** “the optimum satisfies KKT” without a CQ: false in general (classic example $\min x_1$ s.t. $x_2 \leq x_1^3, x_2 \geq 0$).

KKT, sufficiency (ch. 9).

Hypotheses: *convex* problem (f and the g_i convex and differentiable, the h_j affine) and (x^*, λ^*, μ^*) satisfies the KKT conditions. **Conclusion:** x^* is a *global* minimum. (Generalized version: f pseudoconvex and g_i quasiconvex is enough.) **Typical mistake:** using KKT as a certificate of optimality in nonconvex problems: without convexity a KKT point can be a saddle or even a maximum.

A.3.5 Duality, envelope and Fenchel

Saddle point (ch. 10).

Hypotheses: (x^*, λ^*) with $\lambda^* \geq 0$ is a saddle point of the Lagrangian: $L(x^*, \lambda) \leq L(x^*, \lambda^*) \leq L(x, \lambda^*)$ for all x and all $\lambda \geq 0$. *No* convexity or differentiability assumptions. **Conclusion:** x^* is a global optimum of the primal, λ^* of the dual, and there is no duality gap. **Typical mistake:** believing the converse without hypotheses: the *existence* of a saddle point does require convexity + Slater; an optimum of a nonconvex problem may admit no saddle point.

Weak duality (ch. 10).

Hypotheses: none beyond the definition of the dual (it holds for any problem, convex or not). **Conclusion:** $d^* \leq p^*$: every feasible dual value bounds the primal from below (minimization convention). **Typical mistake:** reversing the inequality when working with a *maximization* primal (there the dual bounds from above).

Strong duality (ch. 10).

Hypotheses: convex problem (f, g_i convex, h_j affine) + Slater’s condition: there is a feasible $\hat{x}$ with $g_i(\hat{x}) < 0$ for every nonaffine constraint. **Conclusion:** $d^* = p^*$ and the dual *attains* its optimum (there exists λ^* ; the primal may fail to attain it). **Typical mistake:** invoking it without Slater or in nonconvex problems; also forgetting that Slater relaxes to plain feasibility for affine constraints (which is why LP needs no Slater).

Envelope without constraints (ch. 11).

Hypotheses: $V(\theta) = \max_x f(x, \theta)$ with the maximum attained at $x^*(\theta)$ differentiable or, more weakly, with the Milgrom–Segal conditions below. **Conclusion:** $V'(\theta) = \frac{\partial f}{\partial \theta}(x^*(\theta), \theta)$: only the *direct* effect; the effect through x^* vanishes by optimality. **Typical mistake:** also differentiating $x^*(\theta)$ (double counting) or applying the formula where the maximum is not attained.

Envelope with constraints (ch. 11).

Hypotheses: $V(\theta) = \max_x f(x, \theta)$ s.t. $g(x, \theta) \leq 0$, with well defined solutions and multipliers and regularity conditions (for example LICQ and local uniqueness). **Conclusion:** $V'(\theta) = \frac{\partial L}{\partial \theta}(x^*(\theta), \lambda^*(\theta), \theta)$; in particular λ_i^* is the shadow price of relaxing g_i . **Typical**

mistake: differentiating f instead of L when θ appears in the constraints (this loses the shadow price: it is the mistake that breaks Shephard's lemma).

Milgrom–Segal (ch. 11, [11]).

Hypotheses: X an *arbitrary* choice set (no convexity, no topology), $f(x, \cdot)$ differentiable in $t \in [0, 1]$ for each x , the family $\{f_t(x, \cdot)\}$ dominated by an integrable function, and the maximum attained at each t . **Conclusion:** V is absolutely continuous and $V(t) = V(0) + \int_0^t f_t(x^*(s), s) ds$ for any selection $x^*(s)$ of maximizers. **Typical mistake:** requiring uniqueness of the maximizer, interiority or concavity: the point of Milgrom–Segal is that it does *not* ask for them (which is why it works in auctions with discontinuous strategies).

Fenchel–Young (ch. 12).

Hypotheses: f proper; $f^*(p) = \sup_x \{\langle p, x \rangle - f(x)\}$ its conjugate. **Conclusion:** $f(x) + f^*(p) \geq \langle p, x \rangle$ for all x, p , with equality if and only if $p \in \partial f(x)$. **Typical mistake:** stating the equality without the subgradient condition, or forgetting that for the “if and only if” with $x \in \partial f^*(p)$ you need f convex and closed.

Fenchel–Moreau (biconjugation, ch. 12).

Hypotheses: $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ proper, convex and closed (lower semicontinuous). **Conclusion:** $f^{**} = f$. **Typical mistake:** applying $f^{**} = f$ to nonconvex functions: in general f^{**} is the closed convex envelope of f (this, cited properly, is a useful result, not a mistake).

LP duality and complementary slackness (ch. 13).

Hypotheses: a primal–dual pair of linear programs; for strong duality it is enough that *one* of the two is feasible and bounded. **Conclusion:** both have an optimum and $p^* = d^*$; moreover a primal feasible x and a dual feasible y are optimal if and only if they satisfy complementary slackness ($x_j > 0 \Rightarrow$ dual constraint j active; primal slack $i > 0 \Rightarrow y_i = 0$). **Typical mistake:** forgetting the pathological case (primal infeasible and dual infeasible at the same time is possible) or asking for Slater: in LP it is unnecessary.

A.4 Classic exam mistakes [P2]

Careful (common mistake)

Every item on this list has cost marks to generations of students. Reviewing it the day before the exam pays more than any other review of the same length.

1. **Confusing necessary with sufficient.** $\nabla f(x^*) = 0$ is *necessary* at an interior optimum; it is only *sufficient* with convexity (or pseudoconvexity). Likewise, KKT is necessary (with CQ) and sufficient (with convexity): they are two different theorems with different hypotheses, and you must say which one you are using.
2. **Forgetting the constraint qualification.** Writing “by KKT there exist multipliers” without checking LICQ, Mangasarian–Fromovitz, affineness or Slater. In the exam: one line like “the constraints are affine, so the CQ holds” earns marks; its absence takes them

away.

3. **PSD Hessian for quasiconcavity.** Quasiconcavity is *not* checked with the definiteness of the Hessian: the correct differential criterion uses the *bordered Hessian* with the alternating signs of its bordered principal minors (ch. 6). Cobb–Douglas with increasing returns is the canonical example: a Hessian that is not negative semidefinite and still quasiconcave.
4. **Strict separation without compactness.** “Two disjoint closed convex sets can be strictly separated” is false: one of them must be compact. Counterexample to memorize: $C = \{(x, y) : y \leq 0\}$ and $D = \{(x, y) : x > 0, y \geq 1/x\}$.
5. **Adding or transforming quasiconcave functions.** The sum of quasiconcave functions is not quasiconcave in general (the sum of concave ones is). On the other hand quasiconcavity survives increasing monotone transformations and concavity does not: $e^{u(x)}$ preserves the quasiconcavity of u , not its concavity. Confusing which property is ordinal and which is cardinal is a serious mistake in consumer theory.
6. **Ignoring corner cases.** In KKT you must discuss *all* the complementary slackness patterns ($\lambda_i = 0$ versus g_i active), not only the interior case. The consumer with quasilinear utility and low income consumes zero of one good: if that branch is not examined, the solution given is incomplete.
7. **Multiplier signs and the min/max convention.** With $\min f$ s.t. $g \leq 0$ and Lagrangian $L = f + \lambda g$, you get $\lambda \geq 0$; with $\max f$ s.t. $g \leq 0$ and $L = f - \lambda g$, also $\lambda \geq 0$. Mixing conventions halfway through a problem produces “impossible” negative multipliers. Practical rule: write the chosen convention in the first line and do not change it.
8. **Strong duality for free.** Writing $p^* = d^*$ without checking convexity + Slater (or linearity). In nonconvex problems the duality gap can be strictly positive, and weak duality $d^* \leq p^*$ is all you have.
9. **Maxima of convex functions.** Looking for the maximum of a convex function by setting the gradient to zero. The maximum of a convex function over a polytope is attained at a vertex: it is a combinatorial problem, not a first order one.
10. **Applying Weierstrass without compactness.** “The problem has a solution by Weierstrass” requires exhibiting compactness (or coercivity of f , which replaces it for minima). With zero prices or unbounded constraints, you first argue the effective boundedness of the relevant set.

A.5 How to write proofs and top mark solutions [P2]

A.5.1 Structure of a proof

Exam strategy

A three layer skeleton that graders appreciate:

1. **Hypotheses and goal.** First line: “Let C be nonempty closed convex and $x_0 \notin C$. We want to prove that there exists $p \neq 0$ such that. . .”. Rewriting the goal in symbols removes half the mistakes.
2. **Chained claims.** Each step is a verifiable claim with its justification in parentheses: definition used, theorem cited (with its hypotheses checked!) or computation. If a step requires a theorem, check *in writing* that its hypotheses hold before using it.
3. **Closing.** An explicit last line: “which proves the claim”; in inequalities, indicate where each hypothesis was used (if a hypothesis was never used, suspect an error).

A.5.2 Standard phrases of mathematical writing

Writing formulas worth having on autopilot: “Let $x \in C$ be arbitrary”; “without loss of generality we may assume”; “suppose, for contradiction, that”; “by convexity of f applied to x, y with $\lambda \in [0, 1]$ ”; “taking the limit as $k \rightarrow \infty$ and using the continuity of f ”; “by the separation theorem (ch. 4), whose hypotheses hold since C is closed, convex and $x_0 \notin C$ ”; “the inequality is strict because”; “conversely”; “which contradicts the choice of x_0 and concludes the proof”. Avoid: “it is obvious that”, “clearly” (if it were, it costs one line to prove it) and implicit quantifiers (“ $f(x) \leq f(y)$ ” without saying for which x, y).

A.5.3 Checking a KKT solution before handing it in

Chapter summary

A 6 point checklist for any KKT solution:

1. **Primal feasibility:** $g_i(x^*) \leq 0$ and $h_j(x^*) = 0$, checked numerically, not “by construction”.
2. **Dual sign:** $\lambda_i^* \geq 0$ for all inequalities (the μ_j^* of equalities are free).
3. **Complementary slackness:** $\lambda_i^* g_i(x^*) = 0$ for each i ; list which constraints are active.
4. **Stationarity:** plug the numbers into $\nabla f + \sum \lambda_i \nabla g_i + \sum \mu_j \nabla h_j = 0$ and check component by component.
5. **CQ:** one line justifying the qualification used.
6. **Global optimality:** one line invoking sufficiency (convexity of the problem) or, if there is none, comparing values across the KKT candidates.

If the six points are written down, the solution is airtight.

A.5.4 Time management in the exam

With a 3 hour exam and 4–5 questions: (i) first 10 minutes, read everything and classify each question as *direct* (definition + known theorem), *computational* (KKT, duals) or *creative* (a new proof); (ii) solve the direct and computational ones first (they secure the base of your mark); (iii) budget minutes per points (a 20% question $\approx$ 35 minutes) and *temporarily abandon* it when the budget runs out; (iv) in the creative ones, if the full proof does not come out, write the precise statement of the relevant theorem, the verification of its hypotheses and the plan of the proof: that captures partial credit; (v) keep 10 final minutes to reread multiplier signs and quantifiers.

A.6 Strategy for the four assignments (30%) [P2]

The assignments are written and averaged: consistency is worth more than one brilliant assignment and three weak ones. Golden rules:

- **Rigor in writing.** Each assignment is written as a small article: statement, notation, solution with the structure of sec. A.5, conclusion. Every theorem cited comes with a name, a reference to the notebook or the bibliography and an explicit check of hypotheses: in a convex analysis course, precision about hypotheses is exactly what is being graded.
- **Numerical verification in Python.** Whenever the problem allows it, pair the analytical solution with a numerical check (`numpy + scipy.optimize` or `cvxpy`): solve the problem, compare the optimum with the KKT point found by hand and report the solver multipliers against the analytical ones. A two line paragraph (“the solver confirms $x^* = (2, 1)$ and $\lambda^* = 0.5$ with error 10^{-8} ”) sets an excellent assignment apart. If they do not match, there is an error: find it before the grader does.
- **Drawn counterexamples.** When the assignment asks you to discuss hypotheses, include

the counterexample with a figure (even a schematic one): separation without closedness, quasiconcave functions that do not add up, a CQ that fails.

- **Mathematical spelling.** Equations punctuated as part of the sentence, symbols defined before being used, and no numerical result without units or context. Reread the whole assignment one day after finishing it, before handing it in.

A.7 Strategy for the final presentation (40%: the heaviest mark) [P2]

A.7.1 Likely criteria of the panel

The predictable criteria of any panel are: (i) *mathematical correctness*: statements with complete hypotheses and at least one nontrivial proof carried out fluently; (ii) *command*: answering questions of the type “and what if I drop this hypothesis?” (prepare the counterexamples in advance); (iii) *connection with the course*: that the topic visibly uses separation, duality or KKT, not just mentions them; (iv) *added value*: a computational implementation or an economic application that is not in the class notes; (v) *clarity*: a talk that is understood from start to finish is worth more than a rushed encyclopedia.

A.7.2 Six developed topic proposals

1. **Primal–dual SVM with economic data.** *Content*: hard and soft margin, full derivation of the dual, support vectors as active constraints, complementary slackness as a classifier. *Chapters*: 9, 10, 14. *Implementation*: classify households (for example credit default or poverty with survey data) solving primal and dual in `cvxpy` and showing that the nonzero multipliers are exactly the support vectors. *Why it shines*: it is the perfect example of the whole KKT–duality machinery working end to end, with a striking final plot.
2. **The augmented Lagrangian and ADMM as a Walrasian auctioneer.** *Content*: dual decomposition, interpretation of the price as a multiplier adjusted by excess demand (tatonnement = dual ascent), why the quadratic penalty stabilizes. *Chapters*: 10, 15, 16. *Implementation*: an exchange economy with concave utilities solved by ADMM, plus a Markowitz portfolio solved by augmented Lagrangian, plotting the convergence of the “prices”. *Why it shines*: it translates a modern algorithm into the most classical economic metaphor; it combines theory and computation in equal parts.
3. **Milgrom–Segal and revenue equivalence in auctions.** *Content*: the classical envelope theorem versus [11], why the strategy sets of an auction are neither convex nor compact, and how the integral formula gives the equilibrium utility of the bidder and the revenue equivalence theorem. *Chapters*: 5, 11. *Implementation*: simulate first and second price auctions with uniform values and check numerically the equality of expected revenues. *Why it shines*: it connects directly with mechanism design ([9] ch. 23) and shows that you have read a seminal paper.

4. **The second welfare theorem via separation, and where it fails.** *Content:* proof of the theorem with separation of convex sets (the aggregate of preferred sets versus the production set), the exact role of each hypothesis, and two nonconvex counterexamples (preferences with indivisibilities; increasing returns) where no price supports the Pareto optimum. *Chapters:* 2, 4, plus [9] ch. 16. *Implementation:* an Edgeworth box plotted with nonconvex indifference curves showing the hyperplane that does not exist. *Why it shines:* it is *the* bridge between the course and general equilibrium theory ([9] chs. 15–18), and discussing hypotheses is exactly what the panel rewards.
5. **Fenchel duality and recovering technologies.** *Content:* the cost function as the conjugate (support) of the production set, Shephard's lemma via Fenchel–Young, the cost–technology duality theorem: from $c(w, y)$ you recover the closed convex envelope of the technology (Fenchel–Moreau). *Chapters:* 4, 11, 12. *Implementation:* start from a nonconvex technology, compute c numerically and reconstruct c^* showing what is lost (exactly the nonconvex part). *Why it shines:* elegant mathematics with a deep economic reading: cost data only identify the “convexification” of the technology.
6. **Generalized concavity and demand theory.** *Content:* why utility only needs to be quasiconcave (ordinality), the concave–pseudo–quasi hierarchy, the bordered Hessian, conditions under which a quasiconcave function admits a concave transformation (concavifiability) and what is known when it does not. *Chapters:* 5, 6, 9, with [4] and the specialized literature on generalized concavity. *Implementation:* a map of indifference curves of a nonconcavifiable quasiconcave function and numerical verification of the bordered Hessian. *Why it shines:* it is a live research area directly connected with consumer theory; citing and understanding its results is a high return bet (and a high risk one: master the details before choosing it).

A.7.3 Structure of the 20 minutes

Exam strategy

A tested script: **min 0–2**, the problem in one sentence and why it matters (economic motivation); **min 2–6**, the tools of the course you will use, stated with exact hypotheses (one slide per theorem); **min 6–13**, the central result with *one* complete proof or derivation done at a firm pace (better one full proof than three half ones); **min 13–17**, computational implementation or application with one leading plot; **min 17–19**, what happens if hypotheses are dropped (prepared counterexample); **min 19–20**, a three line summary. Rehearse twice with a clock; prepare two appendix slides for the panel's predictable questions.

A.8 Complete bridge to Microeconomics [P2]

The Microeconomics course relies on this material part by part. The tables follow the canonical organization of Mas-Colell, Whinston and Green [9], in five parts, and summarize what to review from this notebook for each one.

Part of the Microeconomics course	Notebook chapters	Specific results you will use
Individual decision making (MWG 1–6: preference and choice, demand and duality, production, uncertainty)	1, 5, 6, 9, 11, 12	Weierstrass (existence of demand); quasiconcavity of utility and well defined demand; KKT in the consumer problem and in expenditure minimization; the <i>envelope</i> as the source of Roy's identity, Shephard's lemma and Hotelling's lemma; the Fenchel conjugate and the cost function as the support of the technology; concavity of Bernoulli utility = risk aversion and Jensen's inequality.
Game theory (MWG 7–9: Nash in pure and mixed strategies, dynamic, Bayesian and repeated games)	3, 10, 13	The simplex of mixed strategies as the convex hull of the pure ones and Carathéodory's theorem; <i>saddle point and the minimax theorem</i> in zero sum games; the primal–dual pair of linear programs associated with a matrix game and value of the game = common value of both programs; compactness and convexity of the strategy set as hypotheses of the existence theorems.

Part of the Microeconomics course	Notebook chapters	Specific results you will use
Market equilibrium and market failure (MWG 10–14: externalities, market power, adverse selection and signaling, principal–agent)	5, 9, 10, 11, 13	Concavity of Bernoulli utility = risk aversion; the principal’s problem as a program with IC and IR constraints: multipliers as the price of incentives; validity of the first order approach (concavity of the agent’s problem); the envelope for comparative statics of the optimal contract; KKT in screening and in price discrimination; with finite types and a fixed allocation, the problem in transfers is a linear program and complementary slackness says which incentive constraints bind (ch. 13); concavity of aggregate surplus.
General equilibrium (MWG 15–18: Edgeworth, Pareto, welfare, core)	1, 2, 4, 6, 9, 13	Weierstrass (existence of demand); convexity of preferences and of the production set; <i>second welfare theorem by separation</i> ; supporting hyperplane = prices; quasiconcavity and well defined demand; KKT in the characterization of Pareto optima (maximize the utility of agent 1 subject to the utilities of the others); Koopmans activity analysis, where the prices that value resources are the dual variables of a linear program (ch. 13).
Welfare and incentives (MWG 21–23: social choice, welfare economics, mechanism design and implementation)	9, 10, 11, 12, 13	<i>Milgrom–Segal</i> [11] for equilibrium payoffs and revenue equivalence; monotonicity + envelope = characterization of implementability; the revelation principle and programs with a continuum of IC constraints (Lagrangian relaxation); complementary slackness to know which ICs bind (“downward adjacent”); the assignment problem between buyers and objects as a linear transportation program, whose dual delivers the prices that support the efficient allocation (ch. 13); saddle points in designer problems with constraints in expectation; conjugates in dual recursive formulations

Intuition

Weekend advice: on the weekend before each new part starts, review in this order: (1) the chapters in the table, rereading only definitions, theorems and the summary boxes; (2) the corresponding entry of the cheat sheet (sec. A.3); (3) redo *one* representative exercise (individual decision making: Roy's identity and Shephard's lemma by envelope; game theory: the value of a 2×2 matrix game by the primal–dual pair; market failure: principal–agent with two effort levels by KKT; general equilibrium: the second welfare theorem in a 2×2 Edgeworth box; mechanism design: the envelope in the first price auction and which IC constraints bind in a three type model). Two well chosen hours on the weekend save ten hours of confusion on Monday.

A.9 English–Spanish glossary [P2]

The core bibliography ([2], [6], [5], [8], [9], [10], [11]) is in English, and much of the graduate teaching of these topics in Latin America and Spain is in Spanish. These tables, organized by topic, cover the vocabulary of the course in both languages, so you can read either literature and search for references with the right term.

Topology and preliminaries (ch. 1)

English	Spanish
open / closed set	conjunto abierto / cerrado
bounded set	conjunto acotado
compact set	conjunto compacto
closure	clausura (cerradura)
interior	interior
boundary	frontera
neighborhood	entorno (vecindad)
convergent sequence	sucesión convergente
limit point, accumulation point	punto de acumulación
continuous function	función continua
lower semicontinuous (l.s.c.)	semicontinua inferiormente (s.c.i.)
coercive function	función coerciva
inner product	producto interno
norm	norma
unit ball	bola unitaria

Convex sets and cones (chs. 2–4)

English	Spanish
convex set	conjunto convexo
convex combination	combinación convexa
convex hull	cápsula (envolvente) convexa
affine hull	cápsula afín
relative interior	interior relativo
cone	cono
conic hull	cápsula cónica
polar / dual cone	cono polar / dual
normal cone	cono normal
tangent cone	cono tangente
recession cone	cono de recesión
halfspace	semiespacio
hyperplane	hiperplano
supporting hyperplane	hiperplano soporte (de apoyo)
separating hyperplane	hiperplano separador
strict / proper separation	separación estricta / propia
polyhedron	poliedro
polytope	politopo
extreme point	punto extremo
vertex	vértice
face	cara
simplex	símpex (simplejo)
Minkowski sum	suma de Minkowski

Concave, convex and generalized functions (chs. 5–7)

English	Spanish
concave / convex function	función cóncava / convexa
strictly convex	estrictamente convexa
strongly convex	fuertemente convexa
quasiconcave / quasiconvex	cuasicóncava / cuasiconvexa
pseudoconcave	pseudocóncava
concavifiable	concavificable
epigraph	epígrafo
hypograph	hipógrafo
effective domain	dominio efectivo
proper function	función propia
closed function	función cerrada
level set	conjunto de nivel
upper contour (level) set	conjunto de contorno superior
sublevel set	conjunto de subnivel
indicator function	función indicatriz
support function	función soporte
gradient	gradiente
Hessian (matrix)	Hessiana
bordered Hessian	Hessiana orlada
positive / negative (semi)definite	(semi)definida positiva / negativa
Jensen's inequality	desigualdad de Jensen
subgradient	subgradiente
subdifferential	subdiferencial
directional derivative	derivada direccional
Lipschitz continuous	Lipschitz-continua
smooth (L -smooth)	suave (L -suave)
increasing monotone transformation	transformación monótona creciente

Optimization and KKT (chs. 8–9)

English	Spanish
objective function	función objetivo
feasible set	conjunto factible
feasible point	punto factible
equality / inequality constraint	restricción de igualdad / desigualdad
active (binding) constraint	restricción activa (que satura, liga)
inactive (slack) constraint	restricción inactiva (holgada)
constraint qualification (CQ)	cualificación de restricciones
first order condition (FOC)	condición de primer orden (CPO)
second-order condition (SOC)	condición de segundo orden
stationary point	punto estacionario
critical point	punto crítico
local / global minimum	mínimo local / global
maximizer / minimizer	maximizador / minimizador
interior / corner solution	solución interior / de esquina
Lagrange multiplier	multiplicador de Lagrange
Lagrangian	lagrangiano
KKT conditions	condiciones KKT
complementary slackness	holgura complementaria
(orthogonal) projection	proyección (ortogonal)
variational inequality	desigualdad variacional
unconstrained problem	problema sin restricciones

Duality, envelope and Fenchel (chs. 10–12)

English	Spanish
primal / dual problem	problema primal / dual
(Lagrangian) dual function	función dual (lagrangiana)
weak / strong duality	dualidad débil / fuerte
duality gap	brecha de dualidad
Slater's condition	condición de Slater
saddle point	punto silla
shadow price	precio sombra
value function	función valor
envelope theorem	teorema de la envolvente
comparative statics	estática comparativa
(Fenchel) conjugate function	función conjugada (de Fenchel)
biconjugate	biconjugada
Fenchel–Young inequality	desigualdad de Fenchel–Young
closed convex envelope (hull)	envolvente convexa cerrada
perturbation	perturbación
Lagrangian relaxation	relajación lagrangiana

Linear programming and algorithms (chs. 13–16)

English	Spanish
linear program (LP)	programa lineal
standard form	forma estándar
simplex method	método símplex
basic feasible solution	solución básica factible
degenerate	degenerada
sensitivity analysis	análisis de sensibilidad
unbounded ray	rayo de no acotación
support vector machine (SVM)	máquina de vectores soporte
(hard / soft) margin	margen (duro / blando)
hinge loss	pérdida bisagra
gradient descent	descenso de gradiente
step size	tamaño (longitud) de paso
line search	búsqueda lineal
convergence rate	tasa de convergencia
projected gradient	gradiente proyectado
proximal operator	operador proximal
penalty method	método de penalización
augmented Lagrangian	lagrangiano aumentado
method of multipliers	método de multiplicadores
stopping criterion	condición de parada

Economic terms (for [9], [10])

English	Spanish
budget set	conjunto presupuestario
utility function	función de utilidad
indirect utility	utilidad indirecta
expenditure function	función de gasto
cost function	función de costos
profit function	función de beneficios
production set	conjunto de producción
demand correspondence	correspondencia de demanda
Walrasian equilibrium	equilibrio walrasiano
Pareto optimal (efficient)	óptimo de Pareto (eficiente)
welfare theorems	teoremas del bienestar
core	núcleo (de la economía)
Edgeworth box	caja de Edgeworth
risk aversion	aversión al riesgo
expected utility	utilidad esperada
moral hazard	riesgo moral
adverse selection	selección adversa
signaling	señalización
mechanism design	diseño de mecanismos
incentive compatibility (IC)	compatibilidad de incentivos (CI)
participation (IR) constraint	restricción de participación
revelation principle	principio de revelación
(first- / second-price) auction	subasta (de primer / segundo precio)
revenue equivalence	equivalencia de ingresos
price discrimination	discriminación de precios
Shephard's lemma / Roy's identity	lema de Shephard / identidad de Roy

Chapter summary

Closing of the notebook. If only five ideas were to remain from all this: (1) the whole of convex analysis hangs from the separation theorem; (2) a theorem cited without its hypotheses is a false statement; (3) KKT is two theorems (necessity with CQ, sufficiency with convexity) and complementary slackness forces you to discuss cases; (4) quasiconcavity, not concavity, is the ordinal property of consumer theory, and it is checked with the bordered Hessian; (5) multipliers, shadow prices and supporting hyperplanes are the same thing seen from three different chapters, and that identity is the heart of the microeconomics that comes next. Good luck in the midterm, in the assignments and in the presentation.

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